SIGNAL PROCESSING OPTIMIZATION LEARNING

Mathematical Foundations of Data Sciences

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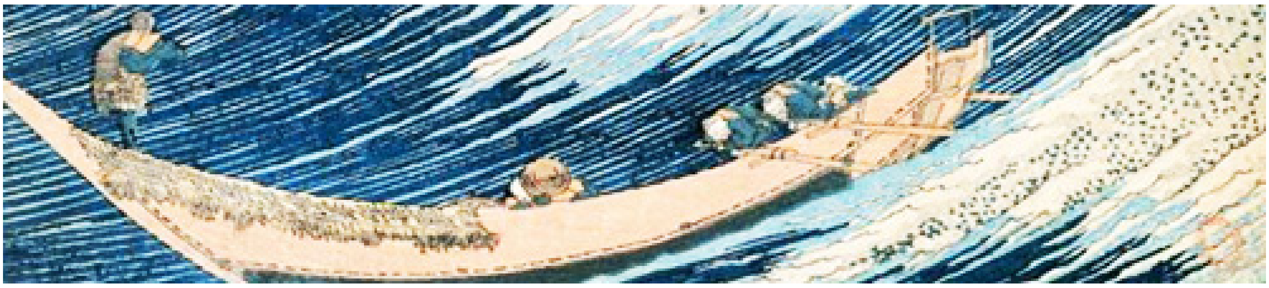


https://mathematical-tours.github.io
https://www.numerical-tours.com

Preface

This book presents the mathematical and numerical foundations of modern data science. It covers signal and image processing (Fourier analysis, wavelets, denoising, and compression), imaging science (inverse problems, sparsity, and compressed sensing), and machine learning (regression, classification, and deep learning). The chapters develop the underlying tools—linear operators, nonlinear approximation, probability, and convex optimization—and show how these tools lead to efficient algorithms. The book also serves as a theoretical companion to the Numerical Tours[1], which provide practical implementations in MATLAB, Python, Julia, and R.

[1] https://www.numerical-tours.com

Contents

CHAPTER 1

Shannon Sampling Theory

Digital measurements retain only samples of a continuous signal, so reconstruction depends on what information sampling preserves. Understanding these limits guides the choice of sampling rate and the filtering applied before acquisition. Starting from continuous and discrete signal models, we use Fourier analysis to prove a sampling theorem, explain aliasing, and distinguish exact interpolation from the additional error introduced by quantization.

Shannon's foundational work of 1948–1949 addresses three closely related topics:

1. Sampling determines when a continuous signal can be recovered from discrete measurements. Quantization then replaces the measured values by symbols from a finite alphabet.
2. Source coding uses symbol probabilities to represent a sequence of symbols compactly as a binary string.
3. Channel coding adds redundancy to protect the coded sequence against transmission errors, such as flipped bits. It is also known as error-correcting coding.

Chapter 3 develops source coding theory. We do not cover channel coding. The main reference for this chapter is [23].

1.1 Continuous and Discrete Signals

To analyze signal-processing methods, we often begin with a continuous model of the physical input to a sensor. Examples include microphones, digital cameras, and medical imaging devices. For example, a one-dimensional signal can be modeled by $f_0 \in L^2([0,1])$, where $[0,1]$ is the acquisition interval, often representing time. A grayscale image can be modeled by $f_0 \in L^2([0,1]^2)$ on the unit square.

Although sound and natural images provide many of our examples, most of the methods extend to multidimensional data. After normalizing the signal values, these data can be represented by mappings

$$f_0 : [0,1]^d \to [0,1]^s$$

Here d is the dimension of the acquisition domain and s is the number of channels. Grayscale images have $(d,s) = (2,1)$, grayscale videos have $(d,s) = (3,1)$, and RGB color images have $(d,s) = (2,3)$. Multispectral images have $d = 2$ and many channels, each recording a different wavelength range. Figures 1.1 and 1.2 illustrate these types of data.

1.1.1 Acquisition and Sampling

Signal acquisition maps a continuous signal to a finite-dimensional vector of measurements. A microphone records samples along a time axis, whereas a digital camera records samples on a two-dimensional pixel grid. Abstractly, an acquisition operator maps

$$f_0 \in L^2([0,1]^d) \mapsto f \in \mathbb{C}^N$$

Figure 1.3 shows examples of discretized signals.

1.1.2 Linear Translation-Invariant Sampling

To model translation-invariant sampling, extend the signal outside its acquisition domain according to the chosen boundary convention, convolve it with a fixed impulse response h, and sample the result:

$$f_n = \int_{\mathbb{R}} f_0(x)h(n/N - x)\,\mathrm{d}x = (f_0 \star h)(n/N). \tag{1.1}$$

The impulse response $h(x)$ depends on the device. It is typically a smooth low-pass kernel concentrated near $x = 0$. Its effective width S controls spatial resolution: a wide kernel blurs the signal, whereas a narrow

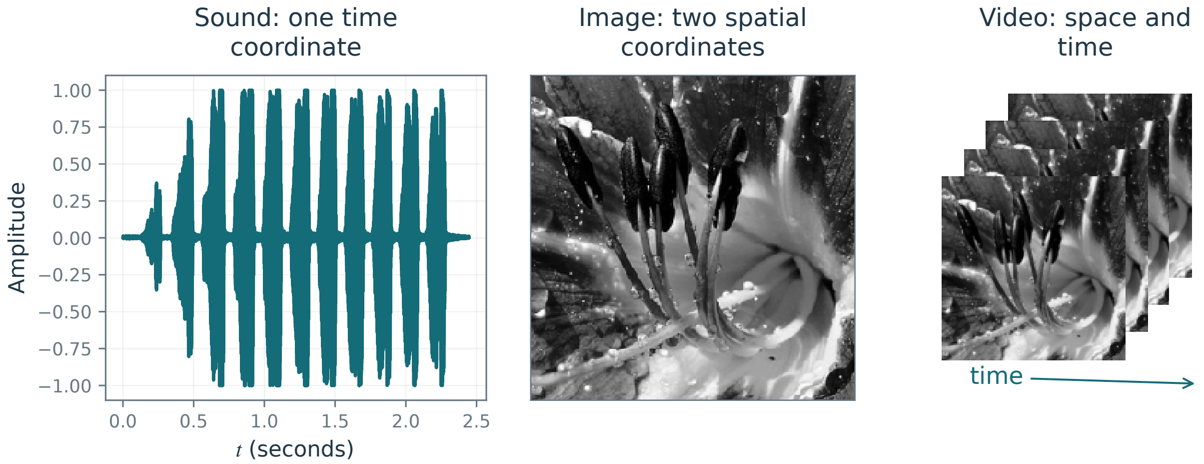


Figure 1.1. Examples of sound ($d = 1$), an image ($d = 2$), and a video ($d = 3$).

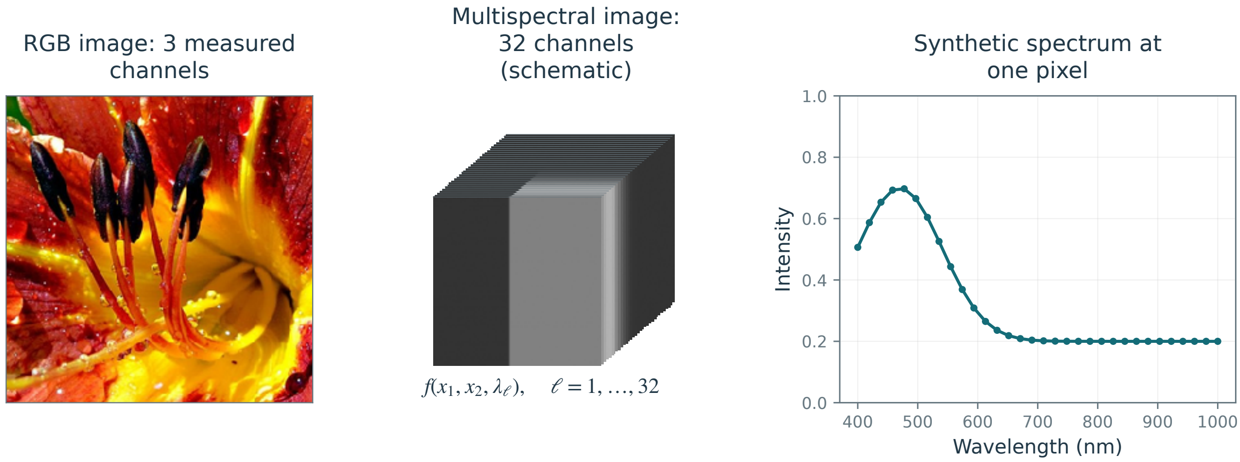


Figure 1.2. A color flower image and a synthetic multispectral image with 32 channels, with an example spectrum.

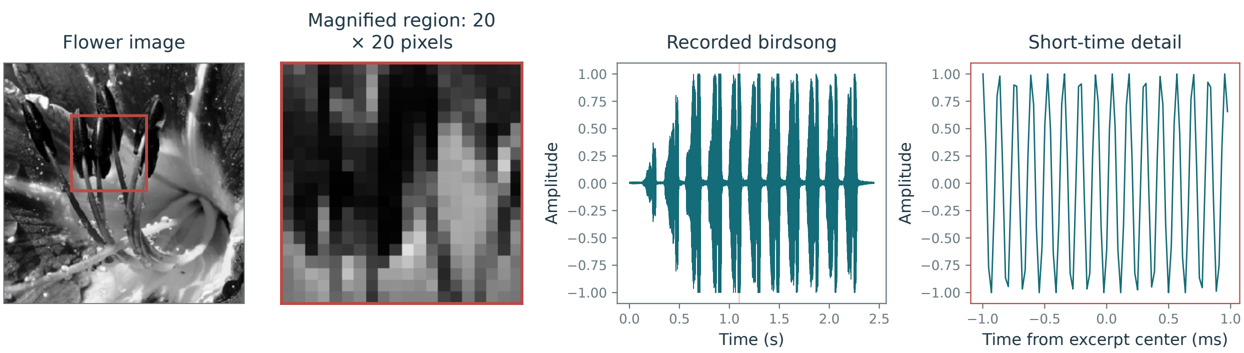


Figure 1.3. Spatial and temporal detail: a flower image with a pixelated crop, and a recorded birdsong waveform with a short-time enlargement. Matching red boxes identify the spatial crop.

kernel may fail to suppress frequencies that cause aliasing. A width of order $1/N$ balances these effects at the sampling scale.

Section 1.2 explains when a bandlimited signal can be reconstructed from its samples. Smoothness alone does not guarantee exact recovery.

1.2 Shannon Sampling Theorem

The Fourier transform. For $f \in L^1(\mathbb{R})$, its Fourier transform is defined as

$$\forall\, \omega \in \mathbb{R}, \quad \hat{f}(\omega) \overset{\text{def.}}{=} \int_{\mathbb{R}} f(x) e^{-\mathrm{i}x\omega} \mathrm{d}x. \tag{1.2}$$

For $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, Plancherel's identity gives $\|\hat{f}\|_2^2 = 2\pi \|f\|_2^2$. The map $f \mapsto \hat{f}$ therefore extends continuously to $L^2(\mathbb{R})$. This extension is the L^2 limit, as $T \to +\infty$, of the truncated integrals $\int_{-T}^{T} f(x) e^{-\mathrm{i}x\omega} \mathrm{d}x$. When $\hat{f} \in L^1(\mathbb{R})$, Fourier inversion gives

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\omega) e^{\mathrm{i}x\omega} \mathrm{d}\omega, \tag{1.3}$$

where the equality holds almost everywhere. The right-hand side defines the continuous representative of f, which vanishes at $\pm\infty$. We use this representative whenever pointwise values are needed.

The Fourier transform $\mathcal{F} : f \mapsto \hat{f}$ exchanges regularity and decay. For an integer $p \geqslant 0$, if $f \in W^{p,1}(\mathbb{R})$, then $\mathcal{F}(f^{(p)})(\omega) = (\mathrm{i}\omega)^p \hat{f}(\omega)$ so that $|\hat{f}(\omega)| = O(1/|\omega|^p)$. Conversely,

$$\int_{\mathbb{R}} (1 + |\omega|)^p |\hat{f}(\omega)| \mathrm{d}\omega < +\infty \quad \Longrightarrow \quad f \in C^p(\mathbb{R}). \tag{1.4}$$

For example, the decay estimate $\hat{f}(\omega) = O(1/|\omega|^{p+1+\varepsilon})$ for $\varepsilon > 0$ implies $f \in C^p(\mathbb{R})$.

A related measure of smoothness is the squared Sobolev norm $\|f\|_{H^1}^2 = (2\pi)^{-1} \int_{\mathbb{R}} (1 + |\omega|^2) |\hat{f}(\omega)|^2 \,\mathrm{d}\omega = \|f\|_2^2 + \|f'\|_2^2$, where f' is the weak derivative. Sobolev spaces provide a natural setting for weak solutions of PDEs. Later chapters use Sobolev regularity to control approximation and denoising errors.

Fourier series. Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ denote the circle of length 2π. A function $f \in L^2(\mathbb{T})$ is identified with a 2π-periodic function on $\mathbb{R}$. Its Fourier coefficients are

$$\forall\, k \in \mathbb{Z}, \quad \hat{f}_k \overset{\text{def.}}{=} \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-\mathrm{i}xk} \mathrm{d}x.$$

Equivalently, $\hat{f}_k = \langle f,\, e_k \rangle$, where $\langle f,\, g \rangle \overset{\text{def.}}{=} \frac{1}{2\pi} \int_{\mathbb{T}} f(x) \bar{g}(x) \mathrm{d}x$ and $e_k(x) \overset{\text{def.}}{=} e^{\mathrm{i}xk}$. The functions $(e_k)_{k \in \mathbb{Z}}$ form an orthonormal basis for this inner product, giving the expansion

$$f = \sum_{k \in \mathbb{Z}} \langle f,\, e_k \rangle e_k \tag{1.5}$$

which means $\|f - \sum_{k=-N}^{N} \langle f,\, e_k \rangle e_k\|_{L^2(\mathbb{T})} \to 0$ for $N \to +\infty$. For a differentiable function, the series (1.5) converges pointwise at every $x \in \mathbb{T}$. If f is of class C^2 on $\mathbb{T}$, then $\hat{f}_k = O(1/k^2)$, which gives normal convergence and hence uniform convergence. For a piecewise C^1 function, the Fourier series converges at a jump to the average of the one-sided limits. Gibbs oscillations prevent uniform convergence across the jump.

Poisson formula. The Poisson formula relates sampling in the signal domain to periodization in the Fourier domain. For a function $h(\omega)$ on $\mathbb{R}$, define its periodization by

$$h_P(\omega) \overset{\text{def.}}{=} \sum_n h(\omega - 2\pi n). \tag{1.6}$$

If $h \in L^1(\mathbb{R})$, the series converges absolutely almost everywhere and $\|h_P\|_{L^1(\mathbb{T})} \leqslant \|h\|_{L^1(\mathbb{R})}$, with equality for nonnegative functions. Here the norm on $\mathbb{T}$ uses unnormalized Lebesgue measure over one period. Indeed,

$$|h_P(x)| \leqslant (|h|_P)(x), \quad \Longrightarrow \quad \|h_P\|_{L^1} \leqslant \||h|_P\|_{L^1} = \|h\|_{L^1}.$$

For $h = \hat{f}$, Proposition 1.1 states that the following diagram

$$\begin{array}{cccc} \text{sampling} & f(x) & \xrightarrow{\mathcal{F}} & \hat{f}(\omega) & \text{periodization} \\ & \downarrow & & \downarrow & \\ & (f(n))_n & \xrightarrow{\text{Fourier series}} & \sum_n f(n)e^{-\mathrm{i}\omega n} & \end{array}$$

is commutative. The minus sign in $\sum_n f(n)e^{-\mathrm{i}\omega n}$ is opposite to the sign in Fourier series synthesis; this accounts for the index reversal in the proof below.

Proposition 1.1 (Poisson formula). *Assume that $f \in C(\mathbb{R})$, that $\hat{f}$ has compact support, and that $|f(x)| \leqslant C(1+|x|)^{-3}$ for some C. Then*

$$\forall\, \omega \in \mathbb{R}, \quad \sum_n f(n)e^{-\mathrm{i}\omega n} = \hat{f}_P(\omega). \tag{1.7}$$

Proof. The decay of f makes $\hat{f}$ continuously differentiable. Its compact support makes the periodization locally a finite sum, so $(\hat{f})_P$ is also C^1. It therefore equals its Fourier series:

$$(\hat{f})_P(\omega) = \sum_k c_k e^{\mathrm{i}k\omega}, \tag{1.8}$$

where

$$c_k = \frac{1}{2\pi}\int_0^{2\pi} (\hat{f})_P(\omega)e^{-\mathrm{i}k\omega}\mathrm{d}\omega = \frac{1}{2\pi}\int_0^{2\pi} \sum_n \hat{f}(\omega - 2\pi n)e^{-\mathrm{i}k\omega}\mathrm{d}\omega.$$

Absolute integrability follows from

$$\int_0^{2\pi} \sum_n |\hat{f}(\omega - 2\pi n)e^{-\mathrm{i}k\omega}|\mathrm{d}\omega = \int_{\mathbb{R}} |\hat{f}|$$

which is finite because $\hat{f}$ is continuous and compactly supported. We may therefore exchange the sum and integral:

$$c_k = \sum_n \frac{1}{2\pi}\int_0^{2\pi} \hat{f}(\omega - 2\pi n)e^{-\mathrm{i}k\omega}\mathrm{d}\omega = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)e^{-\mathrm{i}k\omega}\mathrm{d}\omega = f(-k)$$

where we used the inverse Fourier transform formula (1.3), which is valid because $\hat{f} \in L^1(\mathbb{R})$. □

Shannon's sampling theorem. Shannon's sampling theorem gives a sufficient condition for reconstructing a signal from the samples $f \mapsto (f(ns))_n$ at spacing $s > 0$. The bandlimiting condition is $\mathrm{supp}(\hat{f}) \subset [-\pi/s, \pi/s]$. By Fourier inversion (1.3), such a signal is C^∞ and has an entire extension to the complex plane. Shannon's 1949 paper placed the sampling theorem at the heart of communication theory, while acknowledging earlier interpolation results and Nyquist's work on transmission rates. Figure 1.4 illustrates the reconstruction argument and the aliasing that occurs when spectral replicas overlap; see also Figure 1.7.

Theorem 1.2. *If $f \in C(\mathbb{R})$, $|f(x)| \leqslant C(1+|x|)^{-3}$ for some C and $\mathrm{supp}(\hat{f}) \subset [-\pi/s, \pi/s]$, then one has*

$$\forall\, x \in \mathbb{R}, \quad f(x) = \sum_n f(ns)\,\mathrm{sinc}(x/s - n) \quad \textit{where} \quad \mathrm{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \quad \mathrm{sinc}(0) = 1 \tag{1.9}$$

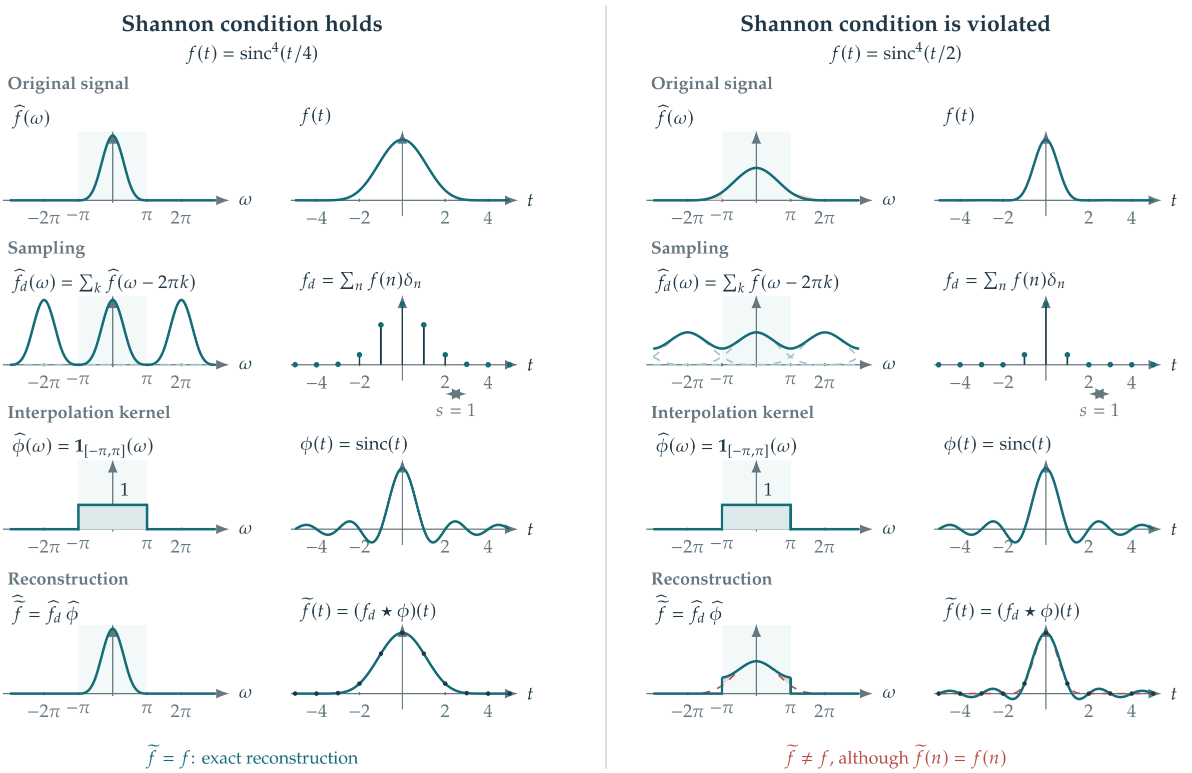
with uniform convergence.

Proof. Set $g \stackrel{\text{def.}}{=} f(s\cdot)$. Then $\hat{g} = 1/s\,\hat{f}(\cdot/s)$, as follows from the substitution $z = sx$:

$$\hat{g}(\omega) = \int f(sx)e^{-\mathrm{i}\omega x}\mathrm{d}x = \frac{1}{s}\int f(z)e^{-\mathrm{i}(\omega/s)z}\mathrm{d}z = \hat{f}(\omega/s)/s,$$

It is therefore enough to prove the result for unit sample spacing:

$$g(x) = \sum_n g(n)\,\mathrm{sinc}(x - n),$$



$$\mathrm{sinc}(t) = \frac{\sin(\pi t)}{\pi t}, \qquad \widetilde{f}(t) = \sum_{n \in \mathbb{Z}} f(n)\,\mathrm{sinc}(t - n)$$

Pale band: $[-\pi, \pi]$. Dashed spectral curves: replicas. Red shading: aliased mass.
Dashed red in the reconstruction row: the original signal and spectrum. Dots: the shared samples.

Figure 1.4. Sampling and Shannon interpolation in the frequency and time domains, with unit sample spacing. Left: the bandlimiting condition of Theorem 1.2 holds. Right: overlapping spectral replicas cause aliasing; the reconstructed signal still passes through every sample. The rows show the original signal, sampling, the interpolation kernel, and reconstruction.

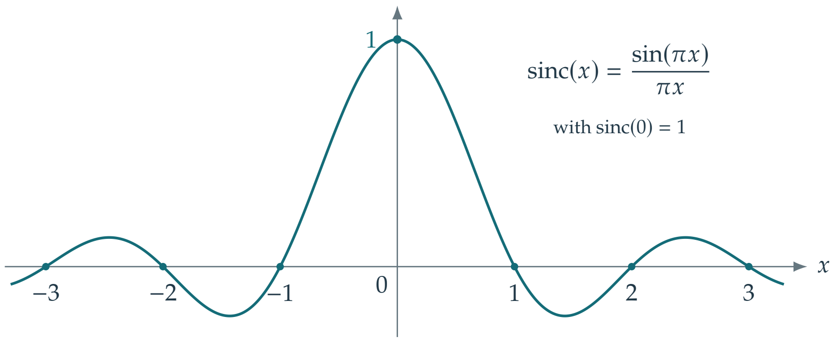
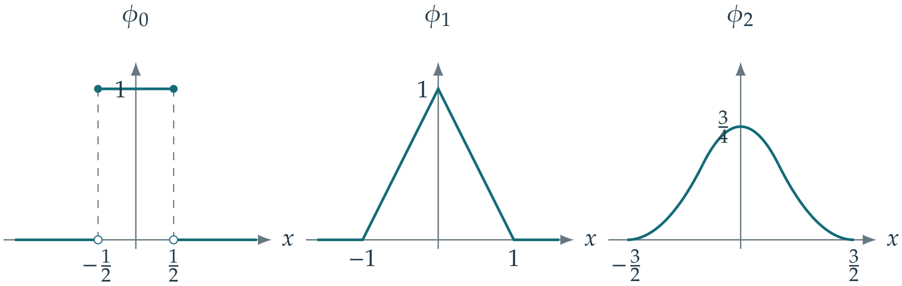


Figure 1.5. The sinc interpolation kernel.



$$\phi_0 = \mathbf{1}_{[-1/2,1/2]}, \qquad \phi_k = \phi_{k-1} * \phi_0$$

Figure 1.6. Cardinal splines as basis functions for interpolation.

For the rest of the proof, write f in place of g. The compact support hypothesis implies $\hat{f}(\omega) = 1_{[-\pi,\pi]}(\omega)\hat{f}_P(\omega)$. Combining the inversion formula (1.3) with the Poisson formula (1.7) gives

$$f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \hat{f}_P(\omega)e^{\mathrm{i}\omega x}\mathrm{d}\omega = \frac{1}{2\pi}\int_{-\pi}^{\pi}\sum_n f(n)e^{\mathrm{i}\omega(x-n)}\mathrm{d}\omega.$$

The decay of f gives $\int_{-\pi}^{\pi}\sum_n |f(n)e^{\mathrm{i}\omega(x-n)}|\mathrm{d}\omega = 2\pi\sum_n |f(n)| < +\infty$. We may therefore exchange summation and integration:

$$f(x) = \sum_n f(n)\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\mathrm{i}\omega(x-n)}\mathrm{d}\omega = \sum_n f(n)\operatorname{sinc}(x-n).$$

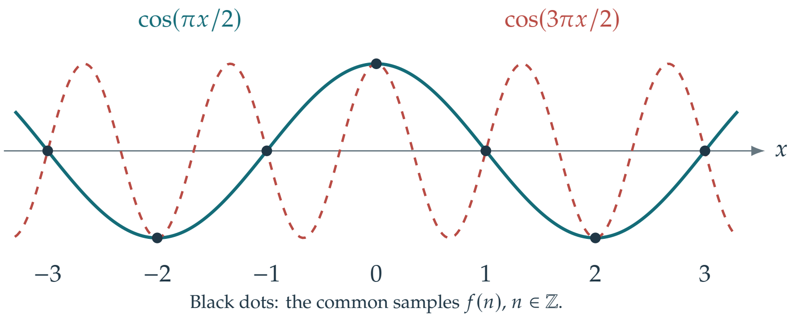
The series converges uniformly by the Weierstrass test, since $|\operatorname{sinc}(x-n)| \leqslant 1$ and $\sum_n |f(n)| < \infty$. □

The sinc kernel decays slowly and oscillates, making it inconvenient for local interpolation. Oversampling gives more flexibility: if $\operatorname{supp}(\hat{f}) \subset [-\pi/s', \pi/s']$ with $s' > s$, the spectrum occupies a strict subinterval of the Nyquist interval. One can then choose a reconstruction kernel with a smooth, compactly supported Fourier transform. The resulting kernel decays faster than every inverse power of time. Smooth compact spectral support does not, in general, imply exponential decay.

For local interpolation, cardinal B-splines offer another useful family of kernels. Define $\varphi_0 = 1_{[-1/2,1/2]}$ and $\varphi_k = \varphi_{k-1} \star \varphi_0$. The function φ_k is piecewise polynomial of degree k, with support $[-(k+1)/2, (k+1)/2]$. For $k \geqslant 1$, it belongs to C^{k-1} and has a bounded weak derivative of order k. The reconstruction takes the form $f \approx \tilde{f} \stackrel{\text{def.}}{=} \sum_n a_n\varphi(\cdot - n)$, with $\varphi = \varphi_k$. The interpolation conditions $\tilde{f}(n) = f(n)$ define a linear system for the coefficients $(a_n)_n$ in terms of the samples $(f(n))_n$. For general sample data, the identity $a_n = f(n)$ holds only for $k \in \{0, 1\}$, corresponding to piecewise constant and piecewise affine interpolation. A common choice is cubic spline interpolation, which corresponds to $k = 3$.

`Associated code: test_sampling.m`

When the spectral replicas overlap, sampling creates aliasing: high-frequency components become indistinguishable from lower-frequency components. A low-pass filter applied before sampling attenuates



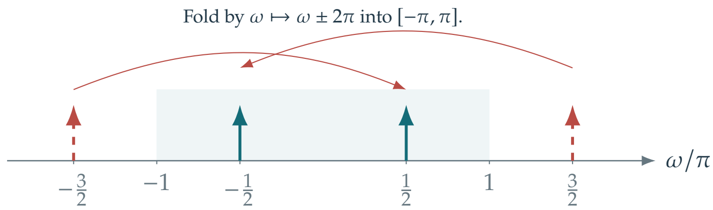


Figure 1.7. Aliasing of a sine wave. This periodic example is interpreted through its discrete spectral lines rather than as an $L^2(\mathbb{R})$ signal.

frequencies beyond the Nyquist interval $[-\pi/s, \pi/s]$. This reduces aliasing at the cost of losing high-frequency information; see Figure 1.7.

Quantization. To store or transmit a sampled signal digitally, one quantizes its values to finite precision. Section 6.1 develops transform coding, in which an invertible linear transform, usually orthogonal, is applied before quantization. Transforming the data can make the coefficients easier to compress. Directly quantizing the sampled values is a simpler alternative.

For example, a step size $s = 1/N$ gives the finite vector $(u_n \stackrel{\text{def.}}{=} f(n/N))_{n=1}^N \in \mathbb{R}^N$. This records only finitely many samples, which do not determine a general bandlimited signal. Exact recovery from these data requires additional assumptions, such as a specified finite-dimensional signal model.

For a quantization step $T > 0$, the quantizer $v_n = Q_T(u_n) \in \mathbb{Z}$ records the index of the nearest multiple of T, i.e.

$$v = Q_T(u) \quad \Leftrightarrow \quad v - \frac{1}{2} \leqslant u/T < v + \frac{1}{2},$$

as illustrated by the quantizer diagram. Dequantization replaces each index by the cell midpoint $D_T(v) \stackrel{\text{def.}}{=} Tv$, which minimizes the worst-case error within that cell. The resulting error is bounded by $T/2$: $|D_T(Q_T(u)) - u| \leqslant T/2$. Quantization introduces the loss relative to the sampled vector. Source coding and decoding then preserve the quantized symbols exactly.

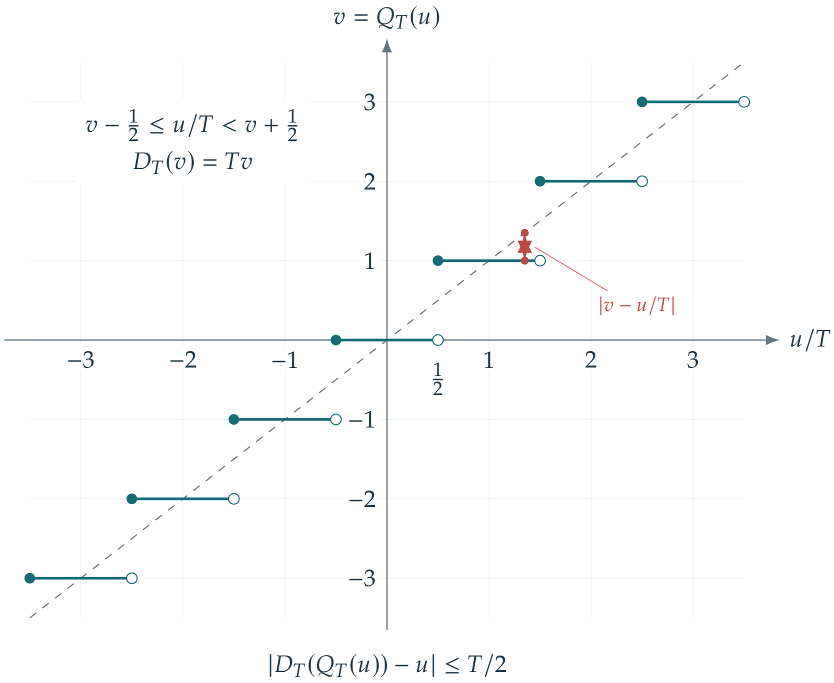


Figure 1.8. A uniform scalar quantizer.

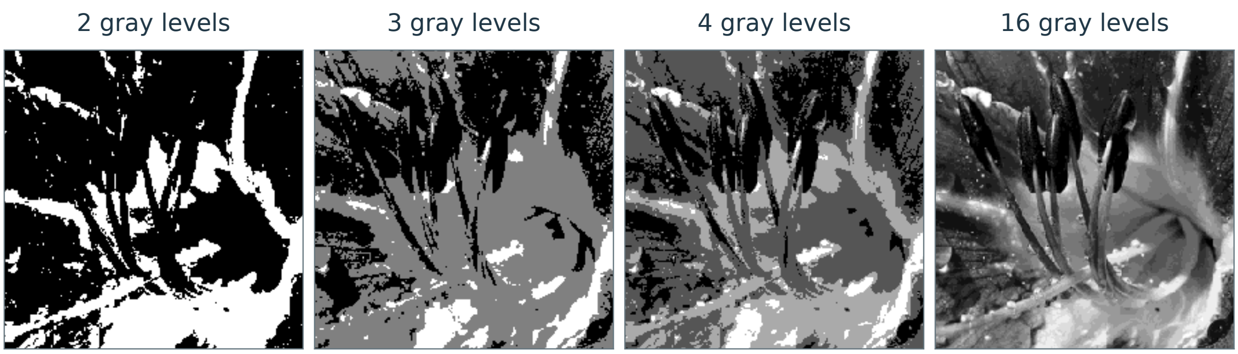


Figure 1.9. Quantizing an image with $K \in \{2, 3, 4, 16\}$ gray levels. The normalized intensities lie in $[0, 1)$; finite-level quantizers also specify clipping or endpoint conventions.

CHAPTER 2

Fourier and Convolution

Convolution combines shifted copies of a signal, but evaluating it directly can be costly and obscure the structure of the operation. Fourier coordinates turn convolution into multiplication, providing both efficient algorithms and a way to solve differential equations. We develop this connection on continuous and finite domains, study sampling and the fast Fourier transform, and extend the viewpoint to groups, surfaces, and graphs.

The main reference for this chapter is [23].

2.1 Hilbert Spaces and Fourier Transforms

2.1.1 Orthonormal Bases

Many methods in data science begin by expanding the input signal in a basis. An orthonormal basis gives a simple reconstruction formula and preserves energy, which simplifies the analysis. We first describe these properties in a Hilbert space, such as $\mathcal{H} = L^2([0,1]^d)$ for signals on a continuous domain or $\mathcal{H} = \mathbb{R}^N$ for discrete signals.

A complex Hilbert space $(\mathcal{H}, \langle\cdot,\cdot\rangle)$ is complete for the norm induced by its Hermitian inner product. We take this inner product to be linear in the first argument and conjugate-linear in the second. A separable Hilbert space admits an orthonormal basis $(\varphi_k)_k$ (finite in finite dimension), so every $f \in \mathcal{H}$ has the expansion

$$f = \sum_k \langle f, \varphi_k \rangle \varphi_k$$

where the norm is determined by $\|f\|^2 \stackrel{\text{def.}}{=} \langle f, f \rangle$. Convergence means that $\|f - \sum_{k=0}^N \langle f, \varphi_k \rangle \varphi_k\| \to 0$ as $N \to +\infty$. Parseval's identity expresses conservation of energy:

$$\|f\|^2 = \sum_k |\langle f, \varphi_k \rangle|^2.$$

The Gram–Schmidt procedure constructs such a basis from a linearly independent family $(\bar{\varphi}_k)_k$ with dense span. Set $\varphi_0 = \bar{\varphi}_0/\|\bar{\varphi}_0\|$ and $\varphi_k = \tilde{\varphi}_k/\|\tilde{\varphi}_k\|$, where $\tilde{\varphi}_k$ is the orthogonal residual $\tilde{\varphi}_k = \bar{\varphi}_k - \sum_{i<k} a_i \varphi_i$. Orthogonality requires $a_i = \langle \bar{\varphi}_k, \varphi_i \rangle$.

On $L^2([-1,1])$ with the usual inner product, orthogonalizing the monomials gives the Legendre polynomials. Their conventional normalization differs from unit L^2 norm:

$$\varphi_0(x) = 1, \quad \varphi_1(x) = x, \quad \varphi_2(x) = \frac{1}{2}(3x^2 - 1), \quad \text{etc.}$$

On $L^2(\mathbb{R}, e^{-x^2/2}\,\mathrm{d}x)$, this construction gives the probabilists' Hermite polynomials $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = x^2 - 1$, up to normalization. Multiplying normalized polynomials by $e^{-x^2/4}$ instead gives orthonormal Hermite functions for Lebesgue measure. A degree-k orthogonal polynomial has exactly k distinct real zeros in the interior of the support of the measure.

The Shannon interpolation theorem gives another example: $(\mathrm{sinc}(x-k))_k$ is an orthonormal basis of the subspace $\left\{f \in L^2(\mathbb{R}) \,;\, \mathrm{supp}(\hat{f}) \subset [-\pi, \pi]\right\}$. Using the continuous representative of f, the reconstruction formula $f = \sum_k f(k)\,\mathrm{sinc}(x-k)$ identifies the coefficients as $\langle f, \mathrm{sinc}(x-k) \rangle = f(k)$. The series converges in L^2 and pointwise; the previous chapter proves the pointwise formula under additional decay assumptions.

Orthogonal bases can also arise as complete families of eigenvectors of self-adjoint operators. Below, we explain how translation-invariant operators (convolutions) characterize Fourier bases.

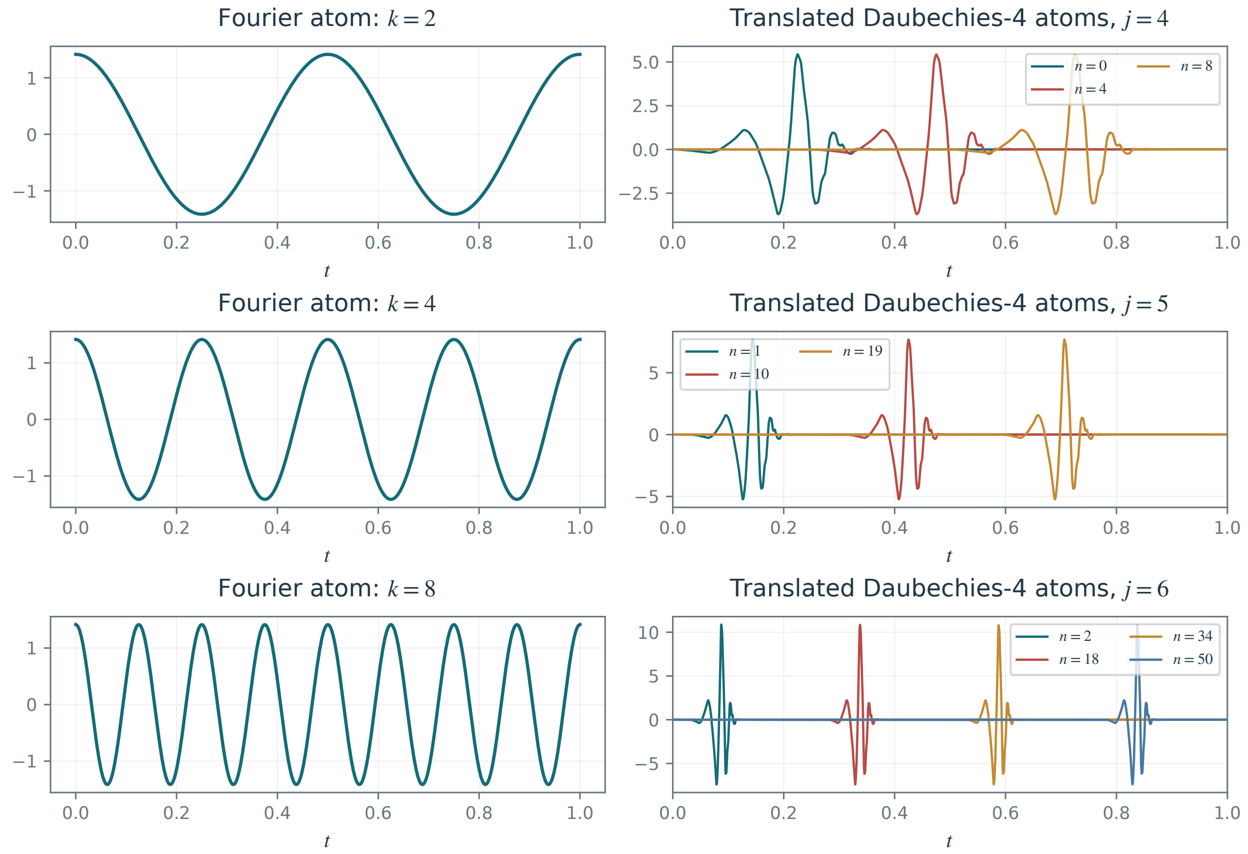


Figure 2.1. Fourier modes (left) and several translated Daubechies wavelets at each scale (right).

2.1.2 Fourier Basis on $\mathbb{R}/2\pi\mathbb{Z}$

Fourier analysis takes different forms on different domains. On the circle, it gives a countable orthonormal basis. On the real line, frequencies form a continuum, and the Fourier transform is not an expansion in a countable basis of exponential functions.

On $L^2(\mathbb{T})$ where $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, equipped with $\langle f,\, g\rangle \overset{\text{def.}}{=} \frac{1}{2\pi}\int_{\mathbb{T}} f(x)\bar{g}(x)\mathrm{d}x$, one can use the Fourier basis

$$\varphi_k(x) \overset{\text{def.}}{=} e^{\mathrm{i}kx} \quad \text{for} \quad k \in \mathbb{Z}. \tag{2.1}$$

One thus has

$$f = \sum_{k\in\mathbb{Z}} \hat{f}_k e^{\mathrm{i}k\cdot} \quad \text{where} \quad \hat{f}_k \overset{\text{def.}}{=} \frac{1}{2\pi}\int_0^{2\pi} f(x)e^{-\mathrm{i}kx}\mathrm{d}x, \tag{2.2}$$

with convergence in $L^2(\mathbb{T})$. Pointwise convergence is delicate; see Section 1.2.

Figure 2.1, left, shows examples of the real parts of Fourier atoms.

We recall that for $f \in L^1(\mathbb{R})$, its Fourier transform is defined as

$$\forall\, \omega \in \mathbb{R}, \quad \hat{f}(\omega) \overset{\text{def.}}{=} \int_{\mathbb{R}} f(x)e^{-\mathrm{i}x\omega}\mathrm{d}x.$$

The transform extends to $L^2(\mathbb{R})$ by density.

The following diagram relates the Fourier transform on $\mathbb{R}$ to Fourier coefficients on $\mathbb{T}$:

$$\text{sampling} \quad \begin{array}{ccc} f(x) & \xrightarrow{\mathcal{F}} & \hat{f}(\omega) \\ \downarrow & & \downarrow \\ (f(n))_n & \xrightarrow{\text{Fourier series}} & \sum_n f(n)e^{-\mathrm{i}\omega n} \end{array} \quad \text{periodization} \; .$$

The two routes through the diagram give the identity

$$\sum_n f(n)e^{-\mathrm{i}\omega n} = \sum_n \hat{f}(\omega - 2\pi n) \tag{2.3}$$

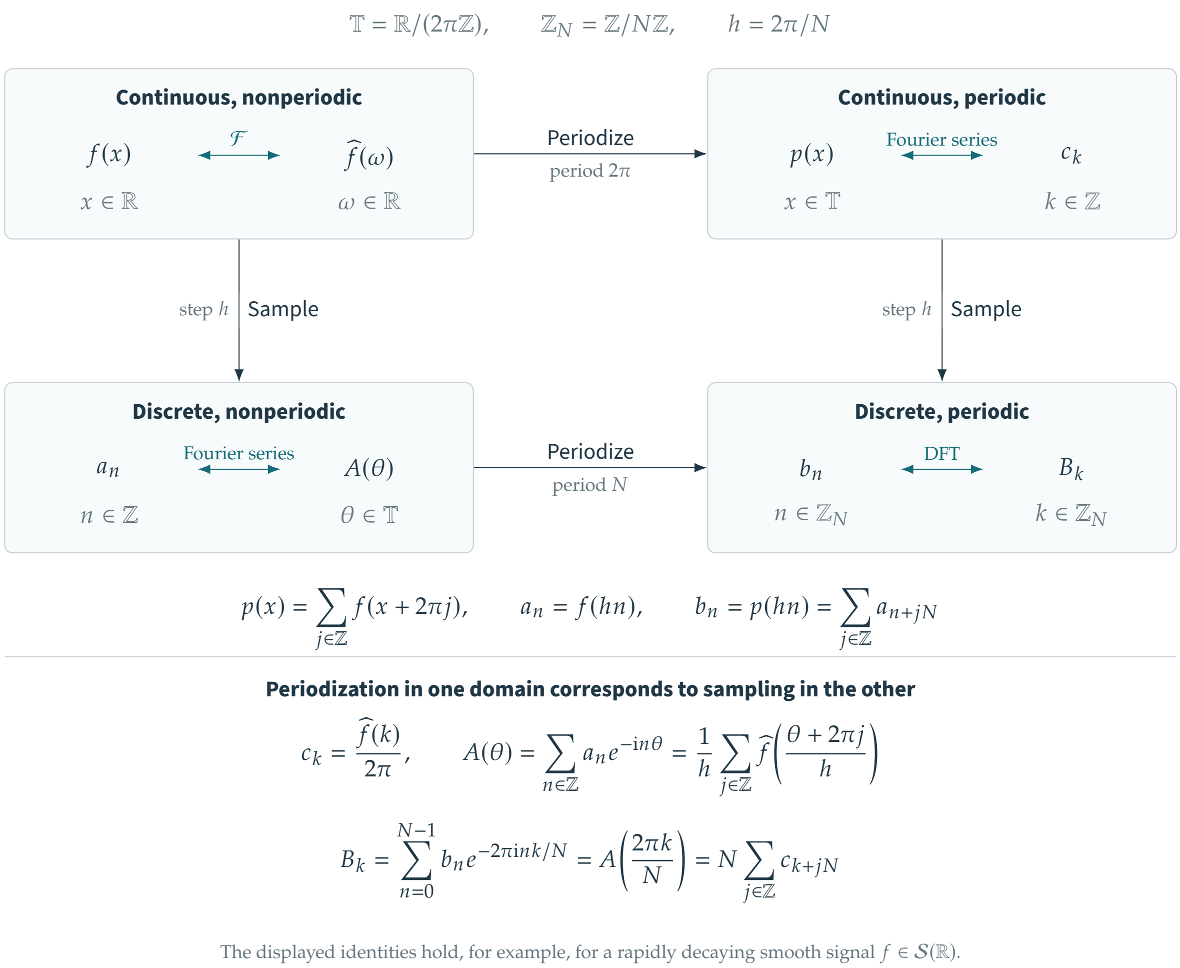


Figure 2.2. Four settings for Fourier analysis, linked by sampling and periodization.

which is Poisson's summation formula (Proposition 1.1).

2.2 Convolution on $\mathbb{R}$ and $\mathbb{T}$

2.2.1 Convolution

On $\mathbb{X} = \mathbb{R}$ or $\mathbb{T}$, one defines

$$f \star g(x) = \int_{\mathbb{X}} f(t)g(x-t)\mathrm{d}t. \tag{2.4}$$

Here and throughout this section, integrals use Lebesgue measure on $\mathbb{R}$ and normalized Haar measure $\mathrm{d}t/(2\pi)$ on $\mathbb{T}$. With this convention, Young's inequality states that, for $1 \leqslant p, q, r \leqslant \infty$ and $(f, g) \in L^p(\mathbb{X}) \times L^q(\mathbb{X})$,

$$\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r} \quad \Longrightarrow \quad f \star g \in L^r(\mathbb{X}) \quad \text{and} \quad \|f \star g\|_{L^r(\mathbb{X})} \leqslant \|f\|_{L^p(\mathbb{X})}\|g\|_{L^q(\mathbb{X})}. \tag{2.5}$$

In particular, convolution by $f \in L^1(\mathbb{X})$ defines a bounded linear map from $L^p(\mathbb{X})$ to itself. When $r = \infty$ and $1 < p, q < \infty$, the convolution $f \star g$ has a bounded continuous representative, illustrating its regularizing effect. On the circle, $p < q \implies L^q(\mathbb{T}) \subset L^p(\mathbb{T})$, so $L^\infty(\mathbb{X})$ is contained in every space in this scale.

Convolution can regularize functions. For instance, if $f \in L^1(\mathbb{X})$ and $g \in C^1(\mathbb{X})$ is bounded with bounded derivative, then $f \star g$ is differentiable and $(f \star g)' = f \star g'$. To construct an approximate identity, choose a smooth compactly supported $\rho \geqslant 0$ with $\int_{\mathbb{R}} \rho = 1$ and set $\rho_\varepsilon(x) = \varepsilon^{-1}\rho(x/\varepsilon)$. Then $f \star \rho_\varepsilon \to f$ in $L^p(\mathbb{R})$ for $1 \leqslant p < \infty$, and uniformly for bounded uniformly continuous f. On the circle, use $2\pi \sum_{k\in\mathbb{Z}} \rho_\varepsilon(\cdot - 2\pi k)$, whose integral against normalized Haar measure is one. This smoothing effect is also useful for denoising signals and images.

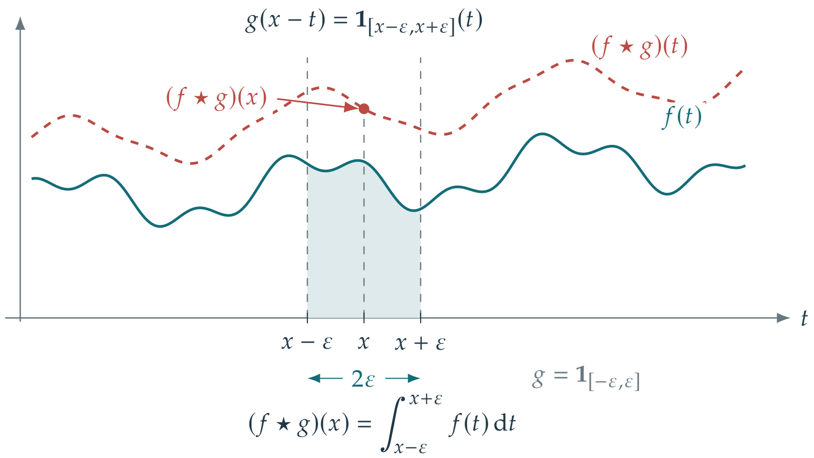


Figure 2.3. Convolution on $\mathbb{R}$. The dashed curve is $f \star g$; the marked value $(f \star g)(x)$ equals the shaded integral.

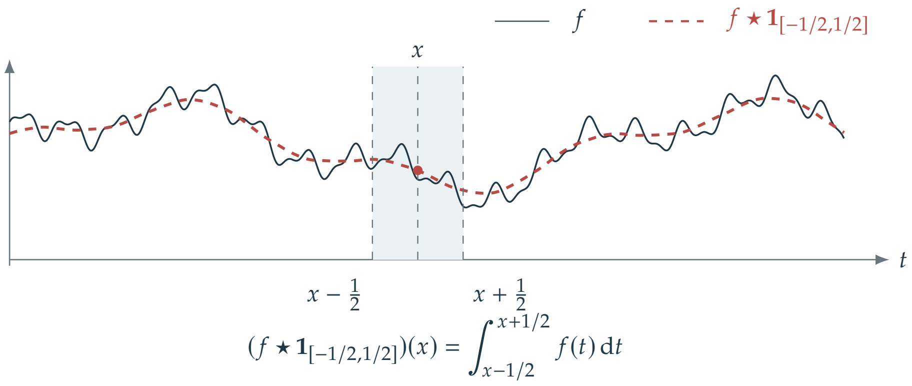


Figure 2.4. Signal filtering with a box filter (running average).

Proposition 2.1. *If $f, g \in L^1(\mathbb{X})$, then*

$$\mathcal{F}(f \star g) = \hat{f} \odot \hat{g}. \tag{2.6}$$

For $\mathbb{X} = \mathbb{R}$, $\odot$ denotes pointwise multiplication of the continuous Fourier transforms. For $\mathbb{X} = \mathbb{T}$, $\odot$ acts on bounded Fourier-coefficient sequences (pointwise multiplication of sequences).

This means that $\mathcal{F}$ is an algebra homomorphism. For instance, if $\mathbb{X} = \mathbb{R}$, its range lies in the algebra of continuous functions that vanish at $\pm\infty$.

As shown in Figure 2.7, successive convolutions of a box function produce cardinal splines: piecewise polynomials of increasing smoothness.

Convolution also describes the sum of independent random variables. Writing f_X for the probability density of a random vector X with respect to Lebesgue measure, independence of X and Y gives $f_{X+Y} = f_X \star f_Y$. This identity is fundamental in probability and statistics.

`Associated code: fourier/test_denoising.m and fourier/test_fft2.m`

2.2.2 Translation Invariant Operators

Translation-invariant operators commute with translations. In signal and image processing, they apply the same rule at every location.

The following propositions characterize translation-invariant operators as convolutions against kernels that may be distributions. The term "translation equivariant" is more precise: translating the input translates the output by the same amount. The kernel's regularity depends on the topologies of the input and output spaces, so the proof differs between the two settings below. We begin with operators whose outputs are continuous functions.

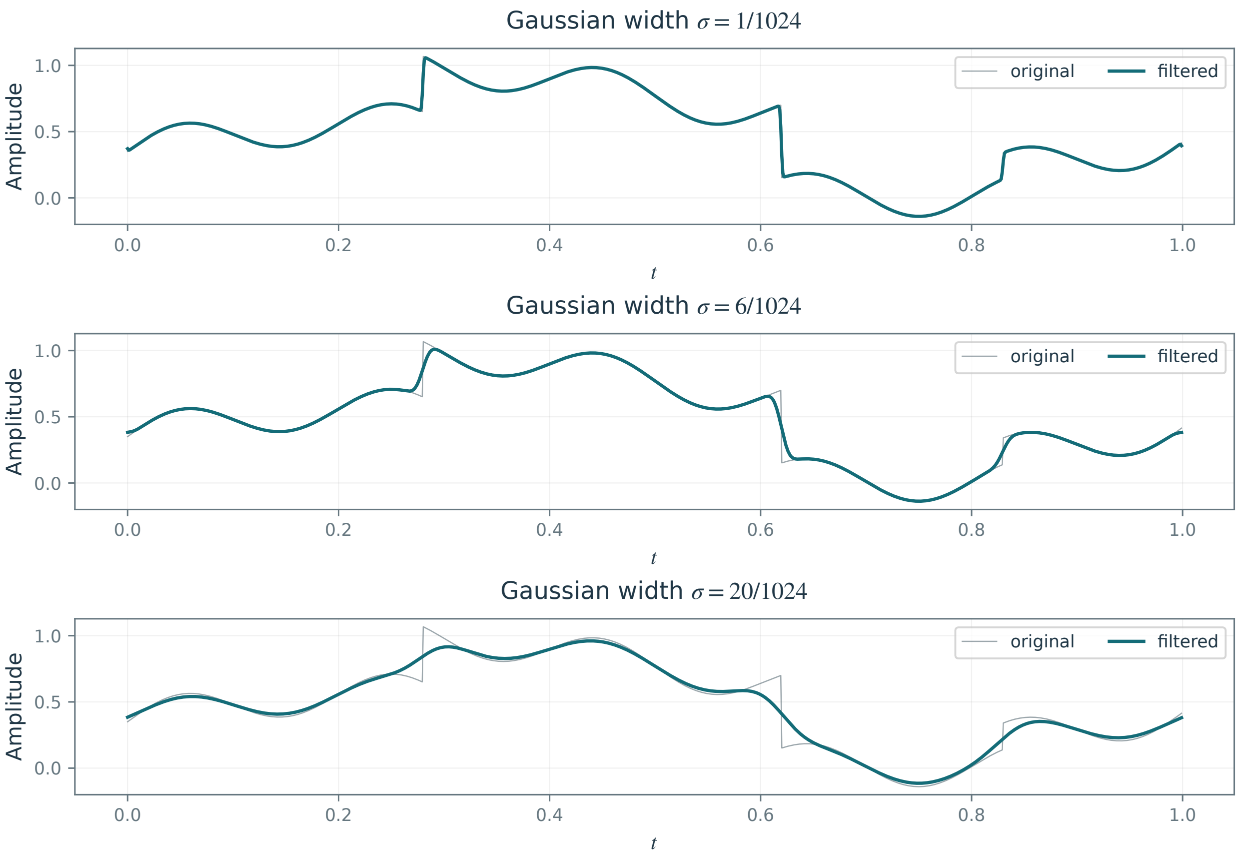
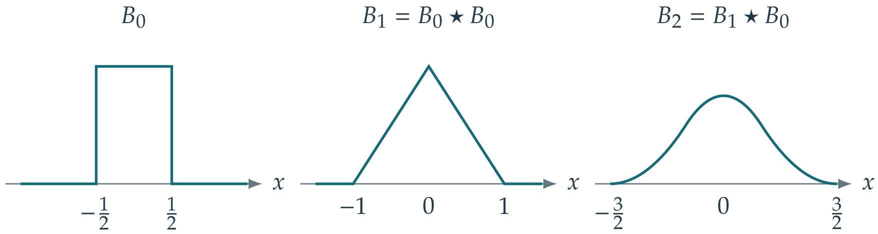


Figure 2.5. Filtering an irregular signal with Gaussian filters of increasing width σ.

$$\begin{tikzcd}
(f, g) \arrow[r, "\star"] \arrow[d, "\mathcal{F} \times \mathcal{F}"'] & f \star g \arrow[d, "\mathcal{F}"] \\
(\widehat{f}, \widehat{g}) \arrow[r, "\cdot"'] & \widehat{f}\,\widehat{g}
\end{tikzcd}$$

Figure 2.6. How the Fourier transform converts convolution into multiplication.



$$B_0 = \mathbf{1}_{[-1/2,1/2]}, \qquad B_m = B_0^{\star(m+1)}$$

Figure 2.7. Cardinal splines are defined by successive convolutions.

$$\begin{tikzcd} f \arrow[r, "H"] \arrow[d, "T_\tau"'] & Hf \arrow[d, "T_\tau"] \\ T_\tau f \arrow[r, "H"'] & T_\tau(Hf) \end{tikzcd}$$

$$T_\tau f(x) = f(x-\tau)$$

Figure 2.8. Commutative diagram for translation invariance.

Proposition 2.2. *We define $T_\tau f = f(\cdot - \tau)$. A bounded linear operator $H : (L^2(\mathbb{X}), \|\cdot\|_2) \longrightarrow (C_b(\mathbb{X}), \|\cdot\|_\infty)$ commutes with every translation, meaning that for all τ, $H \circ T_\tau = T_\tau \circ H$, if and only if*

$$\forall f \in L^2(\mathbb{X}), \quad H(f) = f \star g$$

with $g \in L^2(\mathbb{X})$.

Formally, the identity $f = f \star \delta$ suggests that $Hf = f \star H\delta$ for a translation-invariant operator. Since $H\delta$ need not be defined under the present hypotheses, the following argument uses bounded evaluation functionals instead.

Proof. By (2.5) and the continuity result for $r = \infty$, the convolution operator $H : f \mapsto f \star g$ with $g \in L^2(\mathbb{X})$ is bounded from $L^2(\mathbb{X})$ to $C_b(\mathbb{X})$. Moreover,

$$(H \circ T_\tau)(f) = \int_{\mathbb{X}} f((\cdot - \tau) - y) g(y) \mathrm{d}y = (f \star g)(\cdot - \tau) = T_\tau(Hf),$$

so that such an H is translation-invariant.

Conversely, we define $\ell : f \mapsto H(f)(0) \in \mathbb{C}$, which is well defined since $H(f) \in C^0(\mathbb{X})$. Continuity of H gives a constant C such that $\|Hf\|_\infty \leqslant C\|f\|_2$, hence $|\ell(f)| \leqslant C\|f\|_2$. Thus ℓ is a continuous linear functional on the Hilbert space $L^2(\mathbb{X})$. By the Fréchet–Riesz theorem, there exists $h \in L^2(\mathbb{X})$ such that $\ell(f) = \langle f, h\rangle$. Hence, $\forall x \in \mathbb{X}$,

$$H(f)(x) = T_{-x}(Hf)(0) = H(T_{-x}f)(0) = \ell(T_{-x}f) = \langle T_{-x}f, h\rangle = \int_{\mathbb{X}} f(y+x)\overline{h(y)}\,\mathrm{d}y = (f \star g)(x),$$

where $g(x) \stackrel{\text{def.}}{=} \overline{h(-x)} \in L^2(\mathbb{X})$. ☐

On $\mathbb{T}$, an operator need not produce continuous outputs. Its convolution kernel may then be a distribution, and Fourier coefficients provide a convenient way to define the operator.

Proposition 2.3. *A bounded linear operator $H : L^2(\mathbb{T}) \longrightarrow L^2(\mathbb{T})$ commutes with every translation, meaning that for all τ, $H \circ T_\tau = T_\tau \circ H$, if and only if*

$$\forall f \in L^2(\mathbb{T}), \quad \mathcal{F}(H(f)) = \hat{f} \odot c$$

where $c \in \ell^\infty(\mathbb{Z})$ (a bounded sequence).

Proof. We denote $\varphi_n \stackrel{\text{def.}}{=} e^{\mathrm{i}n\cdot}$. One has

$$H(\varphi_n) = H(T_\tau T_{-\tau}\varphi_n) = H(T_\tau e^{\mathrm{i}n\tau}\varphi_n) = e^{\mathrm{i}n\tau}H(T_\tau(\varphi_n)) = e^{\mathrm{i}n\tau}T_\tau(H(\varphi_n)).$$

Thus, for all n,

$$\langle H(\varphi_n), \varphi_m\rangle = e^{\mathrm{i}n\tau}\langle T_\tau \circ H(\varphi_n), \varphi_m\rangle = e^{\mathrm{i}n\tau}\langle H(\varphi_n), T_{-\tau}(\varphi_m)\rangle = e^{\mathrm{i}(n-m)\tau}\langle H(\varphi_n), \varphi_m\rangle$$

So for $n \neq m$, $\langle H(\varphi_n), \varphi_m\rangle = 0$, and we define $c_n \stackrel{\text{def.}}{=} \langle H(\varphi_n), \varphi_n\rangle$. We thus have $H(\varphi_n) = c_n\varphi_n$. Since H is continuous, $\|Hf\|_{L^2(\mathbb{T})} \leqslant C\|f\|_{L^2(\mathbb{T})}$ for some constant C, and thus by Cauchy–Schwarz

$$|c_n| = |\langle H(\varphi_n), \varphi_n\rangle| \leqslant \|H(\varphi_n)\|\|\varphi_n\| \leqslant C$$

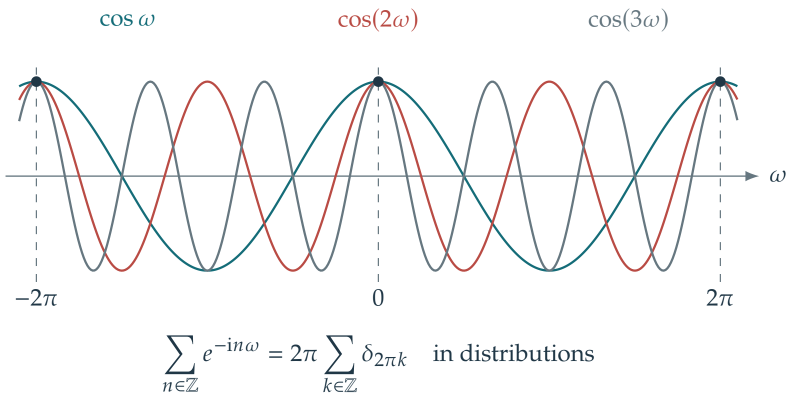


Figure 2.9. Sine waves summed in the Poisson formula.

because $\|\varphi_n\| = 1$, so that $c \in \ell^\infty(\mathbb{Z})$. By continuity of H, recalling that by definition $\hat{f}_n \overset{\text{def.}}{=} \langle f, \varphi_n \rangle$,

$$H(f) = \lim_N H\Big(\sum_{n=-N}^{N} \hat{f}_n \varphi_n\Big) = \lim_N \sum_{n=-N}^{N} \hat{f}_n H(\varphi_n) = \lim_N \sum_{n=-N}^{N} c_n \hat{f}_n \varphi_n = \sum_{n\in\mathbb{Z}} c_n \hat{f}_n \varphi_n$$

where we used $H(\varphi_n) = c_n \varphi_n$. This proves the desired representation. □

Conversely, by Parseval's identity, a bounded sequence c defines a bounded Fourier multiplier, which commutes with translations. Translation-invariant operators are therefore diagonal in the Fourier basis.

2.2.3 Poisson's Formula and Distributions

Informally, the Fourier series

$$\sum_n f(n) e^{-\mathrm{i}\omega n}$$

can be viewed as the Fourier transform $\mathcal{F}(\Pi_1 \odot f)$ of the discrete distribution

$$\Pi_1 \odot f = \sum_n f(n)\delta_n \quad \text{where} \quad \Pi_s = \sum_n \delta_{sn}$$

for $s > 0$, where δ_a is the Dirac mass at location $a \in \mathbb{R}$, i.e. the distribution such that $\int f \,\mathrm{d}\delta_a = f(a)$ for any continuous f. A measure can be multiplied by a continuous function; a general distribution can be multiplied by a smooth function. For a tempered distribution μ, its Fourier transform is defined by duality on Schwartz test functions:

$$\forall\, g \in \mathcal{S}(\mathbb{R}), \quad \int_{\mathbb{R}} g(x)\mathrm{d}\mathcal{F}(\mu) = \int_{\mathbb{R}} \mathcal{F}(g)\mathrm{d}\mu, \quad \text{where} \quad \mathcal{F}(g) \overset{\text{def.}}{=} \int_{\mathbb{R}} g(x)e^{-\mathrm{i}x\cdot}\,\mathrm{d}x,$$

Here $\mathcal{S}(\mathbb{R})$ is the Schwartz space; the integral notation denotes the distributional pairing.

The Poisson formula (2.3) can thus be interpreted as

$$\mathcal{F}(\Pi_1 \odot f) = \sum_n \hat{f}(\cdot - 2\pi n) = \int_{\mathbb{R}} \hat{f}(\cdot - \omega)\mathrm{d}\Pi_{2\pi}(\omega) = \hat{f} \star \Pi_{2\pi}$$

Since $\mathcal{F}^{-1} = \frac{1}{2\pi}\mathcal{S} \circ \mathcal{F}$ where $\mathcal{S}(f) = f(-\cdot)$, applying this operator to both sides gives

$$\Pi_1 \odot f = \frac{1}{2\pi}\mathcal{S} \circ \mathcal{F}(\hat{f} \star \Pi_{2\pi}) = \mathcal{S}\Big(\frac{1}{2\pi}\mathcal{F}(\hat{f}) \odot \hat{\Pi}_{2\pi}\Big) = \mathcal{S}\Big(\frac{1}{2\pi}\mathcal{F}(\hat{f})\Big) \odot \mathcal{S}(\hat{\Pi}_{2\pi}) = \hat{\Pi}_{2\pi} \odot f.$$

This can be interpreted as the relation

$$\hat{\Pi}_{2\pi} = \Pi_1 \quad \Longrightarrow \quad \hat{\Pi}_1 = 2\pi\Pi_{2\pi}.$$

For intuition, consider a finite Fourier series

$$\sum_{n=-N}^{N} e^{-\mathrm{i}n\omega} = \frac{\sin((N+1/2)\omega)}{\sin(\omega/2)}$$

which is the Dirichlet kernel, with removable singularities at multiples of 2π. Its value there is $2N+1$; in the sense of distributions it converges to $2\pi\Pi_{2\pi}$.

2.3 Finite Fourier Transform and Convolution

2.3.1 Discrete Orthonormal Bases

A discrete signal is a finite-dimensional vector $f \in \mathbb{C}^N$, where N is the number of samples and f_n is the value at a specified location. For a 2-D image $f \in \mathbb{C}^N \simeq \mathbb{C}^{N_0 \times N_0}$, we have $N = N_0 \times N_0$, with N_0 pixels in each direction.

The discrete inner product is the analogue of the continuous L^2 inner product:

$$\langle f, g \rangle = \sum_{n=0}^{N-1} f_n \bar{g}_n.$$

The corresponding squared Euclidean distance is

$$\|f - g\|^2 = \sum_{n=0}^{N-1} |f_n - g_n|^2.$$

Exactly as in the continuous case, a discrete orthonormal basis $\{\psi_k\}_{0\leqslant k<N}$ of $\mathbb{C}^N$ satisfies

$$\langle \psi_k, \psi_{k'} \rangle = \delta_{k-k'}. \tag{2.7}$$

A signal expands in this orthonormal basis as

$$f = \sum_{k=0}^{N-1} \langle f, \psi_k \rangle \psi_k.$$

It preserves energy

$$\|f\|^2 = \sum_{n=0}^{N-1} |f_n|^2 = \sum_{k=0}^{N-1} |\langle f, \psi_k \rangle|^2$$

Computing the set of all inner products $\{\langle f, \psi_k \rangle\}_{0\leqslant k<N}$ directly requires $O(N^2)$ operations. This cost is prohibitive when N reaches millions of samples. Structured bases permit faster decompositions: Fourier and wavelet transforms require $O(N\log(N))$ and $O(N)$ operations, respectively.

2.3.2 Discrete Fourier transform

We denote $f = (f_n)_{n=0}^{N-1} \in \mathbb{R}^N$, and regard its entries as indexed by $n \in \mathbb{Z}/N\mathbb{Z}$, which is a finite abelian group under addition. This corresponds to using periodic boundary conditions.

The discrete Fourier transform is defined as

$$\forall\, k = 0, \ldots, N-1, \quad \hat{f}_k \overset{\text{def.}}{=} \sum_{n=0}^{N-1} f_n e^{-\frac{2\mathrm{i}\pi}{N}kn} = \langle f, \varphi_k \rangle \quad \text{where} \quad \varphi_k \overset{\text{def.}}{=} (e^{\frac{2\mathrm{i}\pi}{N}kn})_{n=0}^{N-1} \in \mathbb{C}^N \tag{2.8}$$

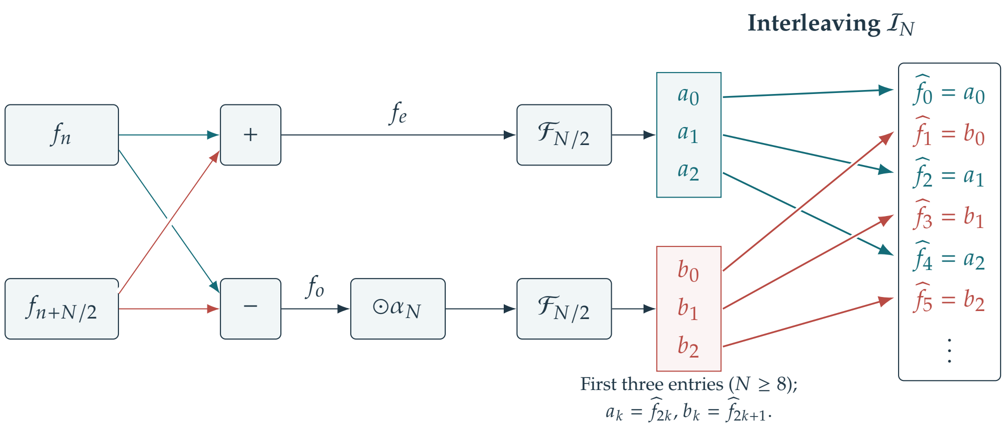
where the canonical inner product on $\mathbb{C}^N$ is $\langle u, v \rangle = \sum_{n=0}^{N-1} u_n \bar{v}_n$ for $(u, v) \in (\mathbb{C}^N)^2$. This definition is motivated by sampling the Fourier basis $x \mapsto e^{\mathrm{i}kx}$ on $\mathbb{R}/2\pi\mathbb{Z}$ at equally spaced points $(\frac{2\pi}{N}n)_{n=0}^{N-1}$. The next proposition establishes the orthogonality of the discrete Fourier atoms and the associated reconstruction formula.

Proposition 2.4. *One has*

$$\langle \varphi_k, \varphi_\ell \rangle = \begin{cases} N & \text{if} \quad k = \ell, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, this implies

$$\forall\, n = 0, \ldots, N-1, \quad f_n = \frac{1}{N} \sum_k \hat{f}_k e^{\frac{2\mathrm{i}\pi}{N}kn}. \tag{2.9}$$



$$0 \leq n < N/2, \quad f_{e,n} = f_n + f_{n+N/2}, \quad f_{o,n} = f_n - f_{n+N/2}$$

$$\alpha_{N,n} = e^{-2\pi \mathrm{i} n/N}, \qquad 0 \leq k < N/2$$

Figure 2.10. Diagram of one radix-two FFT step. Both halves of the input enter the sum and difference; the outputs of the two smaller transforms are interleaved into even and odd frequencies.

Proof. One has, if $k \neq \ell$

$$\langle \varphi_k, \varphi_\ell \rangle = \sum_n e^{\frac{2\mathrm{i}\pi}{N}(k-\ell)n} = \frac{1 - e^{\frac{2\mathrm{i}\pi}{N}(k-\ell)N}}{1 - e^{\frac{2\mathrm{i}\pi}{N}(k-\ell)}} = 0$$

which is the sum of a geometric series (equivalently, the sum of equispaced points on a circle). The inversion formula is simply $f = \sum_k \langle f, \varphi_k \rangle \frac{\varphi_k}{\|\varphi_k\|^2}$. ◻

2.3.3 Fast Fourier transform

Assuming $N = 2N'$, one has

$$\hat{f}_{2k} = \sum_{n=0}^{N'-1} (f_n + f_{n+N/2}) e^{-\frac{2\mathrm{i}\pi}{N'} kn}$$

$$\hat{f}_{2k+1} = \sum_{n=0}^{N'-1} e^{-\frac{2\mathrm{i}\pi}{N} n} (f_n - f_{n+N/2}) e^{-\frac{2\mathrm{i}\pi}{N'} kn}.$$

For the second line, we used the computation

$$e^{-\frac{2\mathrm{i}\pi}{N}(2k+1)(n+N/2)} = e^{-\frac{2\mathrm{i}\pi}{N}(2kn+kN+n+N/2)} = -e^{-\frac{2\mathrm{i}\pi}{N} n} e^{-\frac{2\mathrm{i}\pi}{N'} kn}.$$

Let $\mathcal{F}_N(f) = \hat{f}$ denote the discrete Fourier transform on $\mathbb{C}^N$. Introduce the vectors $f_e = (f_n + f_{n+N/2})_n \in \mathbb{C}^{N'}$ and $f_o = (f_n - f_{n+N/2})_n \in \mathbb{C}^{N'}$ and the twiddle factors $\alpha_N = (e^{-2\mathrm{i}\pi n/N})_{n=0}^{N'-1} \in \mathbb{C}^{N'}$. The recursion becomes

$$\mathcal{F}_N(f) = \mathcal{I}_N(\mathcal{F}_{N/2}(f_e), \mathcal{F}_{N/2}(f_o \odot \alpha_N))$$

where $\mathcal{I}_N$ is the "interleaving" operator, defined by $\mathcal{I}_N(a, b) \stackrel{\text{def.}}{=} (a_0, b_0, a_1, b_1, \ldots, a_{N'-1}, b_{N'-1})$. Recursing yields the radix-two fast Fourier transform (FFT) when N is a power of two. Other FFT factorizations handle general lengths. Zero-padding to a power of two is also useful, although it changes the transform length and the sampled frequencies.

This algorithm can also be interpreted as a factorization of the Fourier matrix into a product of $O(\log N)$ sparse matrices.

Let $C(N)$ denote the number of elementary operations required to compute $\hat{f}$. The recursion gives

$$C(N) = 2C(N/2) + NK \tag{2.10}$$

where KN accounts for N complex additions and $N/2$ multiplications. With the change of variable

$$\ell \stackrel{\text{def.}}{=} \log_2(N) \quad \text{and} \quad T(\ell) \stackrel{\text{def.}}{=} \frac{C(N)}{N}$$

i.e. $C(N) = 2^\ell T(\ell)$, the relation (2.10) becomes

$$2^\ell T(\ell) = 2 \times 2^{\ell-1} T(\ell-1) + 2^\ell K \quad \Longrightarrow \quad T(\ell) = T(\ell-1) + K \quad \Longrightarrow \quad T(\ell) = T(0) + K\ell$$

and using the fact that $T(0) = C(1)/1 = 0$, one obtains

$$C(N) = KN \log_2(N).$$

Direct evaluation instead requires $O(N^2)$ operations to compute the N coefficients (2.8), each involving a sum of size N.

2.3.4 Finite convolution

For $(f,g) \in (\mathbb{R}^N)^2$, one defines $f \star g \in \mathbb{R}^N$ as

$$\forall\, n = 0, \ldots, N-1, \quad (f \star g)_n \stackrel{\text{def.}}{=} \sum_{k=0}^{N-1} f_k g_{n-k} = \sum_{k+\ell=n} f_k g_\ell \tag{2.11}$$

where $+$ and $-$ are interpreted modulo N (vectors are defined on the group $\mathbb{Z}/N\mathbb{Z}$, or equivalently, one uses periodic boundary conditions). This defines a commutative algebra structure $(\mathbb{R}^N, +, \star)$, with neutral element the "Dirac" $\delta_0 \stackrel{\text{def.}}{=} (1, 0, \ldots, 0)^\top \in \mathbb{R}^N$. The following proposition shows that $\mathcal{F} : f \mapsto \hat{f}$ is an algebra isomorphism and an isometry (up to a scaling by $\sqrt{N}$ of the norm) from $(\mathbb{C}^N, +, \star)$ onto $(\mathbb{C}^N, +, \odot)$ with neutral element $\mathbb{1}_N = (1, \ldots, 1) \in \mathbb{R}^N$.

Proposition 2.5. *One has* $\mathcal{F}(f \star g) = \hat{f} \odot \hat{g}$.

Proof. We denote $T : g \mapsto f \star g$. One has

$$(T\varphi_\ell)_n = (f \star \varphi_\ell)_n = \sum_p f_p e^{\frac{2i\pi}{N}\ell(n-p)} = e^{\frac{2i\pi}{N}\ell n} \hat{f}_\ell = \hat{f}_\ell (\varphi_\ell)_n.$$

Thus $(\varphi_\ell)_\ell$ form a basis of N eigenvectors of T, with eigenvalues $\hat{f}_\ell$. The operator T is diagonal in this basis. Let $F = (e^{-\frac{2i\pi}{N}kn})_{k,n}$ be the Fourier transform matrix. The inversion formula (2.9) gives $F^{-1} = \frac{1}{N} F^*$, where $F^* = \bar{F}^\top$ is the adjoint (conjugate transpose). The diagonalization of T is therefore

$$T = F^{-1} \operatorname{diag}(\hat{f}) F \quad \Longrightarrow \quad \mathcal{F}(Tg) = \operatorname{diag}(\hat{f}) F g \quad \Longrightarrow \quad \mathcal{F}(f \star g) = \operatorname{diag}(\hat{f}) \hat{g}.$$

□

This proposition gives an $O(N \log N)$ convolution algorithm:

$$f \star g = \mathcal{F}^{-1}(\hat{f} \odot \hat{g}).$$

This is more efficient than directly evaluating formula (2.11) for filters with large support. When $|\operatorname{Supp}(g)| = P$ is small, direct evaluation costs $O(PN)$ and may be faster. An example is $g = [1, 1, 0, \ldots, 0, 1]/3$, the moving average, where

$$(f \star g)_n = \frac{f_{n-1} + f_n + f_{n+1}}{3}$$

needs $3N$ operations.

Convolution as translation invariant operator. Define translation on $\mathbb{Z}/N\mathbb{Z}$ by $(T_\tau f)_n \stackrel{\text{def.}}{=} f_{n-\tau}$, with $n - \tau$ computed modulo N. The next proposition is the finite counterpart of the results on $\mathbb{R}$ and $\mathbb{R}/2\pi\mathbb{Z}$. On $\mathbb{Z}/N\mathbb{Z}$, no convergence issues arise, and the impulse response can be defined directly.

Proposition 2.6. *Let $H : \mathbb{R}^N \to \mathbb{R}^N$ be linear. Then $H \circ T_\tau = T_\tau \circ H$ for all τ if and only if there exists $h \in \mathbb{R}^N$ (often called the impulse response) so that $Hf = f \star h$.*

Proof. For the convolution-to-invariance implication, we first note that $T_\tau f = f \star \delta_\tau$ is a convolution against a translated Dirac $\delta_\tau = T_\tau(\delta)$ where $\delta = (1, 0, \ldots, 0)$. If $Hf = f \star h$, commutativity gives

$$H \circ T_\tau(f) = h \star \delta_\tau \star f = \delta_\tau \star h \star f = T_\tau \circ H(f).$$

Conversely, we define $h \stackrel{\text{def.}}{=} H(\delta)$. Then

$$H(f) = H(\sum_i f_i (T_i \delta)) = \sum_i f_i H(T_i \delta) = \sum_i f_i T_i H(\delta) = \sum_i f_i h_{\cdot - i} = f \star h.$$

□

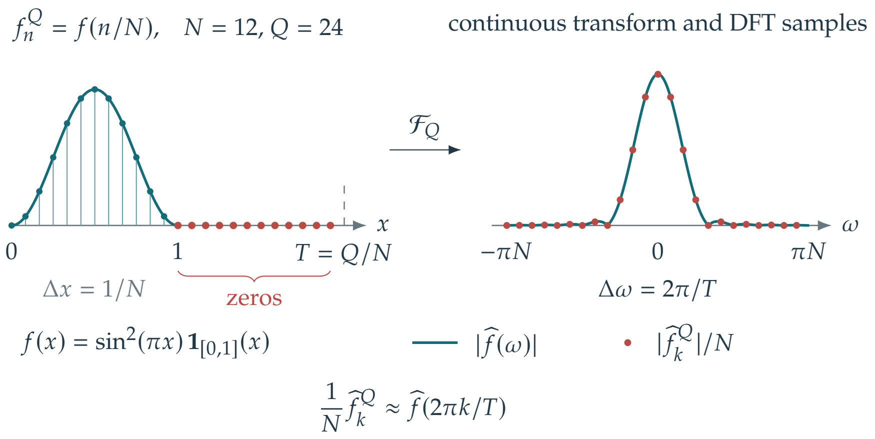


Figure 2.11. Fourier transform approximation by zero-padding in the spatial domain.

Polynomial multiplication. The FFT can multiply large polynomials. It can also multiply large integers by interpreting their digit expansions in a fixed base as polynomials. Indeed,

$$(\sum_{i=0}^{A} a_i X^i)(\sum_{j=0}^{B} b_j X^j) = \sum_{k=0}^{A+B}(\sum_{i+j=k} a_i b_j)X^k$$

Setting aside the carry operation, which can be performed in linear time, one can write $\sum_{i+j=k} a_i b_j = (\bar{a} \star \bar{b})_k$ when one defines $\bar{a}, \bar{b} \in \mathbb{R}^{A+B+1}$ by zero padding.

2.4 Discretization Issues

The FFT also approximates the continuous Fourier transform and its inverse. Reversing the roles of space and frequency leads to an efficient spectral interpolation method.

2.4.1 Fourier approximation via spatial zero padding.

The discrete Fourier transform (2.8) approximates samples of the continuous Fourier transform (1.2). For a sufficiently smooth f supported on $[0,1]$, consider the vector $f^Q \overset{\text{def.}}{=} (f(n/N))_{n=0}^{Q-1} \in \mathbb{R}^Q$. Taking $Q \geqslant N$ pads the original samples with zeros, since continuity and the support condition give $f(n/N) = 0$ for $n \geqslant N$. For signed frequency indices $-\lfloor Q/2 \rfloor \leqslant k \leqslant \lceil Q/2 \rceil - 1$ and $T = Q/N$,

$$\begin{aligned}\frac{1}{N}\hat{f}_k^Q &= \frac{1}{N}\sum_{n=0}^{N-1} f(n/N)e^{-2\pi \mathrm{i} nk/Q} \\ &\approx \int_0^1 f(x)e^{-2\pi \mathrm{i} kx/T}\,\mathrm{d}x = \hat{f}(2\pi k/T).\end{aligned}$$

For a fixed physical frequency, the Riemann-sum error is $O(1/N)$ for $f \in C^1$. The bound need not be uniform over frequencies growing with N. Increasing Q at fixed N samples the same spectrum more densely; it does not increase the Nyquist frequency.

2.4.2 Fourier interpolation via spectral zero padding.

One can reverse the roles of space and frequency in the previous construction. Given N uniform discrete samples $f^N = (f_n^N)_{n=0}^{N-1}$, one can compute their discrete Fourier transform $\mathcal{F}(f^N) = \hat{f}^N$ (in $O(N\log(N))$ operations with the FFT),

$$\hat{f}_k^N \overset{\text{def.}}{=} \sum_{n=0}^{N-1} f_n^N e^{-\frac{2\mathrm{i}\pi}{N}nk},$$

and then zero-pad the spectrum to obtain a vector of length $Q \geqslant N$. For simplicity, assume that $N = 2N' + 1$ is odd. Even lengths admit a similar, slightly more involved construction. With frequency indices $-N' \leqslant k \leqslant N'$,

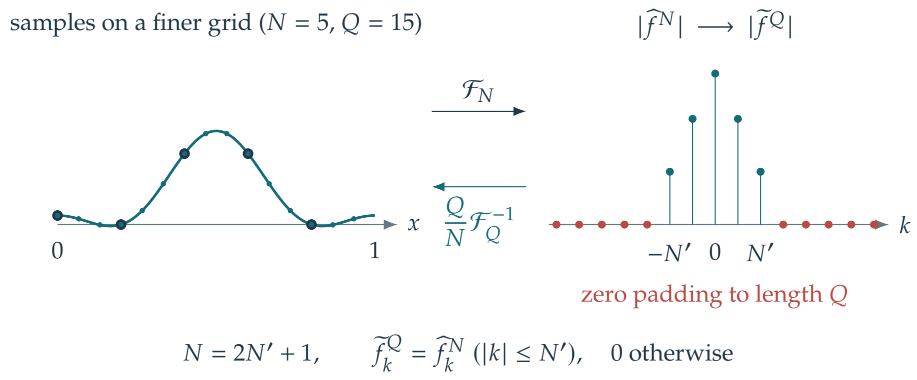


Figure 2.12. Interpolation by zero-padding in the frequency domain.

define the padded spectrum by

$$\tilde{f}_k^Q = \hat{f}_k^N \quad (-N' \leqslant k \leqslant N'), \qquad \tilde{f}_k^Q = 0 \quad \text{otherwise}, \qquad \tilde{f}^Q \in \mathbb{C}^Q$$

The signed frequency indices are interpreted modulo Q when calling an FFT routine. An inverse transform of size Q, multiplied by Q/N, then gives in $O(Q \log Q)$ operations

$$\begin{aligned}
\frac{Q}{N}\mathcal{F}^{-1}(\tilde{f}^Q)_\ell &= \frac{Q}{N} \times \frac{1}{Q} \sum_{k=-N'}^{N'} \hat{f}_k^N e^{\frac{2i\pi}{Q}\ell k} = \frac{1}{N} \sum_{k=-N'}^{N'} \sum_{n=0}^{N-1} f_n^N e^{-\frac{2i\pi}{N} nk} e^{\frac{2i\pi}{Q}\ell k} \\
&= \sum_{n=0}^{N-1} f_n^N \frac{1}{N} \sum_{k=-N'}^{N'} e^{2i\pi\left(-\frac{n}{N}+\frac{\ell}{Q}\right)k} = \sum_{n=0}^{N-1} f_n^N \frac{\sin\left[\pi N\left(\frac{\ell}{Q} - \frac{n}{N}\right)\right]}{N \sin\left[\pi\left(\frac{\ell}{Q} - \frac{n}{N}\right)\right]} \\
&= \sum_{n=0}^{N-1} f_n^N \operatorname{sinc}_N\left(\frac{\ell}{T} - n\right) \quad \text{where} \quad T \overset{\text{def.}}{=} \frac{Q}{N} \quad \text{and} \quad \operatorname{sinc}_N(u) \overset{\text{def.}}{=} \frac{\sin(\pi u)}{N \sin(\pi u/N)}.
\end{aligned}$$

We use the geometric-series identity with $\rho = e^{i\omega}$, $a = -b$, $\omega = 2\pi\left(-\frac{n}{N} + \frac{\ell}{Q}\right)$,

$$\sum_{i=a}^{b} \rho^i = \frac{\rho^{a-\frac{1}{2}} - \rho^{b+\frac{1}{2}}}{\rho^{-\frac{1}{2}} - \rho^{\frac{1}{2}}} = \frac{\sin((b + \frac{1}{2})\omega)}{\sin(\omega/2)}.$$

This gives a discrete counterpart of the Shannon interpolation formula (1.9). It evaluates the trigonometric interpolant of the samples exactly on a grid of size Q, at cost $O(Q \log(Q))$. Furthermore, $\operatorname{sinc}_N(u) \to \operatorname{sinc}(u)$ for fixed u as $N \to \infty$, with removable singularities filled by continuity.

2.5 Fourier in Multiple Dimensions

Tensor products extend the one-dimensional Fourier transform to any finite dimension $d > 1$.

2.5.1 On Continuous Domains

On $\mathbb{R}^d$. Tensor products retain a useful feature of Fourier atoms: a product of elementary 1-D complex exponentials is a plane wave

$$\prod_{\ell=1}^{d} e^{ix_\ell \omega_\ell} = e^{i\langle x, \omega \rangle}$$

whose constant-phase hyperplanes are orthogonal to the wave vector $\omega = (\omega_\ell)_{\ell=1}^d \in \mathbb{R}^d$ when this vector is nonzero. Here $\langle x, \omega \rangle = \sum_\ell x_\ell \omega_\ell$ is the canonical inner product on $\mathbb{R}^d$.

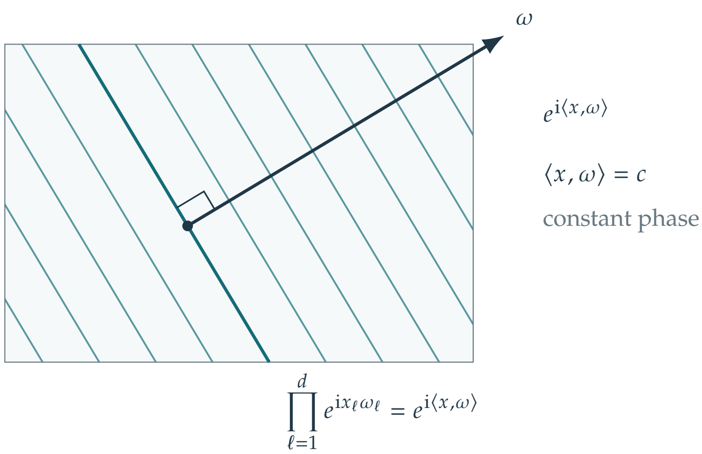


Figure 2.13. 2-D sine wave.

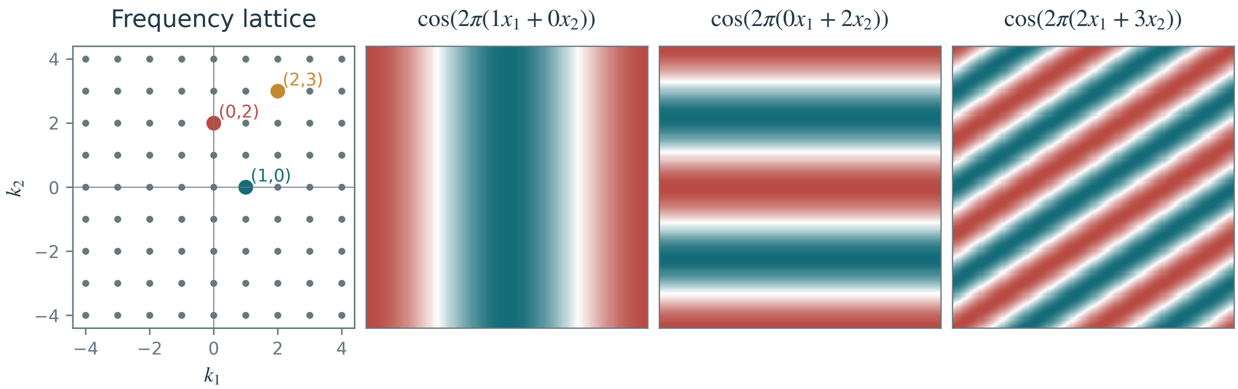


Figure 2.14. 2D Fourier orthogonal bases.

The Fourier transform and its inverse are defined by

$$\forall\, \omega \in \mathbb{R}^d, \quad \hat{f}(\omega) \stackrel{\text{def.}}{=} \int_{\mathbb{R}^d} f(x) e^{-\mathrm{i}\langle x,\,\omega\rangle} \mathrm{d}x,$$
$$\forall\, x \in \mathbb{R}^d, \quad f(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \hat{f}(\omega) e^{\mathrm{i}\langle x,\,\omega\rangle}\, \mathrm{d}\omega,$$

under the same integrability assumptions as in one dimension.

On $(\mathbb{R}/2\pi\mathbb{Z})^d$**.** Given an orthonormal basis $(\varphi_{n_1})_{n_1\in\mathbb{N}}$ of $L^2(\mathbb{X})$, one constructs an orthonormal basis of $L^2(\mathbb{X}^d)$ by tensorization

$$\forall\, k = (k_1, \ldots, k_d) \in \mathbb{N}^d, \quad \forall\, x \in \mathbb{X}^d, \quad \varphi_k(x) = \varphi_{k_1}(x_1) \ldots \varphi_{k_d}(x_d). \tag{2.12}$$

Fubini's theorem gives orthogonality. Completeness gives convergence of the rectangular partial sums $\sum_{\|k\|_\infty \leqslant N} \langle f,\, \varphi_k\rangle \varphi_k \to f$ in $L^2(\mathbb{X}^d)$.

For the multidimensional torus $(\mathbb{R}/2\pi\mathbb{Z})^d$, using the Fourier basis (2.1), this gives the basis

$$\forall\, k \in \mathbb{Z}^d, \quad \varphi_k(x) = e^{\mathrm{i}\langle x,\,k\rangle}$$

which is indeed a Hilbertian orthonormal basis for the inner product $\langle f,\, g\rangle \stackrel{\text{def.}}{=} \frac{1}{(2\pi)^d} \int_{\mathbb{T}^d} f(x)\bar{g}(x)\mathrm{d}x$. This defines the Fourier transform and the reconstruction formula on $L^2(\mathbb{T}^d)$

$$\hat{f}_k \stackrel{\text{def.}}{=} \frac{1}{(2\pi)^d} \int_{\mathbb{T}^d} f(x) e^{-\mathrm{i}\langle x,\,k\rangle} \mathrm{d}x \quad \text{and} \quad f = \sum_{k\in\mathbb{Z}^d} \hat{f}_k e^{\mathrm{i}\langle x,\,k\rangle}.$$

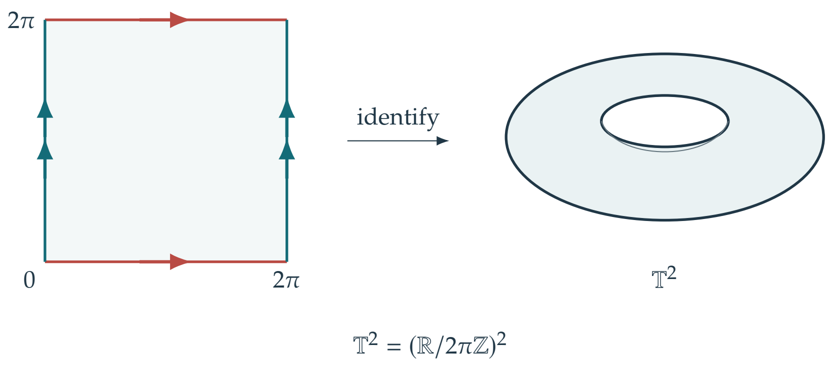


Figure 2.15. The two-dimensional torus $\mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$.

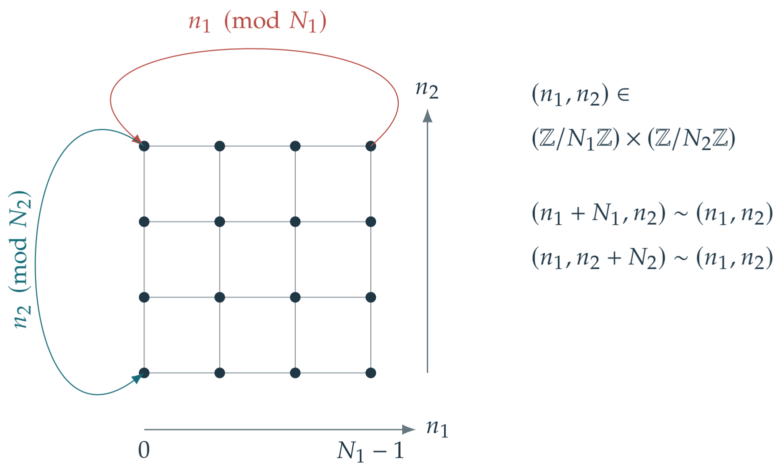


Figure 2.16. Discrete 2-D torus.

2.5.2 On Discrete Domains

Discrete Fourier Transform. On a d-dimensional discrete domain of the form

$$n = (n_1, \dots, n_d) \in \mathbb{Y}_d \overset{\text{def.}}{=} [\![1, N_1]\!] \times \dots \times [\![1, N_d]\!]$$

(we denote $[\![a, b]\!] \overset{\text{def.}}{=} \{i \in \mathbb{Z} \,;\, a \leqslant i \leqslant b\}$) of $N = N_1 \dots N_d$ points, with periodic boundary conditions, one defines an orthogonal basis $(\varphi_k)_k$ by the same tensor product formula as (2.12) but using the 1-D discrete Fourier basis (2.8)

$$\forall\, (k, n) \in \mathbb{Y}_d^2, \quad \varphi_k(n) = \varphi_{k_1}(n_1) \dots \varphi_{k_d}(n_d) = \prod_{\ell=1}^{d} e^{\frac{2\mathrm{i}\pi}{N_\ell} k_\ell n_\ell} = e^{2\mathrm{i}\pi \langle k, n \rangle_{\mathbb{Y}_d}} \tag{2.13}$$

where we used the (rescaled) inner product

$$\langle k, n \rangle_{\mathbb{Y}_d} \overset{\text{def.}}{=} \sum_{\ell=1}^{d} \frac{k_\ell n_\ell}{N_\ell}. \tag{2.14}$$

The pairing in (2.14) acts on frequency and spatial index vectors. The basis $(\varphi_k)_k$ is orthonormal for the signal inner product $N^{-1} \sum_n f_n \bar{g}_n$. As in one dimension, we define the Fourier coefficients without normalizing by $(N_\ell)_\ell$:

$$\forall\, k \in \mathbb{Y}_d, \quad \hat{f}_k \overset{\text{def.}}{=} \sum_{n \in \mathbb{Y}_d} f_n e^{-2\pi\mathrm{i} \langle k, n \rangle_{\mathbb{Y}_d}},$$
$$\forall\, n \in \mathbb{Y}_d, \quad f_n = \frac{1}{N} \sum_{k \in \mathbb{Y}_d} \hat{f}_k e^{2\pi\mathrm{i} \langle k, n \rangle_{\mathbb{Y}_d}}.$$

Fast Fourier Transform. We describe the algorithm for $d = 2$; the same construction works in any finite dimension. A fast algorithm for orthogonal decompositions in two 1-D bases $(\varphi^1_{k_1})_{k_1=1}^{N_1}$, $(\varphi^2_{k_2})_{k_2=1}^{N_2}$ also gives a fast decomposition in the tensor product basis $(\varphi^1_{k_1} \otimes \varphi^2_{k_2})_{k_1,k_2}$: apply the algorithm first to the rows and then to the columns of the matrix $(f_n)_{n=(n_1,n_2)} \in \mathbb{R}^{N_1 \times N_2}$. Either order gives the same result. Indeed

$$\forall\, k = (k_1, k_2), \quad \langle f, \varphi^1_{k_1} \otimes \varphi^2_{k_2} \rangle = \sum_{n=(n_1,n_2)} f_n \overline{\varphi^1_{k_1}(n_1)}\, \overline{\varphi^2_{k_2}(n_2)} = \sum_{n_1} \left(\sum_{n_2} f_{n_1,n_2} \overline{\varphi^2_{k_2}(n_2)} \right) \overline{\varphi^1_{k_1}(n_1)}.$$

If $C(N_1)$ is the cost of the 1-D algorithm on $\mathbb{R}^{N_1}$, the resulting 2-D transform costs $N_2 C(N_1) + N_1 C(N_2)$. For the FFT, this is $O(N_1 N_2 \log(N_1 N_2)) = O(N \log(N))$, where $N = N_1 N_2$.

Represent $f \in \mathbb{R}^{N_1 \times N_2}$ as a matrix, and write $F_N = (e^{-2\mathrm{i}\pi kn/N})_{k,n}$ for the Fourier matrix, whose rows are the φ_k^*. The 2-D transform can then be expressed as matrix products:

$$\hat{f} = F_{N_1} \times f \times F_{N_2}^{\top} \in \mathbb{C}^{N_1 \times N_2}.$$

In practice, the FFT performs these operations without explicit matrix multiplication.

`Associated code: coding/test_fft2.m`

2.5.3 Shannon sampling theorem.

Theorem 1.2 extends to sampling on a uniform Cartesian grid in $\mathbb{R}^d$ by tensorization. Use the continuous representative of the bandlimited function f. In two dimensions, if $\mathrm{supp}(\hat{f}) \subset [-\pi/s_1, \pi/s_1] \times [-\pi/s_2, \pi/s_2]$ and f decays sufficiently fast,

$$\forall\, x \in \mathbb{R}^2, \quad f(x) = \sum_{n \in \mathbb{Z}^2} f(n_1 s_1, n_2 s_2) \,\mathrm{sinc}(x_1/s_1 - n_1) \,\mathrm{sinc}(x_2/s_2 - n_2) \quad \text{where} \quad \mathrm{sinc}(u) = \frac{\sin(\pi u)}{\pi u}.$$

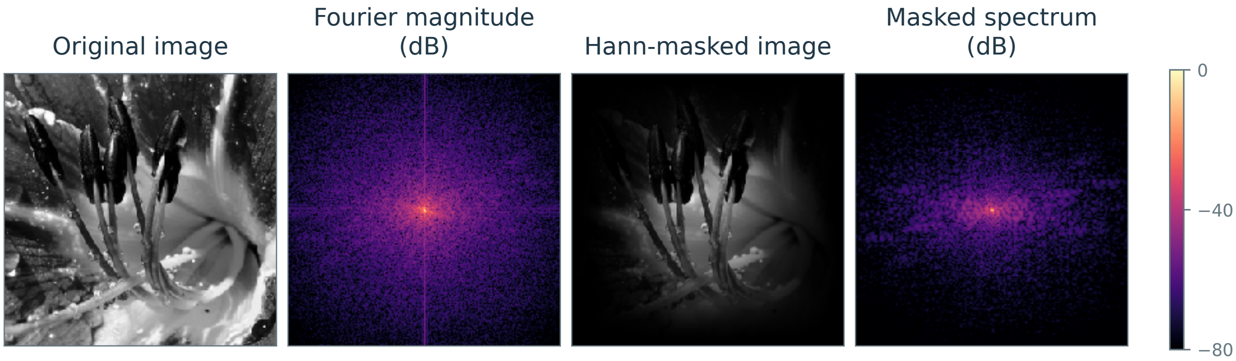


Figure 2.17. 2-D Fourier analysis of an image (left), and attenuation of the periodicity artifact using masking (right).

2.5.4 Convolution in higher dimension.

Convolution on $\mathbb{X}^d$, for $\mathbb{X} = \mathbb{R}$ or $\mathbb{X} = \mathbb{R}/2\pi\mathbb{Z}$, is defined as in one dimension:

$$f \star g(x) = \int_{\mathbb{X}^d} f(t)g(x-t)\mathrm{d}t.$$

Similarly, finite discrete convolution of vectors $f \in \mathbb{R}^{N_1 \times N_2}$ extends formula (2.11) as

$$\forall\, n \in [\![0, N_1-1]\!] \times [\![0, N_2-1]\!], \quad (f \star g)_n \overset{\text{def.}}{=} \sum_{k_1=0}^{N_1-1} \sum_{k_2=0}^{N_2-1} f_k g_{n-k}$$

where additions and subtractions of vectors are performed modulo (N_1, N_2).

The Fourier-convolution identity $\mathcal{F}(f \star g) = \hat{f} \odot \hat{g}$ holds in each of these settings. In the finite case, it yields an $O(N\log(N))$ convolution algorithm even when f and g have large support.

2.6 Application to ODEs and PDEs

2.6.1 On Continuous Domains

We give the main calculations, leaving analytic details of existence and regularity aside.

On $\mathbb{X} = \mathbb{R}$ or $\mathbb{T}$, one has

$$\mathcal{F}(f^{(k)})(\omega) = (\mathrm{i}\omega)^k \hat{f}(\omega).$$

Intuitively, $f^{(k)} = f \star \delta^{(k)}$ where $\delta^{(k)}$ is a distribution with Fourier transform $\hat{\delta}^{(k)}(\omega) = (\mathrm{i}\omega)^k$. Similarly on $\mathbb{X} = \mathbb{R}^d$ (see Section 2.5 for the definition of the Fourier transform in dimension d), one has

$$\mathcal{F}(\Delta f)(\omega) = -\|\omega\|^2 \hat{f}(\omega) \tag{2.15}$$

The same identity holds on $\mathbb{T}^d$, with ω replaced by $n \in \mathbb{Z}^d$. Fourier transforms and Fourier coefficients provide a powerful tool for studying linear differential equations with constant coefficients by turning them into algebraic equations.

As a typical example, we consider the heat equation

$$\frac{\partial f_t}{\partial t} = \Delta f_t \quad \Longrightarrow \quad \forall\, \omega, \quad \frac{\partial \hat{f}_t(\omega)}{\partial t} = -\|\omega\|^2 \hat{f}(\omega).$$

Hence $\hat{f}_t(\omega) = \hat{f}_0(\omega)e^{-\|\omega\|^2 t}$. Applying the inverse Fourier transform and the convolution theorem gives

$$f_t = G_t \star f_0 \quad \text{where} \quad G_t = \frac{1}{(4\pi t)^{d/2}} e^{-\frac{\|x\|^2}{4t}}$$

On $\mathbb{R}^d$, G_t is a Gaussian density with standard deviation $\sqrt{2t}$ in each direction. On the torus, periodize this kernel and adjust its mass to the chosen Haar normalization.

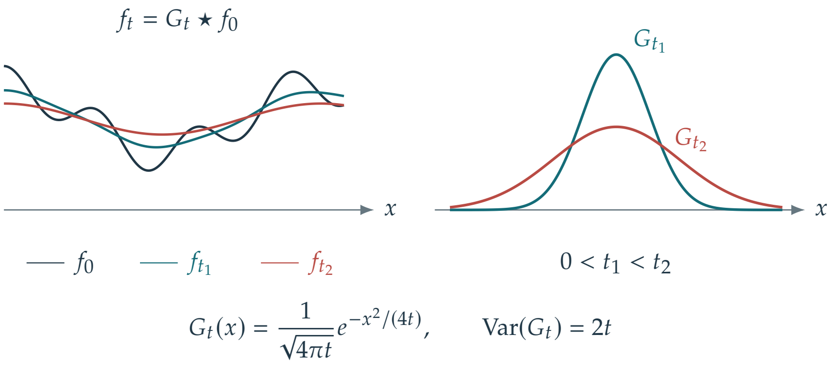


Figure 2.18. Heat diffusion as a convolution.

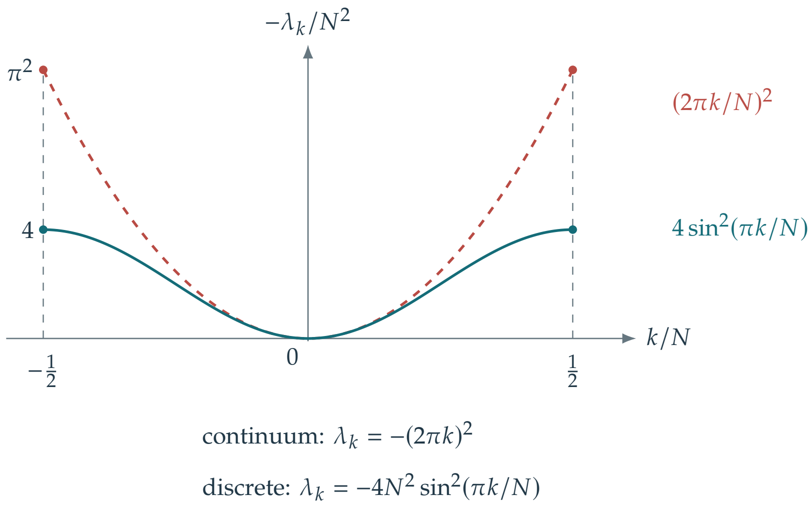


Figure 2.19. Comparison of the spectra of Δ and D_2.

2.6.2 Finite Domain and Discretization

On $\mathbb{Z}/N\mathbb{Z}$, corresponding to a discrete periodic domain, consider the forward first difference and the centered second difference:

$$D_1 f \stackrel{\text{def.}}{=} N(f_{n+1} - f_n)_n = f \star d_1 \quad \text{where} \quad d_1 = N[-1, 0, \ldots, 0, 1]^\top \in \mathbb{R}^N, \tag{2.16}$$

$$D_2 f = -D_1^\top D_1 f \stackrel{\text{def.}}{=} N^2(f_{n+1} + f_{n-1} - 2f_n)_n = f \star d_2 \quad \text{where} \quad d_2 = -d_1 \star \bar{d}_1 = N^2[-2, 1, 0, \ldots, 0, 1]^\top \in \mathbb{R}^N. \tag{2.17}$$

Thanks to Proposition 2.5, one can equivalently compute

$$\mathcal{F}(D_2 f) = \hat{d}_2 \odot \hat{f} \quad \text{where} \quad (\hat{d}_2)_k = N^2(e^{\frac{2i\pi}{N}k} + e^{-\frac{2i\pi}{N}k} - 2) = -4N^2 \sin\left(\frac{\pi k}{N}\right)^2. \tag{2.18}$$

Here $\bar{d}_1[n] = d_1[-n]$ denotes periodic reflection. For fixed nonzero k and $N \to \infty$, one has $(\hat{d}_2)_k \sim -(2\pi k)^2$ which matches the scaling of (2.15).

2.7 A Bit of Group Theory

The reference for this section is [26].

2.7.1 Characters

For $(G, +)$ a commutative group, a character is a group morphism $\chi : (G, +) \to (\mathbb{C}^*, \cdot)$, i.e. it satisfies

$$\forall (n, m) \in G, \quad \chi(n + m) = \chi(n)\chi(m).$$

The characters form the dual group $(\hat{G}, \odot)$ under pointwise multiplication $(\chi_1 \odot \chi_2)(n) \overset{\text{def.}}{=} \chi_1(n)\chi_2(n)$. The inverse of a character χ is $\chi^{-1}(n) = \chi(-n)$.

For a finite group G with $|G| = N$, we have $N \times n = 0$ for every $n \in G$. Consequently, $\chi(n)^N = \chi(Nn) = \chi(0) = 1$: characters take values among the roots of unity,

$$\chi(n) \in \left\{ e^{\frac{2i\pi}{N}k} \; ; \; 0 \leqslant k \leqslant N-1 \right\}. \tag{2.19}$$

Thus $\hat{G}$ is a finite group (since there are finitely many maps between two finite sets) and $\chi^{-1} = \bar{\chi}$. In the case of a cyclic group, the dual is simple to describe.

Proposition 2.7. *For $G = \mathbb{Z}/N\mathbb{Z}$, the dual is $\hat{G} = (\varphi_k)_{k=0}^{N-1}$, where $\varphi_k = (e^{\frac{2i\pi}{N}nk})_n$. The map $k \mapsto \varphi_k$ defines a noncanonical isomorphism $G \sim \hat{G}$.*

Proof. The φ_k are indeed characters.

Conversely, for any $\chi \in \hat{G}$, according to (2.19), $\chi(1) = e^{\frac{2i\pi}{N}k}$ for some k. Then

$$\chi(n) = \chi(1)^n = e^{\frac{2i\pi}{N}kn} = \varphi_k(n).$$

These characters are distinct because the values $\varphi_k(1)$ are distinct. Hence $|G| = |\hat{G}|$, and the displayed map is an isomorphism. □

This proposition thus shows that characters of cyclic groups are exactly the orthogonal discrete Fourier basis defined in (2.8).

Commutative groups. The structure theorem for finitely generated abelian groups expresses every such group as a product of cyclic groups:

$$G \sim (\mathbb{Z}/N_1\mathbb{Z}) \times \ldots \times (\mathbb{Z}/N_d\mathbb{Z}) \times \mathbb{Z}^Q. \tag{2.20}$$

If G is finite, then $Q = 0$ and $N = N_1 \cdots N_d$. In this case, G is simply a discrete d-dimensional "rectangle" with periodic boundary conditions.

For two finite groups (G_1, G_2) one has

$$\widehat{G_1 \times G_2} = \hat{G}_1 \otimes \hat{G}_2 = \left\{ \chi_1 \otimes \chi_2 \; ; \; (\chi_1, \chi_2) \in \hat{G}_1 \times \hat{G}_2 \right\}. \tag{2.21}$$

Here $\otimes$ is the tensor product of two functions

$$\forall (n_1, n_2) \in G_1 \times G_2, \quad (\chi_1 \otimes \chi_2)(n_1, n_2) \overset{\text{def.}}{=} \chi_1(n_1)\chi_2(n_2).$$

The tensor product $\chi_1 \otimes \chi_2$ is a character. Conversely, every character has the factorization $\chi = \chi(\cdot, 0) \otimes \chi(0, \cdot)$, since $(n_1, n_2) = (n_1, 0) + (0, n_2)$.

Combining this factorization with the structure theorem proves the following isomorphism.

Proposition 2.8. *If G is commutative and finite then $\hat{G} \sim G$.*

Proof. The structure theorem (2.20) with $Q = 0$ and the product-duality identity (2.21) give

$$\hat{G} \sim \hat{G}_1 \otimes \ldots \otimes \hat{G}_d$$

where $G_\ell \overset{\text{def.}}{=} \mathbb{Z}/N_\ell\mathbb{Z}$. One then observes that $\hat{G}_1 \otimes \hat{G}_2 \sim \hat{G}_1 \times \hat{G}_2$. Proposition 2.7 then completes the proof, since $\hat{G}_k \sim G_k$. □

The isomorphism $\hat{G} \sim G$ is noncanonical: it depends on a choice of generators and corresponding roots of unity. As in vector-space duality, the double-dual isomorphism $\hat{\hat{G}} \sim G$ is canonical and is given by evaluation:

$$g \in G \longmapsto e_g \in \hat{\hat{G}} \quad \text{where} \quad \left(e_g : \chi \in \hat{G} \mapsto \chi(g) \in \mathbb{C}^*. \right)$$

Discrete Fourier transform from the Character Viewpoint. This construction also identifies the Fourier basis explicitly. The characters in $\hat{G}_\ell$ are the discrete Fourier atoms (2.8), of the form

$$(e^{\frac{2i\pi}{N_\ell} k_\ell n_\ell})_{n_\ell=0}^{N_\ell-1} \quad \text{for some} \quad 0 \leqslant k_\ell < N_\ell.$$

Identifying G and $G_1 \times \ldots \times G_d$, their tensor products show that the characters in $\hat{G}$ form exactly the orthogonal multidimensional Fourier basis (2.13).

2.7.2 More General cases

Infinite groups. For an infinite finitely generated abelian group, the structure theorem has $Q > 0$. For locally compact abelian groups, the dual consists of continuous characters with values on the unit circle; invariant Haar measure determines integration. For $G = \mathbb{Z}$, the characters have a continuous frequency parameter,

$$\hat{\mathbb{Z}} = \left\{\varphi_\omega : n \mapsto e^{in\omega} \in \mathbb{C}^{\mathbb{Z}} \;;\; \omega \in \mathbb{R}/2\pi\mathbb{Z}\right\}$$

so that $\hat{\mathbb{Z}} \sim \mathbb{R}/2\pi\mathbb{Z}$. The case $G = \mathbb{Z}^Q$ follows by tensorization. The $(\varphi_\omega)_\omega$ are "orthogonal" in the sense that $\langle \varphi_\omega, \varphi_{\omega'} \rangle_{\mathbb{Z}} = 2\pi\delta_{2\pi}(\omega - \omega')$ can be understood as a Dirac kernel (this is similar to the Poisson formula), where $\langle u, v \rangle_{\mathbb{Z}} \stackrel{\text{def.}}{=} \sum_n u_n \bar{v}_n$. Expanding a sequence $(c_n)_{n\in\mathbb{Z}}$ in characters amounts to forming the Fourier series $\sum_n c_n e^{-in\omega}$.

Similarly, for $G = \mathbb{R}/2\pi\mathbb{Z}$, one has $\hat{G} = \mathbb{Z}$, with orthonormal characters $\varphi_n = e^{i \cdot n}$, so that the decomposition of functions in $L^2(G)$ is the computation of Fourier coefficients. On a compact group, normalized invariant measure provides the natural L^2 inner product.

Non-commutative groups. For a noncommutative group, one-dimensional characters do not recover G. For example, the symmetric group Σ_N on $N \geqslant 2$ letters has only the characters $\hat{G} = \{1, \varepsilon\}$, where $\varepsilon(\sigma) = (-1)^q$ is the signature and q is the number of transpositions in a decomposition of $\sigma \in \Sigma_N$.

To study noncommutative groups, replace homomorphisms $\chi : G \to \mathbb{C}^*$ by homomorphisms $\rho : G \to \mathrm{GL}(\mathbb{C}^{n_\rho})$ for some n_ρ. These are called representations of G. For $(g, g') \in G$, writing the group operation multiplicatively on G, the defining identity is $\rho(gg') = \rho(g) \circ \rho(g')$. When $n_\rho = 1$, the identification $\mathrm{GL}(\mathbb{C}) \sim \mathbb{C}^*$ recovers the one-dimensional characters. For any representation ρ, the function $\chi(g) \stackrel{\text{def.}}{=} \mathrm{tr}(\rho(g))$, where tr denotes trace, is called its character. This trace function is generally not a group homomorphism.

For a finite group G, every subspace V invariant under all $\rho(g)$ admits an invariant complement W, so that $\mathbb{C}^{n_\rho} = V \oplus W$ with W also invariant. In a basis adapted to these subspaces, the representation matrices are block diagonal. To construct the complement, use the inner product

$$\langle x, y \rangle \stackrel{\text{def.}}{=} \sum_{g\in G} \langle \rho(g)x, \rho(g)y \rangle_{\mathbb{C}^{n_\rho}}$$

and take the orthogonal complement $V^\perp$ for this inner product. In an orthonormal basis of $\mathbb{C}^{n_\rho}$ for this product, all matrices $\rho(g)$ are unitary. A representation is irreducible if it has no nonzero proper invariant subspace. These are the elementary representations from which larger ones are built by block-diagonal assembly. We identify irreducible representations up to isomorphism: (ρ, ρ') are isomorphic if $\rho'(g) = U^{-1}\rho(g)U$ for every g, with a single change-of-basis matrix $U \in \mathrm{GL}(\mathbb{C}^d)$. Their isomorphism classes form the unitary dual, denoted $\hat{G}$, which is generally a set rather than a group.

For the symmetric group, there is an explicit description of the set of irreducible representations. For example, $G = \Sigma_3$ has $|G| = 6$ elements. It has two one-dimensional representations, the trivial representation and the signature, and one two-dimensional irreducible representation obtained by identifying G with the isometries of an equilateral triangle in $\mathbb{R}^2$. Their dimensions satisfy $6 = |G| = \sum n_\rho^2 = 1 + 1 + 2^2$.

The dimensions n_ρ of the irreducible representations $\rho \in \hat{G}$ satisfy $\sum_{\rho\in\hat{G}} n_\rho^2 = N$. After choosing unitary representatives, their matrix entries form an orthogonal basis for functions $f : G \to \mathbb{C}$. This basis is noncanonical because it depends on a choice of basis within each representation. The associated Fourier transform of a function $f : G \to \mathbb{C}$ is defined as

$$\forall\, \rho \in \hat{G}, \quad \hat{f}(\rho) \stackrel{\text{def.}}{=} \sum_{g\in G} f(g)\rho(g) \in \mathbb{C}^{n_\rho \times n_\rho}.$$

With this sign convention, these are pairings with conjugated matrix entries. Their orthogonality makes the transform invertible. The next proposition gives the inverse transform.

Proposition 2.9. *One has*

$$\forall\, g \in G, \quad f(g) = \frac{1}{|G|} \sum_{\rho \in \hat{G}} n_\rho \operatorname{tr}(\hat{f}(\rho)\rho(g^{-1})).$$

Proof. The proof uses the regular-character identity, valid for every $g \in G$

$$A_g \stackrel{\text{def.}}{=} \sum_{\rho \in \hat{G}} n_\rho \operatorname{tr}(\rho(g)) = |G|\delta_g$$

where $\delta_g = 0$ for $g \neq \mathrm{Id}_G$ and $\delta_{\mathrm{Id}_G} = 1$. A proof is given in the book by Diaconis, p. 13: decompose the character of the regular representation into irreducible characters and use their orthogonality. The regular representation acts by permuting, according to G, the canonical basis of $\mathbb{R}^{|G|}$. One then has

$$\begin{aligned}\frac{1}{|G|} \sum_{\rho \in \hat{G}} n_\rho \operatorname{tr}(\hat{f}(\rho)\rho(g^{-1})) &= \frac{1}{|G|} \sum_{\rho \in \hat{G}} n_\rho \operatorname{tr}(\sum_u f(u)\rho(u)\rho(g^{-1})) \\ &= \sum_u f(u) \frac{1}{|G|} \sum_{\rho \in \hat{G}} n_\rho \operatorname{tr}(\rho(ug^{-1})) = \sum_u f(u) A_{ug^{-1}}/|G| = f(g).\end{aligned}$$

□

One can define the convolution of two functions $f, h : G \to \mathbb{C}$ as

$$(f \star h)(a) \stackrel{\text{def.}}{=} \sum_{bc=a} f(b)h(c)$$

Convolution need not be commutative on this group. The Fourier transform block-diagonalizes the convolution operators, as stated next.

Proposition 2.10. *Denoting* $\mathcal{F}(f) = \hat{f}$

$$\mathcal{F}(f \star h)(\rho) = \hat{f}(\rho) \times \hat{h}(\rho)$$

where $\times$ *denotes matrix multiplication.*

Proof.

$$\mathcal{F}(f \star h)(\rho) = \sum_x \sum_y f(y)h(y^{-1}x)\rho(yy^{-1}x) = \sum_x \sum_y f(y)\rho(y)h(y^{-1}x)\rho(y^{-1}x) = \sum_y f(y)\rho(y) \sum_z h(z)\rho(z)$$

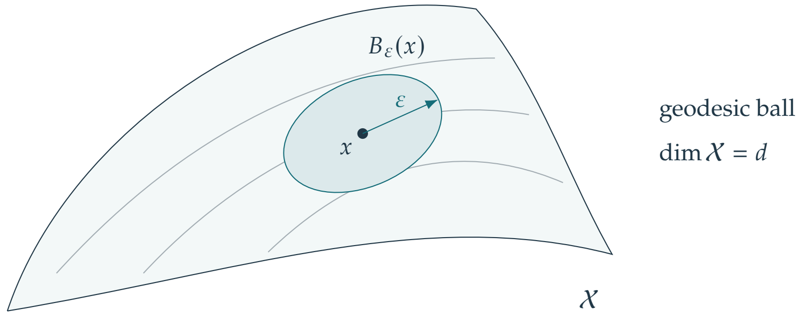
where we made the change of variable $z = y^{-1}x$. □

Fast algorithms for computing $\hat{f}$ exist for some groups, including permutation groups. Their structure is considerably more involved than for $G = \mathbb{Z}/N\mathbb{Z}$.

For infinite compact groups, the theory uses a discrete infinite family of irreducible representations. The rotation group SO(3) is a useful example. Its representations can be described explicitly through the spherical harmonics introduced in Section 2.8.2. There is one representation of dimension $2\ell + 1$ for each frequency index ℓ.

2.8 A Bit of Spectral Theory

On a general domain $\mathbb{X}$, the group-theoretic approach applies when $\mathbb{X} = G$ is a group or when a group acts transitively on $\mathbb{X}$. Another approach defines Fourier-like basis functions as eigenfunctions of a differential operator, following the characterization in Section 2.6. The Laplacian is particularly useful: it is second order, rotation invariant, and has natural counterparts on surfaces and graphs.



$$\frac{1}{\mathrm{Vol}(B_\varepsilon(x))}\int_{B_\varepsilon(x)} f(y)\,\mathrm{d}\mu(y) = f(x) + \frac{\varepsilon^2}{2(d+2)}\Delta f(x) + o(\varepsilon^2)$$

Figure 2.20. Computing the Laplacian on a surface.

2.8.1 On a Surface or a Manifold

Let $\mathbb{X}$ be a smooth compact connected Riemannian manifold of dimension d, without boundary. The Laplace–Beltrami operator can be characterized, for smooth f, by the small-ball mean-value expansion

$$(\Delta f)(x) = \lim_{\varepsilon\to 0}\frac{2(d+2)}{\varepsilon^2}\left(\frac{1}{\mathrm{Vol}(B_\varepsilon(x))}\int_{B_\varepsilon(x)} f(y)\,\mathrm{d}\mu(y) - f(x)\right).$$

Here μ is Riemannian volume and $B_\varepsilon(x)$ is the geodesic ball centered at x. The factor ε^{-2} is essential.

The operator Δ is unbounded and self-adjoint on its natural domain in $L^2(\mathbb{X})$; its resolvent is compact. It has a complete orthonormal family of eigenfunctions $(\varphi_n)_{n\geqslant 0}$, with eigenvalues, counted with multiplicity,

$$0 = \lambda_0 > \lambda_1 \geqslant \lambda_2 \geqslant \cdots \longrightarrow -\infty.$$

The constant eigenfunction is $\varphi_0 = \mathrm{Vol}(\mathbb{X})^{-1/2}$, and the inner product is $\langle f,\, g\rangle_{\mathbb{X}} = \int_{\mathbb{X}} f(x)\overline{g(x)}\,\mathrm{d}\mu(x)$. Boundary conditions must be specified if the manifold has a boundary.

On the flat unit torus $\mathbb{X} = (\mathbb{R}/\mathbb{Z})^d$,

$$\Delta f = \sum_{s=1}^{d}\frac{\partial^2 f}{\partial x_s^2}, \qquad \varphi_n(x) = e^{2\pi\mathrm{i}\langle n,\, x\rangle}, \qquad \lambda_n = -4\pi^2\|n\|^2, \quad n\in\mathbb{Z}^d.$$

2.8.2 Spherical Harmonics

Consider the $(d-1)$-dimensional sphere $\mathbb{S}^{d-1} = \left\{x\in\mathbb{R}^d\ ;\ \|x\|_{\mathbb{R}^d} = 1\right\}$. The Laplacian has explicit eigenfunctions. For $d = 3$, they are indexed by $n = (\ell, m)$:

$$\forall\,\ell\in\mathbb{N},\quad \forall\, m = -\ell,\dots,\ell,\quad \varphi_{\ell,m}(\theta,\varphi) = e^{\mathrm{i}m\varphi}P_\ell^m(\cos(\theta))$$

The corresponding eigenvalue is $\lambda_{\ell,m} = -\ell(\ell+1)$. Here P_ℓ^m are associated Legendre functions (polynomials multiplied by $(1-x^2)^{|m|/2}$). Normalization makes the eigenfunctions orthonormal. We use spherical coordinates $x = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)\in\mathbb{S}^2$ for $(\theta,\varphi)\in[0,\pi]\times[0,2\pi]$. The index ℓ plays the role of the magnitude of a two-dimensional Fourier frequency. For a fixed ℓ, the space $V_\ell = \mathrm{span}(\varphi_{\ell,m})$ is an eigenspace of Δ, and is also invariant under rotation.

2.8.3 On a Graph

Let $\mathbb{X}$ be a graph with N vertices indexed by $\{1,\dots,N\}$. Its geometry is encoded by the weight matrix $W = (w_{i,j})_{1\leqslant i,j\leqslant N}$. The notation $i\sim j$ means that (i,j) is an edge, for $(i,j)\in\mathbb{X}^2$. Set the weights to zero on nonedges and assume nonnegative, symmetric weights: $w_{i,j} = w_{j,i}$.

The graph Laplacian $\Delta:\mathbb{R}^N\to\mathbb{R}^N$ sums weighted differences between neighboring values

$$\forall\, f\in\mathbb{R}^N,\quad (\Delta f)_i \overset{\text{def.}}{=} \sum_{j\sim i} w_{i,j}f_j - \Big(\sum_{j\sim i} w_{i,j}\Big)f_i \quad\Longrightarrow\quad \Delta = W - D$$

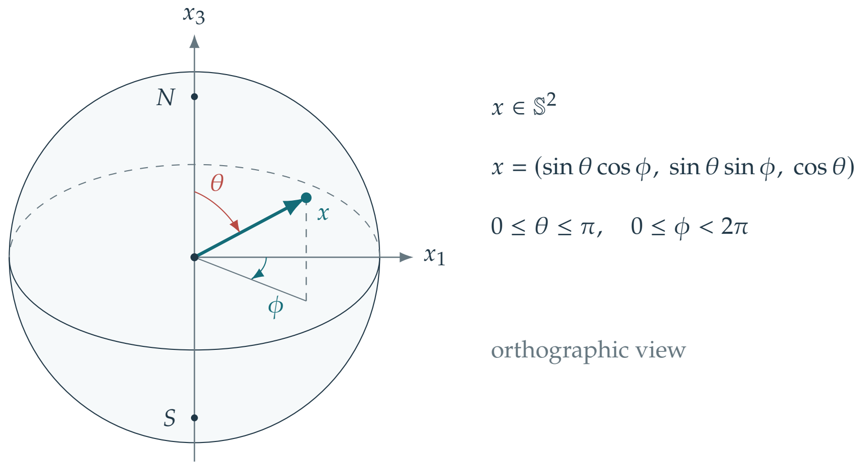


Figure 2.21. Spherical coordinates.

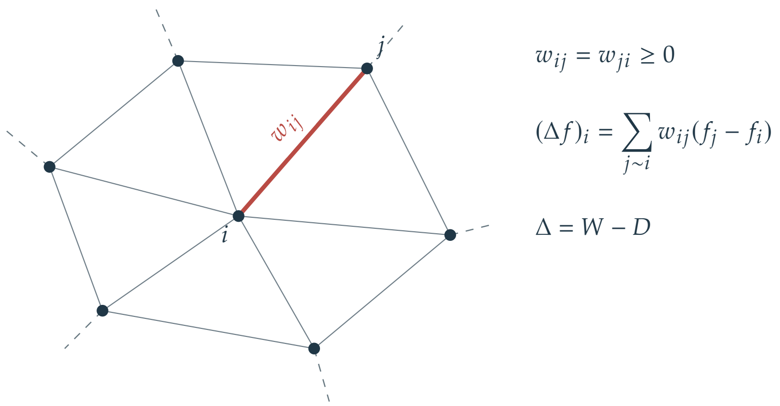


Figure 2.22. Weighted graph.

where $D \overset{\text{def.}}{=} \operatorname{diag}_i(\sum_{j\sim i} w_{i,j})$. In particular, note that $\Delta\mathbb{1} = 0$.

For instance, if $\mathbb{X} = \mathbb{Z}/N\mathbb{Z}$ with the graph $i \sim i-1$ and $i \sim i+1$ (modulo N), then Δ is the finite difference Laplacian operator $\Delta = N^{-2}D_2$ for unit edge weights defined in (2.17). This extends to any dimension by tensorization.

Proposition 2.11. *Define the graph-gradient operator* $G : f \in \mathbb{R}^N \mapsto (\sqrt{w_{i,j}}(f_i - f_j))_{(i,j)\in E}$*, with one component per undirected edge. Then*

$$-\Delta = G^\top G \quad \Longrightarrow \quad \forall f \in \mathbb{R}^N, \quad \langle \Delta f,\, f\rangle_{\mathbb{R}^N} = -\langle Gf,\, Gf\rangle_{\mathbb{R}^P}.$$

where P is the number of undirected edges $E = \{(i,j)\,;\, i\sim j, i<j\}$.

Proof. By symmetry of W and counting each undirected edge once,

$$\begin{aligned}\|Gf\|^2 &= \sum_{i<j} w_{ij}(f_i - f_j)^2 \\ &= \sum_i f_i^2 \sum_j w_{ij} - \sum_{i,j} w_{ij} f_i f_j \\ &= \langle Df,\, f\rangle - \langle Wf,\, f\rangle = -\langle \Delta f,\, f\rangle.\end{aligned}$$

Equality of these quadratic forms proves $G^\top G = -\Delta$. □

Thus Δ is symmetric negative semidefinite and has an orthonormal eigenbasis. One can take $\varphi_1 = N^{-1/2}\mathbb{1}$, with $0 = \lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_N$. The multiplicity of zero is the number of connected components of the graph formed by the positive-weight edges; for a connected graph with $N > 1$, $\lambda_2 < 0$. On a uniform periodic grid, the Fourier basis is an eigenbasis.

The quadratic energy $\sum_{i<j} w_{ij}(f_i - f_j)^2$ is the discrete analogue of the Dirichlet energy and provides a natural measure of smoothness on a graph.

2.8.4 Further Applications

Polynomial multiplication provides another application (`Code text_polymult.m`). Analogous transforms can also be defined over suitable finite fields.

CHAPTER 3

Shannon Coding Theory

A sequence of symbols can often be stored using fewer bits when its probabilities and dependencies are taken into account. The challenge is to reduce the expected length while keeping every message unambiguously recoverable. Prefix codes and entropy lead to Shannon's source coding bound; Huffman's construction and block coding show how to approach it. Empirical block counts connect these bounds to finite texts, while models of dependent symbols reveal further compression gains.

3.1 Source Coding

Uniform coding. We consider an alphabet $(s_1, \dots, s_K)$ of K symbols. For signal samples $0 \leqslant f_0 < 1$ quantized into K bins, the symbols label the bins. For text, they may represent letters, punctuation marks, and spaces. A fixed-length binary code uses $\lceil \log_2(K) \rceil$ bits per symbol. For example, if $K = 4$ and the symbols are $\{0, 1, 2, 3\}$, one possible code is $(c_0 = 00, c_1 = 01, c_2 = 10, c_3 = 11)$.

When some symbols are much more frequent than others, variable-length codes can reduce the average number of bits per symbol by assigning shorter codewords to frequent symbols.

Prefix coding. A binary code assigns each symbol s_k a finite word $c_k = c(s_k) \in \bigcup_{\ell \geqslant 0} \{0, 1\}^\ell$ of length $|c_k|$. An empty word is allowed only for a one-symbol alphabet, with the message length supplied separately. A code is a prefix code if no codeword is a prefix of another. Equivalently, the codewords correspond to leaves of a binary tree T: starting at the root, traversing an edge to the left appends 0, and traversing an edge to the right appends 1. We write $c = \mathrm{Leaves}(T)$ for this tree representation.

To decode a binary stream, follow its bits down the tree. Upon reaching a leaf, output the corresponding symbol and return to the root. The prefix property makes this decoding unambiguous.

Probabilistic modeling. We seek a code with the smallest expected length. Let $p = (p_1, \dots, p_K) \in \mathbb{R}_+^K$ be the symbol probability vector, with $\sum_k p_k = 1$. In practice, one often estimates p_k from the relative frequency of s_k in the data. The decoder must also know the resulting code, either from a shared model or from a transmitted description.

The entropy of the probability distribution p is

$$H(p) \overset{\text{def.}}{=} -\sum_k p_k \log_2(p_k)$$

with the convention $0 \log_2(0) = 0$.

For $h(u) = -u \log_2 u$ and $u > 0$, one has $h'(u) = -(\ln u + 1)/\ln 2$ and $h''(u) = -1/(u \ln 2) < 0$. Thus H is strictly concave on the probability simplex.

For a probability density f with respect to a reference measure $\mathrm{d}x$, such as Lebesgue measure on $\mathbb{R}^d$, differential entropy is defined by $H(f) \overset{\text{def.}}{=} -\int f(x) \log_2(f(x))\, \mathrm{d}x$, whenever this integral is defined. Unlike discrete entropy, it can be negative and depends on the reference measure.

Lemma 3.1. *For every probability vector p,*

$$0 = H(\delta_i) \leqslant H(p) \leqslant H(1/K) = \log_2(K)$$

where δ_i is the point mass at symbol i, and $1/K$ denotes the uniform probability vector.

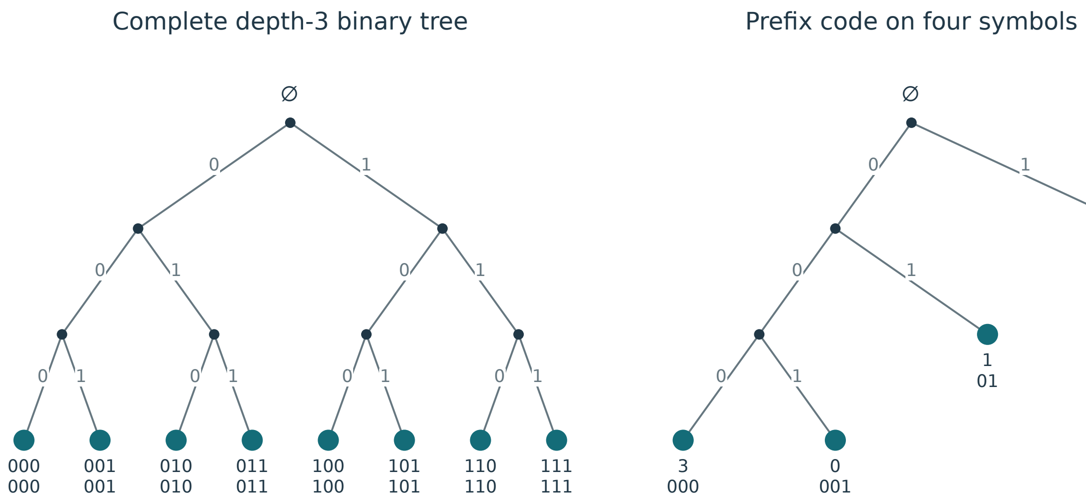


Figure 3.1. Left: the complete tree of binary words of length 3. Right: a prefix code.

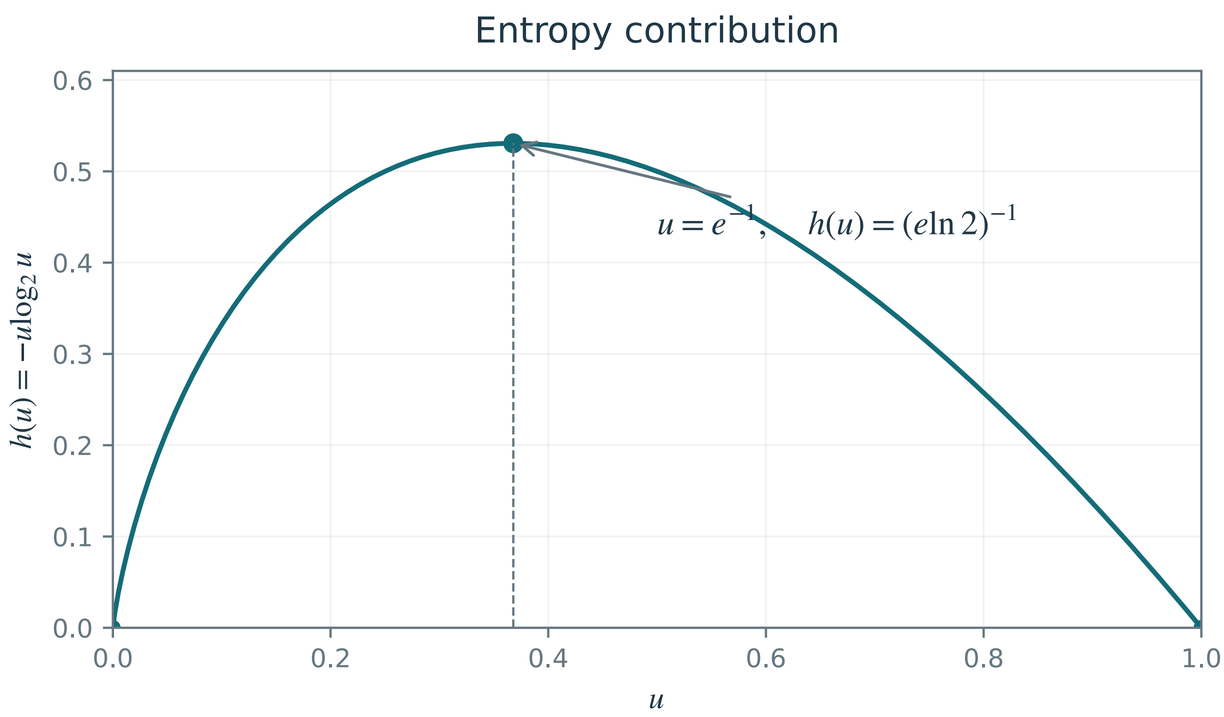


Figure 3.2. The concave function $h(u) = -u \log_2 u$ underlying entropy.

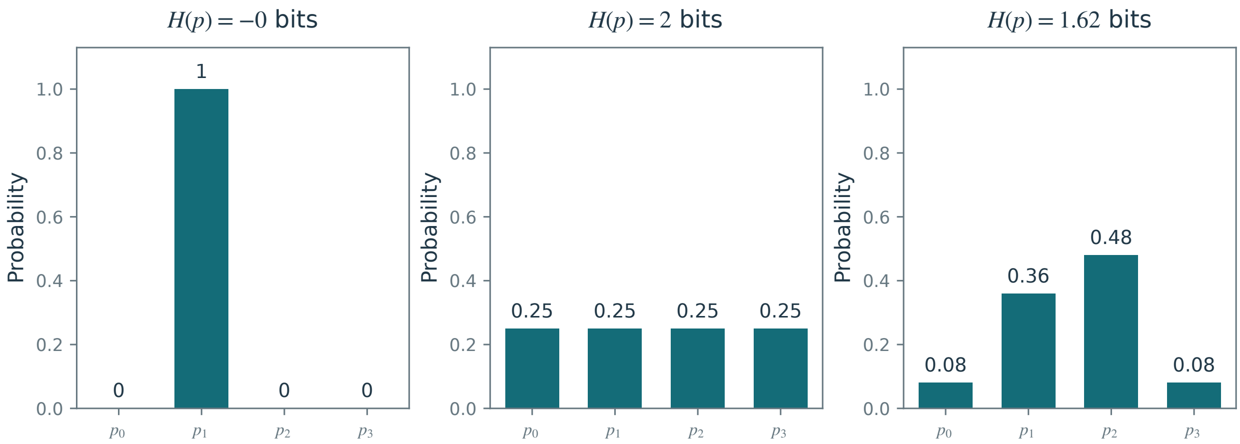


Figure 3.3. Three examples of probability distributions with corresponding entropies in bits.

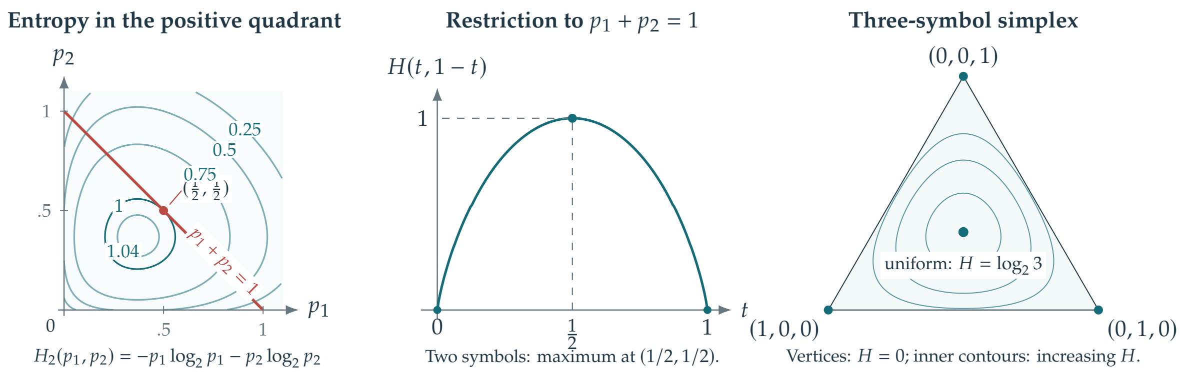


Figure 3.4. Entropy is minimized at a point mass and maximized at the uniform distribution. Left: level sets of the entropy formula in the positive quadrant and the constraint $p_1 + p_2 = 1$. Middle: its restriction to that constraint. Right: entropy level sets on the three-symbol probability simplex.

Proof. For $u \in [0,1]$, $-u \log_2 u \geqslant 0$, so $H(p) \geqslant 0$, with equality precisely for a point mass. For probability vectors p, q with $q_i > 0$ whenever $p_i > 0$, the inequality $\ln u \leqslant u - 1$ gives

$$\sum_{i:p_i>0} p_i \log_2 \frac{q_i}{p_i} \leqslant \frac{1}{\ln 2}\left(\sum_{i:p_i>0} q_i - 1\right) \leqslant 0.$$

Consequently $H(p) \leqslant -\sum_i p_i \log_2 q_i$. Taking $q_i = 1/K$ yields $H(p) \leqslant \log_2 K$, with equality precisely when p is uniform. □

The quantity $\mathrm{KL}(p \mid q) \overset{\text{def.}}{=} \sum_i p_i \log_2(p_i/q_i) \geqslant 0$ is called the Kullback–Leibler divergence, or relative entropy between p and q. The conventions are $0 \log_2(0/q_i) = 0$ and $p_i \log_2(p_i/0) = +\infty$ for $p_i > 0$. Although it vanishes exactly when $p = q$, it is not a metric: it is generally asymmetric and does not satisfy the triangle inequality. Minimizing the cross-entropy between an empirical distribution and a model is equivalent to maximizing the likelihood, and to minimizing their relative entropy. Cross-entropy is therefore a common objective in classification.

3.2 Shannon Source Coding Theorem

For a symbol distribution p, the expected codeword length of a code c is

$$L(c) \overset{\text{def.}}{=} \sum_k p_k |c_k|.$$

We seek a prefix code minimizing $L(c)$. Shannon's source coding theorem bounds the optimal expected length in terms of entropy.

Theorem 3.2. *Assume $p_k > 0$ for every symbol in the alphabet. Symbols of zero probability may be omitted. (i) If $c = \mathrm{Leaves}(T)$ for some tree T, then*

$$L(c) \geqslant H(p).$$

(ii) Conversely, there exists a code c with $c = \mathrm{Leaves}(T)$ such that

$$L(c) < H(p) + 1.$$

Kraft's inequality characterizes the possible codeword lengths of a prefix code and is the key to the proof.

Lemma 3.3 (Kraft inequality)**.** *(i) For a code c, if there exists a tree T such that $c = \mathrm{Leaves}(T)$ then*

$$\sum_k 2^{-|c_k|} \leqslant 1. \tag{3.1}$$

(ii) Conversely, if $(\ell_k)_k$ are nonnegative integers such that

$$\sum_k 2^{-\ell_k} \leqslant 1 \tag{3.2}$$

then there exists a code $c = \mathrm{Leaves}(T)$ such that $|c_k| = \ell_k$.

Proof. $\Rightarrow$ Suppose $c = \mathrm{Leaves}(T)$ and put $m = \max_k |c_k|$. Complete T to a full binary tree of depth m. The subtree rooted at c_k has height $m - |c_k|$ and $2^{m-|c_k|}$ leaves; see Figure 3.5, panel 1. These subtrees are disjoint, and the full tree has 2^m leaves, so

$$\sum_k 2^{m-|c_k|} \leqslant 2^m,$$

which gives (3.1) after division by 2^m.

$\Leftarrow$ Conversely, assume (3.2) holds and order the lengths so that $\ell_1 \leqslant \ldots \leqslant \ell_K =: m$. We construct full subtrees of height $m - \ell_k$, each with $2^{m-\ell_k}$ leaves, and place their leaf blocks consecutively from left to right. The inequality $\sum_k 2^{m-\ell_k} \leqslant 2^m$ ensures that all blocks fit among the 2^m leaves. Because the lengths ℓ_k are nondecreasing, the block sizes are nonincreasing powers of two. Each block therefore begins at a multiple of its own size and is the leaf set of a subtree; see Figure 3.5, panel 2. The roots of these disjoint subtrees form the required prefix code. □

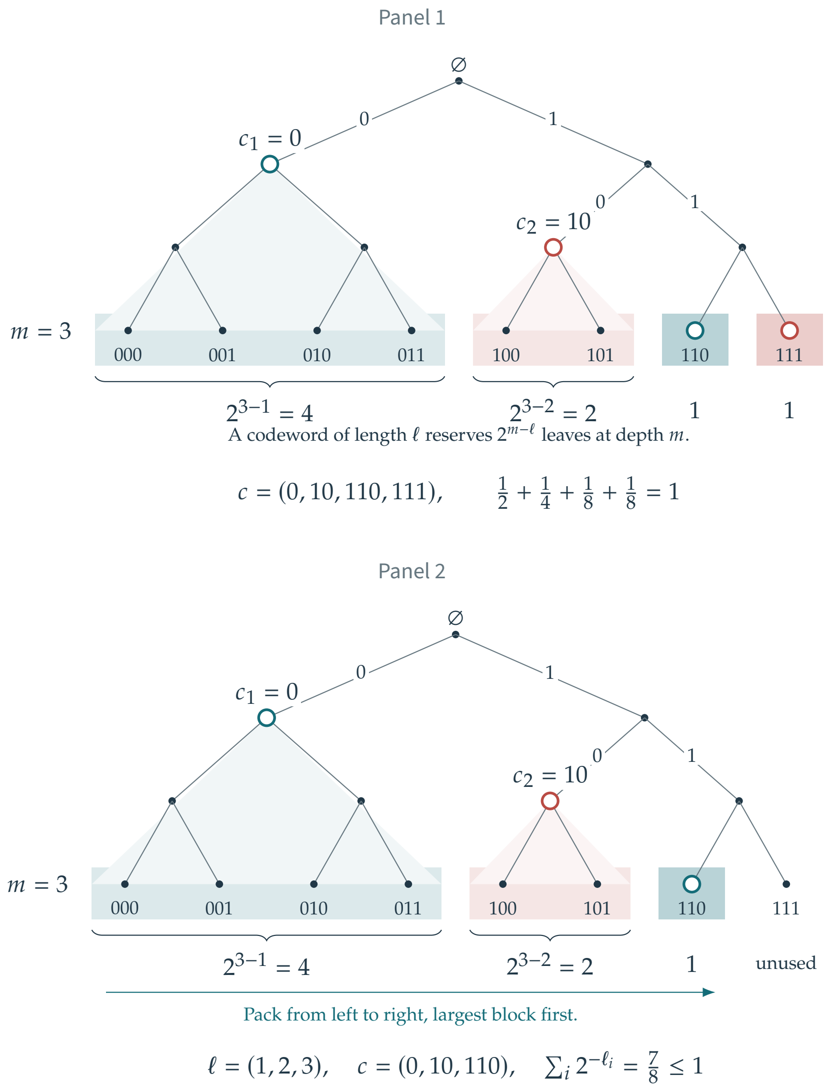


Figure 3.5. Panel 1: full binary tree obtained by completing the tree associated with the code ($c_1 = 0, c_2 = 10, c_3 = 110, c_4 = 111$). Panel 2: subtrees with the prescribed codeword lengths packed from left to right.

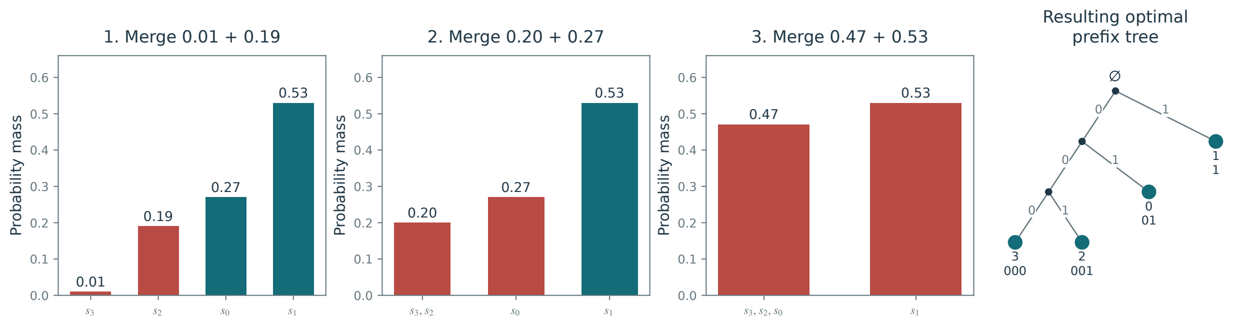


Figure 3.6. Successive merges in Huffman's coding algorithm.

Proof of Shannon's theorem. (i) Put $\ell_k = |c_k|$, $Z = \sum_k 2^{-\ell_k} \leqslant 1$, and $q_k = 2^{-\ell_k}/Z$. The nonnegativity of relative entropy gives

$$L(c) - H(p) = \mathrm{KL}(p \mid q) - \log_2 Z \geqslant 0. \tag{3.3}$$

Equality requires $Z = 1$ and $q = p$, identifying the ideal real-valued code lengths $-\log_2 p_k$.

(ii) Choose $\ell_k = \lceil -\log_2 p_k \rceil$. Then $\sum_k 2^{-\ell_k} \leqslant \sum_k p_k = 1$, so Lemma 3.3 constructs a prefix code with these lengths. Moreover,

$$L(c) = \sum_k p_k \lceil -\log_2 p_k \rceil < \sum_k p_k(-\log_2 p_k + 1) = H(p) + 1.$$

For a one-symbol alphabet, the empty word has length zero and the same conclusions hold. □

This proof is constructive: rounding the ideal lengths and using Kraft's lemma yields a Shannon code. It need not minimize $L(c)$. Huffman's greedy algorithm constructs an optimal binary prefix code by repeatedly merging the two least probable symbols. With a priority queue, its cost is $O(K \log K)$. Figure 3.6 illustrates the construction.

`Associated code: coding/test_text.m`

3.3 Probabilistic Modeling

3.3.1 Block Coding

Symbol-by-symbol prefix coding may be inefficient when one symbol has probability close to one: its ideal length $-\log_2 p_k$ is then below one bit, whereas a nontrivial prefix code uses at least one bit per symbol. A remedy is to group r consecutive symbols into blocks and code the resulting alphabet of size K^r.

For independent symbols with common distribution p, the block distribution is

$$P_{k_1,\dots,k_r} = p_{k_1} \cdots p_{k_r}, \qquad H(P) = rH(p).$$

Shannon's theorem applied to blocks gives an expected length per original symbol in the interval $[H(p), H(p)+1/r)$. This is an exact expectation bound under the model; no large-sample limit is needed. The code is prefix-free on blocks, so it need not use an integer number of bits per original symbol on average.

For dependent symbols whose marginal distributions are all p, entropy subadditivity gives

$$H(P) \leqslant rH(p),$$

with equality if and only if the symbols within the block are independent. For N divisible by r, the expected total length is at most

$$\frac{N}{r}(H(P) + 1) \leqslant N\left(H(p) + \frac{1}{r}\right).$$

Modeling dependence can therefore reduce the expected length further. For a stationary source on a finite alphabet, the block entropies are subadditive, so the entropy rate $\lim_{r\to\infty} H(P)/r$ exists. This rate is the asymptotic lower bound on the expected number of bits per symbol.

3.3.2 Empirical Block Coding

The same entropy bound has a finite-text counterpart. Fix a text $x_1 \cdots x_N$ over the alphabet $\mathcal{A} = \{s_1, \dots, s_K\}$, and assume $N = mr$ with $m, r \geqslant 1$. Split it into m non-overlapping blocks of length r:

$$B_s = (x_{(s-1)r+1}, \dots, x_{sr}) \in \mathcal{A}^r, \qquad s = 1, \dots, m.$$

Define the empirical symbol and block distributions by counting occurrences:

$$\hat{p}(a) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{\{x_i = a\}}, \qquad a \in \mathcal{A}, \tag{3.4}$$

$$\hat{P}(w) = \frac{1}{m} \sum_{s=1}^{m} \mathbf{1}_{\{B_s = w\}}, \qquad w \in \mathcal{A}^r. \tag{3.5}$$

These distributions describe the fixed text; no assumption on how it was generated is needed.

For each position $t \in \{1, \dots, r\}$ within a block, let

$$\hat{q}_t(a) = \frac{1}{m} \sum_{s=1}^{m} \mathbf{1}_{\{x_{(s-1)r+t} = a\}} = \sum_{w : w_t = a} \hat{P}(w).$$

Thus $\hat{q}_t$ is the t-th marginal of $\hat{P}$. The position marginals need not coincide, but every symbol occurrence contributes to exactly one of them. Consequently,

$$\hat{p} = \frac{1}{r} \sum_{t=1}^{r} \hat{q}_t. \tag{3.6}$$

Proposition 3.4 (Empirical block entropy)**.** *For every fixed text and every block length r dividing N,*

$$H(\hat{P}) \leqslant \sum_{t=1}^{r} H(\hat{q}_t) \leqslant rH(\hat{p}). \tag{3.7}$$

Proof. Consider the product of the position marginals,

$$Q = \hat{q}_1 \otimes \cdots \otimes \hat{q}_r, \qquad Q(w) = \prod_{t=1}^{r} \hat{q}_t(w_t).$$

If $\hat{P}(w) > 0$, then $\hat{q}_t(w_t) > 0$ for every t, so $Q(w) > 0$. Relative entropy is therefore finite. Expanding $\log_2 Q(w)$ and summing over each coordinate gives

$$\begin{aligned} 0 \leqslant \mathrm{KL}(\hat{P} \mid Q) &= \sum_{w : \hat{P}(w) > 0} \hat{P}(w) \log_2 \frac{\hat{P}(w)}{Q(w)} \\ &= -H(\hat{P}) + \sum_{t=1}^{r} H(\hat{q}_t). \end{aligned}$$

This proves the first inequality. For the second, concavity of entropy and the counting identity (3.6) yield

$$\frac{1}{r} \sum_{t=1}^{r} H(\hat{q}_t) \leqslant H\left(\frac{1}{r} \sum_{t=1}^{r} \hat{q}_t \right) = H(\hat{p}).$$

□

The first inequality is an equality exactly when $\hat{P} = Q$; the second is an equality exactly when all position marginals equal $\hat{p}$. Hence $H(\hat{P}) = rH(\hat{p})$ holds exactly when $\hat{P} = \hat{p}^{\otimes r}$.

Example 3.5 (Position frequencies and dependence). For the binary text 01010101 and $r = 2$, the symbol distribution is $\hat{p} = (1/2, 1/2)$, while $\hat{q}_1 = (1, 0)$ and $\hat{q}_2 = (0, 1)$. Every block is 01, so $H(\hat{P}) = 0$ although $rH(\hat{p}) = 2$. This shows why one must average the position marginals rather than identify each of them with $\hat{p}$.

For the text 0011 and $r = 2$, both position marginals equal $(1/2, 1/2)$, but only the blocks 00 and 11 occur. The dependence within each block gives $H(\hat{P}) = 1 < 2 = H(\hat{q}_1) + H(\hat{q}_2)$.

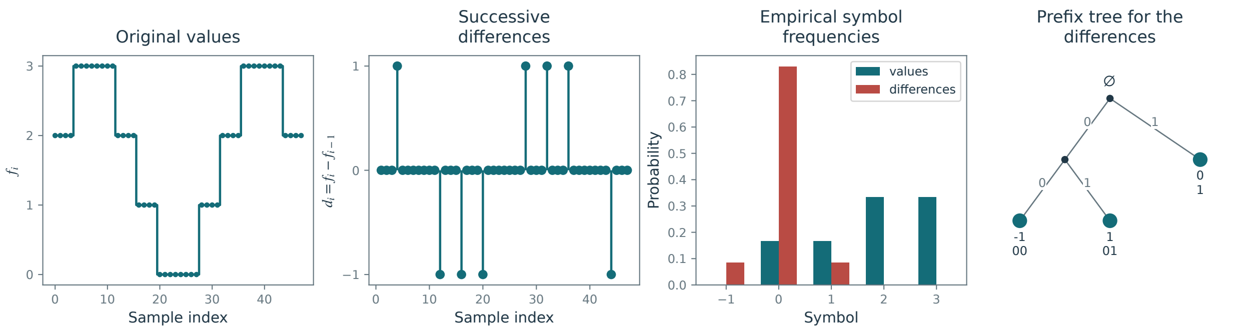


Figure 3.7. Difference coding. From left to right: the signal, its first differences, the corresponding histograms, and a code tree for the differences.

Encoded length of the fixed text. For a prefix code c on the observed blocks, the actual number of encoded bits is

$$\ell_{\text{text}}(c) = \sum_{s=1}^{m} |c(B_s)| = m \sum_{w:\hat{P}(w)>0} \hat{P}(w)|c(w)|.$$

Applying Theorem 3.2 to $\hat{P}$ shows that every such code satisfies $\ell_{\text{text}}(c) \geqslant mH(\hat{P})$, and that some code satisfies

$$\frac{\ell_{\text{text}}(c)}{N} < \frac{H(\hat{P})}{r} + \frac{1}{r} \leqslant H(\hat{p}) + \frac{1}{r}. \tag{3.8}$$

This bounds the actual average number of bits per symbol for the given text, at every admissible block length. The decoder is assumed to know the block code, r, and N; any transmitted codebook or message-length description adds to this length. In particular, when only one distinct block occurs, its empty codeword can be decoded because the number of blocks is known.

3.3.3 Mismatched Probability Models

For ideal real-valued lengths based on a model q, the excess expected length is the relative entropy, also called the Kullback–Leibler divergence. For $q_k > 0$ whenever $p_k > 0$, this excess is

$$\mathrm{KL}(p|q) = -\sum_k p_k \log_2 q_k - H(p) = \sum_k p_k \log_2 \frac{p_k}{q_k} \geqslant 0.$$

After lengths are rounded to integers, the difference between the expected lengths of two prefix codes need not equal this divergence. Relative entropy is jointly convex in (p, q) and is a central tool for comparing probability distributions in information theory and statistics.

3.4 Exploiting Statistical Dependence

A memoryless code based on the marginal distribution cannot have expected length below the marginal entropy. Dependent data, however, can be coded more efficiently by modeling their joint distribution. An invertible transformation of the discrete symbols can make the dependence easier to encode while preserving their joint entropy. For example, consecutive samples of a smooth signal often have small differences. Figure 3.7 illustrates coding these differences instead of the original values. The initial sample must also be retained to make the transformation invertible.

Another way to exploit temporal redundancy is run-length coding: each maximal run of identical symbols is represented by the symbol and the run length, which can itself be entropy-coded. Its performance depends on how the run lengths and transitions are modeled.

For a stationary finite-state Markov chain with stationary probabilities π_i and transition probabilities P_{ij}, the entropy rate is $-\sum_{i,j} \pi_i P_{ij} \log_2 P_{ij}$. Coding with the conditional transition probabilities, or coding sufficiently long blocks, can approach this rate. Run-length coding alone does not guarantee this performance.

CHAPTER 4

Wavelets

Signals contain broad trends and localized details that occur at different scales. Separating these components produces representations suited to compression and denoising, while allowing each scale to be processed efficiently. We build orthonormal wavelet bases from nested approximation spaces, derive fast decomposition and reconstruction algorithms in one and two dimensions, and explain how filter design controls localization, smoothness, and cancellation of polynomials.

The reference for this chapter is [23].

4.1 Multiresolution Approximation Spaces

A multiresolution approximation of $L^2(\mathbb{R})$ consists of nested closed subspaces $(V_j)_j$

$$L^2(\mathbb{R}) \supset \ldots \supset V_{j-1} \supset V_j \supset V_{j+1} \supset \ldots \supset \{0\} \tag{4.1}$$

related by dyadic scaling and invariant under translations on the corresponding dyadic grid:

$$f \in V_j \iff f(\cdot/2) \in V_{j+1} \quad \text{and} \quad f \in V_j \iff \forall\, n \in \mathbb{Z},\ f(\cdot + n2^j) \in V_j$$

Large j corresponds to coarse approximation spaces, and 2^j is often called the "scale".

At the fine-scale limit on the left of (4.1), $\cup_j V_j$ is dense in $L^2(\mathbb{R})$. Equivalently, $P_{V_j}(f) \to f$ as $j \to -\infty$, where P_V denotes orthogonal projection onto V:

$$P_V(f) = \underset{f' \in V}{\operatorname{argmin}}\ \|f - f'\|.$$

The limit on the right of (4.1) means that $\cap_j V_j = \{0\}$, or equivalently that $P_{V_j}(f) \to 0$ as $j \to +\infty$.

The first example consists of piecewise constant functions on dyadic intervals

$$V_j = \left\{ f \in L^2(\mathbb{R})\ ;\ \forall n, f \text{ is constant on } [2^j n, 2^j(n+1)[\right\}, \tag{4.2}$$

A second example is the space used for Shannon interpolation of bandlimited signals

$$V_j = \left\{ f\ ;\ \operatorname{Supp}(\hat{f}) \subset [-2^{-j}\pi, 2^{-j}\pi] \right\} \tag{4.3}$$

whose elements are recovered exactly from the samples $f(2^j n)$. As for piecewise constant signals, the sampling grid has spacing 2^j. Here, however, orthogonal projection is a bandlimiting operation:

$$P_{V_j}(f) = \mathcal{F}^{-1}(\hat{f} \odot 1_{[-2^{-j}\pi, 2^{-j}\pi]}).$$

Scaling functions. We also require a scaling function $\varphi \in L^2(\mathbb{R})$ such that

$$\{\varphi(\cdot - n)\}_n \text{ is a Hilbertian orthonormal basis of } V_0.$$

By the dilation property, this implies that

$$\{\varphi_{j,n}\}_n \text{ is a Hilbertian orthonormal basis of } V_j \quad \text{where} \quad \varphi_{j,n} \overset{\text{def.}}{=} \frac{1}{2^{j/2}} \varphi\left(\frac{\cdot - 2^j n}{2^j}\right).$$

The normalization ensures $\|\varphi_{j,n}\| = 1$.

Note that one then has

$$P_{V_j}(f) = \sum_n \langle f, \varphi_{j,n} \rangle \varphi_{j,n}.$$

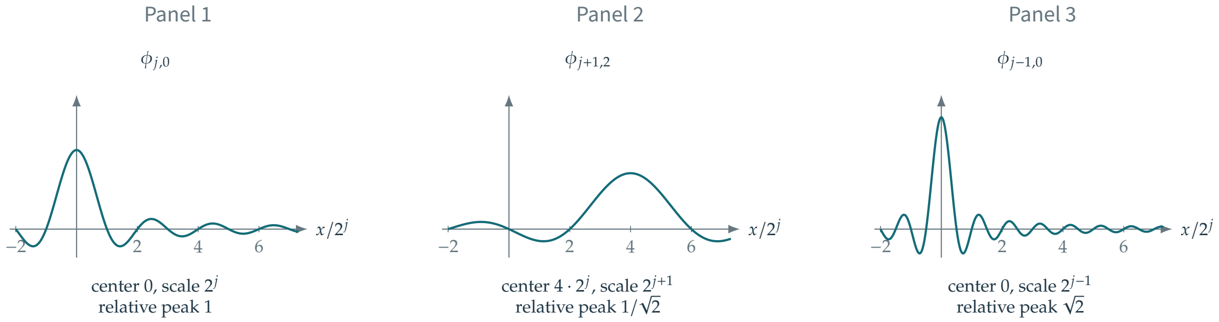


Figure 4.1. Translations and dilations of a scaling function generate the approximation spaces. The Shannon scaling function $\varphi(t) = \sin(\pi t)/(\pi t)$ is illustrated, with horizontal coordinate $x/2^j$ and heights relative to $2^{-j/2}$.

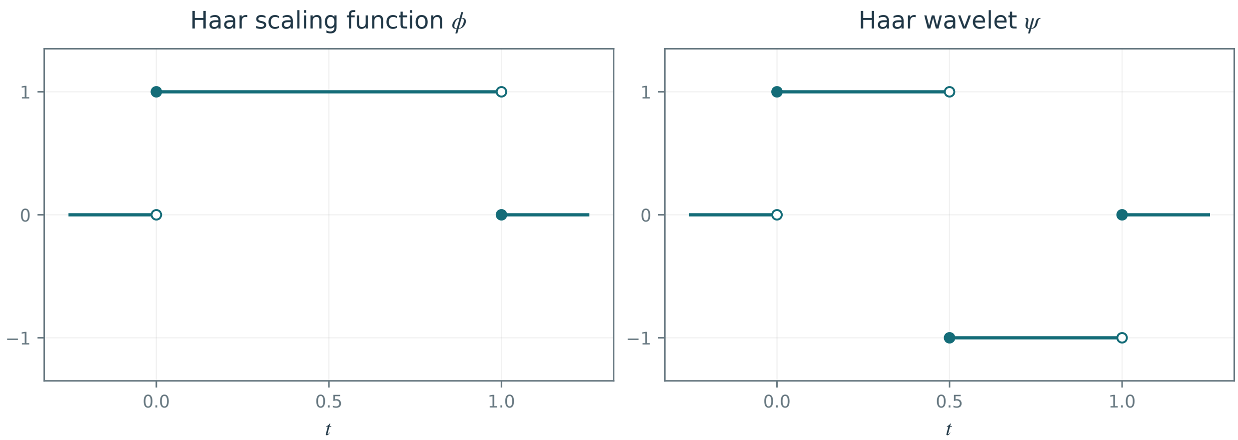


Figure 4.2. Haar scaling and wavelet functions. Both have amplitude $2^{-j/2}$ and unit L^2 norm.

Figure 4.1 illustrates the effects of translation and scaling.

For the case of piecewise constant signals (4.2), one can use

$$\varphi = 1_{[0,1[} \quad \text{and} \quad \varphi_{j,n} = 2^{-j/2} 1_{[2^j n, 2^j(n+1)[}.$$

For the case of Shannon multiresolution (4.3), one can use $\varphi(t) = \sin(\pi t)/(\pi t)$ and one verifies

$$\langle f, \varphi_{j,n} \rangle = 2^{j/2} f(2^j n) \quad (f \in V_j) \quad \text{and} \quad f = \sum_n 2^{j/2} f(2^j n) \varphi_{j,n} \quad (f \in V_j).$$

Spectral orthogonalization. In many applications, V_0 is generated by translates of a function that are not orthogonal: $V_0 = \overline{\mathrm{Span}}\{\theta(\cdot - n)\}_{n\in\mathbb{Z}}$. The next proposition orthogonalizes this family in the Fourier domain. Figure 4.3 shows the orthogonal cardinal spline functions obtained by this construction.

Proposition 4.1. *Let $\theta \in L^2(\mathbb{R})$. Its translates $\{\theta(\cdot - n)\}_{n\in\mathbb{Z}}$ are orthonormal if and only if*

$$A(\omega) \stackrel{\text{def.}}{=} \sum_k |\hat{\theta}(\omega - 2\pi k)|^2 = 1 \quad \textit{for almost every } \omega.$$

If there exist $0 < a \leqslant b < +\infty$ such that $a \leqslant A(\omega) \leqslant b$ almost everywhere, then φ defined by

$$\hat{\varphi}(\omega) = \frac{\hat{\theta}(\omega)}{\sqrt{A(\omega)}}$$

is such that $\{\varphi(\cdot - n)\}_{n\in\mathbb{Z}}$ is an orthonormal basis of $\overline{\mathrm{Span}}\{\theta(\cdot - n)\}_{n\in\mathbb{Z}}$.

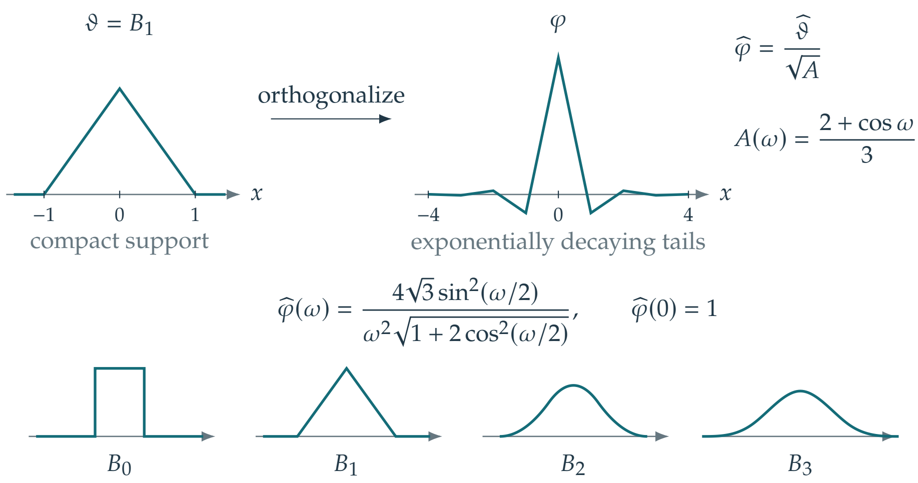


Figure 4.3. Orthogonalization of B-splines to define cardinal orthogonal spline functions.

Proof. The family $\{\theta(\cdot - n)\}_{n\in\mathbb{Z}}$ is orthonormal if and only if

$$\langle \theta,\, \theta(\cdot - n)\rangle = (\theta \star \bar{\theta})(n) = \delta_{0,n} \overset{\text{def.}}{=} \begin{cases} 1 & \text{if} \quad n = 0, \\ 0 & \text{otherwise.} \end{cases}$$

where $\bar{\theta}(x) = \overline{\theta(-x)}$ and δ_0 is the discrete Dirac vector. By Tonelli's theorem and Plancherel's identity, $A \in L^1(\mathbb{T})$. Its Fourier coefficients are

$$\begin{aligned} \frac{1}{2\pi}\int_0^{2\pi} A(\omega)e^{-in\omega}\,\mathrm{d}\omega &= \frac{1}{2\pi}\int_{\mathbb{R}} |\hat{\theta}(\omega)|^2 e^{-in\omega}\,\mathrm{d}\omega \\ &= \langle \theta,\, \theta(\cdot + n)\rangle. \end{aligned}$$

Thus the translates are orthonormal exactly when the Fourier coefficients of A are those of the constant function 1. Uniqueness of Fourier coefficients in $L^1(\mathbb{T})$ gives $A = 1$ almost everywhere. Since A is 2π-periodic, normalization by the bounded function $1/\sqrt{A(\omega)}$ gives $\sum_k |\hat{\varphi}(\omega - 2\pi k)|^2 = 1$. It remains to check equality of the closed spans. Approximate the periodic multipliers $1/\sqrt{A}$ and $\sqrt{A}$ by trigonometric polynomials in $L^2(\mathbb{T})$. The bounds $a \leqslant A \leqslant b$ show that the corresponding products with $\hat{\theta}$ and $\hat{\varphi}$ converge in $L^2(\mathbb{R})$. Inverting the Fourier transform expresses each generator as an L^2 limit of finite linear combinations of translates of the other. □

Spline interpolation provides a typical application. For example, cubic splines are generated by translates of a box-spline function θ, which is a compactly supported piecewise polynomial.

4.2 Multiresolution Detail Spaces

The detail spaces are the orthogonal complements associated with the inclusions $V_j \subset V_{j-1}$. Since these subspaces are closed,

$$\forall\, j, \quad W_j \text{ is such that } V_{j-1} = V_j \oplus^{\perp} W_j.$$

The following diagram shows these nested approximation spaces and their orthogonal complements:

$$\begin{tikzcd} \cdots \arrow[r] \arrow[dr] & V_{j-1} \arrow[r] \arrow[dr] & V_j \arrow[r] \arrow[dr] & V_{j+1} \arrow[r] & \cdots \\ & W_{j-1} & W_j & W_{j+1} & \end{tikzcd}$$

$$V_{j-1} = V_j \oplus^{\perp} W_j$$

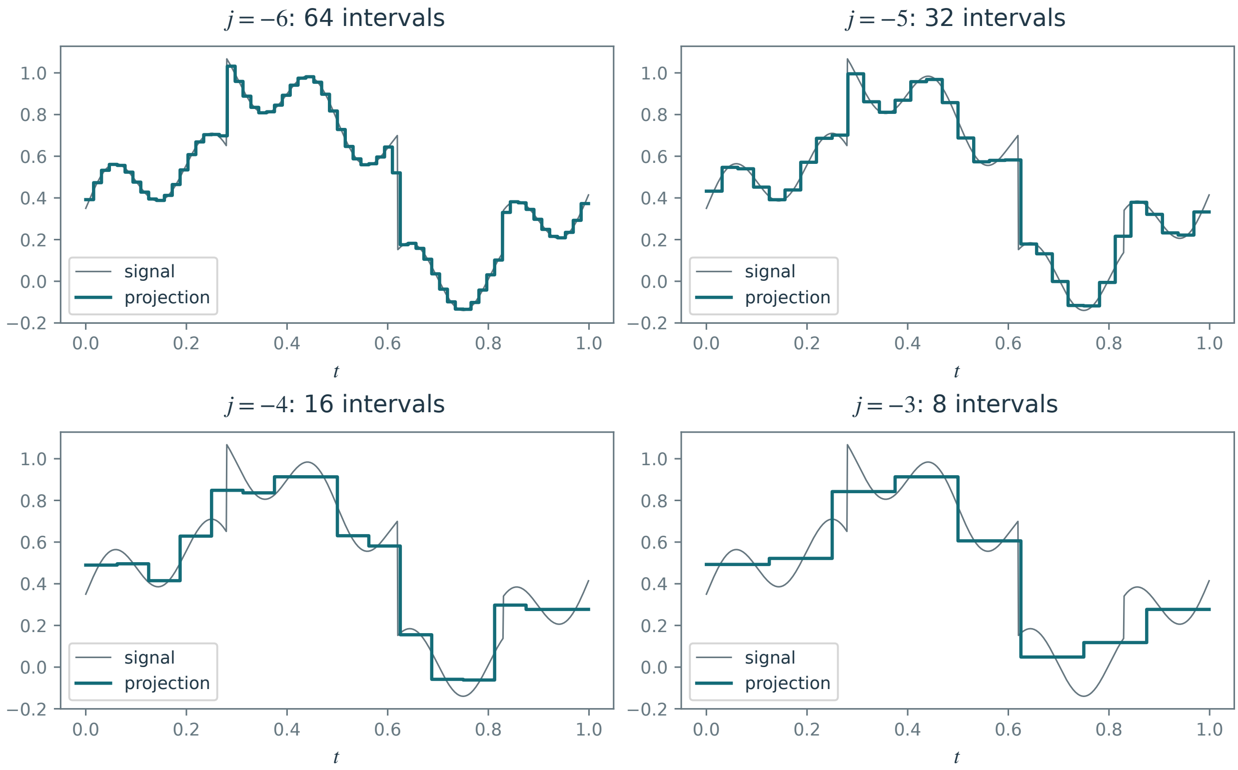


Figure 4.4. 1-D Haar multiresolution projection $P_{V_j} f$ of a function f.

Suppose that W_0 admits an orthonormal basis of translates $\{\psi(\cdot - n)\}_n$. Scaling then gives

$$\{\psi_{j,n}\}_n \text{ is a Hilbertian orthonormal basis of } W_j \quad \text{where} \quad \psi_{j,n} \stackrel{\text{def.}}{=} \frac{1}{2^{j/2}}\psi\left(\frac{\cdot - 2^j n}{2^j}\right).$$

Orthogonal decomposition across all scales yields

$$L^2(\mathbb{R}) = \bigoplus_{j=-\infty}^{j=+\infty} W_j = V_{j_0} \oplus^{\perp} \bigoplus_{j \leqslant j_0} W_j.$$

For every $f \in L^2(\mathbb{R})$, the corresponding expansion converges in $L^2(\mathbb{R})$:

$$f = \lim_{(j_-,j_+)\to(-\infty,+\infty)} \sum_{j=j_-}^{j_+} P_{W_j} f = \lim_{j_- \to -\infty} \left(P_{V_{j_0}} f + \sum_{j=j_-}^{j_0} P_{W_j} f \right).$$

This decomposition shows that

$$\left\{\psi_{j,n} \ ; \ (j,n) \in \mathbb{Z}^2\right\}$$

is an orthonormal basis of $L^2(\mathbb{R})$, which is called a wavelet basis. Stopping at a coarsest scale gives the orthonormal basis

$$\left\{\psi_{j,n} \ ; \ j \leqslant j_0, n \in \mathbb{Z}\right\} \cup \left\{\varphi_{j_0,n} \ ; \ n \in \mathbb{Z}\right\}.$$

The forward wavelet transform computes all inner products of a function f with the elements of this basis.

Haar wavelets. For the Haar multiresolution (4.2), one has

$$W_j = \left\{ f \ ; \ \forall n \in \mathbb{Z}, \ f \text{ constant on } [2^{j-1}n, 2^{j-1}(n+1)) \quad \text{and} \quad \int_{n2^j}^{(n+1)2^j} f = 0 \right\}. \tag{4.4}$$

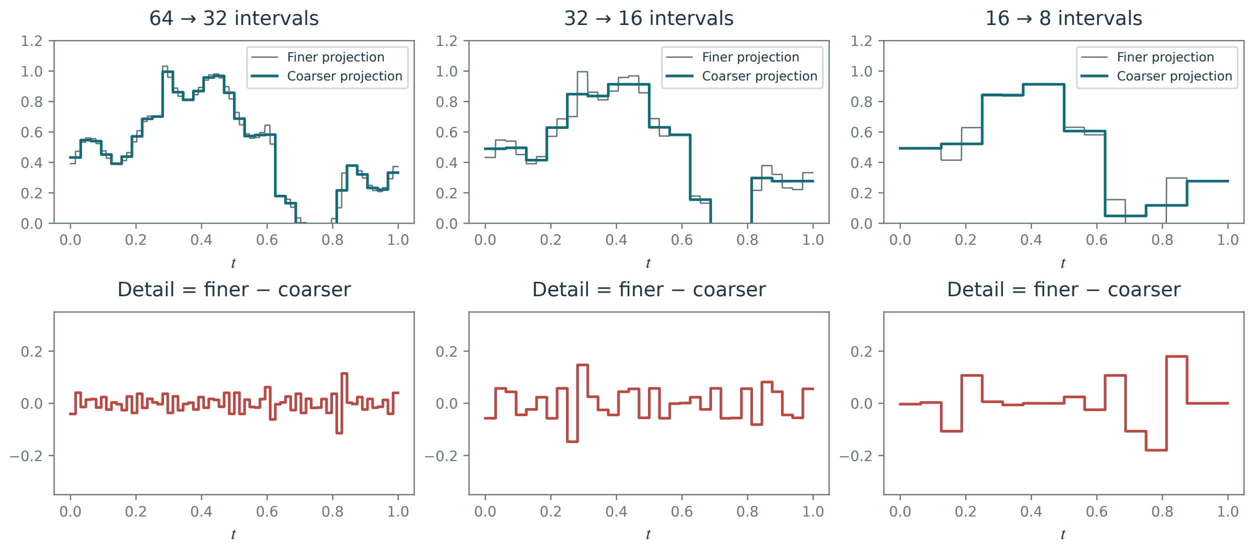


Figure 4.5. Three successive Haar approximations (top) and their extracted details (bottom). Coarser approximations use fewer intervals.

One choice of mother wavelet is

$$\psi(t) = \begin{cases} 1 & \text{for} \quad 0 \leqslant t < 1/2, \\ -1 & \text{for} \quad 1/2 \leqslant t < 1, \\ 0 & \text{otherwise,} \end{cases}$$

as shown in Figure 4.2.

Figure 4.5 illustrates how subtracting consecutive approximation-space projections isolates the details at each scale.

Shannon and splines. Figure 4.6 compares the sharp frequency partitions of Shannon wavelets with the smooth transitions of spline wavelets. Shannon wavelets can be viewed as a limiting case as the spline degree increases.

4.3 On Bounded Domains

On the periodic domain $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, we use period 1 rather than the 2π-period convention of the Fourier chapter. To obtain an orthonormal wavelet basis of $L^2(\mathbb{T})$, periodize the wavelets and restrict their translation points $2^j n$ to $[0, 1]$, choosing $0 \leqslant n < 2^{-j}$. As in (1.6), the periodization of $f \in L^1(\mathbb{R})$ is

$$f^P = \sum_{n \in \mathbb{Z}} f(\cdot - n) \in L^1(\mathbb{T}).$$

For an integer coarsest level $j_0 \leqslant 0$, this gives the family

$$\left\{\psi^P_{j,n} \ ; \ j \leqslant j_0, 0 \leqslant n < 2^{-j}\right\} \cup \left\{\varphi^P_{j_0,n} \ ; \ 0 \leqslant n < 2^{-j_0}\right\}.$$

which is an orthonormal basis of $L^2(\mathbb{T})$, illustrated in Figure 4.7. One can also construct wavelet bases using Neumann (mirror) boundary conditions, but this is more involved.

4.4 Fast Wavelet Transform

4.4.1 Discretization

We now work over $\mathbb{R}/\mathbb{Z}$. We assume access to a discrete signal $a_J \in \mathbb{R}^N$ with $N = 2^{-J}$ at a fixed scale 2^J, whose entries are inner products with the scaling functions, i.e.

$$\forall\, n \in \{0, \dots, N-1\}, \quad a_{J,n} = \langle f, \varphi^P_{J,n} \rangle \approx 2^{J/2} f(2^J n), \tag{4.5}$$

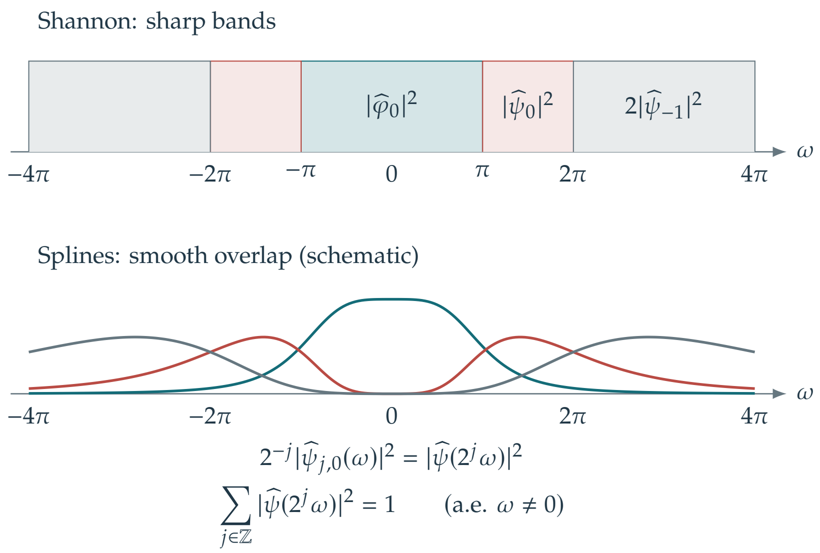


Figure 4.6. Spline and Shannon wavelet segmentation of the frequency axis.

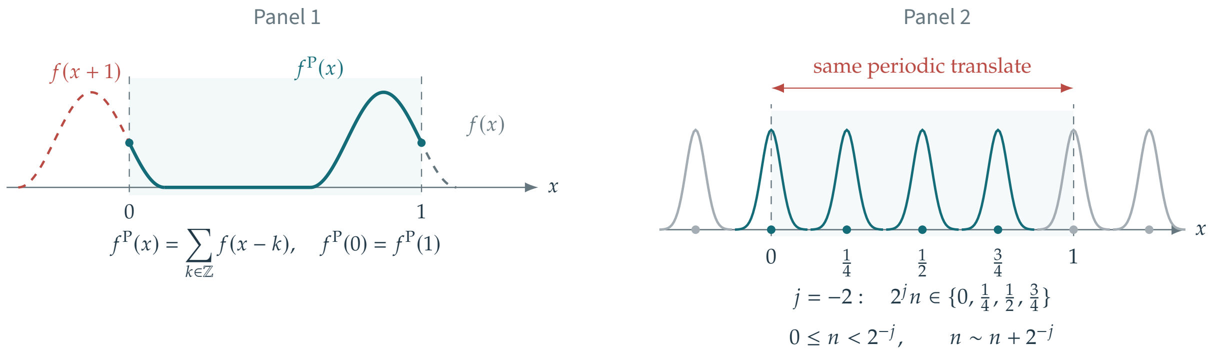


Figure 4.7. Periodized basis functions. Only the translations $0 \leqslant n < 2^{-j}$ are distinct, and $j_0 = 0$ is a common choice of coarsest scale.

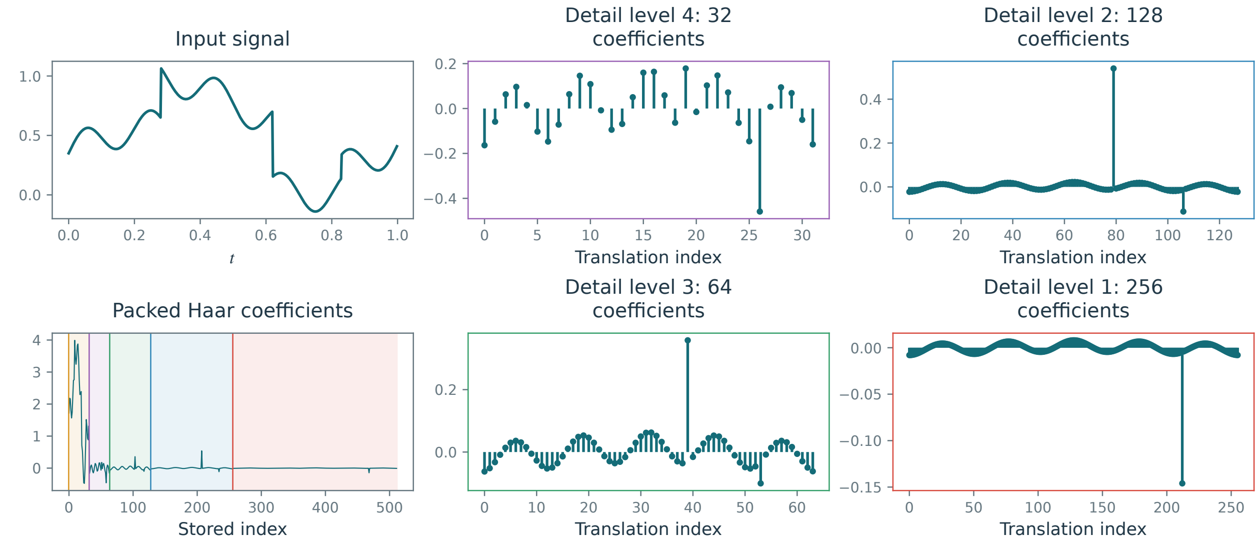


Figure 4.8. Wavelet coefficients. The first column shows the input signal and its packed decomposition; the other panels show individual detail scales.

for the underlying continuous function f. The approximation uses the normalization $\int \varphi = 1$. The equality assumes that acquisition gives exact access to $P_{V_J} f$, much as Shannon sampling assumes bandlimiting. This model is reasonable when the scaling functions $(\varphi_{J,n})_n$ approximate the acquisition device's point-spread function. Preprocessing the measurements can sometimes improve agreement with (4.5).

Starting from a_J, the discrete wavelet transform computes the coefficients

$$\forall\, j \in \{J+1, J+2, \dots, 0\}, \quad \forall\, n \in [\![0, 2^{-j}-1]\!], \quad a_{j,n} \overset{\text{def.}}{=} \langle f, \varphi^P_{j,n} \rangle, \quad \text{and} \quad d_{j,n} \overset{\text{def.}}{=} \langle f, \psi^P_{j,n} \rangle$$

in order of increasing j. After an approximation vector a_j has been used to compute the next scale, it can be discarded; the detail vectors d_j are retained.

The forward discrete wavelet transform on a bounded domain is thus the orthogonal finite-dimensional map

$$a_J \in \mathbb{R}^N \longmapsto \left\{ d_{j,n} \;;\; J < j \leqslant 0,\; 0 \leqslant n < 2^{-j} \right\} \cup \{ a_0 \in \mathbb{R} \}.$$

By orthogonality, the inverse transform is the adjoint.

Figure 4.8 shows examples of wavelet coefficients. For each scale 2^j, there are 2^{-j} coefficients.

4.4.2 Forward Fast Wavelet Transform (FWT)

The algorithm applies a sequence of elementary operators

$$\forall\, j = J+1, \dots, 0, \quad (a_j, d_j) = \mathcal{W}_j(a_{j-1}) \quad \text{where} \quad \mathcal{W}_j : \mathbb{R}^{2^{-j+1}} \to \mathbb{R}^{2^{-j}} \times \mathbb{R}^{2^{-j}} \tag{4.6}$$

There are $|J| = \log_2(N)$ steps. Each $\mathcal{W}_j$ changes coordinates between orthonormal bases and is therefore orthogonal.

To describe the algorithm for $\mathcal{W}_j$, we introduce the filter coefficients $h, g \in \mathbb{R}^{\mathbb{Z}}$

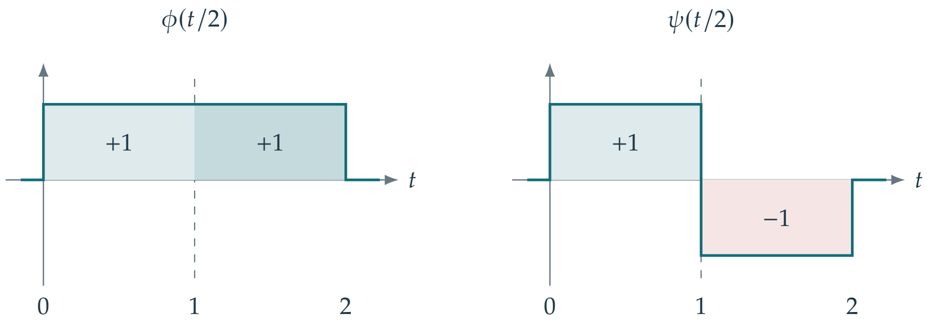
$$h_n \overset{\text{def.}}{=} \frac{1}{\sqrt{2}} \langle \varphi(\cdot/2), \varphi(\cdot - n) \rangle \quad \text{and} \quad g_n \overset{\text{def.}}{=} \frac{1}{\sqrt{2}} \langle \psi(\cdot/2), \varphi(\cdot - n) \rangle. \tag{4.7}$$

Figure 4.9 illustrates the computation of these weights for the Haar system.

We denote by $\downarrow_2 : \mathbb{R}^K \to \mathbb{R}^{K/2}$ the subsampling operator by a factor of 2, i.e.

$$u \downarrow_2 \overset{\text{def.}}{=} (u_0, u_2, u_4, \dots, u_{K-2}).$$

Throughout the following derivation, the scaling function, wavelet, and filters are real, and the filters decay sufficiently fast. The notation $\bar{h}_n = h_{-n}$ and $\bar{g}_n = g_{-n}$ denotes reflection. Complex filters would require conjugation as well.



$\langle \phi(\cdot/2), \phi(\cdot - n)\rangle = 1$ for $n = 0, 1$ $\qquad$ $\langle \psi(\cdot/2), \phi(\cdot - n)\rangle = (-1)^n$ for $n = 0, 1$

$h_0 = h_1 = 1/\sqrt{2}$ $\qquad$ $g_0 = 1/\sqrt{2}, \quad g_1 = -1/\sqrt{2}$

$\phi = \mathbf{1}_{[0,1)}, \quad \psi = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}; \qquad h_n = g_n = 0$ for $n \notin \{0, 1\}$.

Figure 4.9. Haar filter weights as inner products.

Proposition 4.2. *One has*

$$\mathcal{W}_j(a_{j-1}) = ((\bar{h} \star a_{j-1}) \downarrow_2, (\bar{g} \star a_{j-1}) \downarrow_2), \tag{4.8}$$

where $\star$ *denotes periodic convolutions on* $\mathbb{R}^{2^{-j+1}}$.

Proof. We note that $\varphi(\cdot/2)$ and $\psi(\cdot/2)$ belong to V_0, so one has the decompositions

$$\frac{1}{\sqrt{2}}\varphi(t/2) = \sum_n h_n \varphi(t-n) \quad \text{and} \quad \frac{1}{\sqrt{2}}\psi(t/2) = \sum_n g_n \varphi(t-n) \tag{4.9}$$

Making the substitution $t \mapsto \frac{t-2^j p}{2^{j-1}}$ in (4.9), one obtains

$$\frac{1}{\sqrt{2}}\varphi\left(\frac{t-2^j p}{2^j}\right) = \sum_n h_n \varphi\left(\frac{t}{2^{j-1}} - (n+2p)\right)$$

(similarly for ψ) and then reindexing $n \mapsto n - 2p$, one obtains

$$\varphi_{j,p} = \sum_{n\in\mathbb{Z}} h_{n-2p}\varphi_{j-1,n} \quad \text{and} \quad \psi_{j,p} = \sum_{n\in\mathbb{Z}} g_{n-2p}\varphi_{j-1,n}.$$

For periodized functions $(\varphi^P_{j,n}, \psi^P_{j,p})$, these identities remain valid after periodizing the filters modulo 2^{-j+1}. Taking inner products with f on both sides gives the fundamental recursion below; the decay of h, g justifies exchanging the sum and inner product.

$$a_{j,p} = \sum_{n\in\mathbb{Z}} h_{n-2p} a_{j-1,n} = (\bar{h} \star a_{j-1})_{2p} \quad \text{and} \quad d_{j,p} = \sum_{n\in\mathbb{Z}} g_{n-2p} a_{j-1,n} = (\bar{g} \star a_{j-1})_{2p} \tag{4.10}$$

where $\bar{u}_n \overset{\text{def.}}{=} u_{-n}$. On the periodic domain $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, the same recursion holds with $\star$ interpreted as convolution on $\mathbb{Z}/2^{-j+1}\mathbb{Z}$. ∎

Figure 4.10 shows two successive decomposition steps, each separating a coarse approximation from its details.

The FWT thus operates as follows. The same real filters also act on complex-valued input signals:

– **Input:** signal $f \in \mathbb{C}^N$.

– **Initialization:** $a_J = f$.

– **For** $j = J, \ldots, j_0 - 1$.

$$a_{j+1} = (a_j \star \bar{h}) \downarrow 2 \quad \text{and} \quad d_{j+1} = (a_j \star \bar{g}) \downarrow 2$$

– **Output:** the coefficients $\{d_j\}_{J<j\leqslant j_0} \cup \{a_{j_0}\}$.

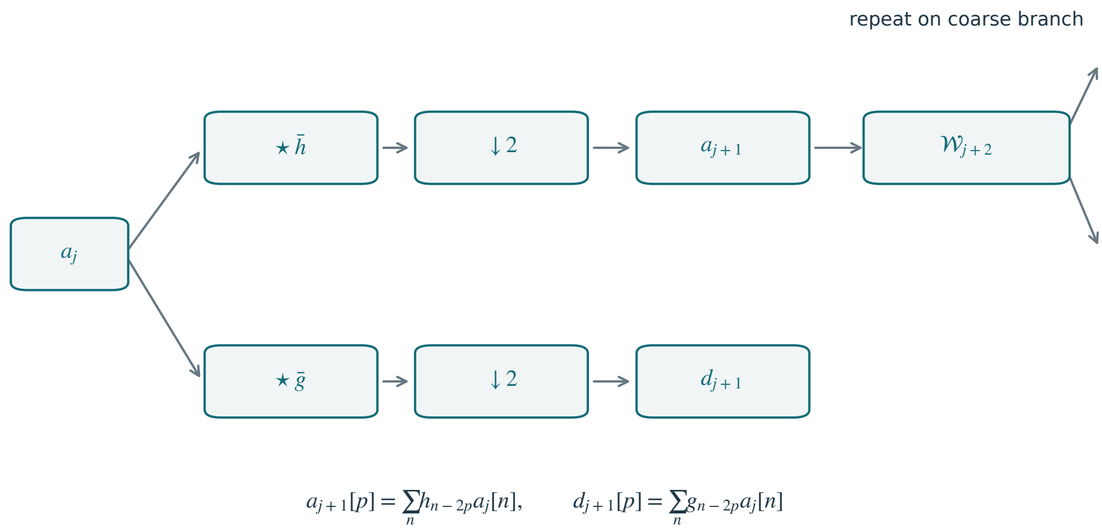


Figure 4.10. Forward filter bank decomposition.

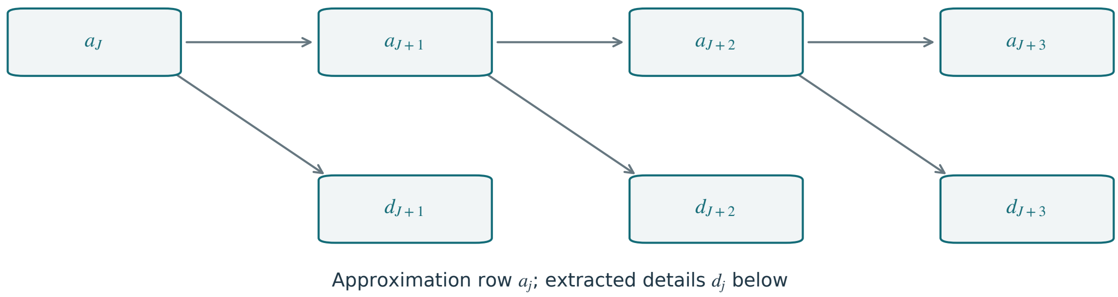


Figure 4.11. Pyramidal computation of wavelet coefficients: approximation vectors a_j on the top row and detail vectors d_j below.

If the supports of h and g each contain at most C indices, then evaluating each $\mathcal{W}_j$ requires $(2C)2^{-j}$ operations, so that the complexity of the whole wavelet transform is

$$\sum_{j=J+1}^{0} (2C)2^{-j} = 2C(2^{-J} - 1) < 2CN.$$

This shows that the fast wavelet transform is a linear-time algorithm. Figure 4.11 shows the iterative extraction of the wavelet coefficients. Figure 4.12 illustrates the storage layout: at each step, the new vectors a_j and d_j occupy the left portion of the output array.

Fast Haar transform. For the Haar wavelets, one has

$$\varphi_{j,n} = \frac{1}{\sqrt{2}}(\varphi_{j-1,2n} + \varphi_{j-1,2n+1}),$$

$$\psi_{j,n} = \frac{1}{\sqrt{2}}(\varphi_{j-1,2n} - \varphi_{j-1,2n+1}).$$

This corresponds to the filters

$$h = [\ldots,\ 0,\ h[0] = \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ 0, \ldots],$$

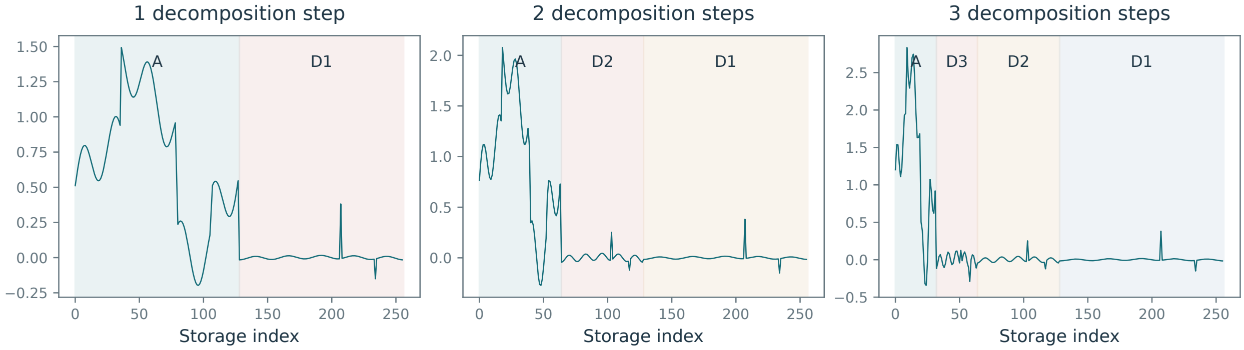


Figure 4.12. Packed coefficient arrays after one, two, and three wavelet decomposition steps.

$$g = [\dots,\ 0,\ g[0] = \frac{1}{\sqrt{2}},\ -\frac{1}{\sqrt{2}},\ 0, \dots].$$

The Haar transform iterates normalized sums and differences:
– **Input:** signal $f \in \mathbb{C}^N$.
– **Initialization:** $a_J = f$.
– **For** $j = J, \dots, j_0 - 1$.

$$a_{j+1,n} = \frac{1}{\sqrt{2}}(a_{j,2n} + a_{j,2n+1}) \quad \text{and} \quad d_{j+1,n} = \frac{1}{\sqrt{2}}(a_{j,2n} - a_{j,2n+1}).$$

– **Output:** the coefficients $\{d_j\}_{J<j\leqslant j_0} \cup \{a_{j_0}\}$.

4.4.3 Inverse Fast Transform (iFWT)

The inverse algorithm proceeds by inverting each step (4.6)

$$\forall\, j = 0, -1, \dots, J+1, \quad a_{j-1} = \mathcal{W}_j^{-1}(a_j, d_j) = \mathcal{W}_j^*(a_j, d_j), \tag{4.11}$$

where $\mathcal{W}_j^*$ is the adjoint for the canonical inner product on $\mathbb{R}^{2^{-j+1}}$; in matrix form, it is the transpose.

Let $\uparrow_2 \colon \mathbb{R}^{K/2} \to \mathbb{R}^K$ denote up-sampling by inserting zeros:

$$a \uparrow_2 = (a_0, 0, a_1, 0, \dots, a_{K/2-1}, 0) \in \mathbb{R}^K.$$

Proposition 4.3. *One has*

$$\mathcal{W}_j^{-1}(a_j, d_j) = (a_j \uparrow_2) \star h + (d_j \uparrow_2) \star g.$$

Proof. Since $\mathcal{W}_j$ is orthogonal, $\mathcal{W}_j^{-1} = \mathcal{W}_j^*$. We write the whole transform as

$$\mathcal{W}_j = S_2 \circ C_{\bar{h},\bar{g}} \circ \mathcal{D} \quad \text{where} \quad \begin{cases} \mathcal{D}(a) = (a, a), \\ C_{\bar{h},\bar{g}}(a, b) = (\bar{h} \star a, \bar{g} \star b), \\ S_2(a, b) = (a \downarrow_2, b \downarrow_2). \end{cases}$$

The adjoints are

$$\mathcal{D}^*(a, b) = a + b, \quad C^*_{\bar{h},\bar{g}}(a, b) = (h \star a, g \star b), \quad \text{and} \quad S_2^*(a, b) = (a \uparrow_2, b \uparrow_2).$$

To verify the convolution adjoint, assume the sequences are summable; in the periodic setting, the sums are finite:

$$\langle f \star \bar{h},\ g \rangle = \sum_n (f \star \bar{h})_n g_n = \sum_n \sum_k f_k h_{k-n} g_n = \sum_k f_k \sum_n h_{k-n} g_n = \langle f,\ h \star g \rangle,$$

For the copying operator,

$$\langle \mathcal{D}(a),\ (u, v) \rangle = \langle a,\ u \rangle + \langle a,\ v \rangle = \langle a,\ u + v \rangle = \langle a,\ \mathcal{D}^*(u, v) \rangle,$$

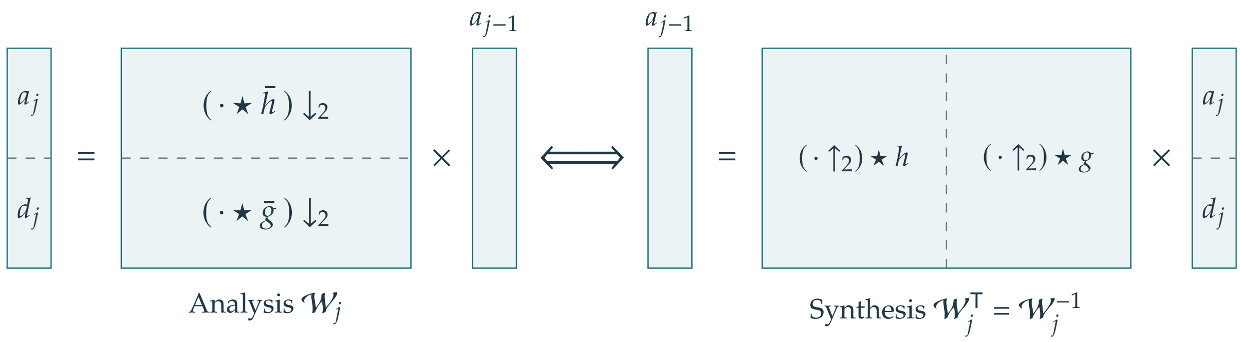


Figure 4.13. Analysis and inverse wavelet transforms in block-matrix form. Synthesis uses the transpose of the orthogonal analysis operator.

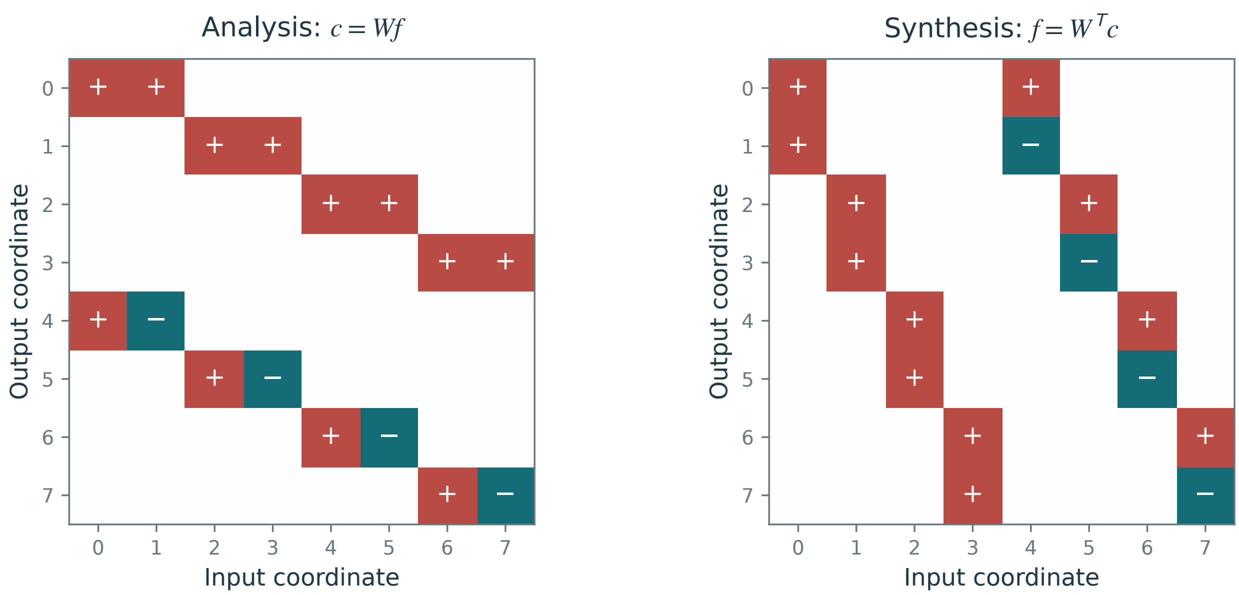


Figure 4.14. One-step Haar analysis matrix W and synthesis matrix W^{T} for eight samples.

and for down-sampling,

$$\langle f \downarrow_2, g \rangle = \sum_n f_{2n} g_n = \sum_n (f_{2n} g_n + f_{2n+1} 0) = \langle f, g \uparrow_2 \rangle.$$

This is shown using matrix notation in Figure 4.13. Composing these adjoints in reverse order gives the reconstruction formula. □

Figure 4.14 makes this transpose relation concrete for one Haar step.

The inverse fast wavelet transform iteratively applies this elementary step

– **Input:** $\{d_j\}_{J<j\leqslant j_0} \cup \{a_{j_0}\}$.

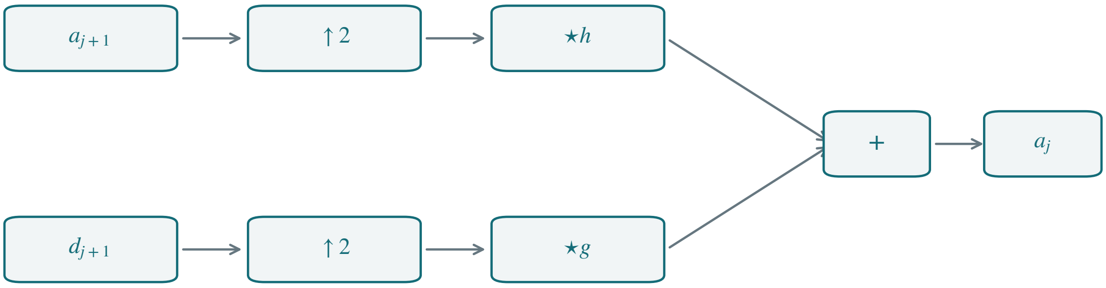
– **For** $j = j_0, j_0 - 1, \ldots, J + 1$.

$$a_{j-1} = (a_j \uparrow 2) \star h + (d_j \uparrow 2) \star g.$$

– **Output:** $f = a_J$.

The block diagram in Figure 4.15 reverses the operations of the forward transform in Figure 4.10.

4.5 2-D Wavelets



$$a_j = (a_{j+1} \uparrow 2) \star h + (d_{j+1} \uparrow 2) \star g$$

Figure 4.15. Inverse filter-bank reconstruction.

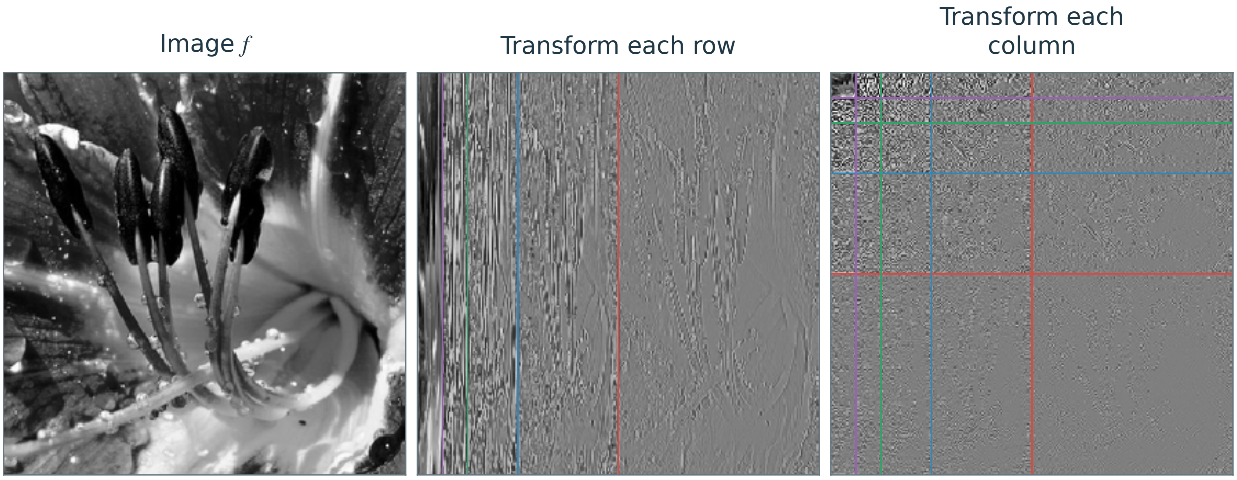


Figure 4.16. Steps of the anisotropic wavelet transform.

4.5.1 Anisotropic Wavelets

The anisotropic basis on $\mathbb{T}^2$ is the tensor product of the complete one-dimensional periodized basis $\mathcal{B}$ (including its coarsest scaling functions):

$$\{b_1 \otimes b_2 : b_1, b_2 \in \mathcal{B}\}.$$

Its wavelet–wavelet elements have the form

$$\psi_{(j_1,j_2),(n_1,n_2)}(x_1, x_2) = \psi^P_{j_1,n_1}(x_1)\psi^P_{j_2,n_2}(x_2). \tag{4.12}$$

The basis also includes wavelet–scaling, scaling–wavelet, and scaling–scaling products. The anisotropic 2-D transform follows the same separable construction as the 2-D FFT in Section 2.5.2. Treat the image $a_J \in \mathbb{R}^{2^{-J} \times 2^{-J}}$ as a matrix, apply the 1-D FWT to each row, and then apply it to each column. The total cost is $O(N)$ for $N = 2^{-2J}$ pixels.

4.5.2 Isotropic Wavelets

The anisotropic atoms (4.12) have independent scales in the two coordinate directions. For compactly supported wavelets, their supports lie in axis-aligned rectangles of size proportional to $2^{j_1} \times 2^{j_2}$. This directional bias can produce visible artifacts in denoising and compression when it does not match the image geometry.

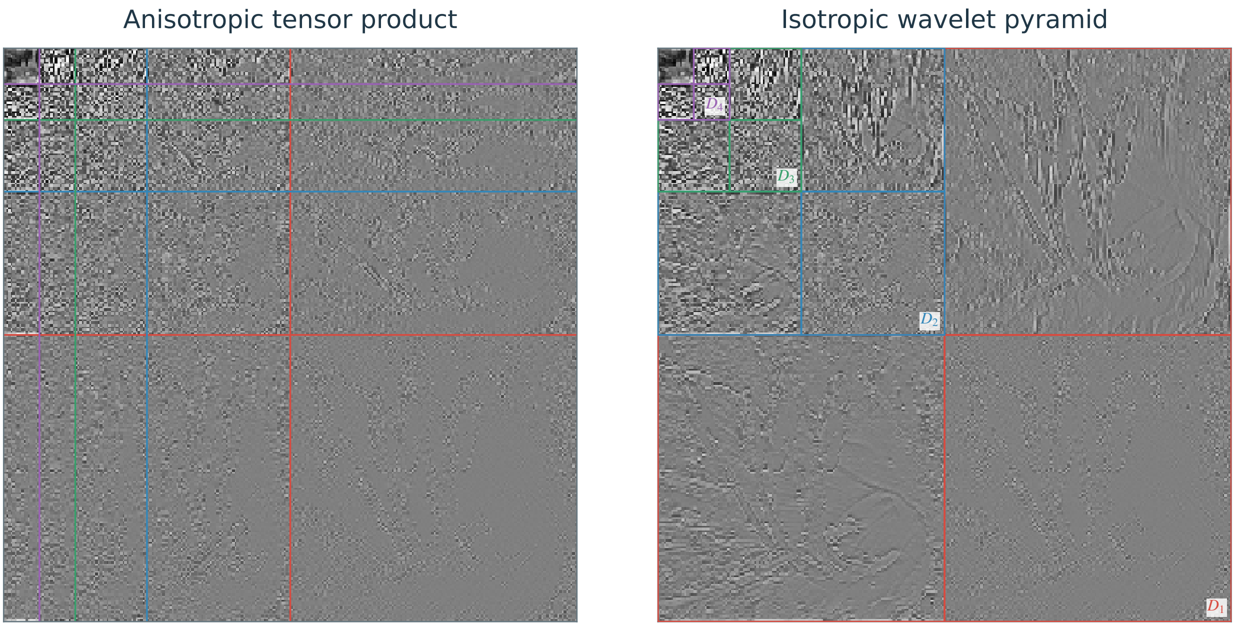


Figure 4.17. Anisotropic (left) versus isotropic (right) wavelet coefficients.

An alternative uses a common scale in both directions. Tensor products of one-dimensional approximation spaces give a two-dimensional multiresolution

$$L^2(\mathbb{R}^2) \supset \ldots \supset V_{j-1} \otimes V_{j-1} \supset V_j \otimes V_j \supset V_{j+1} \otimes V_{j+1} \supset \ldots \supset \{0\}.$$

We write

$$V_j^O \stackrel{\text{def.}}{=} V_j \otimes V_j$$

for this isotropic 2-D approximation space.

Recall that the tensor product of two closed subspaces $V_1, V_2 \subset L^2(\mathbb{R})$ is

$$V_1 \otimes V_2 = \text{Closure}\left(\text{Span}\left\{f_1(x_1)f_2(x_2) \in L^2(\mathbb{R}^2)\ ;\ f_1 \in V_1, f_2 \in V_2\right\}\right).$$

If $(\varphi_k^s)_k$ are orthonormal bases of V_s, their tensor products $(\varphi^1_{k_1}(x_1)\varphi^2_{k_2}(x_2))_{k_1,k_2}$ form an orthonormal basis of $V_1 \otimes V_2$.

The tensor product distributes over these orthogonal sums

$$(V_j \oplus^\perp W_j) \otimes (V_j \oplus^\perp W_j) = V_j^O \oplus^\perp W_j^V \oplus^\perp W_j^H \oplus^\perp W_j^D \quad \text{where} \quad \begin{cases} W_j^V \stackrel{\text{def.}}{=} (V_j \otimes W_j), \\ W_j^H \stackrel{\text{def.}}{=} (W_j \otimes V_j), \\ W_j^D \stackrel{\text{def.}}{=} (W_j \otimes W_j). \end{cases}$$

The labels $\{V, H, D\}$ stand for *Vertical, Horizontal, Diagonal* detail spaces. The following diagram summarizes the nested spaces and their detail components:

$$\begin{tikzcd}
\cdots \arrow[r] \arrow[dr] & V_{j-1} \otimes V_{j-1} \arrow[r] \arrow[dr] & V_j \otimes V_j \arrow[r] \arrow[dr] & V_{j+1} \otimes V_{j+1} \arrow[r] & \cdots \\
& W_{j-1}^{(2)} & W_j^{(2)} & W_{j+1}^{(2)} &
\end{tikzcd}$$

$$V_{j-1} \otimes V_{j-1} = (V_j \otimes V_j) \oplus^\perp W_j^{(2)}$$

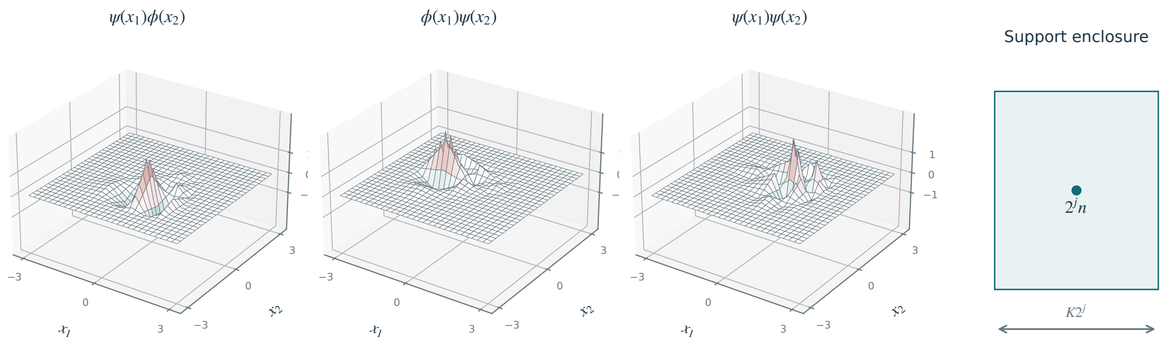


Figure 4.18. 2-D wavelets and a schematic support centered at $(2^j n_1, 2^j n_2)$, with width $K2^j$ (right).

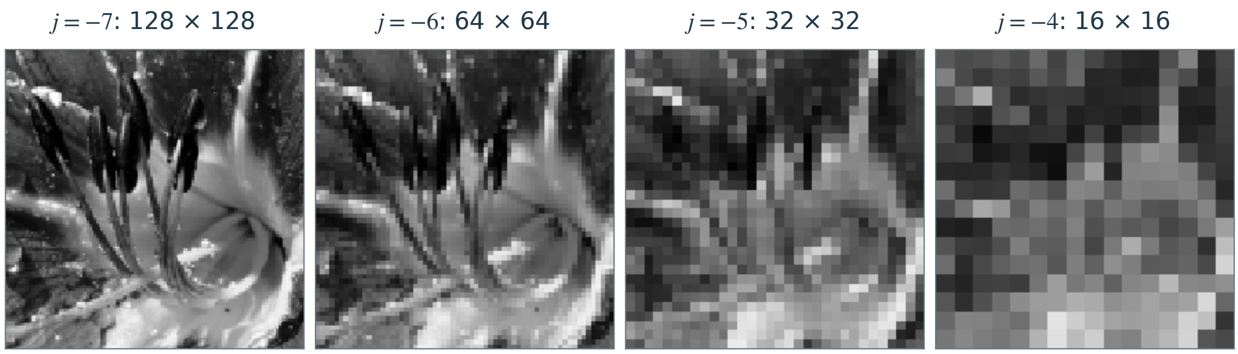


Figure 4.19. 2-D Haar approximation $P_{V_j^O} f$ for increasing j.

For $j \in \mathbb{Z}$, translates of a mother wavelet span each detail space. We index them by $n = (n_1, n_2) \in \mathbb{Z}^2$, or by indices in $\{0, \ldots, 2^{-j} - 1\}^2$ on the torus $\mathbb{T}^2$:

$$\forall\, \omega \in \{V, H, D\}, \quad W_j^\omega = \mathrm{Span}\{\psi^\omega_{j,n_1,n_2}\}_{n_1,n_2}$$

where

$$\forall\, \omega \in \{V, H, D\}, \quad \psi^\omega_{j,n_1,n_2}(x) = \frac{1}{2^j}\psi^\omega\left(\frac{x_1 - 2^j n_1}{2^j}, \frac{x_2 - 2^j n_2}{2^j}\right)$$

and where the three mother wavelets are

$$\psi^H(x) = \psi(x_1)\varphi(x_2), \quad \psi^V(x) = \varphi(x_1)\psi(x_2), \quad \text{and} \quad \psi^D(x) = \psi(x_1)\psi(x_2).$$

Here the span is closed in L^2 on an unbounded domain; on the torus, the wavelets are periodized. Figure 4.18 displays examples of these wavelets.

Haar 2-D multiresolution. The 2-D Haar construction gives piecewise constant approximations: functions in $V_j \otimes V_j$ are constant on squares of size $2^j \times 2^j$. Figure 4.19 shows the corresponding projections of an image.

Discrete 2-D wavelet coefficients. As in (4.5), assume that acquisition provides inner products of the continuous signal f with scaling functions at scale 2^J, with $N = 2^{-J}$ samples per direction:

$$\forall\, n \in \{0, \ldots, N-1\}^2, \quad a_{J,n} = \langle f, (\varphi_{J,n_1} \otimes \varphi_{J,n_2})^P \rangle$$

Discrete wavelet coefficients are defined as

$$\forall\, \omega \in \{V, H, D\},\ \forall\, J < j \leqslant 0,\ \forall\, 0 \leqslant n_1, n_2 < 2^{-j}, \quad d^\omega_{j,n} = \langle f, \psi^\omega_{j,n} \rangle.$$

Here the wavelets are periodized. Approximation coefficients are defined as

$$a_{j,n} = \langle f, (\varphi_{j,n_1} \otimes \varphi_{j,n_2})^P \rangle.$$

Figure 4.20 shows how the wavelet coefficients are arranged in an image of N^2 pixels. Figure 4.21 shows other examples of wavelet decompositions.

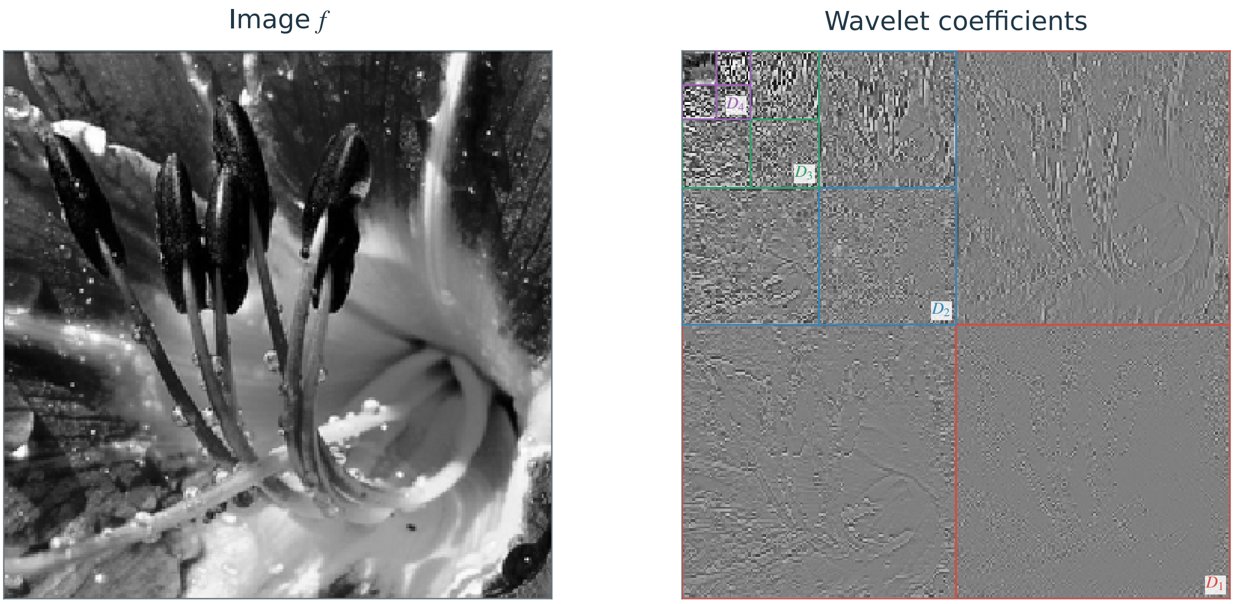


Figure 4.20. 2-D wavelet coefficients.

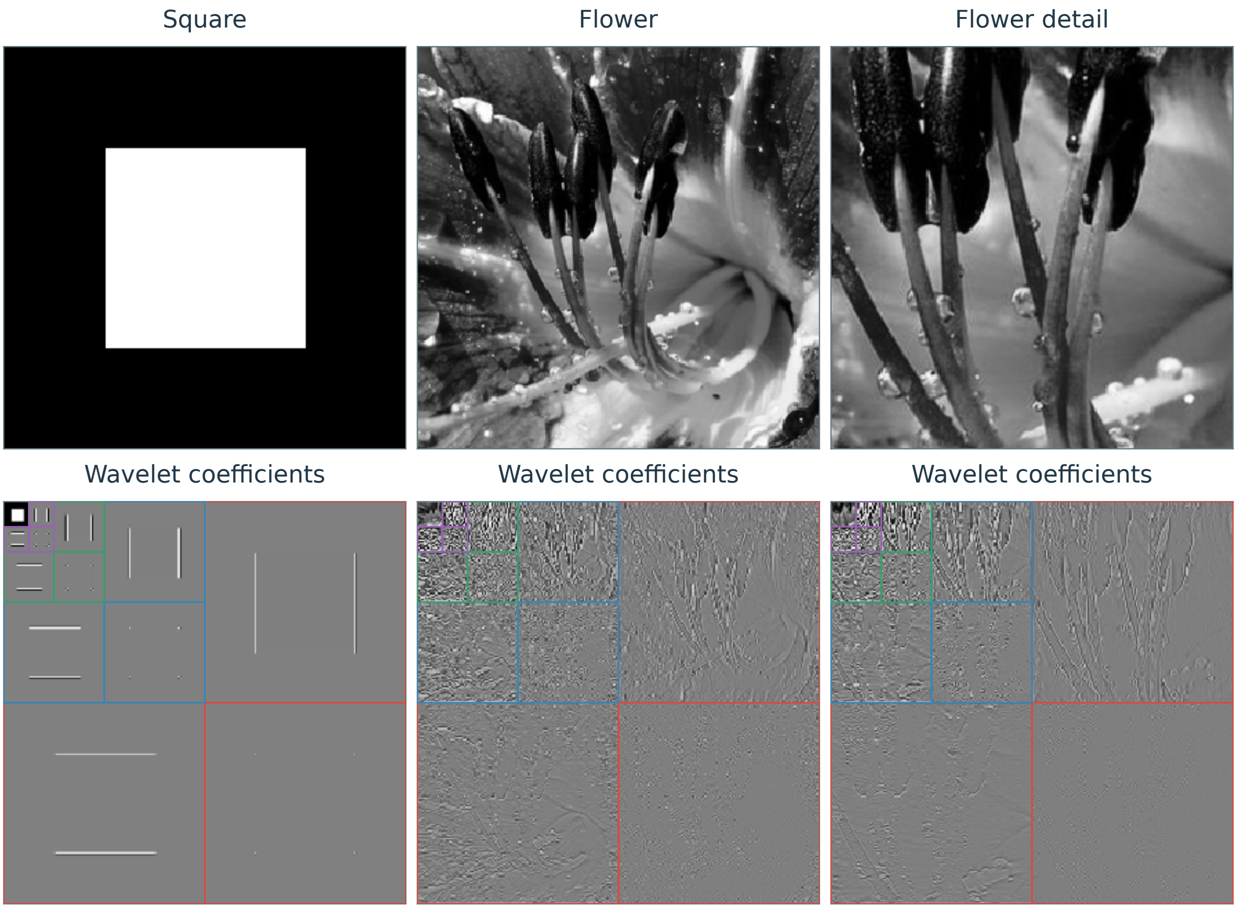


Figure 4.21. Images (top) and their wavelet coefficients (bottom).

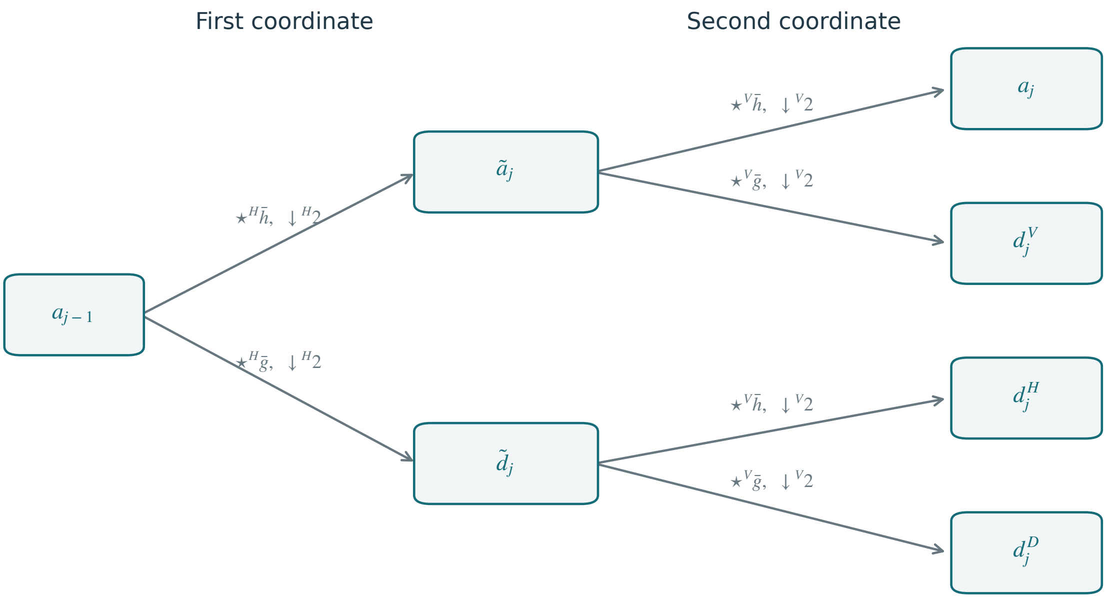


Figure 4.22. Forward 2-D filterbank step.

One step of the forward 2-D transform. Each step separates fine-scale approximation coefficients into three detail arrays and a coarse approximation:

$$a_{j-1} \longmapsto (a_j, d_j^H, d_j^V, d_j^D).$$

As in one dimension, this map is orthogonal. It is implemented by applying the filtering and sub-sampling formula (4.8) in each coordinate direction.

One first applies 1-D horizontal filtering and sub-sampling

$$\tilde{a}_j = (a_{j-1} \star^H \bar{h}) \downarrow^H 2$$
$$\tilde{d}_j = (a_{j-1} \star^H \bar{g}) \downarrow^H 2,$$

where $\star^H$ denotes convolution along the first coordinate (called horizontal here), applied to each column of the array

$$a \star^H b_{n_1,n_2} = \sum_{m_1=0}^{P-1} a_{n_1-m_1,n_2} b_{m_1}$$

Here $a \in \mathbb{C}^{P\times P}$ is a matrix and $b \in \mathbb{C}^P$ is a filter. The notation $\downarrow^H 2$ denotes sub-sampling in the horizontal direction

$$(a \downarrow^H 2)_{n_1,n_2} = a_{2n_1,n_2}.$$

One then applies 1-D vertical filtering and sub-sampling to $\tilde{a}_j$ and $\tilde{d}_j$ to obtain

$$\begin{aligned} a_j &= (\tilde{a}_j \star^V \bar{h}) \downarrow^V 2, & d_j^H &= (\tilde{d}_j \star^V \bar{h}) \downarrow^V 2, \\ d_j^V &= (\tilde{a}_j \star^V \bar{g}) \downarrow^V 2, & d_j^D &= (\tilde{d}_j \star^V \bar{g}) \downarrow^V 2, \end{aligned}$$

where the vertical operators are defined analogously to the horizontal operators, acting on rows.

These two forward steps are shown in the block diagram in Figure 4.22. The updates can be performed in place, storing all coefficients in an image of N^2 pixels, as shown in Figure 4.23. This produces the familiar multiscale arrangement used in Figure 4.21.

Fast 2-D wavelet transform. The 2-D FWT algorithm iterates these steps through the scales:

– **Input:** image $f \in \mathbb{C}^{N\times N}$.

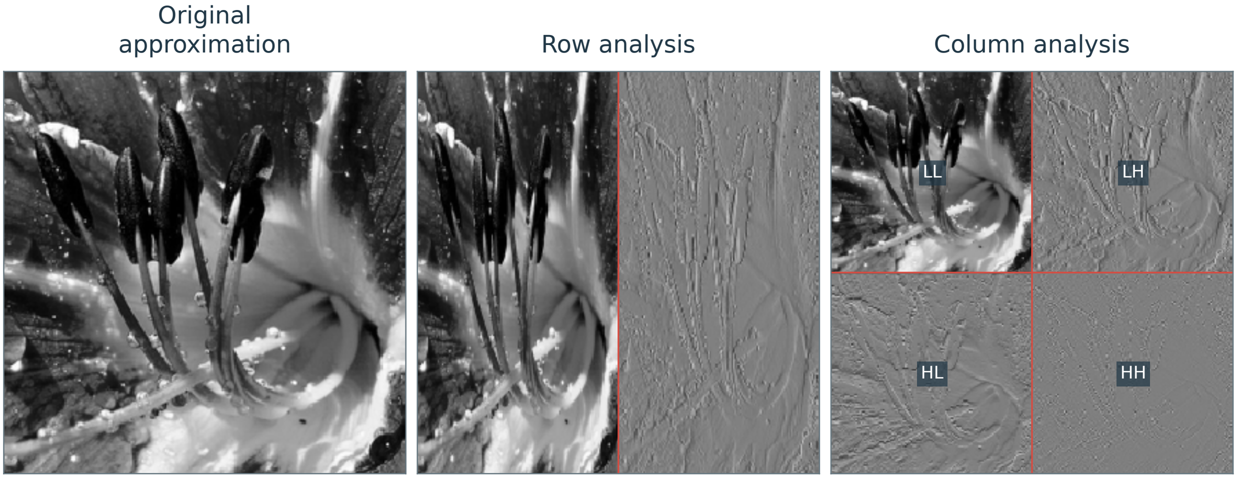


Figure 4.23. One step of the 2-D wavelet transform algorithm.

– **Initialization:** $a_J = f$.
– **For** $j = J+1, \ldots, j_0$.

$$\begin{aligned} \tilde{a}_j &= (a_{j-1} \star^H \bar{h}) \downarrow^H 2, & d_j^V &= (\tilde{a}_j \star^V \bar{g}) \downarrow^V 2, \\ \tilde{d}_j &= (a_{j-1} \star^H \bar{g}) \downarrow^H 2, & d_j^H &= (\tilde{d}_j \star^V \bar{h}) \downarrow^V 2, \\ a_j &= (\tilde{a}_j \star^V \bar{h}) \downarrow^V 2, & d_j^D &= (\tilde{d}_j \star^V \bar{g}) \downarrow^V 2. \end{aligned}$$

– **Output:** the coefficients $\{d_j^\omega\}_{J<j\leqslant j_0,\omega} \cup \{a_{j_0}\}$.

Fast 2-D inverse wavelet transform. The inverse transform undoes the horizontal and vertical filtering steps. The first step computes

$$\begin{aligned} \tilde{a}_j &= (a_j \uparrow^V 2) \star^V h + (d_j^V \uparrow^V 2) \star^V g, \\ \tilde{d}_j &= (d_j^H \uparrow^V 2) \star^V h + (d_j^D \uparrow^V 2) \star^V g, \end{aligned}$$

where the vertical up-sampling is

$$(a \uparrow^V 2)_{n_1,n_2} = \begin{cases} a_{n_1,k} & \text{if} \quad n_2 = 2k, \\ 0 & \text{if} \quad n_2 = 2k+1. \end{cases}$$

The second inverse step computes

$$a_{j-1} = (\tilde{a}_j \uparrow^H 2) \star^H h + (\tilde{d}_j \uparrow^H 2) \star^H g.$$

Figure 4.24 gives the inverse filter-bank diagram corresponding to Figure 4.22.

The inverse fast wavelet transform iteratively applies these elementary steps

– **Input:** $\{d_j^\omega\}_{J<j\leqslant j_0,\omega} \cup \{a_{j_0}\}$.
– **For** $j = j_0, j_0 - 1, \ldots, J+1$.

$$\begin{aligned} \tilde{a}_j &= (a_j \uparrow^V 2) \star^V h + (d_j^V \uparrow^V 2) \star^V g, \\ \tilde{d}_j &= (d_j^H \uparrow^V 2) \star^V h + (d_j^D \uparrow^V 2) \star^V g, \\ a_{j-1} &= (\tilde{a}_j \uparrow^H 2) \star^H h + (\tilde{d}_j \uparrow^H 2) \star^H g. \end{aligned}$$

– **Output:** $f = a_J$.

4.6 Wavelet Design

Implementing the FWT requires the filters h and g, rather than explicit formulas for the scaling function φ and wavelet ψ. Most useful wavelets have no elementary closed form; their refinement equations define them implicitly, and the cascade algorithm approximates them.

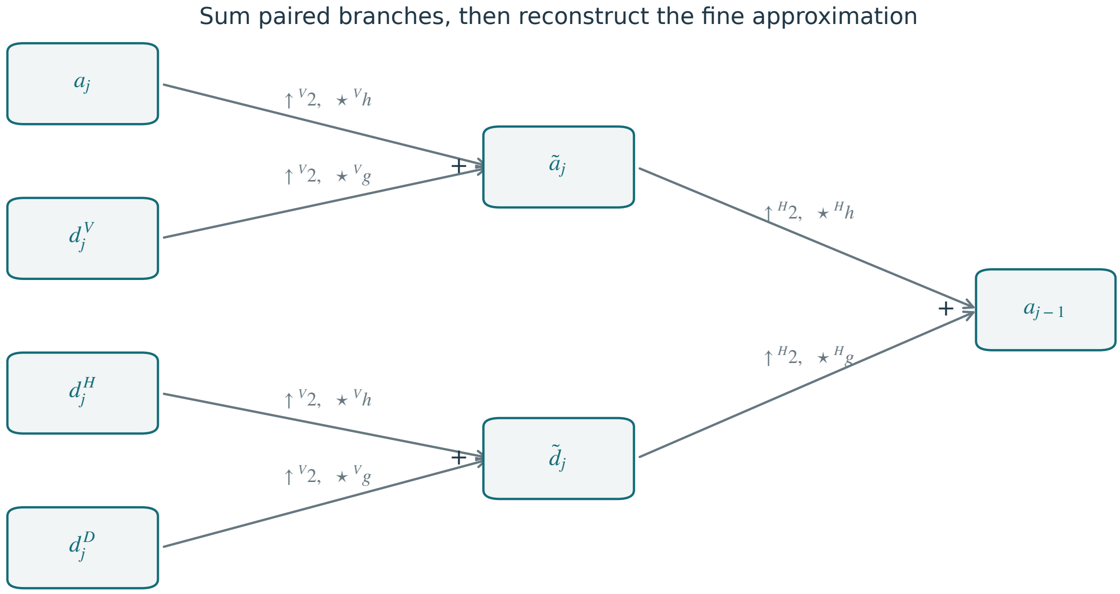


Figure 4.24. Backward 2-D filterbank step.

This section derives constraints on the filters and gives examples. Under the usual quadrature-mirror construction, the low-pass filter h determines the high-pass filter g.

4.6.1 Low-pass Filter Constraints

We introduce the following three conditions on a filter h

$$\hat{h}(0) = \sqrt{2} \tag{C_1}$$

$$|\hat{h}(\omega)|^2 + |\hat{h}(\omega+\pi)|^2 = 2, \tag{C_2}$$

$$\inf_{\omega\in[-\pi/2,\pi/2]} |\hat{h}(\omega)| > 0. \tag{C^*}$$

These conditions use the Fourier series of the filter $h \in \mathbb{R}^{\mathbb{Z}}$:

$$\forall \omega \in \mathbb{R}/2\pi\mathbb{Z}, \quad \hat{h}(\omega) \overset{\text{def.}}{=} \sum_{n\in\mathbb{Z}} h_n e^{-\mathrm{i}n\omega}. \tag{4.13}$$

If $h \in \ell^1(\mathbb{Z})$, this defines a continuous periodic function $\hat{h} \in C^0(\mathbb{R}/2\pi\mathbb{Z})$, and the definition extends to $h \in \ell^2(\mathbb{Z})$ to give $\hat{h} \in L^2(\mathbb{R}/2\pi\mathbb{Z})$.

Theorem 4.4. *If an integrable real scaling function φ defines a multiresolution analysis and its filter h is summable, then (C_1) and (C_2) hold for h defined in (4.7). Conversely, if a finitely supported real filter h satisfies (C_1), (C_2), and (C^*), then there exists a φ defining multi-resolution approximation spaces whose associated filter is h as defined in (4.7).*

Proof. We prove the forward implication. The converse requires a separate construction.

We now prove condition (C_1). The refinement equation convolves a function on the real line with a discrete filter, represented by a sum of Dirac masses:

$$\frac{1}{\sqrt{2}}\varphi\left(\frac{t}{2}\right) = \sum_{n\in\mathbb{Z}} h_n \varphi(t-n). \tag{4.14}$$

Writing $h \star \varphi$ for this convolution and assuming $h \in \ell_1(\mathbb{Z})$ and $\varphi \in L^1(\mathbb{R})$, Fubini's theorem shows that

$h \star \varphi \in L^1(\mathbb{R})$ and gives

$$\mathcal{F}(\sum_{n\in\mathbb{Z}} h_n \varphi(t-n))(\omega) = \int_{\mathbb{R}} \sum_{n\in\mathbb{Z}} h_n \varphi(t-n) e^{-\mathrm{i}\omega t} \mathrm{d}t = \sum_{n\in\mathbb{Z}} h_n \int_{\mathbb{R}} \varphi(t-n) e^{-\mathrm{i}\omega t} \mathrm{d}t$$
$$= \sum_{n\in\mathbb{Z}} h_n e^{-in\omega} \int_{\mathbb{R}} \varphi(x) e^{-\mathrm{i}\omega x} \mathrm{d}x = \hat{\varphi}(\omega)\hat{h}(\omega)$$

where we substituted $x = t - n$. Here $\hat{h}(\omega)$ is the 2π-periodic Fourier series of the filter, defined in (4.13), while $\hat{\varphi}(\omega)$ is the continuous Fourier transform of the scaling function. The product therefore combines a 2π-periodic factor $\hat{h}$ with a generally nonperiodic factor $\hat{\varphi}$. We recall that $\mathcal{F}(f(\cdot/s)) = s\hat{f}(s\cdot)$. In the Fourier domain, equation (4.14) thus reads

$$\hat{\varphi}(2\omega) = \frac{1}{\sqrt{2}}\hat{h}(\omega)\hat{\varphi}(\omega). \tag{4.15}$$

One can show that $\hat{\varphi}(0) \neq 0$ (actually, $|\hat{\varphi}(0)| = 1$), so that this relation implies the first condition (C_1).

We now prove condition (C_2). The orthogonality of $\{\varphi(\cdot - n)\}_n$ can be expressed as a continuous convolution (see also Proposition 4.1)

$$\forall\, n \in \mathbb{Z}, \quad (\varphi \star \bar{\varphi})(n) = \delta_{n,0}$$

where $\bar{\varphi}(x) = \varphi(-x)$, and thus in the Fourier domain, using (4.15) which shows $\hat{\varphi}(\omega) = \frac{1}{\sqrt{2}}\hat{h}(\omega/2)\hat{\varphi}(\omega/2)$

$$1 = \sum_k |\hat{\varphi}(\omega + 2k\pi)|^2 = \frac{1}{2}\sum_k |\hat{h}(\omega/2 + k\pi)|^2 |\hat{\varphi}(\omega/2 + k\pi)|^2.$$

Since $\hat{h}$ is 2π-periodic, one can separate even and odd indices k and obtain

$$2 = |\hat{h}(\omega/2)|^2 \sum_k |\hat{\varphi}(\omega/2 + 2k\pi)|^2 + |\hat{h}(\omega/2 + \pi)|^2 \sum_k |\hat{\varphi}(\omega/2 + 2k\pi + \pi)|^2$$

To obtain condition (C_2), apply $\sum_k |\hat{\varphi}(\omega + 2k\pi)|^2 = 1$ at $\omega' = \omega/2$ instead of ω:

$$|\hat{h}(\omega')|^2 + |\hat{h}(\omega' + \pi)|^2 = 2.$$

The converse requires constructing a scaling function φ from the filter h; we do not prove it here. Iterating (4.15) suggests the infinite-product construction

$$\hat{\varphi}(\omega) = \prod_{k=1}^{\infty} \frac{\hat{h}(\omega/2^k)}{\sqrt{2}}. \tag{4.16}$$

Condition (C^*) can be shown to imply that this infinite product converges and defines a (non-periodic) function in $L^2(\mathbb{R})$. □

Condition (C^*) prevents zeros of $\hat{h}$ on the central half-period. It is a sufficient nondegeneracy condition for the converse construction used here.

4.6.2 High-pass Filter Constraints

We now introduce the following two conditions on a pair of filters (g, h)

$$|\hat{g}(\omega)|^2 + |\hat{g}(\omega + \pi)|^2 = 2 \tag{C_3}$$
$$\hat{g}(\omega)\hat{h}(\omega)^* + \hat{g}(\omega + \pi)\hat{h}(\omega + \pi)^* = 0. \tag{C_4}$$

Figure 4.25 illustrates these filter constraints.

Theorem 4.5. *If (φ, ψ) defines a multi-resolution analysis, then (C_3) and (C_4) hold for (h, g) defined in (4.7). Conversely, if the finitely supported real low-pass filter satisfies (C_1)–(C^*), and a summable real filter g satisfies (C_3)–(C_4), then there is a multiresolution wavelet pair (φ, ψ) whose associated filters are (h, g) as defined in (4.7). Furthermore,*

$$\hat{\psi}(\omega) = \frac{1}{\sqrt{2}}\hat{g}(\omega/2)\hat{\varphi}(\omega/2). \tag{4.17}$$

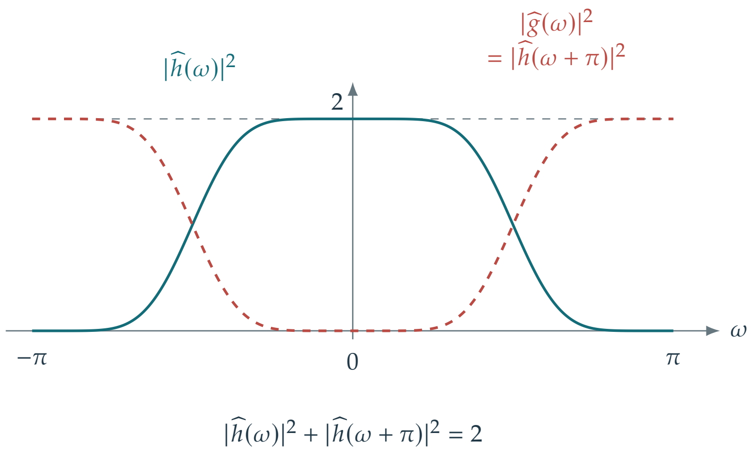


Figure 4.25. Constraints on low-pass and high-pass filters.

Proof. We prove the forward implication, beginning with condition (C_3). The refinement equation for the wavelet reads

$$\frac{1}{\sqrt{2}}\psi\left(\frac{t}{2}\right) = \sum_{n\in\mathbb{Z}} g_n \varphi(t-n)$$

and thus in the Fourier domain

$$\hat{\psi}(2\omega) = \frac{1}{\sqrt{2}}\hat{g}(\omega)\hat{\varphi}(\omega). \tag{4.18}$$

Orthogonality of the translates $\{\psi(\cdot - n)\}_n$ gives

$$\forall\, n \in \mathbb{Z}, \quad (\psi \star \bar{\psi})(n) = \delta_{n,0}$$

and thus in the Fourier domain (using Poisson's formula, see also Proposition 4.1)

$$\sum_k |\hat{\psi}(\omega + 2k\pi)|^2 = 1.$$

Apply the Fourier refinement equation (4.18) and separate the even and odd terms, as in the proof of condition (C_2) in Theorem 4.4. This gives condition (C_3). Figure 4.25 shows the squared Fourier magnitudes of a pair of filters satisfying this complementarity condition.

We now prove condition (C_4). The orthogonality between $\{\psi(\cdot - n)\}_n$ and $\{\varphi(\cdot - n)\}_n$ is written as

$$\forall\, n \in \mathbb{Z}, \quad \psi \star \bar{\varphi}(n) = 0$$

and hence in the Fourier domain (using Poisson's formula, similarly to Proposition 4.1)

$$\sum_k \hat{\psi}(\omega + 2k\pi)\hat{\varphi}^*(\omega + 2k\pi) = 0.$$

Using the Fourier domain refinement equations (4.15) and (4.18), this is equivalent to condition (C_4). □

Quadrature mirror filters. Quadrature mirror filters (QMF) define g as a function of h so that the conditions of Theorem 4.5 are automatically satisfied. Most wavelet constructions implicitly use this choice (other phase choices can yield different wavelets spanning the same detail spaces).

Proposition 4.6. *For a real filter $h \in \ell^1(\mathbb{Z})$ satisfying (C_2), the filter $g \in \ell^2(\mathbb{Z})$ defined by*

$$\forall\, n \in \mathbb{Z}, \quad g_n = (-1)^{1-n} h_{1-n} \tag{4.19}$$

satisfies conditions (C_3) and (C_4).

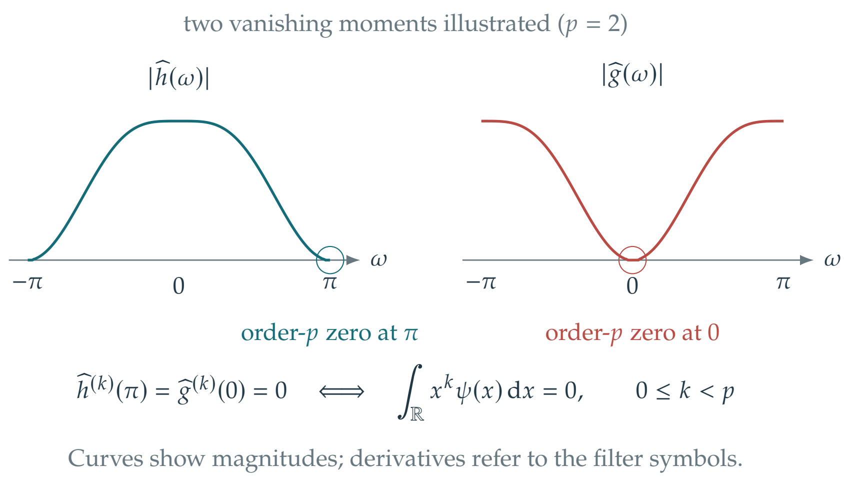


Figure 4.26. Vanishing moments correspond to vanishing derivatives of the Fourier transform.

Proof. Taking the Fourier series of the defining relation gives

$$\hat{g}(\omega) = e^{-i\omega}\hat{h}(\omega + \pi)^*, \tag{4.20}$$

so that

$$|\hat{g}(\omega)|^2 + |\hat{g}(\omega+\pi)|^2 = |e^{-i\omega}\hat{h}(\omega+\pi)^*|^2 + |e^{-i(\omega+\pi)}\hat{h}(\omega+2\pi)^*|^2 = |\hat{h}(\omega+\pi)|^2 + |\hat{h}(\omega+2\pi)|^2 = 2.$$

where we used the fact that $\hat{h}$ is 2π-periodic, and also

$$\begin{aligned}\hat{g}(\omega)\hat{h}(\omega)^* + \hat{g}(\omega+\pi)\hat{h}(\omega+\pi)^* &= e^{-i\omega}\hat{h}(\omega+\pi)^*\hat{h}(\omega)^* + e^{-i(\omega+\pi)}\hat{h}(\omega+2\pi)^*\hat{h}(\omega+\pi)^* \\ &= (e^{-i\omega} + e^{-i(\omega+\pi)})\hat{h}(\omega+\pi)^*\hat{h}(\omega)^* = 0.\end{aligned}$$

□

4.6.3 Wavelet Design Constraints

One can construct a multiresolution wavelet pair by designing a finite filter h satisfying (C_1)–(C^*). The scaling function is obtained through the infinite product (4.16). Except in special cases such as Haar wavelets, it has no elementary closed form. The QMF relation (4.19) then defines g, and (4.17) defines ψ.

Fourier atoms are fixed exponentials, whereas mother wavelets can be designed to meet several requirements:

– Size of the support.
– Cancellation of polynomials, measured by the number p of vanishing moments.
– Symmetry (for compactly supported scalar orthogonal wavelets, the Haar system is the exceptional symmetric case).
– Smoothness (number of derivatives).

We now explain how these design requirements interact with conditions (C_1)–(C_4).

Vanishing moments. A wavelet ψ has p vanishing moments if

$$\forall\, k = 0, \ldots, p-1, \quad \int_{\mathbb{R}} \psi(x)\, x^k \mathrm{d}x = 0. \tag{4.21}$$

Vanishing moments make $\langle f, \psi_{j,n}\rangle$ small when f has regularity C^α with $\alpha < p$ on $\mathrm{Supp}(\psi_{j,n})$. A Taylor expansion of f about $2^j n$ on the wavelet's support explains this cancellation. The vanishing-moment condition has the following equivalent Fourier characterization; see Figure 4.26.

Proposition 4.7. *Assuming the required moments of ψ and derivatives of the filter symbols exist, a wavelet obtained by the QMF construction (4.19) has p vanishing moments if and only if*

$$\forall\, k = 0, \ldots, p-1, \quad \frac{\mathrm{d}^k\hat{h}}{\mathrm{d}\omega^k}(\pi) = \frac{\mathrm{d}^k\hat{g}}{\mathrm{d}\omega^k}(0) = 0. \tag{4.22}$$

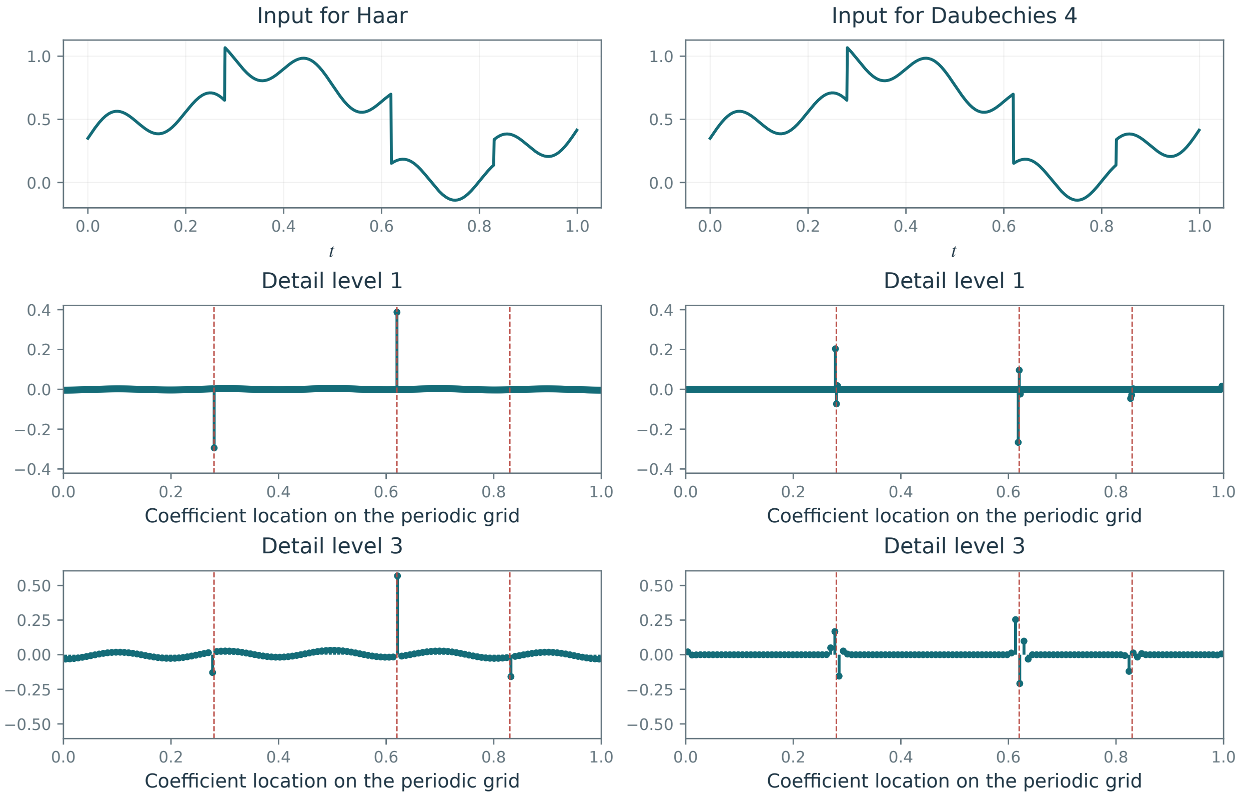


Figure 4.27. Location of large coefficients for Haar (left) and Daubechies-4 (right) wavelets applied to the same piecewise-smooth signal.

Proof. Integrability of the moments allows differentiation under the Fourier integral: $\hat{\psi}^{(k)} = \mathcal{F}((-\mathrm{i}\cdot)^k \psi(\cdot))$. Thus

$$(-\mathrm{i})^k \int_{\mathbb{R}} x^k \psi(x) \mathrm{d}x = \hat{\psi}^{(k)}(0).$$

Relations (4.17) and (4.19) imply

$$\hat{\psi}(2\omega) = \hat{h}(\omega+\pi)^* \rho(\omega) \quad \text{where} \quad \rho(\omega) \stackrel{\text{def.}}{=} \frac{1}{\sqrt{2}} e^{-i\omega} \hat{\varphi}(\omega).$$

and differentiating this relation shows

$$2\hat{\psi}^{(1)}(2\omega) = \hat{h}(\omega+\pi)^* \rho^{(1)}(\omega) + \hat{h}^{(1)}(\omega+\pi)^* \rho(\omega)$$

Since $\hat{h}(\pi) = 0$ and $\rho(0) = \hat{\varphi}(0)/\sqrt{2} \neq 0$, this shows that $\hat{\psi}^{(1)}(0) = 0 \Leftrightarrow \hat{h}^{(1)}(\pi)^* = 0$. Induction gives the higher-order equivalence $\hat{\psi}^{(k)}(0) = 0 \Leftrightarrow \hat{h}^{(k)}(\pi)^* = 0$, provided all derivatives of lower order vanish. □

Conditions (C_1) and (C_2) give $\hat{h}(\pi) = 0$, so every admissible wavelet has at least 1 vanishing moment: $\int \psi = 0$. By (4.22), additional vanishing moments require the Fourier transform $\hat{h}$ to vanish to higher order at $\omega = \pi$.

Support. Figure 4.27 shows the wavelet coefficients of a piecewise smooth signal. The cancellation provided by ψ suppresses coefficients in smooth regions, leaving the largest coefficients near singularities.

Choosing ψ with short support limits how many wavelets intersect each singularity. One can show that the size of the support of ψ is proportional to the size of the support of h. Short support competes with the vanishing-moment requirement (4.21). Indeed, one can prove that for an orthogonal wavelet basis with p vanishing moments

$$|\mathrm{Supp}(\psi)| \geqslant 2p - 1,$$

where the left-hand side denotes the length of the smallest closed interval containing the support of a compactly supported orthogonal mother wavelet.

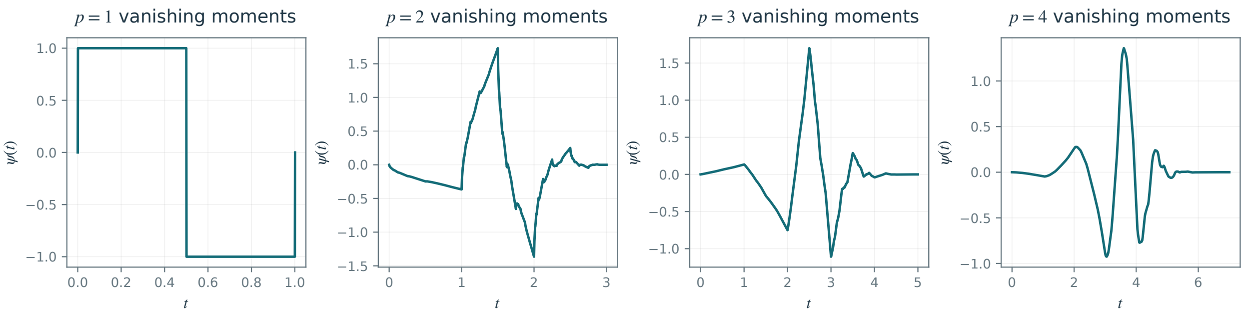


Figure 4.28. Daubechies mother wavelets ψ with increasing numbers p of vanishing moments.

Chapter 5 studies in detail the tradeoff between support size and vanishing moments in nonlinear approximation of piecewise smooth signals.

Smoothness. In compression or denoising applications, an approximate signal is recovered from a partial set I_M of coefficients,

$$f_M = \sum_{(j,n)\in I_M} \langle f, \psi_{j,n} \rangle \psi_{j,n}.$$

This finite sum is at least as smooth as ψ.

A smooth wavelet ψ helps avoid visible reconstruction artifacts. Smoothness controls the regularity of the reconstructed signal, while vanishing moments and localization primarily determine the approximation rates studied below. These properties are often linked: in many families, including the Daubechies wavelets introduced next, more vanishing moments also bring greater smoothness.

4.6.4 Daubechies Wavelets

To build a wavelet ψ with a fixed number p of vanishing moments, one designs the filter h, and uses the quadrature mirror filter relation (4.19) to compute g. One therefore looks for h such that

$$|\hat{h}(\omega)|^2 + |\hat{h}(\omega + \pi)|^2 = 2, \quad \hat{h}(0) = \sqrt{2}, \quad \text{and} \quad \forall\, k = 0, \dots, p-1, \ \frac{\mathrm{d}^k \hat{h}}{\mathrm{d}\omega^k}(\pi) = 0.$$

These conditions give algebraic relations between the coefficients of h, which can be solved explicitly using the Euclidean division algorithm for polynomials.

The resulting Daubechies wavelets have p vanishing moments and attain the minimum support length $2p - 1$ among compactly supported orthogonal wavelets.

The following filter coefficients are rounded to four decimal places. For $p = 1$, the construction gives the Haar system (up to the overall sign of the wavelet), with

$$h = [h_0 = 0.7071; 0.7071].$$

For $p = 2$, one obtains the celebrated Daubechies 4 filter

$$h = [0.4830; h_0 = 0.8365; 0.2241; -0.1294],$$

and for $p = 3$,

$$h = [0; 0.3327; 0.8069; h_0 = 0.4599; -0.1350; -0.0854; 0.0352].$$

Wavelet display. Figure 4.28 compares Daubechies mother wavelets with increasing numbers of vanishing moments. Each continuous wavelet is approximated by a discrete wavelet $\bar{\psi}_{j,n}$ computed on a fine grid of N samples. Applying the inverse transform to the following coefficient vector produces that discrete wavelet: $d_{j',n'} = \delta_{j-j'}\delta_{n-n'}$.

CHAPTER 5

Linear and Nonlinear Approximation

When only a limited number of coefficients can be retained, the choice of representation determines which features of a signal survive. Approximation rates quantify this tradeoff and help explain the performance of compression and denoising methods. We compare fixed and adaptive coefficient selection, relate Fourier and wavelet errors to smoothness and edges, and examine triangulations and curvelets for images with smooth contours.

The analysis concerns functions $f \in L^2([0,1]^d)$ on a continuous domain, with $d = 1, 2$.

5.1 Approximation

5.1.1 Approximation in an Orthonormal Basis

Let $\mathcal{B} = \{\psi_m\}_m$ be an orthonormal basis of $L^2([0,1]^d)$, with $d = 1$ for signals or $d = 2$ for images. The complete expansion in this basis,

$$f = \sum_{m \in \mathbb{Z}} \langle f, \psi_m \rangle \psi_m$$

reconstructs the signal exactly. Approximation and processing methods modify the coefficients $\langle f, \psi_m \rangle$, introducing a reconstruction error.

The simplest approximation reconstructs the signal from a subset $I_M \subset \mathbb{Z}$ of M coefficients:

$$f_M \overset{\text{def.}}{=} \sum_{m \in I_M} \langle f, \psi_m \rangle \psi_m, \quad \text{where} \quad M = |I_M|.$$

The reconstructed signal f_M is the orthogonal projection of f onto the space

$$V_M \overset{\text{def.}}{=} \text{Span}\{\psi_m \; ; \; m \in I_M\}.$$

If the selected space V_M depends on f, the overall approximation map $f \mapsto f_M$ may be nonlinear, even though projection onto each fixed space is linear.

Since the basis is orthogonal, the approximation error is

$$\|f - f_M\|^2 = \sum_{m \notin I_M} |\langle f, \psi_m \rangle|^2.$$

The central choice is the retained index set I_M, which may depend on the signal f.

5.1.2 Linear Approximation

Fixing I_M independently of the input gives a linear approximation: the same coefficients are retained for every f. The map $f \mapsto f_M$ is then orthogonal projection onto a fixed space V_M, and it satisfies

$$(f + g)_M = f_M + g_M$$

For the Fourier basis and an even coefficient budget, one usually selects the low-frequency atoms

$$I_M = \{-M/2 + 1, \ldots, M/2\}.$$

For a 1-D wavelet basis, one usually selects the coarse wavelets

$$I_M = \{\text{coarsest scaling atoms and wavelets } (j, n) \text{ with } j \geqslant j_0\}$$

where j_0 is selected such that $|I_M| = M$.

Figure 5.1, center, illustrates linear wavelet approximation. Discarding all fine-scale details blurs singularities, where the approximation is least accurate.

Figure 5.1. Linear versus nonlinear wavelet approximation.

5.1.3 Nonlinear Approximation

Allowing I_M to depend on f gives a nonlinear approximation. To choose I_M that minimizes $\|f - f_M\|$, orthogonality reduces the problem to retaining the M coefficients of largest magnitude:

$$I_M = \{M \text{ largest coefficients } |\langle f, \psi_m \rangle|\}.$$

When there are no ties at the cutoff, this can be obtained by thresholding

$$I_M = \{m \; ; \; |\langle f, \psi_m \rangle| > T\}$$

where T depends on the number of coefficients M,

$$M = \# \{m \; ; \; |\langle f, \psi_m \rangle| > T\} .$$

Computation of the threshold. Order the coefficient magnitudes in nonincreasing order. For a finite vector, write

$$d_1 \geqslant \cdots \geqslant d_N \geqslant 0, \qquad \{d_m\}_{m=1}^N = \{|\langle f, \psi_m \rangle|\}_{m=1}^N. \tag{5.1}$$

If $d_M > d_{M+1}$, any threshold $d_{M+1} \leqslant T < d_M$ retains exactly M coefficients. At a tie, a deterministic rule must select among equal coefficients. Thus the threshold T and coefficient count M do not determine each other uniquely. Figure 5.2 illustrates the relationship.

The following proposition shows that the decay of the ordered coefficients governs the decay of the nonlinear approximation error.

Proposition 5.1. *For $\alpha > 0$, with $(d_m)_{m \geqslant 1}$ in nonincreasing order, one has*

$$d_m = O(m^{-\frac{\alpha+1}{2}}) \quad \Longleftrightarrow \quad \|f - f_M\|^2 = O(M^{-\alpha}). \tag{5.2}$$

Proof. One has

$$\|f - f_M\|^2 = \sum_{m>M} d_m^2$$

The forward implication follows from $\sum_{m>M} m^{-\alpha-1} \leqslant \alpha^{-1} M^{-\alpha}$. For the converse, taking M even (integer rounding changes only constants),

$$d_M^2 \leqslant \frac{2}{M} \sum_{m=M/2+1}^{M} d_m^2 \leqslant \frac{2}{M} \sum_{m>M/2} d_m^2 = \frac{2}{M} \|f - f_{M/2}\|^2.$$

□

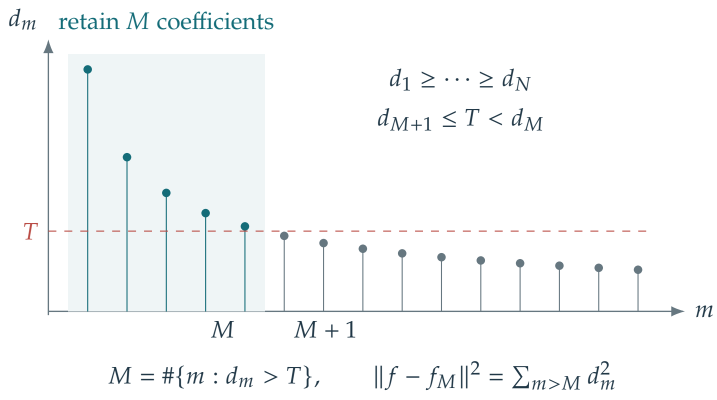


Figure 5.2. Ordered coefficient magnitudes and the threshold used for nonlinear approximation.

Hard thresholding. For the coefficient set selected by a threshold T, the nonlinear approximation can be written as

$$f_M = \sum_{|\langle f,\psi_m\rangle|>T} \langle f, \psi_m\rangle \psi_m = \sum_m S_T(\langle f, \psi_m\rangle)\psi_m, \tag{5.3}$$

where

$$S_T(x) = \begin{cases} x & \text{if} \quad |x| > T \\ 0 & \text{if} \quad |x| \leqslant T \end{cases} \tag{5.4}$$

is the hard thresholding operator displayed in Figure 5.3.

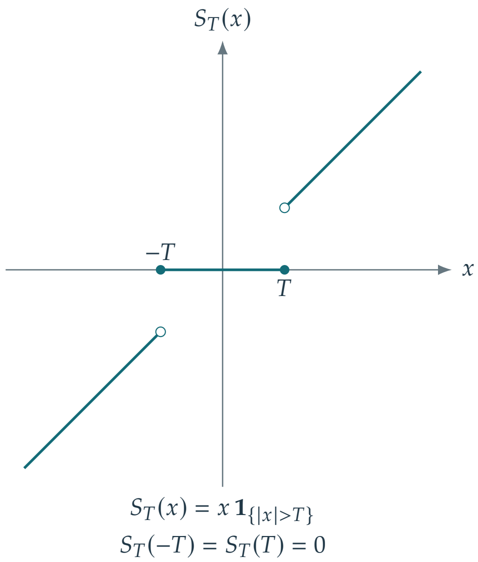


Figure 5.3. Hard thresholding.

5.2 Signal and Image Modeling

A signal model specifies a constraint $f \in \Theta$, where $\Theta \subset \mathrm{L}^2([0,1]^d)$ is the class of signals of interest. Figure 5.4 shows different image models, which we describe below.

5.2.1 Uniformly Smooth Signals and Images

Signals with derivatives. The simplest model consists of uniformly smooth signals with bounded derivatives

$$\Theta = \left\{ f \in \mathrm{L}^2([0,1]^d) \; ; \; \|f\|_{C^\alpha} \leqslant C \right\}, \tag{5.5}$$

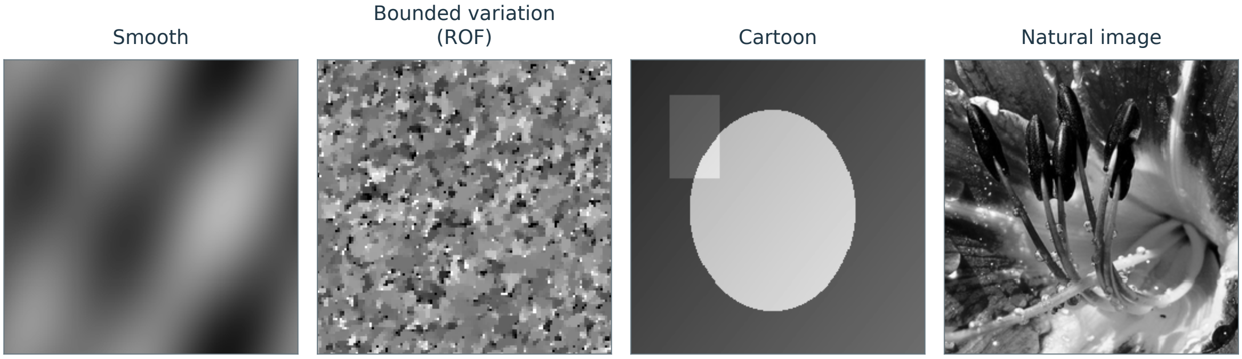


Figure 5.4. Examples of image models. The bounded-variation example is generated by ROF total-variation denoising of white noise.

where $C > 0$ is fixed. For integer $\alpha \geqslant 1$, in one dimension

$$\|f\|_{C^\alpha} \stackrel{\text{def.}}{=} \max_{0 \leqslant k \leqslant \alpha} \left\| \frac{d^k f}{dt^k} \right\|_\infty .$$

In higher dimensions, take the maximum over all partial derivatives of total order at most α. For noninteger $\alpha = m + \beta$, $0 < \beta < 1$, use the usual Hölder norm: derivatives through order m are bounded, and order-m derivatives are β-Hölder continuous.

Sobolev smooth signals and images. For integer $\alpha \geqslant 1$, a C^α signal in the model (5.5) has derivatives with bounded energy:

$$\frac{d^\alpha f}{dt^\alpha}(t) = f^{(\alpha)}(t) \in L^2([0,1]).$$

For integer α and periodic boundary conditions, use the identity

$$\hat{f}_m^{(\alpha)} = (2i\pi m)^\alpha \hat{f}_m$$

Here $\hat{f}$ denotes Fourier coefficients as in (2.2), now on $\mathbb{R}/\mathbb{Z}$ rather than $\mathbb{R}/2\pi\mathbb{Z}$:

$$\hat{f}_n \stackrel{\text{def.}}{=} \int_0^1 e^{-2i\pi n x} f(x) dx,$$

This motivates the homogeneous Sobolev seminorm

$$\|f\|_{\mathrm{Sob}(\alpha)}^2 = \sum_{m \in \mathbb{Z}} |2\pi m|^{2\alpha} |\langle f, e_m \rangle|^2, \tag{5.6}$$

so that $\|f\|_{\mathrm{Sob}(\alpha)} = \|f^{(\alpha)}\|$ for smooth periodic functions. The same seminorm applies when the derivatives exist only in the distributional sense and belong to $L^2([0,1])$.

This definition extends to distributions and signals $f \in L^2([0,1]^d)$ of arbitrary dimension d as

$$\|f\|_{\mathrm{Sob}(\alpha)}^2 = \sum_{m \in \mathbb{Z}^d} (2\pi\|m\|)^{2\alpha} |\langle f, e_m \rangle|^2, \tag{5.7}$$

The periodic Sobolev model

$$\Theta = \left\{ f \in L^2([0,1]^d) \; ; \; \|f\|_2^2 + \|f\|_{\mathrm{Sob}(\alpha)}^2 \leqslant C \right\} \tag{5.8}$$

includes the corresponding periodic C^α model (5.5) for integer α, after adjusting the radius of the ball. For noninteger α, Hölder regularity gives Sobolev regularity of every order $s < \alpha$, but need not give regularity of order α itself.

Figure 5.5 shows progressively smoother images obtained by Gaussian filtering of the same white-noise realization. Each result is rescaled to the same mean and variance, so the comparison emphasizes spatial regularity rather than a loss of contrast.

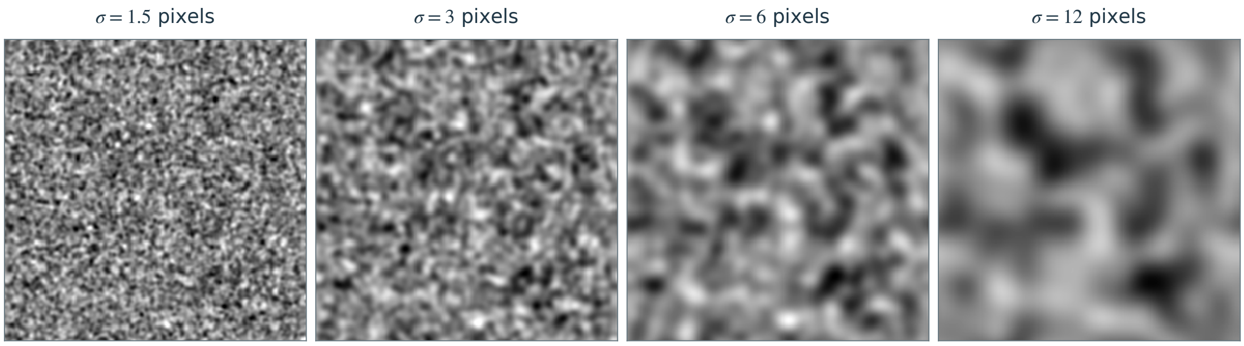


Figure 5.5. Progressive Gaussian filtering of one white-noise image. Smoothing widths increase from left to right; the displayed images have the same mean and variance.

5.2.2 Piecewise Regular Signals and Images

Piecewise smooth signals. A piecewise smooth 1-D signal $f \in \mathrm{L}^2([0,1])$ is C^α on the intervals separated by at most K interior singular points:

$$\Theta = \left\{ f \in \mathrm{L}^2([0,1]) \; ; \; \exists\, 0 = t_0 < t_1 < \cdots < t_{K+1} = 1, \quad \max_{0 \leqslant i \leqslant K} \|f|_{(t_i, t_{i+1})}\|_{\mathrm{C}^\alpha} \leqslant C \right\} \tag{5.9}$$

Here $f|_{(t_i,t_{i+1})}$ denotes the restriction of f to the open interval (t_i, t_{i+1}).

Piecewise smooth images. A piecewise smooth image is a function $f \in \mathrm{L}^2([0,1]^2)$ with C^α regularity away from at most K rectifiable curves:

$$\Theta = \left\{ f \in \mathrm{L}^2([0,1]^2) \; ; \; \exists\, \Gamma = \bigcup_{i=0}^{K-1} \gamma_i, \; \|f\|_{\mathrm{C}^\alpha(\Gamma^c)} \leqslant C_1 \quad \text{and} \quad |\gamma_i| \leqslant C_2 \right\} \tag{5.10}$$

Here $|\gamma_i|$ denotes curve length. The C^α bound applies separately within each smooth region, with uniform one-sided regularity up to its edges.

Segmentation methods such as the one proposed by Mumford and Shah [24] implicitly assume such a piecewise smooth image model.

5.2.3 Bounded Variation Signals and Images

Bounded variation provides a more flexible model for signals with edges

$$\Theta = \left\{ f \in \mathrm{L}^2([0,1]^d) \; ; \; \|f\|_\infty \leqslant C_1 \quad \text{and} \quad \|f\|_{\mathrm{TV}} \leqslant C_2 \right\}. \tag{5.11}$$

For $d = 1$ and $d = 2$, this model includes the piecewise smooth signal and image models (5.9) and (5.10) when $\alpha \geqslant 1$, after adjusting the constants. Bounds on the derivatives, jump sizes, and total edge length then control the variation.

The total variation of a smooth function is

$$\int \|\nabla f(x)\| \mathrm{d}x$$

where

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_i} \right)_{i=0}^{d-1} \in \mathbb{R}^d$$

is the gradient at x. Total variation also extends to discontinuous images, including those with jumps across edges. The coarea formula expresses the total variation of a piecewise smooth image as an integral of the perimeters of its superlevel sets:

$$\|f\|_{\mathrm{TV}} = \int_{-\infty}^{\infty} \mathrm{Per}(\{f > t\}; \Omega)\, \mathrm{d}t, \qquad \Omega = (0,1)^d. \tag{5.12}$$

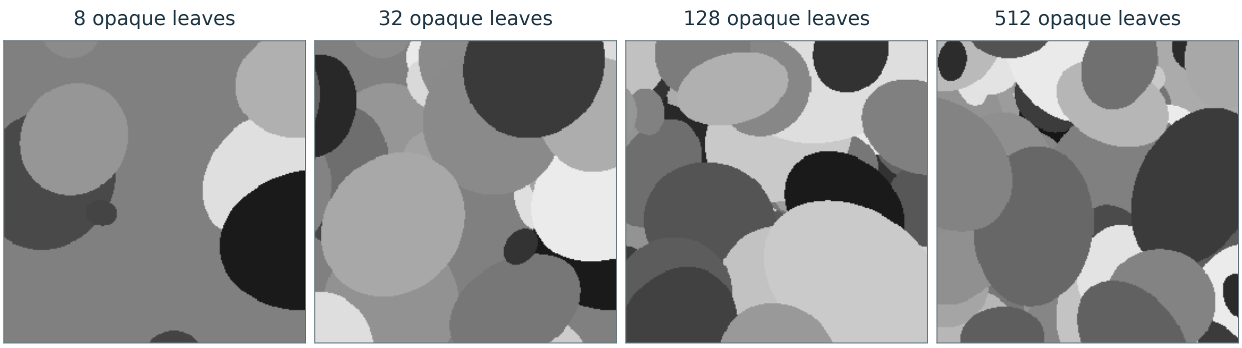


Figure 5.6. Progressive dead-leaves construction: later opaque shapes cover earlier ones, producing a family of cartoon images.

Here the perimeter is relative to the image domain. For smooth functions in two dimensions it agrees, for almost every t, with the length of the level curve. For a bounded set $\Omega \subset \mathbb{R}^2$ with piecewise smooth boundary, taking total variation on all of $\mathbb{R}^2$ gives

$$\|1_\Omega\|_{\mathrm{TV}} = |\partial\Omega|.$$

The model of bounded variation was introduced in image processing by Rudin, Osher and Fatemi [28].

5.2.4 Cartoon Images

The bounded-variation model (5.11) controls total perimeter but does not require individual edges to be smooth. For images with smooth contours, incorporating that additional geometry can improve approximation and processing.

The C^α cartoon model consists of 2-D functions that are C^α away from at most K smooth edge curves γ_i:

$$\Theta = \left\{ f \in \mathrm{L}^2([0,1]^2)\ ;\ \exists \Gamma = \bigcup_{i=0}^{K-1} \gamma_i,\ \|f\|_{\mathrm{C}^\alpha(\Gamma^c)} \leqslant C_1 \quad \text{and} \quad \|\gamma_i\|_{\mathrm{C}^\alpha} \leqslant C_2 \right\} \tag{5.13}$$

Each curve is parameterized by arc length on $[0, A_i]$, with a uniform bound on A_i and its C^α norm. Figure 5.6 illustrates a progressive dead-leaves construction: opaque shapes are placed successively, producing smooth visible contour segments and occlusion boundaries.

Optical diffraction can blur these edges, motivating the blurred cartoon model

$$\tilde{\Theta} = \left\{ \tilde{f} = f \star h \in \mathrm{L}^2([0,1]^2)\ ;\ f \in \Theta \quad \text{and} \quad h \in \mathcal{H} \right\} \tag{5.14}$$

where Θ is the sharp-image model (5.13) and $\mathcal{H}$ specifies the admissible blur kernels. For example, $h \geqslant 0$ may be required to be smooth and localized in both space and frequency. Unknown blur makes it difficult to first detect the edges Γ and then process the smooth regions separately.

Figure 5.7 shows examples of images in Θ and $\tilde{\Theta}$.

5.3 Efficient Approximation

5.3.1 Decay of Approximation Error

To process signals in Θ efficiently, we seek an orthonormal basis in which the nonlinear approximation error $\|f - f_M\|$ tends to 0 as rapidly as possible when M grows.

Polynomial error decay. This error decay is measured using a power law

$$\forall f \in \Theta,\ \forall M, \quad \|f - f_M\|^2 \leqslant C_f M^{-\alpha} \tag{5.15}$$

Here α is common to all signals in the model; larger values mean faster decay. The exponent depends on the basis and the model Θ, and measures how efficiently the basis represents that class. The constant C_f may depend on the particular signal f.

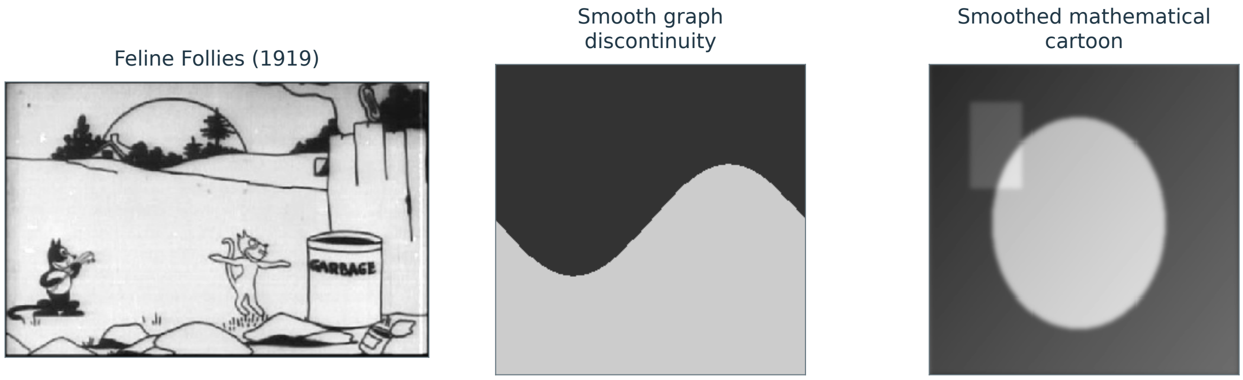


Figure 5.7. A grayscale frame from *Feline Follies* (1919), a mathematical jump along a smooth graph, and a Gaussian-smoothed cartoon. Film frame: Pat Sullivan, via Wikimedia Commons, public domain.

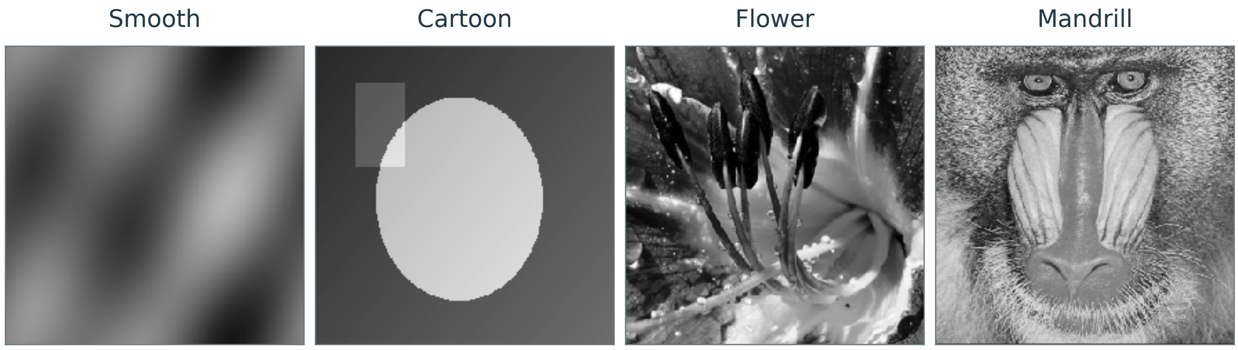


Figure 5.8. Test images with different structures: smooth, cartoon, flower, and mandrill.

Relevance for compression, denoising and inverse problems. Approximation rates have practical consequences. Section 6.1 relates compression error to nonlinear approximation error, explaining why an effective approximation basis also supports efficient compression.

Chapter 7 develops a similar connection for thresholding denoisers. Thresholding forms a nonlinear approximation of the noisy image, and the expected denoising error is controlled by how well the basis approximates the clean signal.

Chapter 10 uses compressibility in a chosen basis to address inverse problems such as super-resolution and inpainting. Efficient approximation of the unknown signal helps recover missing information, provided the measurement operator retains enough information about sparse combinations of basis atoms. Recovery performance therefore depends on both the basis and the measurements.

Comparison of signals. With the basis fixed, the decay of $\|f - f_M\|$ compares the approximation complexity of different images. Figure 5.9 illustrates how intricate geometry and textures slow down wavelet approximation.

To reveal approximate power-law behavior, the error curves are displayed on logarithmic axes. An exact power law has the form

$$\log(\|f - f_M\|^2) = \mathrm{cst} - \alpha \log(M)$$

and appears as a straight line with slope $-\alpha$. For sampled data, this behavior is generally studied when $M \ll N$; as $M \approx N$, retaining nearly all coefficients drives the error to zero.

5.3.2 Comparison of bases.

For a fixed image f, the decay of $\|f - f_M\|$ compares the efficiency of different bases. Figure 5.11 makes this comparison for an image containing contours and textures. The Fourier basis of Section 2.5 performs poorly here: periodic boundaries create artifacts, and globally supported atoms do not follow local contours. The cosine basis removes the periodic boundary discontinuity through symmetric extension, but its atoms remain global. Local DCT bases use cosine atoms on small square patches, improving localization. At small

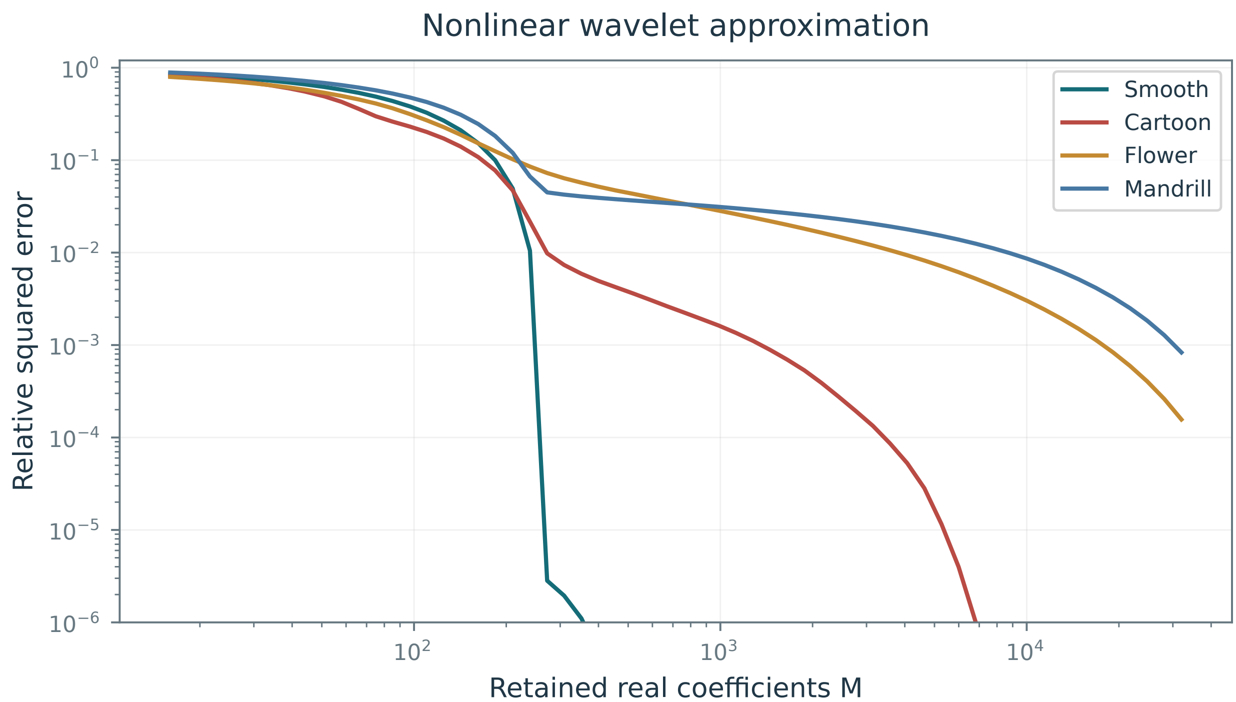


Figure 5.9. Wavelet approximation error for the images in Figure 5.8.

coefficient budgets M, however, their block structure produces visible artifacts. The isotropic wavelet basis of Section 4.5.2 performs best in this example by combining localized atoms with a multiresolution organization.

Figure 5.12 summarizes the approximation rates developed in the following sections for different data models.

5.4 Fourier Linear Approximation of Smooth Functions

For integer α, the smooth model (5.5) bounds all continuous derivatives through that order. Greater regularity of f, measured by a larger α, permits faster approximation. Figure 5.5 illustrates the related Sobolev measure of smoothness.

5.4.1 1-D Fourier Approximation

A 1-D signal $f \in \mathrm{L}^2([0,1])$ is associated with a 1-periodic function $f(t+1) = f(t)$ defined for $t \in \mathbb{R}$.

Low pass approximation. For even M, consider the linear Fourier approximation that retains frequencies up to $M/2$:

$$f_M^{\text{lin}} = \sum_{m=-M/2}^{M/2} \langle f, e_m \rangle e_m$$

where we use the 1-D Fourier atoms

$$\forall\, m \in \mathbb{Z}, \quad e_m(t) \overset{\text{def.}}{=} e^{2i\pi m t}.$$

This symmetric cutoff retains $M+1$ Fourier atoms; the extra coefficient does not affect the asymptotic rates below.

Figure 5.13 shows these approximations for increasing M. The jumps in f produce substantial error and ringing near the singularities.

This low-pass approximation is a convolution, since

$$f_M = \sum_{m=-M/2}^{M/2} \langle f, e_m \rangle e_m = f \star h_M \quad \text{where} \quad \hat{h}_M \overset{\text{def.}}{=} 1_{[-M/2, M/2]}.$$

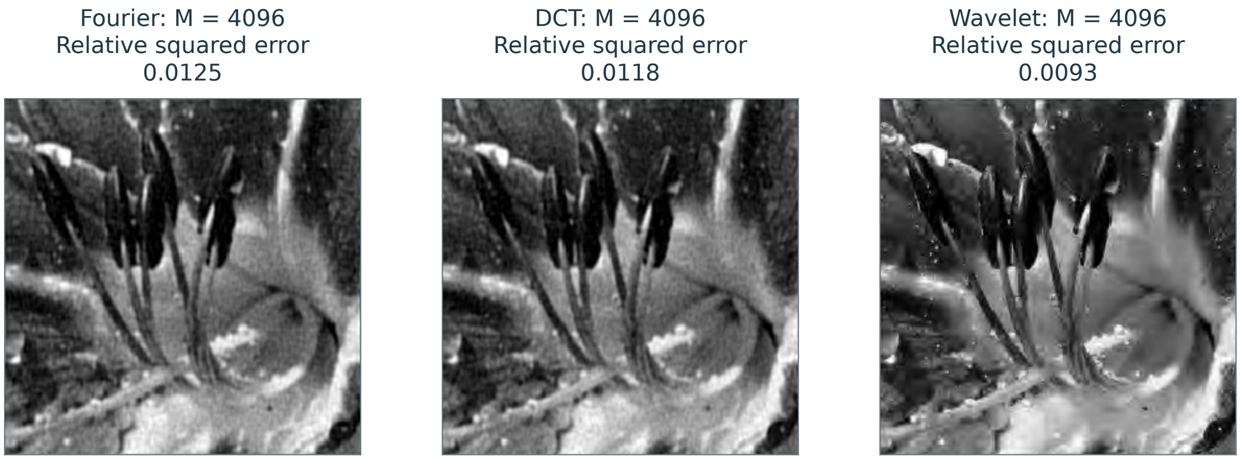


Figure 5.10. Fourier, DCT, and wavelet approximations of the same 256×256 flower image, each retaining $M = 4096$ real orthonormal coefficients.

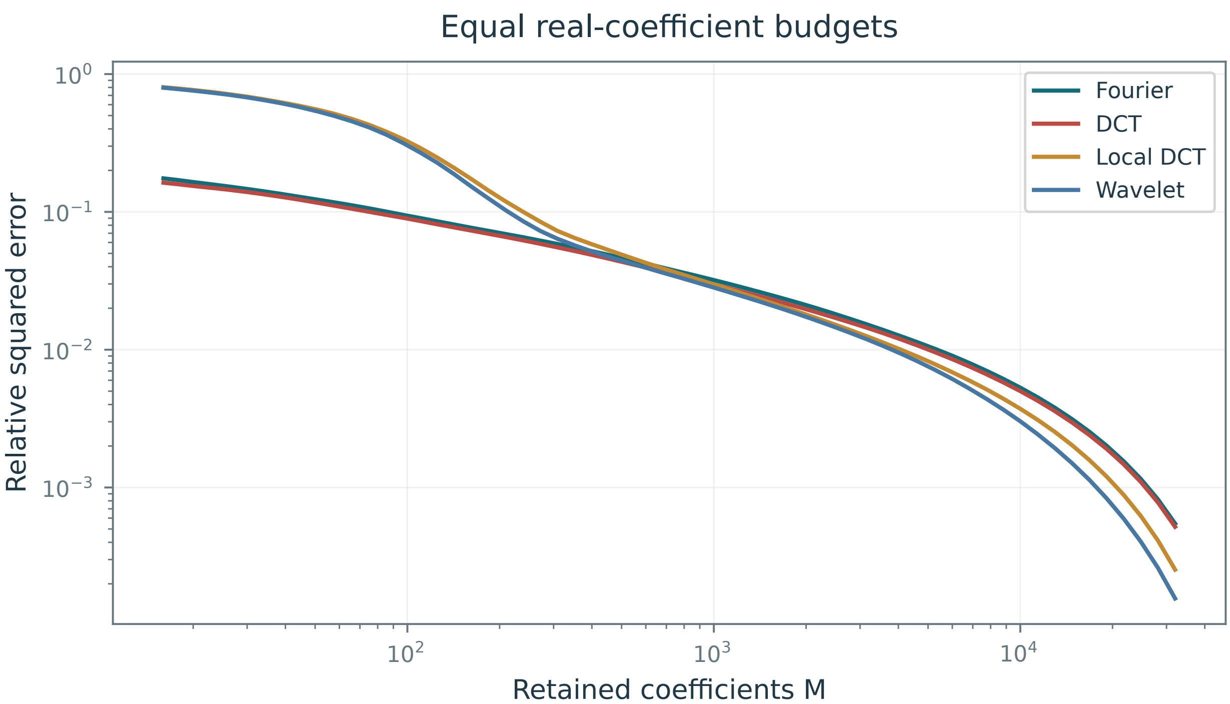


Figure 5.11. Comparison of approximation error decay for different bases.

$\|f - f_M\|_2^2 \lesssim$	Sobolev 1-D	piecewise 1-D	Sobolev 2-D	BV 2-D	cartoon 2-D
Fourier, L	$M^{-2\alpha}$	M^{-1}	$M^{-\alpha}$	$M^{-1/2}$	$M^{-1/2}$
Fourier, NL	$M^{-2\alpha}$	M^{-1}	$M^{-\alpha}$	$M^{-1/2}$	$M^{-1/2}$
Wavelets, L	$M^{-2\alpha}$	M^{-1}	$M^{-\alpha}$	$M^{-1/2}$	$M^{-1/2}$
Wavelets, NL	$M^{-2\alpha}$	$M^{-2\alpha}$	$M^{-\alpha}$	M^{-1}	M^{-1}

Cartoon images: adaptive triangulations $O(M^{-2})$; curvelets $O(M^{-2}(\log M)^3)$.
L: linear; NL: nonlinear. Sobolev columns: H^α, $\alpha \geq 1$.
Use compatible wavelet bases and boundary conditions.
The BV class also has the uniform bound assumed in the chapter.

Figure 5.12. Summary of linear and nonlinear approximation rates $\|f - f_M\|^2$ for different classes of 1-D signals and images.

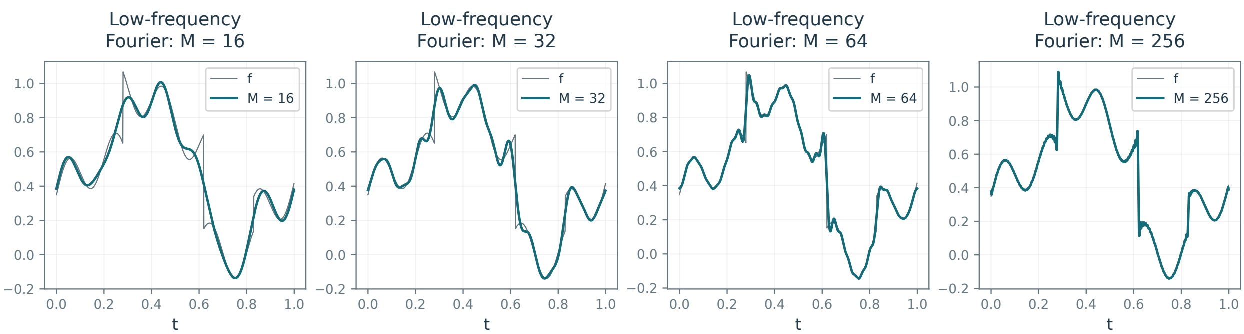


Figure 5.13. Fourier approximation of a signal.

The kernel h_M is the Dirichlet kernel introduced in the Fourier chapter.

The following elementary bound shows that this approximation error decays for C^α signals.

Proposition 5.2. *For integer $\alpha \geqslant 1$ and $f \in C^\alpha(\mathbb{R}/\mathbb{Z})$*

$$\|f - f_M^{lin}\|^2 = O(M^{-2\alpha+1}).$$

Proof. Using integration by parts, one shows that for a C^α function f in the smooth signal model (5.5), $\mathcal{F}(f^{(\ell)})_m = (2im\pi)^\ell \hat{f}_m$, so that Cauchy–Schwarz gives

$$|\langle f, e_m \rangle| \leqslant |2\pi m|^{-\alpha} \|f^{(\alpha)}\|_2.$$

This implies

$$\|f - f_M^{\text{lin}}\|^2 = \sum_{|m|>M/2} |\langle f, e_m \rangle|^2 \leqslant \|f^{(\alpha)}\|_2^2 \sum_{|m|>M/2} |2\pi m|^{-2\alpha} = O(M^{-2\alpha+1}).$$

□

Using the full energy of $f^{(\alpha)}$ gives a sharper bound under the same assumptions. The argument also applies to the larger Sobolev class.

Proposition 5.3. *For $\alpha > 0$ and a periodic signal in the Sobolev model (5.8), one has*

$$\|f - f_M^{lin}\|^2 \leqslant C\|f\|_{\text{Sob}(\alpha)}^2 M^{-2\alpha}. \tag{5.16}$$

Proof. One has

$$\|f\|_{\text{Sob}(\alpha)}^2 = \sum_m |2\pi m|^{2\alpha} |\langle f, e_m \rangle|^2 \geqslant \sum_{|m|>M/2} |2\pi m|^{2\alpha} |\langle f, e_m \rangle|^2 \tag{5.17}$$

$$\geqslant (\pi M)^{2\alpha} \sum_{|m|>M/2} |\langle f, e_m \rangle|^2 \geqslant (\pi M)^{2\alpha} \|f - f_M^{\text{lin}}\|^2. \tag{5.18}$$

□

This rate is optimal uniformly over a Sobolev ball. The best M-term Fourier approximation satisfies the same upper bound because it is at least as accurate as the low-pass approximation; individual functions may have faster decay.

For signals with piecewise Lipschitz regularity in the model (5.9), such as the example in Figure 5.13, the general bound for both linear and nonlinear Fourier approximation errors is

$$\|f - f_M\|^2 \leqslant C_f M^{-1} \tag{5.19}$$

and Fourier atoms are no longer optimal for approximation. For example, the interval indicator $f = 1_{[a,b]}$ has coefficients $\hat{f}_m = (e^{-2\pi i m a} - e^{-2\pi i m b})/(2\pi i m)$ for $m \neq 0$, and hence $\|f - f_M\|^2 \asymp M^{-1}$ when $0 < b - a < 1$ for both linear and nonlinear approximations.

5.4.2 Sobolev Images

The same argument applies to images and higher-dimensional data through the Sobolev seminorm (5.7) with $d > 1$.

For an α-regular Sobolev image, the linear and nonlinear approximation errors satisfy

$$\|f - f_M\|^2 \leqslant C\|f\|_{\text{Sob}(\alpha)}^2 M^{-\alpha}.$$

For d-dimensional data $f : [0,1]^d \to \mathbb{R}$, the corresponding error bound is $O(M^{-2\alpha/d})$.

For an image with piecewise Lipschitz regularity and finite-length edges in the model (5.10), the linear and nonlinear errors satisfy the slower general bound

$$\|f - f_M\|^2 \leqslant C_f M^{-1/2}, \tag{5.20}$$

and Fourier atoms are no longer optimal for approximation.

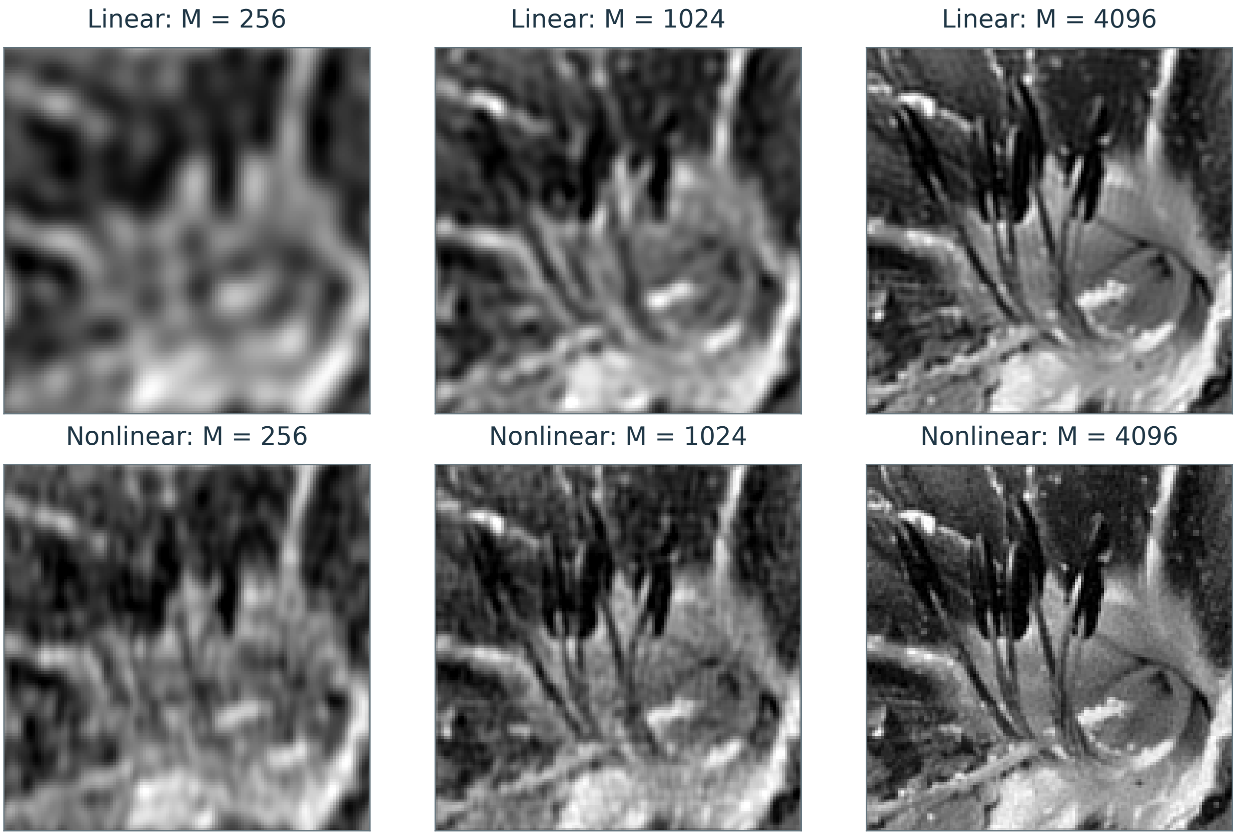


Figure 5.14. Linear (top row) and nonlinear (bottom row) Fourier approximation.

5.5 Wavelet Approximation of Piecewise Smooth Functions

Wavelets improve approximation near singularities because their support is localized.

5.5.1 Decay of Wavelet Coefficients

To approximate smooth regions efficiently, choose wavelets with sufficiently many vanishing moments, denoted by p:

$$\forall\, k = 0, \dots, p-1, \quad \int \psi(x) x^k \mathrm{d}x = 0.$$

Choose $p \geqslant \alpha$, where α is the regularity of the signal outside singularities (for instance jumps or kinks).

To quantify the approximation error decay for piecewise smooth signals (5.9) and images (5.10), one treats wavelets supported in smooth regions separately from those crossing singularities. Figure 5.15 locates the singular and regular regions of a signal and an image.

Proposition 5.4. *Assume a compactly supported wavelet with $p \geqslant \alpha$ vanishing moments (all moments of total degree less than p in dimension d). If f admits a C^α extension to a neighborhood of the support, one has*

$$|\langle f,\, \psi_{j,n} \rangle| \leqslant C_f 2^{j(\alpha+d/2)} \int |t|^\alpha |\psi(t)|\, \mathrm{d}t. \tag{5.21}$$

Without the smoothness assumption, every bounded f satisfies

$$|\langle f,\, \psi_{j,n} \rangle| \leqslant \|f\|_\infty \|\psi\|_1 2^{j\frac{d}{2}}. \tag{5.22}$$

Proof. Under these assumptions, one can perform a Taylor expansion of f around the point $2^j n$

$$f(x) = P(x - 2^j n) + R(x - 2^j n) = P(2^j t) + R(2^j t)$$

where $\deg(P) < \alpha$ and

$$|R(x)| \leqslant C_f \|x\|^\alpha.$$

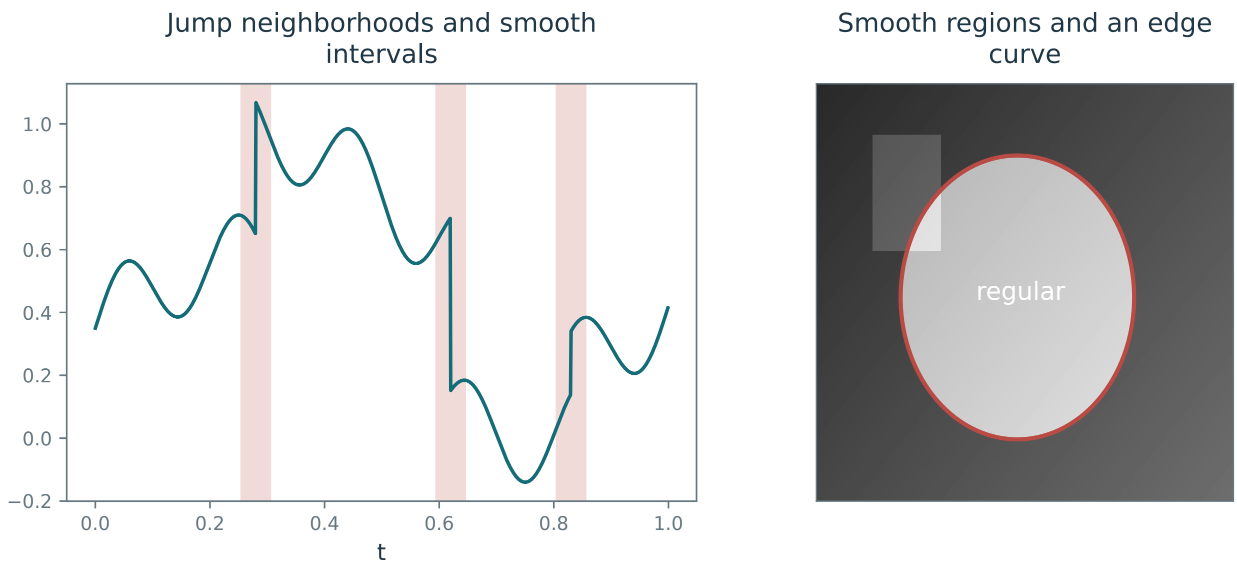


Figure 5.15. Singular and regular regions of a signal (left) and an image (right).

One then bounds the wavelet coefficient

$$\langle f, \psi_{j,n} \rangle = \frac{1}{2^{j\frac{d}{2}}} \int f(x)\psi\left(\frac{x - 2^j n}{2^j}\right) \mathrm{d}x = 2^{j\frac{d}{2}} \int R(2^j t)\psi(t)\mathrm{d}t$$

where we substituted $t = \frac{x-2^j n}{2^j}$. This proves (5.21). The bound (5.22) follows directly by bounding the function in the inner product. □

5.5.2 1-D Piecewise Smooth Approximation

For a piecewise smooth 1-D signal in the model (5.9), the large coefficients $\langle f, \psi_{j,n} \rangle$ concentrate near singularities. Denote the finite set of singular points by $\mathcal{S} \subset [0, 1]$.

Theorem 5.5. *For $\alpha > 0$, let f belong to the piecewise C^α model (5.9). With compactly supported orthonormal wavelets having at least α vanishing moments (and a compatible boundary construction), the best M-term error satisfies*

$$\|f - f_M\|^2 = O(M^{-2\alpha}). \tag{5.23}$$

Proof. The proof is split into several parts.

Step 1. Separating regular and singular coefficients. At scale 2^j, define the singular index set by the wavelets whose support crosses a singularity:

$$C_j = \left\{n \; ; \; \mathrm{supp}(\psi_{j,n}) \cap \mathcal{S} \neq \emptyset\right\} \tag{5.24}$$

Compact support and dyadic spacing imply a bound on its cardinality that is uniform in j:

$$|C_j| \leqslant K|\mathcal{S}| = \text{constant}.$$

Using (5.21) for $d = 1$, the decay of regular coefficients is bounded as

$$\forall\, n \in C_j^c, \quad \left|\langle f, \psi_{j,n} \rangle\right| \leqslant C2^{j(\alpha+1/2)}.$$

Using (5.22) for $d = 1$, the decay of singular coefficients is bounded as

$$\forall\, n \in C_j, \quad \left|\langle f, \psi_{j,n} \rangle\right| \leqslant C2^{j/2}.$$

For a sufficiently small threshold T, define two cutoff scales, one for regular coefficients and one for singular coefficients, as functions of T:

$$2^{j_1} = (T/C)^{\frac{1}{\alpha+1/2}} \quad \text{and} \quad 2^{j_2} = (T/C)^2.$$

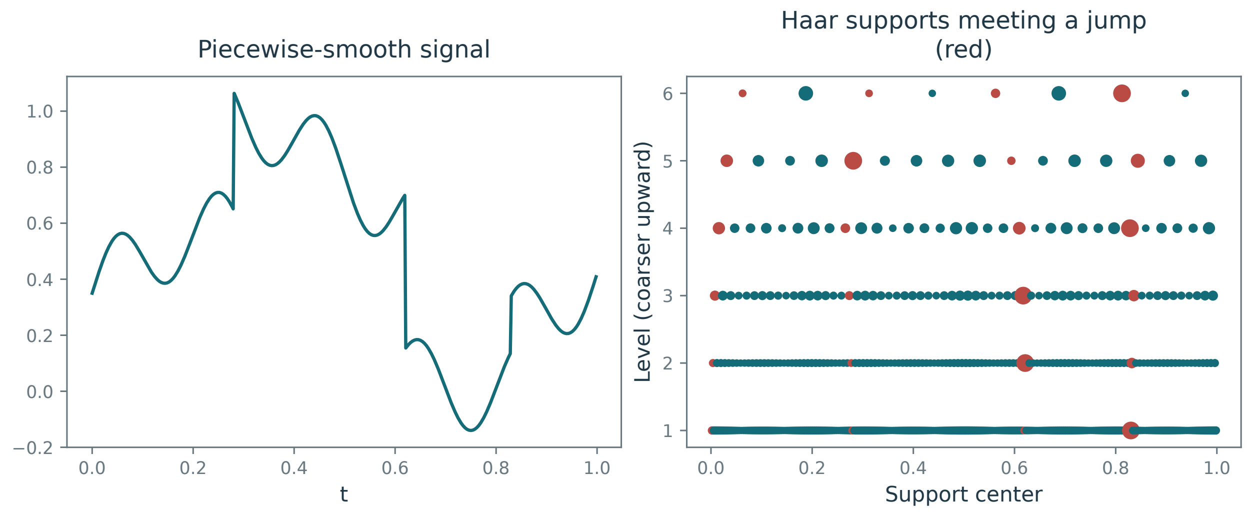


Figure 5.16. Segmentation of the wavelet coefficients into regular and singular parts.

Round these cutoff scales to integers; this changes constants only. All subsequent scale sums are bounded above by the fixed coarsest scale $j_0 = 0$, and the finitely many scaling coefficients are retained. Figure 5.16 shows a schematic segmentation of the set of wavelet coefficients into regular and singular parts, together with the cut-off scales.

Step 2. Bounding the error. These cut-off scales allow us to define a comparison approximation

$$\tilde{f}_M = P_{V_0} f + \sum_{j_2 \leqslant j \leqslant 0} \sum_{n \in C_j} \langle f, \psi_{j,n} \rangle \psi_{j,n} + \sum_{j_1 \leqslant j \leqslant 0} \sum_{n \in C_j^c} \langle f, \psi_{j,n} \rangle \psi_{j,n}. \tag{5.25}$$

The error of the comparison M-term approximation $\tilde{f}_M$ bounds the best M-term error from above:

$$\|f - f_M\|^2 \leqslant \|f - \tilde{f}_M\|^2 \leqslant \sum_{j<j_2, n \in C_j} |\langle f, \psi_{j,n} \rangle|^2 + \sum_{j<j_1, n \in C_j^c} |\langle f, \psi_{j,n} \rangle|^2 \tag{5.26}$$

$$\leqslant \sum_{j<j_2} (K|\mathcal{S}|) \times C^2 2^j + \sum_{j<j_1} 2^{-j} \times C^2 2^{j(2\alpha+1)} \tag{5.27}$$

$$= O(2^{j_2} + 2^{2\alpha j_1}) = O(T^2 + T^{\frac{2\alpha}{\alpha+1/2}}) = O(T^{\frac{2\alpha}{\alpha+1/2}}). \tag{5.28}$$

Step 3. Counting the Coefficients. The number of coefficients needed to build the approximating signal $\tilde{f}_M$ is

$$M \leqslant 1 + \sum_{j_2 \leqslant j \leqslant 0} |C_j| + \sum_{j_1 \leqslant j \leqslant 0} |C_j^c| \leqslant 1 + \sum_{j_2 \leqslant j \leqslant 0} K|\mathcal{S}| + \sum_{j_1 \leqslant j \leqslant 0} 2^{-j} \tag{5.29}$$

$$= O(|\log(T)| + T^{\frac{-1}{\alpha+1/2}}) = O(T^{\frac{-1}{\alpha+1/2}}). \tag{5.30}$$

Step 4. Putting everything together. Given a coefficient budget M, choose T to be a sufficiently large constant times $M^{-(\alpha+1/2)}$. The coefficient-count bound then uses at most M terms, and the error bound is $O(M^{-2\alpha})$. □

For piecewise Lipschitz signals, the wavelet rate is faster than the jump-limited $O(M^{-1})$ Fourier bound in (5.19). It also matches the smooth-signal rate (5.16): finitely many singularities do not worsen the asymptotic exponent in one dimension. The rate (5.23) is asymptotically optimal for this class.

Figure 5.17 shows examples of wavelet approximation of singular signals.

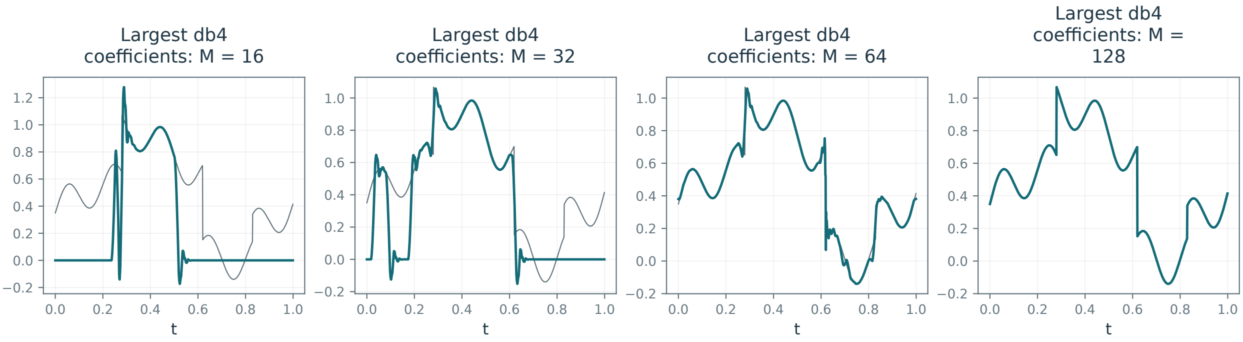


Figure 5.17. 1-D wavelet approximation.

5.5.3 2-D Piecewise Smooth Approximation

We now give the counterpart of Theorem 5.5 for 2-D functions.

Theorem 5.6. *Let f belong to the piecewise smooth image model (5.10), with $\alpha \geqslant 1$ and finitely many rectifiable edge curves. For compactly supported orthonormal wavelets with at least α vanishing moments and compatible boundary treatment, the best M-term error satisfies*

$$\|f - f_M\|^2 = O(M^{-1}). \tag{5.31}$$

Proof. For an image in the model (5.10), define the singular index set C_j as in (5.24). Unlike the 1-D case, its cardinality grows as the scale 2^j decreases:

$$|C_j^\omega| \leqslant 2^{-j} K |\mathcal{S}|,$$

where $|\mathcal{S}|$ is the total length of the singular curves $\mathcal{S}$, and $\omega \in \{V, H, D\}$ is the wavelet orientation. Using (5.21) for $d = 2$, the decay of regular coefficients is bounded as

$$\forall n \in (C_j^\omega)^c, \quad \left|\langle f, \psi_{j,n}^\omega \rangle\right| \leqslant C 2^{j(\alpha+1)}.$$

Using (5.22) for $d = 2$, the decay of singular coefficients is bounded as

$$\forall n \in C_j^\omega, \quad \left|\langle f, \psi_{j,n}^\omega \rangle\right| \leqslant C 2^j.$$

After fixing T, the cut-off scales are defined as

$$2^{j_1} = (T/C)^{\frac{1}{\alpha+1}} \quad \text{and} \quad 2^{j_2} = T/C.$$

We define, as in (5.25), a comparison approximation. Similarly to (5.26), we bound the approximation error as

$$\|f - f_M\|^2 \leqslant \|f - \tilde{f}_M\|^2 = O(2^{j_2} + 2^{2\alpha j_1}) = O(T + T^{\frac{2\alpha}{\alpha+1}}) = O(T)$$

and the number of coefficients as

$$M = O(T^{-1} + T^{-2/(\alpha+1)}) = O(T^{-1}).$$

Since $\alpha \geqslant 1$, choose T of order M^{-1} with a sufficiently large constant. This meets the coefficient budget and gives the asserted error. For $0 < \alpha < 1$, the same argument gives $O(M^{-\alpha})$ instead. □

The wavelet rate improves on the $O(M^{-1/2})$ Fourier rate (5.20), but it still fails to exploit the full C^α regularity away from the edge curves.

This rate also holds for the broader bounded-variation model (5.11), for which wavelet approximation is asymptotically optimal.

Figure 5.18 shows wavelet approximations of a bounded variation image.

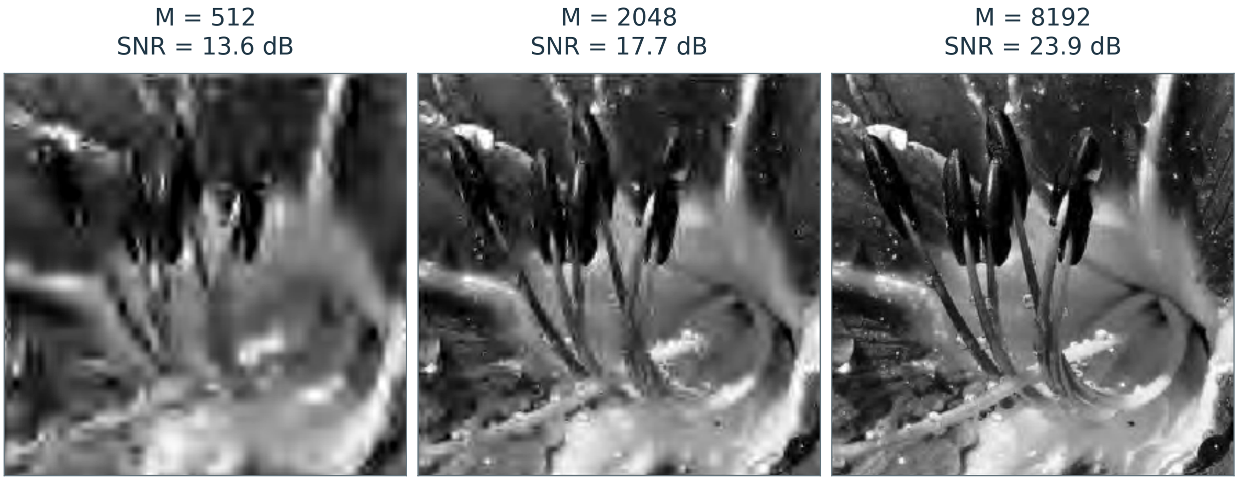


Figure 5.18. 2-D wavelet approximation.

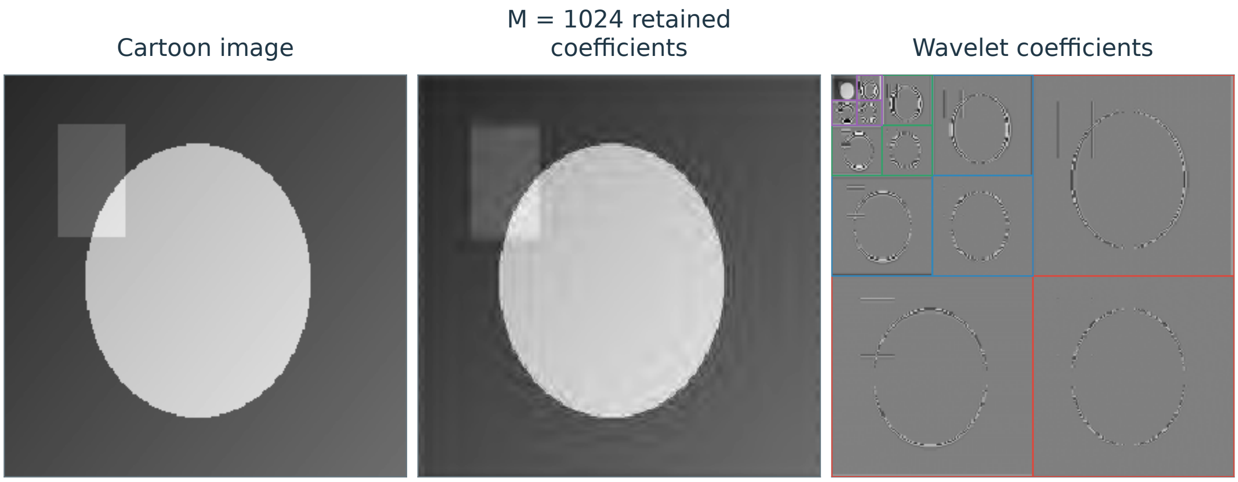


Figure 5.19. Wavelet approximation of a cartoon image.

5.6 Approximation of Cartoon Images

The isotropic, compactly supported wavelets considered here occupy squares that do not follow the smooth contours of geometric images (5.13). These contours have more structure than the finite-length level sets allowed by the bounded-variation model (5.11).

5.6.1 Wavelet Approximation of Cartoon Images

The bound (5.31) gives a wavelet approximation rate of $O(M^{-1})$ for cartoon images (5.13). Even the simple indicator $f = 1_\Omega$, where Ω is a disk, can attain this slow rate: isotropic wavelets do not efficiently follow its curved boundary. The smoothed cartoon model (5.14) can behave similarly when the coefficient budget M is too small to resolve the blur.

Figure 5.19 shows that many large coefficients are located near edge curves, and retaining only a small number leads to a poor approximation with visually unpleasant artifacts.

5.6.2 Finite Element Approximation

An adaptive triangulation can improve the approximation by using elongated triangles aligned with the edges. Figure 5.20 shows an example of such a triangulation.

Select M points in the image domain $[0,1]^2$ and connect them into a triangulation. Define $\tilde{f}_M$ by piecewise affine interpolation on the resulting triangles.

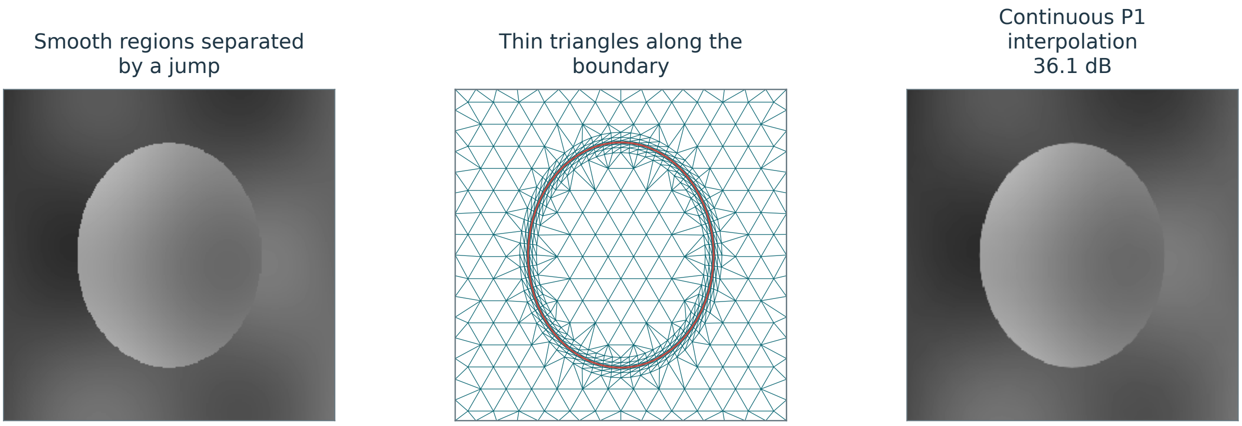


Figure 5.20. A cartoon with smooth variation on both sides of an elliptical jump, a triangulation elongated along that boundary, and the continuous P_1 interpolant.

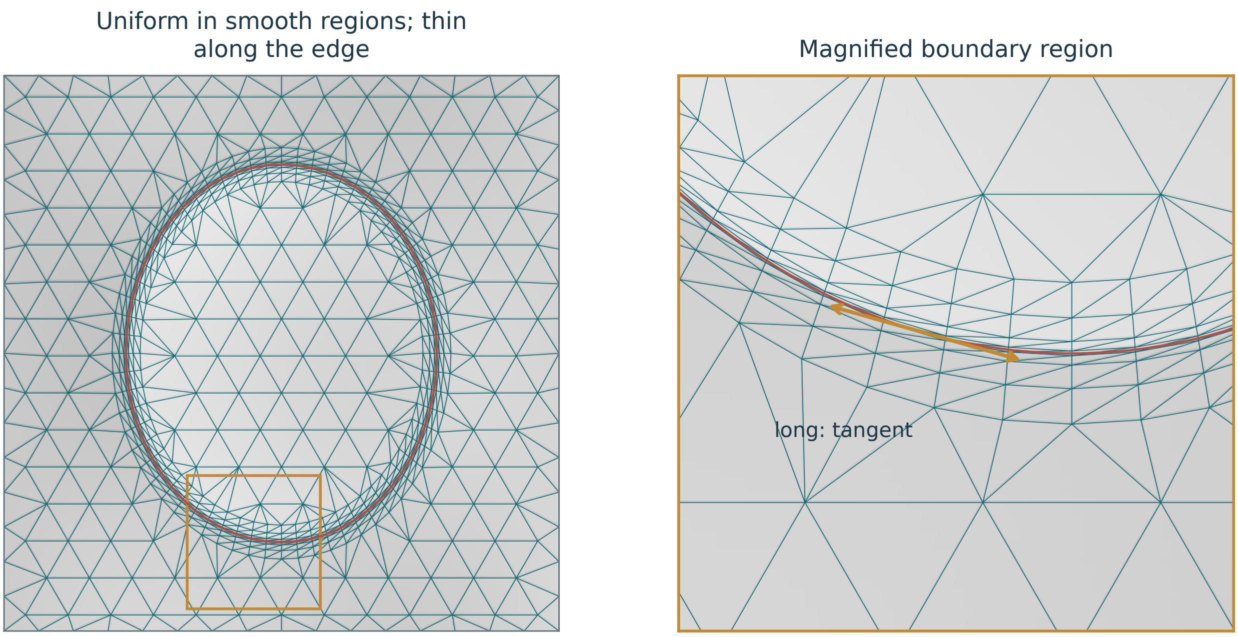


Figure 5.21. The same mesh over a lightened P_1 interpolant (left), with the boxed boundary region enlarged (right). Thin triangles follow the red singularity curve; the arrow indicates its tangent. Smooth regions use nearly equilateral elements.

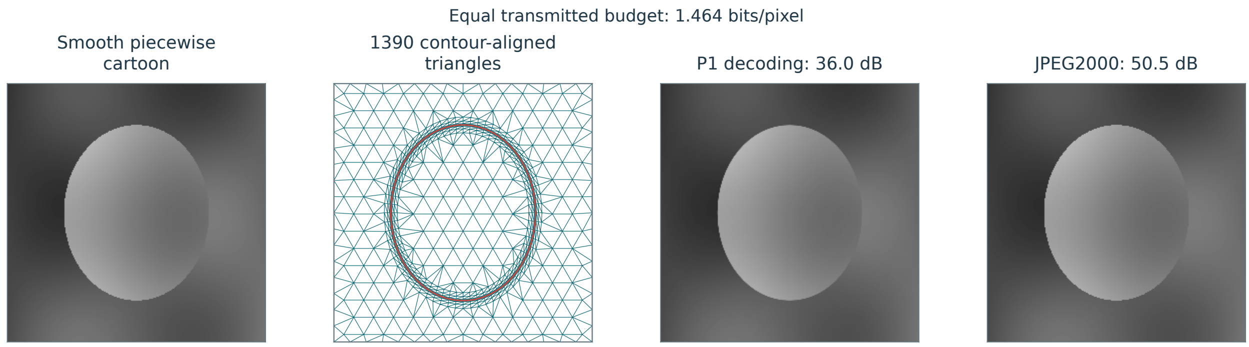


Figure 5.22. Compression of a piecewise-smooth cartoon using a contour-aligned mesh and JPEG-2000 at the same transmitted bit budget. From left to right: original, mesh, decoded P_1 image, and JPEG-2000 image. Mesh costs include coordinates, nodal values, connectivity, and headers.

Figure 5.21 illustrates a construction for C^2 cartoon images (5.13) with $\alpha = 2$. In smooth regions, use $\approx M/2$ nearly equilateral triangles of width $\approx M^{-1/2}$. Near the edges, exploit the C^2 contour regularity by placing $\approx M/2$ elongated triangles of length $\approx M^{-1}$ along the contours and width $\approx M^{-2}$ across them. Such a triangulation yields the error bound

$$\|f - f_M\|^2 = O(M^{-2}), \tag{5.32}$$

which improves over the wavelet approximation error decay (5.31).

Implementing this construction requires estimating the edges. This is particularly difficult for blurred cartoon images with an unknown smoothing kernel h.

Practical methods use heuristics or greedy rules to select sample points and triangulate them from discrete or noisy data. Figure 5.22 uses a known synthetic contour to isolate the effect of mesh geometry. Both regions have smooth, nonaffine intensities. The displayed P_1 image is decoded from an explicit mesh stream whose bit count includes vertex coordinates, values, connectivity, and headers; this example does not assume an optimal mesh code.

5.6.3 Curvelet Approximation

A fixed family of oriented, anisotropic atoms offers an alternative to adaptive triangulations. Candès and Donoho introduced the curvelet frame for this purpose [5].

Curvelets. At fine scales $j \leqslant 0$, the geometry can be illustrated by applying parabolic scaling to a horizontally oriented mother function c,

$$c_{2^j}(x_1, x_2) \approx 2^{-3j/4} c(2^{-j/2} x_1, 2^{-j} x_2),$$

followed by translation and rotation,

$$c^{\theta}_{2^j,u}(x_1, x_2) = c_{2^j}(R_{-\theta}(x - u))$$

where R_θ is the rotation of angle θ.

The atom $c^{\theta}_{2^j,u}$ is localized near u and oriented at angle θ. Its dimensions obey "width $\approx$ length^2", the same relation used for adaptive triangles near edges. This scaling matches the contour geometry of the cartoon model (5.13) with $\alpha = 2$.

Figure 5.23 illustrates this localization in space and frequency.

Parameter discretization. To build an image representation, one needs to sample the position u and orientation θ. The angular sampling interval depends on the scale:

$$\forall\, 0 \leqslant k < 2^{-\lceil j/2 \rceil + 2}, \quad \theta^{(j)}_k = k\pi 2^{\lceil j/2 \rceil - 1}$$

and the spatial grid depends on both scale and orientation:

$$\forall\, m = (m_1, m_2) \in \mathbb{Z}^2, \quad u^{(j,\theta)}_m = R_\theta(2^{j/2} m_1, 2^j m_2).$$

Figure 5.24 shows this sampling pattern.

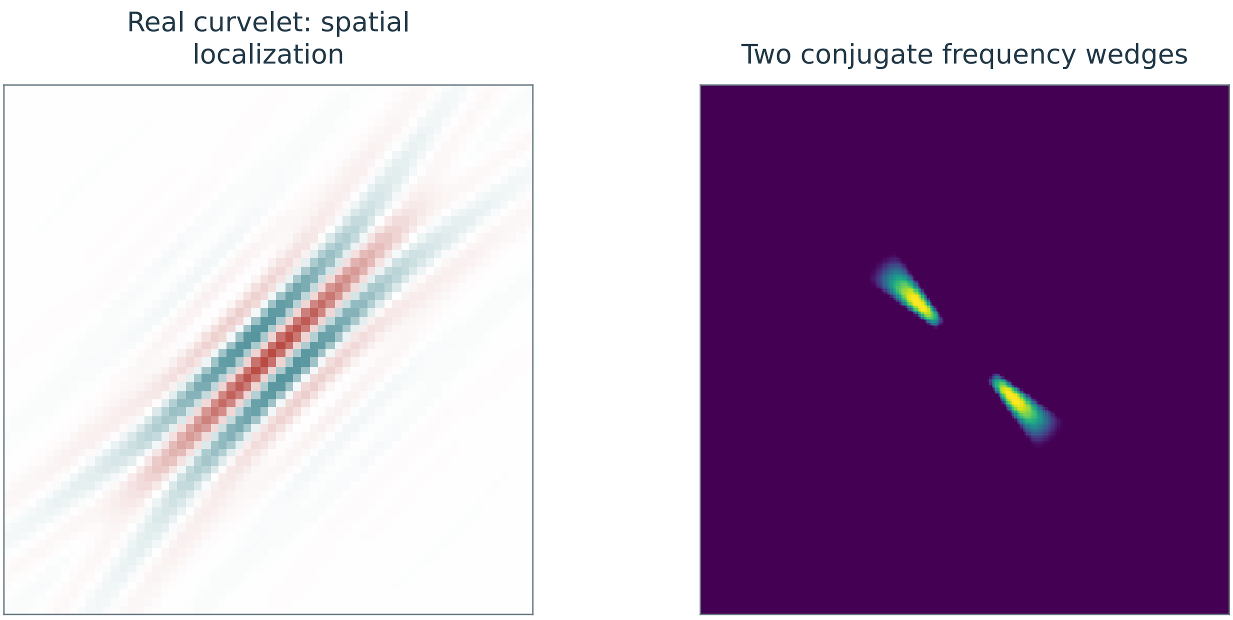


Figure 5.23. A curvelet c_m in space (left) and its localization in the Fourier domain (right).

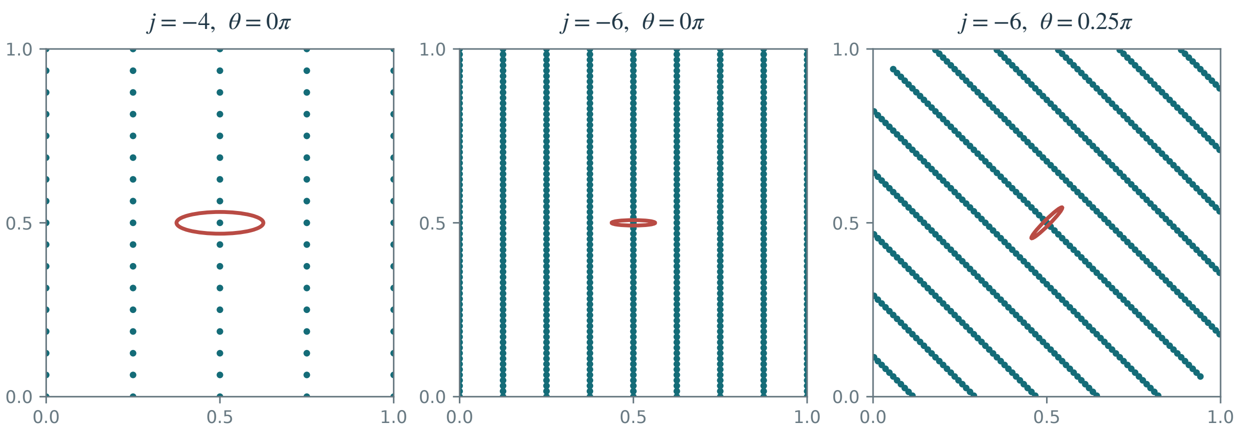


Figure 5.24. Sampling pattern for the curvelet positions.

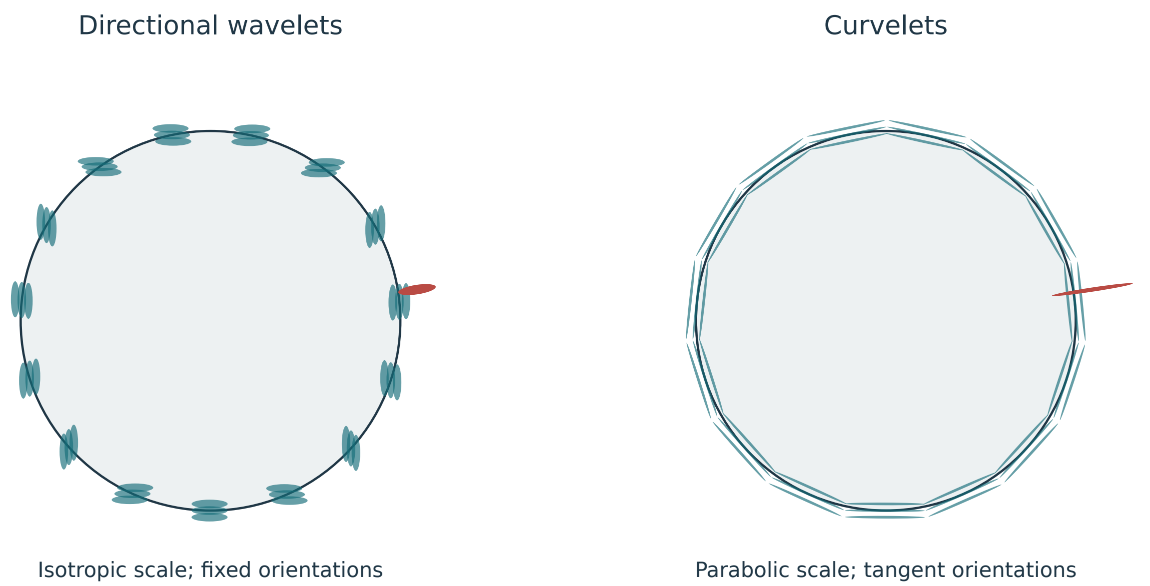


Figure 5.25. Wavelet (left) and curvelet (right) approximation of a cartoon image.

Curvelet tight frame. The scaling and sampling formulas above describe the geometry of curvelets. An exact tight frame additionally requires carefully chosen frequency windows and a low-frequency family; an arbitrary mother function c does not suffice. With this construction, write the complete family as $(C_\lambda)_{\lambda\in\Lambda}$, where Λ includes the fine-scale indices (j, m, k) and the low-frequency indices. Then

$$\|f\|^2 = \sum_{\lambda\in\Lambda} |\langle f, C_\lambda\rangle|^2, \qquad f = \sum_{\lambda\in\Lambda} \langle f, C_\lambda\rangle C_\lambda.$$

The reconstruction converges in L^2. These are the Parseval-frame identities; the family is redundant and its atoms are not mutually orthogonal.

One discrete construction for an image with N pixels uses $\approx 5N$ atoms [7].

Curvelet approximation. A nonlinear M-term approximation in curvelets is defined as

$$f_M = \sum_{\lambda\in\Lambda:\, |\langle f, C_\lambda\rangle|>T} \langle f, C_\lambda\rangle C_\lambda$$

The complete index set Λ includes the low-frequency family, and the threshold T determines the number M of retained coefficients. As before, ties require a selection rule if an exact coefficient count is prescribed. Because the frame is not orthogonal, f_M need not be the best M-term curvelet approximation.

Far from an edge, at locations $u_m^{(j,\theta)}$, cancellation makes the coefficients $\langle f, C_{j,m,k}\rangle$ small. When $u_m^{(j,\theta)}$ lies near an edge with tangent direction $\tilde{\theta}$, the coefficient $\langle f, C_{j,m,k}\rangle$ rapidly decreases as the orientation mismatch $|\theta - \tilde{\theta}|$ grows. Figure 5.25 compares this adaptation to edges with the square supports of directional wavelets.

Combining spatial cancellation, directional selectivity, and scale-dependent sampling gives the approximation bound

$$\|f - f_M\|^2 = O(\log^3(M) M^{-2})$$

for images in the cartoon model (5.13) for $\alpha = 2$. This nearly matches the adaptive-triangulation rate (5.32), while allowing f_M to be computed in $O(N \log(N))$ operations for an image of N pixels.

Redundancy complicates the use of curvelets for compression, but their approximation properties are useful for denoising geometric images and textures. Figure 5.26 compares curvelet and wavelet denoising using the thresholding method of Section 7.3.1.

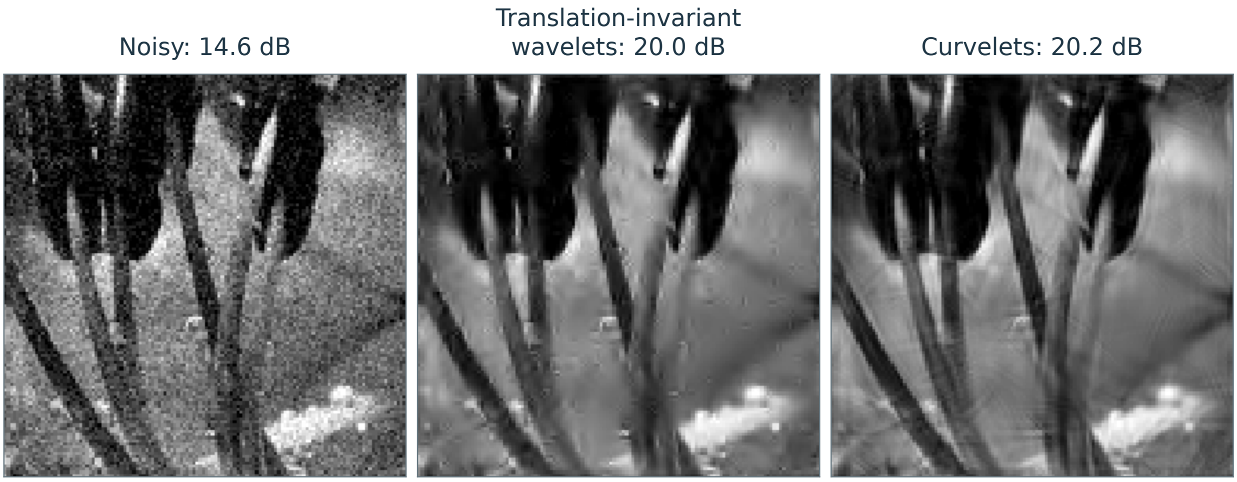


Figure 5.26. Comparison of translation-invariant wavelet denoising and curvelet denoising.

CHAPTER 6

Compression

Compression must turn a small set of transform coefficients into a short binary description without introducing excessive reconstruction error. This requires controlling both the loss from quantization and the cost of recording coefficient values and locations. We derive error and bit-rate bounds for transform coding, explain how probability models reduce coding costs, and describe the wavelet and entropy-coding stages of JPEG-2000.

6.1 Transform Coding

6.1.1 Coding

Choose an orthonormal basis, such as a wavelet basis, and write the coefficients as

$$a_m = \langle f, \psi_m \rangle \in \mathbb{R}.$$

Quantization maps each coefficient to an integer using a step size $T > 0$:

$$q_m = Q_T(a_m) \in \mathbb{Z} \quad \text{where} \quad Q_T(x) = \operatorname{sign}(x) \left\lfloor \frac{|x|}{T} \right\rfloor.$$

This quantizer has a zero bin of width $2T$: coefficients in $(-T, T)$ are set to zero.

Like hard thresholding (5.4), quantization discards coefficients smaller than T in magnitude. It also rounds the retained coefficients, introducing an additional error; see Figure 6.1.

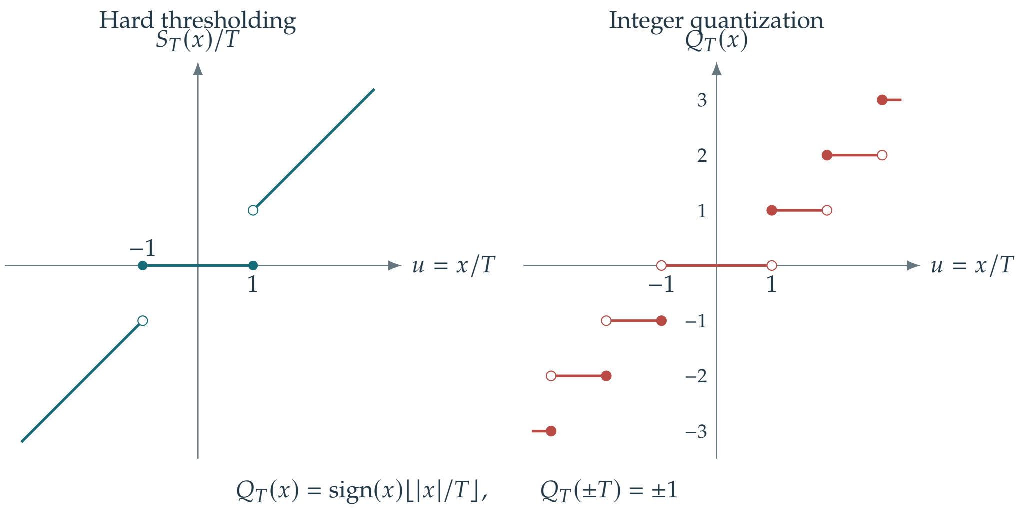


Figure 6.1. Hard thresholding and quantization as functions of the input coefficient.

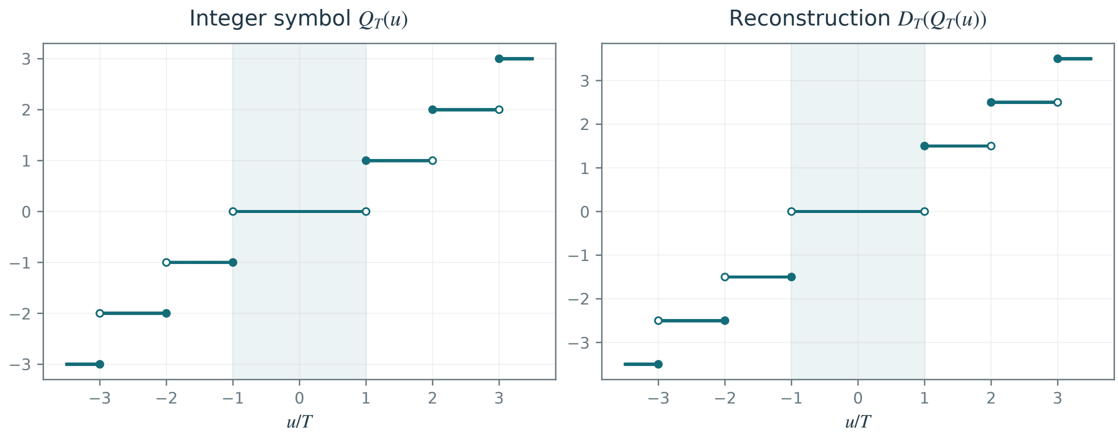
The integer coefficients $(q_m)_m$ are encoded in a binary stream of length R bits. Sections 6.1.3 and 6.2 describe two approaches to encoding these integers. The goal is to minimize R for an acceptable distortion.

6.1.2 Decoding

The decoder retrieves the integers q_m from the binary file and reconstructs coefficient values using

$$\tilde{a}_m = \operatorname{sign}(q_m) \left(|q_m| + \frac{1}{2} \right) T. \tag{6.1}$$

Each nonzero symbol is reconstructed at the center of its quantization bin; zero is reconstructed as zero:



The decoded signal or image is reconstructed as

$$Q_T(f) \stackrel{\text{def.}}{=} \sum_{m\in I_T} \tilde{a}_m \psi_m = \sum_m D_T(Q_T(\langle f, \psi_m \rangle))\psi_m, \qquad I_T = \{m : |a_m| \geqslant T\}, \tag{6.2}$$

where $D_T(q) = T\,\mathrm{sign}(q)(|q| + 1/2)$ and $\mathrm{sign}(0) = 0$. This produces a reconstruction error $\|f - Q_T(f)\|$.

Let $M = \#I_T$ and let f_M retain the coefficients indexed by I_T without quantizing them. This is the approximation (5.3), with equality at the threshold retained here. Orthogonality gives $\|f - f_M\| \leqslant \|f - Q_T(f)\|$. The following bound separates the approximation error from the additional error caused by rounding.

Proposition 6.1. *With this choice of f_M,*

$$\|f - Q_T(f)\|^2 \leqslant \|f - f_M\|^2 + MT^2/4 \quad \textit{where} \quad M = \#\{m\ ;\ \tilde{a}_m \neq 0\}\,. \tag{6.3}$$

Proof. The reconstruction rule (6.1) gives, for $|a_m| \geqslant T$,

$$|a_m - \tilde{a}_m| \leqslant \frac{T}{2}.$$

By orthogonality,

$$\|f - Q_T(f)\|^2 = \sum_m (a_m - \tilde{a}_m)^2 \leqslant \sum_{|a_m|<T} |a_m|^2 + \sum_{|a_m|\geqslant T} \left(\frac{T}{2}\right)^2,$$

The first sum is $\|f - f_M\|^2$, and the second contains M terms, proving the bound. □

6.1.3 Support Coding

To express the error bound (6.3) in terms of a bit budget, we must relate M and T to the cost of encoding the coefficients.

At high compression ratios, only a small number M of the N coefficients remain nonzero. Their support

$$I_M = \{m\ ;\ \tilde{a}_m \neq 0\}$$

can be encoded separately from the nonzero values $q_m \neq 0$ for $m \in I_M$.

The following theorem gives a rate bound for this support-and-value coding strategy.

Theorem 6.2. *Assume that $f \in L^2$ has best M-term approximation error $\|f - f_M\|^2 \leqslant CM^{-\alpha}$ for some $\alpha > 0$. Suppose that, for each sufficiently small $T > 0$, all coefficients of magnitude at least T occur among a known first $N(T)$ basis elements, with $N(T) \leqslant C_0 T^{-\gamma}$ for some $\gamma > 0$. Then $Q_T(f)$ admits a binary description of length $R(T)$ such that, as $R(T) \to \infty$,*

$$\|f - Q_T(f)\|^2 = O\left(\left(\frac{\log R(T)}{R(T)}\right)^{\alpha}\right).$$

The decoder is assumed to know the basis and quantization step; describing these parameters adds any required finite header.

Proof. **Coefficient and quantization bounds.** By Proposition 5.1, the ordered magnitudes satisfy

$$d_m \leqslant C_1 m^{-(\alpha+1)/2}. \tag{6.4}$$

Put $K = T^{-2/(\alpha+1)}$. At most $M = O(K)$ coefficients have magnitude at least T. Summing the squared errors of the zero and nonzero quantization cells gives

$$\|f - Q_T(f)\|^2 \leqslant \sum_{m \geqslant 1} \min(d_m^2, T^2) \leqslant \lceil K \rceil T^2 + \sum_{m > \lceil K \rceil} d_m^2 = O(K^{-\alpha}). \tag{6.5}$$

An upper approximation bound is enough for this estimate.

Finite computation. The finite-access assumption is

$$\forall m \geqslant N(T), \qquad |\langle f, \psi_m \rangle| < T. \tag{6.6}$$

For a bounded signal on the unit cube with periodic boundary conditions, using periodized compactly supported d-dimensional wavelets, the coefficient bound is $O(2^{jd/2})$ at fine scales $j \to -\infty$. There are $O(2^{-Jd})$ coefficients at scales coarser than J. Retaining scales down to J with $2^{Jd/2}$ proportional to T therefore suffices, and one may choose

$$N(T) = O(T^{-2}). \tag{6.7}$$

More generally, the hypothesis gives

$$N(T) = O(K^{\gamma(\alpha+1)/2}). \tag{6.8}$$

The ordering here is a fixed computable ordering of basis elements, not the unknown ordering by coefficient magnitude.

Support and value coding. Encode each of the M nonzero indices with $\lceil \log_2 N \rceil$ bits:

$$R_{\text{ind}} = M \lceil \log_2 N \rceil = O(K \log K). \tag{6.9}$$

Since $|a_m| \leqslant \|f\|_2$, every quantized value lies in $[-A, A] \cap \mathbb{Z}$ with $A \leqslant \|f\|_2 / T$. Thus

$$R_{\text{val}} \leqslant M \lceil \log_2(2A+1) \rceil = O(K \log K). \tag{6.10}$$

A header encoding M, N, and the integer amplitude bound A has logarithmic size and is absorbed in this bound. Consequently,

$$R = O(K \log K). \tag{6.11}$$

Because the inverse of $x \mapsto x \log x$ is asymptotic to $r / \log r$, this implies

$$K \geqslant cR / \log R \tag{6.12}$$

for large R. Combining this with (6.5) proves the claim. □

The theorem links compression rates to nonlinear approximation rates and motivates choosing a basis adapted to the signal model Θ. A practical compression algorithm operates on a discrete signal or image of size N. It computes N transform coefficients $\{\langle f, \psi_m \rangle\}_{0 \leqslant m < N}$ from these data. Section 4.4 describes the discrete wavelet transform and gives a compatibility condition (4.5) under which these discrete coefficients coincide with the continuous inner products.

6.2 Entropy Coding

Entropy coding exploits the uneven distribution of the quantized coefficients to reduce the expected file size: zeros are frequent, while large values are rare. Shannon's coding theory [30] quantifies the resulting savings.

We refer to Section 3.1 for the theoretical foundations of the methods described below.

Probabilistic modeling. The quantized coefficients $q_m \in \{-A, \dots, A\}$ take values in an alphabet of $Q = 2A+1$ elements. A code maps the coefficient sequence to a binary string:

$$\{q_m\}_m \longmapsto \{0, 1, 1, \dots, 0, 1\} \in \{0, 1\}^R.$$

We first model the q_m as independent draws from a known probability distribution:

$$\mathbb{P}(q_m = i) = p_i \in [0, 1].$$

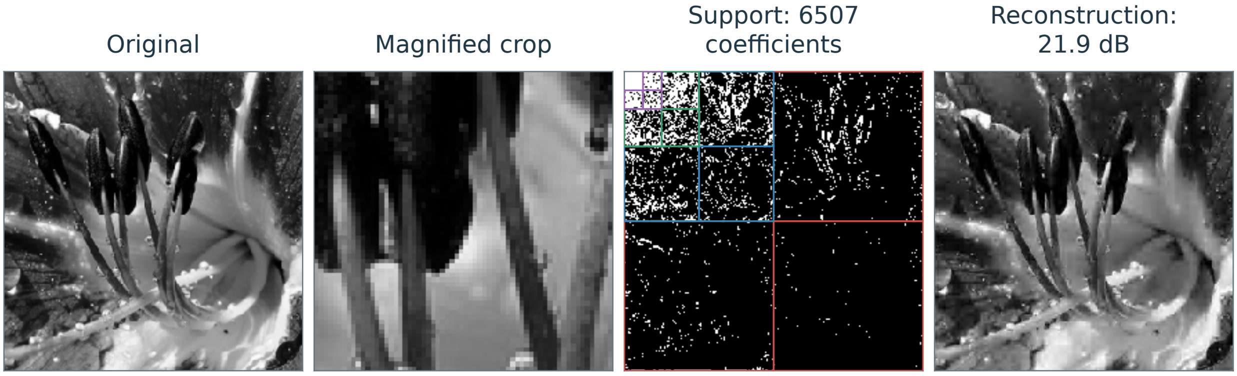


Figure 6.2. Image compression using wavelet support coding.

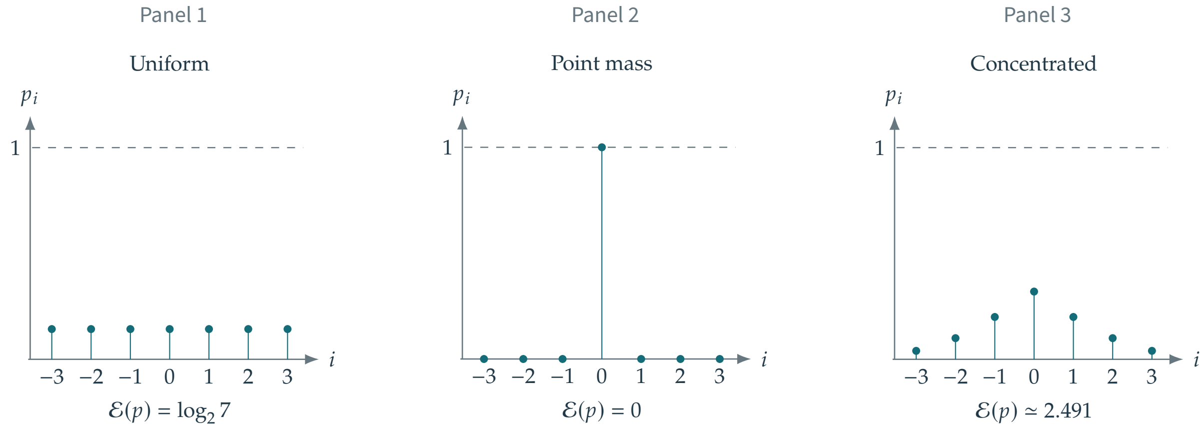


Figure 6.3. Three different probability distributions on the same alphabet of $Q = 7$ symbols. Entropy is measured in bits.

Huffman code. A Huffman code assigns a variable-length binary string to each symbol:

$$q_m = i \in \{-A, \ldots, A\} \longmapsto c_i \in \{0, 1\}^{|c_i|}$$

where $|c_i|$ is the codeword length. Codeword lengths are assigned so that a less probable symbol receives no shorter a codeword than a more probable symbol. Huffman coding minimizes the expected length among binary prefix codes. Shannon's construction gives lengths $\lceil -\log_2 p_i \rceil$ for positive-probability symbols, so optimality of Huffman's code implies

$$\mathbb{E}R = N \sum_i p_i |c_i| \leqslant N(\mathcal{E}(p) + 1),$$

where $\mathcal{E}$ is the entropy of the distribution, defined as

$$\mathcal{E}(p) = -\sum_i p_i \log_2(p_i).$$

Figure 6.3 illustrates how entropy depends on the distribution. Concentrated distributions have low entropy, as is often the case for quantized wavelet coefficients because many are zero.

Symbol-by-symbol Huffman coding uses an integer number of bits per symbol. This constraint can cause a substantial gap above entropy when one symbol is very probable. Arithmetic coding encodes whole sequences and can achieve an expected length close to $N\mathcal{E}(p)$, with a small overhead, under the assumed probability model.

6.3 JPEG-2000

JPEG-2000 is a family of still-image compression standards based on wavelet transform coding and adaptive entropy coding. Its irreversible transform uses the biorthogonal CDF 9/7 filters; its reversible transform uses

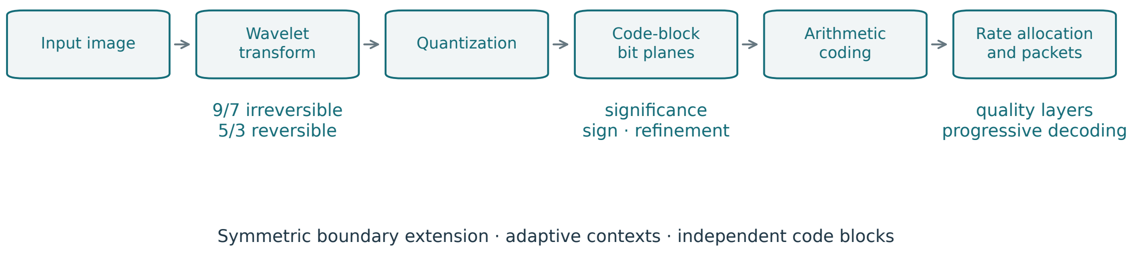


Figure 6.4. JPEG-2000 coding architecture.

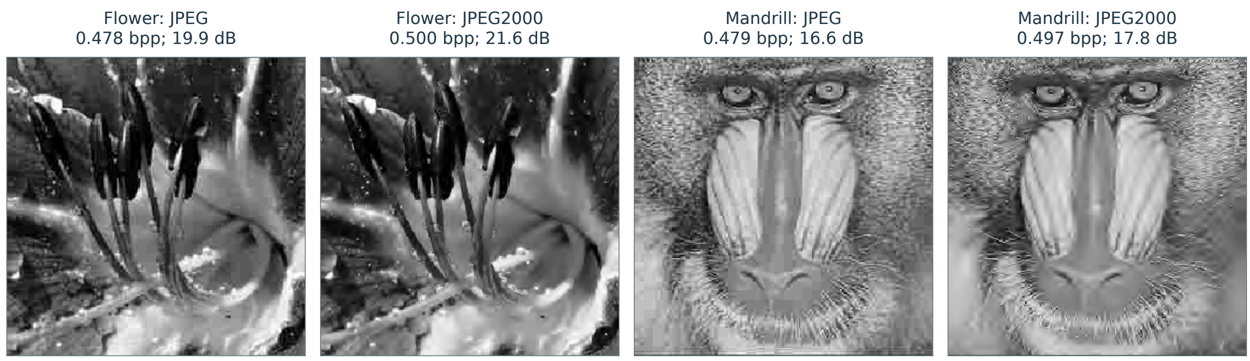


Figure 6.5. JPEG and JPEG-2000 reconstructions of the flower (first pair) and mandrill (second pair), at matched target bit rates.

integer-to-integer 5/3 lifting for lossless coding. Symmetric boundary extension limits edge artifacts. Because the transform is biorthogonal, the orthonormal energy identities above are replaced by bounds involving the synthesis operator.

Figure 6.4 outlines the JPEG-2000 architecture, and Figure 6.5 compares its reconstructions with those of JPEG. JPEG uses a blockwise DCT and can exhibit visible block boundaries at low bit rates. JPEG-2000 uses a wavelet transform and supports features such as regions of interest, which allow selected parts of an image to be encoded more accurately.

Progressive coefficient refinement. Bit-plane coding progressively reveals the binary digits of the quantized coefficient magnitudes. We describe this refinement schematically using thresholds $T_i = 2^{-i}T_0$: each new bit plane resolves coefficient values at a finer scale.

Rate allocation and quality layers. The coder processes coefficient blocks S_k independently, producing embedded streams c_i^k. The encoder selects coding-pass truncation points within these streams to balance rate against estimated distortion, then organizes the retained data into quality layers. Decoding additional layers improves image quality. This construction supports progressive transmission, with truncation at valid codestream boundaries. The allocation uses an additive distortion model; it does not guarantee the smallest reconstruction error for every possible bit budget.

Bit-plane coding. At threshold T_i, consider a coefficient $d_j^\omega[n]$ at scale j, orientation $\omega \in \{V, H, D\}$, and position $n \in S_k$. Three types of bits describe its significance, sign, and magnitude. We suppress (j, ω) in the notation for these bits.

– If $|d_j^\omega[n]| < T_{i-1}$, the coefficient was not significant at bit-plane $i-1$. The encoder writes a significance bit $b_i^1[n]$ indicating whether $|d_j^\omega[n]| \geqslant T_i$.

– If $b_i^1[n] = 1$, the coefficient has just become significant, and the encoder writes its sign as a bit $b_i^2[n]$.

– For every position n that was previously significant, meaning $|d_j^\omega[n]| \geqslant T_{i-1}$, the encoder writes a refinement bit $b_i^3[n]$ giving the next binary digit of the coefficient magnitude, namely $\lfloor |d_j^\omega[n]|/T_i \rfloor$ mod 2.

Context modeling. The bits $\{b_i^s[n]\}_{s=1}^3$ for $n \in S_k$ are compressed using context-adaptive arithmetic coding to form the streams c_i^k. Contexts exploit spatial dependence, especially near edges, where large wavelet coefficients tend to cluster. The scan proceeds through stripes four samples high, visiting each stripe column by column.

For a coefficient at position $n \in S_k$, the context index $v_i^s[n]$ summarizes its coding state and those of coefficients in its 3×3 neighborhood

$$w_n = \{(n_1 + \varepsilon_1, n_2 + \varepsilon_2) : \varepsilon_1, \varepsilon_2 \in \{-1, 0, 1\}\}.$$

The relevant states include whether neighboring coefficients have become significant, their known signs, and the coefficient's refinement history. Only information already available to the decoder may enter the context. The precise context rules differ for significance, sign, and refinement bits.

An adaptive arithmetic coder estimates the conditional probability $\mathbb{P}(b_i^s[n] | v_i^s[n])$ and uses it to encode $b_i^s[n]$. Informative contexts can reduce the conditional entropy and hence the expected number of bits.

CHAPTER 7

Denoising

Denoising estimates a clean signal from measurements corrupted by noise. Its central difficulty is to suppress random fluctuations without erasing edges, textures, or other features of the underlying signal. We model common acquisition noise, compare linear filters with nonlinear thresholding, analyze how regularity and sparsity govern reconstruction error, and adapt the methods to Poisson and multiplicative noise.

7.1 Noise Modeling

7.1.1 Noise in Images

Figure 7.1 illustrates the variety of imaging noise. Different acquisition devices produce different noise statistics, while the images themselves vary in smoothness and geometry.

The model should reflect both the acquisition process and the noise statistics. The regularity and geometry of the clean signal then guide the choice of a basis for its representation.

Since noise perturbs discrete measurements from acquisition devices, we consider only finite-dimensional real signals $f \in \mathbb{R}^N$.

7.1.2 Image Formation

Figure 7.2 summarizes the model: a clean image f_0 combines with noise w to produce the observation $Y := f_0 \oplus w$. The operation $\oplus$ may be addition or multiplication. The observed image Y is modeled as a random vector.

In the statistical model, w is a random vector with a specified distribution. A numerical experiment uses one realization, also written w.

Additive Noise. The simplest image formation model adds noise to a clean signal f_0

$$Y := f_0 + w$$

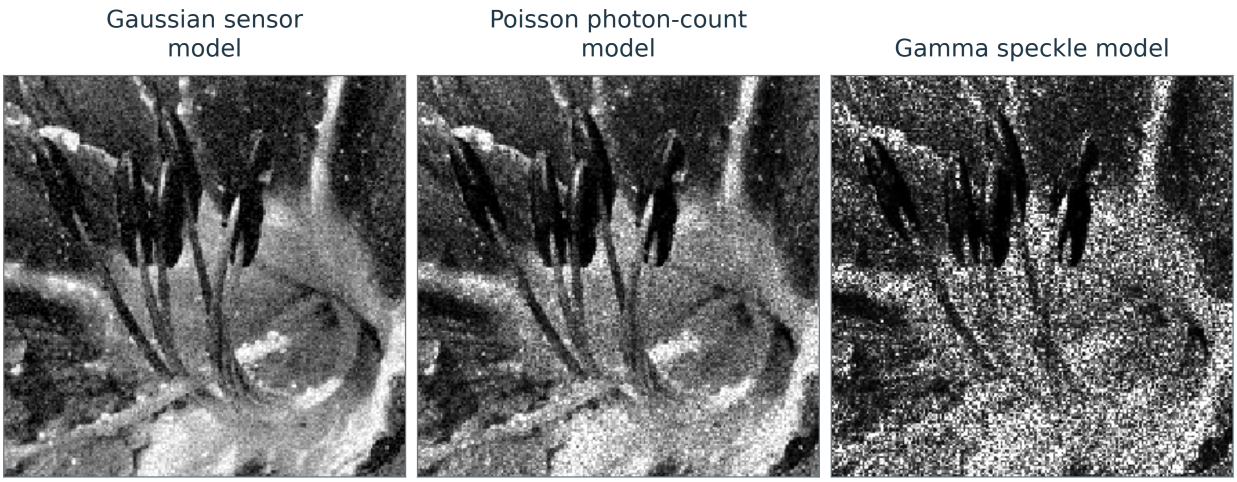


Figure 7.1. Gaussian, Poisson, and multiplicative Gamma noise simulated on the same flower image.

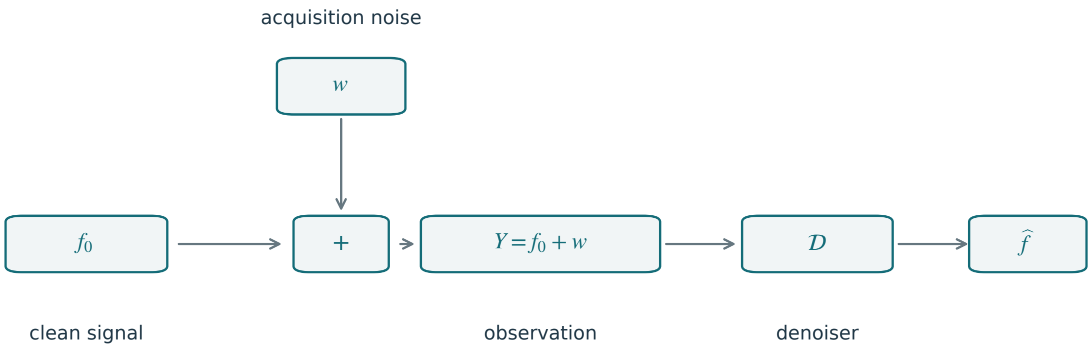


Figure 7.2. Image acquisition, noise, and reconstruction by denoising.

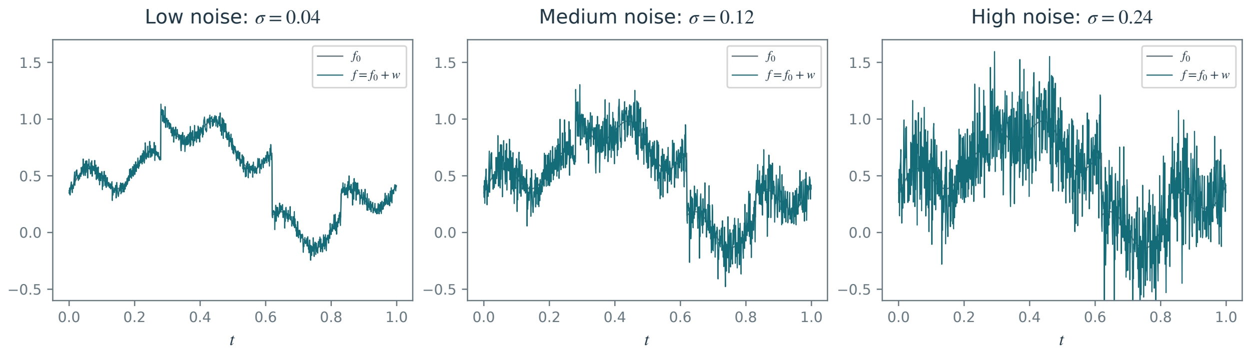


Figure 7.3. One-dimensional additive Gaussian noise at low, medium, and high levels, with a common clean signal and normalized noise realization.

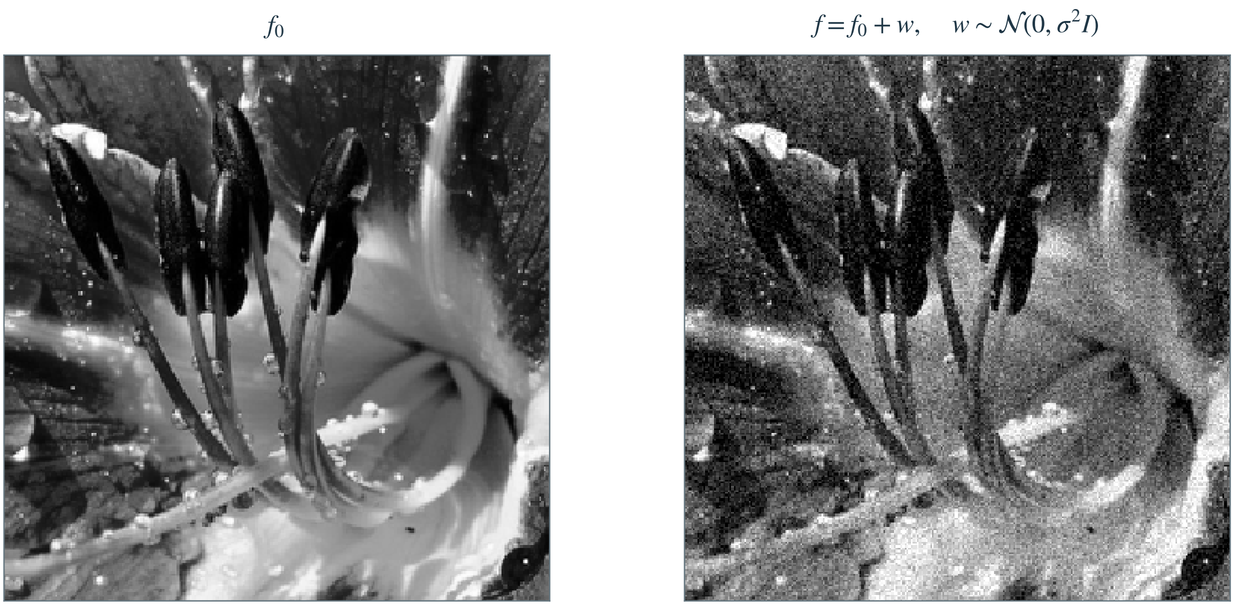


Figure 7.4. Two-dimensional additive noise: the clean image f_0 and the observation $f = f_0 + w$, with $w \sim \mathcal{N}(0, \sigma^2 I)$.

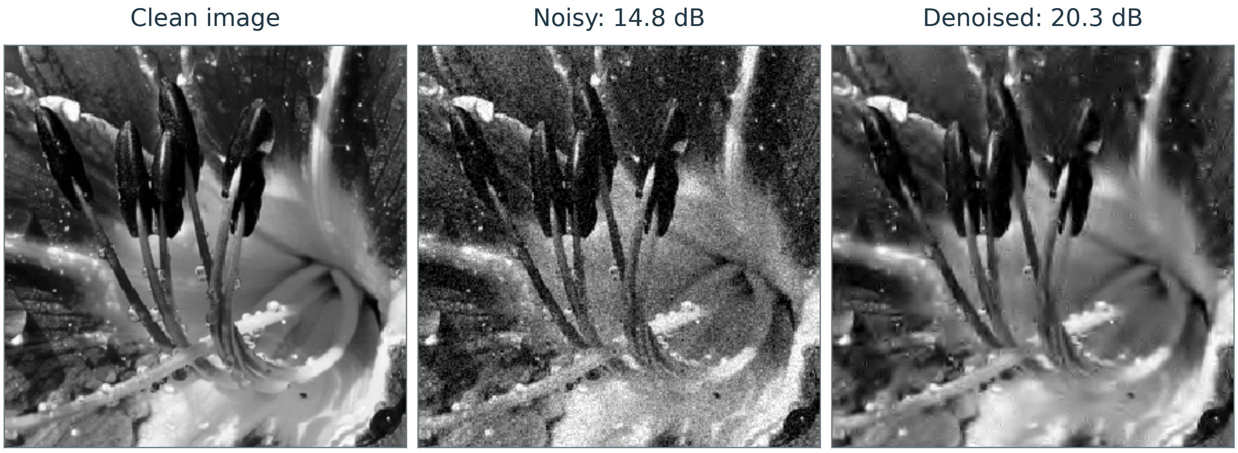


Figure 7.5. Left: clean image, center: noisy image, right: denoised image.

where w is the noise residual. Both w and Y are random; an acquired image is one realization of Y, denoted by y or f below. Figures 7.3 and 7.4 illustrate additive noise on a signal and an image.

The simplest model assumes independent centered Gaussian entries w_n with variance σ^2: $w \sim \mathcal{N}(0, \sigma^2 \mathrm{Id}_N)$. This is Gaussian white noise.

Depending on the image acquisition device, other noise distributions may be appropriate, including uniform noise $w_n \in [-a, a]$ or heavy-tailed generalized Gaussian noise with density

$$p_w(u) \propto e^{-|u/s|^\alpha}, \qquad s > 0, \quad 0 < \alpha < 2.$$

In many situations, the noise is not additive; its intensity may, for instance, depend on the signal intensity. This is the case with the Poisson and multiplicative noise models considered in Section 7.4.

7.1.3 Denoiser

A denoiser produces an estimate $\tilde{f} = \mathcal{D}(Y)$ of f_0 from the observation Y alone. The map $\mathcal{D}$ is deterministic, but its output is random because it depends on the noise w. With centered additive noise, Y has mean f_0, so denoising estimates that mean from a single realization. Figure 7.5 illustrates the result.

The quality of a denoiser is measured using the mean squared risk $\mathbb{E}_w \|f_0 - \tilde{f}\|^2$, where $\mathbb{E}_w$ averages over the noise w. This risk is a theoretical quantity because f_0 is unknown in practice. In experiments with known clean data, we instead evaluate one realization $y \sim Y = f_0 + w$ using the signal-to-noise ratio $\mathrm{SNR}(\mathcal{D}(y), f_0)$, defined by

$$\mathrm{SNR}(f, g) := -20 \log_{10}(\|f - g\| / \|g\|).$$

The SNR is measured in decibels (dB). Computing it requires the clean signal f_0, so it serves as a benchmark in controlled experiments. Because an ℓ^2 error does not fully describe perceptual quality, visual inspection should accompany the numerical comparison.

7.2 Linear Denoising by Filtering

7.2.1 Translation Invariant Estimators

A linear estimator $\mathcal{D}(Y) = \tilde{f}$ of f_0 preserves addition, $\mathcal{D}(f + g) = \mathcal{D}(f) + \mathcal{D}(g)$, and scalar multiplication. A translation-invariant estimator commutes with translations: $\mathcal{D}(f_\tau) = \mathcal{D}(f)_\tau$, where $f_\tau(t) = f(t - \tau)$. We assume periodic boundary conditions, in either one or two dimensions. An estimator with both properties is a convolution filter:

$$\mathcal{D}(f) = f \star h$$

where $h \in \mathbb{R}^N$ is a filter. Denoising filters are usually low-pass. To preserve constant signals, one imposes

$$\sum_n h_n = \hat{h}_0 = 1$$

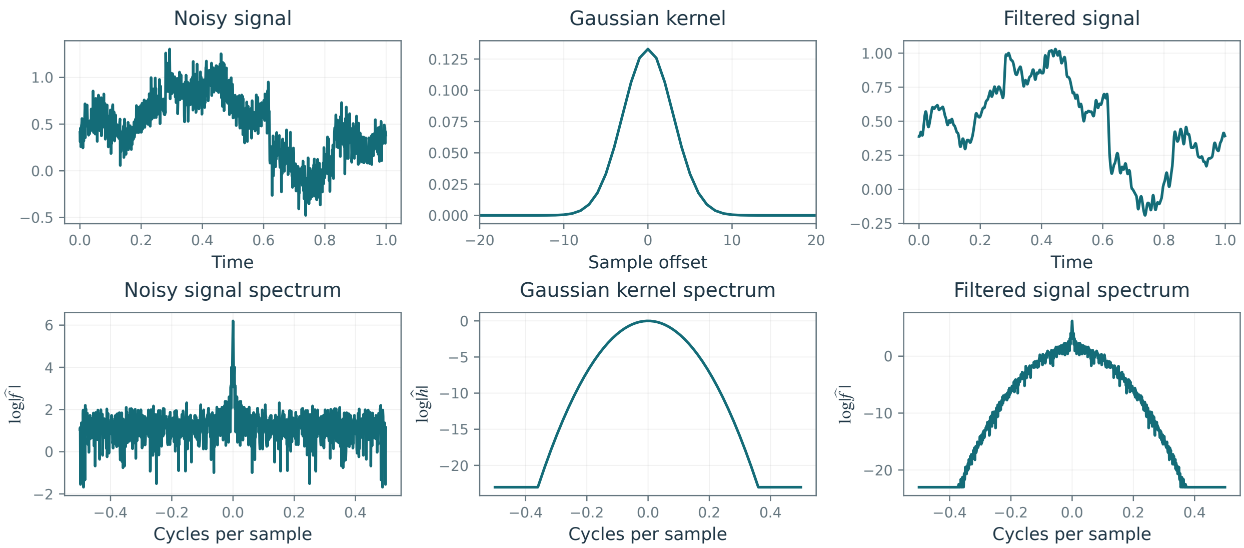


Figure 7.6. Gaussian filtering: noisy signal, kernel, and filtered signal (top), with their Fourier magnitudes below.

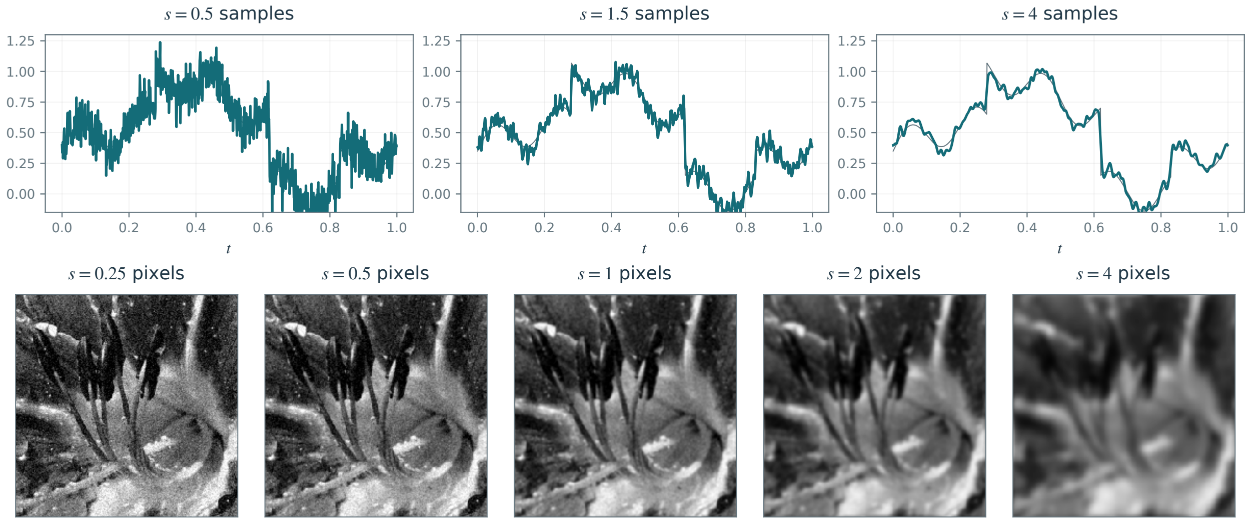


Figure 7.7. Gaussian filtering with increasing width: three signals (top) and five images (bottom).

where $\hat{h}$ is the discrete Fourier transform.

Figure 7.6 illustrates denoising with a low-pass filter.

The filtering strength is usually controlled by the width s of h. A typical example is the (discretized) Gaussian filter

$$\forall -N/2 < i \leqslant N/2, \quad h_{s,i} = \frac{1}{Z_s} \exp\left(-\frac{i^2}{2s^2}\right) \tag{7.1}$$

where Z_s normalizes the filter so that $\sum_i h_{s,i} = 1$, preserving constant signals. Figure 7.6 shows the effect of Gaussian filtering in the spatial and Fourier domains.

Figure 7.7 shows the effect of increasing the filter width s. Linear filtering works well on smooth signals, but removing more noise also blurs edges and other singularities.

7.2.2 Optimal Filter Selection and Bias-Variance Tradeoff

The best filter depends on both the unknown signal f_0 and the noise level σ. Rather than optimize over all filters, we can tune a width parameter s within a family such as the Gaussian filters in (7.1).

The triangle inequality separates the smoothing error from the remaining noise:

$$\|\tilde{f} - f_0\| \leqslant \|h_s \star f_0 - f_0\| + \|h_s \star w\|$$

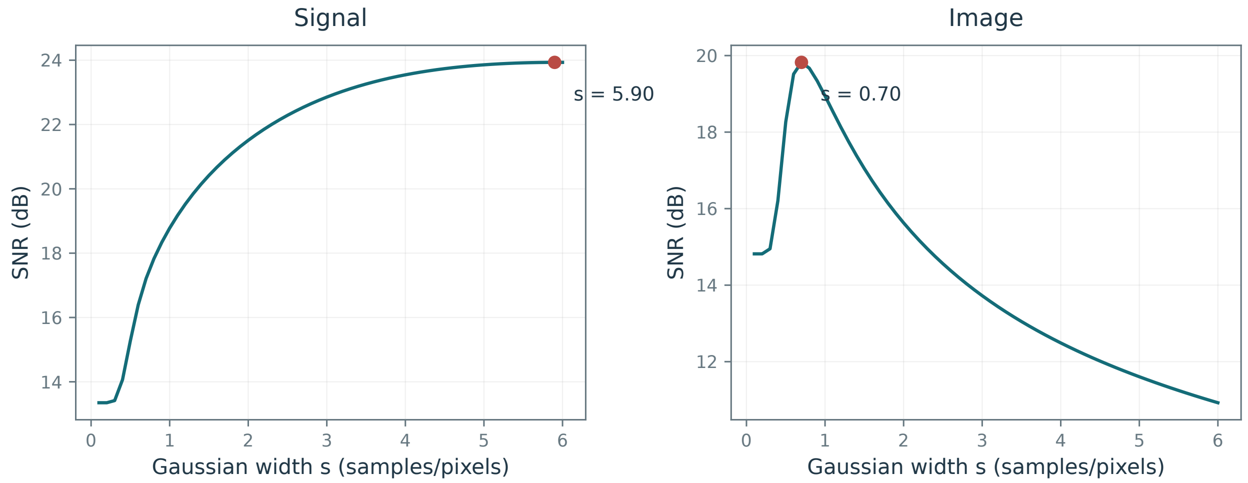


Figure 7.8. SNR as a function of filter width for a signal (left) and an image (right).

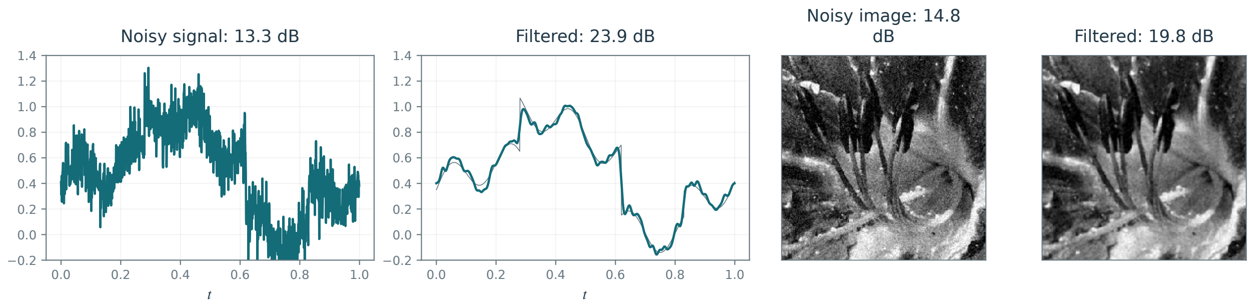


Figure 7.9. Gaussian filtering at the SNR-optimal widths. From left to right: noisy signal, filtered signal, noisy image, and filtered image.

The filter width s balances noise removal against excessive smoothing of singularities. A wider filter typically reduces $\|h_s \star w\|$ while increasing $\|h_s \star f_0 - f_0\|$.

Figure 7.8 shows the SNR as a function of filter width, evaluated using the known clean signal.

Figure 7.9 uses the width $s^\star$ that maximizes SNR for the displayed observation. This choice uses the known clean signal to assess the best achievable result within the filter family.

Some noise remains at the SNR-optimal width. Increasing the filter width introduces strong blur and reduces the SNR, although the result may look more pleasant.

7.2.3 Oracle Denoiser

To study the best possible choice of h, first suppose that an oracle knows the clean signal f_0. Recall the model $Y = f_0 + w$, where w is Gaussian white noise: $w_k \sim \mathcal{N}(0, \sigma^2)$ and the w_k are independent and identically distributed. The clean signal f_0 is fixed. Choose an orthonormal basis $(\psi_k)_k$, typically the Fourier basis, and restrict attention to linear estimators that are diagonal in it:

$$\tilde{f} := \sum_k \lambda_k \langle Y, \psi_k \rangle \, \psi_k. \tag{7.2}$$

For the Fourier basis, this means $\tilde{f} = Y \star h$ where $\hat{h} = \lambda$. Note that $\tilde{f}$ is a random vector. The oracle may use f_0 to select the gains λ, but the estimator must still be linear in Y and have the displayed diagonal form. It cannot simply return f_0.

Denoting $c_k := \langle f_0, \psi_k \rangle$ (for the normalized discrete Fourier basis, $c = \hat{f}_0/\sqrt{N}$), the expected risk of this denoiser is

$$\mathbb{E}_w \|\tilde{f} - f_0\|^2 = \sum_k \mathbb{E}_w \left|\langle \tilde{f} - f_0, \psi_k \rangle\right|^2 = \sum_k \mathbb{E}_w \left|\lambda_k \left(c_k + \langle w, \psi_k \rangle\right) - c_k\right|^2.$$

For a real orthonormal basis, the transformed noise is again independent Gaussian noise with variance σ^2. For a complex unitary Fourier basis applied to real noise, each coefficient still has second moment σ^2, which suffices here, although conjugate-frequency coefficients are dependent. By expanding the square and using $\mathbb{E}(\langle w, \psi_k\rangle) = 0$, we obtain

$$\mathbb{E}_w \|\tilde{f} - f_0\|^2 = \sum_k \left(|c_k|^2 |\lambda_k - 1|^2 + |\lambda_k|^2 \sigma^2\right).$$

Knowing c_k, the oracle minimizes $\mathbb{E}_w\left(\|\tilde{f} - f_0\|^2\right)$ over λ. The minimization separates over the indices k:

$$\min_{\lambda_k} \left(|c_k|^2 |\lambda_k - 1|^2 + |\lambda_k|^2 \sigma^2\right).$$

The solution thus satisfies

$$|c_k|^2(\lambda_k - 1) + \lambda_k \sigma^2 = 0,$$

i.e.,

$$\lambda_k = \frac{|c_k|^2}{|c_k|^2 + \sigma^2}.$$

If $\sigma = 0$, choosing $\lambda = 1$ gives the identity map. When $|c_k|$ decays, increasing σ suppresses the weaker coefficients more strongly, concentrating the gains on the dominant signal components.

7.2.4 Wiener Filter

We now replace access to the clean signal by a known probability model for it. Suppose both the noise $w \sim W$ and the clean signal $f_0 \sim F$ have known distributions. We seek a filter minimizing the error averaged over both sources of randomness, assuming W and F are independent. We assume $\mathcal{D}$ has the form (7.2).

The optimal h thus minimizes

$$\mathbb{E}_{w,F} \|\mathcal{D}(F + w) - F\|^2$$

Independence of W and F allows the same risk calculation:

$$\mathbb{E}_{w,F} \|\mathcal{D}(F + w) - F\|^2 = \sum_k \left(\alpha_k |\lambda_k - 1|^2 + |\lambda_k|^2 \sigma^2\right) \quad \text{where} \quad \alpha_k := \mathbb{E}_F |\langle F, \psi_k\rangle|^2$$

The optimal filter thus has coefficients

$$\lambda_k = \frac{\alpha_k}{\alpha_k + \sigma^2}. \tag{7.3}$$

Relative to the oracle filter, the individual energy $|c_k|^2$ is replaced by the expected coefficient energy $\alpha_k \geqslant 0$, the signal power spectrum in the Fourier basis. This filter is known as the Wiener filter.

For the normalized discrete Fourier basis, $\alpha_k = N^{-1}\mathbb{E}|\hat{F}_k|^2$. Define the averaged autocorrelation

$$\eta_j = \frac{1}{N}\mathbb{E}\sum_{n=0}^{N-1} F_n \overline{F_{n-j}}, \qquad \alpha_k = \hat{\eta}_k.$$

Here indices are periodic and $\hat{\eta}$ uses the unnormalized DFT of Chapter 2. This normalization is needed because the Fourier atoms used in the risk calculation have unit norm.

So far, $\mathcal{D}$ has been restricted to the basis ψ_m through the diagonal form (7.2). If F is stationary, its law is unchanged by shifts: $F(\cdot - \tau)$ has the same law as F. In that case, the Fourier Wiener filter (7.3) is optimal among all linear denoisers, even without imposing a diagonal form in advance.

An empirical estimate based on one noisy realization is $N^{-1}|\hat{y}_k|^2$. Its expectation is $\alpha_k + \sigma^2$, and its variance can be large. Empirical Wiener methods therefore need noise subtraction, nonnegativity constraints, and usually additional smoothing or a signal model.

7.2.5 Denoising and Linear Approximation

Approximation theory, developed in Chapter 5, provides a different route to denoising bounds. It requires a class of clean signals rather than a probability distribution on that class. Fix an orthonormal basis $\mathcal{B} = (\psi_m)_m$ of $\mathbb{R}^N$. A simple estimator retains only the first M coefficients of the noisy observation in $\mathcal{B}$:

$$\mathcal{D}(f) = \tilde{f} \stackrel{\text{def.}}{=} \sum_{m=1}^{M} \langle f, \psi_m\rangle \psi_m. \tag{7.4}$$

This is a linear projection onto the space spanned by $(\psi_m)_{m=1}^M$. This denoising scheme is parameterized by the number $1 \leqslant M \leqslant N$ of retained coefficients. Increasing M retains more signal detail but also more noise. For the discrete Fourier basis ordered by increasing frequency, this is ideal low-pass filtering by convolution with a discrete Dirichlet kernel. This is the special case of (7.2) in which the gains λ_m are binary, $\lambda_m \in \{0, 1\}$. Although restrictive, this choice yields useful rates under the approximation assumptions of Theorem 7.1. It requires a regularity model for f_0, but does not assume a random signal distribution F.

The proof is finite-dimensional, but its bound has no explicit dependence on N. Extending it to a continuum model requires interpreting white noise as a generalized random process.

Theorem 7.1. *Assume $\sigma > 0$, $\beta > 0$, and*

$$\|f_0 - f_{0,M}^{\mathrm{lin}}\|^2 \leqslant CM^{-2\beta}, \qquad f_{0,M}^{\mathrm{lin}} = \sum_{m=1}^{M} \langle f_0,\, \psi_m \rangle \psi_m, \quad 1 \leqslant M \leqslant N.$$

The projection denoiser (7.4) has risk

$$\mathbb{E}\|f_0 - \tilde{f}\|^2 = M\sigma^2 + \|f_0 - f_{0,M}^{\mathrm{lin}}\|^2.$$

If $0 < \sigma^2 \leqslant C$, choose

$$t = (C/\sigma^2)^{1/(2\beta+1)}, \qquad M = \min\{N, \lceil t \rceil\}. \tag{7.5}$$

Then

$$\mathbb{E}\|f_0 - \tilde{f}\|^2 \leqslant 3C^{1/(2\beta+1)}\sigma^{4\beta/(2\beta+1)}.$$

Proof. Orthogonality separates retained noise coefficients from discarded signal coefficients. Each retained noise coefficient has variance σ^2, which proves the risk identity. If $t \leqslant N$, then $t \leqslant M \leqslant 2t$ and

$$M\sigma^2 + CM^{-2\beta} \leqslant 2t\sigma^2 + Ct^{-2\beta} = 3C^{1/(2\beta+1)}\sigma^{4\beta/(2\beta+1)}.$$

If $t > N$, the estimator retains all coefficients and its risk is $N\sigma^2 \leqslant t\sigma^2$, which obeys the same bound. □

Several observations help interpret this result:

- The approximation assumption bounds the denoising error $\mathbb{E}(\|f_0 - \tilde{f}\|^2)$ independently of the sample count N, even though the total input noise energy $\mathbb{E}(\|w\|^2) = N\sigma^2$ grows with N.
- A larger approximation exponent β gives faster decay of the error as σ tends to zero. The exponent approaches that of the parametric rate σ^2 as β increases.
- The proof uses a finite sample count N. White noise with constant positive variance in every coordinate has infinite energy in an infinite-dimensional Hilbert space, although other random vectors may have finite energy. The balancing choice applies when $N \geqslant t$; otherwise the estimator is the identity. A continuum interpretation as $\sigma \to 0$ also requires increasing the sampling resolution.
- Section 5.3.1 bounds approximation errors for continuous signal and image models. To transfer these bounds to sampled data, choose N large enough that discretization error is smaller than denoising error.

The Sobolev model in Section 5.2.1 provides a useful example. For discrete signals with an increasing sample count N, especially as σ decreases, a compatible weighted coefficient bound gives the same approximation argument as Proposition 5.3.

Proposition 7.2. *Let $\alpha > 0$. If*

$$\sum_{m=1}^{N} m^{2\alpha} |\langle f_0,\, \psi_m \rangle|^2 \leqslant C \tag{7.6}$$

then

$$\forall\, M \in \{1, \ldots, N\}, \quad \|f_0 - f_{0,M}^{lin}\|^2 \leqslant CM^{-2\alpha}$$

Proof.

$$C \geqslant \sum_{m=1}^{N} m^{2\alpha} |\langle f_0,\, \psi_m \rangle|^2 \geqslant \sum_{m>M} m^{2\alpha} |\langle f_0,\, \psi_m \rangle|^2 \geqslant M^{2\alpha} \sum_{m>M} |\langle f_0,\, \psi_m \rangle|^2 \geqslant M^{2\alpha} \|f_0 - f_{0,M}^{\mathrm{lin}}\|^2.$$

□

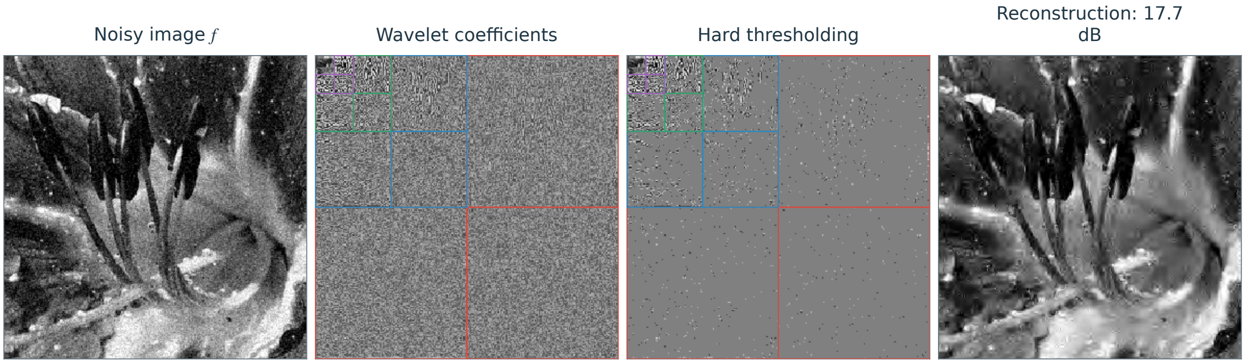


Figure 7.10. Denoising using thresholding of wavelet coefficients.

For normalized Fourier atoms ordered by increasing physical frequency, an appropriately scaled discrete weighted-energy bound approximates the periodic Sobolev model. The sample normalization, frequency weights, and boundary convention must match those of the continuous signal. Under this compatibility, low-frequency projection efficiently denoises smooth signals and images.

7.3 Nonlinear Denoising by Thresholding

7.3.1 Hard Thresholding

We consider a real orthonormal basis $\{\psi_m\}_m$ of $\mathbb{R}^N$, for instance a discrete wavelet basis. The noisy coefficients satisfy

$$\langle f, \psi_m\rangle = \langle f_0, \psi_m\rangle + \langle w, \psi_m\rangle. \tag{7.7}$$

Orthogonal transforms preserve Gaussian white noise, so the coefficients $\langle w, \psi_m\rangle$ remain independent centered Gaussians with variance σ^2. If the basis $\{\psi_m\}_m$ represents f_0 sparsely, most clean coefficients $\langle f_0, \psi_m\rangle$ are small, leaving many noisy coefficients dominated by noise; see Figure 7.10. Donoho and Johnstone systematically developed thresholding estimators based on this observation [17].

Hard thresholding retains only coefficients whose magnitude exceeds the threshold T:

$$\tilde{f} = \sum_{|\langle f, \psi_m\rangle|>T} \langle f, \psi_m\rangle\psi_m = \sum_m S_T(\langle f, \psi_m\rangle)\psi_m.$$

where S_T is defined in (5.4). Retaining the M largest noisy coefficients gives the approximation $\tilde{f} = f_M$ of f. Figure 7.10 illustrates how a suitable T removes much of the noise while preserving sharp features.

7.3.2 Soft Thresholding

We recall that the hard thresholding operator is defined as

$$S_T(x) = S_T^0(x) = \begin{cases} x & \text{if} \quad |x| > T, \\ 0 & \text{if} \quad |x| \leqslant T. \end{cases} \tag{7.8}$$

Hard thresholding makes a discontinuous keep-or-discard decision, which can create artifacts. Soft thresholding is a continuous alternative:

$$S_T^1(x) = \max(1 - T/|x|, 0)x. \tag{7.9}$$

Set $S_T^1(0) = 0$ by continuity. Figure 7.11 compares the two nonlinear maps.

For $q = 0$ and $q = 1$, these thresholding rules define two different estimators

$$\tilde{f}^q = \sum_m S_T^q(\langle f, \psi_m\rangle)\psi_m \tag{7.10}$$

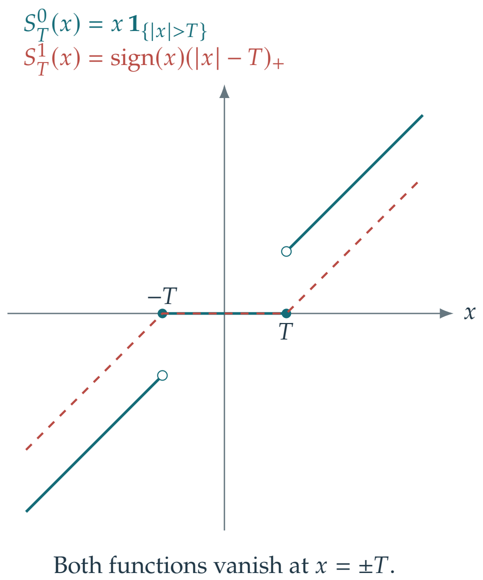


Figure 7.11. Hard and soft thresholding functions.

Preserving coarse-scale coefficients. Soft thresholding S_T^1 biases retained coefficients by reducing their magnitudes. Shrinking coarse wavelet coefficients can introduce unwanted low-frequency artifacts. At a coarsest scale 2^{j_0}, it is therefore common to retain the approximation coefficients unchanged. In one dimension, this gives

$$\tilde{f}^1 = \sum_{0 \leqslant n < 2^{-j_0}} \langle f, \varphi_{j_0,n} \rangle \varphi_{j_0,n} + \sum_{j=J+1}^{j_0} \sum_{0 \leqslant n < 2^{-j}} S_T^1(\langle f, \psi_{j,n} \rangle) \psi_{j,n}.$$

Empirical choice of the threshold. Figure 7.12 plots SNR against the threshold T for a fixed natural image f_0. In this experiment, hard thresholding performs best near $T \approx 3\sigma$, and soft thresholding near $T \approx 3\sigma/2$. The best soft-thresholded estimate has a slightly higher SNR.

These values describe the displayed experiments; the best threshold depends on the image, basis, noise level, and loss function.

7.3.3 Minimax Optimality of Thresholding

Estimating sparse coefficients. To analyze the estimator and guide the choice of T, we first assume that the coefficients

$$a_{0,m} = \langle f_0, \psi_m \rangle \in \mathbb{R}, \qquad a_0 \in \mathbb{R}^N$$

are sparse: most entries $a_{0,m}$ vanish, so the ℓ^0 count

$$\|a_0\|_0 = \#\{m \,;\, a_{0,m} \neq 0\}$$

is small. As shown in (7.7), noisy coefficients

$$\langle f, \psi_m \rangle = a_m = a_{0,m} + z_m$$

are perturbed by additive Gaussian white noise of variance σ^2. Figure 7.14 shows an example of such a noisy sparse signal.

Universal threshold value. When all nonzero clean coefficients are well separated from the noise level, choosing a threshold comparable to the largest pure-noise coefficient suppresses false detections:

$$T \approx \tau_N, \qquad \tau_N = \max_{0 \leqslant m < N} |z_m|.$$

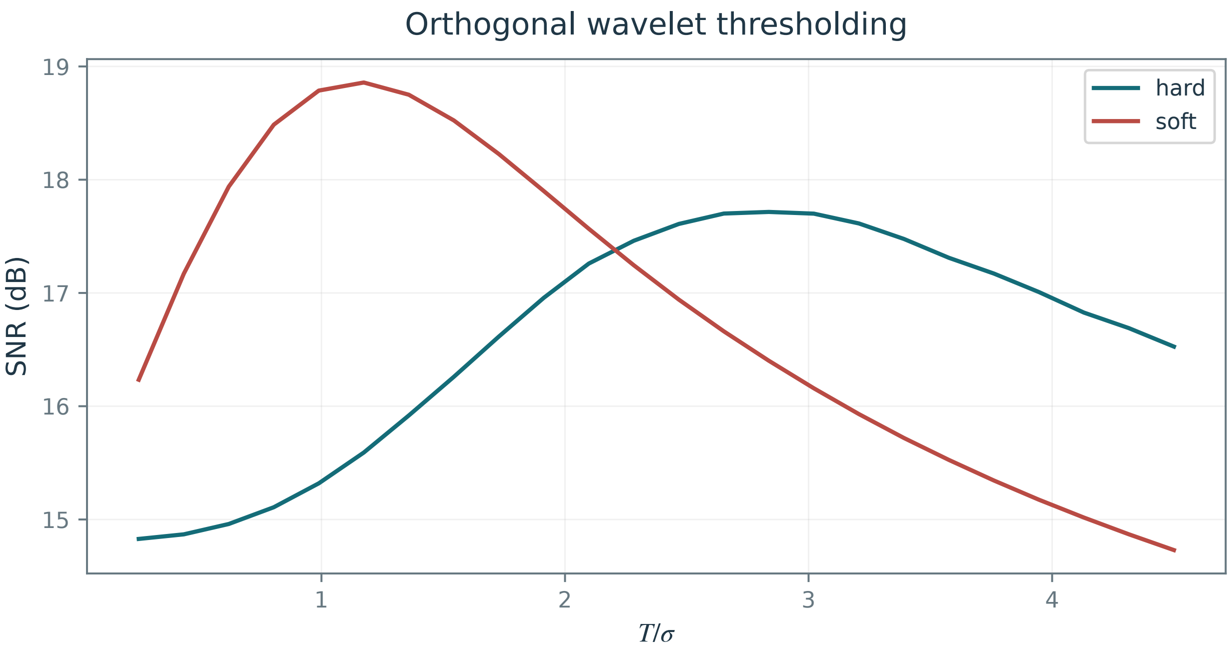


Figure 7.12. SNR as a function of T/σ for hard and soft thresholding.

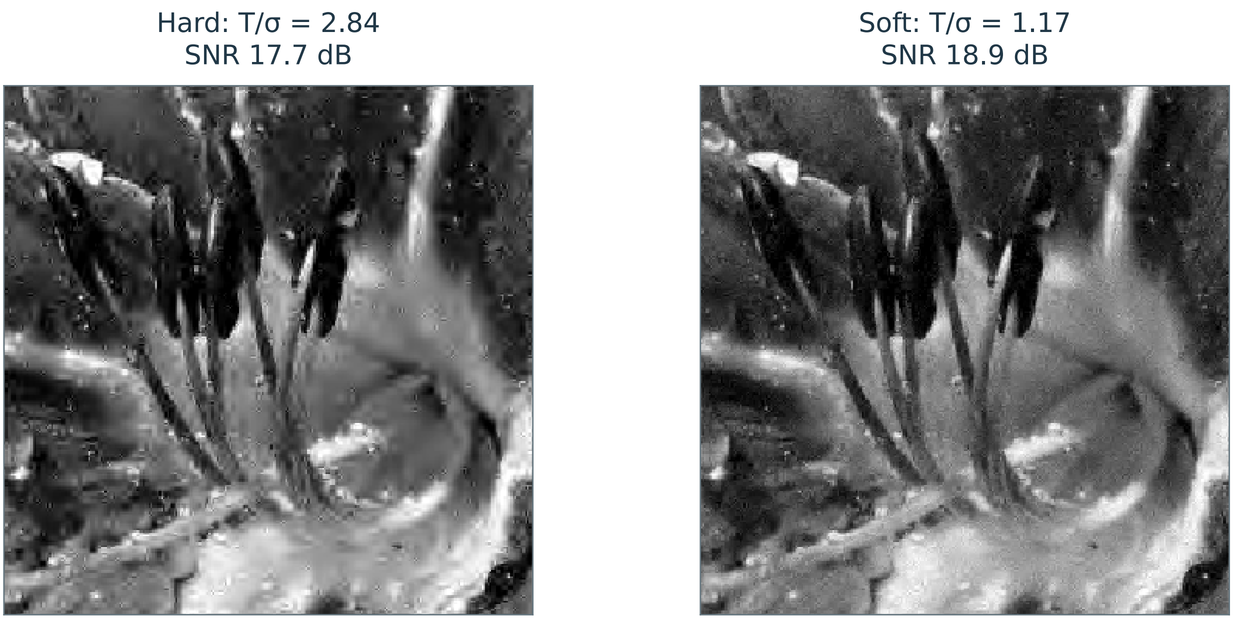


Figure 7.13. Comparison of hard (left) and soft (right) thresholding.

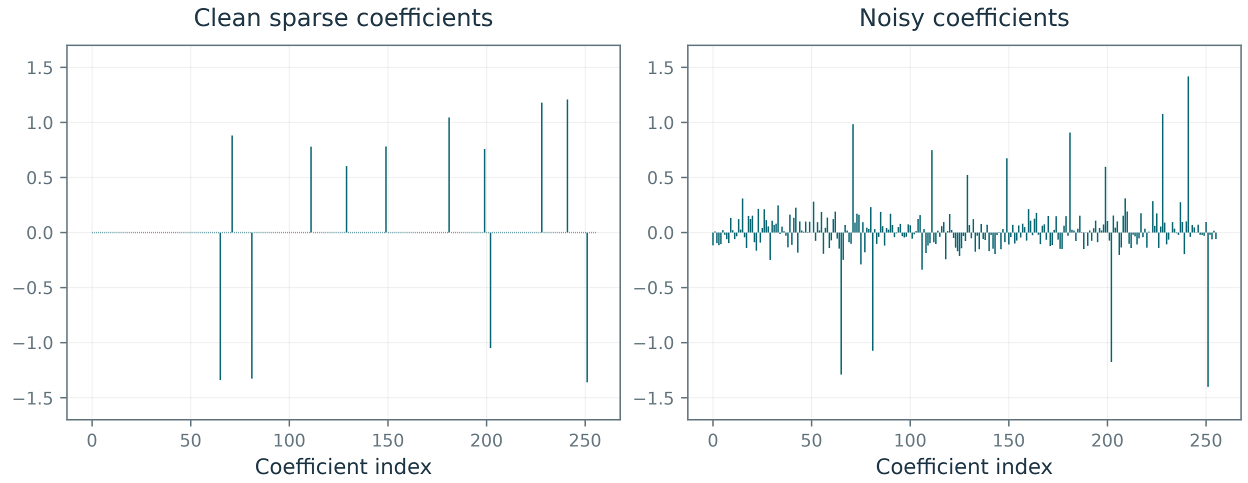


Figure 7.14. Left: clean sparse coefficients a_0; right: noisy coefficients a.

This motivates a deterministic threshold based on the typical size of τ_N; it is not an exact minimization formula for the realized denoising error. The random maximum τ_N depends on N. For fixed σ, its mean is asymptotic to $\sigma\sqrt{2\log N}$ as $N \to \infty$. Its variance tends to zero with N, so τ_N concentrates increasingly near its mean. Figure 7.15 illustrates these trends numerically.

Asymptotic optimality. Donoho and Johnstone [17] showed that the universal threshold $T = \sigma\sqrt{2\log(N)}$ gives a risk bound for signals that admit accurate nonlinear approximations in $\{\psi_m\}_m$. The risk is within a logarithmic factor of an ideal coefficient-selection risk.

Theorem 7.3. *Let $N \geqslant 2$ and $Y = f_0 + w$, with $w \sim \mathcal{N}(0, \sigma^2 I_N)$, in a real orthonormal basis. Hard or soft thresholding* (7.10) *with*

$$T = \sigma\sqrt{2\ln N}$$

satisfies an oracle inequality of the form

$$\mathbb{E}\|f_0 - \tilde{f}^q\|^2 \leqslant C_0(1+\ln N)\left[\sigma^2 + \min_{0\leqslant M\leqslant N}\left(\|f_0 - f_{0,M}^{\mathrm{nlin}}\|^2 + M\sigma^2\right)\right],$$

where C_0 is an absolute constant and $f_{0,M}^{\mathrm{nlin}}$ retains the M largest clean coefficients. In particular, if $\|f_0 - f_{0,M}^{\mathrm{nlin}}\|^2 \leqslant CM^{-2\beta}$ for $1 \leqslant M \leqslant N$, with $\beta > 0$, then for $0 < \sigma \leqslant 1$,

$$\mathbb{E}\|f_0 - \tilde{f}^q\|^2 \leqslant C'(1+\ln N)\sigma^{4\beta/(2\beta+1)},$$

where C' depends on C and β.

The oracle inequality is a standard Gaussian thresholding bound; see [17]. The rate follows by balancing $CM^{-2\beta}$ and $M\sigma^2$, with integer rounding and truncation at N as in Theorem 7.1. The added σ^2 term accounts for residual noise even at the zero signal.

The universal threshold $T = \sigma\sqrt{2\ln N}$ is conservative: it suppresses nearly all pure-noise coefficients. The smaller thresholds illustrated in Figure 7.13 often give better SNR in these experiments.

7.3.4 Translation Invariant Thresholding Estimators

Translation invariance. Let $f \mapsto \tilde{f} = \mathcal{D}(f)$ be a denoising method, and $f_\tau(x) = f(x-\tau)$ be a translated signal or image for $\tau \in \mathbb{R}^d$ ($d = 1$ or $d = 2$). The denoiser is translation invariant on the shift lattice Δ if

$$\forall\, \tau \in \Delta, \quad \mathcal{D}(f) = \mathcal{D}(f_\tau)_{-\tau}$$

where Δ is a lattice of $\mathbb{R}^d$. A denser shift lattice imposes equivariance at finer spatial resolution. This corresponds to the fact that $\mathcal{D}$ commutes with the translation operator.

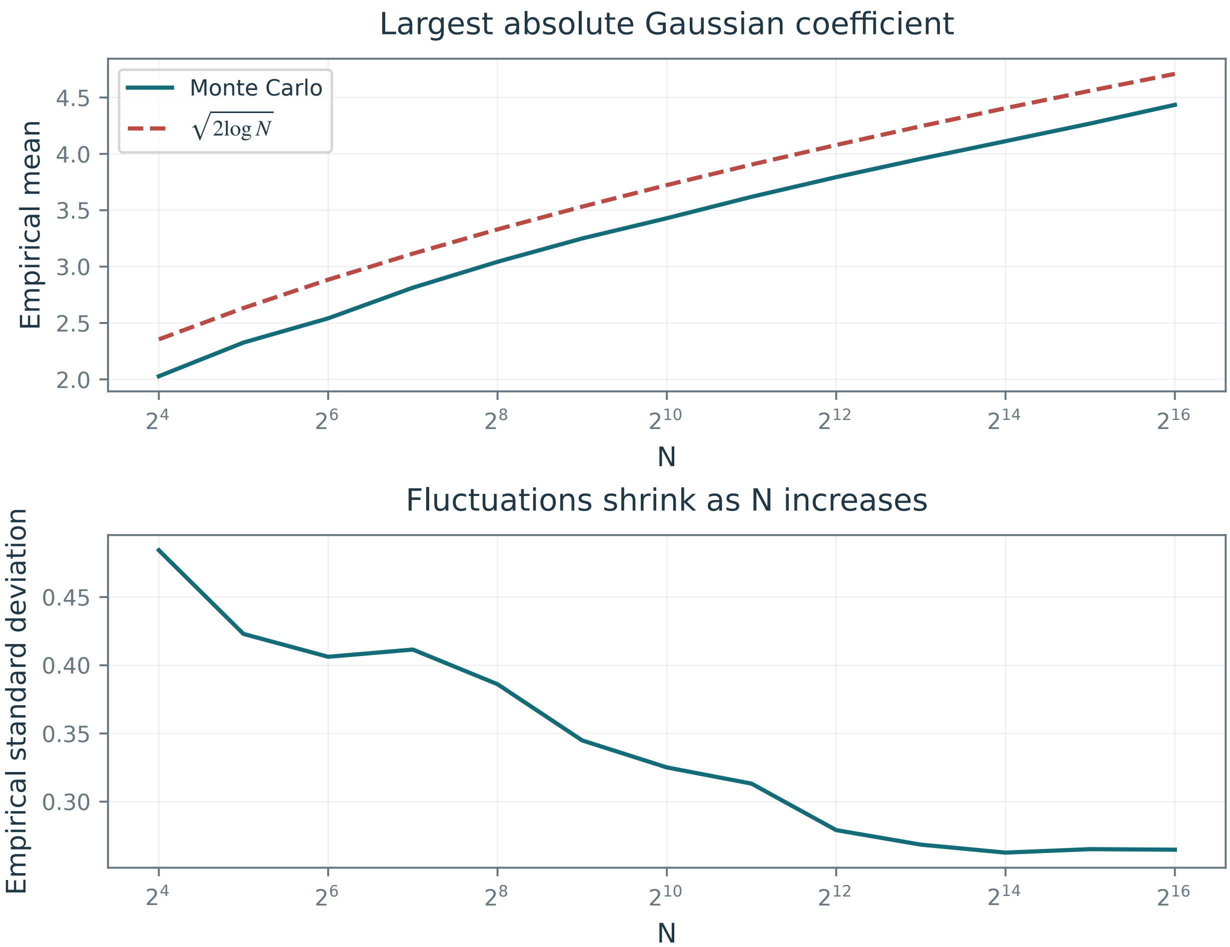


Figure 7.15. Empirical mean (top) and standard deviation (bottom) of the largest absolute Gaussian noise coefficient.

$$\begin{tikzcd}
f \arrow[r, "\mathcal{D}"] \arrow[d, "T_\tau"'] & \mathcal{D}(f) \arrow[d, "T_\tau"] \\
T_\tau f \arrow[r, "\mathcal{D}"'] & \mathcal{D}(T_\tau f) = T_\tau \mathcal{D}(f)
\end{tikzcd}$$

Translation invariance on a sufficiently fine set Δ prevents the denoiser from favoring particular feature locations. Without it, the same feature may be reconstructed differently depending on where it appears, producing visible artifacts.

For denoising by thresholding

$$\mathcal{D}(f) = \sum_m S_T(\langle f, \psi_m \rangle)\psi_m.$$

a sufficient condition is invariance of the basis $\{\psi_m\}_m$ under shifts in Δ, up to unit-modulus factors:

$$\forall\, m', \ \forall\, \tau \in \Delta, \ \exists m, \ \exists \lambda \in \mathbb{C}, \qquad (\psi_{m'})_\tau = \lambda \psi_m$$

where $|\lambda| = 1$.

On the periodic domain, the Fourier basis is invariant under every shift $\Delta = \mathbb{R}^d$ of $[0,1]^d$. The discrete Fourier basis is invariant under all integer shifts $\Delta = \{0, \ldots, N_0 - 1\}^d$, where $N = N_0$ counts samples in one dimension and $N = N_0 \times N_0$ counts pixels in two dimensions.

Unfortunately, an orthogonal wavelet basis

$$\{\psi_m = \psi_{j,n}\}_{j,n}$$

is generally not invariant under arbitrary shifts in either the continuous or discrete setting. For instance, in 1-D,

$$(\psi_{j',n'})_\tau \notin \{\psi_{j,n}\} \quad \text{for} \quad \tau = 2^{j'}/2.$$

Cycle spinning. For a finite group Δ of periodic shifts, average the shifted, denoised, and realigned outputs of $\mathcal{D}$ to obtain a translation-invariant estimator:

$$\mathcal{D}_{\text{inv}}(f) = \frac{1}{|\Delta|} \sum_{\tau \in \Delta} \mathcal{D}(f_\tau)_{-\tau}. \tag{7.11}$$

One checks that

$$\forall\, \tau \in \Delta, \quad \mathcal{D}_{\text{inv}}(f) = \mathcal{D}_{\text{inv}}(f_\tau)_{-\tau}$$

To obtain translation invariance at pixel precision for data with N samples, one should use a set of $|\Delta| = N$ translation vectors. For wavelets, this requires $O(N^2)$ operations.

Figure 7.16 applies cycle spinning to hard thresholding in an orthogonal wavelet basis, using the following shifts: $\Delta = \{0, 1/N_0, 2/N_0, 3/N_0\}^2$ on an $N_0 \times N_0$ image, corresponding to shifts by $\{0, 1, 2, 3\}^2$ pixels. This requires 16 forward and inverse wavelet-transform pairs. Averaging this subset of shifts does not in general guarantee exact pixel-shift equivariance. In this example, partial averaging reduces sensitivity to shifts, substantially improves the SNR, and suppresses oscillatory artifacts. These artifacts move as τ changes, so averaging reduces them.

In Figure 7.17, translation-invariant hard thresholding slightly outperforms translation-invariant soft thresholding, reversing their ranking in the orthogonal-basis experiment.

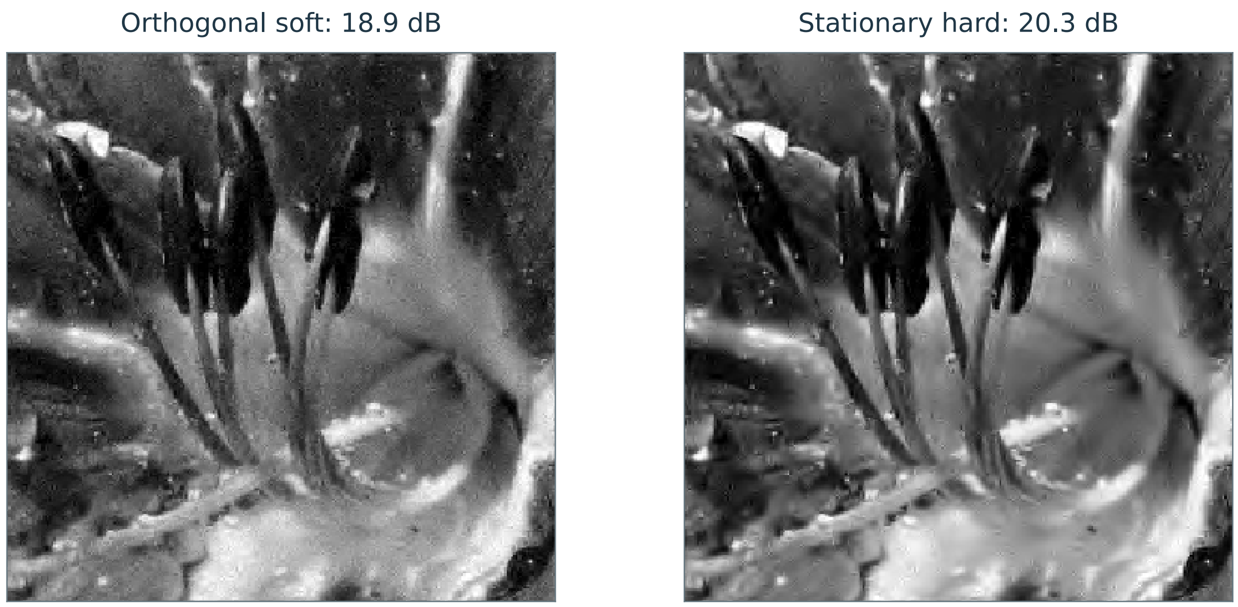


Figure 7.16. Soft thresholding in an orthogonal wavelet basis (left) and translation-invariant hard thresholding (right).

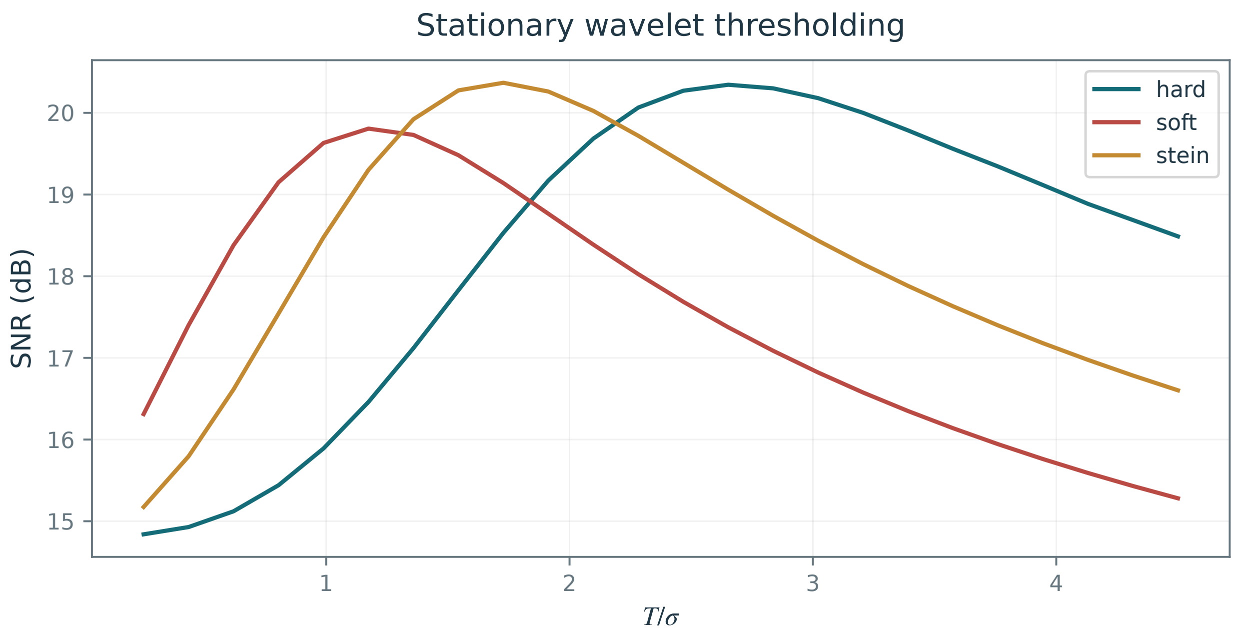


Figure 7.17. SNR as a function of T/σ for translation-invariant thresholding.

Translation invariant wavelet frame. Cycle spinning also has a frame interpretation: replace the orthogonal basis $\mathcal{B} = \{\psi_m\}$ by its translated copies,

$$\mathcal{B}_{\text{inv}} = \{(\psi_m)_\tau\}_{m,\tau\in\Delta}. \tag{7.12}$$

Although $\mathcal{B}_{\text{inv}}$ is no longer an orthogonal basis, orthonormality of each translated copy gives the energy and reconstruction identities

$$\|f\|^2 = \frac{1}{|\Delta|}\sum_{m,\tau\in\Delta} |\langle f, (\psi_m)_\tau\rangle|^2 \quad \text{and} \quad f = \frac{1}{|\Delta|}\sum_{m,\tau\in\Delta} \langle f, (\psi_m)_\tau\rangle(\psi_m)_\tau.$$

Such redundant families are called tight frames.

One can then define a translation invariant thresholding denoiser

$$\mathcal{D}_{\text{inv}}(f) = \frac{1}{|\Delta|}\sum_{m,\tau\in\Delta} S_T(\langle f, (\psi_m)_\tau\rangle)(\psi_m)_\tau. \tag{7.13}$$

This estimator equals the cycle spinning estimator defined in (7.11).

Counted with multiplicity, the translated frame has $|\Delta||\mathcal{B}|$ elements. For a basis of signals with N samples and a lattice of $|\Delta| = N$ shifts, this gives up to N^2 vectors in $\mathcal{B}_{\text{inv}}$. In a hierarchical construction such as a wavelet basis, however, different shifts can produce the same atom:

$$(\psi_m)_\tau = (\psi_{m'})_{\tau'} \quad \text{for} \quad m \neq m' \quad \text{and} \quad \tau \neq \tau',$$

The number of distinct vectors in $\mathcal{B}_{\text{inv}}$ can therefore be much smaller than N^2. For instance, for an orthogonal wavelet basis, one has

$$(\psi_{j,n})_{k2^j} = \psi_{j,n+k},$$

so many translated atoms coincide. For a signal of length N, an undecimated transform has $\log_2 N$ detail arrays of length N, plus a coarse array. For an $N_0 \times N_0$ image ($N = N_0^2$), it has $3\log_2 N_0$ detail arrays of N entries, plus a coarse array. Appropriate multiplicity weights must be retained when replacing the full translated family by distinct atoms.

The fast undecimated wavelet transform, often called the “à trous” transform, computes these coefficients in $O(N \log N)$ operations. Each detail array is indexed directly by every pixel location:

$$d_j^\omega[n] = \langle f, (\psi_{j,0}^\omega)_n\rangle, \qquad n \in \{0,\dots,N_0-1\}^2, \quad \omega \in \{V,H,D\},$$

where the wavelets and shifts are discrete and appropriately normalized. Each array is a band-pass filtered image, as illustrated in Figure 7.18.

Figure 7.19 illustrates hard thresholding of these translated coefficients.

7.3.5 Other Shrinkage Rules

Other shrinkage rules offer different compromises between retaining large coefficients and suppressing noise. We describe two examples whose performance depends on the coefficient distribution of f in the chosen basis.

Semi-soft thresholding. Semi-soft thresholding interpolates between hard and soft thresholding through a parameter $\mu > 1$:

$$S_T^\theta(x) = g_{\frac{1}{1-\theta}}(x) \quad \text{where} \quad g_\mu(x) = \begin{cases} 0 & \text{if} \quad |x| < T \\ x & \text{if} \quad |x| > \mu T \\ \text{sign}(x)\frac{\mu(|x|-T)}{\mu-1} & \text{otherwise.} \end{cases}$$

For $0 < \theta < 1$, set $\mu = 1/(1-\theta)$. The limits $\theta \downarrow 0$ and $\theta \uparrow 1$ give hard and soft thresholding, respectively, away from the threshold boundary. Figure 7.20 shows an example of this nonlinearity.

Figure 7.21 shows an SNR improvement for a suitable choice of μ, although the visual difference from hard and soft thresholding is small.

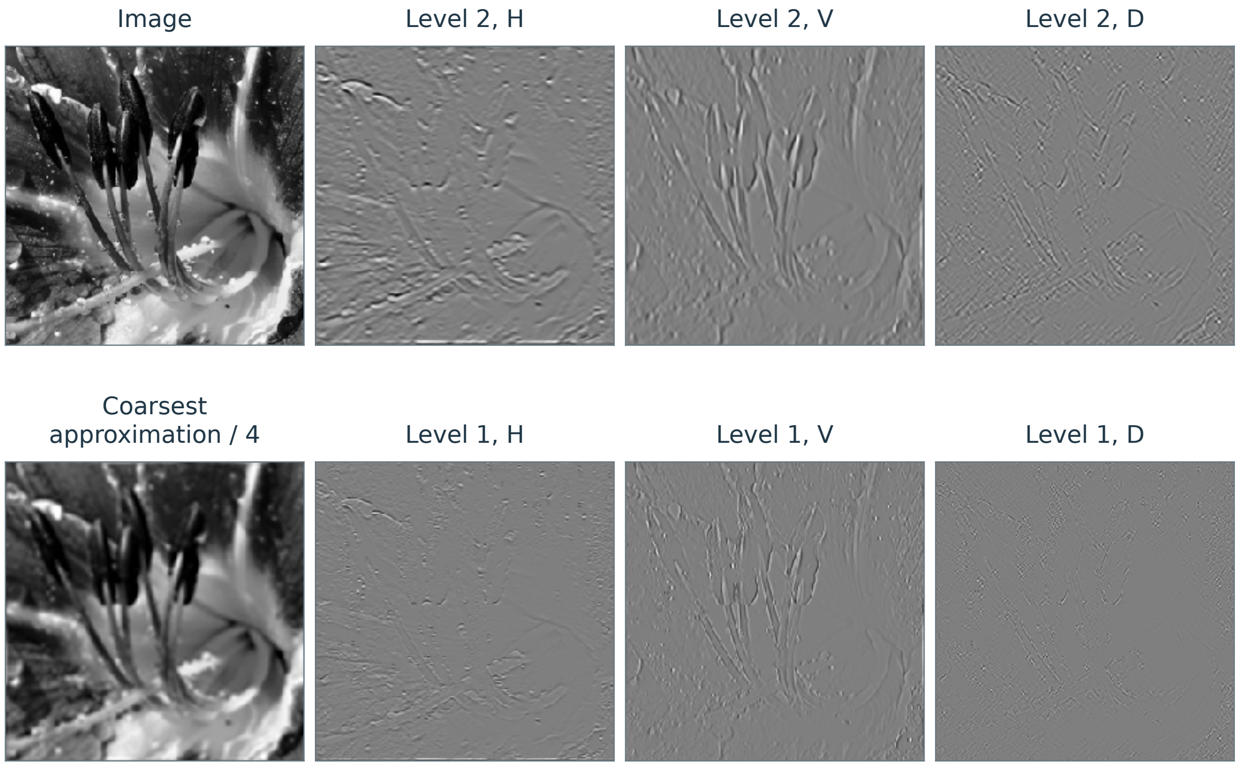


Figure 7.18. Translation invariant wavelet coefficients.

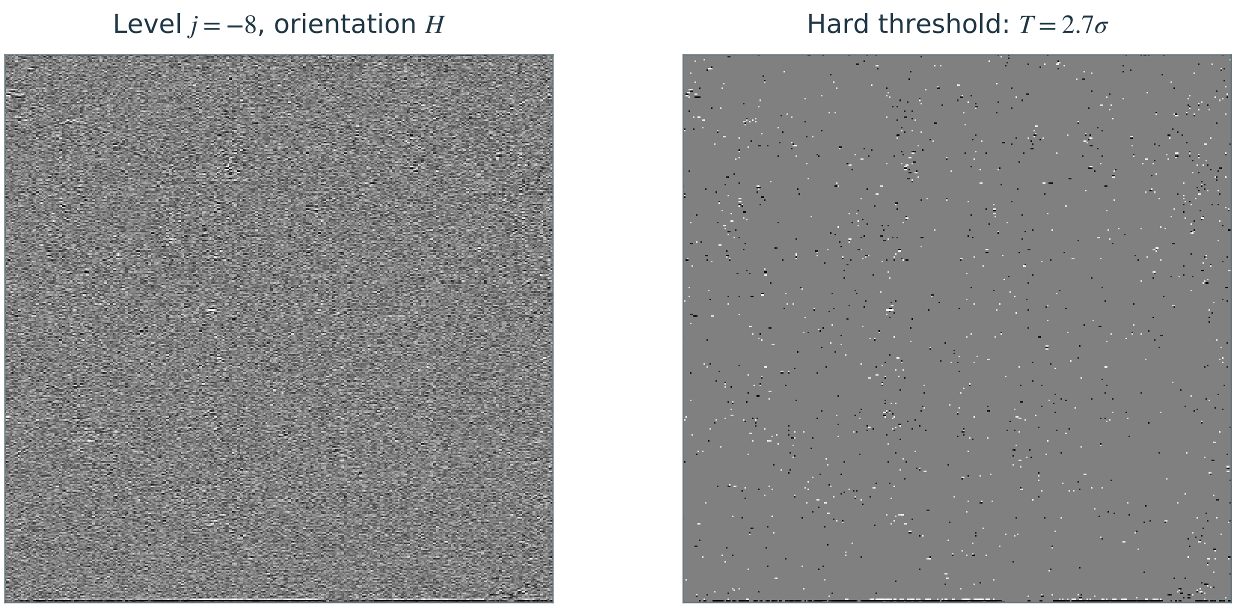


Figure 7.19. Left: translation invariant wavelet coefficients, for $j = -8, \omega = H$, right: thresholded coefficients.

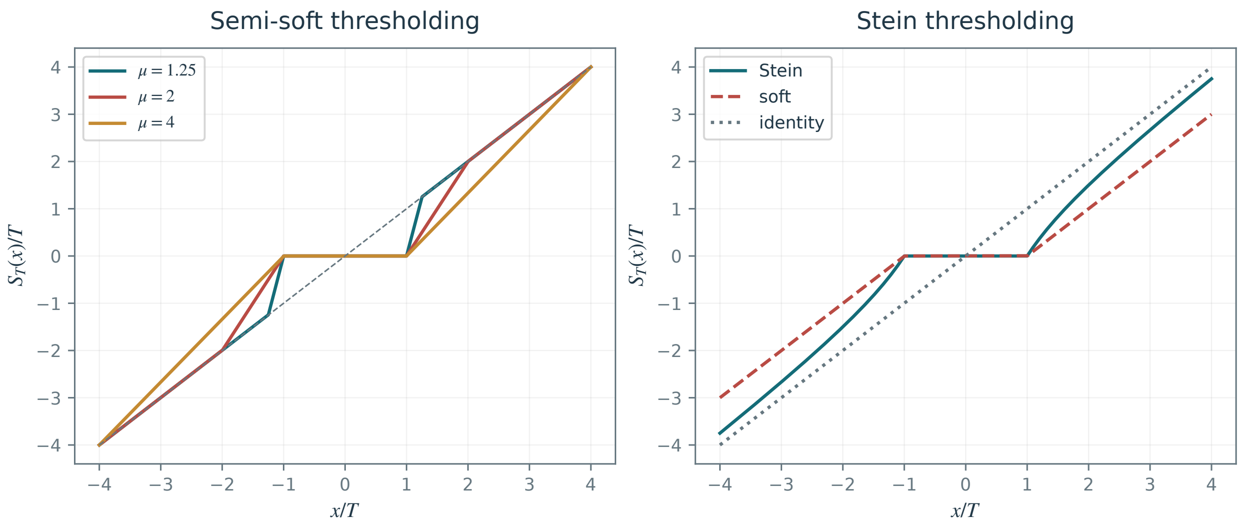


Figure 7.20. Left: semi-soft thresholder, right: Stein thresholder.

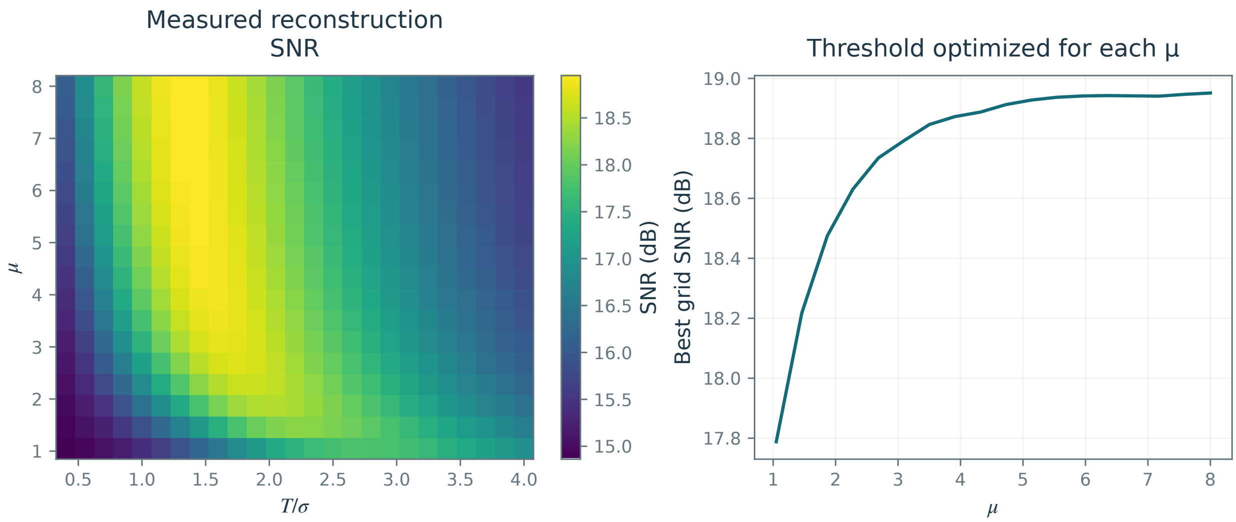


Figure 7.21. Left: SNR as a function of μ and T/σ. Right: SNR versus μ, with T/σ optimized separately for each μ.

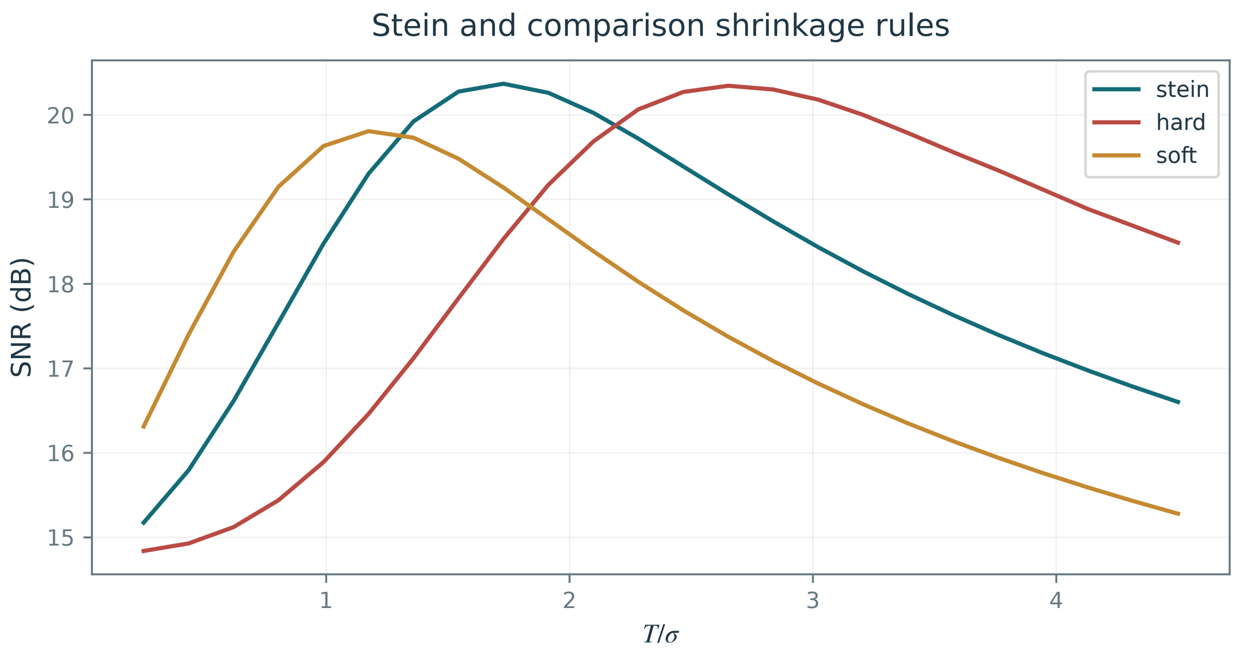


Figure 7.22. SNR as a function of T/σ for Stein thresholding.

Stein thresholding. Stein thresholding is defined using a quadratic attenuation of large coefficients

$$S_T^{\text{Stein}}(x) = \max\left(1 - \frac{T^2}{|x|^2}, 0\right) x.$$

The value at $x = 0$ is defined to be zero. This should be compared with the linear attenuation of soft thresholding

$$S_T^1(x) = \max\left(1 - \frac{T}{|x|}, 0\right) x.$$

For large coefficients, Stein thresholding has the following shrinkage behavior:

$$|S_T^{\text{Stein}}(x) - x| \longrightarrow 0 \quad \text{whereas} \quad |S_T^1(x) - x| \longrightarrow T,$$

as $x \longrightarrow \pm\infty$. Its deterministic shrinkage tends to zero, whereas soft thresholding retains a fixed shrinkage. This limiting property does not imply unbiased estimation at finite amplitudes.

Stein and hard thresholding give similar results in the translation-invariant natural-image experiments shown here.

7.3.6 Block Thresholding

The nonlinear thresholding methods above are diagonal estimators, since they attenuate each coefficient separately

$$\tilde{f} = \sum_m A_T^q(\langle f, \psi_m \rangle) \langle f, \psi_m \rangle \psi_m$$

where

$$A_T^q(x) = \begin{cases} \max(1 - T^2/|x|^2, 0) & \text{for} \quad q = \text{Stein} \\ \max(1 - T/|x|, 0) & \text{for} \quad q = 1 \text{ (soft)} \\ 1_{|x|>T} & \text{for} \quad q = 0 \text{ (hard)} \end{cases}$$

Define every attenuation factor to be zero at $x = 0$ when $T > 0$. Block thresholding shares an attenuation factor across a group of coefficients, exploiting their statistical dependence. This is useful for natural images, where edges create clusters of large wavelet coefficients. Grouping also helps suppress isolated noise coefficients in otherwise smooth regions.

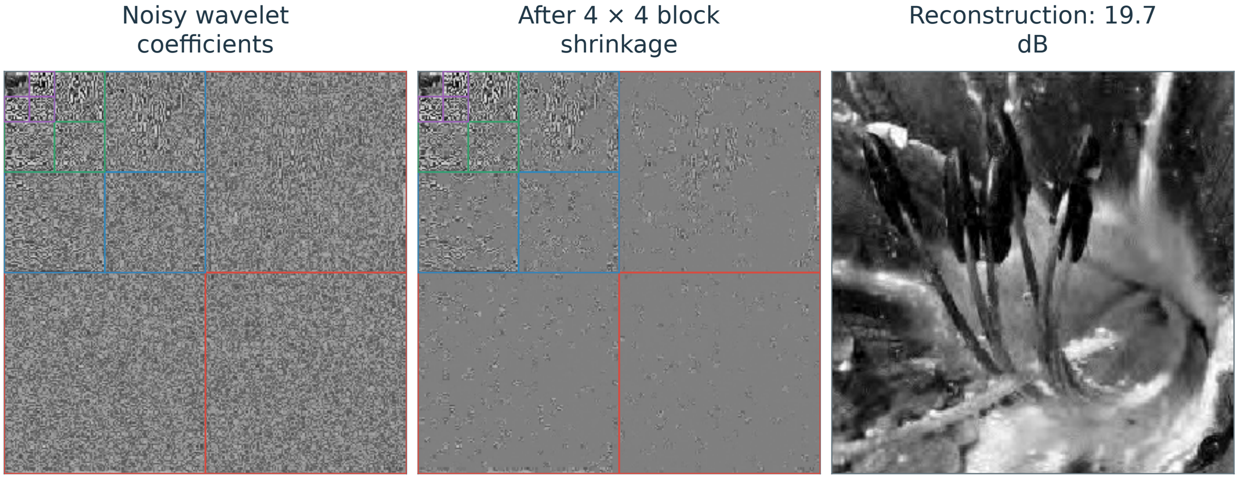


Figure 7.23. Left: wavelet coefficients, center: block thresholded coefficients, right: denoised image.

Partition the coefficients into disjoint blocks; for example, in a 2-D wavelet transform,

$$\{(j, n, \omega)\}_{j,n,\omega} = \bigcup_k B_k,$$

where each B_k contains $s \times s$ neighboring coefficients at a fixed scale and orientation. The block size s is a parameter of the method. Figure 7.23 shows an example of such a block.

The mean squared coefficient magnitude in a block is

$$E_k = \frac{1}{|B_k|} \sum_{m \in B_k} |\langle f, \psi_m \rangle|^2.$$

The block estimator

$$\tilde{f} = \sum_m S_T^{\text{block},q}(\langle f, \psi_m \rangle)\psi_m$$

applies the same attenuation to every coefficient in a block,

$$\forall\, m \in B_k, \quad S_T^{\text{block},q}(\langle f, \psi_m \rangle) = A_T^q(\sqrt{E_k})\langle f, \psi_m \rangle,$$

for $q \in \{0, 1, \text{stein}\}$. Figure 7.23 shows the attenuated blocks and the reconstructed image.

Figure 7.24, left, compares the rules for $q \in \{0, 1, \text{stein}\}$. Stein block thresholding gives the highest SNR in this comparison. Figure 7.24, right, compares block sizes for Stein thresholding. The choice $s = 4$ performs well in the displayed experiment.

Figure 7.25 compares the reconstructions. Stein block thresholding in an orthogonal wavelet basis approaches the quality of translation-invariant hard thresholding here, at lower computational cost. The block thresholding strategy can also be applied to wavelet coefficients in a translation invariant tight frame, which gives the best result among the methods compared in this experiment.

The same block attenuation can be implemented directly after the wavelet transform.

More advanced denoisers use richer statistical models to improve on the methods presented here; see, for instance, [27].

7.4 Signal-Dependent Noise

In many imaging systems, the variance of the noise affecting $f_{0,n}$ depends on the intensity $f_{0,n}$ itself. We now study Poisson counts and multiplicative noise. Both can be expressed as additive residuals, but their residual distributions depend on the signal.

7.4.1 Poisson Noise

Many imaging devices measure intensity by counting photons. Examples include digital cameras, confocal microscopy, PET and SPECT tomography.

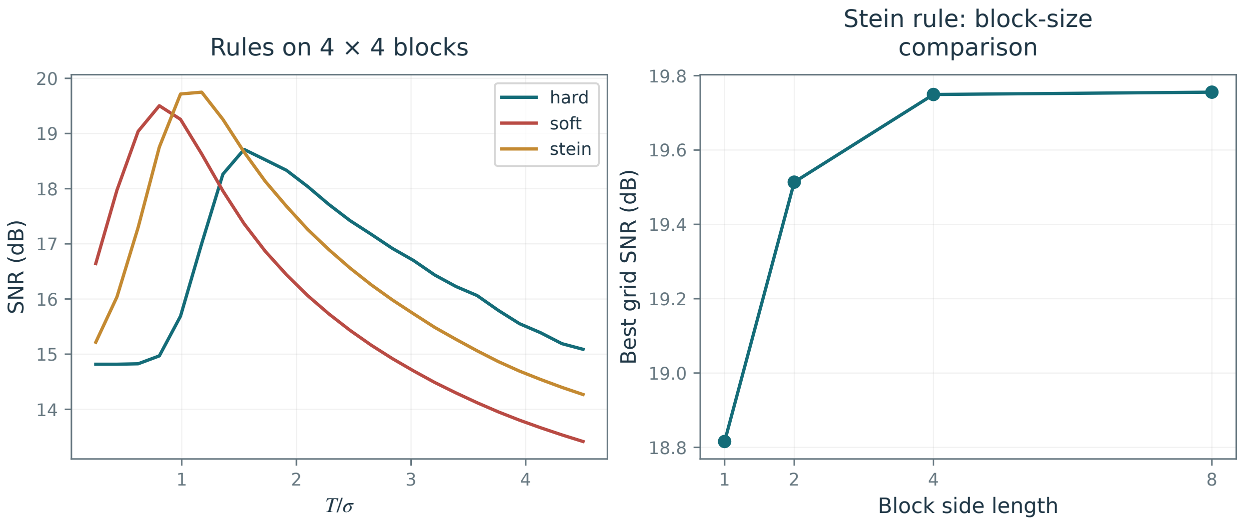


Figure 7.24. SNR as a function of T/σ (left) and comparison of different block sizes (right).

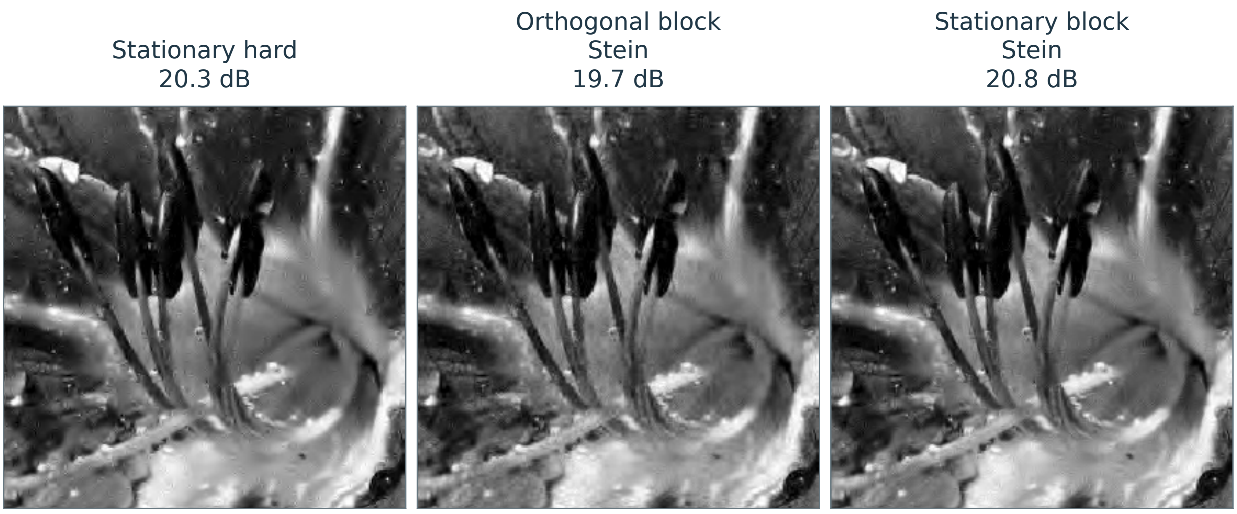


Figure 7.25. Left: translation invariant wavelet hard thresholding, center: block orthogonal Stein thresholding, right: block translation invariant Stein thresholding.

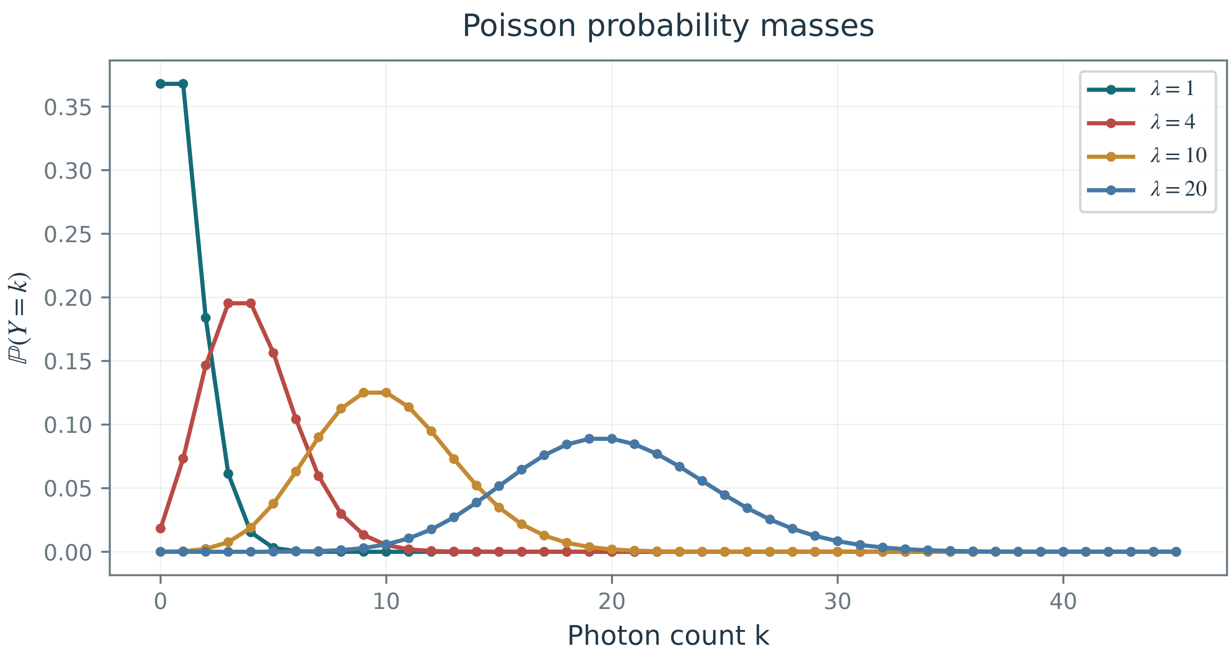


Figure 7.26. Poisson distributions for various λ.

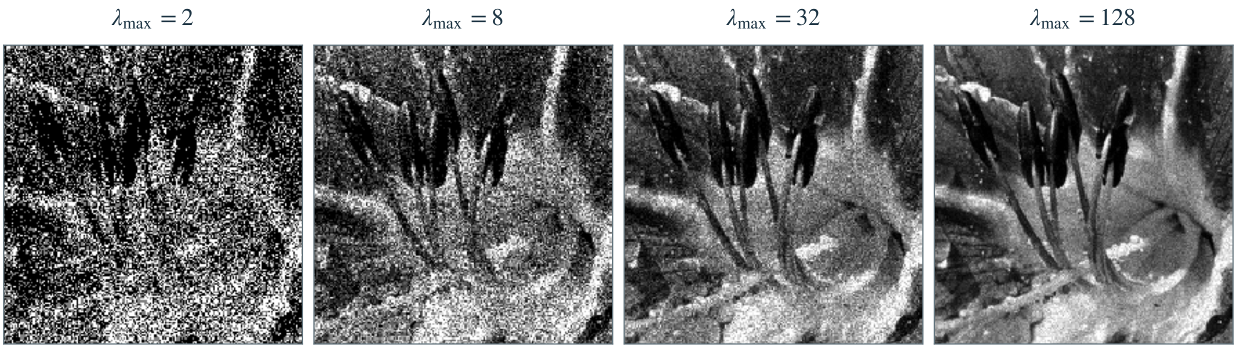


Figure 7.27. Poisson-corrupted flower images at increasing photon-count scales $\lambda_{\max}$.

Poisson model. For an unknown mean intensity $f_{0,n} \geqslant 0$, a Poisson model describes the observed photon count:

$$f_n \sim \mathcal{P}(\lambda) \quad \text{where} \quad \lambda = f_{0,n} \in [0, \infty),$$

The Poisson distribution $\mathcal{P}(\lambda)$ has probability mass function

$$\mathbb{P}(f_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

and its parameter varies across pixels with the underlying intensity. Figure 7.26 shows several Poisson distributions.

One has

$$\mathbb{E}(f_n) = \lambda = f_{0,n} \quad \text{and} \quad \operatorname{Var}(f_n) = \lambda = f_{0,n}$$

Thus denoising again estimates a mean from one observation, but the variance now varies with intensity. The absolute noise level increases as more photons reach the sensor. For positive intensity, the relative standard deviation $f_{0,n}^{-1/2}$ instead decreases.

Figure 7.27 shows Poisson-corrupted versions of a clean image f_0 at different maximum mean intensities $\lambda_{\max}$.

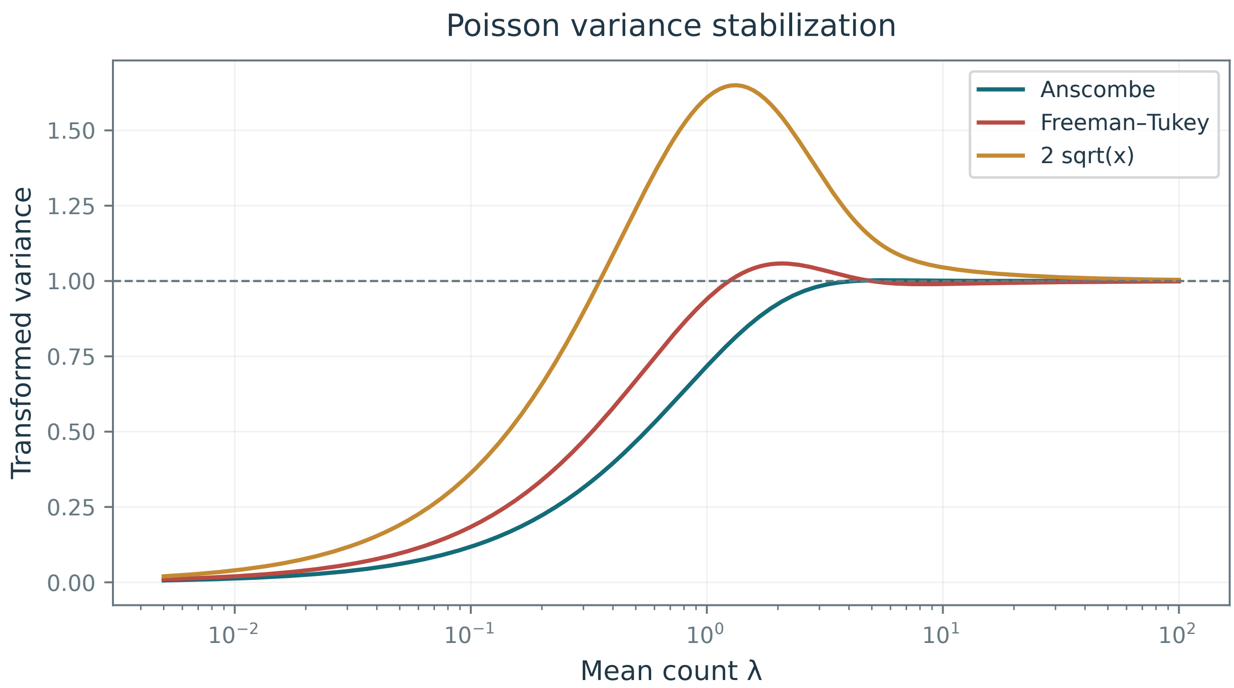


Figure 7.28. Exact Poisson variances of the Anscombe, Freeman–Tukey, and $2\sqrt{x}$ transforms as functions of the mean intensity.

Variance stabilization. Applying a thresholding estimator

$$\mathcal{D}(f) = \sum_m S_T^q(\langle f, \psi_m \rangle)\psi_m$$

directly to f can perform poorly because a single threshold T cannot adapt to spatially varying noise. A variance-stabilizing transform $\varphi : [0, +\infty) \to \mathbb{R}$, applied pixelwise, maps the counts to $\varphi(f)$. An additive Gaussian white-noise model can then be a useful approximation:

$$\varphi(f) \approx \varphi(f_0) + w \tag{7.14}$$

where $w_n \sim \mathcal{N}(0, \sigma^2)$ are independent centered Gaussians with constant variance σ^2.

The model (7.14) is approximate, and its accuracy depends on the intensity range of $f_{0,n}$. Two common variance-stabilizing transforms for Poisson noise are the Anscombe transform

$$\varphi(x) = 2\sqrt{x + 3/8}$$

and the Freeman–Tukey transform

$$\varphi(x) = \sqrt{x+1} + \sqrt{x}.$$

Figure 7.28 compares the variance of $\varphi(f)$ after these transformations.

A variance-stabilized denoiser is defined as

$$\mathcal{D}^{\mathrm{stab},q}(f) = \varphi^{-1}\Big(\sum_m S_T^q(\langle \varphi(f), \psi_m \rangle)\psi_m\Big)$$

where φ^{-1} is applied after projecting the denoised values onto its domain. The direct inverse generally introduces bias, especially at low counts; bias-corrected inverse transforms can improve the result.

At the moderate count levels in Figure 7.29, variance stabilization improves the denoising result.

7.4.2 Multiplicative Noise

Multiplicative image formation. A multiplicative noise model assumes that

$$f_n = f_{0,n} w_n$$

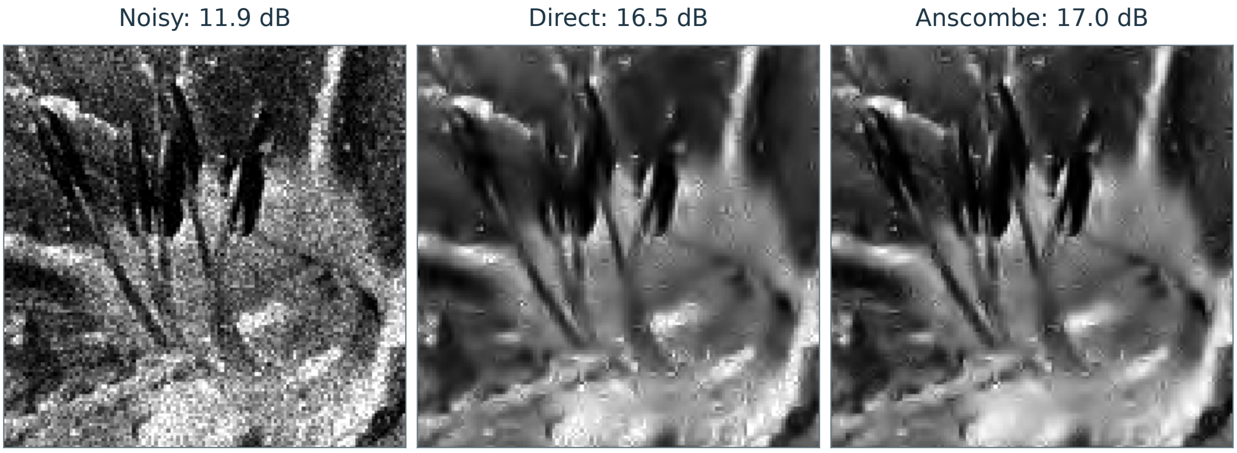


Figure 7.29. Left: noisy image, center: denoising without variance stabilization, right: denoising after variance stabilization.

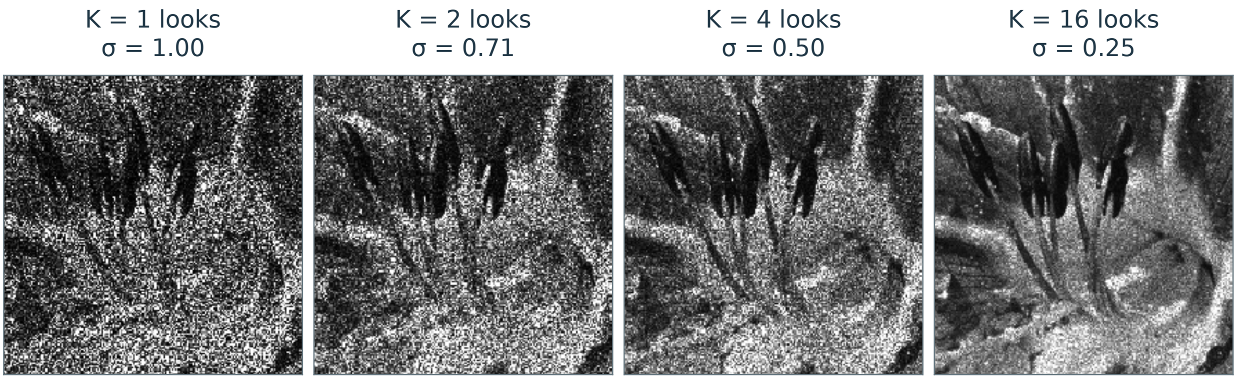


Figure 7.30. Multiplicative Gamma noise for increasing numbers K of averaged looks, with standard deviation $\sigma = K^{-1/2}$.

where w is a random multiplier, observed through one realization, with $\mathbb{E}(w_n) = 1$. Once again, the noise level depends on the pixel value

$$\mathbb{E}(f_n) = f_{0,n} \quad \text{and} \quad \mathrm{Var}(f_n) = f_{0,n}^2\sigma^2 \quad \text{where} \quad \sigma^2 = \mathrm{Var}(w_n).$$

In SAR imaging, a common model forms f by averaging K independent observations, called looks:

$$\forall\, 1 \leqslant s \leqslant K, \quad f_n^{(s)} = f_{0,n} w_n^{(s)} + r_n^{(s)}$$

where $r^{(s)}$ is additive Gaussian white noise and the multiplier $w_n^{(s)}$ has an exponential distribution:

$$p_{w_n^{(s)}}(x) = e^{-x}\mathbf{1}_{x>0}.$$

Averaging K independent looks divides the additive-noise variance by K. When the remaining additive component is negligible, the average is approximately

$$f_n = \frac{1}{K}\sum_{s=1}^{K} f_n^{(s)} \approx f_{0,n} w_n$$

where the multiplicative factor is distributed according to a Gamma distribution

$$w_n \sim \mathrm{Gamma}(\text{shape} = K, \text{rate} = K), \qquad p_{w_n}(x) = \frac{K^K}{\Gamma(K)} x^{K-1} e^{-Kx}\mathbf{1}_{x>0}.$$

Its mean is one and its variance is $1/K$, so increasing K reduces the relative noise level.

Figure 7.30 illustrates how the noise changes with the number $K = 1/\sigma^2$ of averaged looks.

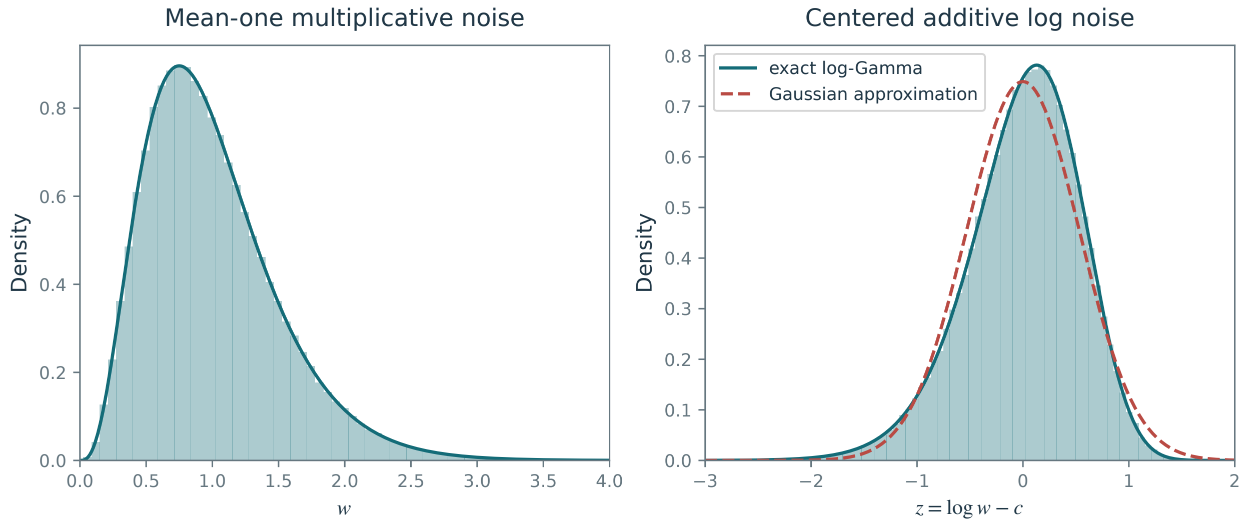


Figure 7.31. Histogram of multiplicative noise before (left) and after (right) stabilization.

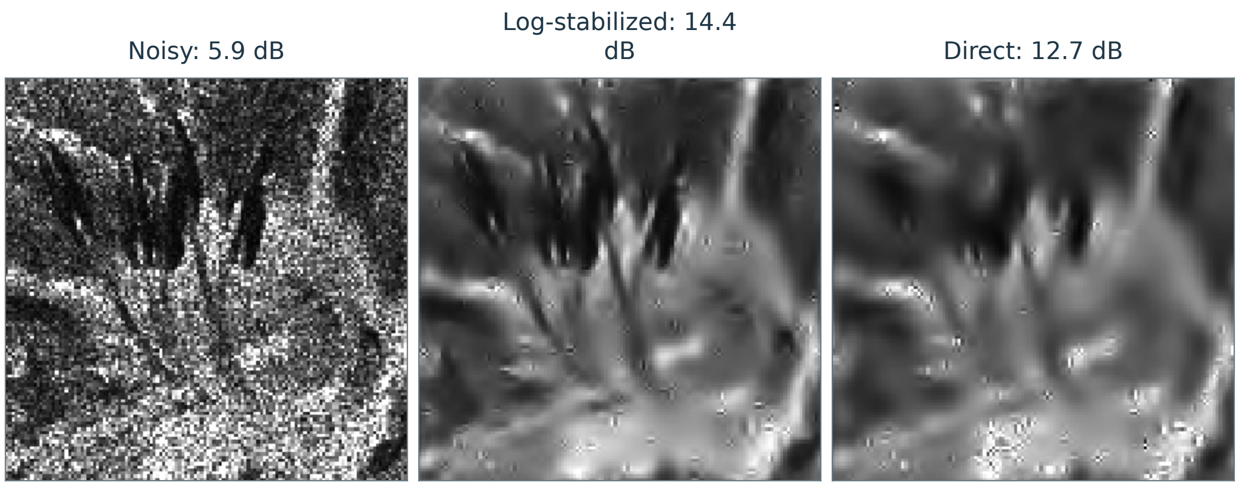


Figure 7.32. Left: noisy image, center: denoising after variance stabilization, right: denoising without variance stabilization.

A simple variance stabilization transform is

$$\varphi(x) = \log(x) - c$$

where

$$c = \mathrm{E}(\log(w)) = \psi(K) - \log(K) \quad \text{where} \quad \psi(x) = \Gamma'(x)/\Gamma(x)$$

Here Γ is the Gamma function and ψ is its logarithmic derivative, the digamma function. The transformed observation satisfies

$$\varphi(f)_n = \log f_{0,n} + z_n,$$

for strictly positive intensities, where $z_n = \log(w_n) - c$ is centered additive noise with variance $\psi^{(1)}(K)$; $\psi^{(1)}$ is the derivative of the digamma function. Thus the transformed target is $\log f_0$, not $\varphi(f_0)$. Exponentiating an estimate of $\log f_0$ introduces a further statistical bias and may require correction.

Figure 7.31 shows the effect of this variance stabilization on the distributions of w and z.

Figure 7.32 compares the two approaches at moderate noise levels σ, where variance stabilization improves the displayed result.

CHAPTER 8

Variational Priors and Regularization

Variational methods reconstruct images by balancing agreement with measured data against a penalty describing expected image structure. The choice of penalty controls the tradeoff between noise removal and preservation of sharp edges. Starting from Sobolev and total variation energies, we develop their discrete operators, derive the associated diffusion flows, and compare stopping a flow early with minimizing an energy that includes a data fidelity term.

8.1 Sobolev and Total Variation Priors

The simplest priors integrate local differential quantities over the image domain. They measure smoothness through seminorms on function spaces. We study two widely used examples: the Sobolev and total variation priors.

8.1.1 Continuous Priors

In this section we define priors for functions $f \in L^2([0,1]^2)$ on a continuous domain. Section 8.1.2 develops their counterparts for discrete images $f \in \mathbb{R}^N$.

Sobolev prior. A prior energy assigns small values to images in the target class Θ and larger values, possibly $+\infty$, to images that depart from that model. The smoothness classes described in Section 5.2.1 motivate Sobolev priors. The simplest such energy is

$$J_{\mathrm{Sob}}(f) = \frac{1}{2}|f|_{H^1}^2 = \frac{1}{2}\int \|\nabla f(x)\|^2 \mathrm{d}x, \tag{8.1}$$

where ∇f is the distributional gradient and the integral is over $[0,1]^2$. The energy is one half of the squared H^1 seminorm and vanishes on constant functions. We set it to $+\infty$ when the gradient is not square integrable.

Total variation prior. To accommodate discontinuities in images, we use the total variation energy introduced in Section 5.2.3. Rudin, Osher, and Fatemi introduced its use in image denoising [28].

The total variation of a smooth image f is defined as

$$J_{\mathrm{TV}}(f) = \|f\|_{\mathrm{TV}} = \int \|\nabla_x f\| \mathrm{d}x. \tag{8.2}$$

This energy extends to functions of bounded variation $f \in \mathrm{BV}([0,1]^2)$, which may be discontinuous. This class includes indicator functions $f = 1_\Omega$ of sets Ω with finite perimeter $|\partial\Omega|$.

The total variation seminorm can also be computed using the coarea formula (5.12), which shows in particular that $\|1_\Omega\|_{\mathrm{TV}} = |\partial\Omega|$.

8.1.2 Discrete Priors

An acquisition device discretizes a continuous-domain image $f \in L^2([0,1]^2)$ into a pixel array $f \in \mathbb{R}^N$. To process these data, we need priors defined on finite-dimensional image spaces.

Discrete gradient. Discrete Sobolev and total variation priors use finite-difference approximations to derivatives. For example, the forward differences are

$$\begin{aligned} \delta_1 f_{n_1,n_2} &= f_{n_1+1,n_2} - f_{n_1,n_2} \\ \delta_2 f_{n_1,n_2} &= f_{n_1,n_2+1} - f_{n_1,n_2}, \end{aligned}$$

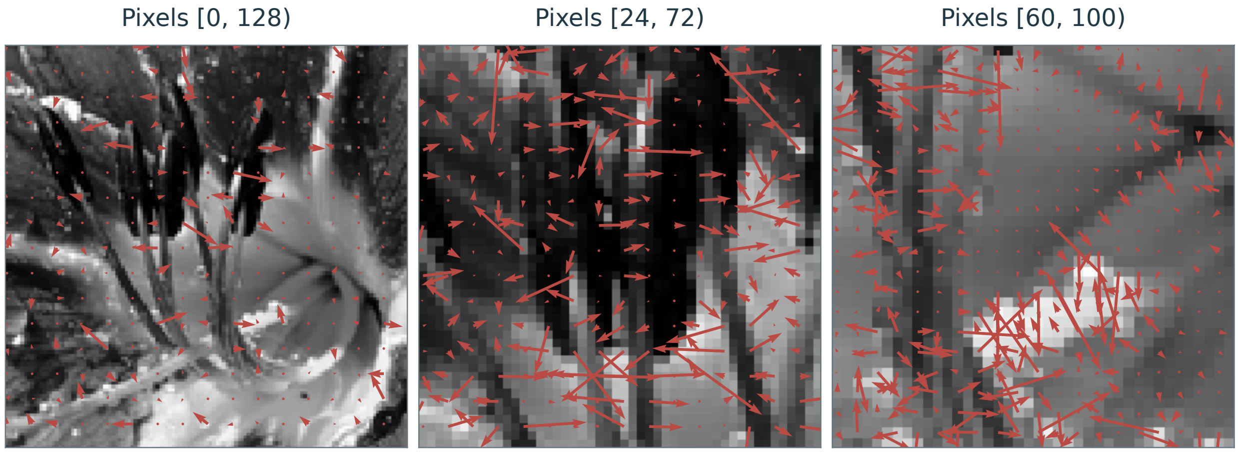


Figure 8.1. Discrete gradient vectors.

where the image has $n \times n$ pixels and $N = n^2$. Higher-order schemes give more accurate approximations for smooth functions. Boundary conditions require care. For simplicity, we use periodic boundary conditions, computing the indices $n_i + 1$ modulo n. Symmetric boundary conditions can also be used to avoid boundary artifacts.

With the grid-spacing factor absorbed into the energy scaling, the discrete gradient at each pixel is

$$\nabla f_n = (\delta_1 f_n, \delta_2 f_n) \in \mathbb{R}^2$$

and the gradient operator maps images to vector fields:

$$\nabla : \mathbb{R}^N \longrightarrow \mathbb{R}^{N \times 2}.$$

Figure 8.1 shows discrete gradient vectors approximating the local direction of steepest intensity increase. Figure 8.2 displays the signed gradient components and their Euclidean magnitude. A shared nonlinear display curve enhances weak red and blue components without changing the numerical gradient. Large magnitudes indicate rapid intensity variations, typically at edges or in textured regions.

Discrete divergence. One can also use backward differences,

$$\tilde{\delta}_1 f_{n_1,n_2} = f_{n_1,n_2} - f_{n_1-1,n_2}$$
$$\tilde{\delta}_2 f_{n_1,n_2} = f_{n_1,n_2} - f_{n_1,n_2-1}.$$

The adjoint relation between backward and forward differences is

$$\delta_i^* = -\tilde{\delta}_i,$$

which means that

$$\forall f, g \in \mathbb{R}^N, \quad \langle \delta_i f, g \rangle = -\langle f, \tilde{\delta}_i g \rangle,$$

This is the discrete counterpart of integration by parts,

$$\int_0^1 f' g = -\int_0^1 f g'$$

for smooth periodic functions on $[0, 1]$.

We define the discrete divergence using backward differences:

$$\operatorname{div}(v)_n = \tilde{\delta}_1 v_{1,n} + \tilde{\delta}_2 v_{2,n},$$

This operator maps vector fields to images:

$$\operatorname{div} : \mathbb{R}^{N \times 2} \longrightarrow \mathbb{R}^N.$$

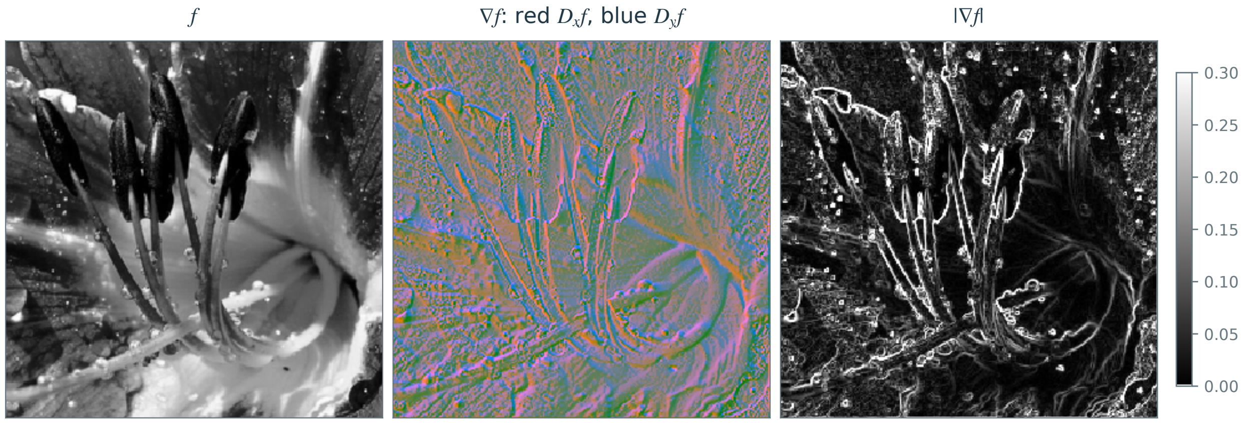


Figure 8.2. Discrete gradient of the flower image. Red and blue encode the signed horizontal and vertical components with enhanced display contrast; neutral gray represents zero. The last panel shows the Euclidean gradient magnitude.

It is the negative adjoint of the gradient:

$$\mathrm{div} = -\nabla^*$$

which means that

$$\forall f \in \mathbb{R}^N,\ \forall v \in \mathbb{R}^{N\times 2}, \quad \langle \nabla f, v\rangle_{\mathbb{R}^{N\times 2}} = -\langle f, \mathrm{div}(v)\rangle_{\mathbb{R}^N}$$

This identity is a discrete version of the divergence theorem.

Discrete Laplacian. The Laplacian is the divergence of the gradient:

$$\Delta f = \mathrm{div}(\nabla f),$$

This operator is negative semidefinite because $\Delta = -\nabla^*\nabla$.

With the discrete gradient and divergence defined above, the Laplacian is the local high-pass filter

$$\Delta f_n = \sum_{p\in V_4(n)} f_p - 4 f_n, \tag{8.3}$$

which, after scaling by the grid spacing, approximates the continuous Laplacian:

$$\frac{\partial^2 f}{\partial x_1^2}(x) + \frac{\partial^2 f}{\partial x_2^2}(x) \approx n^2 \Delta f_k \quad \text{for} \quad x = k/n.$$

Laplacian operators thus act as filters. With the Fourier convention $\hat{f}(\omega) = \int f(x) e^{-\mathrm{i}\langle x,\omega\rangle}\,\mathrm{d}x$, the continuous Laplacian is diagonal in the Fourier domain:

$$g = \Delta f \quad \Longrightarrow \quad \hat{g}(\omega) = -\|\omega\|^2 \hat{f}(\omega)$$

while the discrete Laplacian (8.3) has Fourier representation

$$g = \Delta f \quad \Longrightarrow \quad \hat{g}_\omega = -\rho_\omega^2 \hat{f}_\omega \quad \text{where} \quad \rho_\omega^2 = 4\sin\left(\frac{\pi}{n}\omega_1\right)^2 + 4\sin\left(\frac{\pi}{n}\omega_2\right)^2. \tag{8.4}$$

Discrete energies. The discrete Sobolev energy is one half of the squared ℓ^2 norm of the gradient field:

$$J_{\mathrm{Sob}}(f) = \frac{1}{2}\sum_n \left((\delta_1 f_n)^2 + (\delta_2 f_n)^2\right) = \frac{1}{2}\|\nabla f\|^2. \tag{8.5}$$

The isotropic discrete TV energy instead sums the Euclidean magnitudes of the gradient vectors:

$$J_{\mathrm{TV}}(f) = \sum_n \sqrt{(\delta_1 f_n)^2 + (\delta_2 f_n)^2} = \|\nabla f\|_1 \tag{8.6}$$

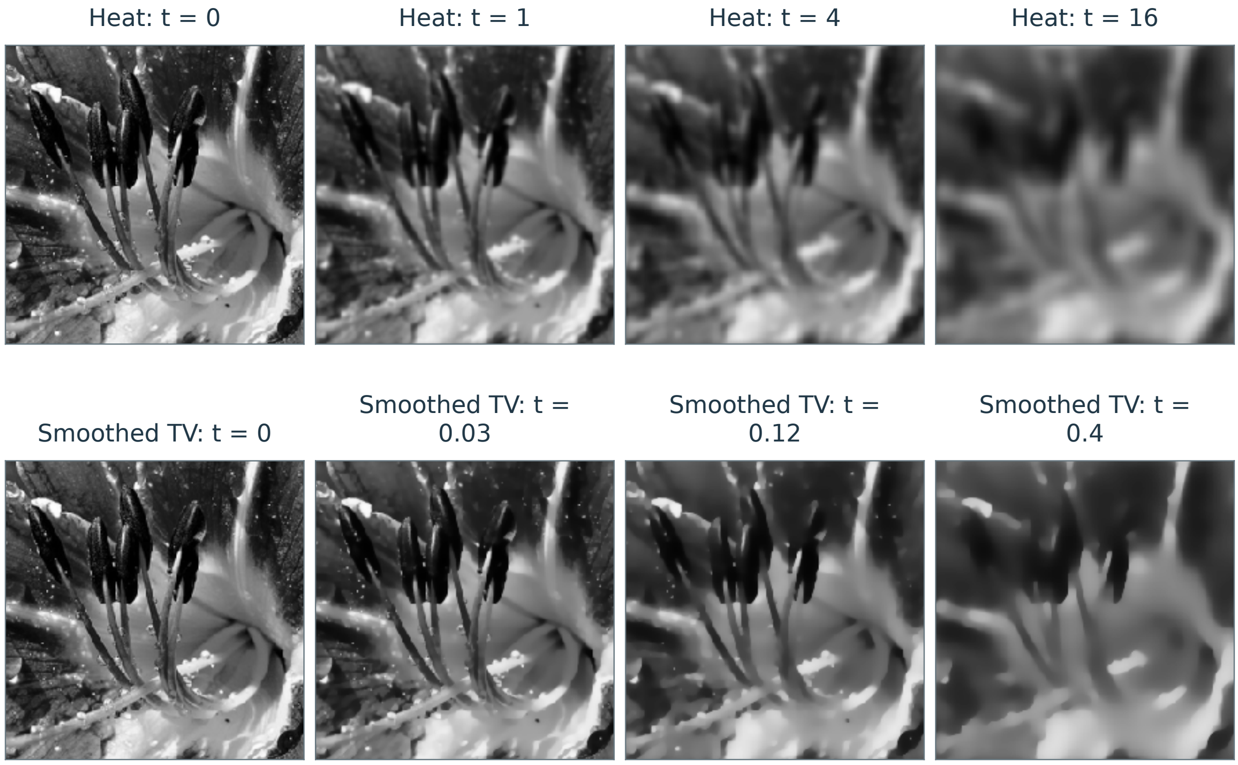


Figure 8.3. Heat flow (top) and TV flow (bottom): images f_t at increasing times t.

Here the notation $\|v\|_1$ for a vector field $v \in \mathbb{R}^{N\times 2}$ means

$$\|v\|_1 = \sum_n \|v_n\| \tag{8.7}$$

where $v_n \in \mathbb{R}^2$.

8.2 PDE and Energy Minimization

Minimizing a smooth prior by gradient descent produces an image-smoothing flow.

8.2.1 General Flows

The gradient of a differentiable prior $J : \mathbb{R}^N \to \mathbb{R}$ is the vector $\operatorname{grad} J(f)$ that describes its local first-order variation:

$$J(f+\varepsilon) = J(f) + \langle \varepsilon, \operatorname{grad} J(f)\rangle + o(\|\varepsilon\|).$$

Gradient descent seeks to decrease a smooth discrete energy by iterating

$$f^{(k+1)} = f^{(k)} - \tau \operatorname{grad} J(f^{(k)}), \tag{8.8}$$

where the step size τ is chosen to ensure descent. The quadratic and smoothed TV energies below admit explicit sufficient bounds on τ.

To pass to continuous time, associate iteration k with time $t = k\tau$ and let τ tend to zero. The limiting flow

$$t > 0 \longmapsto f_t \in \mathbb{R}^N$$

satisfies the following differential equation, an ODE for a finite-dimensional image and a PDE when space is continuous:

$$\frac{\partial f_t}{\partial t} = -\operatorname{grad} J(f_t) \quad \text{and} \quad f_0 = f. \tag{8.9}$$

Gradient descent is an explicit time discretization of this differential equation at times $t_k = k\tau$.

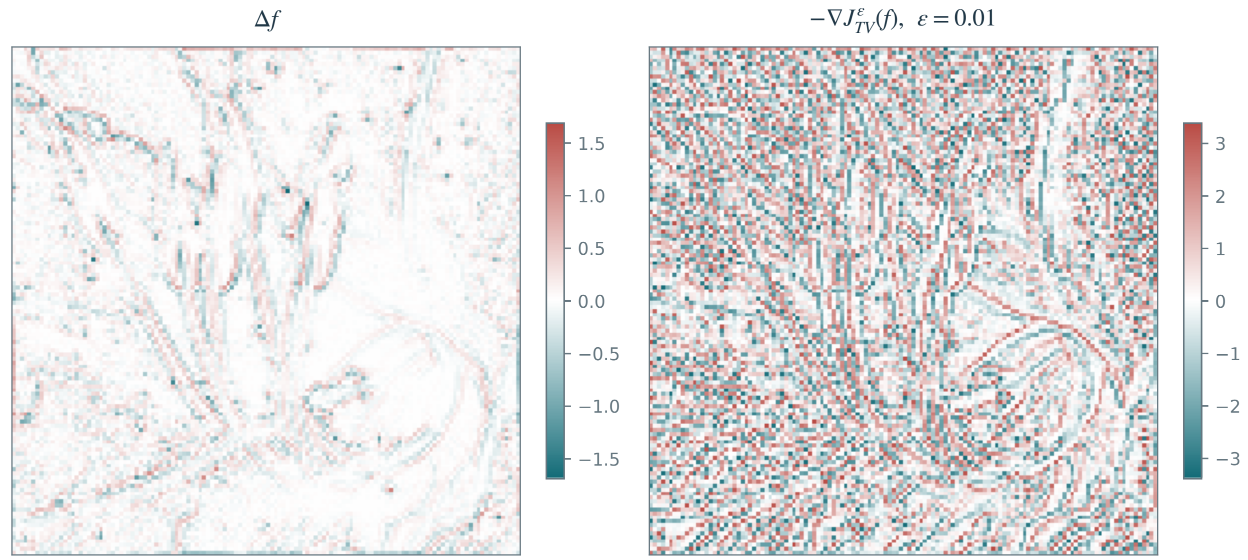


Figure 8.4. Discrete Laplacian (left) and negative discrete TV gradient (right).

8.2.2 Heat Flow

Heat flow results from applying (8.9) to the Sobolev energy $J_{\mathrm{Sob}}(f)$, defined in (8.1) for functions and in (8.5) for discrete images.

Expanding the quadratic energy gives

$$J(f+\varepsilon) = \frac{1}{2}\|\nabla f + \nabla \varepsilon\|^2 = J(f) - \langle \Delta f,\, \varepsilon\rangle + \frac{1}{2}\|\nabla \varepsilon\|^2,$$

so that

$$\operatorname{grad} J_{\mathrm{Sob}}(f) = -\Delta f.$$

Figure 8.4, left, shows an image Laplacian. Its magnitude is typically large near edges, with either sign.

The heat flow is thus

$$\frac{\partial f_t}{\partial t}(x) = -(\operatorname{grad} J(f_t))(x) = \Delta f_t(x) \quad \text{and} \quad f_0 = f. \tag{8.10}$$

Continuous in space. For continuous images on the unbounded domain $\mathbb{R}^2$, the PDE (8.10) has an explicit solution: convolution with a Gaussian kernel whose variance grows with time

$$f_t = f \star h_t \quad \text{where} \quad h_t(x) = \frac{1}{4\pi t} e^{-\frac{\|x\|^2}{4t}}. \tag{8.11}$$

Convolution with this widening kernel progressively smooths the image.

Discrete in space. The gradient flow of the discrete Sobolev energy (8.5) is

$$\frac{\partial f_{n,t}}{\partial t} = -(\operatorname{grad} J(f_t))_n = (\Delta f_t)_n.$$

Discretizing time as in (8.8) gives the fully discrete iteration

$$f_n^{(k+1)} = f_n^{(k)} + \tau\Big(\sum_{p\in V_4(n)} f_p^{(k)} - 4 f_n^{(k)}\Big) = (f^{(k)} \star h)_n$$

where $V_4(n)$ denotes the four neighbors of pixel n. Each iteration convolves the current image with the same filter:

$$f^{(k)} = f \star h \star \ldots \star h = f \star^k h.$$

The filter has coefficients $h_0 = 1 - 4\tau$ and $h_{\pm e_1} = h_{\pm e_2} = \tau$; all other entries vanish.

This iteration is stable and converges if $0 < \tau < 1/4$.

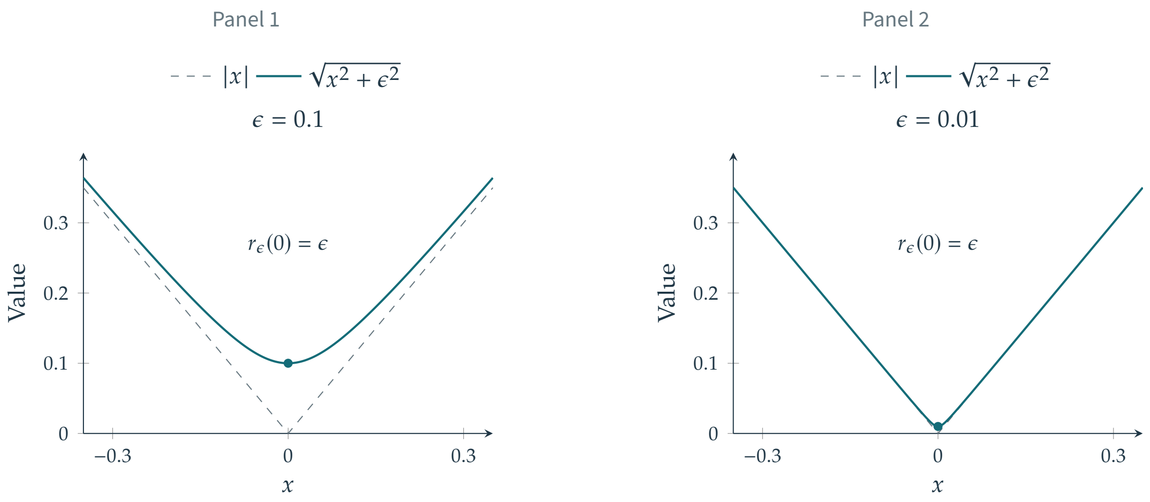


Figure 8.5. Regularized absolute value $x \mapsto \sqrt{x^2 + \varepsilon^2}$.

8.2.3 Total Variation Flows

Total variation gradient. The total variation energy J_{TV} is nonsmooth, both in its continuous form (8.2) and in its discrete form (8.6). In the discrete case, J_{TV} is nondifferentiable at any image f with a pixel n where $\nabla f_n = 0$.

If $\nabla f_n \neq 0$ at every pixel, the gradient is

$$(\operatorname{grad} J(f))_n = -\operatorname{div}\left(\frac{\nabla f}{\|\nabla f\|}\right)_n$$

The displayed quotient is undefined where the spatial gradient vanishes. Figure 8.4, right, shows the negative TV gradient; normalization by small gradient magnitudes makes it sensitive to perturbations in smooth regions.

Nondifferentiability prevents the use of classical gradient descent at such images. A TV flow can nevertheless be defined through the subgradient inclusion $\partial_t f_t \in -\partial J_{\mathrm{TV}}(f_t)$. Here we first study a smooth approximation.

Regularized total variation. To obtain a differentiable energy, we smooth the TV prior as follows:

$$J_{\mathrm{TV}}^{\varepsilon}(f) = \sum_n \sqrt{\varepsilon^2 + \|\nabla f_n\|^2} \tag{8.12}$$

where $\varepsilon > 0$ controls the smoothing. Figure 8.5 illustrates the corresponding smoothing of the absolute value function.

The smoothed TV energy has gradient

$$\operatorname{grad} J_{\mathrm{TV}}^{\varepsilon}(f) = -\operatorname{div}\left(\frac{\nabla f}{\sqrt{\varepsilon^2 + \|\nabla f\|^2}}\right). \tag{8.13}$$

This smoothing interpolates between TV and a rescaled Sobolev energy. For fixed f,

$$\operatorname{grad} J_{\mathrm{TV}}^{\varepsilon}(f) = -\Delta f/\varepsilon + O(\varepsilon^{-3}) \quad \text{when} \quad \varepsilon \longrightarrow +\infty.$$

Figure 8.6 displays the regularized spatial-gradient magnitude for several smoothing parameters.

Regularized total variation flow. The smoothed total variation flow is then defined as

$$\frac{\partial f_t}{\partial t} = \operatorname{div}\left(\frac{\nabla f_t}{\sqrt{\varepsilon^2 + \|\nabla f_t\|^2}}\right). \tag{8.14}$$

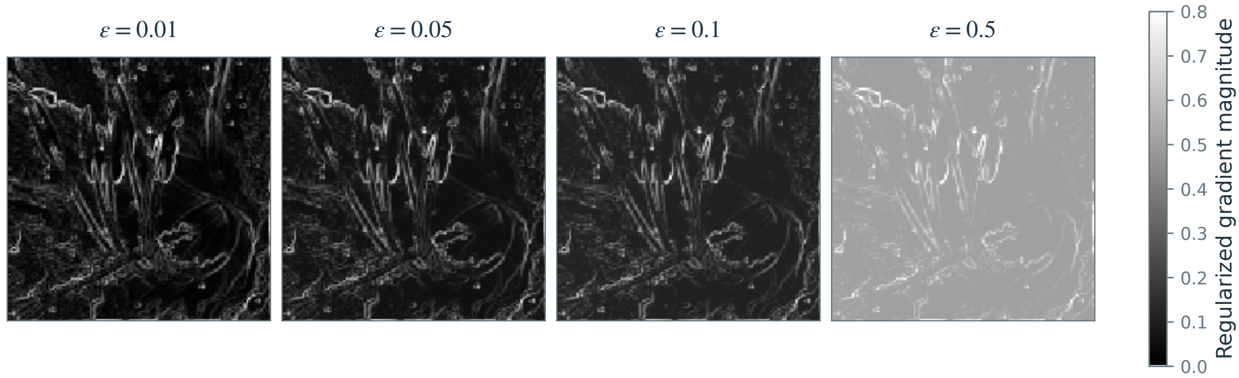


Figure 8.6. Regularized gradient norm $\sqrt{\|\nabla f(x)\|^2 + \varepsilon^2}$.

Choosing a small ε makes the flow approximate total variation minimization more closely, but requires smaller time steps in explicit discretizations.

In practice, gradient descent (8.8) discretizes this flow in time. For the smoothed total variation flow to converge, a sufficient condition is $0 < \tau < \varepsilon/4$. Thus, a closer approximation to the TV energy requires smaller time steps and more iterations.

Figure 8.3 compares heat flow with smoothed TV flow for a small ε. TV flow preserves edges better than heat diffusion, consistent with the ability of the TV energy to represent sharp transitions.

8.2.4 PDE Flows for Denoising

PDE flows can be used to remove noise from an observation $f = f_0 + w$. As detailed in Section 7.1.2, a simple noise model assumes that each pixel is corrupted by Gaussian noise $w_n \sim \mathcal{N}(0, \sigma^2)$, and that these perturbations are independent (white noise).

To denoise, we initialize the PDE flow at the observed image f at time $t = 0$:

$$\frac{\partial f_t}{\partial t} = -\operatorname{grad}_{f_t} J \quad \text{and} \quad f_{t=0} = f.$$

Stopping the flow defines an estimator $\tilde{f} = f_{t_0}$, whose quality depends on the stopping time t_0. Figure 8.7 illustrates denoising by Sobolev and TV flows.

On a connected periodic grid, f_t converges to a constant image when $t \to +\infty$, so the choice of t_0 balances noise removal against excessive smoothing of image edges. Selecting this stopping time is often difficult. In simulations, if one has access to the clean image f_0, one can monitor the denoising error $\|f_0 - f_t\|$ and choose the $t = t_0$ that minimizes this error. Figure 8.8, top row, shows reconstructions obtained with such an oracle choice of stopping time.

8.3 Regularization for Denoising

The flow-based approach in Section 8.2.4 defines the estimator by choosing a stopping time. Alternatively, we can define it as the minimizer of an objective that balances data fidelity and a prior. This formulation also accommodates nonsmooth priors such as total variation J_{TV} and sparsity J_1. Chapter 9 extends it to general inverse problems.

8.3.1 Regularization

For a noisy image $f = f_0 + w$ with N pixels and a prior J, consider

$$f_\lambda^\star \in \operatorname*{argmin}_{g \in \mathbb{R}^N} \frac{1}{2}\|f - g\|^2 + \lambda J(g), \tag{8.15}$$

where the regularization parameter $\lambda > 0$ balances the data fidelity term $\|f - g\|^2$ against the penalty $J(g)$.

If a clean reference image f_0 is available, one can choose the parameter by minimizing the denoising error $\|f_\lambda^\star - f_0\|$. Such a reference is rarely available in practice, so parameter selection must account for the noise level and the regularity of the unknown image f_0.

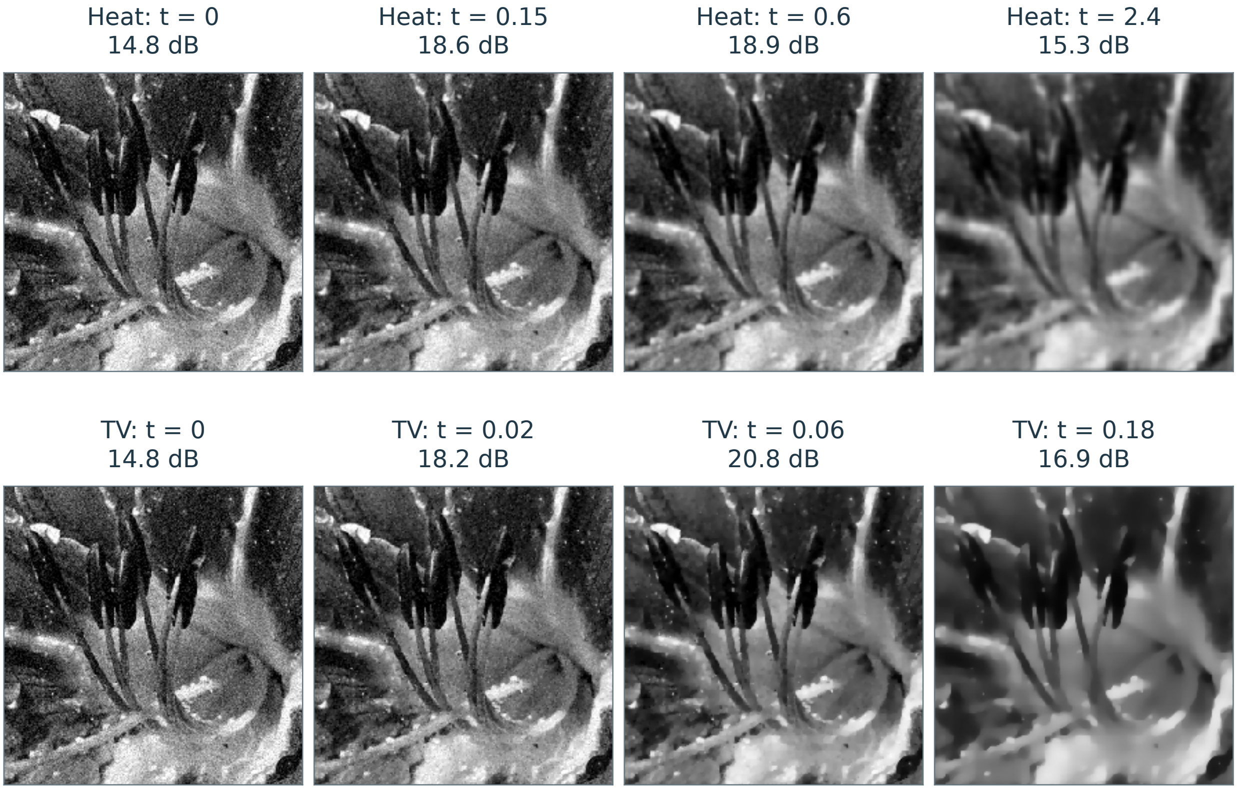


Figure 8.7. Denoised images f_t at several times t: Sobolev flow (top) and TV flow (bottom).

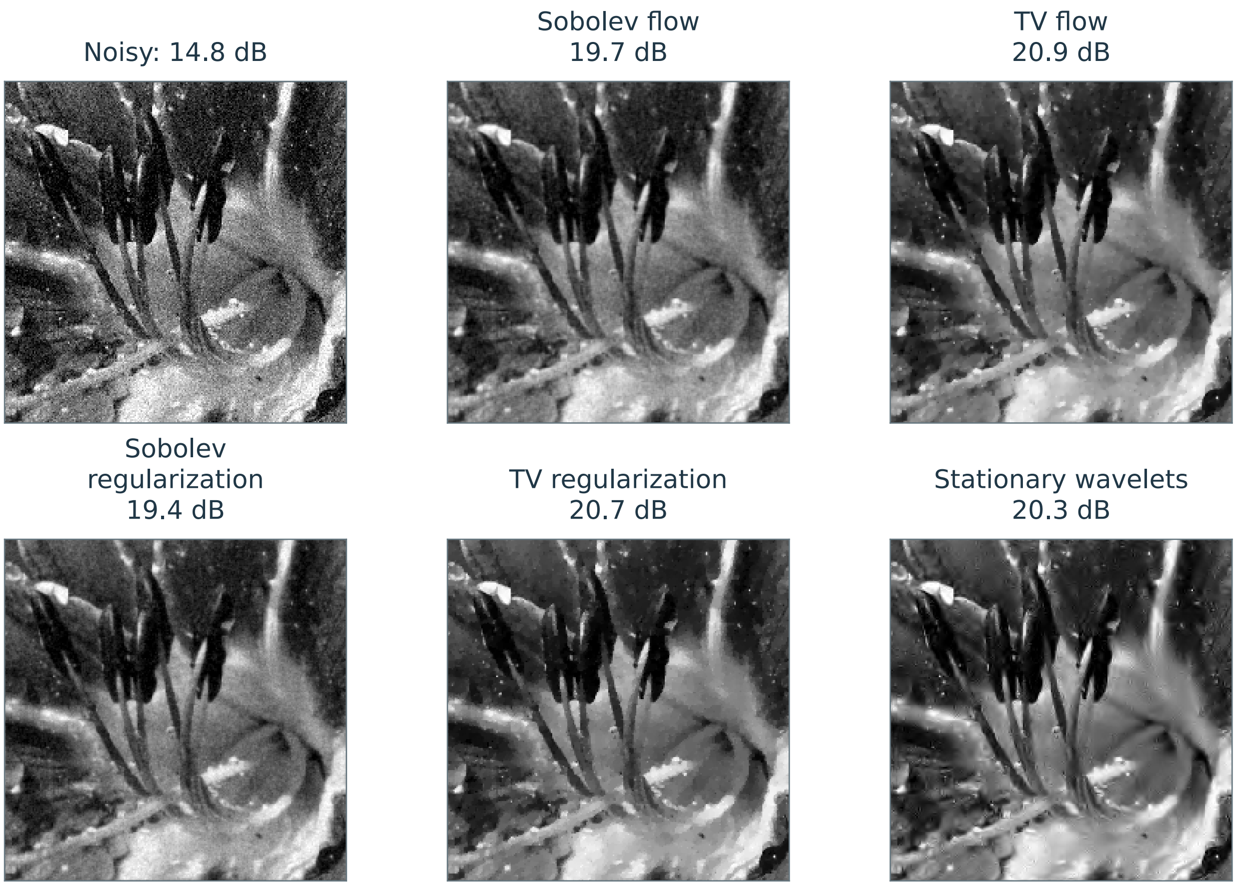


Figure 8.8. Denoising using PDE flows and regularization.

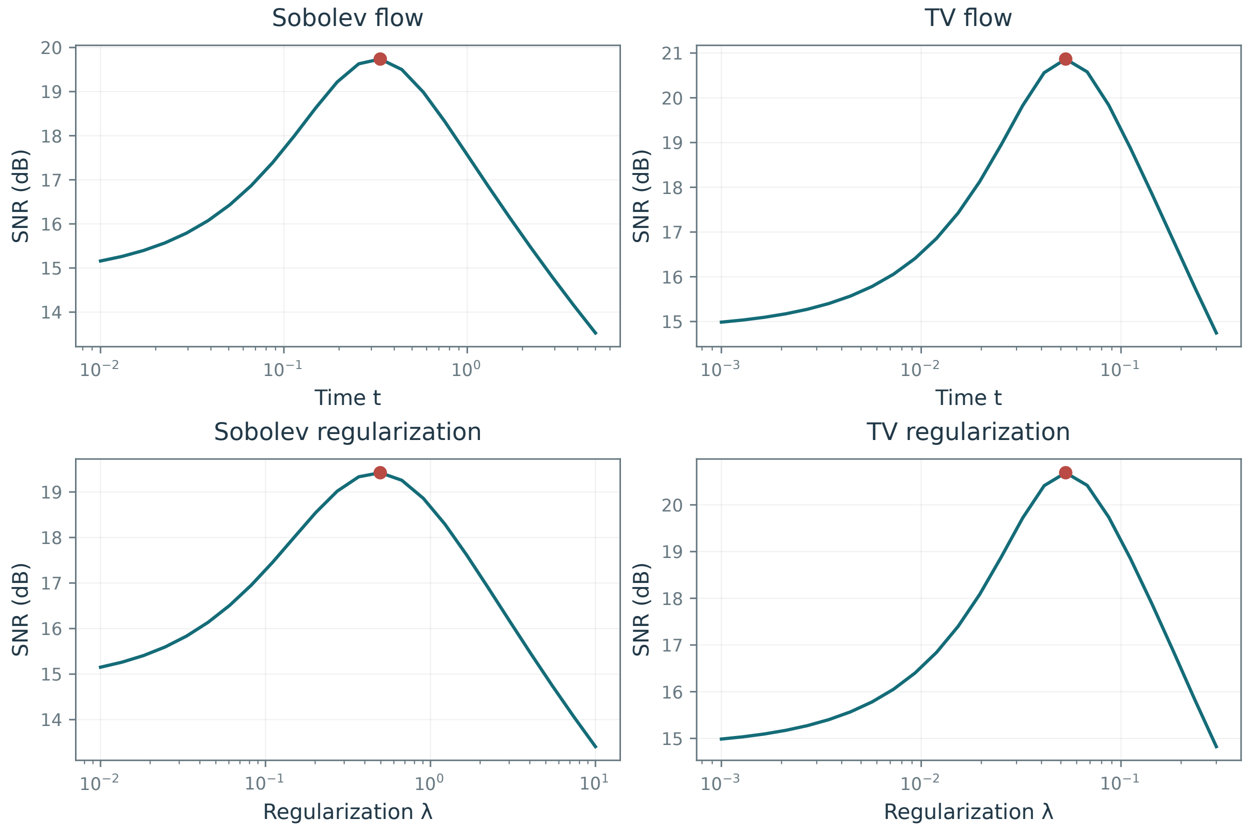


Figure 8.9. SNR as a function of time t for flows (top) and λ for regularization (bottom).

For a proper lower semicontinuous convex prior J, the quadratic fidelity makes the objective strongly convex, so it has a unique minimizer. For nonconvex priors, minimizers need not be unique; properness, lower semicontinuity, and a lower bound for J suffice for existence.

The resulting estimator is

$$\tilde{f} = f_\lambda^\star$$

with λ chosen according to the noise level and prior.

For a differentiable convex prior J, gradient descent for (8.15) takes the form

$$f^{(k+1)} = f^{(k)} - \tau \left(f^{(k)} - f + \lambda \operatorname{grad} J(f^{(k)}) \right) \tag{8.16}$$

where $\tau > 0$ is chosen according to the smoothness of the objective. The iteration adds a data fidelity force to the time-discretized flow (8.8), counteracting the prior's tendency to smooth the image toward a constant.

Total variation J_{TV} and the sparsity prior J_1 are nondifferentiable; the ideal sparsity prior J_0 is also nonconvex. Classical gradient descent therefore cannot be applied directly to (8.15). Computing $f_\lambda^\star$ then requires smoothing the prior or using algorithms adapted to nonsmooth objectives, as developed later in the book.

8.3.2 Sobolev Regularization

The discrete Sobolev prior defined in (8.5) is differentiable, and the gradient descent (8.16) reads

$$f^{(k+1)} = (1-\tau) f^{(k)} + \tau f + \tau \lambda \Delta f^{(k)}.$$

Since $J(f) = \frac{1}{2}\|\nabla f\|^2$ is quadratic, one can also solve the associated linear system by conjugate gradients.

The solution $f_\lambda^\star$ has the explicit linear-system representation

$$f_\lambda^\star = (\mathrm{Id}_N - \lambda \Delta)^{-1} f,$$

Thus, for this prior, the estimator defined by (8.15) depends linearly on the observations f.

If the differential operators are computed with periodic boundary conditions, this linear system can be solved exactly in the Fourier domain

$$(\hat{f}^{\star}_{\lambda})_{\omega} = \frac{1}{1 + \lambda \rho^2_{\omega}} \hat{f}_{\omega} \tag{8.17}$$

where ρ_{ω} is determined by the discrete Laplacian; see (8.4).

Equation (8.17) shows that denoising using Sobolev regularization corresponds to low-pass filtering with strength controlled by λ. Compare this with the solution (8.11) of the heat equation, which uses a Gaussian low-pass kernel with variance $2t$ in each coordinate.

Sobolev denoising belongs to the class of linear estimators studied in Section 7.2. Choosing λ is therefore a filter-selection problem, as in Section 7.2.2, restricted here to a one-parameter family.

8.3.3 TV Regularization

The total variation prior J_{TV} in (8.6) is nondifferentiable. We can either smooth it or use an algorithm designed for nonsmooth optimization.

The approximation $J^{\varepsilon}_{\mathrm{TV}}(g)$ defined in (8.12) is differentiable in g. Substituting its gradient (8.13) into (8.16) gives

$$f^{(k+1)} = (1 - \tau) f^{(k)} + \tau f + \lambda \tau \operatorname{div} \left(\frac{\nabla f^{(k)}}{\sqrt{\varepsilon^2 + \|\nabla f^{(k)}\|^2}} \right). \tag{8.18}$$

For $0 < \tau < 2/(1 + 8\lambda/\varepsilon)$, these iterates converge to the unique minimizer of (8.15) with $J = J^{\varepsilon}_{\mathrm{TV}}$.

Section 19.5.1 presents an algorithm for the original nonsmooth TV penalty.

CHAPTER 9

Inverse Problems

Inverse problems seek to reconstruct a signal from measurements that blur, omit, or mix its values. Direct inversion can amplify noise or leave several signals indistinguishable, so reconstruction must incorporate assumptions about the unknown signal. We study how spectral and variational regularization stabilize inversion, derive error bounds for quadratic penalties, and develop methods for deconvolution, inpainting, and tomography.

The main references for this chapter are [23, 29, 19].

9.1 Regularization of Inverse Problems

We extend the convex regularization introduced in Chapter 8 to linear measurement operators.

We consider a bounded linear map $\Phi : \mathcal{S} \to \mathcal{H}$, where the signal space $\mathcal{S}$ is a Hilbert space or, more generally, a Banach space. The data space $\mathcal{H}$ is a Hilbert space. The operator models acquisition: an unknown high-resolution signal $f_0 \in \mathcal{S}$ gives rise to the noisy observation

$$y = \Phi f_0 + w \in \mathcal{H}$$

where $w \in \mathcal{H}$ represents acquisition noise. Here the noise is deterministic; we assume only that $\|w\|_{\mathcal{H}}$ is bounded.

An acquisition device records finitely many observations, so applications usually take $\mathcal{H} = \mathbb{R}^P$. The number P may be small relative to the desired reconstruction dimension. For numerical implementation, we discretize the signal space as $\mathcal{S} = \mathbb{R}^N$, where N is the number of grid points. Usually the grid is much finer than the measurement set, so $N \gg P$. Infinite-dimensional function spaces remain useful for modeling the unknown f_0 and analyzing recovery, particularly when choosing the signal space $\mathcal{S}$.

Direct inversion may be impossible because Φ is not injective or the data do not lie in its range. Even when an inverse exists on the range, it may have a large norm or be unbounded. Applying it to noisy data amplifies the perturbation: formally, $\Phi^{-1} y = f_0 + \Phi^{-1} w$.

We now give a few representative examples of forward operators Φ.

Denoising. The identity operator $\Phi = \mathrm{Id}_{\mathcal{S}}$ with $\mathcal{S} = \mathcal{H}$ gives the denoising problem studied in Chapters 7 and 8.

De-blurring and super-resolution. For a general operator Φ, recovering f_0 requires both inversion and denoising. These goals often conflict because inversion amplifies noise. For example, in deblurring, Φ is a translation-invariant operator corresponding to low-pass filtering with a kernel h

$$\Phi f = f \star h. \tag{9.1}$$

A periodic model takes $\mathcal{S} = \mathcal{H} = L^2(\mathbb{T}^d)$; see Proposition 2.3. In practice, the blurred signal is sampled on a grid: $\Phi f = \{(f \star h)(x_k) \; ; \; 0 \leqslant k < P\}$. Figure 9.1, middle, shows the resulting low-resolution image Φf_0. Inverting such operators is useful for increasing the resolution of digital photographs and videos.

Interpolation and inpainting. Inpainting reconstructs missing pixels. We model the masking operation by an operator that is diagonal in the spatial domain:

$$(\Phi f)(x) = \begin{cases} 0 & \text{if} \quad x \in \Omega, \\ f(x) & \text{if} \quad x \notin \Omega. \end{cases} \tag{9.2}$$

Here Ω is the missing region, either a subset of $[0,1]^d$ in the continuous model or a set of pixel indices in the discrete model. Figure 9.1, right, shows an example of a damaged image Φf_0.

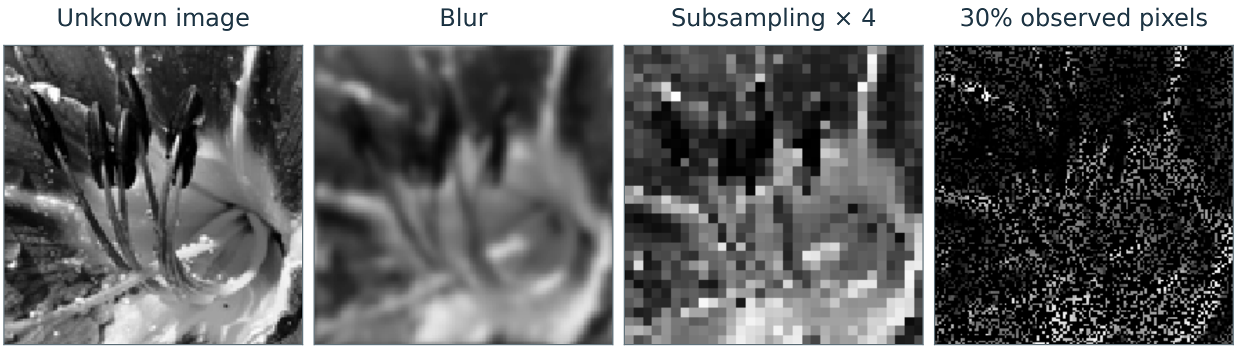


Figure 9.1. Examples of forward operators in inverse problems.

Medical imaging. Many medical imaging devices provide indirect access to the signal of interest, with acquisition processes that can be approximated by a linear operator Φ. For a tomographic scanner, the Radon transform models acquisition. The Fourier slice theorem relates these measurements to Fourier data along radial lines. An idealized magnetic resonance imaging (MRI) model also uses partial Fourier measurements:

$$\Phi f = \left\{ \hat{f}(x) \; ; \; x \in \Omega \right\}. \tag{9.3}$$

Here Ω consists of radial lines for tomography or acquisition trajectories such as spirals for MRI.

Electroencephalography (EEG) and magnetoencephalography (MEG) also infer internal activity from measurements at sensors outside the sources. Abstractly, such a linearized model can be written as $(\Phi f)(s) = \int K(s,x) f(x)\,\mathrm{d}x$, where the kernel depends on the geometry and physical model. Unlike a simple image blur, this kernel need not be translation invariant.

Regression for supervised learning. Linear supervised learning is closely related to the imaging problems studied here, although it uses different notation. Section 13.3 develops the connection between regression and inverse problems. In statistical learning, the data are n pairs $(x_i, y_i)_{i=1}^n$ with feature vectors $x_i \in \mathbb{R}^p$. A linear prediction model has the form $y_i = \langle \beta, \, x_i \rangle$, with unknown parameter $\beta \in \mathbb{R}^p$. Placing the vectors x_i in the rows of $X \in \mathbb{R}^{n \times p}$ gives the approximate system $X\beta \approx y$. This corresponds to the inverse problem $\Phi f = y$ under the substitutions $\Phi \mapsto X$ and $f \mapsto \beta$, with dimensions $(P, N) \to (n, p)$. In statistical learning, the model need not be correctly specified, and the design matrix X is often random. Its sampling fluctuations must then be controlled as $n \to \infty$.

Ridge regression estimates the parameter by solving the normalized problem

$$\min_{\beta} \frac{1}{2n} \|X\beta - y\|^2 + \lambda \|\beta\|^2$$

The corresponding empirical quantities are therefore $\frac{1}{n} X^* X \sim \Phi^* \Phi$ for the covariance and $\frac{1}{n} X^* y \sim \Phi^* y$ for the data term. For independent observations in fixed dimension and with suitable finite moments, these empirical quantities fluctuate around their expectations at order $n^{-1/2}$. In growing dimension, the relevant norm bounds also depend on the dimension and tail assumptions.

9.2 Warmup: Oracle Linear Inversion

We first seek an oracle estimator among inversion methods that act diagonally in a fixed basis, allowing the gains to depend on the unknown clean signal. The optimization then separates into one scalar problem per coefficient. Later we will obtain such estimators from a single optimization problem on the signal space.

Assume that the forward operator is diagonal in an orthonormal basis (ψ_k):

$$\Phi f = \sum_k \varphi_k \langle f, \, \psi_k \rangle \, \psi_k,$$

where the coefficients φ_k typically represent attenuation. The observation model is

$$Y = \Phi f_0 + w$$

where w is Gaussian white noise with variance σ^2. In a real orthonormal basis, its coefficients are independent centered Gaussians. In a complex Fourier basis they still have mean zero and second moment σ^2, which suffices for the risk calculation below. We work in finite dimension; infinite-dimensional white noise requires a generalized random-process interpretation. For instance, if $(\psi_k)_k$ is the Fourier basis, this corresponds to a deconvolution problem $\Phi f = h \star f$ where $\hat{h}_k = \varphi_k$. We consider the same diagonal estimator $\tilde{f}$ as in the denoising case:

$$\tilde{f} := \sum_k \lambda_k \langle Y, \psi_k \rangle \psi_k.$$

We minimize the mean squared error with respect to λ:

$$\mathbb{E}_w \|\tilde{f} - f_0\|^2 = \sum_k \mathbb{E} \left| \langle \tilde{f} - f_0, \psi_k \rangle \right|^2 = \sum_k \mathbb{E} \left| \lambda_k (\varphi_k c_k + \langle w, \psi_k \rangle) - c_k \right|^2.$$

Writing $c_k = \langle f_0, \psi_k \rangle$ and using the zero mean and variance of the noise,

$$\mathbb{E}_w \|\tilde{f} - f_0\|^2 = \sum_k |c_k|^2 |\varphi_k \lambda_k - 1|^2 + |\lambda_k|^2 \sigma^2.$$

The optimal coefficient of the oracle estimator solves

$$\min_{\lambda_k} |c_k|^2 |\varphi_k \lambda_k - 1|^2 + |\lambda_k|^2 \sigma^2,$$

i.e., it satisfies

$$|c_k|^2 \bar{\varphi}_k (\varphi_k \lambda_k - 1) + \sigma^2 \lambda_k = 0,$$

which gives

$$\lambda_k = \frac{|c_k|^2 \bar{\varphi}_k}{|c_k|^2 |\varphi_k|^2 + \sigma^2}.$$

For $\varphi_k \neq 0$ and $c_k \neq 0$, the oracle gains approach direct inversion as the noise vanishes:

$$\lambda_k \longrightarrow \frac{1}{\varphi_k} \qquad \text{as } \sigma \longrightarrow 0.$$

As σ increases, inversion is attenuated. For a simple Fourier-domain example, let $k \geqslant 1$, $|c_k|^2 = k^{-\alpha}$ and $\varphi_k = k^{-\beta/2}$, with $\alpha, \beta > 0$. Then

$$\lambda_k = \frac{k^{\beta/2}}{1 + \sigma^2 k^{\alpha+\beta}}.$$

At low frequencies this filter approximately compensates for attenuation, whereas at high frequencies it suppresses noise. Treating k as a positive real variable, its maximum occurs at

$$k^\star = \left(\frac{\beta}{(2\alpha + \beta)\sigma^2} \right)^{1/(\alpha+\beta)}.$$

On the discrete range $k \geqslant 1$, the maximum is attained near this value, or at the lowest frequency if $k^\star < 1$.

9.3 Theoretical Study of Quadratic Regularization

We outline a standard approach to recovery guarantees in Hilbert spaces. The emphasis is on modeling assumptions expressed through source conditions, and on limitations of the squared-norm penalty, notably saturation of convergence rates and loss of components in $\ker(\Phi)$.

9.3.1 Singular Value Decomposition

Finite dimension. We begin with the finite-dimensional case $\Phi \in \mathbb{R}^{P \times N}$ so that $\mathcal{S} = \mathbb{R}^N$ and $\mathcal{H} = \mathbb{R}^P$ are Hilbert spaces. In this case, the singular value decomposition (SVD) provides a precise description of the operator and of linear inversion.

Proposition 9.1 (SVD). *There exist matrices $(U, V) \in \mathbb{R}^{P\times R} \times \mathbb{R}^{N\times R}$, with $R = \text{rank}(\Phi) = \dim(\text{Im}(\Phi))$, satisfying $U^\top U = V^\top V = \text{Id}_R$. Their orthonormal columns $(u_m)_{m=1}^R \subset \mathbb{R}^P$, $(v_m)_{m=1}^R \subset \mathbb{R}^N$ and singular values $(\sigma_m)_{m=1}^R$, with $\sigma_m > 0$, satisfy*

$$\Phi = U \operatorname{diag}_m(\sigma_m) V^\top = \sum_{m=1}^{R} \sigma_m u_m v_m^\top. \tag{9.4}$$

Proof. To motivate the construction, suppose that $\Phi = U\Sigma V^\top$ with $\Sigma = \operatorname{diag}_m(\sigma_m)$. Then $\Phi\Phi^\top = U\Sigma^2 U^\top$, which determines the left singular vectors, and $V^\top = \Sigma^{-1}U^\top\Phi$ determines the right ones. Since $\Phi\Phi^\top$ is symmetric and positive semidefinite, its eigendecomposition on its range is $\Phi\Phi^\top = U\Sigma^2U^\top$, where $\Sigma = \operatorname{diag}_{m=1}^R(\sigma_m)$ with $\sigma_m > 0$. Define $V \overset{\text{def.}}{=} \Phi^\top U \Sigma^{-1}$. One then verifies that

$$V^\top V = (\Sigma^{-1}U^\top\Phi)(\Phi^\top U\Sigma^{-1}) = \Sigma^{-1}U^\top(U\Sigma^2U^\top)U\Sigma^{-1} = \text{Id}_R \quad \text{and} \quad U\Sigma V^\top = U\Sigma\Sigma^{-1}U^\top\Phi = \Phi.$$

□

The theorem also holds for complex matrices, with $^\top$ replaced by *. Expression (9.4) describes Φ as a sum of rank-1 matrices $u_m v_m^\top$. One usually orders the singular values $(\sigma_m)_m$ in decreasing order $\sigma_1 \geqslant \ldots \geqslant \sigma_R$. If the positive singular values are distinct, the real reduced SVD is unique up to simultaneous sign changes of each pair (u_m, v_m). In the complex case, simultaneous multiplication by a unit-modulus scalar replaces the sign change.

The columns of U form an orthonormal basis of $\text{Im}(\Phi)$, while the columns of V form an orthonormal basis of $\text{Im}(\Phi^\top) = \ker(\Phi)^\perp$. The decomposition (9.4) is called the reduced SVD because it retains only the R nonzero singular values. The full SVD completes the columns of U and V to orthonormal bases of $\mathbb{R}^P$ and $\mathbb{R}^N$, respectively. The diagonal factor Σ then has size $P \times N$.

For a periodic discrete convolution $\Phi f = h \star f$, let F be the unitary discrete Fourier matrix. Then

$$\Phi = F^* \operatorname{diag}(\hat{h}_m) F, \tag{9.5}$$

where $\hat{h}_m$ denotes the convolution multiplier (the unnormalized DFT of h for the usual discrete convolution). If $q_m = F^* e_m$ and $\hat{h}_m \neq 0$, one may take $v_m = q_m$, $u_m = (\hat{h}_m/|\hat{h}_m|)q_m$, and $\sigma_m = |\hat{h}_m|$.

Computing the SVD of a dense matrix $\Phi \in \mathbb{R}^{N\times N}$ typically costs $O(N^3)$ operations.

Compact operators. The SVD extends to compact operators $\Phi : \mathcal{S} \to \mathcal{H}$ between separable Hilbert spaces. Compactness means that ΦB_1 is relatively compact, where $B_1 = \{s \in \mathcal{S} \; ; \; \|s\| \leqslant 1\}$ is the unit ball. Equivalently, every sequence $(\Phi s_k)_k$ with $s_k \in B_1$ has a convergent subsequence. In infinite dimension, the identity operator $\Phi : \mathcal{S} \to \mathcal{S}$ is not compact.

Equivalently, compact operators Φ admit an expansion analogous to (9.4):

$$\Phi = \sum_{m=1}^{+\infty} \sigma_m \langle \cdot , v_m \rangle u_m \tag{9.6}$$

Here (u_m) and (v_m) are orthonormal systems in $\mathcal{H}$ and $\mathcal{S}$, respectively, and the positive singular values decrease to zero; finite-rank operators give a finite sum. Convergence in (9.6) holds in the operator norm induced by the norms on $\mathcal{S}$ and $\mathcal{H}$

$$\|\Phi\|_{\mathcal{L}(\mathcal{S},\mathcal{H})} \overset{\text{def.}}{=} \sup_{\|u\|_{\mathcal{S}} \leqslant 1} \|\Phi u\|_{\mathcal{H}}.$$

For a nonzero operator Φ with the expansion (9.6), $\|\Phi\|_{\mathcal{L}(\mathcal{S},\mathcal{H})} = \sigma_1$.

If only R singular values are positive, then Φ has finite rank $R = \dim(\text{Im}(\Phi))$. Squared-norm regularization restricts the reconstruction to $\ker(\Phi)^\perp$, a space of dimension R, so the problem reduces to finite dimension. Nonlinear methods can also recover components in $\ker(\Phi)$ under suitable priors; this is sometimes called super-resolution. Between Hilbert spaces, compact operators are exactly the operator-norm limits of finite-rank operators. This approximation property need not hold for arbitrary Banach spaces, although the definition of compactness itself extends to them.

Examples include integral operators on $L^2(\Omega)$ with a continuous kernel $k(x, y)$ defined for $(x, y) \in \Omega \times \Omega$, where Ω is a compact subset of $\mathbb{R}^d$ or the torus $\mathbb{T}^d$:

$$(\Phi f)(x) = \int_\Omega k(x, y) f(y) \mathrm{d}y$$

where $\mathrm{d}y$ is the Lebesgue measure. Convolution $\Phi f = f \star h$ on $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$ generalizes (9.5). Its kernel $k(x,y) = h(x-y)$ is translation invariant, and the Fourier functions $q_m(x) = (2\pi)^{-d/2}e^{\mathrm{i}m\cdot x}$ diagonalize the operator. Its singular values are $\sigma_m = |\hat{h}_m|$, where $\hat{h}_m = \int_{\mathbb{T}^d} h(x)e^{-\mathrm{i}m\cdot x}\,\mathrm{d}x$ denotes the convolution multiplier. Another example on $\Omega = [0,1]$ is the integration operator $(\Phi f)(x) = \int_0^x f(y)\mathrm{d}y$, which has the square-integrable kernel $k(x,y) = 1_{y \leqslant x}$ and is also compact.

Pseudoinverse. We first work in finite dimension, or more generally assume that $\mathrm{Im}(\Phi)$ is closed. Even without noise, $\Phi f = y$ may be inconsistent, and a nontrivial kernel makes a solution nonunique. The Moore–Penrose pseudoinverse selects the least-squares solution of minimum norm

$$f^+ \overset{\text{def.}}{=} \underset{\Phi f = y^+}{\operatorname{argmin}} \|f\|_{\mathcal{S}} \quad \text{where} \quad y^+ = \mathrm{Proj}_{\mathrm{Im}(\Phi)}(y) = \underset{z\in\mathrm{Im}(\Phi)}{\operatorname{argmin}} \|y-z\|_{\mathcal{H}}.$$

The next proposition expresses the pseudoinverse through the SVD. Under injectivity or surjectivity, it also gives formulas involving linear systems with $\Phi\Phi^*$ or $\Phi^*\Phi$.

Proposition 9.2. *One has*

$$f^+ = \Phi^+ y \quad \textit{where} \quad \Phi^+ = V\,\mathrm{diag}_m(1/\sigma_m)U^*. \tag{9.7}$$

If $\mathrm{Im}(\Phi) = \mathcal{H}$, *one has* $\Phi^+ = \Phi^*(\Phi\Phi^*)^{-1}$. *If* $\ker(\Phi) = \{0\}$, *one has* $\Phi^+ = (\Phi^*\Phi)^{-1}\Phi^*$.

Proof. The columns of U form an orthonormal basis of $\mathrm{Im}(\Phi)$, so $y^+ = UU^*y$. Thus $\Phi f = y^+$ becomes $U\Sigma V^* f = UU^*y$, or $V^*f = \Sigma^{-1}U^*y$. In the orthogonal decomposition $f = f_0 + r$, with $f_0 \in \ker(\Phi)^\perp$ and $r \in \ker(\Phi)$, the data fix $f_0 = VV^*f = V\Sigma^{-1}U^*y = \Phi^+y$. Since $\|f\|^2 = \|f_0\|^2 + \|r\|^2$, minimizing the total norm amounts to minimizing $\|r\|$, which gives $r = 0$. If $\mathrm{Im}(\Phi) = \mathcal{H}$, then $R = P$ so that $\Phi\Phi^* = U\Sigma^2U^*$ is the eigendecomposition of an invertible matrix, and $(\Phi\Phi^*)^{-1} = U\Sigma^{-2}U^*$. One then verifies $\Phi^*(\Phi\Phi^*)^{-1} = V\Sigma U^*U\Sigma^{-2}U^*$ which is the desired result. The second case follows similarly. □

In infinite dimension, a compact operator of infinite rank has nonclosed range. Its pseudoinverse is unbounded and is defined only for data satisfying the corresponding square-summability condition on the inverse-weighted SVD coefficients. The projections and inverse formulas above cannot then be used on arbitrary data.

For convolution operators $\Phi f = f \star h$, the spectral formula gives

$$\Phi^+ y = y \star h^+ \quad \text{where} \quad \hat{h}_m^+ = \begin{cases} \hat{h}_m^{-1} & \text{if} \quad \hat{h}_m \neq 0 \\ 0 \quad \text{if} & \hat{h}_m = 0. \end{cases}.$$

9.3.2 Tikhonov Regularization

Regularized inverse. For an ill-conditioned operator, applying the unregularized inverse to noisy data can be unstable, since

$$\Phi^+ y = \Phi^+\Phi f_0 + \Phi^+ w = f_0^+ + \Phi^+ w \quad \text{where} \quad f_0^+ \overset{\text{def.}}{=} \mathrm{Proj}_{\ker(\Phi)^\perp}(f_0),$$

The recovery error is therefore $\|\Phi^+y - f_0^+\| = \|\Phi^+w\|$. It reaches $\|w\|/\sigma_R$ when $w \propto u_R$, so a small singular value produces a large amplification factor $1/\sigma_R$. In the infinite-rank case $R = +\infty$, the compact operator's pseudoinverse is unbounded. To control this amplification, replace Φ^+ by a regularized inverse of the form

$$\Phi_\lambda^+ = V\,\mathrm{diag}_m(\mu_\lambda(\sigma_m))U^* \tag{9.8}$$

where the spectral filter μ_λ depends on a parameter $\lambda > 0$ and satisfies the boundedness and consistency conditions

$$|\mu_\lambda(\sigma)| \leqslant C_\lambda < +\infty \quad \text{and} \quad \lim_{\lambda\to 0} \mu_\lambda(\sigma) = \frac{1}{\sigma}.$$

Figure 9.2, left, illustrates a capped reciprocal filter, which limits the amplification of small singular values.

Variational regularization. A typical regularized inverse is obtained by solving a penalized least-squares problem

$$f_\lambda \overset{\text{def.}}{=} \underset{f\in\mathcal{S}}{\operatorname{argmin}} \|y-\Phi f\|_{\mathcal{H}}^2 + \lambda J(f) \tag{9.9}$$

where J is a regularization functional. For general convex priors, lower semicontinuity, rather than continuity, is the natural assumption; existence also requires compactness or coercivity conditions. The simplest example is the quadratic norm $J = \|\cdot\|_{\mathcal{S}}^2$,

$$f_\lambda \overset{\text{def.}}{=} \underset{f\in\mathcal{S}}{\operatorname{argmin}} \|y-\Phi f\|_{\mathcal{H}}^2 + \lambda \|f\|^2 \tag{9.10}$$

Proposition 9.3 shows that this estimator is a special case of (9.8). Its normal equations give

$$f_\lambda = (\Phi^*\Phi + \lambda \mathrm{Id}_{\mathcal{S}})^{-1}\Phi^* y. \tag{9.11}$$

This shows that $f_\lambda \in \mathrm{Im}(\Phi^*) \subset \ker(\Phi)^\perp$, and that it depends linearly on y.

Proposition 9.3. *The solution of* (9.10) *is* $f_\lambda = \Phi_\lambda^+ y$, *with the regularized inverse* (9.8) *defined by the filter*

$$\forall\, \sigma \geqslant 0, \quad \mu_\lambda(\sigma) = \frac{\sigma}{\sigma^2+\lambda}.$$

Proof. The normal equations are $(\Phi^*\Phi + \lambda\mathrm{Id}_{\mathcal{S}})f_\lambda = \Phi^* y$. Taking components along v_m gives

$$(\sigma_m^2 + \lambda)\langle f_\lambda, v_m\rangle = \sigma_m \langle y, u_m\rangle.$$

The component in $\ker(\Phi)$ vanishes because $\lambda > 0$. Thus

$$f_\lambda = \sum_m \frac{\sigma_m}{\sigma_m^2+\lambda}\langle y, u_m\rangle v_m,$$

which proves the formula.

□

For convolution $\Phi f = f \star h$, the FFT computes the regularized inverse in $O(N\log(N))$ operations:

$$\hat f_{\lambda,m} = \frac{\hat h_m^*}{|\hat h_m|^2+\lambda}\hat y_m.$$

Figure 9.2 compares quadratic regularization (9.11) (right) with a capped reciprocal filter (left).

We choose λ according to the noise level and seek a rate for $f_\lambda \to f_0$. This requires $f_0 = f_0^+$, equivalently $f_0 \in \overline{\mathrm{Im}(\Phi^*)} = \ker(\Phi)^\perp$: squared-norm regularization always produces $f_\lambda \in \mathrm{Im}(\Phi^*) \subset \ker(\Phi)^\perp$, so it cannot recover a kernel component of f_0. Nonlinear priors can recover such components when additional structure makes the signal identifiable.

Source condition. To obtain a convergence rate, we impose a source condition of order β, which reads

$$f_0 \in \mathrm{Im}((\Phi^*\Phi)^{\beta/2}) = \mathrm{Im}(V\,\mathrm{diag}(\sigma_m^\beta)V^*). \tag{9.12}$$

For a fixed bound on the source vector, larger β imposes stronger decay along directions with small singular values and makes stable inversion easier. The condition asserts the existence of $z \in \mathcal{S}$ such that $f_0 = V\,\mathrm{diag}(\sigma_m^\beta)V^* z$. Choose the source vector of minimum norm, $z = V\,\mathrm{diag}(\sigma_m^{-\beta})V^* f_0$, and impose $\|z\| \leqslant \rho$ for some $\rho > 0$. This is equivalent to the coefficient bound

$$\sum_m \sigma_m^{-2\beta}|\langle f_0,\, v_m\rangle|^2 \leqslant \rho^2 < +\infty. \tag{$S_{\beta,\rho}$}$$

The assumptions $\beta > 0$ and $f_0 \in \ker(\Phi)^\perp$ are part of the source condition; the coefficient bound alone leaves kernel components unconstrained. The bound resembles the Sobolev constraint in 7.6. For example, consider a low-pass convolution $\Phi f = f \star h$ where the kernel h has a polynomially decaying multiplier: $|\hat h_m| \sim 1/m^\alpha$ for large m. Since the vectors v_m are Fourier modes, ($S_{\beta,\rho}$) gives, up to constants and the Fourier normalization, a bound of the form

$$\sum_m \|m\|^{2\alpha\beta}|\hat f_m|^2 \leqslant \rho^2 < +\infty.$$

This is a Sobolev-type coefficient bound of smoothness order $\alpha\beta$.

Sublinear convergence speed. The following theorem gives the convergence rate implied by this source condition. With the noise bound $\|w\| \leqslant \delta$, recovery depends on the parameters (δ, ρ, β). Assuming $f_0 \in \ker(\Phi)^\perp$, we study the convergence of f_λ to f_0 for data $y = \Phi f_0 + w$ as $\delta \to 0$. The analysis also determines how λ should depend on δ.

Theorem 9.4. *Let $\delta > 0$. Under the source condition ($S_{\beta,\rho}$) with $0 < \beta \leqslant 2$, the solution of (9.10) for $\|w\| \leqslant \delta$ satisfies*

$$\|f_\lambda - f_0\| \leqslant C\rho^{\frac{1}{\beta+1}}\delta^{\frac{\beta}{\beta+1}}$$

for a constant C which depends only on β, and for a choice

$$\lambda \sim \delta^{\frac{2}{\beta+1}}\rho^{-\frac{2}{\beta+1}}.$$

Proof. The source condition implies $f_0 \in \ker(\Phi)^\perp$. We decompose

$$f_\lambda = \Phi_\lambda^+(\Phi f_0 + w) = f_\lambda^0 + \Phi_\lambda^+ w \quad \text{where} \quad f_\lambda^0 \overset{\text{def.}}{=} \Phi_\lambda^+(\Phi f_0),$$

Thus $f_\lambda = f_\lambda^0 + \Phi_\lambda^+ w$, and for any regularized inverse of the form (9.8)

$$\|f_\lambda - f_0\| \leqslant \|f_\lambda - f_\lambda^0\| + \|f_\lambda^0 - f_0\|. \tag{9.13}$$

The noise-propagation term $\|f_\lambda - f_\lambda^0\|$ measures the effect of measurement noise. For Tikhonov regularization it decreases as λ increases and is often called the variance term. The bias term $\|f_\lambda^0 - f_0\|$ is independent of the noise and measures the smoothing of f_0. For Tikhonov regularization it increases with λ. Choosing λ therefore balances bias against noise amplification. Assuming

$$\forall\, \sigma \geqslant 0, \quad |\mu_\lambda(\sigma)| \leqslant C_\lambda$$

the variance term is bounded as

$$\|f_\lambda - f_\lambda^0\|^2 = \|\Phi_\lambda^+ w\|^2 = \sum_m \mu_\lambda(\sigma_m)^2|w_m|^2 \leqslant C_\lambda^2\|w\|_{\mathcal{H}}^2.$$

Using $\frac{|f_{0,m}|^2}{\sigma_m^{2\beta}} = |z_m|^2$, we bound the bias term as follows:

$$\|f_\lambda^0 - f_0\|^2 = \sum_m (1-\mu_\lambda(\sigma_m)\sigma_m)^2|f_{0,m}|^2 = \sum_m \left((1-\mu_\lambda(\sigma_m)\sigma_m)\sigma_m^\beta\right)^2 \frac{|f_{0,m}|^2}{\sigma_m^{2\beta}} \leqslant D_{\lambda,\beta}^2\rho^2 \tag{9.14}$$

where we assumed

$$\forall\, \sigma \geqslant 0, \quad \left|(1-\mu_\lambda(\sigma)\sigma)\sigma^\beta\right| \leqslant D_{\lambda,\beta}. \tag{9.15}$$

For $\beta > 2$, the supremum over all $\sigma \geqslant 0$ is infinite. On the bounded spectrum of a fixed operator, the bias remains bounded, but the resulting rate saturates. Putting (9.14) and (9.15) together, one obtains

$$\|f_\lambda - f_0\| \leqslant C_\lambda\delta + D_{\lambda,\beta}\rho. \tag{9.16}$$

In the case of the regularization (9.10), one has $\mu_\lambda(\sigma) = \frac{\sigma}{\sigma^2+\lambda}$, and thus $(1-\mu_\lambda(\sigma)\sigma)\sigma^\beta = \frac{\lambda\sigma^\beta}{\sigma^2+\lambda}$. For $\beta \leqslant 2$, one verifies (see Figure 9.2 and 9.3) that

$$C_\lambda = \frac{1}{2\sqrt{\lambda}} \quad \text{and} \quad D_{\lambda,\beta} = c_\beta\lambda^{\frac{\beta}{2}},$$

for some constant c_β. Balancing the two terms in (9.16) gives the correct order; optimizing over λ would improve only the constant. Set $\frac{\delta}{\sqrt{\lambda}} = \lambda^{\frac{\beta}{2}}\rho$, which gives $\lambda = (\delta/\rho)^{\frac{2}{\beta+1}}$. Then

$$\|f_\lambda - f_0\| = O(\delta/\sqrt{\lambda}) = O(\delta(\delta/\rho)^{-\frac{1}{\beta+1}}) = O(\delta^{\frac{\beta}{\beta+1}}\rho^{\frac{1}{\beta+1}}).$$

□

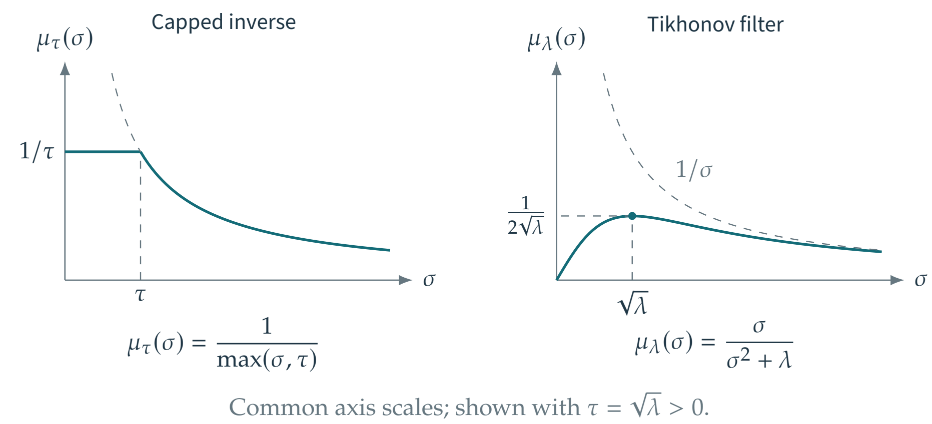


Figure 9.2. Left: capping reciprocal singular values. Right: the Tikhonov bound $\mu_\lambda(\sigma) \leqslant C_\lambda = \frac{1}{2\sqrt{\lambda}}$.

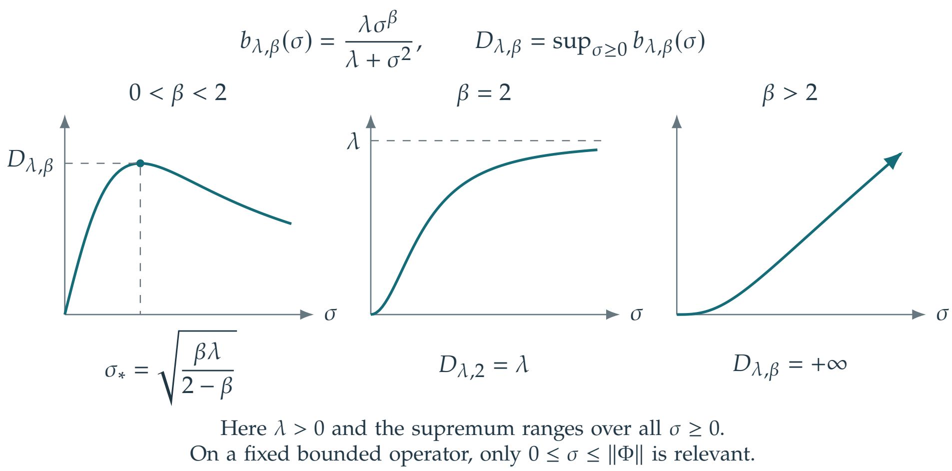


Figure 9.3. Bounding $\lambda\frac{\sigma^\beta}{\lambda+\sigma^2} \leqslant D_{\lambda,\beta}$.

Larger source orders $\beta \leqslant 2$ therefore give faster convergence as $\|w\|$ tends to zero. The rate in Theorem 9.4 saturates: choosing $\beta > 2$ gives the same general worst-case rate as $\beta = 2$. The best rate obtainable in this way is

$$\|f_\lambda - f_0\| = O(\rho^{\frac{1}{3}}\delta^{\frac{2}{3}}).$$

Alternative spectral filters μ_λ, together with sufficiently large β, can give the rate $\|f_\lambda - f_0\| = O(\delta^{1-\kappa})$ for arbitrarily small $\kappa > 0$. The capped inverse in Figure 9.2, left, avoids this finite-order saturation, whereas Tikhonov regularization is limited to source orders $\beta \leqslant 2$. Quadratic regularization is easier to implement: its variational formulation leads to a linear system and does not require an SVD. For compact operators with nonclosed range, no uniform linear rate follows over these source balls from finite-order source conditions alone. Linear rates are possible in finite dimension with a fixed smallest positive singular value, or under different structural assumptions, such as those used for sparse ℓ^1 regularization in Chapter 10.

9.4 Quadratic Regularization

We now turn from infinite-dimensional recovery theory to numerical methods in finite dimension.

Convex regularization. As in (9.9), we reconstruct a high-resolution image $f_0 \in \mathbb{R}^N$ from noisy measurements $y = \Phi f_0 + w \in \mathbb{R}^P$ by minimizing a convex regularized objective:

$$f_\lambda \in \underset{f\in\mathbb{R}^N}{\operatorname{argmin}}\ \mathcal{E}(f) \overset{\text{def.}}{=} \frac{1}{2}\|y - \Phi f\|^2 + \lambda J(f) \tag{9.17}$$

Here $\|y - \Phi f\|^2$ measures data fidelity, with $\|\cdot\|$ denoting the ℓ^2 norm on $\mathbb{R}^P$, and $J(f)$ is a convex penalty on $\mathbb{R}^N$.

The regularization parameter $\lambda > 0$ balances these two terms and can be difficult to choose in practice. In simulations with a known reference signal f_0, the parameter can be calibrated by minimizing the reconstruction error $\|f_0 - \tilde{f}\|$. Such a reference is generally unavailable in applications.

For noiseless data, $w = 0$, we study small values of λ. As $\lambda \to 0$, minimizers of (9.17) approach solutions of a constrained problem under the assumptions below. We may assume $y \in \operatorname{Im}(\Phi)$ without changing the minimizers, since orthogonality gives

$$\|y - \Phi f\|^2 = \|y - \operatorname{Proj}_{\operatorname{Im}(\Phi)}(y)\|^2 + \|\operatorname{Proj}_{\operatorname{Im}(\Phi)}(y) - \Phi f\|^2$$

The first term is independent of f, so replacing y by $\operatorname{Proj}_{\operatorname{Im}(\Phi)}(y)$ in (9.17) changes only an additive constant.

Let us recall that a function J is coercive if

$$\lim_{\|f\|\to+\infty} J(f) = +\infty$$

i.e.

$$\forall K \in \mathbb{R},\ \exists R > 0, \quad \|f\| \geqslant R \quad \Longrightarrow \quad J(f) \geqslant K.$$

Equivalently, every sublevel set $\{f \,;\, J(f) \leqslant c\}$ is bounded. In finite dimension, lower semicontinuity of J then makes these sets compact.

Proposition 9.5. *Assume that $J : \mathbb{R}^N \to \mathbb{R}$ is lower semicontinuous and coercive and that $y \in \operatorname{Im}(\Phi)$. For each λ, let f_λ solve (9.17). The family $(f_\lambda)_\lambda$ is bounded, and every accumulation point $f^\star$ as $\lambda \downarrow 0$ solves*

$$f^\star \in \underset{f\in\mathbb{R}^N}{\operatorname{argmin}}\ \{J(f) \,;\, \Phi f = y\}. \tag{9.18}$$

Proof. Lower semicontinuity and coercivity ensure that J attains its minimum on the nonempty closed constraint set $\{f : \Phi f = y\}$. Let h be such a minimizer. Optimality of f_λ in (9.17) gives

$$\frac{1}{2\lambda}\|\Phi f_\lambda - y\|^2 + J(f_\lambda) \leqslant \frac{1}{2\lambda}\|\Phi h - y\|^2 + J(h) = J(h).$$

The inequality $J(f_\lambda) \leqslant J(h)$ and coercivity of J make $(f_\lambda)_\lambda$ bounded. Choose a convergent subsequence $f_{\lambda_k} \to f^\star$ as $k \to +\infty$. Write $m = \inf J > -\infty$. The estimate $\|\Phi f_{\lambda_k} - y\|^2 \leqslant 2\lambda_k(J(h) - m)$ and the limit $\lambda_k \to 0$ imply $\Phi f^\star = y$, so $f^\star$ is feasible for (9.18). Lower semicontinuity of J, applied to $J(f_{\lambda_k}) \leqslant J(h)$, gives $J(f^\star) \leqslant J(h)$. Thus $f^\star$ solves (9.18). □

Full coercivity of J can be replaced by boundedness of joint sublevel sets of $J(f)$ and $\|\Phi f\|$. For seminorm penalties such as TV, this holds when $\ker(\Phi)$ intersects the nullspace of the seminorm only at zero; in particular, the measurements must detect constant images.

Quadratic Regularization. The simplest prior functionals are quadratic, and can be written as

$$J(f) = \frac{1}{2}\|Gf\|^2_{\mathbb{R}^K} = \frac{1}{2}\langle Lf, f\rangle_{\mathbb{R}^N} \tag{9.19}$$

where $G \in \mathbb{R}^{K\times N}$ and $L = G^*G \in \mathbb{R}^{N\times N}$ is a positive semidefinite matrix. The special case (9.10) is recovered when setting $G = L = \mathrm{Id}_N$.

Writing down the first-order optimality conditions for (9.17) leads to

$$\nabla\mathcal{E}(f) = \Phi^*(\Phi f - y) + \lambda L f = 0,$$

hence, if

$$\ker(\Phi) \cap \ker(G) = \{0\},$$

then (9.17) has a unique minimizer f_λ, which is obtained by solving a linear system

$$f_\lambda = (\Phi^*\Phi + \lambda L)^{-1}\Phi^* y. \tag{9.20}$$

If L is diagonal in a full right-singular basis of Φ, with eigenvalues α_m^2, then, for the positive singular values,

$$\langle f_\lambda, v_m\rangle = \frac{\sigma_m}{\sigma_m^2 + \lambda\alpha_m^2}\langle y, u_m\rangle. \tag{9.21}$$

Example of convolution. Consider the convolution operator

$$\Phi f = h \star f, \tag{9.22}$$

and $G = \nabla$, a discretization of the gradient operator, for example using first-order finite differences (2.16). This corresponds to the discrete Sobolev prior introduced in Section 8.1.2. Such an operator computes, for a d-dimensional signal $f \in \mathbb{R}^N$ (for instance a 1-D signal for $d = 1$ or an image when $d = 2$), an approximation $\nabla f_n \in \mathbb{R}^d$ of the gradient vector at each sample location n. Thus typically, $\nabla : f \mapsto (\nabla f_n)_n \in \mathbb{R}^{N\times d}$ maps to d-dimensional vector fields. Then $-\nabla^* : \mathbb{R}^{N\times d} \to \mathbb{R}^N$ is a discretized divergence operator. In this case, $\Delta = -G^*G$ is a discretization of the Laplacian, which is itself a convolution operator. One then has

$$\hat{f}_{\lambda,m} = \frac{\hat{h}_m^*\hat{y}_m}{|\hat{h}_m|^2 - \lambda\hat{d}_{2,m}}, \tag{9.23}$$

where $\hat{d}_2$ is the Fourier transform of the filter d_2 corresponding to the Laplacian. For instance, in dimension 1, using first-order finite differences, the expression for $\hat{d}_{2,m}$ is given in (2.18).

9.4.1 Solving Linear Systems

If Φ and L are not simultaneously diagonalizable in the required bases, the coefficient formula (9.21) is unavailable. We instead solve (9.20), written as

$$Af = b \quad \text{where} \quad A \overset{\text{def.}}{=} \Phi^*\Phi + \lambda L \quad \text{and} \quad b = \Phi^* y.$$

For moderate N, up to a few thousand, direct methods solve this system at a cost of $O(N^3)$ for a generic A. Since A is symmetric positive definite, Cholesky factorization $A = BB^*$ is a natural choice, with B lower triangular. Reordering the rows and columns can help preserve sparsity in the factors of A.

For large N, iterative solvers access A only through matrix-vector products. In imaging, operator structure often makes these products much cheaper than a dense $O(N^2)$ computation. For a sufficiently sparse matrix A, the cost is typically $O(N)$. When A is symmetric positive definite, as assumed here, the conjugate gradient method solves the equivalent quadratic minimization problem

$$\min_{f\in\mathbb{R}^N} \mathcal{E}(f) \overset{\text{def.}}{=} Q(f) \overset{\text{def.}}{=} \frac{1}{2}\langle Af, f\rangle - \langle f, b\rangle \tag{9.24}$$

This objective agrees with (9.17) up to a constant. Starting from any $f^{(0)}$, conjugate gradients computes $f^{(\ell+1)}$ by minimizing over the affine space generated by all gradients obtained so far, rather than taking a single gradient step as in Section 19.1:

$$f^{(\ell+1)} \stackrel{\text{def.}}{=} \underset{f}{\operatorname{argmin}} \left\{ \mathcal{E}(f) \; ; \; f \in f^{(\ell)} + \operatorname{Span}(\nabla\mathcal{E}(f^{(0)}), \dots, \nabla\mathcal{E}(f^{\ell})) \right\}.$$

This subspace minimization requires only one matrix-vector product per iteration. For $\ell \geqslant 1$, with initialization $d^{(0)} = \nabla\mathcal{E}(f^{(0)}) = Af^{(0)} - b$, the recurrence is

$$f^{(\ell+1)} = f^{(\ell)} - \tau_\ell d^{(\ell)} \quad \text{where} \quad d^{(\ell)} = g^{(\ell)} + \frac{\|g^{(\ell)}\|^2}{\|g^{(\ell-1)}\|^2} d^{(\ell-1)} \quad \text{and} \quad \tau_\ell = \frac{\langle g^{(\ell)}, d^{(\ell)} \rangle}{\langle Ad^{(\ell)}, d^{(\ell)} \rangle} \tag{9.25}$$

where $g^{(\ell)} \stackrel{\text{def.}}{=} \nabla\mathcal{E}(f^{(\ell)}) = Af^{(\ell)} - b$. The iteration stops as soon as $g^{(\ell)} = 0$. It can also be shown that the directions $d^{(\ell)}$ are A-orthogonal (conjugate), so that after at most N iterations in exact arithmetic, the conjugate gradient method computes the unique solution $f^{(\ell)}$ of the linear system $Af = b$. In practice, it is usually used as an approximate solver, with far fewer than N iterations. Nonlinear conjugate-gradient methods extend this idea to smooth objectives, but require a line search and additional safeguards. The linear-system step-size formula and finite-termination property above do not carry over unchanged.

9.5 Non-Quadratic Regularization

9.5.1 Total Variation Regularization

Quadratic regularization (9.19) often produces a blurred reconstruction f_λ. This can poorly approximate a signal f_0 with sharp transitions, such as image edges. In the convolutional case (9.23), the restoration operator is a filter, which tends to smooth sharp features.

For the squared-norm penalty, the restored signal belongs to $\ker(\Phi)^\perp$, so missing components cannot be recovered. This statement does not hold for every quadratic penalty: a different matrix L can couple observed and unobserved components. Nonquadratic priors, particularly nonsmooth ones such as TV and ℓ^1, encode structures such as edges and sparse coefficients that quadratic penalties do not model well.

Total variation. Total variation is a widely used nonquadratic, nonsmooth prior. For smooth functions $f : \mathbb{R}^d \mapsto \mathbb{R}$, it replaces the Sobolev or Dirichlet energy

$$J_{\text{Sob}}(f) \stackrel{\text{def.}}{=} \frac{1}{2} \int_{\mathbb{R}^d} \|\nabla f\|_{\mathbb{R}^d}^2 \mathrm{d}x,$$

where $\nabla f(x) = (\partial_{x_1} f(x), \dots, \partial_{x_d} f(x))^\top$ is the gradient, by the (vectorial) L^1 norm of the gradient

$$J_{\text{TV}}(f) \stackrel{\text{def.}}{=} \int_{\mathbb{R}^d} \|\nabla f\|_{\mathbb{R}^d} \mathrm{d}x.$$

Section 8.1.1 gives further background on these priors. Removing the square ^{2} from the integrand changes which functions have finite energy.

For a nontrivial bounded set Ω with a smooth boundary, $J_{\text{Sob}}(1_\Omega) = +\infty$, whereas $J_{\text{TV}}(1_\Omega) = \operatorname{Per}(\Omega) < +\infty$. More generally, an integrable function f is of bounded variation when its distributional gradient Df is a finite vector-valued Radon measure. Its total variation is

$$J_{\text{TV}}(f) = |Df|(\mathbb{R}^d) = \sup_{\substack{h \in C_c^1(\mathbb{R}^d;\mathbb{R}^d) \\ \|h\|_\infty \leqslant 1}} \int_{\mathbb{R}^d} f(x) \operatorname{div} h(x) \, \mathrm{d}x.$$

For a finite vector measure m, its total mass is equivalently

$$|m|(\mathbb{R}^d) = \sup_{\substack{h \in C_c(\mathbb{R}^d;\mathbb{R}^d) \\ \|h\|_\infty \leqslant 1}} \int_{\mathbb{R}^d} \langle h, \mathrm{d}m \rangle.$$

The coarea formula for $f \in \mathrm{BV}(\mathbb{R}^d)$ reads

$$J_{\mathrm{TV}}(f) = \int_{\mathbb{R}} \mathrm{Per}(\{f > t\})\,\mathrm{d}t.$$

For smooth functions, this agrees, for almost every level, with integrating the $(d-1)$-dimensional Hausdorff measure of the level sets. For general BV functions, superlevel-set perimeters, rather than measures of the pointwise sets $\{f = t\}$, are required.

Discretized Total variation. For discrete data $f \in \mathbb{R}^N$, the construction in Section 8.1.2 generalizes (8.6) to any spatial dimension:

$$J_{\mathrm{TV}}(f) = \sum_n \|\nabla f_n\|_{\mathbb{R}^d}$$

where $\nabla f_n \in \mathbb{R}^d$ approximates the spatial gradient at grid point n.

The discrete total variation prior $J_{\mathrm{TV}}(f)$ defined in (8.6) is a convex but nondifferentiable function of f, since a term of the form $\|\nabla f_n\|$ is nondifferentiable if $\nabla f_n = 0$. Chapters 18 and 19 develop nonsmooth convex optimization methods for such functionals.

To use classical gradient descent, we replace the nonsmooth ℓ^2 norm $\|\cdot\|$ by a differentiable approximation, for example

$$\forall\, u \in \mathbb{R}^d, \quad \|u\|_\varepsilon \stackrel{\text{def.}}{=} \sqrt{\varepsilon^2 + \|u\|^2}.$$

This gives the smoothed TV functional already introduced in (8.12):

$$J^\varepsilon_{\mathrm{TV}}(f) \stackrel{\text{def.}}{=} \sum_n \|\nabla f_n\|_\varepsilon$$

For fixed u, the limits as $\varepsilon \downarrow 0$ and $\varepsilon \to +\infty$ are described by

$$\|u\|_\varepsilon \xrightarrow{\varepsilon \to 0} \|u\| \quad \text{and} \quad \|u\|_\varepsilon = \varepsilon + \frac{1}{2\varepsilon}\|u\|^2 + O(\varepsilon^{-3})$$

Thus $J^\varepsilon_{\mathrm{TV}}$ connects J_{TV} to a rescaled version of J_{Sob}, up to an additive constant.

The resulting regularized inverse problem (9.17) thus reads

$$f_\lambda \stackrel{\text{def.}}{=} \underset{f \in \mathbb{R}^N}{\operatorname{argmin}}\ \mathcal{E}(f) = \frac{1}{2}\|y - \Phi f\|^2 + \lambda J^\varepsilon_{\mathrm{TV}}(f) \tag{9.26}$$

If $\ker(\Phi) \cap \ker(\nabla) = \{0\}$, the objective is coercive and strictly convex, so it has a unique minimizer. Strict convexity of $\|\cdot\|_\varepsilon$ alone does not ensure uniqueness after composition with the gradient, whose nullspace contains constants.

9.5.2 Gradient Descent Method

The optimization program (9.26) is an example of smooth unconstrained convex optimization of the form

$$\min_{f \in \mathbb{R}^N} \mathcal{E}(f) \tag{9.27}$$

where $\mathcal{E} : \mathbb{R}^N \to \mathbb{R}$ is a C^1 function. Recall that the gradient $\nabla\mathcal{E} : \mathbb{R}^N \mapsto \mathbb{R}^N$ of this functional (not to be confused with the discretized gradient $\nabla f \in \mathbb{R}^{N \times d}$ of f) is defined by the following first-order relation

$$\mathcal{E}(f + r) = \mathcal{E}(f) + \langle \nabla\mathcal{E}(f), r \rangle_{\mathbb{R}^N} + o(\|r\|_{\mathbb{R}^N})$$

A locally Lipschitz gradient improves the remainder to $O(\|r\|^2)$. Mere continuity of the gradient does not give this stronger estimate.

For such a function, the gradient descent algorithm is defined as

$$f^{(\ell+1)} \stackrel{\text{def.}}{=} f^{(\ell)} - \tau_\ell \nabla\mathcal{E}(f^{(\ell)}), \tag{9.28}$$

The step size $\tau_\ell > 0$ must balance stability against progress per iteration.

Section 19.1 analyzes convergence and explains how it depends on the step size τ_ℓ.

9.5.3 Examples of Gradient Computation

Note that the gradient of a quadratic function $Q(f)$ of the form (9.24) reads

$$\nabla Q(f) = Af - b.$$

In particular, the first-order optimality condition $\nabla Q(f) = 0$ is equivalent to the linear system $Af = b$.

For the quadratic fidelity term $\mathcal{G}(f) = \frac{1}{2}\|\Phi f - y\|^2$, one thus obtains

$$\nabla \mathcal{G}(f) = \Phi^*(\Phi f - y).$$

In the special case of the regularized TV problem (9.26), the gradient of $\mathcal{E}$ reads

$$\nabla \mathcal{E}(f) = \Phi^*(\Phi f - y) + \lambda \nabla J^{\varepsilon}_{\mathrm{TV}}(f).$$

For differentiable maps $\mathcal{F} : \mathbb{R}^P \to \mathbb{R}$ and $\mathcal{H} : \mathbb{R}^N \to \mathbb{R}^P$, the chain rule gives

$$\nabla(\mathcal{F} \circ \mathcal{H})(f) = D\mathcal{H}(f)^* \nabla \mathcal{F}(\mathcal{H}(f)).$$

Apply it to $J^{\varepsilon}_{\mathrm{TV}} = \|\cdot\|_{1,\varepsilon} \circ \nabla$, where $\|u\|_{1,\varepsilon} = \sum_n \sqrt{\varepsilon^2 + \|u_n\|^2}$. The result is

$$\nabla J^{\varepsilon}_{\mathrm{TV}}(f) = \nabla^* \mathcal{N}^{\varepsilon}(\nabla f) = -\operatorname{div}(\mathcal{N}^{\varepsilon}(\nabla f)), \qquad \mathcal{N}^{\varepsilon}(u)_n = \frac{u_n}{\sqrt{\varepsilon^2 + \|u_n\|^2}}.$$

Thus the gradient of the energy is computed by taking finite differences, normalizing the resulting vector field, and applying minus the divergence.

Since $\operatorname{div} = -\nabla^*$, their operator norms are equal. For unscaled forward differences in d dimensions with periodic boundaries, $\|\nabla\|_{\mathrm{op}} \leqslant 2\sqrt{d}$. The Hessian of $u \mapsto \sqrt{\varepsilon^2 + \|u\|^2}$ is

$$\frac{\mathrm{Id}_d}{\sqrt{\varepsilon^2 + \|u\|^2}} - \frac{uu^*}{(\varepsilon^2 + \|u\|^2)^{3/2}},$$

and is bounded above by $\varepsilon^{-1}\mathrm{Id}_d$. Consequently, a Lipschitz constant for the gradient of $\mathcal{E}$ is

$$L = \|\Phi\|^2_{\mathrm{op}} + \frac{\lambda \|\nabla\|^2_{\mathrm{op}}}{\varepsilon}.$$

The smallest eigenvalue of this pointwise Hessian is $\varepsilon^2/(\varepsilon^2 + \|u\|^2)^{3/2}$, which tends to zero as $\|u\| \to \infty$. Thus smoothed TV is not globally strongly convex. If Φ is injective, the fidelity term supplies a global strong-convexity constant $\sigma_{\min}(\Phi)^2$; otherwise additional analysis on bounded sublevel sets is needed.

A fixed step $0 < \tau < 2/L$ gives convergence of gradient descent when a minimizer exists. As $\varepsilon \downarrow 0$, this bound forces small steps, motivating the nonsmooth methods developed in later chapters. A modest positive ε can also reduce the staircase artifacts of exact TV regularization.

9.6 Examples of Inverse Problems

We now discuss several inverse problems in imaging that can be solved using quadratic regularization or nonlinear TV.

9.6.1 Deconvolution

The blurring operator (9.1) is diagonal in the Fourier domain, so quadratic regularization can be solved efficiently using fast Fourier transforms under periodic boundary conditions. We refer to (9.22) and the corresponding explanations. TV regularization does not reduce to a single diagonal Fourier solve and requires an iterative optimization algorithm.

9.6.2 Inpainting

For the inpainting problem, the operator defined in (9.2) is diagonal in space

$$\Phi = \operatorname{diag}_m(\delta_{\Omega^c}[m]),$$

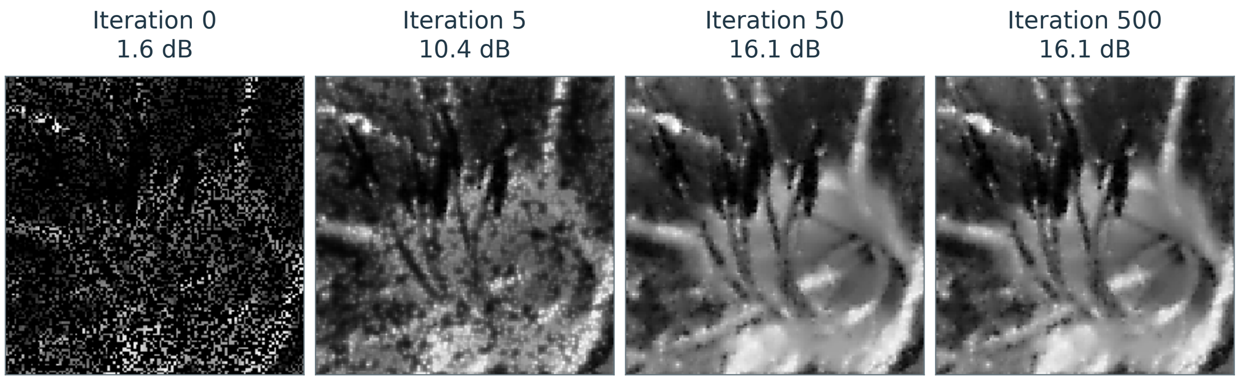


Figure 9.4. Sobolev projected gradient descent algorithm.

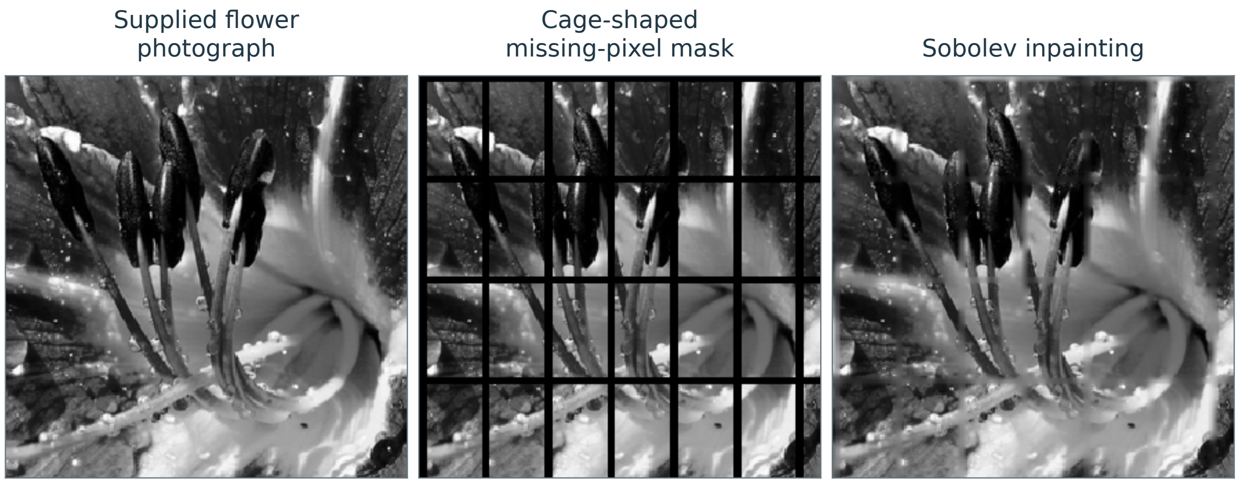


Figure 9.5. Sobolev inpainting of a grid-shaped occlusion on the flower image.

and is an orthogonal projector $\Phi^* = \Phi$.

For noiseless data, we enforce the affine constraint $\{f \in \mathbb{R}^N \ ; \ y = \Phi f\}$ using the orthogonal projector

$$\forall\, x, \quad P_y(f)(x) = \begin{cases} f(x) & \text{if} \quad x \in \Omega, \\ y(x) & \text{if} \quad x \notin \Omega. \end{cases}$$

Projected gradient descent solves (9.18) for a smooth prior. For the Sobolev energy, the algorithm iterates

$$f^{(\ell+1)} = P_y(f^{(\ell)} + \tau \Delta f^{(\ell)}).$$

The iteration converges if $0 < \tau < 1/4$, since $\|\Delta\| \leqslant 8$. Figure 9.4 shows how it progressively fills the missing region.

Figure 9.5 illustrates Sobolev inpainting on a flower image with a known grid-shaped occlusion.

For the smoothed TV prior, the gradient descent reads

$$f^{(\ell+1)} = P_y\left(f^{(\ell)} + \tau\, \mathrm{div}\left(\frac{\nabla f^{(\ell)}}{\sqrt{\varepsilon^2 + \|\nabla f^{(\ell)}\|^2}}\right)\right)$$

which converges if $0 < \tau < \varepsilon/4$.

Figure 9.6 compares Sobolev and TV inpainting on the same observed flower pixels. The displayed reconstructions illustrate their different treatment of smooth regions and edges; their SNR values are measured against the same clean image.

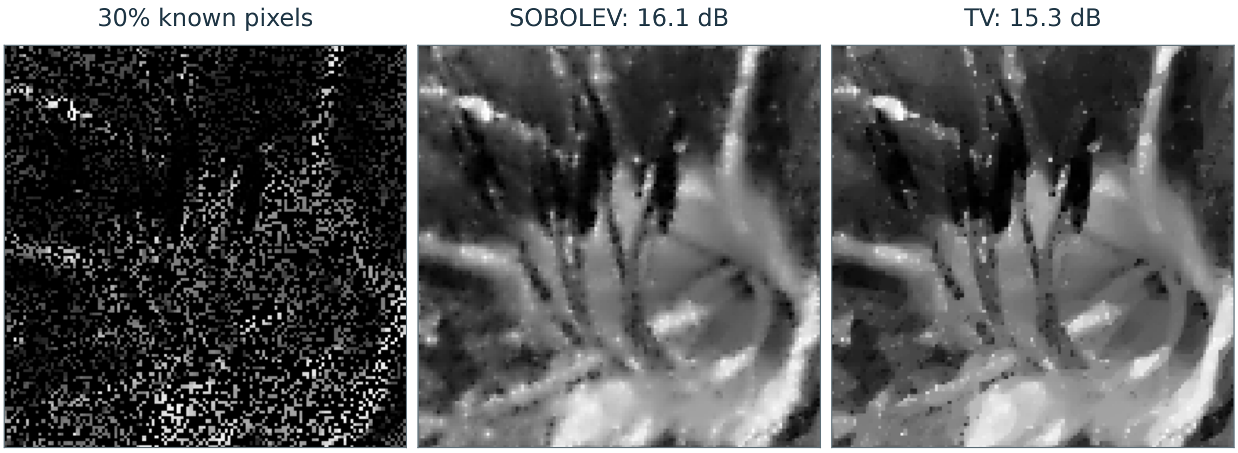


Figure 9.6. Inpainting with Sobolev and TV regularization.

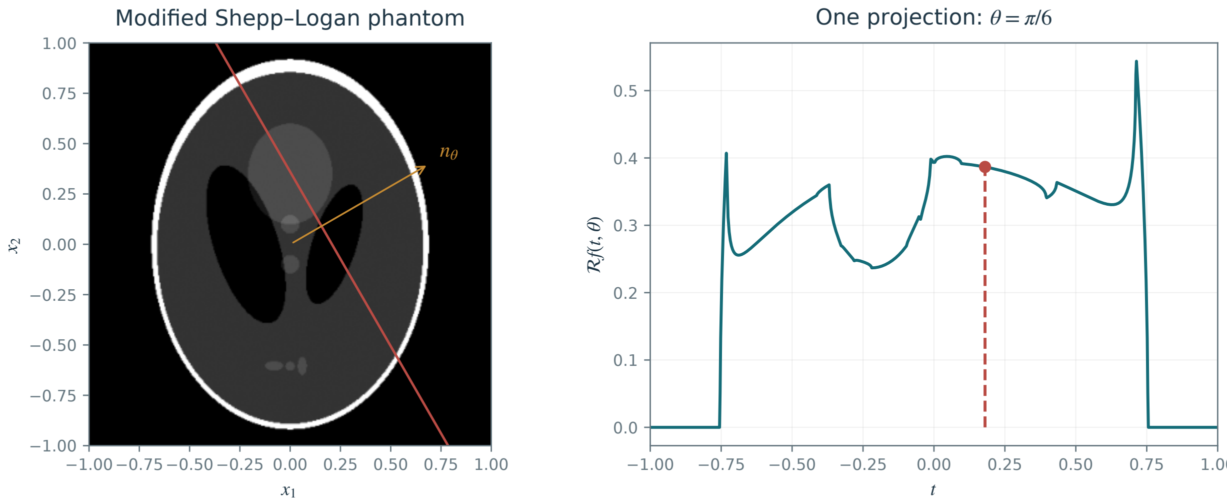


Figure 9.7. The modified Shepp–Logan phantom and one Radon projection at angle $\theta = \pi/6$. The marked line and point correspond to the same detector coordinate.

9.6.3 Tomography Inversion

In an idealized two-dimensional tomography model, each projection integrates the unknown image along a line $\Delta_{t,\theta}$ defined by

$$x \cdot \tau_\theta = x_1 \cos\theta + x_2 \sin\theta = t$$

Here $\tau_\theta = (\cos\theta, \sin\theta)$ is the unit normal to the line.

The resulting Radon transform is

$$\forall\, \theta \in [0,\pi), \forall\, t \in \mathbb{R}, \quad p_\theta(t) = \int_{\Delta_{t,\theta}} f(x)\, ds = \iint f(x)\, \delta(x \cdot \tau_\theta - t)\, dx$$

see Figure 9.7.

With $\hat{f}(\omega) = \int f(x) e^{-\mathrm{i}x\cdot\omega}\, \mathrm{d}x$, the Fourier slice theorem identifies the one-dimensional Fourier transform of each projection with a radial slice of the two-dimensional Fourier transform

$$\forall\, \theta \in [0,\pi)\,,\ \forall\, \xi \in \mathbb{R} \quad \hat{p}_\theta(\xi) = \hat{f}(\xi\cos\theta, \xi\sin\theta). \tag{9.29}$$

Changing to polar coordinates in Fourier inversion gives the filtered-backprojection formula

$$f(x) = \frac{1}{2\pi} \int_0^\pi (p_\theta \star h)(x \cdot \tau_\theta)\, d\theta$$

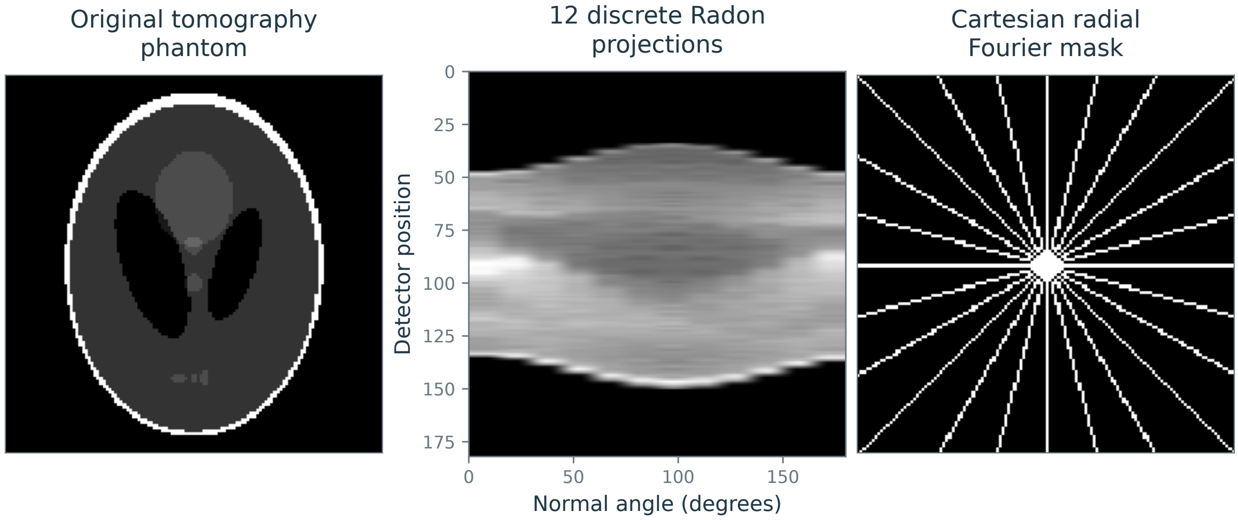


Figure 9.8. Tomography test phantom, discrete Radon projections, and a Cartesian radial Fourier-sampling mask.

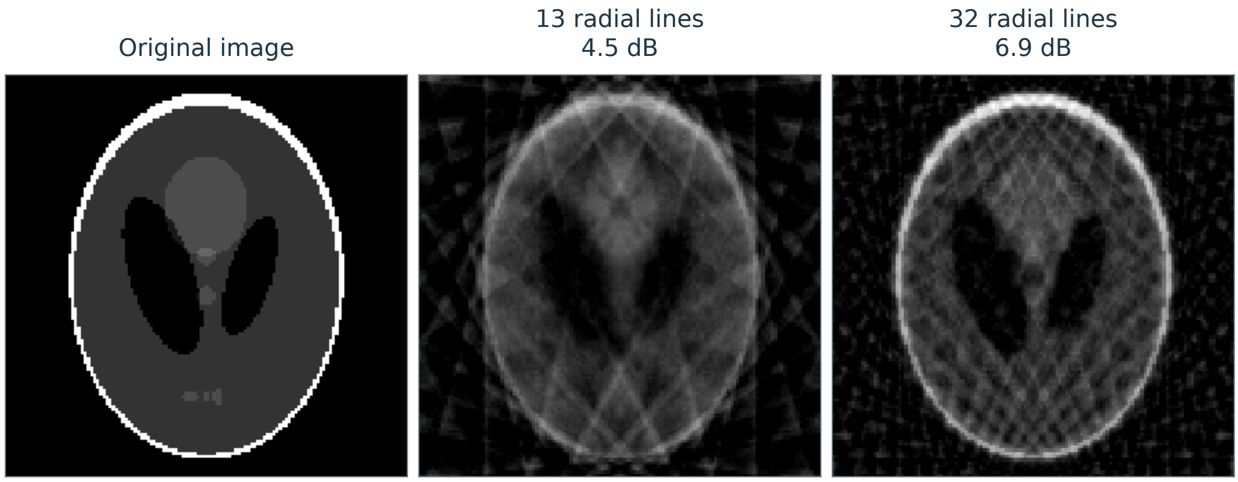


Figure 9.9. Pseudoinverse reconstruction from partial Cartesian Fourier measurements on 13 and 32 radial lines.

with $\hat{h}(\xi) = |\xi|$.

We model acquisition at finitely many equally spaced orientations $\{\theta_k = \pi k/K\}_{0\leqslant k<K}$ by the partial Radon transform

$$Rf = (p_{\theta_k})_{0\leqslant k<K}.$$

By (9.29), knowing Rf is equivalent to knowing the Fourier transform of f along the corresponding radial lines:

$$\{\hat{f}(\xi\cos(\theta_k), \xi\sin(\theta_k))\}_k.$$

For a simplified discrete model, we therefore treat acquisition as direct sampling of the Fourier transform:

$$\Phi f = (\hat{f}[\omega])_{\omega\in\Omega} \in \mathbb{C}^P$$

where Ω consists of discrete radial sampling lines in the Fourier plane; see Figure 9.8, right.

In this idealized discrete model, recovery is an inpainting problem in the Fourier domain. We use the unitary discrete Fourier transform and consider measurements

$$\forall\, \omega \in \Omega, \quad y[\omega] = \hat{f}[\omega] + w[\omega]$$

where $w[\omega]$ denotes measurement noise, modeled here as Gaussian white noise.

Figure 9.10. TV-regularized reconstruction and the pseudoinverse for the same partial Fourier measurements.

For these partial Fourier measurements, the pseudoinverse $f^+ = \Phi^+ y$ from (9.7) is given by

$$\hat{f}^+[\omega] = \begin{cases} y[\omega] & \text{if} \quad \omega \in \Omega, \\ 0 & \text{if} \quad \omega \notin \Omega. \end{cases}$$

Figure 9.9 shows examples of pseudoinverse reconstruction as the size of Ω increases. This reconstruction exhibits pronounced artifacts because missing Fourier frequencies are set to zero.

The total variation regularization (9.17) reads

$$f^\star \in \underset{f}{\operatorname{argmin}} \; \frac{1}{2} \sum_{\omega \in \Omega} |y[\omega] - \hat{f}[\omega]|^2 + \lambda \|f\|_{\mathrm{TV}}.$$

TV regularization suits images with approximately constant regions separated by sharp boundaries, as in the cartoon model of Section 5.2.4. Figure 9.10 compares TV reconstruction and the pseudoinverse on a synthetic image of this type. The example illustrates recovery of sharp features from incomplete Fourier data. Spatial inpainting can be more challenging for TV, as Figure 9.6 illustrates.

CHAPTER 10

Sparse Regularization

Many signals admit accurate approximations using only a small number of coefficients in a suitable basis or dictionary. Sparse regularization turns this observation into a reconstruction principle for noisy or incomplete measurements. We move from counting nonzero coefficients to a convex penalty, distinguish analysis and synthesis models, derive thresholding and iterative algorithms, and apply them to deconvolution and inpainting.

The main references for this chapter are [23, 31, 29].

10.1 Sparsity Priors

10.1.1 Ideal Sparsity Prior

Chapter 5 shows that an orthonormal basis $\mathcal{B} = \{\psi_m\}_m$ adapted to an image class Θ can approximate each $f \in \Theta$ using relatively few atoms.

The ℓ^0 prior measures representation complexity by counting the nonzero coefficients:

$$J_0(f) \overset{\text{def.}}{=} \#\{m \; ; \; \langle f, \psi_m\rangle \neq 0\} \quad \text{where} \quad x_m = \langle f, \psi_m\rangle.$$

This count is also called the ℓ^0 pseudonorm:

$$\|x\|_0 \overset{\text{def.}}{=} J_0(f).$$

It measures sparsity directly, without accounting for the magnitudes of the nonzero coefficients.

Natural images usually have many nonzero coefficients, but can often be approximated by a function f_M with a small count $M = J_0(f_M)$. Retaining coefficients above a threshold gives a best M-term approximation, with M determined by that threshold:

$$f_M = \sum_{|\langle f, \psi_m\rangle| > T} \langle f, \psi_m\rangle \psi_m \quad \text{where} \quad M = \#\{m \; ; \; |\langle f, \psi_m\rangle| > T\}.$$

For suitable wavelet bases and the bounded-variation image classes specified in Section 5.2, the error $\|f - f_M\|$ satisfies quantitative decay estimates. These estimates motivate approximating natural images f by functions with small J_0.

Figure 10.1 displays a natural image in the wavelet basis $\psi_m = \psi^{\omega}_{j,n}$, indexed by $m = (j, n, \omega)$. Most coefficients $\langle f, \psi_m\rangle$ have small magnitude, so retaining the few large coefficients captures much of the image.

10.1.2 Convex Relaxation

The ideal sparsity prior J_0 is nonconvex, which makes optimization difficult. For example, if f and g have disjoint nonempty coefficient supports in $\mathcal{B}$, then $J_0((f+g)/2) = J_0(f) + J_0(g)$, violating convexity of J_0.

In a general inverse problem, minimizing J_0 entails a combinatorial search over possible coefficient supports.

To approximate the ideal prior J_0, consider the ℓ^q family with $q > 0$:

$$J_q(f) = \sum_m |\langle f, \psi_m\rangle|^q.$$

As shown in Figure 10.2, the unit balls in $\mathbb{R}^2$ become concentrated near the coordinate axes as $q \downarrow 0$. For each fixed finite-dimensional vector, $J_q(f) \to J_0(f)$; the limiting shape of these bounded unit balls should not be confused with the unbounded set $\{x : \|x\|_0 \leqslant 1\}$. Small values of q favor sparsity.

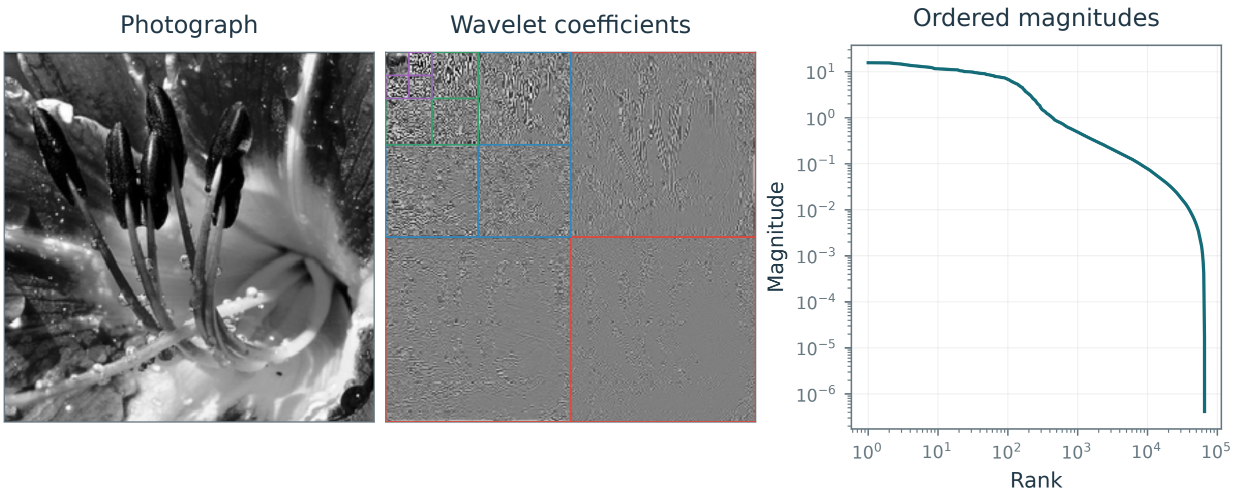


Figure 10.1. Most wavelet coefficients of a natural image have small magnitude.

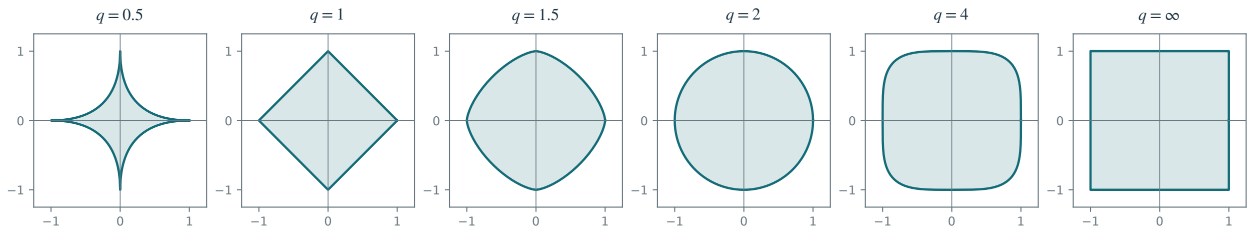


Figure 10.2. ℓ^q balls $\{x \; ; \; J_q(x) \leqslant 1\}$ for varying q.

The prior J_q is convex exactly when $q \geqslant 1$. The smallest exponent yielding a convex prior, $q = 1$, gives the ℓ^1 prior J_1:

$$J_1(f) = \|(\langle f, \psi_m \rangle)\|_1 = \sum_m |\langle f, \psi_m \rangle|. \tag{10.1}$$

In the following, we consider discrete orthonormal bases $\mathcal{B} = \{\psi_m\}_{m=0}^{N-1}$ of $\mathbb{R}^N$.

10.1.3 Sparse Regularization and Thresholding

Given an orthonormal basis $\{\psi_m\}_m$ of $\mathbb{R}^N$, regularization-based denoising (8.15) can be written using the sparsity priors J_0 and J_1 as

$$f^\star \in \operatorname*{argmin}_{g \in \mathbb{R}^N} \frac{1}{2}\|f - g\|^2 + \lambda J_q(g)$$

for $q = 0$ or $q = 1$, writing $J_0 = J_0$. Orthonormality separates the objective into coefficientwise terms:

$$f^\star = \sum_m x_m^\star \psi_m$$

$$\text{where} \quad x^\star \in \operatorname*{argmin}_{y \in \mathbb{R}^N} \sum_m \frac{1}{2}|x_m - y_m|^2 + \lambda |y_m|^q$$

Here $x_m \overset{\text{def.}}{=} \langle f, \psi_m \rangle$ and $y_m \overset{\text{def.}}{=} \langle g, \psi_m \rangle$. For $q = 0$, we adopt the convention

$$\forall\, u \in \mathbb{R}, \quad |u|^0 = \begin{cases} 0 & \text{if} \quad u = 0, \\ 1 & \text{otherwise.} \end{cases}$$

Each coefficient of the denoised image solves a one-dimensional optimization problem

$$x_m^\star \in \operatorname*{argmin}_{u \in \mathbb{R}} \frac{1}{2}|x_m - u|^2 + \lambda |u|^q \tag{10.2}$$

The following proposition gives a closed-form solution by thresholding.

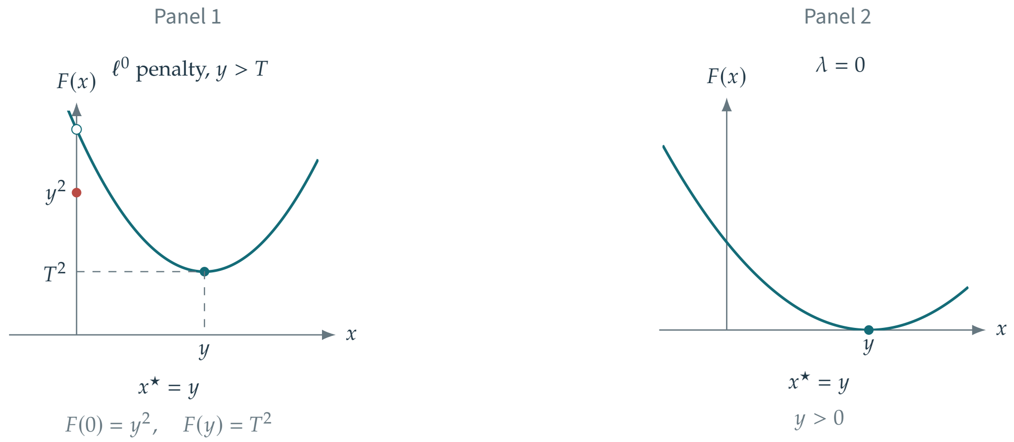


Figure 10.3. Panel 1: the objective $\|\cdot - y\|^2 + T^2\|\cdot\|_0$. Panels 2–5: the scalar objective $F(x) \stackrel{\text{def.}}{=} \frac{1}{2}|x-y|^2 + \lambda|x|$ for increasing λ.

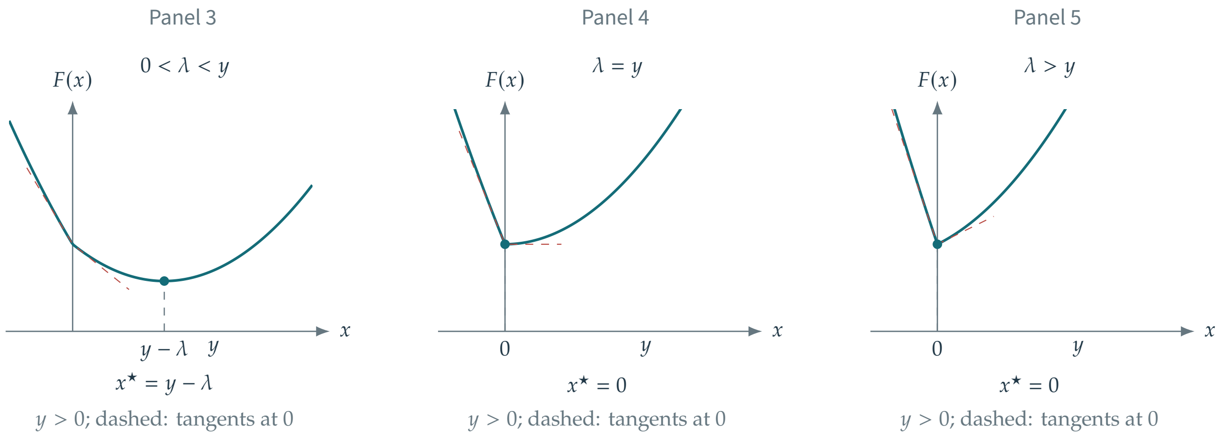


Figure 10.3 (continued). Panel 1: the objective $\|\cdot - y\|^2 + T^2\|\cdot\|_0$. Panels 2–5: the scalar objective $F(x) \stackrel{\text{def.}}{=} \frac{1}{2}|x-y|^2 + \lambda|x|$ for increasing λ.

Proposition 10.1. *One may choose the minimizer*

$$x_m^\star = S_T^q(x_m) \quad \text{where} \quad \begin{cases} T = \sqrt{2\lambda} & \text{for} \quad q = 0, \\ T = \lambda & \text{for} \quad q = 1, \end{cases} \tag{10.3}$$

where

$$\forall\, u \in \mathbb{R}, \quad S_T^0(u) \stackrel{\text{def.}}{=} \begin{cases} 0 & \text{if} \quad |u| < T, \\ u & \text{otherwise} \end{cases} \tag{10.4}$$

is hard thresholding, with the alternative tie convention discussed below, and

$$\forall\, u \in \mathbb{R}, \quad S_T^1(u) \stackrel{\text{def.}}{=} \operatorname{sign}(u)(|u| - T)_+ \tag{10.5}$$

is the soft thresholding introduced in (7.9). For hard thresholding at $|u| = T$, both 0 and u minimize the objective; the displayed rule selects u. Soft thresholding gives the unique minimizer.

Proof. For $q = 0$, a zero input is minimized at zero. For a nonzero input, the best nonzero candidate is $u = x_m$, with cost λ, whereas $u = 0$ has cost $|x_m|^2/2$. Comparing the two costs gives the threshold $\sqrt{2\lambda}$ and the stated tie. For $q = 1$, the optimality condition is $0 \in u - x_m + \lambda\partial|u|$. If $u > 0$, this gives $u = x_m - \lambda$; if $u < 0$, it gives $u = x_m + \lambda$; and $u = 0$ is optimal exactly when $|x_m| \leqslant \lambda$. These cases give soft thresholding. $\square$

Transforming the thresholded coefficients back to the image domain gives

$$f_{\lambda,q} = \sum_m S_T^q(\langle f, \psi_m\rangle)\psi_m.$$

For Gaussian white noise w of variance σ^2, thresholds can be chosen from the noise level; see Section 7.3. The universal threshold $T = \sigma\sqrt{2\log N}$ controls the largest noise coefficients and gives asymptotic risk guarantees under the signal assumptions of Section 7.3.3. Common empirical choices are $T \approx 3\sigma$ for hard thresholding (ℓ^0 regularization) and $T \approx 3\sigma/2$ for soft thresholding (ℓ^1 regularization); see Figure 7.13.

10.2 Sparse Regularization of Inverse Problems

Using the ℓ^1 prior in an orthonormal basis $\{\psi_m\}_m$ of $\mathbb{R}^N$, with J_1 defined in (10.1), gives the convex inverse problem

$$f_\lambda \in \underset{f\in\mathbb{R}^N}{\operatorname{argmin}} \frac{1}{2}\|y - \Phi f\|^2 + \lambda \sum_m |\langle f,\, \psi_m\rangle|. \tag{10.6}$$

Chen, Donoho, and Saunders introduced this basis pursuit denoising formulation in [12]. Section 10.3 derives an iterative thresholding algorithm for (10.6).

Analysis vs. synthesis priors. For a nonorthogonal dictionary $\Psi = \{\psi_m\}_{m=1}^{Q}$, which is redundant if $Q > N$, there are two distinct extensions of (10.6). Each uses a convex prior J in

$$f_\lambda \in \underset{f\in\mathbb{R}^N}{\operatorname{argmin}} \frac{1}{2}\|y - \Phi f\|^2 + \lambda J(f). \tag{10.7}$$

We also use the dictionary symbol for its synthesis operator; its adjoint is the analysis operator:

$$\Psi : x \in \mathbb{R}^Q \mapsto \Psi x = \sum_m x_m \psi_m \quad \text{and} \quad \Psi^* : f \in \mathbb{R}^N \mapsto (\langle f,\, \psi_m\rangle)_{m=1}^{Q} \in \mathbb{R}^Q.$$

The analysis prior penalizes the sum of the magnitudes of correlations with the dictionary atoms:

$$J_1^{\mathrm{A}}(f) \overset{\text{def.}}{=} \sum_m |\langle f,\, \psi_m\rangle| = \|\Psi^* f\|_1. \tag{10.8}$$

The synthesis prior minimizes the coefficient norm over expansions of f in Ψ:

$$J_1^{\mathrm{S}}(f) \overset{\text{def.}}{=} \min_{x\in\mathbb{R}^Q,\Psi x = f} \|x\|_1. \tag{10.9}$$

The two priors agree for an orthonormal basis, but generally differ for redundant dictionaries. Analysis regularization often requires primal–dual algorithms, as detailed in Chapter 19. Its recovery theory depends on the dictionary and on the structure of the vanishing analysis coefficients. The synthesis formulation leads directly to the coefficient-space algorithms studied below. We assume the dictionary spans $\mathbb{R}^N$; otherwise the synthesis prior is $+\infty$ outside its span.

We now focus on synthesis regularization $J = J_1^{\mathrm{S}}$, and rewrite (10.7) as $f_\lambda = \Psi x_\lambda$ where x_λ is any solution of the following basis pursuit denoising problem

$$x_\lambda \in \underset{x\in\mathbb{R}^Q}{\operatorname{argmin}} \frac{1}{2\lambda}\|y - Ax\|^2 + \|x\|_1 \tag{10.10}$$

where the measurement matrix in coefficient space is

$$A \overset{\text{def.}}{=} \Phi\Psi \in \mathbb{R}^{P\times Q}.$$

For consistent data $y \in \operatorname{Im}(A)$, the limit $\lambda \downarrow 0$ leads to the constrained problem

$$x^\star \in \underset{Ax=y}{\operatorname{argmin}} \|x\|_1 \tag{10.11}$$

and the signal is recovered as $f^\star = \Psi x^\star \in \mathbb{R}^N$.

10.3 Iterative Soft Thresholding Algorithm

We now derive iterative methods for (10.10), beginning with a comparison to its constrained counterpart.

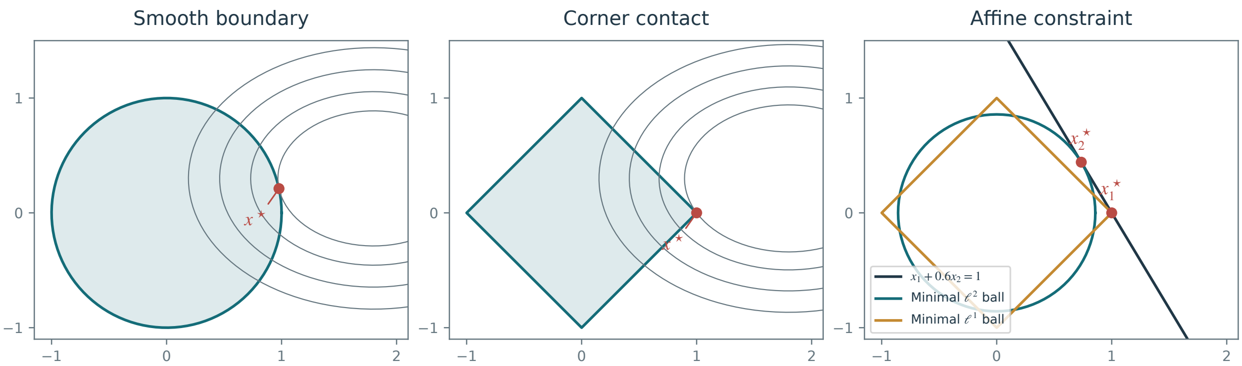


Figure 10.4. Geometry of convex optimization: smooth-boundary and corner contact with objective level sets, followed by the smallest scaled ℓ^2 and ℓ^1 balls touching an affine constraint. Red dots mark the contact points.

10.3.1 Noiseless Recovery as a Linear Program

Before treating $\lambda > 0$, note that (10.11) can be written as a linear program. Split $x = x_+ - x_-$ with $(x_+, x_-) \in (\mathbb{R}_+^Q)^2$ and solve

$$(x_+^\star, x_-^\star) \in \underset{(x_+,x_-)\in(\mathbb{R}_+^Q)^2}{\operatorname{argmin}} \left\{ \langle x_+, \mathbb{1}_Q \rangle + \langle x_-, \mathbb{1}_Q \rangle \,;\, y = A(x_+ - x_-) \right\}. \tag{10.12}$$

Then recover $x^\star = x_+^\star - x_-^\star$. For small or moderate Q, the linear program can be solved by simplex or interior-point methods. For large imaging and learning problems, first-order methods are often preferable. One option is Douglas–Rachford splitting, developed in Section 19.5.2. The penalized problem (10.10) admits a particularly simple splitting algorithm because its fidelity term is smooth. The relative computational cost of the two formulations depends on the solver and on A.

10.3.2 Projected Gradient Descent for ℓ^1

We first solve (10.10) by projected gradient descent, which is analyzed in detail in Section 19.1.3. As in (10.12), we rewrite (10.10) as a constrained minimization problem of the form (19.4), with the nonnegative constraint set C and

$$u = (u_+, u_-) \in (\mathbb{R}^Q)^2, \quad C = (\mathbb{R}_+^Q)^2, \quad \text{and} \quad \mathcal{E}(u) = \frac{1}{2}\|A(u_+ - u_-) - y\|^2 + \lambda\langle u_+, \mathbb{1}_Q\rangle + \lambda\langle u_-, \mathbb{1}_Q\rangle.$$

Projection onto C is easy to compute

$$\mathrm{Proj}_{(\mathbb{R}_+^Q)^2}(u_+, u_-) = ((u_+)_\oplus, (u_-)_\oplus) \quad \text{where} \quad (r)_\oplus \overset{\text{def.}}{=} \max(r, 0),$$

and the gradient reads

$$\nabla\mathcal{E}(u_+, u_-) = (\eta + \lambda\mathbb{1}_Q, -\eta + \lambda\mathbb{1}_Q) \quad \text{where} \quad \eta = A^*(A(u_+ - u_-) - y)$$

Denoting $u^{(\ell)} = (u_+^{(\ell)}, u_-^{(\ell)})$ and $x^{(\ell)} \overset{\text{def.}}{=} u_+^{(\ell)} - u_-^{(\ell)}$, the iterates of projected gradient descent (19.5) are

$$u_+^{(\ell+1)} \overset{\text{def.}}{=} \left(u_+^{(\ell)} - \tau_\ell(\eta^{(\ell)} + \lambda)\right)_\oplus \quad \text{and} \quad u_-^{(\ell+1)} \overset{\text{def.}}{=} \left(u_-^{(\ell)} - \tau_\ell(-\eta^{(\ell)} + \lambda)\right)_\oplus$$

where $\eta^{(\ell)} \overset{\text{def.}}{=} A^*(Ax^{(\ell)} - y)$, and adding the scalar λ means adding $\lambda\mathbb{1}_Q$ componentwise.

Theorem 19.2 ensures that $u^{(\ell)} \to u^\star$, a solution of (19.4), if

$$\forall\, \ell, \quad 0 < \tau_{\min} < \tau_\ell < \tau_{\max} < \frac{1}{\|A\|^2},$$

The smooth objective in (u_+, u_-) has gradient Lipschitz constant $2\|A\|^2$, which explains the step-size bound. Consequently, $x^{(\ell)} \to x^\star = u_+^\star - u_-^\star$, a solution of (10.10).

10.3.3 Iterative Soft Thresholding and Forward Backward

Splitting positive and negative parts requires storing $2Q$ coefficients. The iterative soft thresholding algorithm (ISTA) avoids this duplication while retaining comparable convergence guarantees. For a fixed step $0 < \tau < 2/\|A\|^2$, ISTA converges to a minimizer. The surrogate derivation below uses the stricter bound $\tau \leqslant 1/\|A\|^2$, which gives a direct proof of energy decrease.

ISTA was derived by several authors [20, 16]. It is a special case of forward–backward proximal splitting [14], studied in Section 19.4.2.

We derive ISTA for the ℓ^1 penalty by minimizing a sequence of surrogate objectives. Rescale the objective in (10.10) as

$$\mathcal{E}(x) \overset{\text{def.}}{=} \frac{1}{2}\|y - Ax\|^2 + \lambda\|x\|_1$$

For a fixed reference point x', define

$$\mathcal{E}_\tau(x, x') \overset{\text{def.}}{=} \mathcal{E}(x) - \frac{1}{2}\|Ax - Ax'\|^2 + \frac{1}{2\tau}\|x - x'\|^2.$$

We have $\mathcal{E}_\tau(x, x) = \mathcal{E}(x)$, and the difference between the two objectives is

$$K(x, x') \overset{\text{def.}}{=} -\frac{1}{2}\|Ax - Ax'\|^2 + \frac{1}{2\tau}\|x - x'\|^2 = \frac{1}{2}\langle\left(\frac{1}{\tau}\mathrm{Id}_Q - A^*A\right)(x - x'),\ x - x'\rangle.$$

The correction $K(x, x')$ is nonnegative when $\lambda_{\max}(A^*A) \leqslant 1/\tau$, where the left side is the largest eigenvalue. Equivalently, $\tau \leqslant 1/\|A\|^2_{\mathrm{op}}$, with $\|A\|_{\mathrm{op}} = \sigma_{\max}(A)$ denoting the operator norm. Under this condition, $\mathcal{E}_\tau(x, x')$ is a surrogate that bounds the objective from above and agrees with it at the reference point:

$$\mathcal{E}(x) \leqslant \mathcal{E}_\tau(x, x'), \quad \mathcal{E}_\tau(x', x') = \mathcal{E}(x'), \quad \text{and} \quad \mathcal{E}(\cdot) - \mathcal{E}_\tau(\cdot, x') \text{ is smooth.}$$

Minimizing this surrogate at each iteration gives

$$x^{(\ell+1)} \overset{\text{def.}}{=} \underset{x}{\operatorname{argmin}}\ \mathcal{E}_{\tau_\ell}(x, x^{(\ell)}) \tag{10.13}$$

The surrogate upper bound and equality at the current iterate imply

$$\mathcal{E}(x^{(\ell+1)}) \leqslant \mathcal{E}(x^{(\ell)}).$$

Proposition 10.2. *The iterates $x^{(\ell)}$ defined by (10.13) satisfy*

$$x^{(\ell+1)} = \mathcal{S}^1_{\lambda\tau_\ell}\left(x^{(\ell)} - \tau_\ell A^*(Ax^{(\ell)} - y)\right) \tag{10.14}$$

where $\mathcal{S}^1_\lambda(x) = (S^1_\lambda(x_m))_m$ and $S^1_\lambda(r) = \mathrm{sign}(r)(|r| - \lambda)_\oplus$ is the soft thresholding operator defined in (10.5).

Proof. Expanding the quadratic terms and collecting constants independent of the optimization variable gives

$$\begin{aligned}
\mathcal{E}_\tau(x, x') &= \frac{1}{2}\|Ax - y\|^2 - \frac{1}{2}\|Ax - Ax'\|^2 + \frac{1}{2\tau}\|x - x'\|^2 + \lambda\|x\|_1 \\
&= C + \frac{1}{2}\|Ax\|^2 - \frac{1}{2}\|Ax\|^2 + \frac{1}{2\tau}\|x\|^2 - \langle Ax,\ y\rangle + \langle Ax,\ Ax'\rangle - \frac{1}{\tau}\langle x,\ x'\rangle + \lambda\|x\|_1 \\
&= C + \frac{1}{2\tau}\|x\|^2 + \langle x,\ -A^*y + A^*Ax' - \frac{1}{\tau}x'\rangle + \lambda\|x\|_1 \\
&= C' + \frac{1}{\tau}\left(\frac{1}{2}\|x - (x' - \tau A^*(Ax' - y))\|^2 + \tau\lambda\|x\|_1\right)
\end{aligned}$$

Proposition 10.1 therefore identifies the minimizer of $\mathcal{E}_\tau(x, x')$ as $\mathcal{S}^1_{\lambda\tau}(x' - \tau A^*(Ax' - y))$. □

The iterations (10.14) coincide with the forward–backward iterates (19.20). For the coefficient-space objective (10.10), rescaled by λ, use the splitting

$$\mathcal{F} = \frac{1}{2}\|A \cdot - y\|^2 \quad \text{and} \quad \mathcal{G} = \lambda\|\cdot\|_1. \tag{10.15}$$

In the case (10.15), Proposition 10.1 shows that $\mathrm{Prox}_{\rho\mathcal{G}}$ is soft thresholding at level $\rho\lambda$.

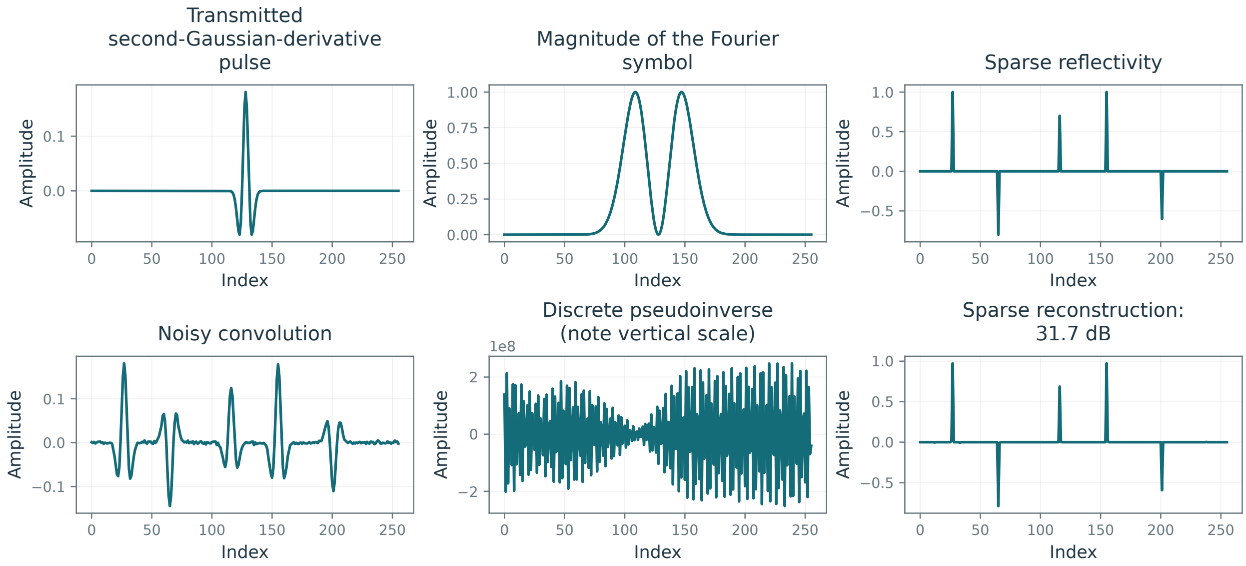


Figure 10.5. Sparse spike recovery by the pseudoinverse and by ℓ^1 regularization.

10.4 Example: Sparse Deconvolution

10.4.1 Sparse Spikes Deconvolution

Sparse spike deconvolution uses sparsity in the spatial domain, which corresponds to the orthonormal basis of Diracs $\psi_m[n] = \delta[n-m]$. In seismic imaging, a simple sparse reflectivity model represents f_0 as a few impulses associated with changes in acoustic impedance.

In a linearized one-dimensional model that neglects multiple reflections, the subsurface reflectivity f_0 produces observations $y = h \star f_0 + w$. Here h is the transmitted pulse, called a seismic wavelet. This use of the word differs from the orthogonal wavelet bases constructed in Chapter 4, although the terminology originated in seismic imaging.

The seismic wavelet h is typically band-pass, balancing spatial and frequency concentration. Figure 10.5 shows a second derivative of a Gaussian and its Fourier transform. The narrow transmitted band illustrates how acquisition suppresses information about f_0.

Using ℓ^1 regularization in the Dirac basis gives

$$f^\star = \operatorname*{argmin}_{f \in \mathbb{R}^N} \frac{1}{2}\|f \star h - y\|^2 + \lambda \sum_m |f_m|.$$

Figure 10.5 shows the result of ℓ^1 minimization with an oracle choice of λ that minimizes the error $\|f^\star - f_0\|$.

ISTA for sparse spike recovery alternates the updates

$$\tilde{f}^{(k)} = f^{(k)} - \tau h^\vee \star (h \star f^{(k)} - y)$$

and

$$f_m^{(k+1)} = S^1_{\lambda\tau}(\tilde{f}_m^{(k)})$$

where $h^\vee[n] = \overline{h[-n]}$ is the adjoint convolution kernel. It equals h for a real symmetric filter. The step size must satisfy

$$0 < \tau < 2/\|\Phi^*\Phi\| = 2/\max_\omega |\hat{h}(\omega)|^2$$

to guarantee convergence. Figure 10.6 shows convergence of both the objective values and the iterates. Objective convergence alone would not ensure convergence of the iterates when minimizers are nonunique; the forward–backward convergence theorem provides the latter guarantee here.

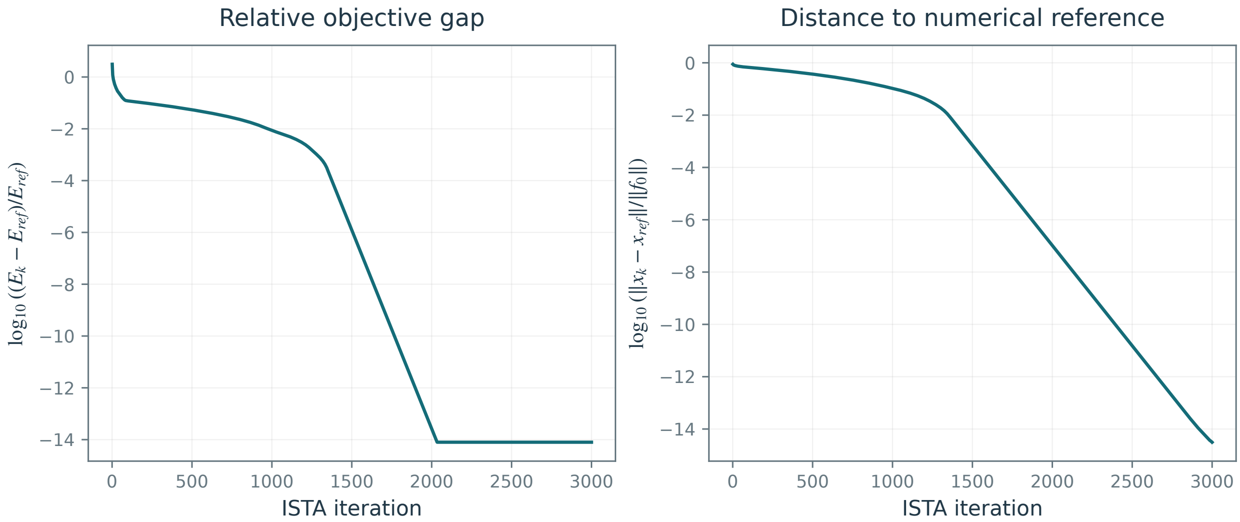


Figure 10.6. Convergence of the energy and iterates under iterative soft thresholding.

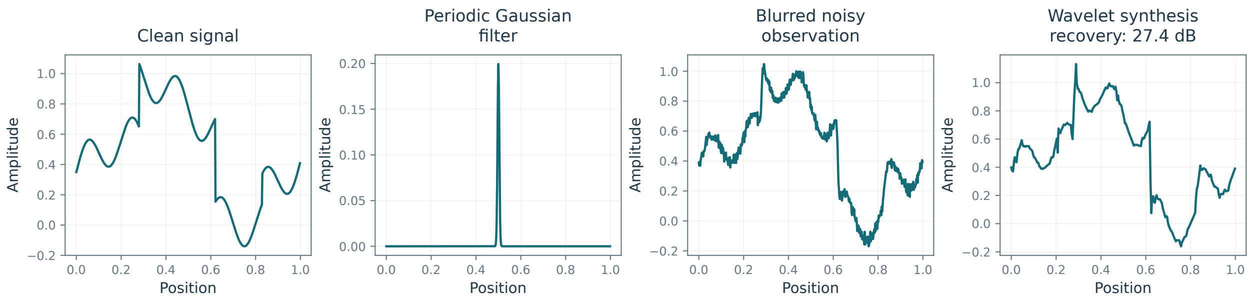


Figure 10.7. One-dimensional deconvolution with sparsity in an orthogonal wavelet basis.

10.4.2 Sparse Wavelets Deconvolution

Camera images can be blurred by defocus, motion during exposure, or diffraction. Assuming spatial invariance, we model the blur operator Φ by convolution:

$$y = f_0 \star h + w.$$

We use a Gaussian filter h of width $\mu > 0$. On a fixed d-dimensional domain, the number of effectively transmitted frequencies scales roughly as μ^{-d}, with a threshold-dependent factor, because higher frequencies are strongly attenuated.

Figures 10.7 and 10.8 show examples of signal and image acquisition with Gaussian blur.

Sobolev regularization (9.23) suppresses high-frequency noise more strongly than the squared-norm penalty (9.10). Both can blur sharp transitions. TV regularization and sparsity in a wavelet basis are better adapted to signals with edges.

Figure 10.7 compares wavelet and Sobolev regularization for a one-dimensional signal; Figure 10.8 shows the corresponding image-deblurring results. Replacing an orthogonal wavelet basis in (10.14) by a translation-invariant tight frame (7.12) can reduce artifacts. For a redundant frame, synthesis ISTA acts in coefficient space with $A = \Phi\Psi$. Simply analyzing, thresholding, and synthesizing is generally not the proximal map of the analysis penalty.

Figure 10.9 plots the SNR against λ. Computing the SNR requires the clean reference image f_0; the reported experiments use the maximizing value of λ.

10.4.3 Sparse Inpainting

This section continues the discussion in Section 9.6.2.

For noiseless inpainting, a small $\lambda > 0$ approximates constrained ℓ^1 minimization. Exact interpolation is obtained in the limit under the assumptions of Proposition 9.5. For an orthonormal basis, the iterative

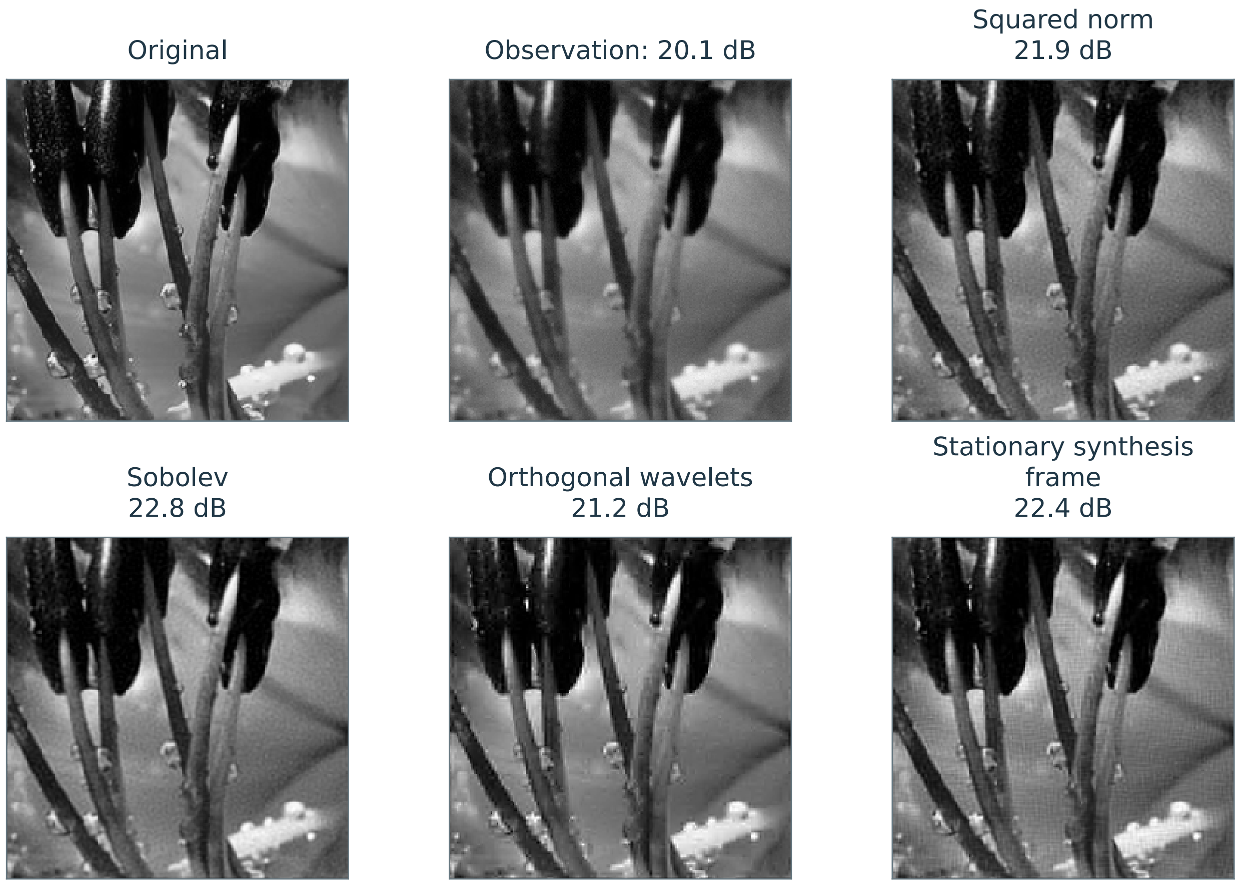


Figure 10.8. Image deconvolution.

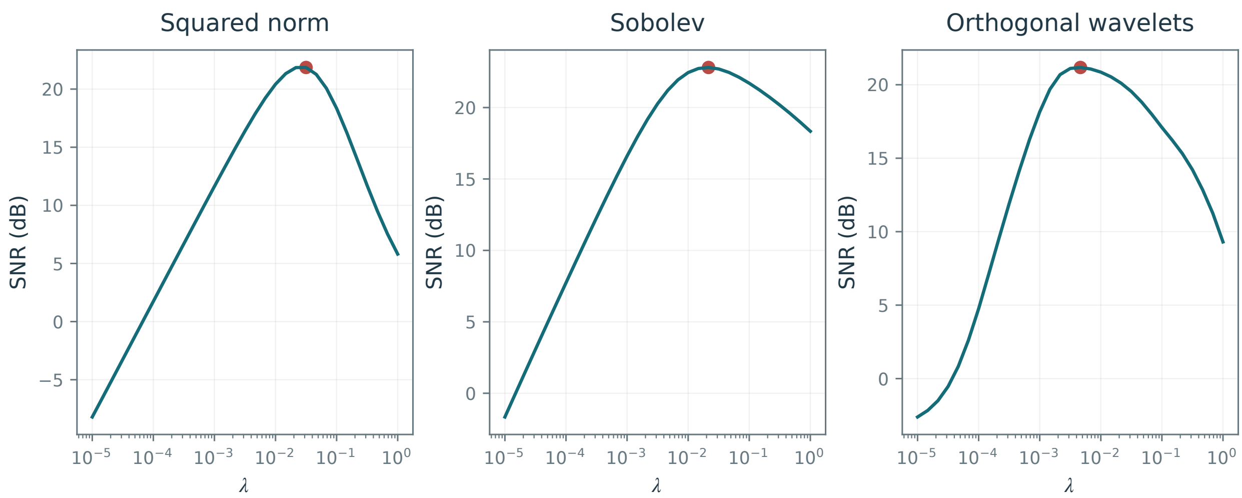


Figure 10.9. Reconstruction SNR as a function of the regularization parameter λ, swept logarithmically from 10^{-5} to 1. Red dots mark the measured maxima.

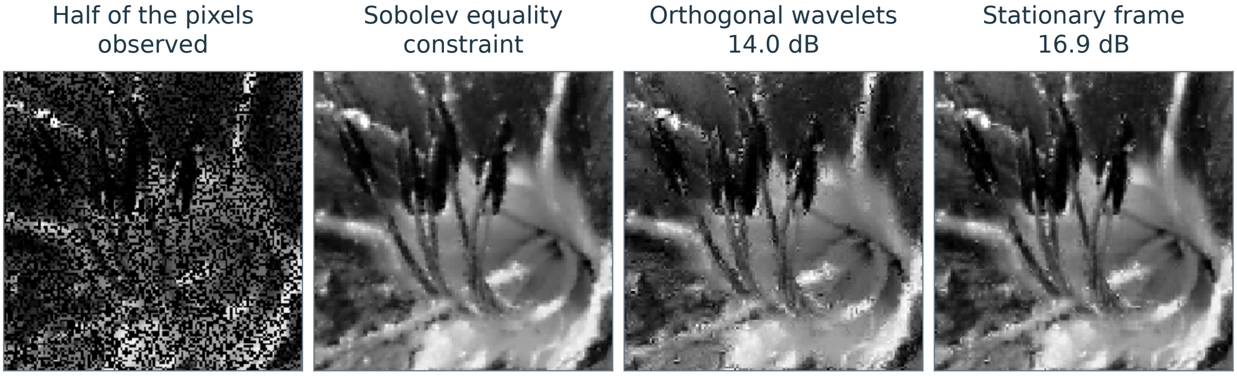


Figure 10.10. Inpainting with Sobolev and wavelet sparsity priors.

thresholding algorithm (10.14) takes the following form for $\tau = 1$:

$$f^{(k+1)} = \sum_m S^1_\lambda(\langle P_y(f^{(k)}),\, \psi_m\rangle)\psi_m$$

Figure 10.10 compares Sobolev inpainting with wavelet sparsity priors, including a translation-invariant frame.

CHAPTER 11

Theory of Sparse Regularization

The theory of sparse recovery asks when a reconstruction is unique, how much noise changes its coefficients, and whether it identifies the correct nonzero locations. A small reconstruction error alone does not guarantee that these locations are recovered. We use convex geometry and dual certificates to establish recovery conditions for the Lasso and its constrained counterpart, then examine their implications for spike deconvolution on increasingly fine grids.

11.1 Existence and Uniqueness

11.1.1 Existence

We study the penalized problem (10.10), with $\lambda > 0$, and its constrained counterpart (10.11):

$$\min_{x \in \mathbb{R}^N} f_\lambda(x) \overset{\text{def.}}{=} \frac{1}{2\lambda}\|y - Ax\|^2 + \|x\|_1 \qquad (\mathcal{P}_\lambda(y))$$

and the limiting problem as $\lambda \to 0$,

$$\min_{Ax=y} \|x\|_1 = \min_x f_0(x) \overset{\text{def.}}{=} \iota_{\mathcal{L}_y}(x) + \|x\|_1. \qquad (\mathcal{P}_0(y))$$

where $A \in \mathbb{R}^{P \times N}$, and $\mathcal{L}_y \overset{\text{def.}}{=} \left\{x \in \mathbb{R}^N \; ; \; Ax = y\right\}$.

Recall that we observe noisy measurements

$$y = Ax_0 + w$$

We seek conditions ensuring that x_0 solves $\mathcal{P}_0(Ax_0)$ in the noiseless case, and quantitative bounds on its distance to solutions of $\mathcal{P}_\lambda(Ax_0 + w)$ when λ is chosen according to the noise level.

The objective in ($\mathcal{P}_\lambda(y)$) is continuous and coercive because $\|\cdot\|_1$ is coercive, so a minimizer exists. Uniqueness is not automatic: A may have a nontrivial kernel, and $\|\cdot\|_1$ is not strictly convex. If $y \in \mathrm{Im}(A)$, the constraint set in ($\mathcal{P}_0(y)$) is nonempty. A constrained minimizer also exists but need not be unique.

Figure 11.1 shows how the recovered coefficients change with λ, illustrating the effect of regularization on sparsity and reconstruction accuracy.

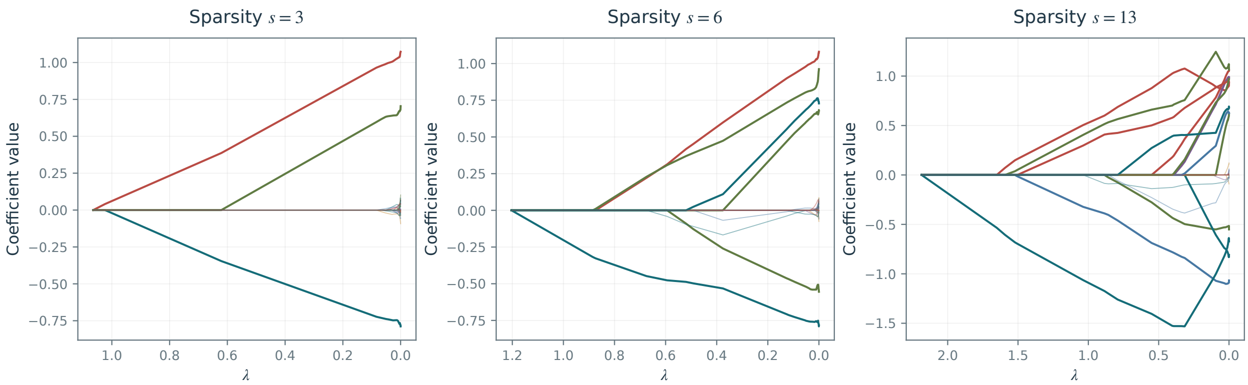


Figure 11.1. Lasso solution paths $\lambda \mapsto x_\lambda$ for sparsities $s = 3, 6, 13$, with the same measurement matrix and noise realization.

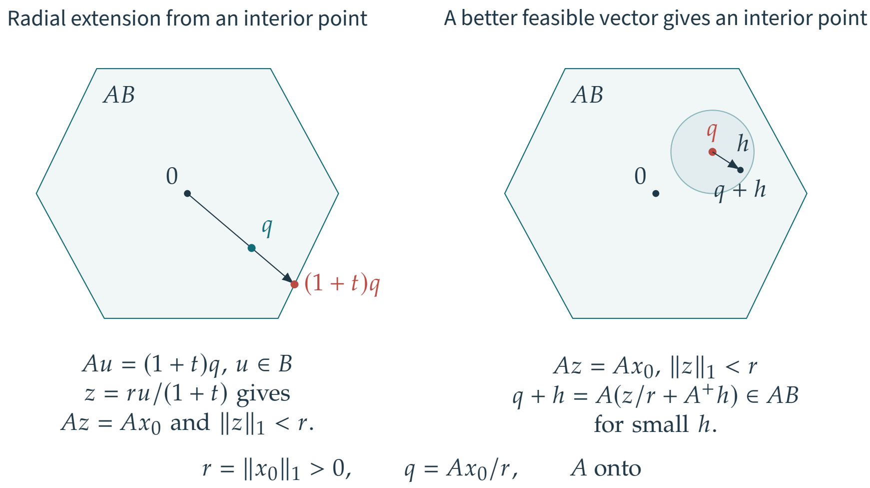


Figure 11.2. Geometry of the polytope characterization of exact recovery.

11.1.2 Polytope Projection for the Constrained Problem

The next proposition characterizes noiseless ℓ^1 recovery geometrically.

Proposition 11.1. *We denote* $B \overset{\text{def.}}{=} \left\{x \in \mathbb{R}^N \;;\; \|x\|_1 \leqslant 1\right\}$. *For* $x_0 \neq 0$, *assuming* $\operatorname{Im}(A) = \mathbb{R}^P$,

$$x_0 \text{ is a solution to } \mathcal{P}_0(Ax_0) \quad \Longleftrightarrow \quad A\frac{x_0}{\|x_0\|_1} \in \partial(AB) \tag{11.1}$$

where "∂" denotes the boundary and $AB = \{Ax \;;\; x \in B\}$.

Proof. Set $r = \|x_0\|_1 > 0$ and $q = Ax_0/r$. If x_0 is not optimal, there is z with $Az = Ax_0$ and $\|z\|_1 < r$. Since A is onto, A^+ is a bounded right inverse. For sufficiently small $h \in \mathbb{R}^P$,

$$\|z/r + A^+h\|_1 \leqslant \|z\|_1/r + \|A^+h\|_1 < 1,$$

and $q + h = A(z/r + A^+h) \in AB$. Thus q is an interior point of AB.

Conversely, if $q \neq 0$ is interior, then $(1+t)q \in AB$ for some $t > 0$. Choose $u \in B$ with $Au = (1+t)q$. The vector $z = ru/(1+t)$ satisfies $Az = Ax_0$ and $\|z\|_1 < r$, so x_0 is not optimal. If $q = 0$, the zero vector is already a better feasible point. These two implications prove the equivalence. □

For a nonzero recovered vector x_0, its measurement Ax_0 lies on the boundary of the scaled polytope $\|x_0\|_1 AB$. When P is much smaller than N, projection can remove faces and prevent ℓ^1 recovery. Chapter 12 studies random projections that preserve enough of the geometry of the ℓ^1 ball in $\mathbb{R}^N$ to allow recovery from $\mathbb{R}^P$.

Identifiability is unchanged by positive scaling: if x_0 is identifiable, so is ρx_0 for every $\rho > 0$. More generally, the relevant face of B, and hence condition (11.1), depends only on $\operatorname{sign}(x_0)$. For this geometric discussion, assume uniqueness and let $\mathcal{A} : y \mapsto x^\star$ map each datum y to the solution of ($\mathcal{P}_0(y)$). By (11.1), A and $\mathcal{A}$ are inverse maps between the cones $C_s = \left\{x \;;\; \operatorname{sign}(x) = s\right\}$ and AC_s for recoverable sign patterns s. Including the lower-dimensional cones and the origin, their images AC_s partition $\mathbb{R}^P$. Suppose that the columns $(a_j)_j$ of A have unit norm. For $P = 3$, intersecting the cones AC_s with the unit sphere in $\mathbb{R}^3$ gives a spherical Delaunay subdivision, which is a triangulation under a general-position assumption; see Figure 11.7. In higher dimensions, simplices replace triangles. The subdivision satisfies the empty-cap property: the spherical cap bounded by a triangle's circumcircle contains no signed column $\pm a_j$ in its interior. Figure 11.3 illustrates these conclusions in $\mathbb{R}^2$ and $\mathbb{R}^3$.

11.1.3 Optimality Conditions

For an index set $I \subset \{1, \ldots, N\}$, write $A = (a_i)_{i=1}^N$ for the columns of A and $A_I \overset{\text{def.}}{=} (a_i)_{i \in I} \in \mathbb{R}^{P \times |I|}$ for the corresponding submatrix. For a vector $x \in \mathbb{R}^N$, write $x_I \overset{\text{def.}}{=} (x_i)_{i \in I} \in \mathbb{R}^{|I|}$ for its restriction.

The following proposition rephrases the first-order optimality conditions in a convenient form.

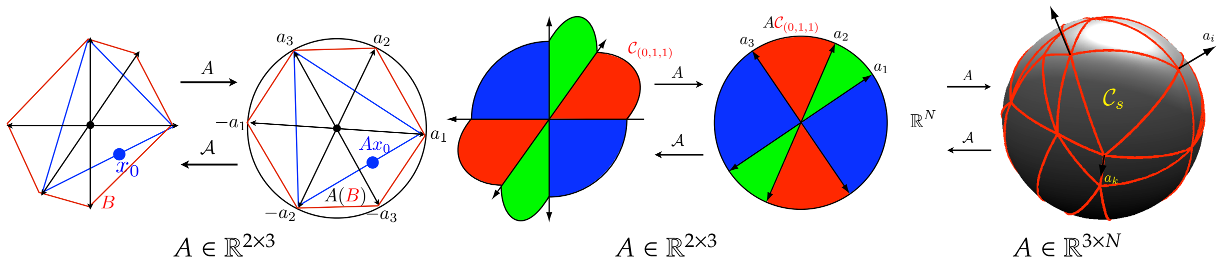


Figure 11.3. The linear map A projects the ℓ^1 ball B; the nonlinear recovery map $\mathcal{A}$ selects solutions of $(\mathcal{P}_0(y))$.

Proposition 11.2. *x_λ is a solution to $(\mathcal{P}_\lambda(y))$ for $\lambda > 0$ if and only if*

$$\eta_{\lambda,I} = \operatorname{sign}(x_{\lambda,I}) \quad \textit{and} \quad \|\eta_{\lambda,I^c}\|_\infty \leqslant 1$$

where we define

$$I \overset{\text{def.}}{=} \operatorname{supp}(x_\lambda) \overset{\text{def.}}{=} \{i \;;\; x_{\lambda,i} \neq 0\}, \quad \textit{and} \quad \eta_\lambda \overset{\text{def.}}{=} \frac{1}{\lambda} A^*(y - Ax_\lambda). \tag{11.2}$$

Proof. The objective in $(\mathcal{P}_\lambda(y))$ is the sum of a differentiable convex function and a continuous convex penalty. The subdifferential sum rule gives

$$\partial f_\lambda(x) = \frac{1}{\lambda} A^*(Ax - y) + \partial\|\cdot\|_1(x).$$

Thus x_λ is a solution to $(\mathcal{P}_\lambda(y))$ if and only if $0 \in \partial f_\lambda(x_\lambda)$, which gives the desired result. □

In particular, $\operatorname{supp}(x_\lambda) \subset \operatorname{sat}(\eta_\lambda)$, where $\operatorname{sat}(\eta) = \{i : |\eta_i| = 1\}$ is the saturation set.

For the constrained case $\lambda = 0$, the next proposition introduces dual certificates arising from the Lagrange multipliers associated with $\mathcal{L}_y$.

Proposition 11.3. *$x^\star$ solves $(\mathcal{P}_0(y))$ if and only if $Ax^\star = y$ and*

$$\exists \eta \in \mathcal{D}_0(y, x^\star) \overset{\text{def.}}{=} \operatorname{Im}(A^*) \cap \partial\|\cdot\|_1(x^\star). \tag{11.3}$$

Proof. The norm penalty in $(\mathcal{P}_0(y))$ is continuous, so the subdifferential sum rule gives

$$\partial f_0(x) = \partial \iota_{\mathcal{L}_y}(x) + \partial\|\cdot\|_1(x).$$

If $x \in \mathcal{L}_y$, then $\partial \iota_{\mathcal{L}_y}(x)$ is the normal space to the affine set $\mathcal{L}_y$, i.e. $\ker(A)^\perp = \operatorname{Im}(A^*)$. □

Every valid certificate satisfies $\operatorname{supp}(x^\star) \subset \operatorname{sat}(\eta)$, for $\eta \in \mathcal{D}_0(y, x^\star)$.

Writing $I = \operatorname{supp}(x^\star)$, one thus has

$$\mathcal{D}_0(y, x^\star) = \left\{\eta = A^* p \;;\; \eta_I = \operatorname{sign}(x^\star_I),\; \|\eta\|_\infty \leqslant 1\right\}.$$

Although the notation $\mathcal{D}_0(y, x^\star)$ involves a particular solution $x^\star$, the set of certificates is the same for every primal solution. The duality argument below explains this independence.

11.1.4 Uniqueness

The next proposition shows that at least one minimizer uses linearly independent columns of A. It does not assert that every minimizer has this property.

Proposition 11.4. *For $\lambda > 0$, there is a solution $x^\star$ to $(\mathcal{P}_\lambda(y))$ whose support I satisfies $\ker(A_I) = \{0\}$. The same holds for $(\mathcal{P}_0(y))$ when it is feasible.*

Proof. Let x be a minimizer and $I = \operatorname{supp}(x)$. If A_I is not injective, choose a nonzero $h_I \in \ker(A_I)$ and extend it by zero outside I. Along the interval containing $t = 0$ on which $x + th$ retains the sign of x, the fidelity is constant and the ℓ^1 norm is affine. Optimality at the interior point $t = 0$ forces its slope to vanish. Move to a finite endpoint of this interval, where at least one coefficient becomes zero. Continuity shows that the endpoint is still a minimizer and has smaller support. Repeating the argument terminates after finitely many steps at a minimizer with injective A_I. □

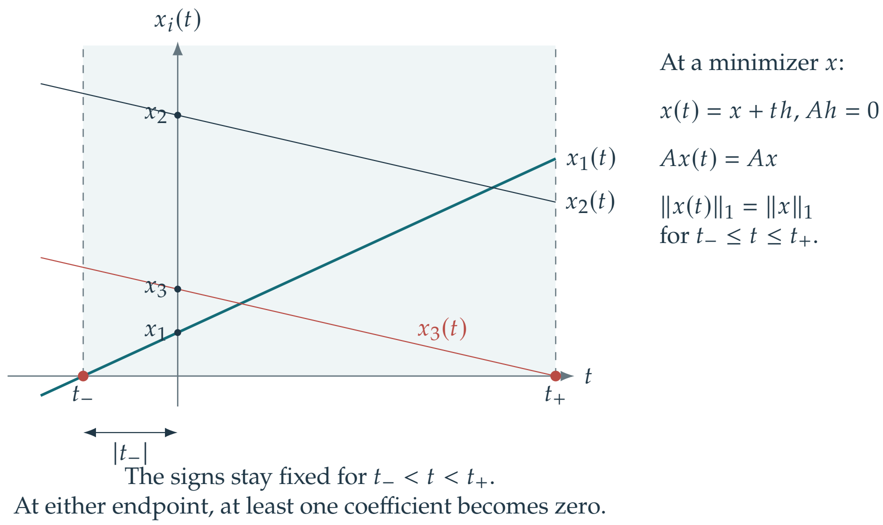


Figure 11.4. Coefficient trajectories along a support-reducing direction.

If the columns of A_I on a minimizer's support are linearly dependent, the proof constructs another minimizer. Thus that solution of $(\mathcal{P}_\lambda(y))$ is nonunique.

For a solution x_λ with $\ker(A_I) = \{0\}$, the optimality condition for $(\mathcal{P}_\lambda(y))$ gives the implicit formula

$$x_{\lambda,I} = A_I^+ y - \lambda (A_I^* A_I)^{-1} \operatorname{sign}(x_{\lambda,I}). \tag{11.4}$$

This expression generalizes soft thresholding (which is recovered when $A = \mathrm{Id}_N$).

Although the coefficient vector x_λ may not be unique, its fitted data Ax_λ, also called the predictor, are unique. Multiplying (11.4) by the restricted measurement matrix gives

$$Ax_\lambda = \operatorname{Proj}_{\operatorname{Im}(A_I)} y - \lambda A_I (A_I^* A_I)^{-1} \operatorname{sign}(x_{\lambda,I}). \tag{11.5}$$

Thus, up to an $O(\lambda)$ bias, this predictor is an orthogonal projection onto a low-dimensional subspace indexed by I.

Lemma 11.5. *For $\lambda > 0$, if x_1, x_2 solve $(\mathcal{P}_\lambda(y))$, then $Ax_1 = Ax_2$. For the feasible constrained problem $(\mathcal{P}_0(y))$, both predictors equal y.*

Proof. For $\lambda = 0$, feasibility gives $Ax_1 = Ax_2 = y$. Otherwise, suppose $Ax_1 \neq Ax_2$ and set $x = (x_1 + x_2)/2$. Convexity gives

$$\|x\|_1 \leqslant \frac{\|x_1\|_1 + \|x_2\|_1}{2} \quad \text{and} \quad \|Ax - y\|^2 < \frac{\|Ax_1 - y\|^2 + \|Ax_2 - y\|^2}{2}$$

where strict convexity of the squared Euclidean norm makes the second inequality strict. Hence

$$\frac{1}{2\lambda}\|Ax - y\|^2 + \|x\|_1 < \frac{1}{2\lambda}\|Ax_1 - y\|^2 + \|x_1\|_1,$$

contradicting the optimality of x_1. ◻

Proposition 11.6. *For $\lambda > 0$, let x_λ be a solution to $(\mathcal{P}_\lambda(y))$ and denote $\eta_\lambda \overset{\text{def.}}{=} \frac{1}{\lambda} A^*(y - Ax_\lambda)$. We define the "extended support" as*

$$J \overset{\text{def.}}{=} \operatorname{sat}(\eta_\lambda) \overset{\text{def.}}{=} \{i \,;\, |\eta_{\lambda,i}| = 1\}.$$

If $\ker(A_J) = \{0\}$ then x_λ is the unique solution of $(\mathcal{P}_\lambda(y))$.

Proof. If $\tilde{x}_\lambda$ is also a minimizer, then by Lemma 11.5, $Ax_\lambda = A\tilde{x}_\lambda$, so that in particular they share the same dual certificate

$$\eta_\lambda = \frac{1}{\lambda} A^*(y - Ax_\lambda) = \frac{1}{\lambda} A^*(y - A\tilde{x}_\lambda).$$

The first-order condition, Proposition 11.2, shows that necessarily $\operatorname{supp}(x_\lambda) \subset J$ and $\operatorname{supp}(\tilde{x}_\lambda) \subset J$. Since $A_J x_{\lambda,J} = A_J \tilde{x}_{\lambda,J}$, and since $\ker(A_J) = \{0\}$, one has $x_{\lambda,J} = \tilde{x}_{\lambda,J}$, and thus $x_\lambda = \tilde{x}_\lambda$ because their supports are included in J. ◻

Proposition 11.7. *Let $x^\star$ be a solution to ($\mathcal{P}_0(y)$). If there exists $\eta \in \mathcal{D}_0(y, x^\star)$ such that $\ker(A_J) = \{0\}$ where $J \overset{\text{def.}}{=} \operatorname{sat}(\eta)$ then $x^\star$ is the unique solution of ($\mathcal{P}_0(y)$).*

Proof. The proof is the same as for Proposition 11.6, replacing η_λ by η. □

For $\lambda > 0$, a joint density for the entries of A on $\mathbb{R}^{P\times N}$ implies that its columns are in general position almost surely. The Lasso solution is then unique for every y. Randomizing only y, while keeping A fixed, does not in general ensure uniqueness.

11.1.5 Duality

The first-order conditions can also be derived from the convex duality theory in Section 18.3. This interpretation explains why certificates characterize optimal recovery.

Theorem 11.8. *For $\lambda > 0$, strong duality holds between ($\mathcal{P}_\lambda(y)$) and the following dual problem. For $\lambda = 0$, the same statement holds with ($\mathcal{P}_0(y)$) as the primal problem, provided $y \in \operatorname{Im}(A)$:*

$$\sup_{p\in\mathbb{R}^P} \left\{ \langle y,\, p\rangle - \frac{\lambda}{2}\|p\|^2 \;;\; \|A^* p\|_\infty \leqslant 1 \right\}. \tag{11.6}$$

A pair $(x^\star, p^\star)$ is primal–dual optimal if and only if it satisfies the following certificate condition together with the residual relation below:

$$A^* p^\star \in \partial\|\cdot\|_1(x^\star) \quad \Leftrightarrow \quad \left(\|A^* p^\star\|_\infty \leqslant 1, \quad I \subset \operatorname{sat}(A^* p^\star) \quad \textit{and} \quad \operatorname{sign}(x_I^\star) = A_I^* p^\star \right), \tag{11.7}$$

where we denote $I = \operatorname{supp}(x^\star)$, and furthermore, for $\lambda > 0$,

$$p^\star = \frac{y - Ax^\star}{\lambda}.$$

while for $\lambda = 0$, $Ax^\star = y$.

Proof. Introduce $z = Ax$ and define $g_\lambda(z) = \|z - y\|^2/(2\lambda)$ when $\lambda > 0$, and $g_0 = \iota_{\{y\}}$. The Lagrangian is

$$\mathcal{L}(x, z, p) = g_\lambda(z) + \|x\|_1 + \langle z - Ax, p\rangle.$$

For $\lambda > 0$, continuity gives strong duality. For $\lambda = 0$, feasibility and polyhedral convex duality give the same conclusion. Minimizing the Lagrangian over x and z yields

$$\sup_{p\in\mathbb{R}^P} \left[-g_\lambda^*(-p) - (\|\cdot\|_1)^*(A^* p) \right].$$

The conjugates are

$$g_\lambda^*(q) = \langle y, q\rangle + \frac{\lambda}{2}\|q\|^2, \qquad (\|\cdot\|_1)^* = \iota_{\{\|\cdot\|_\infty \leqslant 1\}},$$

including $\lambda = 0$, which gives (11.6). This is also a direct application of the Fenchel–Rockafellar formula (18.17).

The optimality conditions are $z^\star = Ax^\star$, $A^* p^\star \in \partial\|\cdot\|_1(x^\star)$, and $-p^\star \in \partial g_\lambda(z^\star)$. For $\lambda > 0$, the last condition gives $p^\star = (y - Ax^\star)/\lambda$. For $\lambda = 0$, it is equivalent to $Ax^\star = y$. The first condition on $p^\star$ gives the support and sign relations in (11.7). □

For $\lambda > 0$, the dual objective (11.6) is strongly concave. Completing the square shows that its maximizer p_λ is an orthogonal projection:

$$p_\lambda \in \operatorname*{argmin}_{p\in\mathbb{R}^P} \left\{ \|p - y/\lambda\| \;;\; \|A^* p\|_\infty \leqslant 1 \right\}.$$

Thus the supremum is attained at a unique point for $\lambda > 0$. For $\lambda = 0$ and feasible data, a dual optimum also exists, though it need not be unique.

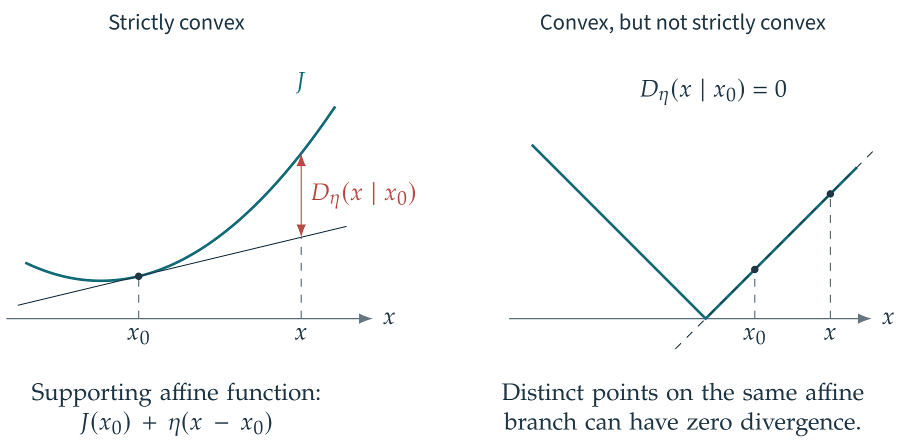


Figure 11.5. Visualization of Bregman divergences.

11.2 Consistency and Sparsistency

11.2.1 Bregman Divergence Rates for General Regularizations

Consider the more general regularized problem

$$\min_{x \in \mathbb{R}^N} \frac{1}{2\lambda}\|Ax - y\|^2 + J(x) \tag{11.8}$$

for a proper convex regularizer J and $\lambda > 0$. We state estimates for any minimizer, without assuming uniqueness.

For x_0 with nonempty subdifferential and a chosen $\eta \in \partial J(x_0)$, define the Bregman divergence

$$D_\eta(x|x_0) \overset{\text{def.}}{=} J(x) - J(x_0) - \langle \eta,\, x - x_0 \rangle.$$

One has $D_\eta(x_0|x_0) = 0$ and, by convexity, $D_\eta(x|x_0) \geqslant 0$; see Figure 11.5.

If J is differentiable, then $\partial J(x_0) = \{\nabla J(x_0)\}$, and the divergence becomes

$$D(x|x_0) \overset{\text{def.}}{=} J(x) - J(x_0) - \langle \nabla J(x_0),\, x - x_0 \rangle.$$

If J is strictly convex, then $D(x|x_0) = 0$ exactly when $x = x_0$. Thus $D(\cdot|\cdot)$ separates points, although it need not be symmetric or satisfy the triangle inequality.

If $J = \|\cdot\|^2$, then $D(x|x_0) = \|x - x_0\|^2$ is the squared Euclidean distance. For the negative-entropy functional $J(x) = \sum_i x_i(\log x_i - 1) + \iota_{\mathbb{R}^N_+}(x)$, one obtains

$$D(x|x_0) = \sum_i x_i \log\left(\frac{x_i}{x_{0,i}}\right) + x_{0,i} - x_i$$

This is the generalized Kullback–Leibler divergence for $x_0 \in \mathbb{R}^N_{++}$ and $x \in \mathbb{R}^N_+$, with the convention $0 \log 0 = 0$. For probability vectors, the linear terms sum to zero.

The following estimate, associated with the work of Burger and Osher, yields a linear noise rate measured by Bregman divergence.

Theorem 11.9. *If there exists*

$$\eta = A^* p \in \operatorname{Im}(A^*) \cap \partial J(x_0), \tag{11.9}$$

then every solution x_λ of (11.8) *satisfies*

$$D_\eta(x_\lambda|x_0) \leqslant \frac{1}{2}\left(\frac{\|w\|}{\sqrt{\lambda}} + \sqrt{\lambda}\|p\|\right)^2. \tag{11.10}$$

The measurement residual also satisfies

$$\|Ax_\lambda - y\| \leqslant \|w\| + 2\|p\|\lambda. \tag{11.11}$$

Proof. The optimality of x_λ for (11.8) implies

$$\frac{1}{2\lambda}\|Ax_\lambda - y\|^2 + J(x_\lambda) \leqslant \frac{1}{2\lambda}\|Ax_0 - y\|^2 + J(x_0) = \frac{1}{2\lambda}\|w\|^2 + J(x_0).$$

Hence, using $\langle \eta,\, x_\lambda - x_0\rangle = \langle p,\, Ax_\lambda - Ax_0\rangle = \langle p,\, Ax_\lambda - y + w\rangle$, one has

$$\begin{aligned} D_\eta(x_\lambda|x_0) = J(x_\lambda) - J(x_0) - \langle \eta,\, x_\lambda - x_0\rangle &\leqslant \frac{1}{2\lambda}\|w\|^2 - \frac{1}{2\lambda}\|Ax_\lambda - y\|^2 - \langle p,\, Ax_\lambda - y\rangle - \langle p,\, w\rangle \\ &= \frac{1}{2\lambda}\|w\|^2 - \frac{1}{2\lambda}\|Ax_\lambda - y + \lambda p\|^2 + \frac{\lambda}{2}\|p\|^2 - \langle p,\, w\rangle \\ &\leqslant \frac{1}{2\lambda}\|w\|^2 + \frac{\lambda}{2}\|p\|^2 + \|p\|\|w\| = \frac{1}{2}\left(\frac{\|w\|}{\sqrt{\lambda}} + \sqrt{\lambda}\|p\|\right)^2. \end{aligned}$$

Since $D_\eta(x_\lambda|x_0) \geqslant 0$, the completed-square bound gives

$$\|Ax_\lambda - y + \lambda p\|^2 \leqslant \|w - \lambda p\|^2 \leqslant (\|w\| + \lambda\|p\|)^2.$$

The triangle inequality therefore yields

$$\|Ax_\lambda - y\| \leqslant \|w\| + 2\lambda\|p\|.$$

□

When $\|w\| > 0$ and $p \neq 0$, choose $\lambda = \|w\|/\|p\|$ in (11.10). The resulting estimate $D_\eta(x_\lambda|x_0) \leqslant 2\|w\|\|p\|$ is linear in the noise level. For the simple case of a quadratic regularizer $J(x) = \|x\|^2/2$, as used in Section 9.3.2, the source condition (11.9) becomes

$$x_0 \in \operatorname{Im}(A^*)$$

This is (9.12) with $\beta = 1$. Under this condition, (11.10) gives the sublinear ℓ^2 estimate

$$\|x_0 - x_\lambda\| \leqslant 2\sqrt{\|w\|\|p\|}.$$

This agrees with Theorem 9.4 under the convention $x_0 \in \operatorname{Im}((A^*A)^{\beta/2})$.

The source condition (11.9) is sufficient for x_0 to minimize J subject to $Ax = Ax_0$. In finite dimension, it is also necessary under a subdifferential sum-rule qualification, for example when J is finite and continuous on $\mathbb{R}^N$. Without such a qualification, existence of a Lagrange multiplier is an additional assumption.

11.2.2 Linear Rates in Norms for ℓ^1 Regularization

For a regularizer J that is not strictly convex, the Bregman bound (11.10) need not control the error in norm. We now examine how additional certificate conditions restore such control for the ℓ^1 penalty $J = \|\cdot\|_1$.

The next lemma quantifies this control through the gap between $|\eta_i|$ and one.

Lemma 11.10. *For $\eta \in \partial\|\cdot\|_1(x_0)$, let $J \overset{\text{def.}}{=} \operatorname{sat}(\eta)$. Then*

$$D_\eta(x|x_0) \geqslant (1 - \|\eta_{J^c}\|_\infty)\|(x - x_0)_{J^c}\|_1. \tag{11.12}$$

Proof. Note that $x_{0,J^c} = 0$ since $\operatorname{supp}(x_0) \subset \operatorname{sat}(\eta)$ by definition of the subdifferential of the ℓ^1 norm. Since the ℓ^1 norm is separable, each term in the sum defining $D_\eta(x|x_0)$ is nonnegative, hence

$$\begin{aligned} D_\eta(x|x_0) &= \sum_i |x_i| - |x_{0,i}| - \eta_i(x_i - x_{0,i}) \geqslant \sum_{i \in J^c} |x_i| - |x_{0,i}| - \eta_i(x_i - x_{0,i}) \\ &= \sum_{i \in J^c} |x_i| - \eta_i x_i \geqslant \sum_{i \in J^c} (1 - |\eta_i|)|x_i| \geqslant (1 - \|\eta_{J^c}\|_\infty) \sum_{i \in J^c} |x_i| = (1 - \|\eta_{J^c}\|_\infty) \sum_{i \in J^c} |x_i - x_{0,i}|. \end{aligned}$$

□

We use the convention $\|\eta_\emptyset\|_\infty = 0$. The margin $1 - \|\eta_{J^c}\|_\infty > 0$ measures how far η lies from saturation on the complementary coordinates. A larger margin gives stronger control of the coefficient error.

This lemma yields a norm convergence rate under the certificate and injectivity assumptions of Proposition 11.7, with $x^\star = x_0$.

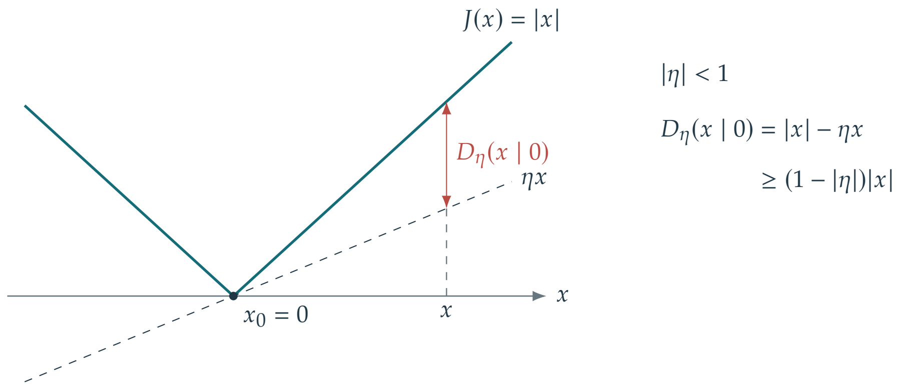


Figure 11.6. Bregman divergence controls the ℓ^1 error on coordinates where η does not saturate.

Theorem 11.11. *If there exists*

$$\eta \in \mathcal{D}_0(Ax_0, x_0) \tag{11.13}$$

and $\ker(A_J) = \{0\}$ *where* $J \overset{\text{def.}}{=} \operatorname{sat}(\eta)$ *then, for fixed* $c > 0$ *and* $\|w\| > 0$*, choosing* $\lambda = c\|w\|$ *gives a constant* C *(depending on* A*, the certificate, and* c*, but not on* w*) such that any solution* x_λ *of* $\mathcal{P}_\lambda(Ax_0 + w)$ *satisfies*

$$\|x_\lambda - x_0\| \leqslant C\|w\|. \tag{11.14}$$

Proof. Write $\eta = A^* p$ and $y = Ax_0 + w$. The optimality of x_λ in ($\mathcal{P}_\lambda(y)$) implies

$$\frac{1}{2\lambda}\|Ax_\lambda - y\|^2 + \|x_\lambda\|_1 \leqslant \frac{1}{2\lambda}\|Ax_0 - y\|^2 + \|x_0\|_1 = \frac{1}{2\lambda}\|w\|^2 + \|x_0\|_1$$

and hence

$$\|Ax_\lambda - y\|^2 \leqslant \|w\|^2 + 2\lambda\|x_0\|_1$$

Using the fact that A_J is injective, one has $A_J^+ A_J = \mathrm{Id}_J$, so that

$$\begin{aligned}
\|(x_\lambda - x_0)_J\|_1 &= \|A_J^+ A_J (x_\lambda - x_0)_J\|_1 \leqslant \|A_J^+\|_{2,1}\|A_J x_{\lambda,J} - y + w\| \leqslant \|A_J^+\|_{2,1}\left(\|A_J x_{\lambda,J} - y\| + \|w\|\right) \\
&\leqslant \|A_J^+\|_{2,1}\left(\|Ax_\lambda - y\| + \|A_{J^c} x_{\lambda,J^c}\| + \|w\|\right) \\
&\leqslant \|A_J^+\|_{2,1}\left(\|Ax_\lambda - y\| + \|A_{J^c}\|_{1,2}\|x_{\lambda,J^c} - x_{0,J^c}\|_1 + \|w\|\right) \\
&\leqslant \|A_J^+\|_{2,1}\left(\|w\| + 2\|p\|\lambda + \|A_{J^c}\|_{1,2}\|x_{\lambda,J^c} - x_{0,J^c}\|_1 + \|w\|\right)
\end{aligned}$$

where we used $x_{0,J^c} = 0$ and (11.11). The mixed operator norm $\|B\|_{p,q}$ maps ℓ^p to ℓ^q, as defined in Remark 11.14. Substituting this bound into the decomposition and using (11.12) and (11.10) gives

$$\begin{aligned}
\|x_\lambda - x_0\|_1 &= \|(x_\lambda - x_0)_J\|_1 + \|(x_\lambda - x_0)_{J^c}\|_1 \\
&\leqslant \|(x_\lambda - x_0)_{J^c}\|_1\left(1 + \|A_J^+\|_{2,1}\|A_{J^c}\|_{1,2}\right) + \|A_J^+\|_{2,1}\left(2\|p\|\lambda + 2\|w\|\right) \\
&\leqslant \frac{D_\eta(x_\lambda|x_0)}{1 - \|\eta_{J^c}\|_\infty}\left(1 + \|A_J^+\|_{2,1}\|A_{J^c}\|_{1,2}\right) + \|A_J^+\|_{2,1}\left(2\|p\|\lambda + 2\|w\|\right) \\
&\leqslant \frac{\frac{1}{2}\left(\frac{\|w\|}{\sqrt{\lambda}} + \sqrt{\lambda}\|p\|\right)^2}{1 - \|\eta_{J^c}\|_\infty}\left(1 + \|A_J^+\|_{2,1}\|A_{J^c}\|_{1,2}\right) + \|A_J^+\|_{2,1}\left(2\|p\|\lambda + 2\|w\|\right).
\end{aligned}$$

Thus setting $\lambda = c\|w\|$, one obtains the constant

$$C \overset{\text{def.}}{=} \frac{\frac{1}{2}\left(\frac{1}{\sqrt{c}} + \sqrt{c}\|p\|\right)^2}{1 - \|\eta_{J^c}\|_\infty}\left(1 + \|A_J^+\|_{2,1}\|A_{J^c}\|_{1,2}\right) + \|A_J^+\|_{2,1}\left(2\|p\|c + 2\right).$$

□

This theorem guarantees uniqueness of the noiseless solution x_0, but does not imply uniqueness of x_λ. The source condition (11.13), combined with the injectivity condition $\ker(A_J) = \{0\}$, gives the ℓ^2 stability estimate (11.14). The estimate (11.14) controls the error linearly relative to the fixed noiseless signal x_0. It does not by itself establish pairwise Lipschitz continuity between arbitrary noisy solutions.

Compare this linear rate with the sublinear rates for quadratic regularization in Theorem 9.4. The source conditions for linear regularization (9.12) and nonlinear regularization (11.13) express different structural assumptions. In the quadratic case with $\beta = 1$, the source condition is $x_0 \in \mathrm{Im}(A^*) = \ker(A)^\perp$ in finite dimension. The signal itself must have no component in $\ker(A)$ because squared-norm regularization cannot recover such a component. The nonlinear condition instead requires a subgradient η to belong to $\mathrm{Im}(A^*)$. It can therefore permit recovery of signal components in $\ker(A)$ when the prior supplies sufficient structure.

11.2.3 Sparsistency for Low Noise

Theorem 11.11 gives an abstract guarantee whose certificate assumptions may be difficult to verify. We now construct a candidate certificate explicitly. When it satisfies a strict off-support bound, it guarantees both a linear rate and support stability.

For any solution x_λ of $(\mathcal{P}_\lambda(y))$, define the dual certificate as in (11.2). Uniqueness of the fitted data makes it independent of the chosen solution:

$$\eta_\lambda \overset{\text{def.}}{=} A^* p_\lambda \quad \text{where} \quad p_\lambda \overset{\text{def.}}{=} \frac{y - Ax_\lambda}{\lambda}.$$

The next proposition shows that p_λ converges to the dual solution of minimum norm for the constrained problem.

Proposition 11.12. *If $y = Ax_0$ where x_0 solves $\mathcal{P}_0(y)$, one has*

$$p_\lambda \longrightarrow p_0 \overset{\text{def.}}{=} \underset{p \in \mathbb{R}^P}{\operatorname{argmin}} \; \{\|p\| \; ; \; A^* p \in \mathcal{D}_0(y, x_0)\}\,. \tag{11.15}$$

The vector $\eta_0 \overset{\text{def.}}{=} A^* p_0$ is called the minimum-norm certificate; the minimized norm is that of the data-space vector p_0.

Proof. Write $\mathcal{D} = \{p : \|A^* p\|_\infty \leqslant 1\}$ and let M be the maximum of $\langle y, p\rangle$ on $\mathcal{D}$. Optimality relative to the minimum-norm dual maximizer p_0 gives $0 \leqslant M - \langle y, p_\lambda\rangle \leqslant \frac{\lambda}{2}(\|p_0\|^2 - \|p_\lambda\|^2)$. Thus $\|p_\lambda\| \leqslant \|p_0\|$, every cluster point is a dual maximizer of no greater norm, and uniqueness of p_0 proves convergence as $\lambda \downarrow 0$. □

The certificates in $\mathcal{D}_0(y, x_0)$ for $\lambda = 0$ may be nonunique, but the limit $\lambda \to 0$ selects one of them. This selected certificate governs support stability when both the noise and λ are small.

The inequality constraint $\|\eta_0\|_\infty \leqslant 1$ makes the minimum-norm problem (11.15) difficult to solve explicitly. Dropping this ℓ^∞ constraint while retaining interpolation on the support gives the minimum-norm precertificate:

$$\eta_F \overset{\text{def.}}{=} A^* p_F \quad \text{where} \quad p_F \overset{\text{def.}}{=} \underset{p \in \mathbb{R}^P}{\operatorname{argmin}} \; \left\{\|p\| \; ; \; A_I^* p = \mathrm{sign}(x_{0,I})\right\}. \tag{11.16}$$

The notation "η_F" refers to the "Fuchs" certificate, named after J.-J. Fuchs, who first used it to study ℓ^1 minimization.

The vector p_F may fail dual feasibility because $\|\eta_F\|_\infty \leqslant 1$ need not hold. This is why the construction gives a precertificate. If the interpolation system is consistent, p_F is its minimum-norm solution: $A_I^* p = \mathrm{sign}(x_{0,I})$. The pseudoinverse gives $p_F = A_I^{*,+} \,\mathrm{sign}(x_{0,I})$; see Proposition 9.2. Under $\ker(A_I) = \{0\}$, this simplifies to

$$p_F = A_I (A_I^* A_I)^{-1} \,\mathrm{sign}(x_{0,I}).$$

With the Gram matrix $C \overset{\text{def.}}{=} A^* A$, the certificate becomes

$$\eta_F = C_{\cdot,I} C_{I,I}^{-1} \,\mathrm{sign}(x_{0,I}). \tag{11.17}$$

The next proposition relates η_F to η_0: whenever η_F is feasible, it equals η_0.

Proposition 11.13. *If $\|\eta_F\|_\infty \leqslant 1$, then $p_F = p_0$ and $\eta_F = \eta_0$.*

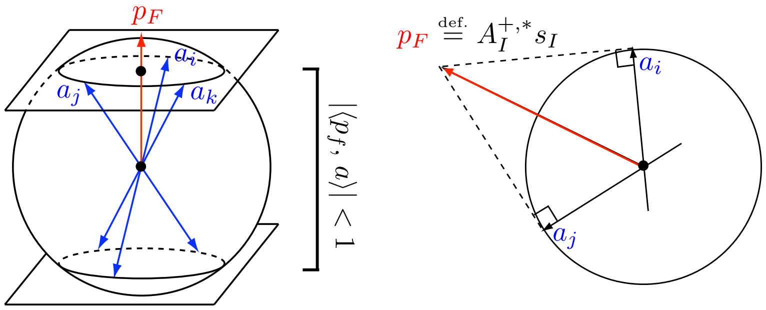


Figure 11.7. For unit-norm columns, the supporting hyperplane defined by a valid certificate determines an empty spherical cap: its interior contains no signed column $\pm a_i$.

The condition $\|\eta_F\|_\infty \leqslant 1$ implies that x_0 is a solution to ($\mathcal{P}_0(y)$). A strict off-support bound on η_F strengthens this recovery condition and gives both a linear error rate and support stability for small noise.

The feasibility condition $\|\eta_F\|_\infty \leqslant 1$ has the empty-cap interpretation shown in Figure 11.7. We thank Charles Dossal for pointing out this connection to spherical Delaunay triangulations.

Remark 11.14 (Operator norm). In the proof, we use the $\ell^p - \ell^q$ matrix operator norm, which is defined as

$$\|B\|_{p,q} \overset{\text{def.}}{=} \max\left\{\|Bu\|_q \;;\; \|u\|_p \leqslant 1\right\}.$$

For $p = q$, we denote $\|B\|_p \overset{\text{def.}}{=} \|B\|_{p,p}$. For $p = q = 2$, $\|B\|_2$ is the maximum singular value, and one has

$$\|B\|_1 = \max_j \sum_i |B_{i,j}| \quad \text{and} \quad \|B\|_\infty = \max_i \sum_j |B_{i,j}|.$$

Theorem 11.15. *Let $I = \operatorname{supp}(x_0) \neq \emptyset$, assume A_I is injective, and suppose $\|\eta_{F,I^c}\|_\infty < 1$. There exist $c_0, c_1 > 0$, depending on A, I, and x_0, such that if*

$$0 < \lambda < c_0, \qquad \|A^* w\|_\infty < c_1 \lambda,$$

then the solution x_λ of ($\mathcal{P}_\lambda(y)$) is unique and has the same support and sign as x_0. It satisfies

$$x_{\lambda,I} = x_{0,I} + A_I^+ w - \lambda (A_I^* A_I)^{-1} \operatorname{sign}(x_{0,I}), \qquad x_{\lambda,I^c} = 0. \tag{11.18}$$

In particular, the error is $O(\|A^* w\|_\infty + \lambda)$. When $\|A^* w\|_\infty > 0$, choosing λ proportional to this quantity with a sufficiently large constant gives an $O(\|w\|)$ error as the noise tends to zero.

Proof. Let $T \overset{\text{def.}}{=} \min_{i \in I} |x_{0,i}|$ be the smallest nonzero signal amplitude, and let $\delta \overset{\text{def.}}{=} \|A^* w\|_\infty$ measure the noise after correlation with the measurement columns. Set $s \overset{\text{def.}}{=} \operatorname{sign}(x_0)$ and use (11.18) to define the candidate

$$\hat{x}_I \overset{\text{def.}}{=} x_{0,I} + A_I^+ w - \lambda (A_I^* A_I)^{-1} s_I, \tag{11.19}$$

and $\hat{x}_{I^c} = 0$. The goal is to show that $\hat{x}$ is indeed the unique solution of ($\mathcal{P}_\lambda(y)$).

Step 1. Sign consistency means $\operatorname{sign}(\hat{x}) = s$. It follows from $\|x_{0,I} - \hat{x}_I\|_\infty < T$, for which a sufficient condition is

$$\|x_{0,I} - \hat{x}_I\|_\infty \leqslant K \|A_I^* w\|_\infty + K\lambda < T \quad \text{where} \quad K \overset{\text{def.}}{=} \|(A_I^* A_I)^{-1}\|_\infty, \tag{11.20}$$

where we used the fact that $A_I^+ = (A_I^* A_I)^{-1} A_I^*$.

Step 2. We verify the strict off-support bound in Proposition 11.6: $\|\hat{\eta}_{I^c}\|_\infty < 1$, where $\lambda \hat{\eta} = A^*(y - A\hat{x})$. Together with sign consistency and injectivity, this proves that $\hat{x}$ is the unique solution of ($\mathcal{P}_\lambda(y)$). Compute

$$\begin{aligned} \lambda \hat{\eta} &= A^*\big(A_I x_{0,I} + w - A_I \hat{x}_I\big) \\ &= A^*(\mathrm{Id} - A_I A_I^+) w + \lambda \eta_F. \end{aligned}$$

The residual formula also gives $\hat{\eta}_I = s_I$. Write $\delta = \|A^* w\|_\infty$, $L = \|A_{I^c}^* A_I\|_\infty$, $R = KL + 1$, and $S = 1 - \|\eta_{F,I^c}\|_\infty > 0$. Then

$$\lambda \|\hat{\eta}_{I^c}\|_\infty \leqslant R\delta + \lambda \|\eta_{F,I^c}\|_\infty < \lambda \quad \text{if} \quad R\delta < S\lambda. \tag{11.21}$$

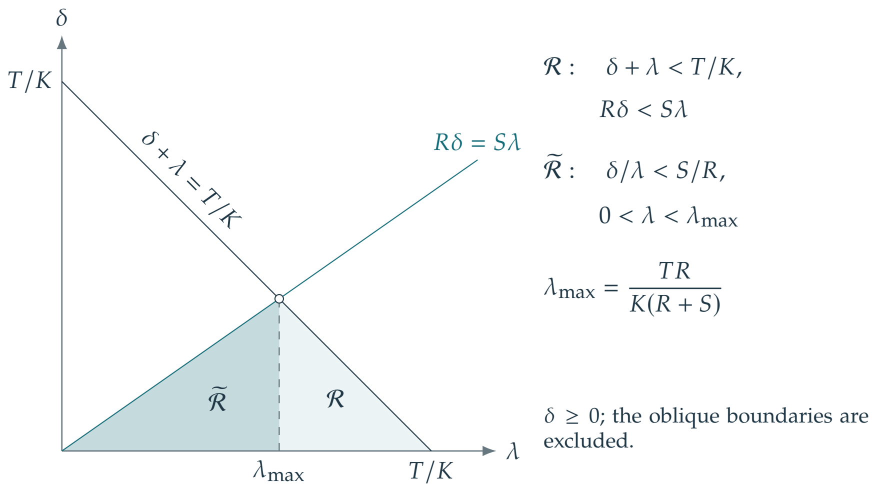


Figure 11.8. Region in the (λ, δ) plane where sign consistency holds.

Together with sign consistency and injectivity of A_I, this strict inequality proves uniqueness.
Conclusion. Combining (11.20) and (11.21), we obtain uniqueness of $\hat{x}$ whenever the parameters (λ, w) satisfy

$$\mathcal{R} = \left\{(\delta, \lambda)\ ;\ \delta + \lambda < \frac{T}{K} \quad \text{and} \quad R\delta - S\lambda < 0\right\}$$

The triangular region $\mathcal{R}$ contains the simpler triangle

$$\tilde{\mathcal{R}} = \left\{(\delta, \lambda)\ ;\ \frac{\delta}{\lambda} < \frac{S}{R} \quad \text{and} \quad \lambda < \lambda_{\max}\right\} \quad \text{where} \quad \lambda_{\max} \overset{\text{def.}}{=} \frac{T(KL+1)}{K(R+S)}. \tag{11.22}$$

□

Theorem 11.15 requires small correlated noise $\|A^* w\|_\infty$ and a small λ, with their ratio controlled, so that regularization does not eliminate the nonzero coefficients of x_0. Theorem 11.11, by contrast, applies at any noise level $\|w\|$, but gives neither support control nor uniqueness of x_λ, and its constants can be less favorable.

The proof provides explicit constants in terms of three parameters K, L, S:

– K accounts for the conditioning of the operator on the support I;
– L is the largest sum of absolute correlations between an off-support atom and all atoms on the support;
– S measures the margin by which η_F avoids saturation outside the support.

The bounds on $\|A^* w\|_\infty / \lambda$ and on λ are given by (11.22). For fixed $\gamma > 0$ and sufficiently small $\delta > 0$, choosing $\lambda = (1+\gamma)R\delta/S$ gives

$$\|x_0 - x_\lambda\|_\infty \leqslant K\left(1 + (1+\gamma)\frac{R}{S}\right)\delta.$$

The strict factor $1+\gamma$ ensures the required strict inequality. Such a choice is primarily theoretical because its constants involve the unknown support.

For a class of signals x_0, analyzing ℓ^1 support recovery therefore starts with testing the Fuchs precertificate η_F: feasibility requires $\|\eta_F\|_\infty \leqslant 1$, and support stability requires a strict bound outside the support. Figure 11.9 illustrates the behavior for random A: the precertificate η_F becomes nondegenerate when P is sufficiently large. Section 12.2 analyzes this phenomenon.

11.2.4 Sparsistency for Arbitrary Noise

The small-noise theorem fixes the sign of the solution in advance. To obtain support containment for arbitrary noise levels, one can instead control all sign patterns on a prescribed support I. This gives the exact recovery coefficient of Tropp [32]:

$$\mathrm{ERC}(I) = 1 - \max_{j \notin I} \|A_I^+ a_j\|_1.$$

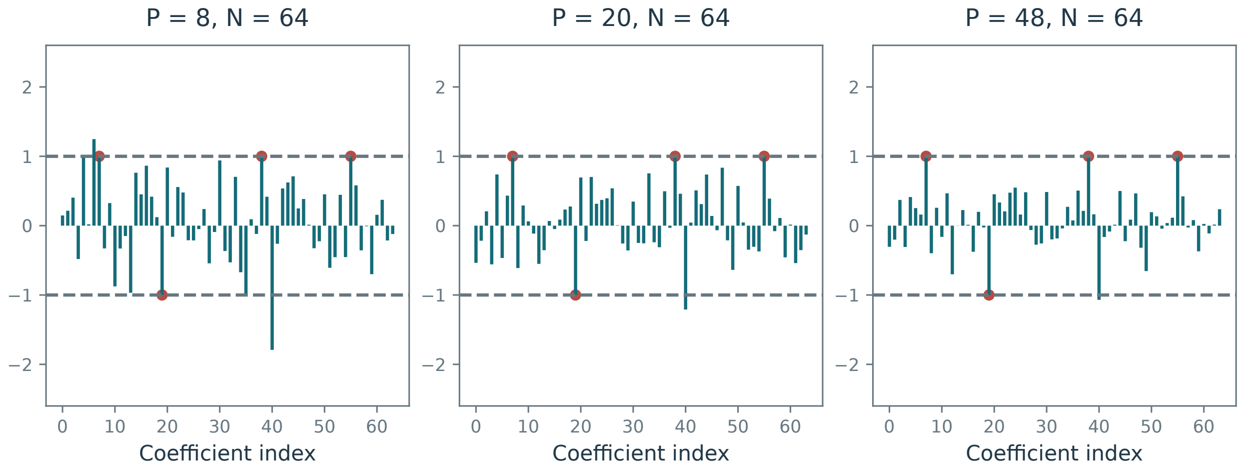


Figure 11.9. Fuchs precertificate η_F for a Gaussian matrix $A \in \mathbb{R}^{P\times N}$ with $N = 64$ and increasing measurement count P.

The maximum over an empty complement is taken to be zero. Assume A_I is injective and $\mathrm{ERC}(I) > 0$. Let $P_I = A_I A_I^+$ and choose $\lambda > 0$ so that

$$\|A_{I^c}^*(\mathrm{Id} - P_I)y\|_\infty < \lambda\, \mathrm{ERC}(I).$$

Then the Lasso has a unique solution supported within I; some coefficients may vanish, so equality of supports is not asserted.

Indeed, minimize the Lasso objective over vectors supported on I. The resulting vector $\hat{x}_I$ is unique, since A_I is injective. Its optimality condition gives $q_I \in \partial\|\cdot\|_1(\hat{x}_I)$ and

$$y - A_I\hat{x}_I = (\mathrm{Id} - P_I)y + \lambda A_I(A_I^* A_I)^{-1}q_I.$$

For $j \notin I$, the corresponding dual coefficient is bounded by

$$\frac{|\langle a_j, (\mathrm{Id} - P_I)y\rangle|}{\lambda} + \|A_I^+ a_j\|_1 < 1.$$

Thus the restricted minimizer satisfies the full optimality conditions with strict inequality outside I, and injectivity proves uniqueness. If $y = Ax_0 + w$ and $\mathrm{supp}(x_0) \subset I$, the orthogonal residual above is $(\mathrm{Id} - P_I)w$. The condition therefore states explicitly how large λ must be relative to the noise.

11.3 Sparse Deconvolution Case Study

For random A, Chapter 12 gives probabilistic conditions under which η_F is a valid certificate.

Super-resolution exposes a limitation of the Fuchs construction. The columns $(a_i)_i$ of A are often samples of a smooth kernel, so nearby columns are highly correlated.

We consider $a_i = \varphi(z_i)$, where $(z_i)_i \subset \mathbb{X}$ is a sampling grid in a domain $\mathbb{X}$ and $\varphi : \mathbb{X} \to \mathcal{H}$ is a smooth map. One has

$$Ax = \sum_i x_i \varphi(z_i).$$

The coefficients of a sparse vector x are the weights of the discrete measure $m_x \overset{\text{def.}}{=} \sum_{i=1}^N x_i \delta_{z_i}$, whose atoms lie on the reconstruction grid.

The matrix A discretizes an operator on Radon measures, $\mathcal{A} : m \in \mathcal{M}(\mathbb{X}) \mapsto y = \mathcal{A}m \in \mathcal{H}$, defined by

$$\mathcal{A}(m) \overset{\text{def.}}{=} \int_{\mathbb{X}} \varphi(x)\mathrm{d}m(x).$$

For a discrete measure, the two representations agree: $\mathcal{A}(m_x) = Ax$.

For example, take $\mathcal{H} = L^2(\mathbb{X})$ on $\mathbb{X} = \mathbb{R}^d$ or $\mathbb{X} = \mathbb{T}^d$ and let $\varphi(z) = \tilde{\varphi}(\cdot - z)$. The resulting operator is convolution:

$$(\mathcal{A}m)(z) = \int \tilde{\varphi}(z - x)\mathrm{d}m(x) = (\tilde{\varphi} \star m)(z).$$

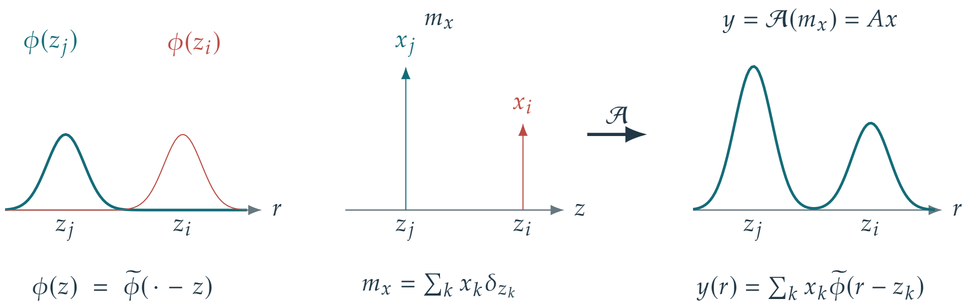


Figure 11.10. Convolution operator.

The observation space $\mathcal{H}$ is infinite-dimensional in this example. Sampling the output instead gives $\varphi(z) = (\tilde{\varphi}(r_j - z))_{j=1}^P$. The observation grid $r \in \mathbb{X}^P$ need not coincide with the reconstruction grid $z \in \mathbb{X}^N$.

A closely related example uses $\varphi(z) = (e^{\mathrm{i}kz})_{k=-f_c}^{f_c}$ on $\mathbb{X} = \mathbb{T}$, so that $\mathcal{A}$ corresponds to computing the $2f_c + 1$ low-frequency coefficients of the Fourier transform of the measure

$$\mathcal{A}(m) = \left(\int_{\mathbb{T}} e^{\mathrm{i}kx} \mathrm{d}m(x) \right)_{k=-f_c}^{f_c}.$$

The operator $\mathcal{A}^*\mathcal{A}$ is a convolution against an ideal low-pass (Dirichlet) kernel. By weighting the Fourier coefficients, one can model low-pass filters on the torus.

On $\mathbb{X} = \mathbb{R}^+$, another example is the Laplace transform:

$$\mathcal{A}(m) = z \mapsto \int_{\mathbb{R}^+} e^{-xz} \mathrm{d}m(x).$$

Define the continuous covariance kernel by

$$\forall (z, z') \in \mathbb{X}^2, \quad C(z, z') \stackrel{\text{def.}}{=} \langle \varphi(z), \varphi(z') \rangle_{\mathcal{H}}.$$

The function C is the kernel of $\mathcal{A}^*\mathcal{A}$. The discrete covariance, defined on the computational grid, is $C = (C(z_i, z_{i'}))_{i,i'} \in \mathbb{R}^{N \times N}$, while its restriction to some support set I is $C_{I,I} = (C(z_i, z_{i'}))_{(i,i') \in I^2} \in \mathbb{R}^{|I| \times |I|}$.

By (11.17), the vector η_F samples the continuous precertificate $\tilde{\eta}_F$ on the grid:

$$\eta_F = (\tilde{\eta}_F(z_i))_{i=1}^N \in \mathbb{R}^N,$$

$$\text{where} \quad \tilde{\eta}_F(x) = \sum_{i \in I} b_i C(x, z_i) \quad \text{where} \quad b_I = C_{I,I}^{-1} \operatorname{sign}(x_{0,I}), \tag{11.23}$$

Thus η_F samples a linear combination of $|I|$ kernel functions $(C(x, z_i))_{i \in I}$.

The question is whether $\|\eta_F\|_{\ell^\infty} \leqslant 1$. Along increasingly dense grids, continuity implies that a uniform bound on the sampled certificate requires $\|\tilde{\eta}_F\|_{L^\infty} \leqslant 1$. A finite grid may miss an overshoot between samples. The construction constrains $\tilde{\eta}_F$ only to interpolate $\operatorname{sign}(x_{0,i})$ at support points z_i; it does not prevent the function from leaving $[-1, 1]$ nearby. Figure 11.11 illustrates this fact.

In one dimension, a certificate bounded in magnitude by one must satisfy $\eta'(z_i) = 0$ at every interior support point $i \in I$, where it already interpolates the sign. Adding these necessary conditions gives the minimum-norm precertificate with vanishing derivatives:

$$\tilde{\eta}_V = \mathcal{A}^* p_V, \qquad p_V = \operatorname*{argmin}_{p \in \mathcal{H}} \left\{ \|p\|_{\mathcal{H}} \; ; \; (\mathcal{A}^* p)(z_i) = \operatorname{sign}(x_{0,i}), \quad (\mathcal{A}^* p)'(z_i) = 0 \quad (i \in I) \right\}. \tag{11.24}$$

Here we restrict to a one-dimensional domain and assume that the interpolation constraints are independent. The adjoint is $(\mathcal{A}^* p)(z) = \langle \varphi(z), p \rangle_{\mathcal{H}}$, and $\eta_V = (\tilde{\eta}_V(z_i))_{i=1}^N$. A derivative constraint alone does not ensure $|\tilde{\eta}_V| \leqslant 1$; this bound must be checked separately. As in (11.23), this precertificate is a linear combination of kernel functions, now with $2|I|$ terms:

$$\tilde{\eta}_V(x) = \sum_{i \in I} b_i C(x, z_i) + c_i \partial_2 C(x, z_i),$$

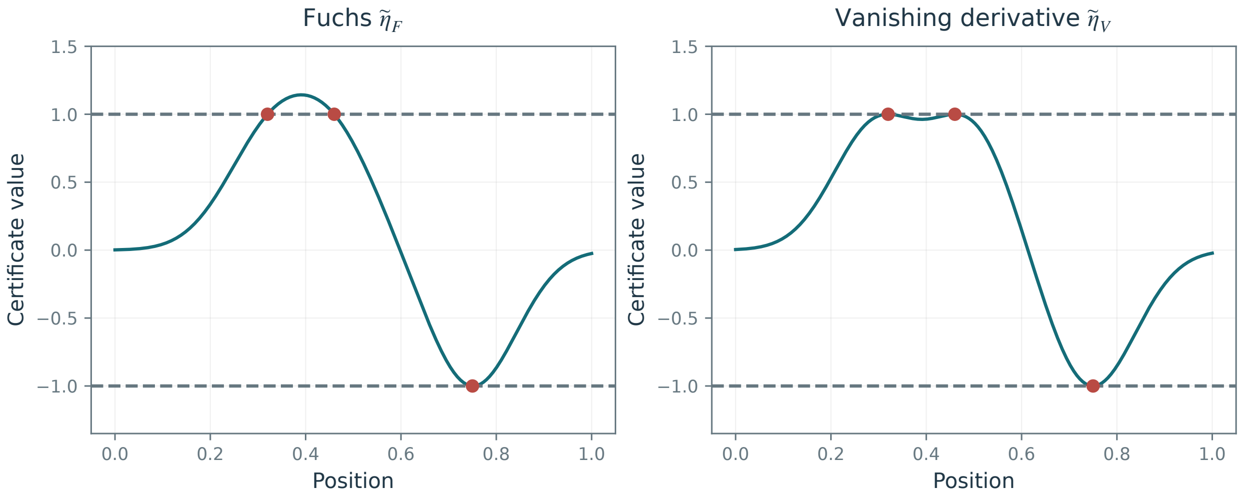


Figure 11.11. Continuous representations of the precertificates η_F and η_V for a convolution operator A.

where $\partial_2 C$ is the derivative of C with respect to the second variable, and (b, c) solve a $2|I| \times 2|I|$ linear system

$$\begin{pmatrix} b \\ c \end{pmatrix} = \begin{pmatrix} (C(z_i, z_{i'}))_{i,i' \in I^2} & (\partial_2 C(z_i, z_{i'}))_{i,i' \in I^2} \\ (\partial_1 C(z_i, z_{i'}))_{i,i' \in I^2} & (\partial_1 \partial_2 C(z_i, z_{i'}))_{i,i' \in I^2} \end{pmatrix}^{-1} \begin{pmatrix} \operatorname{sign}(x_{0,I}) \\ \mathbf{0}_I \end{pmatrix}.$$

Evaluating $\tilde{\eta}_V = \mathcal{A}^* p_V$ on the reconstruction grid gives the discrete precertificate η_V. Figure 11.11 shows better behavior of η_V than of η_F. If $\|\eta_V\|_\infty \leqslant 1$ and A is injective on its saturation set, Theorem 11.11 gives a linear rate in the grid-based ℓ^2 norm. As the grid is refined, however, this coefficient ℓ^2 norm does not provide an intrinsic distance between spike measures, which need not have L^2 densities. When η_V differs from η_F, one cannot directly apply Theorem 11.15: exact support stability on increasingly fine grids can fail in super-resolution problems. Methods without a fixed grid provide a more suitable framework for such stability results. In a grid-free formulation, a valid vanishing-derivative precertificate coincides with the minimum-norm certificate under the corresponding interpolation and nondegeneracy assumptions. This leads to stability results for spike locations and amplitudes, rather than a fixed-grid coefficient norm.

CHAPTER 12

Compressed Sensing

Compressed sensing aims to recover sparse signals from far fewer measurements than their ambient dimension. This can reduce acquisition costs when measurements are expensive, but success depends on how the sensing operator interacts with the signal representation. Motivated by the single-pixel camera, we analyze recovery from random measurements, distinguish guarantees for fixed signals from uniform guarantees, and extend the discussion to structured Fourier sampling.

Hardware constraints often prevent direct implementation of an ideal random matrix. The theory nevertheless guides the design of structured acquisition schemes, whose physical assumptions and computational costs must be considered separately.

The main references for this chapter are [23, 21, 29].

12.1 Motivation and Potential Applications

12.1.1 Single-Pixel Camera

The "single-pixel camera" developed at Rice University [18] motivates compressed acquisition. Figure 12.1 illustrates the reconstruction principle with simulated measurements of a flower image. A central idea is to combine acquisition and compression in a single device. Conventional acquisition first samples the analog scene $\tilde{f}_0$ at high resolution to obtain an image $f_0 \in \mathbb{R}^Q$, then compresses it to M significant coefficients using (5.3). Compressed acquisition instead records a vector $y \in \mathbb{R}^P$ directly. The aim is to keep the measurement count P close to the sparsity level M, typically with a logarithmic overhead, while retaining enough information for accurate reconstruction.

The scene $\tilde{f}_0$ is modeled as a light-intensity function. Its value $\tilde{f}_0(s)$ gives the intensity at position $s \in \mathbb{R}^2$ in the camera's focal plane. Light is focused onto an array of Q micromirrors in the focal plane. Each mirror directs light either toward the sensor or away from it; the mirrors themselves record no information. The device cycles rapidly through P mirror configurations. For $p = 1, \dots, P$, the entry $\Phi_{p,q} \in \{0, 1\}$ indicates whether mirror q is off or on during exposure p. A single sensor integrates the light directed toward it during each exposure, producing one measurement $y_p \in \mathbb{R}$. Let c_q be the region occupied by mirror q in the focal plane, and define its integrated intensity by $f_{0,q} = \int_{c_q} \tilde{f}_0(s)\,\mathrm{d}s$. The discrete image $f_0 \in \mathbb{R}^Q$ and the measurements $y \in \mathbb{R}^P$ are then related by

$$\forall\, p = 1, \dots, P, \quad y_p \approx \sum_{q=1}^{Q} \Phi_{p,q} \int_{c_q} \tilde{f}_0(s)\mathrm{d}s = (\Phi f_0)_p,$$

The symbol $\approx$ accounts for acquisition noise. We therefore use the usual inverse-problem model

$$y = \Phi f_0 + w \in \mathbb{R}^P$$

where w is the noise vector. The intermediate image f_0 is never recorded: the device measures y directly from the analog scene $\tilde{f}_0$. Binary mirror patterns are not centered random variables. Centered signed measurements can be formed by subtracting an appropriate total-intensity measurement or by differencing complementary exposures, followed by normalization. This distinction matters when applying the random-matrix assumptions below.

Recovering the ideal discrete image f_0 from y requires inverting the acquisition model while accounting for the noise w. Figure 12.1 uses nested subsets of signed Walsh–Hadamard patterns with a fixed random pixel permutation. Each reconstruction minimizes the same wavelet-ℓ^1 objective with its own measurements. The examples are noiseless simulations; the signed measurements correspond to centered patterns, rather than individual binary exposures.

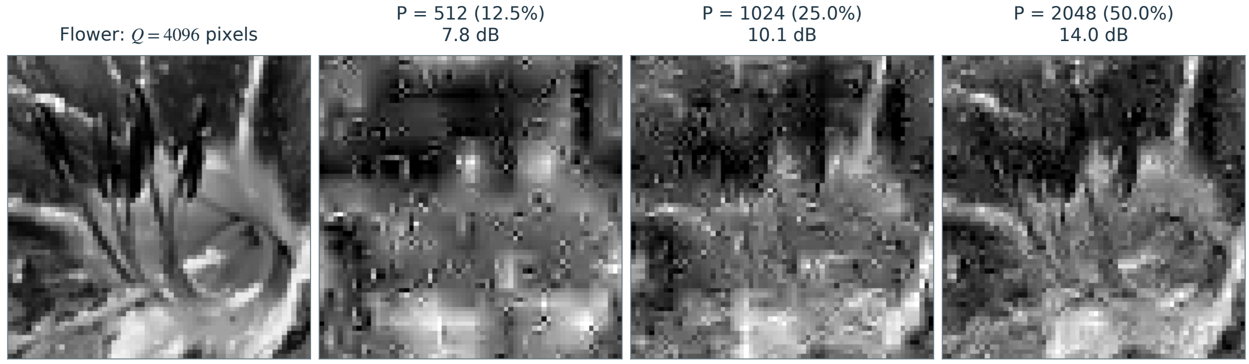


Figure 12.1. Simulated compressed sensing of the flower image, with $Q = 4096$ pixels. The three wavelet-ℓ^1 reconstructions use $P = 512$, 1024, and 2048 orthonormal signed Walsh–Hadamard measurements, respectively.

12.1.2 Sparse Recovery

Following Section 10.2, write $f_0 = \Psi x_0$, where $\Psi \in \mathbb{R}^{Q\times N}$ is a dictionary and x_0 is sparse. Denoting $A \overset{\text{def.}}{=} \Phi\Psi \in \mathbb{R}^{P\times N}$, this leads us to consider the usual ℓ^1 regularized problem (10.10)

$$x_\lambda \in \underset{x\in\mathbb{R}^N}{\operatorname{argmin}} \frac{1}{2\lambda}\|y - Ax\|^2 + \|x\|_1, \qquad (\mathcal{P}_\lambda(y))$$

so that the reconstructed image is $f_\lambda = \Psi x_\lambda$. We also sometimes consider the constrained problem

$$x_\varepsilon \in \underset{\|Ax-y\|\leqslant\varepsilon}{\operatorname{argmin}} \|x\|_1, \qquad (\mathcal{P}^\varepsilon(y))$$

Assume that a noise bound $\|w\| \leqslant \varepsilon$ is known, so that the true coefficient vector is feasible. Convex optimality conditions relate the two formulations through suitable Lagrange multipliers. The parameter correspondence depends on y and need not be one-to-one; a range of parameter values may produce the same solution. The endpoint $\varepsilon = 0$ may require a limiting parameter value.

The distribution of $A = \Phi\Psi$ depends on both the acquisition matrix and the dictionary. If Ψ is square and orthogonal and Φ has independent centered Gaussian entries with a common variance, right orthogonal invariance gives $\Phi\Psi$ the same distribution as Φ. For a general dictionary, this simplification fails. The results below therefore impose distributional assumptions directly on A.

12.2 Dual Certificate Theory and Non-Uniform Guarantees

12.2.1 Random Projection of Polytopes

For noiseless data, $w = 0$, we can use the geometric characterization of $(\mathcal{P}^0(Ax_0)) = (\mathcal{P}_0(Ax_0))$ from Section 11.1.2. Proposition 11.1 relates identifiable s-sparse vectors to $(s-1)$-dimensional faces of the ℓ^1 ball B_1 that survive projection onto AB_1. Recovery thus becomes a problem of counting faces of random polytopes. For Gaussian projections, the asymptotic analysis of Donoho and Tanner distinguishes two thresholds, described by functions C_A and C_M. In the proportional-growth regime, with probability tending to one as the dimensions increase,

– Every vector x_0 with $\|x_0\|_0 \leqslant C_A(P/N)P$ is identifiable.
– A typical support/sign pattern with sparsity at most $C_M(P/N)P$ is identifiable.

For illustration, at $P/N = 1/4$, the weak threshold is roughly one quarter of P. Strong thresholds require recovery for every support/sign pattern; weak thresholds concern a typical pattern. Their precise values depend on the asymptotic regime. Figure 12.5 illustrates the two transitions numerically. Above the weak threshold $C_M(P/N)P$, a typical support/sign pattern fails to be identifiable with high probability. The sharp transition between these regimes is called a phase transition. The function C_M can be computed numerically and has inverse-logarithmic decay, $C_M(r) \asymp 1/\log(1/r)$, as $r \to 0$. Thus, in the high-compression regime, typical sparse vectors remain recoverable when their sparsity is proportional to the number of measurements, up to a logarithmic factor.

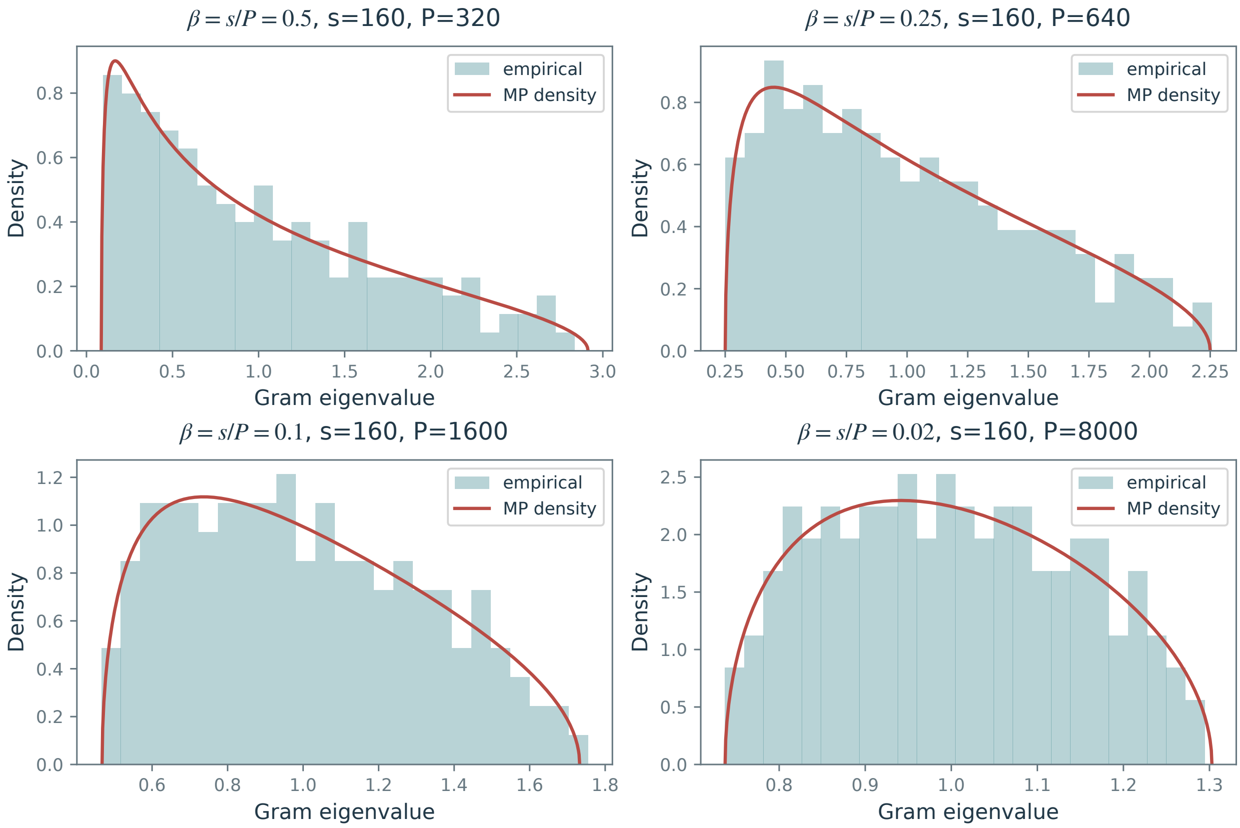


Figure 12.2. Empirical eigenvalue distributions approaching the Marchenko–Pastur law.

12.2.2 Random Matrices

To prove recovery guarantees, we need quantitative control of the singular values of random column submatrices. The Gaussian ensemble provides a useful starting point.

Let I contain s column indices and set $B = A_I \in \mathbb{R}^{P \times s}$. Throughout this subsection, the entries of A are independent $\mathcal{N}(0, 1/P)$ variables. The Gram matrix B^*B is a scaled Wishart matrix; when it is invertible, its inverse has the corresponding inverse-Wishart distribution. We need to control both matrices. For fixed s, the law of large numbers gives $B^*B \to \mathrm{Id}_s$ almost surely as $P \to \infty$. Compressed sensing also requires estimates when s grows with P.

Proportional growth. Suppose $s, P \to \infty$ with $s/P \to \beta \in (0, 1)$. The empirical eigenvalue distribution of B^*B converges almost surely to the Marchenko–Pastur law:

$$\frac{1}{s}\sum_{j=1}^{s} \delta_{\lambda_j(B^*B)} \rightharpoonup f_\beta(\lambda)\,\mathrm{d}\lambda, \qquad f_\beta(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi\beta\lambda} 1_{[\lambda_-,\lambda_+]}(\lambda),$$

where $\lambda_\pm = (1 \pm \sqrt{\beta})^2$. For Gaussian matrices, the extreme eigenvalues also converge to these endpoints. When $\beta < 1$ there is no atom at zero; such an atom appears when $s > P$. Figure 12.2 illustrates this convergence.

Finite-sample concentration. For a fixed support I of size s, Gaussian singular-value concentration gives, for $t > 0$, probability at least $1 - 2e^{-Pt^2/2}$ that

$$1 - \sqrt{s/P} - t \leqslant \sigma_{\min}(A_I) \leqslant \sigma_{\max}(A_I) \leqslant 1 + \sqrt{s/P} + t. \tag{12.1}$$

In particular, writing $a = \sqrt{s/P} + t$, on this event

$$\|A_I^*A_I - \mathrm{Id}_s\|_{\mathrm{op}} \leqslant 2a + a^2.$$

Thus the singular-value bounds also control the deviation of the Gram matrix from the identity. They show how increasing the measurement count P relative to the support size s improves conditioning.

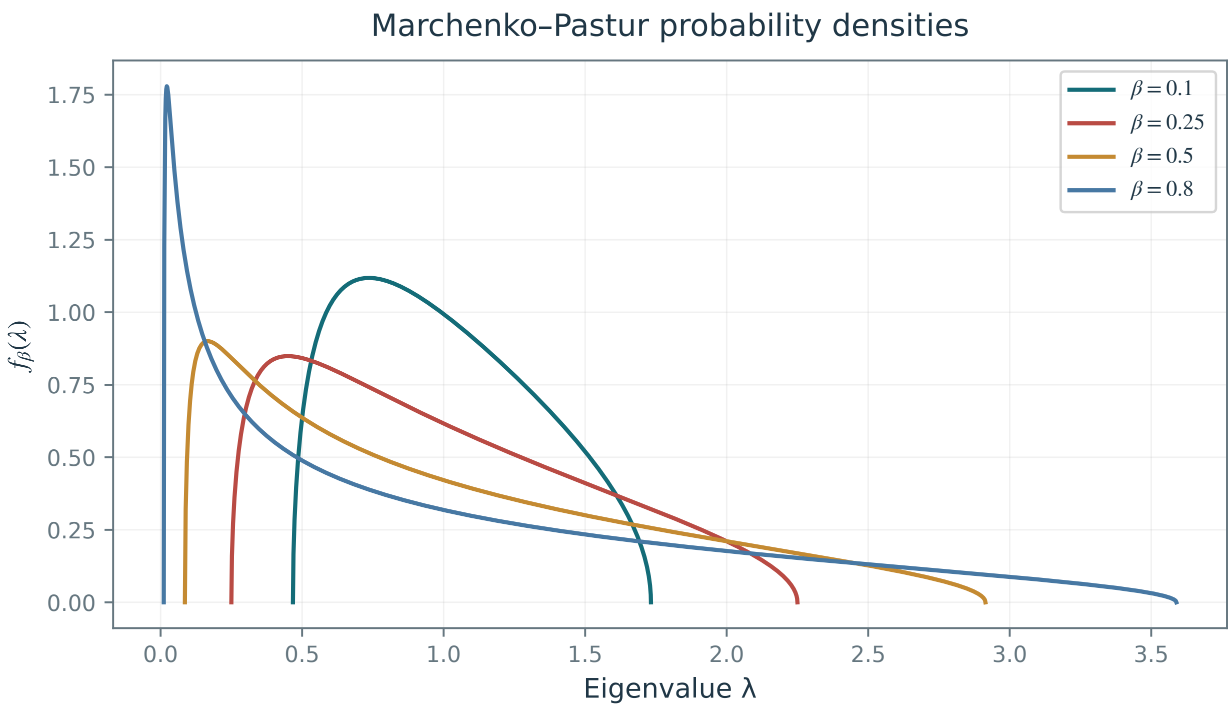


Figure 12.3. Marchenko–Pastur densities f_β for several aspect ratios β.

12.2.3 Dual Certificates

We seek both a small ℓ^2 reconstruction error and exact support recovery at low noise levels. Section 11.2.3 gives a sufficient condition: the Fuchs precertificate (11.16) must be nondegenerate, meaning

$$\|\eta_{F,I^c}\|_\infty < 1 \quad \text{where} \quad \eta_F = A^* A_I (A_I^* A_I)^{-1} \operatorname{sign}(x_{0,I}). \tag{12.2}$$

Figure 11.9 suggests nondegeneracy when P is sufficiently large relative to $\|x_0\|_0$. We now quantify this observation.

Coherence-based analysis. We begin with a bound based on the coherence of $A = (a_j)_{j=1}^N$. Assume that each column $a_j \in \mathbb{R}^P$ has unit norm. The coherence is

$$\mu \stackrel{\text{def.}}{=} \max_{i \neq j} |\langle a_i, a_j \rangle| \tag{12.3}$$

This is an entrywise maximum, not the induced ℓ^∞ matrix norm used below. An orthogonal matrix has coherence 0. In general, coherence is at most 1, so $\mu \in [0, 1]$. Small coherence controls the conditioning of submatrices with few columns and gives a sufficient condition for nondegeneracy of η_F.

Proposition 12.1. *Let $I = \operatorname{supp}(x_0)$ and $s = |I| \geqslant 1$. If $(s-1)\mu < 1$, then*

$$\|\eta_{F,I^c}\|_\infty \leqslant \frac{s\mu}{1-(s-1)\mu} \quad \text{and} \quad \|p_F\|^2 \leqslant \frac{s}{1-(s-1)\mu}. \tag{12.4}$$

In particular, if $\mu > 0$ and $s < \frac{1}{2}\left(1 + \frac{1}{\mu}\right)$, then $\|\eta_{F,I^c}\|_\infty < 1$, so Theorem 11.15 applies. If $\mu = 0$, the columns are orthonormal and the conclusion holds directly.

Proof. We recall that the ℓ^∞ operator norm (see Remark 11.14) is

$$\|B\|_\infty = \max_i \sum_j |B_{i,j}|.$$

Write $s_I = \operatorname{sign}(x_{0,I})$ and $C = A^*A$. Then

$$\|A_{I^c}^* A_I\|_\infty = \max_{j \in I^c} \sum_{i \in I} |C_{i,j}| \leqslant s\mu \quad \text{and} \quad \|\mathrm{Id}_s - A_I^* A_I\|_\infty = \max_{j \in I} \sum_{i \in I, i \neq j} |C_{i,j}| \leqslant (s-1)\mu$$

One also has

$$\begin{aligned}\|(A_I^*A_I)^{-1}\|_\infty &= \|(\mathrm{Id}_s - (\mathrm{Id}_s - A_I^*A_I))^{-1}\|_\infty = \|\sum_{k \geqslant 0} (\mathrm{Id}_s - A_I^*A_I)^k\|_\infty \\ &\leqslant \sum_{k \geqslant 0} \|\mathrm{Id}_s - A_I^*A_I\|_\infty^k \leqslant \sum_{k \geqslant 0} ((s-1)\mu)^k = \frac{1}{1-(s-1)\mu}\end{aligned}$$

The Neumann series converges because $(s-1)\mu < 1$. Using these two bounds, one has

$$\|\eta_{F,I^c}\|_\infty = \|A_{I^c}^* A_I (A_I^*A_I)^{-1} \operatorname{sign}(x_{0,I})\|_\infty \leqslant \|A_{I^c}^* A_I\|_\infty \|(A_I^*A_I)^{-1}\|_\infty \|\operatorname{sign}(x_{0,I})\|_\infty \leqslant (s\mu) \times \frac{1}{1-(s-1)\mu} \times 1.$$

One has

$$\frac{s\mu}{1-(s-1)\mu} < 1 \quad \Longleftrightarrow \quad 2s\mu < 1 + \mu$$

which gives the last statement. One also has

$$\|p_F\|^2 = \langle (A_I^*A_I)^{-1} s_I,\, s_I \rangle \leqslant \|(A_I^*A_I)^{-1}\|_\infty \|s_I\|_1 \leqslant \frac{s}{1-(s-1)\mu}$$

□

The proof applies to every sign pattern on every support satisfying the coherence condition. It also yields support containment for arbitrary noise levels when λ satisfies the noise-dependent ERC bound from the preceding chapter. An arbitrary choice of λ need not preserve the support.

For $N > P$, the Welch bound gives

$$\mu \geqslant \sqrt{\frac{N-P}{P(N-1)}} \tag{12.5}$$

which is equivalent to $1/\sqrt{P}$ for $N \gg P$. For Gaussian matrices with normalized columns $A \in \mathbb{R}^{P \times N}$, a union bound gives, with high probability when $\log N$ is small relative to P,

$$\mu = O\left(\sqrt{\log N / P}\right)$$

Thus Gaussian matrices attain the Welch scale up to a factor of order $\sqrt{\log N}$ when $N \gg P$. Ignoring logarithmic factors, Proposition 12.1 guarantees support stability for sparsities of order $\sqrt{P}$. This coherence bound is conservative: a direct probabilistic analysis permits sparsities of order P, again up to logarithmic factors.

Randomized analysis of the Fuchs certificate. For the probabilistic analysis, we specify independent entries with a common variance and a uniform Gaussian-type tail bound.

Definition 12.2 (sub-Gaussian random matrix)**.** A random matrix $A \in \mathbb{R}^{P \times N}$ is called normalized sub-Gaussian if its entries are independent, $\mathbb{E}(A_{i,j}) = 0$, $\mathbb{E}(A_{i,j}^2) = 1/P$, and, for some constants $\beta, \kappa > 0$ and every $t > 0$,

$$\mathbb{P}(|\sqrt{P} A_{i,j}| \geqslant t) \leqslant \beta e^{-\kappa t^2}. \tag{12.6}$$

The entries may have different distributions, but the tail constants (β, κ) must be uniform over (i, j). The normalization by $\sqrt{P}$ is consistent with unit expected squared column norm, $\mathbb{E}(\|a_j\|^2) = 1$, where a_j denotes a column of A.

Examples include Gaussian matrices with entry variance $1/P$ and matrices with independent entries taking the values $\pm 1/\sqrt{P}$ with equal probability.

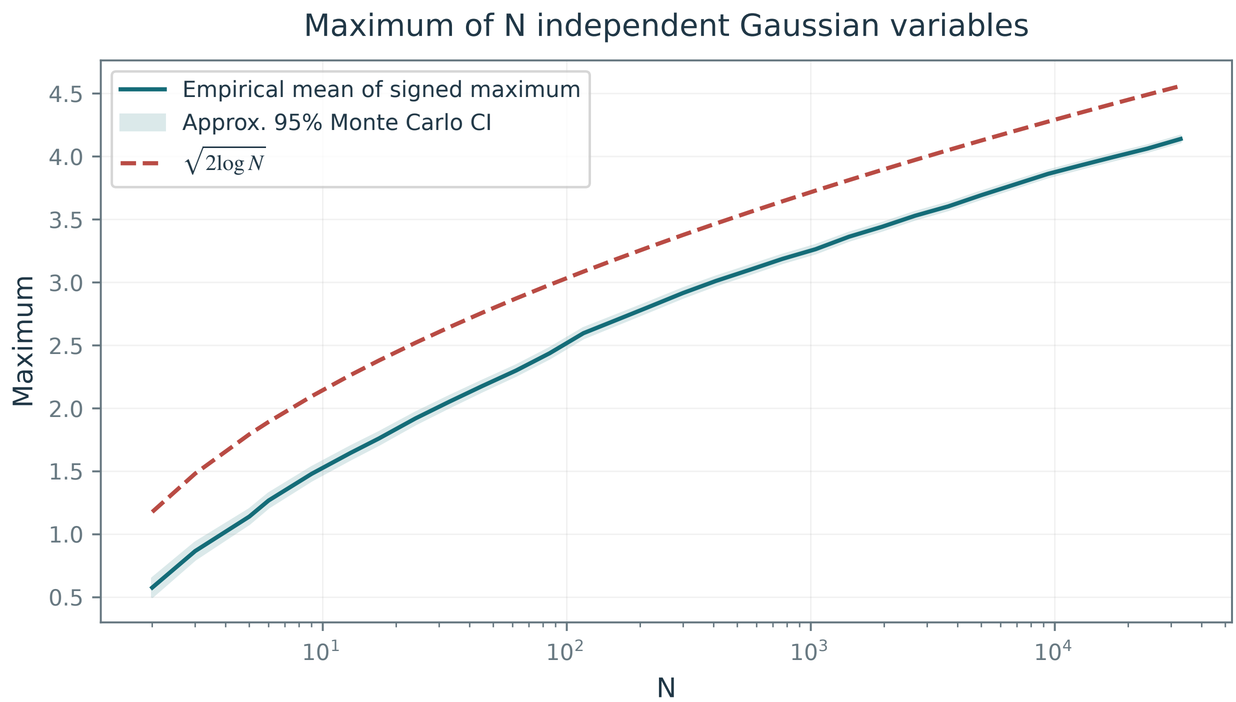


Figure 12.4. Maximum of N independent standard Gaussian variables, compared with the dashed curve $\sqrt{2\log(N)}$.

Theorem 12.3. *Fix a nonzero $x_0 \in \mathbb{R}^N$, independently of A, and write $s = \|x_0\|_0$. For a normalized sub-Gaussian matrix, there is a constant C, depending only on the sub-Gaussian parameters, such that*

$$P \geqslant Cs\log(2N/\varepsilon), \qquad 0 < \varepsilon < 1,$$

implies, with probability at least $1-\varepsilon$, that A_I is injective and $\|\eta_{F,I^c}\|_\infty < 1$. The support-stability conclusion of Theorem 11.15 then holds for sufficiently small λ and sufficiently small $\|A^*w\|_\infty/\lambda$.

Proof. We outline the concentration argument. With probability at least $1-\varepsilon/2$, taking P as in the theorem with a sufficiently large constant gives $\sigma_{\min}(A_I) \geqslant 1/2$, hence

$$\|p_F\|^2 = \langle (A_I^*A_I)^{-1}s_I, s_I\rangle \leqslant 4s, \qquad s_I = \operatorname{sign}(x_{0,I}). \tag{12.7}$$

Conditioning on A_I fixes p_F, while the off-support columns a_j remain independent of this vector. The probability that $|\langle a_j, p_F\rangle|$ reaches or exceeds 1 is bounded by $2\exp(-cP/\|p_F\|^2)$. A union bound over at most N columns gives the stated sample complexity and the strict off-support bound, by excluding $|\langle a_j, p_F\rangle| \geqslant 1$.

For Gaussian entries, conditional on A_I, these inner products are Gaussian with variance $\|p_F\|^2/P$. When s/P is small, $\|p_F\|^2$ is close to s, so the largest of the $N-s$ absolute correlations is approximately $\sqrt{2s\log(N-s)/P}$. This gives the sharper leading-order Gaussian threshold $P \approx 2s\log N$.

Rotational invariance makes this statement more precise:

$$\frac{Ps}{\|p_F\|^2} \sim \chi^2_{P-s+1}.$$

To see this, rotate the s columns so that $s_I/\sqrt{s}$ becomes the first coordinate vector. The inverse of the corresponding diagonal entry of $(G^*G)^{-1}$, for $G = \sqrt{P}A_I$, is the squared distance of the first column of G to the span of the remaining columns. This is a chi-square variable with $P-s+1$ degrees of freedom. Thus $\|p_F\| \leqslant (1+\delta)\sqrt{s}$ with high probability when s/P is sufficiently small and P is large; the factor must exceed one. $\square$

In a Gaussian asymptotic regime with $s/P \to 0$, the leading-order scaling is

$$P \gtrsim 2s\log(2N/\varepsilon), \tag{12.8}$$

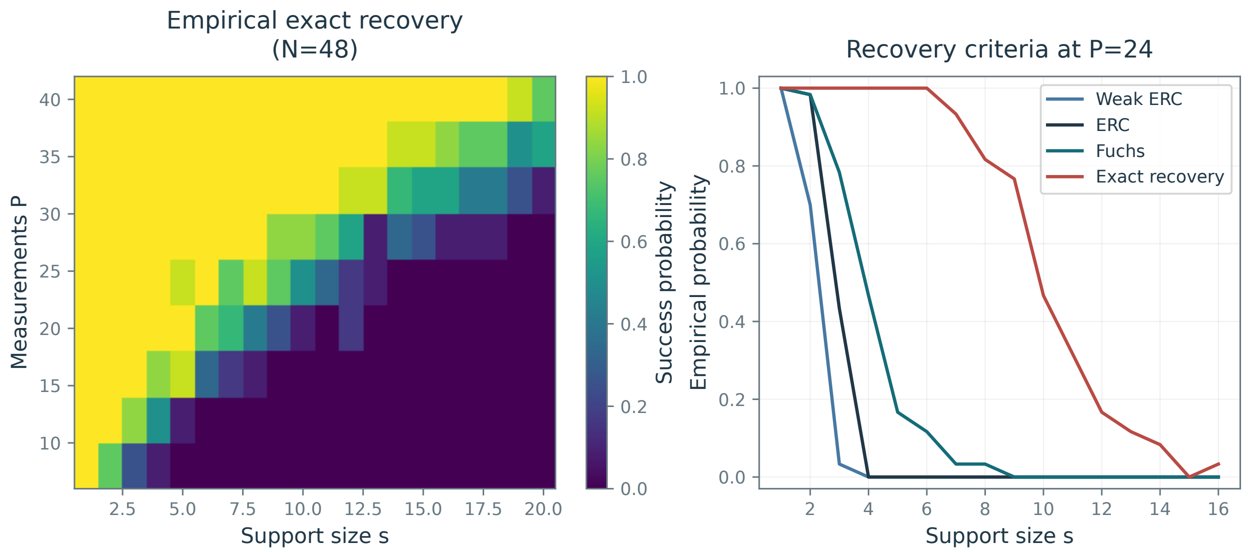


Figure 12.5. Phase transitions. Right: empirical probabilities that recovery criteria hold as sparsity varies. Blue: weak ERC; black: ERC; green: $\|\eta_F\|_\infty \leqslant 1$; red: exact recovery.

with slack needed for a finite-sample probability bound. The unspecified constant in Theorem 12.3 applies to general sub-Gaussian ensembles; it should not be identified with this Gaussian leading constant.

This is a nonuniform guarantee: the vector x_0 is fixed before the matrix A is drawn. The RIP theory in Section 12.3 gives a single event on which recovery holds for all sparse vectors simultaneously.

12.3 RIP Theory for Uniform Guarantees

12.3.1 Restricted Isometry Constants

For a positive integer s, the restricted isometry constant δ_s of $A \in \mathbb{R}^{P\times N}$ is the smallest nonnegative number such that

$$\forall z \in \mathbb{R}^N, \quad \|z\|_0 \leqslant s \quad \Longrightarrow \quad (1-\delta_s)\|z\|^2 \leqslant \|Az\|^2 \leqslant (1+\delta_s)\|z\|^2, \tag{12.9}$$

A small δ_s therefore means that A approximately preserves the Euclidean norm on every coordinate subspace of dimension at most s.

A related quantity is the restricted orthogonality constant $\theta_{s,s'}$: the smallest nonnegative number such that, whenever $\|x\|_0 \leqslant s$, $\|x'\|_0 \leqslant s'$ and the supports are disjoint,

$$|\langle Ax, Ax'\rangle| \leqslant \theta_{s,s'}\|x\|\|x'\|$$

The next lemma relates restricted isometry and restricted orthogonality.

Lemma 12.4. *One has*

$$\theta_{s,s'} \leqslant \delta_{s+s'} \leqslant \theta_{s,s'} + \max(\delta_s, \delta_{s'}).$$

Proof. For the first inequality, normalize nonzero disjointly supported vectors so that $\|z\| = \|z'\| = 1$. RIP applied to $z + z'$ and $z - z'$ and polarization give

$$|\langle Az, Az'\rangle| = \frac{1}{4}\left|\|Az + Az'\|^2 - \|Az - Az'\|^2\right| \leqslant \delta_{s+s'}.$$

Rescaling gives the general claim. For the second inequality, split any vector supported on at most $s + s'$ indices as $x + x'$ with disjoint supports of sizes at most s, s'. Then

$$\left|\|A(x+x')\|^2 - \|x+x'\|^2\right| \leqslant \max(\delta_s, \delta_{s'})(\|x\|^2 + \|x'\|^2) + 2\theta_{s,s'}\|x\|\|x'\|,$$

and use $2\|x\|\|x'\| \leqslant \|x\|^2 + \|x'\|^2$. $\square$

The next theorem bounds the number of measurements needed to keep restricted isometry constants small at sparsity level s.

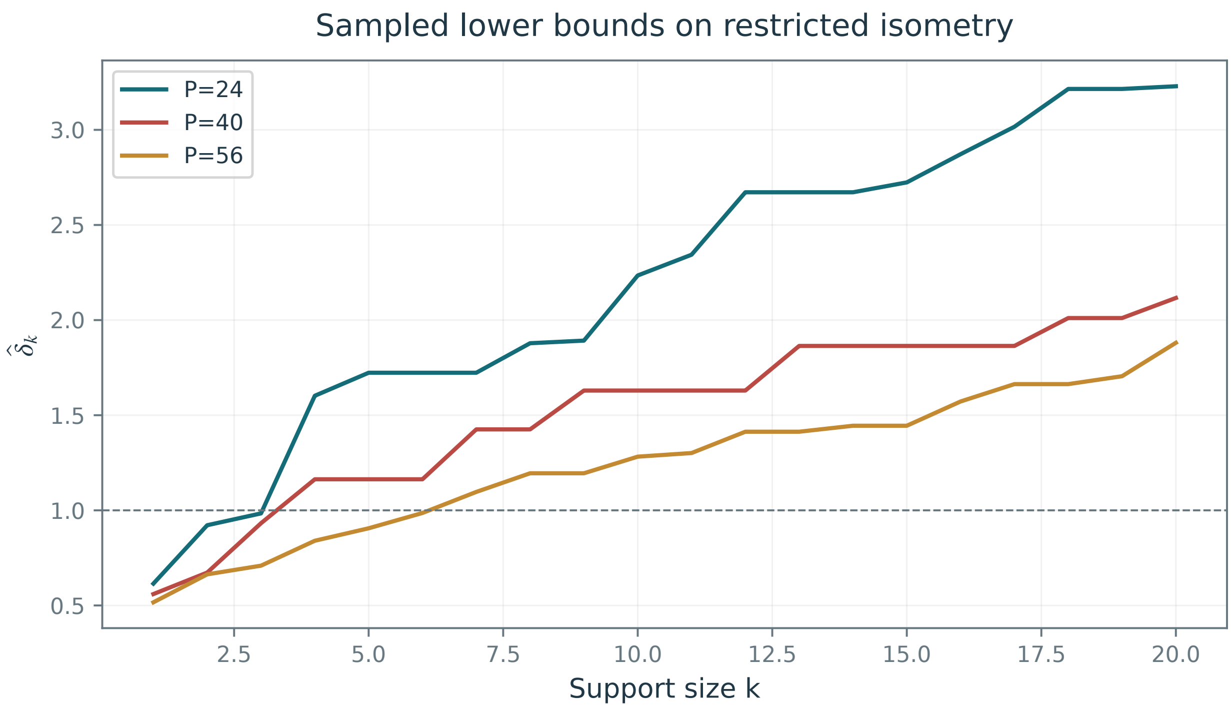


Figure 12.6. Evolution of lower bounds $\hat{\delta}_k$ on the RIP constant.

Theorem 12.5. *For $0 < \delta < 1$ and $1 \leqslant s \leqslant N$, if A is a normalized sub-Gaussian matrix and*

$$P \geqslant C\delta^{-2} s \log(eN/s) \tag{12.10}$$

it satisfies $\delta_s \leqslant \delta$ with probability at least $1 - 2e^{-\delta^2 \frac{P}{2C}}$, where C depends only on the sub-Gaussian parameters in (12.6).

The proof combines concentration on each support with a union bound over supports. The RIP condition (12.9) is equivalent to the spectral bound $\operatorname{eig}(A_I^* A_I) \subset [1 - \delta_s, 1 + \delta_s]$ for every column submatrix A_I with $|I| \leqslant s$. Thus the proof reduces to controlling singular values uniformly over supports. For Gaussian matrices, Section 12.2.2 provides these estimates for $B^* B \in \mathbb{R}^{s \times s}$, where $B \in \mathbb{R}^{P \times s}$ has independent $\mathcal{N}(0, 1/P)$ entries. The concentration bound (12.1) controls δ_s on each support. Analogous nonasymptotic estimates hold under the stated sub-Gaussian assumptions; finite covariance alone would not suffice. For a fixed support, the smallest eigenvalue is close to $(1 - \sqrt{s/P})^2$, a deficit of $2\sqrt{s/P} - s/P$ from 1. Thus a fixed tolerance requires P to scale with s. A union bound over all supports adds the logarithmic factor. This step requires nonasymptotic concentration, which the Marchenko–Pastur limit alone does not provide.

12.3.2 RIP Implies Dual Certificates

The following theorem shows that a sufficiently small restricted isometry constant guarantees a valid dual certificate. Theorem 11.11 then yields stable recovery of sparse signals.

Theorem 12.6. *Assume $\delta_{2s} + \theta_{s,s} + \theta_{s,2s} < 1$. For every x_0 with $\|x_0\|_0 \leqslant s$, there is a certificate $\eta = A^* p$ satisfying $\eta_I = \operatorname{sign}(x_{0,I})$, where $I = \operatorname{supp}(x_0)$, and*

$$\|\eta_{I^c}\|_\infty \leqslant \frac{\theta_{s,s}}{1 - \delta_{2s} - \theta_{s,2s}} < 1.$$

Moreover, if $Q = \theta_{s,2s}/(1 - \delta_{2s})$, one may choose

$$\|p\| \leqslant \frac{\sqrt{s}}{\sqrt{1 - \delta_s}} + \frac{\theta_{s,s}\sqrt{s}}{(1 - \delta_s)\sqrt{1 - \delta_{2s}}(1 - Q)}.$$

Lemma 12.4 gives

$$\delta_{2s} + \theta_{s,s} + \theta_{s,2s} \leqslant \delta_{2s} + \delta_{2s} + \delta_{3s} \leqslant 3\delta_{3s}$$

so that condition $\delta_{2s} + \theta_{s,s} + \theta_{s,2s} < 1$ is implied by $\delta_{3s} < 1/3$. Sharper recovery conditions are available, for example $\delta_{2s} < \sqrt{2} - 1$ for robust recovery [6]. These sufficient bounds on restricted isometry constants are commonly referred to as restricted isometry properties (RIP). The numerical constants in elementary RIP bounds can be conservative, so the resulting sufficient sample sizes may exceed the observed phase-transition thresholds.

The iterative construction below follows [8], under a slightly stronger hypothesis than the sharpest certificate results. Rather than requiring the Fuchs precertificate η_F in (12.2) to be feasible, we correct its large off-support entries. At each step, an interpolant cancels the largest s entries outside I while preserving the values on the target support. To construct these corrections to η_F, we first bound minimum-norm interpolants using restricted isometry and restricted orthogonality.

Lemma 12.7. *We assume $\delta_s < 1$. Let $c \in \mathbb{R}^N$ with $\|c\|_0 \leqslant s$ and $\operatorname{supp}(c) = J$, and let $s' \geqslant 1$ be an integer. Let p be the minimum-norm solution of $A_J^* p = c_J$ and set $\eta = A^* p$. Explicitly,*

$$p \overset{\text{def.}}{=} A_J^{*,+} c_J = A_J (A_J^* A_J)^{-1} c_J$$

(note that A_J is full rank because $\delta_s < 1$). Then, letting $K \subset J^c$ index the $\min(s', |J^c|)$ largest entries in magnitude of η_{J^c}, one has

$$\|\eta_{(J\cup K)^c}\|_\infty \leqslant \frac{\theta_{s,s'}\|c\|}{(1-\delta_s)\sqrt{s'}} \quad \textit{and} \quad \|\eta_K\| \leqslant \frac{\theta_{s,s'}\|c\|}{1-\delta_s} \quad \textit{and} \quad \|p\| \leqslant \frac{\|c\|}{\sqrt{1-\delta_s}}$$

Proof. Since $|J| \leqslant s$, the smallest eigenvalue satisfies $\lambda_{\min}(A_J^* A_J) \geqslant 1 - \delta_s$. Hence

$$\|(A_J^* A_J)^{-1}\| \leqslant \frac{1}{1-\delta_s}$$

One has

$$\|p\|^2 = \langle A_J(A_J^*A_J)^{-1}c_J,\, A_J(A_J^*A_J)^{-1}c_J\rangle = \langle (A_J^*A_J)^{-1}c_J,\, c_J\rangle \leqslant \frac{\|c\|^2}{1-\delta_s}.$$

Let L be any set with $L \cap J = \emptyset$ and $|L| \leqslant s'$. One has

$$\|\eta_L\|^2 = |\langle \eta_L,\, \eta_L\rangle| = |\langle A_J(A_J^*A_J)^{-1}c_J,\, A_L\eta_L\rangle| \leqslant \theta_{s,s'}\|(A_J^*A_J)^{-1}c_J\|\|\eta_L\| \leqslant \frac{\theta_{s,s'}}{1-\delta_s}\|c_J\|\|\eta_L\|.$$

If $\eta_L \neq 0$, divide by $\|\eta_L\|$; if $\eta_L = 0$, the bound is immediate. In either case,

$$\|\eta_L\| \leqslant \frac{\theta_{s,s'}}{1-\delta_s}\|c_J\|. \tag{12.11}$$

Let us denote $\bar{K} = \{k \in J^c \;;\; |\eta_k| > T\}$ where $T \overset{\text{def.}}{=} \frac{\theta_{s,s'}}{(1-\delta_s)\sqrt{s'}}\|c_J\|$. One necessarily has $|\bar{K}| \leqslant s'$, otherwise one would have, taking $K \subset \bar{K}$ the s' largest entries (in fact any s' entries)

$$\|\eta_K\| > \sqrt{s'T^2} = \frac{\theta_{s,s'}}{1-\delta_s}\|c_J\|$$

This contradicts (12.11). Consequently, all entries of η_{J^c} beyond rank s' have magnitude at most T. □

We can now prove Theorem 12.6.

Proof. For $x_0 = 0$, take $p = 0$. Otherwise, let $I_0 = \operatorname{supp}(x_0)$ and apply Lemma 12.7 with $J = I_0$, $c_{I_0} = \operatorname{sign}(x_{0,I_0})$, and $s' = s$. It gives $\eta_1 = A^* p_1$ and a set $I_1 \subset I_0^c$, $|I_1| \leqslant s$, such that

$$\eta_{1,I_0} = \operatorname{sign}(x_{0,I_0}), \quad \|\eta_{1,I_1}\| \leqslant b\sqrt{s}, \quad \|\eta_{1,(I_0\cup I_1)^c}\|_\infty \leqslant b, \qquad b = \frac{\theta_{s,s}}{1-\delta_s}.$$

Also $\|p_1\| \leqslant \sqrt{s}/\sqrt{1-\delta_s}$. Given η_n and I_n, apply the lemma on $J = I_0 \cup I_n$ to the interpolation values $(0, \eta_{n,I_n})$, using $(2s, s)$ in place of (s, s'). This produces $\eta_{n+1} = A^* p_{n+1}$ and I_{n+1} disjoint from $I_0 \cup I_n$, with

$$\eta_{n+1,I_0} = 0, \quad \eta_{n+1,I_n} = \eta_{n,I_n}, \quad \|\eta_{n+1,I_{n+1}}\| \leqslant b\sqrt{s}\,Q^n,$$

$$\|\eta_{n+1,(I_0\cup I_n\cup I_{n+1})^c}\|_\infty \leqslant bQ^n, \qquad Q = \frac{\theta_{s,2s}}{1-\delta_{2s}} < 1. \tag{12.12}$$

The denominator involves δ_{2s} because the interpolation support can have $2s$ indices. Furthermore,

$$\|p_{n+1}\| \leqslant \frac{b\sqrt{s}}{\sqrt{1-\delta_{2s}}} Q^{n-1}.$$

Thus $p = \sum_{n\geqslant 1}(-1)^{n-1}p_n$ converges and $\eta = A^* p$ interpolates the signs on I_0.

Fix $j \notin I_0$. Whenever $j \in I_n$, the two contributions $\eta_{n,j}$ and $\eta_{n+1,j}$ cancel. Since consecutive I_n are disjoint, these cancellations do not overlap. Every remaining term obeys the pointwise geometric bound, including the initial bound for $n = 1$. Hence

$$|\eta_j| \leqslant \sum_{n\geqslant 1} bQ^{n-1} = \frac{b}{1-Q} \leqslant \frac{\theta_{s,s}}{1-\delta_{2s}-\theta_{s,2s}} < 1.$$

Summing the norm bounds on p_1 and the remaining p_n gives the stated bound on $\|p\|$. □

12.3.3 RIP Implies Stable Recovery

For uniform stability, it is convenient to use the noise-constrained formulation ($\mathcal{P}^\varepsilon(y)$). The standard robust RIP estimate [6], together with Theorem 12.5, gives the following statement.

Theorem 12.8. *Let $1 \leqslant s \leqslant N$ and let A be a normalized sub-Gaussian matrix. There are constants $C, c, C_0, C_1 > 0$, depending only on its sub-Gaussian parameters, such that if*

$$P \geqslant Cs\log(eN/s), \tag{12.13}$$

then, with probability at least $1-2e^{-cP}$, the following holds simultaneously for every $x_0 \in \mathbb{R}^N$, every $\varepsilon \geqslant 0$, every noise vector w with $\|w\| \leqslant \varepsilon$, and every minimizer x_ε of ($\mathcal{P}^\varepsilon(y)$) with $y = Ax_0 + w$:

$$\|x_\varepsilon - x_0\| \leqslant C_0\varepsilon + C_1\frac{\|x_0 - x_{0,s}\|_1}{\sqrt{s}}.$$

Here $x_{0,s}$ is a best s-term approximation. In particular, exactly s-sparse vectors are recovered with error at most $C_0\varepsilon$, and uniquely when $\varepsilon = 0$.

The penalized formulation also admits linear noise-error bounds: combine Theorem 12.6 with Theorem 11.11 and choose λ accordingly. The bounds hold for every minimizer, but do not assert uniqueness of noisy Lasso solutions. For approximately sparse vectors, the constrained estimate measures the ℓ^1 approximation tail divided by $\sqrt{s}$, without an additional factor $\|A\|$.

The order of the quantifiers distinguishes uniform from nonuniform recovery. In the noiseless case, this theorem bounds the probability of a single event on which every s-sparse vector is recovered. Theorem 12.3 instead bounds the recovery probability separately for each vector fixed independently of A.

For uniform norm recovery, RIP analysis replaces $\log N$ by $\log(eN/s)$. The Fuchs analysis using η_F addresses an additional question: stability of the support. The guarantees also differ in their constants. Elementary RIP estimates can be conservative, while Gaussian fixed-signal calculations give a sharp leading constant in their asymptotic regime.

The uniform certificate constructed above need not coincide with the Fuchs precertificate η_F.

12.3.4 RIP for Fourier Sampling

Fully independent random measurements can be difficult to implement. A practical alternative is to sample a random subset of coefficients in an orthonormal basis $\Xi = (\xi_\omega)_{\omega=1}^N$ of $\mathbb{R}^N$ or $\mathbb{C}^N$:

$$Ax \overset{\text{def.}}{=} \sqrt{N/P}\,(\langle x,\, \xi_\omega\rangle)_{\omega\in\Omega} \tag{12.14}$$

The factor $\sqrt{N/P}$ makes the expected Gram matrix the identity. For a complex basis, the measurements are complex. The index set $\Omega \subset \{1,\dots,N\}$ is sampled uniformly without replacement, with $|\Omega| = P$. The following theorem gives an RIP guarantee when the basis vectors are sufficiently spread out, as quantified by

$$\rho(\Xi) \overset{\text{def.}}{=} \sqrt{N}\max_{1\leqslant\omega\leqslant N}\|\xi_\omega\|_\infty.$$

Theorem 12.9 (Rudelson–Vershynin)**.** *For every fixed* $0 < c < 1$ *and failure exponent* $\gamma > 0$*, there is a constant* $C = C(c, \gamma)$ *such that, provided*

$$P \geqslant C\rho(\Xi)^2 s \log(2N)^4 \tag{12.15}$$

the normalized operator (12.14) *satisfies* $\delta_{2s} \leqslant c$ *with probability at least* $1 - N^{-\gamma}$.

One always has $1 \leqslant \rho(\Xi)^2 \leqslant N$. The worst case is the Dirac basis, $\Xi = \mathrm{Id}_N$, for which $\rho(\Xi)^2 = N$. Indeed, uniform recovery of even one-sparse vectors then requires observing every coordinate, so $P = N$. Fourier atoms $\xi_\omega = (N^{-1/2} e^{\frac{2i\pi}{N}\omega n})_{n=1}^N \in \mathbb{C}^N$ attain the smallest possible value, $\rho(\Xi) = 1$; Hadamard bases do so as well. Up to logarithmic factors, the sample count (12.15) then matches the sub-Gaussian scaling (12.13).

Theorem 12.9 assumes that x_0 is sparse in the coordinate basis. If instead f_0 is sparse in an orthonormal basis $\Psi = (\psi_m)_{m=1}^N$, use the coordinates $x = \Psi^* f$, so that x contains the coefficients of f in Ψ. The factor $\rho(\Xi)$ in (12.15) is then replaced by the mutual coherence of the sampling and sparsity bases:

$$\rho(\Psi^*\Xi) \overset{\text{def.}}{=} \sqrt{N} \max_{1 \leqslant \omega, m \leqslant N} |\langle \psi_m, \xi_\omega \rangle|.$$

Small mutual coherence therefore yields stronger measurement bounds. Fourier and Dirac bases are maximally incoherent. Fourier and wavelet bases can have large correlations, whereas specially constructed "noiselet" bases have low coherence with suitable wavelets. Unlike the Gaussian ensemble, these structured measurements are not universal: their recovery guarantees depend on the sparsity basis Ψ.

CHAPTER 13

Basics of Machine Learning

Machine learning seeks useful structure in data and predictors that remain accurate on new observations. We begin with dimensionality reduction and clustering, then develop regression and classification through empirical risk minimization, regularization, and kernel methods. Connections with inverse problems explain the role of model assumptions, while the final analysis of generalization balances approximation error against the statistical complexity of the chosen model class.

Imaging problems often concern the reconstruction or processing of an individual signal or image, whereas machine learning studies patterns across collections of observations. Their goals and performance measures differ, but both use tools such as linear models, regularization, and convex optimization.

13.1 Unsupervised Learning

In unsupervised learning, we observe n points $(x_i)_{i=1}^n$. The aim is to identify structure in these observations, for example to visualize them or group them into clusters. We assume for simplicity that the observations lie in Euclidean space, $x_i \in \mathbb{R}^p$, where p is the number of features. We store the observations as the rows of a matrix $X \in \mathbb{R}^{n\times p}$.

13.1.1 Dimensionality Reduction and PCA

Dimensionality reduction produces a compact representation of the data for visualization or subsequent analysis. By retaining informative features, it can also reduce computational cost and improve prediction. Principal component analysis (PCA) projects the data orthogonally onto the principal axes of the covariance matrix.

Presentation of the method. The empirical mean is defined as

$$\hat{m} \stackrel{\text{def.}}{=} \frac{1}{n}\sum_{i=1}^n x_i \in \mathbb{R}^p$$

and the empirical covariance as

$$\hat{C} \stackrel{\text{def.}}{=} \frac{1}{n}\sum_{i=1}^n (x_i - \hat{m})(x_i - \hat{m})^* \in \mathbb{R}^{p\times p}. \tag{13.1}$$

With the centered data matrix $\tilde{X} \stackrel{\text{def.}}{=} X - \mathbb{1}_n \hat{m}^*$, the covariance is $\hat{C} = \tilde{X}^*\tilde{X}/n$.

Suppose the points $(x_i)_i$ are i.i.d. with a finite second moment, and let $\mathbf{x}$ denote a random variable with their common distribution. The law of large numbers gives the following almost sure limits as $n \to +\infty$:

$$\hat{m} \to m \stackrel{\text{def.}}{=} \mathbb{E}(\mathbf{x}) \quad \text{and} \quad \hat{C} \to C \stackrel{\text{def.}}{=} \mathbb{E}((\mathbf{x}-m)(\mathbf{x}-m)^*). \tag{13.2}$$

Let μ be the distribution on $\mathbb{R}^p$ of $\mathbf{x}$. Then these limits can also be written as

$$m = \int_{\mathbb{R}^p} x \mathrm{d}\mu(x) \quad \text{and} \quad C = \int_{\mathbb{R}^p} (x-m)(x-m)^* \mathrm{d}\mu(x).$$

The PCA basis consists of the right singular vectors of the centered data matrix. Proposition 9.1 gives the reduced singular value decomposition

$$\tilde{X} = \sqrt{n} U \operatorname{diag}(\sigma) V^*$$

where $U \in \mathbb{R}^{n\times r}$ and $V \in \mathbb{R}^{p\times r}$ have orthonormal columns, and $r = \operatorname{rank}(\tilde{X}) \leqslant \min(n,p)$. Write $V = (v_k)_{k=1}^r$ for the orthonormal columns, which are eigenvectors of $\hat{C} = V \operatorname{diag}(\sigma^2) V^\top$, with $v_k \in \mathbb{R}^p$. These vectors

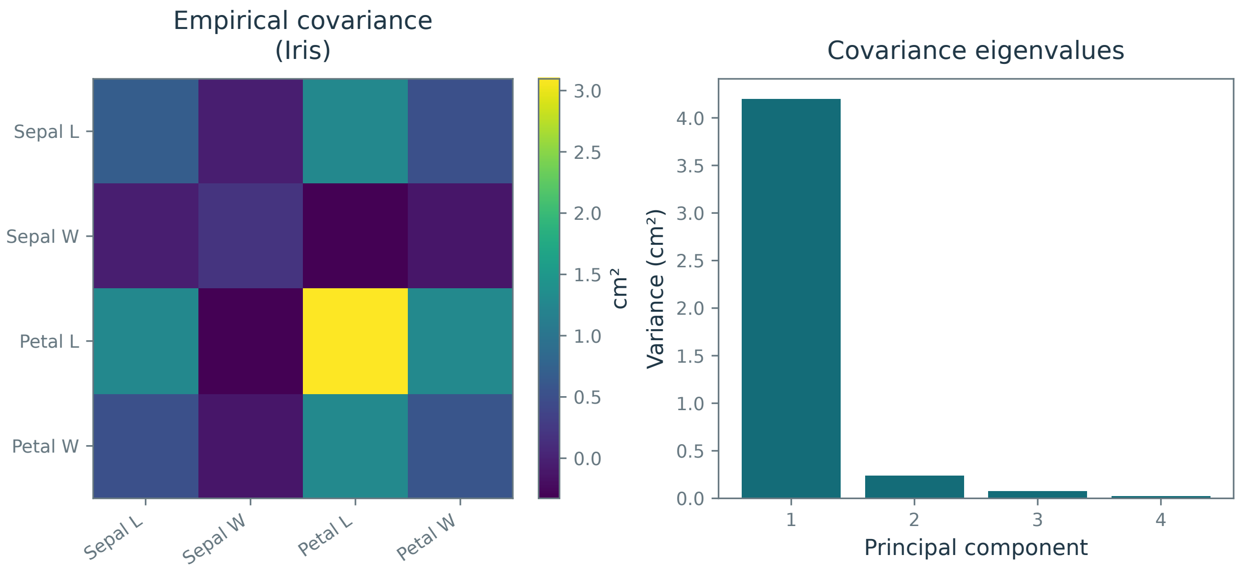
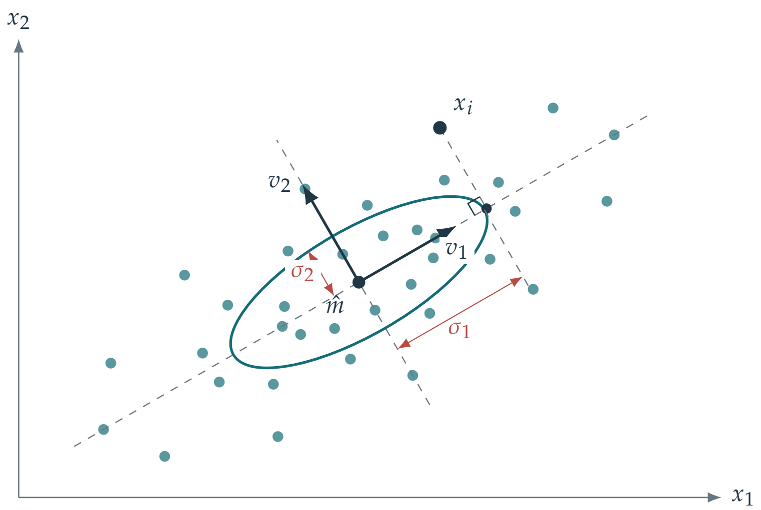


Figure 13.1. Empirical covariance of the Iris measurements and its eigenvalues.



$$\widehat{C} v_j = \sigma_j^2 v_j, \qquad \|v_j\| = 1$$

$$\frac{1}{n} \sum_i \langle x_i - \hat{m}, v_j \rangle^2 = \sigma_j^2$$

Project first; the variance is the mean squared coordinate along that direction.

Figure 13.2. Principal directions of variation.

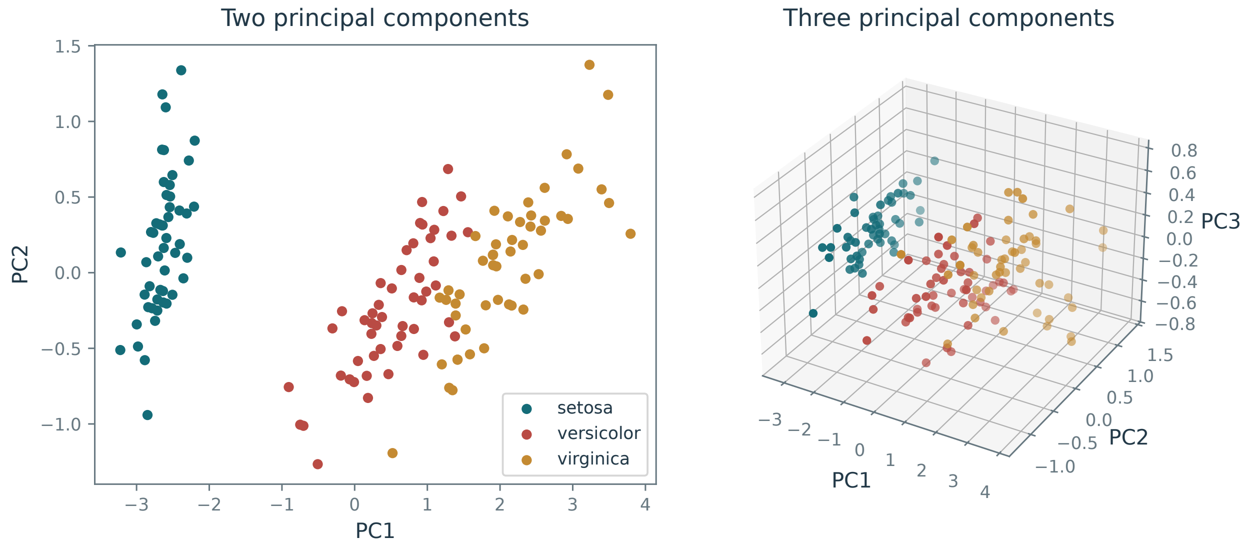


Figure 13.3. PCA projections of the data into two and three dimensions.

describe the principal directions of variation in the point cloud $(x_i)_i$ in $\mathbb{R}^p$. We order the singular values as $\sigma_1 \geqslant \ldots \geqslant \sigma_r$, so the leading components account for the greatest variance.

Figure 13.1 displays an empirical covariance matrix and its spectrum $(\sigma_k^2)_k$ for the Iris dataset[1] introduced by Fisher. It contains 50 samples from each of three iris species: Iris setosa, Iris virginica, and Iris versicolor. The four features are sepal length, sepal width, petal length, and petal width.

To obtain a PCA embedding $x_i \in \mathbb{R}^p \mapsto z_i \in \mathbb{R}^d$ of dimension $d \leqslant p$, project onto the first d right singular vectors, extending the basis if $d > r$:

$$z_i \stackrel{\text{def.}}{=} (\langle x_i - \hat{m}, v_k \rangle)_{k=1}^d \in \mathbb{R}^d. \tag{13.3}$$

The corresponding reconstruction is

$$\tilde{x}_i \stackrel{\text{def.}}{=} \hat{m} + \sum_{k=1}^d z_{i,k} v_k \in \mathbb{R}^p. \tag{13.4}$$

Thus $\tilde{x}_i = \mathrm{Proj}_{\tilde{T}}(x_i)$, where $\tilde{T} \stackrel{\text{def.}}{=} \hat{m} + \mathrm{Span}_{k=1}^d(v_k)$ is the affine reconstruction space.

Figure 13.3 shows an example of PCA for 2-D and 3-D visualization.

Optimality analysis. We prove that PCA minimizes the ℓ^2 reconstruction error among linear dimensionality-reduction methods. An optimal affine reconstruction space can be chosen to contain the empirical mean: for any fixed direction space, translating it to the mean cannot increase squared reconstruction error. After centering, we therefore assume $\hat{m} = 0$, hence $X = \tilde{X}$.

We recall that $X = \sqrt{n} U \operatorname{diag}(\sigma) V^\top$ with observations in rows, and $\hat{C} = X^\top X / n = V \operatorname{diag}(\sigma_i^2) V^\top$. We assume $1 \leqslant k \leqslant p$ and extend V to an orthogonal basis of $\mathbb{R}^p$ when necessary.

For the centered data, compare linear compression maps into dimension k followed by reconstruction:

$$\min_{R,S} \left\{ f(R,S) \stackrel{\text{def.}}{=} \sum_i \|x_i - RS^\top x_i\|_{\mathbb{R}^p}^2 \ ; \ R, S \in \mathbb{R}^{p \times k} \right\} \tag{13.5}$$

Although $f(\cdot, S)$ and $f(R, \cdot)$ are convex separately, f is not jointly convex. Alternating minimization in R and S therefore need not reach a global minimizer. Nevertheless, a global minimizer has a simple explicit form.

Theorem 13.1. *A solution of* (13.5) *is* $S = R = V_{1:k} \stackrel{\text{def.}}{=} [v_1, \ldots, v_k]$.

The compressor $S^\top$ and decompressor $R = S$ give precisely the PCA maps in (13.3) and (13.4).

We first prove that one can restrict attention to orthogonal projection matrices.

[1] https://en.wikipedia.org/wiki/Iris_flower_data_set

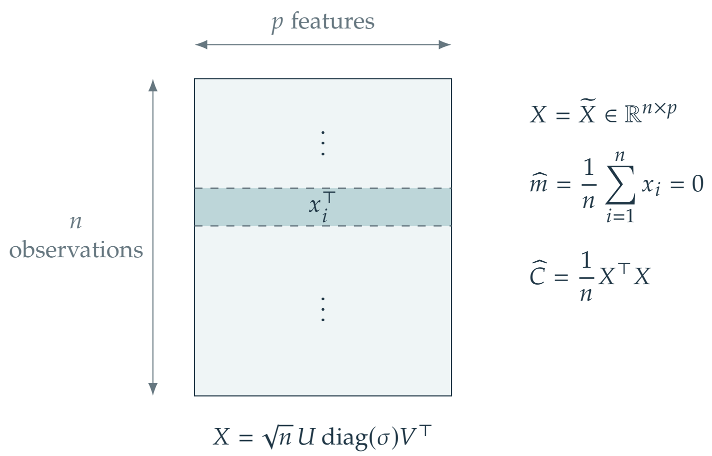


Figure 13.4. Rows of the centered data matrix are observations; columns are features.

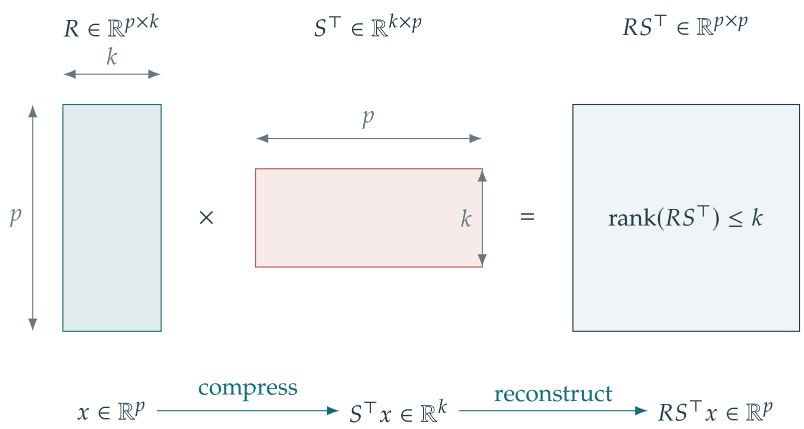


Figure 13.5. Linear compression followed by reconstruction.

Lemma 13.2. *The minimization may be restricted to $S = R$ with $S^\top S = \mathrm{Id}_{k\times k}$.*

Proof. Fix an arbitrary pair (R, S). The matrix $RS^\top$ has rank $k' \leqslant k$. Choose $W \in \mathbb{R}^{p\times k'}$ whose columns form an orthonormal basis of $\mathrm{Im}(RS^\top)$, so that $W^\top W = \mathrm{Id}_{k'\times k'}$. Orthogonal projection gives

$$\operatorname*{argmin}_z \|x - Wz\|^2 = W^\top x$$

because the first-order optimality condition is $W^\top(Wz - x) = 0$. Hence, denoting $RS^\top x_i = Wz_i$ for some $z_i \in \mathbb{R}^{k'}$

$$f(R, S) = \sum_i \|x_i - RS^\top x_i\|^2 = \sum_i \|x_i - Wz_i\|^2 \geqslant \sum_i \|x_i - WW^\top x_i\|^2 \geqslant f(\tilde{W}, \tilde{W}).$$

where W has been extended to a matrix $\tilde{W} \in \mathbb{R}^{p\times k}$ with orthonormal columns and $\tilde{W}_{1:k'} = W$. □

Lemma 13.3. *Denoting $C \overset{\text{def.}}{=} X^\top X = \sum_i x_i x_i^\top \in \mathbb{R}^{p\times p}$, an optimal S is obtained by solving*

$$\max_{S\in\mathbb{R}^{p\times k}} \left\{ \mathrm{tr}(S^\top CS) \; ; \; S^\top S = \mathrm{Id}_k \right\}.$$

Proof. By the previous lemma, restrict to $R = S$ with $S^\top S = \mathrm{Id}_k$. Expanding the residual gives

$$f(S, S) = \sum_i \|x_i - SS^\top x_i\|^2 = \sum_i \left(\|x_i\|^2 - 2x_i^\top SS^\top x_i + x_i^\top S(S^\top S)S^\top x_i \right).$$

Using that $S^\top S = \mathrm{Id}_k$, one has

$$f(S, S) = \sum_i \|x_i\|^2 - \sum_i \mathrm{tr}(S^\top x_i x_i^\top S) = \|X\|_{\mathrm{Fro}}^2 - \mathrm{tr}(S^\top CS).$$

The first term is independent of S, giving the stated maximization problem. □

The next lemma bounds the objective by a linear program. We then show that PCA attains this bound, proving optimality.

Lemma 13.4. *Denoting $C = V\Lambda V^\top = n\hat{C}$ with eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_p \geqslant 0$ (including zeros), one has*

$$\mathrm{tr}(S^\top CS) \leqslant \max_{\beta\in\mathbb{R}^p} \left\{ \sum_{i=1}^p \lambda_i \beta_i \; ; \; 0 \leqslant \beta \leqslant 1, \sum_i \beta_i \leqslant k \right\} = \sum_{i=1}^k \lambda_i \tag{13.6}$$

with a maximizing vector $\beta = (1, \ldots, 1, 0, \ldots, 0)$.

Proof. Use the full orthogonal eigenbasis $V \in \mathbb{R}^{p\times p}$ introduced above, including eigenvectors corresponding to zero eigenvalues. Thus $V^\top V = VV^\top = \mathrm{Id}_p$. One has

$$\mathrm{tr}(S^\top CS) = \mathrm{tr}(S^\top V\Lambda V^\top S) = \mathrm{tr}(B^\top \Lambda B) = \mathrm{tr}(\Lambda BB^\top) = \sum_{i=1}^p \lambda_i \|b_i\|^2 = \sum_i \lambda_i \beta_i$$

Here $B \overset{\text{def.}}{=} V^\top S \in \mathbb{R}^{p\times k}$. The vectors $(b_i)_{i=1}^p$, with $b_i \in \mathbb{R}^k$, are the rows of B, and $\beta_i \overset{\text{def.}}{=} \|b_i\|^2 \geqslant 0$. One has

$$B^\top B = S^\top VV^\top S = S^\top S = \mathrm{Id}_k,$$

so the columns of B are orthonormal. Consequently,

$$\sum_i \beta_i = \sum_i \|b_i\|^2 = \|B\|_{\mathrm{Fro}}^2 = \mathrm{tr}(B^\top B) = k.$$

We extend the k columns of B into an orthogonal basis $\tilde{B} \in \mathbb{R}^{p\times p}$ such that $\tilde{B}\tilde{B}^\top = \tilde{B}^\top \tilde{B} = \mathrm{Id}_p$, so that

$$0 \leqslant \beta_i = \|b_i\|^2 \leqslant \|\tilde{b}_i\|^2 = 1$$

Thus $(\beta_i)_{i=1}^p$ is feasible for the linear program, so $\mathrm{tr}(S^\top CS)$ cannot exceed its maximum.

For any feasible β, use $\lambda_i \leqslant \lambda_k$ for $i > k$ and $\sum_i \beta_i \leqslant k$ to obtain

$$\sum_i \lambda_i \beta_i \leqslant \sum_{i\leqslant k} \lambda_i \beta_i + \lambda_k \left(k - \sum_{i\leqslant k} \beta_i \right) \leqslant \sum_{i\leqslant k} \lambda_i.$$

The choice $\beta_i = 1$ for $i \leqslant k$ and $\beta_i = 0$ otherwise attains this value. Figure 13.6 illustrates the argument. □

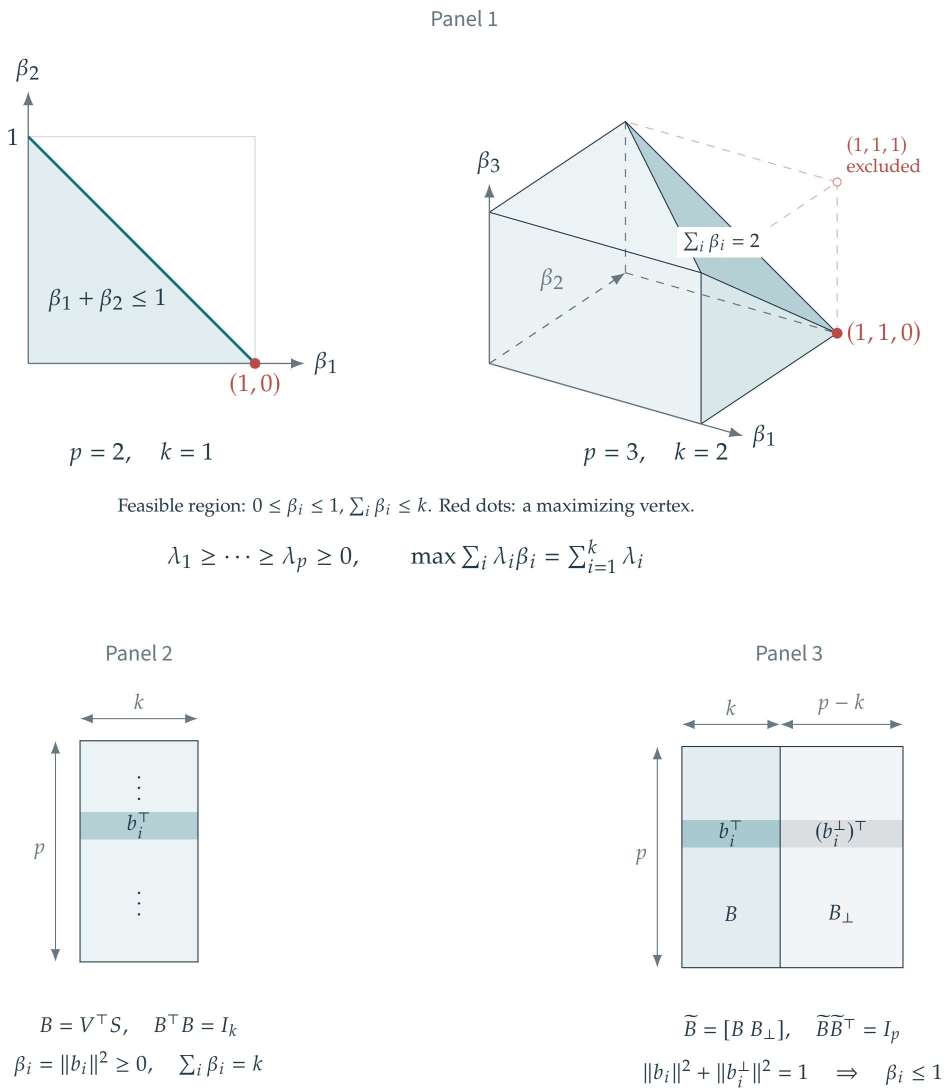
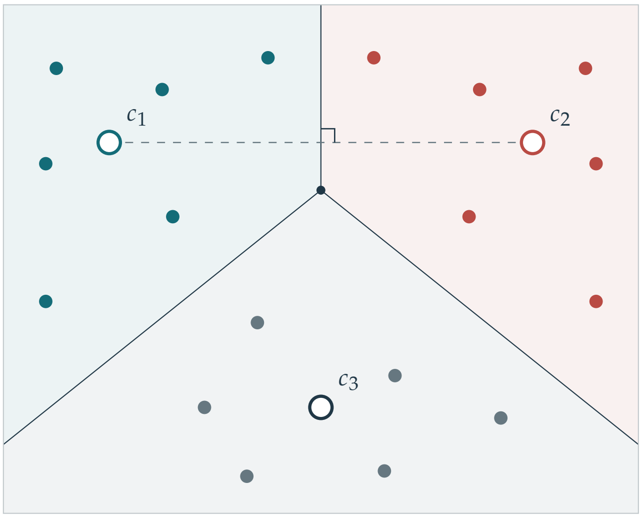


Figure 13.6. Panel 1: the linear-program bound in (13.6). Panel 2: matrix B. Panel 3: its orthogonal extension $\tilde{B}$.



$V_j = \{x : \|x - c_j\| \leq \|x - c_\ell\| \text{ for every } \ell\}$

Dashed segment: two centroids. Its perpendicular bisector is their common cell edge.

Figure 13.7. k-means clusters according to Voronoi cells.

Proof of PCA optimality. For $S = V_{1:k} = [v_1, \dots, v_k]$, the eigenvector identities give $CS = V\Lambda V^\top V_{1:k} = V_{1:k}\,\mathrm{diag}(\lambda_i)_{i=1}^k$ and hence

$$\mathrm{tr}(S^\top CS) = \mathrm{tr}(S^\top S\,\mathrm{diag}(\lambda_i)_{i=1}^k) = \mathrm{tr}(\mathrm{Id}_k\,\mathrm{diag}(\lambda_i)_{i=1}^k) = \sum_{i=1}^k \lambda_i.$$

This attains the upper bound in Lemma 13.4, proving that S is optimal. □

13.1.2 Clustering and k-means

Clustering assigns a label $y_i \in \{1, \dots, k\}$ to each observation x_i. Observations with the same label form a cluster.

k-means One approach is to seek compact clusters by minimizing a measure of within-cluster dispersion. In Euclidean space, the k-means objective measures distances from observations to their centroids $c = (c_\ell)_{\ell=1}^k$, with $c_\ell \in \mathbb{R}^p$. Related formulations are possible in general metric spaces, although the centroid computations may be more difficult. The optimization problem is

$$\min_{(y,c)} \mathcal{E}(y,c) \overset{\text{def.}}{=} \sum_{\ell=1}^k \sum_{i:y_i=\ell} \|x_i - c_\ell\|^2.$$

The k-means algorithm alternates between updating the labels and updating the centroids, a form of block coordinate minimization. Initialize the centroids c, for example by choosing well-separated observations. We discuss a more systematic initialization below. For fixed centroids c, minimize $y \mapsto \mathcal{E}(y,c)$ by assigning each observation to its nearest centroid:

$$\forall\, i \in \{1, \dots, n\}, \quad y_i \leftarrow \underset{1 \leqslant \ell \leqslant k}{\mathrm{argmin}}\, \|x_i - c_\ell\|. \tag{13.7}$$

For fixed labels y, minimize $c \mapsto \mathcal{E}(y,c)$ by setting each centroid to the mean of its cluster:

$$\forall\, \ell \in \{1, \dots, k\}, \quad c_\ell \leftarrow \frac{\sum_{i:y_i=\ell} x_i}{|\{i\,;\, y_i = \ell\}|} \tag{13.8}$$

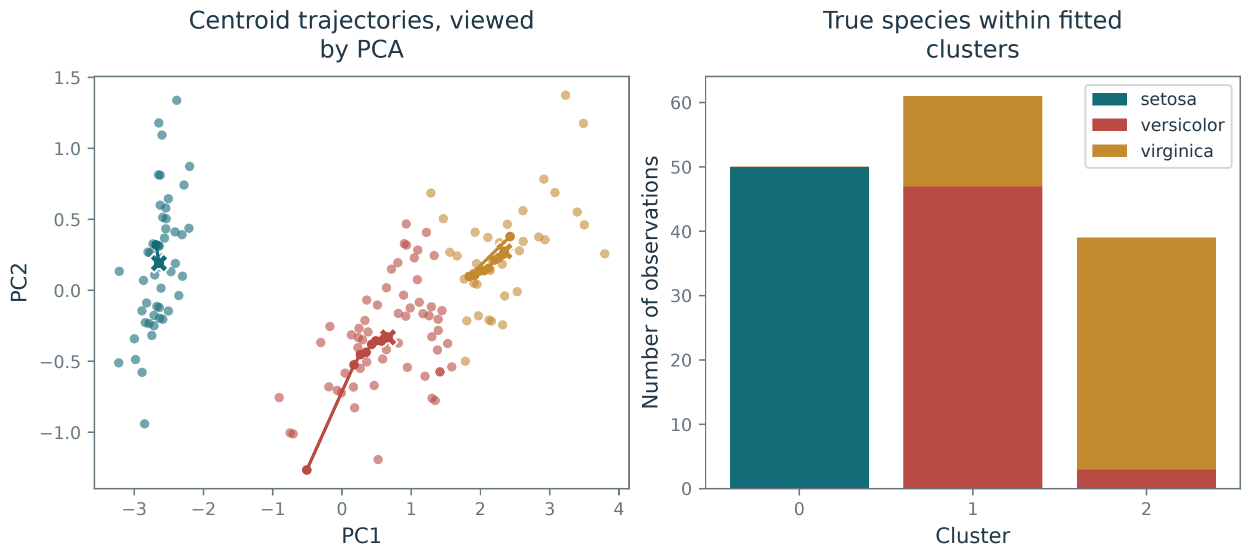


Figure 13.8. Left: iterations of the k-means algorithm. Right: class histograms after the k-means optimization.

If a cluster becomes empty, its centroid does not contribute to the objective for the current labels. One may leave it in place or reseed it at a data point before the next assignment step. The centroid formula applies only to nonempty clusters.

Each step decreases or preserves $\mathcal{E}$. With a fixed tie-breaking rule that retains existing assignments at ties, and without indefinite reseeding, a changed assignment strictly decreases the objective. There are finitely many label assignments, so the algorithm terminates after finitely many such changes.

Because the objective is nonconvex, k-means need not find globally optimal clusters. A reseeding heuristic moves centroids c_ℓ from redundant or low-contribution clusters to regions with high distortion. This can help the iterations escape a poor local configuration.

Figure 13.8 shows an example of k-means iterations on the Iris dataset.

k-means++ The result of k-means depends strongly on initialization. Well-separated initial centers tend to cover the data more effectively than tightly grouped centers.

The randomized k-means++ initialization has an approximation guarantee even before Lloyd iterations are applied. In practice, Lloyd iterations usually improve the resulting centers. First choose c_1 uniformly among the sample points. After choosing $c_1, \dots, c_\ell$, select $c_{\ell+1}$ from the sample according to a probability $\pi^{(\ell)}$ on $\{1, \dots, n\}$ proportional to the squared distance from the nearest selected center

$$\forall\, i \in \{1, \dots, n\}, \quad \pi_i^{(\ell)} \overset{\text{def.}}{=} \frac{d_i^2}{\sum_{j=1}^n d_j^2} \quad \text{where} \quad d_i \overset{\text{def.}}{=} \min_{1 \leqslant r \leqslant \ell} \|x_i - c_r\|.$$

Points far from the previously seeded centers are more likely to be chosen. If every $d_i = 0$, the current centers already achieve zero distortion and the procedure can stop.

The following theorem, due to David Arthur and Sergei Vassilvitskii, bounds the expected seeding cost by a logarithmic factor times the optimal cost. Computing a global optimum is NP-hard in general.

Theorem 13.5. *Let $c^\star$ be the centroids selected by k-means++ and let $y^\star$ be their nearest-centroid labels, as in (13.7). Then*

$$\mathbb{E}(\mathcal{E}(y^\star, c^\star)) \leqslant 8(2 + \log(k)) \min_{(y,c)} \mathcal{E}(y, c),$$

where the expectation is over the random choices made during initialization.

Lloyd algorithm and continuous densities. The k-means iterations are also known as Lloyd's algorithm and are used in vector quantization for compression. The same alternating procedure applies to a probability

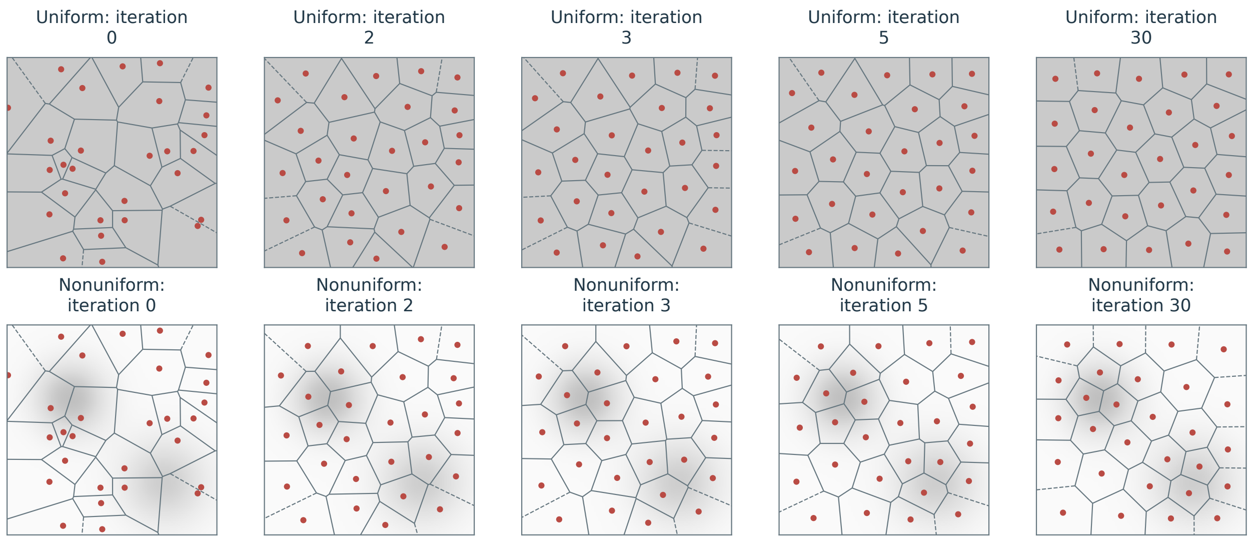


Figure 13.9. Continuous k-means (Lloyd) iterations $0, 2, 3, 5, 30$. The top row uses a uniform density; the bottom row uses the nonuniform density shown in grayscale.

measure μ on $\mathbb{R}^p$ with finite second moment. The energy to minimize becomes

$$\min_{(\mathcal{V},c)} \sum_{\ell=1}^{k} \int_{\mathcal{V}_\ell} \|x - c_\ell\|^2 \mathrm{d}\mu(x)$$

where $(\mathcal{V}_\ell)_\ell$ is a partition of the domain. Step (13.7) is replaced by the computation of a Voronoi cell

$$\forall\, \ell \in \{1, \dots, k\}, \quad \mathcal{V}_\ell \overset{\text{def.}}{=} \{x \,;\, \forall\, \ell' \neq \ell, \|x - c_\ell\| \leqslant \|x - c_{\ell'}\|\}\,.$$

For distinct centroids, the Voronoi cells are polyhedra bounded by perpendicular bisectors. Resolve distance ties consistently to obtain a partition; this matters if μ assigns positive mass to a boundary. In low dimensions, computational geometry provides efficient algorithms for constructing the cells. Step (13.8) is then replaced by

$$\forall\, \ell \in \{1, \dots, k\}, \quad c_\ell \leftarrow \frac{\int_{\mathcal{V}_\ell} x \mathrm{d}\mu(x)}{\int_{\mathcal{V}_\ell} \mathrm{d}\mu(x)}.$$

The centroid update applies to cells of positive μ-mass; zero-mass cells require a separate convention, as in the discrete algorithm. For a uniform planar density, hexagonal Voronoi patterns are characteristic of high-resolution quantization away from the boundary. They need not be exact finite-k minimizers on a bounded domain. Figure 13.9 displays two examples of Lloyd iterations on 2-D densities on a square domain.

13.2 Empirical Risk Minimization

We first describe empirical risk minimization, a framework shared by regression and classification. For classification, predictions may represent class probabilities rather than labels.

To make fitting $y_i \approx f(x_i)$ computationally tractable and obtain useful predictions, we restrict the admissible functions to a class of controlled complexity. The class may become richer as the sample size n grows, but its complexity must remain compatible with the available data.

13.2.1 Empirical Risk

Let $\mathcal{F}_n$ be a class of functions that may depend on the sample size. A common learning procedure is empirical risk minimization (ERM)

$$\hat{f} \in \operatorname*{argmin}_{f \in \mathcal{F}_n} \frac{1}{n} \sum_{i=1}^{n} L(f(x_i), y_i). \tag{13.9}$$

The loss function $L : \mathcal{Y}^2 \to \mathbb{R}^+$ measures prediction error for the chosen task. Classification and regression typically require different loss functions. We sometimes write $\hat{f}_n$ to emphasize the estimator's dependence on the sample size n.

13.2.2 Prediction and Consistency

For the analysis, assume the training pairs (x_i, y_i) are i.i.d. with distribution π on $\mathcal{X} \times \mathcal{Y}$. The corresponding population optimization problem is

$$\bar{f} \in \underset{f \in \mathcal{F}_\infty}{\operatorname{argmin}} \int_{\mathcal{X} \times \mathcal{Y}} L(f(x), y) \mathrm{d}\pi(x, y) = \mathbb{E}_{(\mathbf{x},\mathbf{y}) \sim \pi}\big(L(f(\mathbf{x}), \mathbf{y})\big). \tag{13.10}$$

The desired consistency property is $\hat{f}_n \to \bar{f}$ as $n \to +\infty$. One way to express this is through expected prediction error over the training samples $(x_i, y_i)_i$:

$$E_n \overset{\text{def.}}{=} \mathbb{E}(\tilde{L}(\hat{f}_n(\mathbf{x}), \bar{f}(\mathbf{x}))) \longrightarrow 0.$$

Here the expectation includes both the test input $\mathbf{x}$, drawn from the marginal $\pi_{\mathcal{X}}$, and the n i.i.d. training pairs $(x_i, y_i) \sim \pi$ that determine $\hat{f}_n$. The comparison loss $\tilde{L}$ measures discrepancies between predictions in $\mathcal{Y}$; for example, we may take $\tilde{L} = L$. One can also study convergence in probability, i.e.

$$\forall\, \varepsilon > 0, \quad E_{\varepsilon,n} \overset{\text{def.}}{=} \mathbb{P}(\tilde{L}(\hat{f}_n(\mathbf{x}), \bar{f}(\mathbf{x})) > \varepsilon) \to 0.$$

These properties define consistency in expectation and in probability, respectively. A convergence rate gives an explicit upper bound on the decay of E_n or $E_{\varepsilon,n}$.

For $\tilde{L}(y, y') = |y - y'|^r$, convergence in expectation implies convergence in probability by Markov's inequality:

$$E_{\varepsilon,n} = \mathbb{P}(|\hat{f}_n(\mathbf{x}) - \bar{f}(\mathbf{x})|^r > \varepsilon) \leqslant \frac{1}{\varepsilon}\mathbb{E}(|\hat{f}_n(\mathbf{x}) - \bar{f}(\mathbf{x})|^r) = \frac{E_n}{\varepsilon}.$$

13.2.3 Parametric Approaches and Regularization

Instead of specifying $\mathcal{F}_n$ through a hard constraint, we can favor simple or regular functions through a penalty. A typical way to achieve this is by using a parametric model $y \approx f(x, \beta)$ where $\beta \in \mathcal{B}$ parametrizes the function $f(\cdot, \beta) : \mathcal{X} \to \mathcal{Y}$. The empirical risk minimization procedure (13.9) now becomes

$$\hat{\beta} \in \underset{\beta \in \mathcal{B}}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} L(f(x_i, \beta), y_i) + \lambda_n J(\beta). \tag{13.11}$$

where J is a regularizer. For example, $J = \|\cdot\|_2^2$ controls parameter magnitude, while $J = \|\cdot\|_1$ promotes sparse coefficients and, in a linear feature model, feature selection. Here $\lambda_n > 0$ is a regularization parameter, and its asymptotic scaling must balance approximation and estimation; consistency often requires $\lambda_n \to 0$ at a controlled rate.

The population counterpart of (13.10) defines the parameter $\bar{\beta}$. A limiting estimator as $n \to +\infty$ then takes the form $\bar{f} = f(\cdot, \bar{\beta})$, with $\bar{\beta}$ satisfying

$$\bar{\beta} \in \underset{\beta}{\operatorname{argmin}} \int_{\mathcal{X} \times \mathcal{Y}} L(f(x, \beta), y) \mathrm{d}\pi(x, y) = \mathbb{E}_{(\mathbf{x},\mathbf{y}) \sim \pi}\big(L(f(\mathbf{x}, \beta), \mathbf{y})\big). \tag{13.12}$$

Prediction vs. estimation risks. We may also ask how accurately $\hat{\beta}$ estimates $\bar{\beta}$, as measured by the parameter error $\|\hat{\beta} - \bar{\beta}\|$ for a chosen norm $\|\cdot\|$. Parameter estimation is generally more demanding than prediction: different parameters can produce similar predictions, especially when the model is poorly identified.

13.2.4 Validation, Test Sets, and Cross-validation

The errors E_n and $E_{\varepsilon,n}$ cannot be evaluated directly because the population optimum $\bar{f}$ is unknown. To tune a parameter such as the regularization strength λ, we estimate the prediction risk $\mathbb{E}(L(\hat{f}(\mathbf{x}), \mathbf{y}))$ on data held out from training.

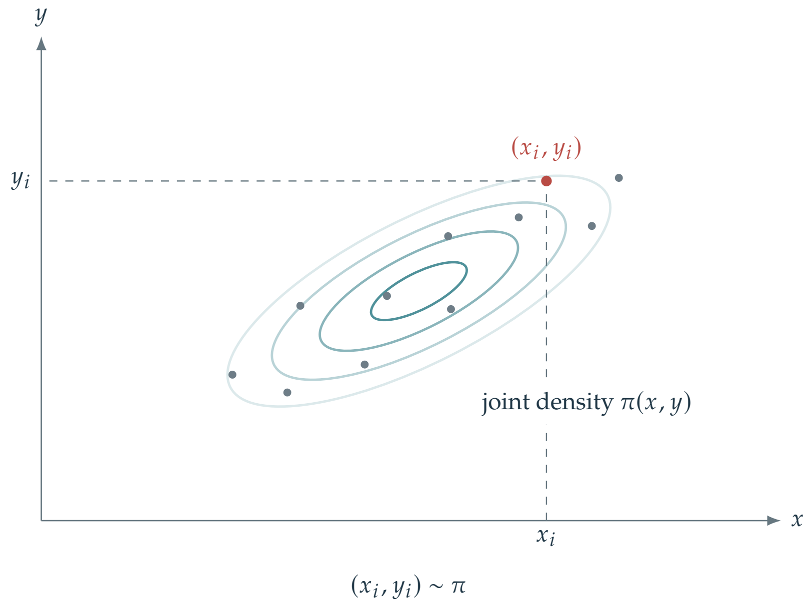


Figure 13.10. Probabilistic modeling.

Use a second sample $(\bar{x}_j, \bar{y}_j)_{j=1}^{\bar{n}}$, called a validation set. In the statistical model, these observations are i.i.d. with distribution π and independent of the training sample. The validation risk is

$$R_{\bar{n}} = \frac{1}{\bar{n}} \sum_{j=1}^{\bar{n}} L(\hat{f}(\bar{x}_j), \bar{y}_j) \tag{13.13}$$

which converges to $\mathbb{E}(L(\hat{f}(\mathbf{x}), \mathbf{y}))$ for large $\bar{n}$. Minimizing $R_{\bar{n}}$ over hyperparameters (such as λ_n) is holdout validation. Cross-validation repeats training and validation across folds. A separate test set, untouched by hyperparameter selection, estimates the performance of the final selected procedure.

13.3 Supervised Learning: Regression

In supervised learning, the training data consist of pairs $(x_i, y_i) \in \mathcal{X} \times \mathcal{Y}$, with $\mathcal{X} = \mathbb{R}^p$ here for simplicity. We seek a function $f : \mathcal{X} \to \mathcal{Y}$ that captures the relationship $y_i \approx f(x_i)$ and predicts an output $f(x)$ for a new input x.

When $\mathcal{Y}$ is finite and discrete, the task is supervised classification, studied in Section 13.4. Binary classification uses $\mathcal{Y} = \{0, 1\}$; for example, in a medical application, $y_i = 0$ may indicate a healthy subject and $y_i = 1$ a subject with the condition of interest. When $\mathcal{Y}$ is continuous, typically $\mathcal{Y} = \mathbb{R}$, the task is regression.

13.3.1 Linear Regression

For regression, we specialize empirical risk minimization to $\mathcal{Y} = \mathbb{R}$ with the quadratic loss $L(y, y') = \frac{1}{2}|y - y'|^2$.

Nonlinear regression can be formulated using a dictionary of features, such as polynomials. This lifts the data into a higher-dimensional space and leads to the kernel methods of Section 13.5.

Least squares and conditional expectation. If f ranges over measurable functions and $\mathbb{E}(\mathbf{y}^2) < \infty$, a population minimizer $\bar{f}$ in (13.10) is the conditional mean of the response given the input. It is defined up to equality almost everywhere under the input distribution. Suppose for simplicity that π has density $\frac{d\pi}{dxdy}$ with respect to a product measure $dxdy$, such as Lebesgue measure. Then

$$\forall\, x \in \mathcal{X}, \quad \bar{f}(x) = \mathbb{E}(\mathbf{y}|\mathbf{x} = x) = \frac{\int_{\mathcal{Y}} y \frac{d\pi}{dxdy}(x, y)dy}{\int_{\mathcal{Y}} \frac{d\pi}{dxdy}(x, y)dy}$$

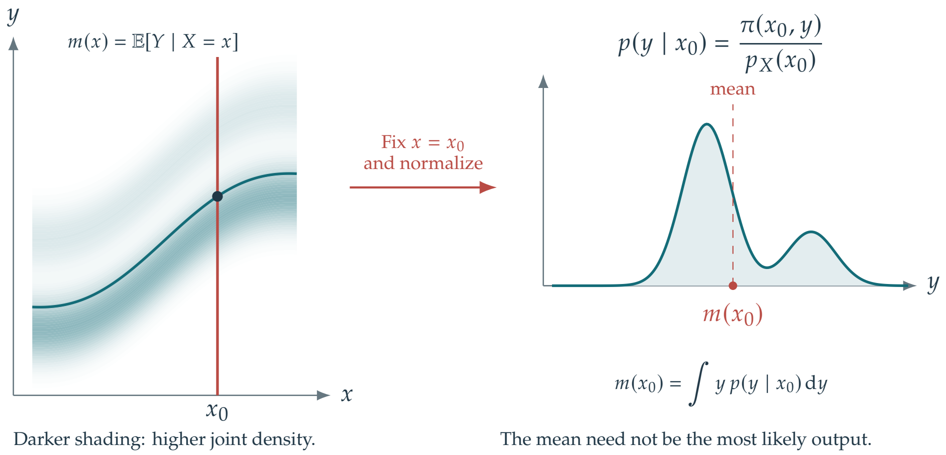


Figure 13.11. Conditional expectation.

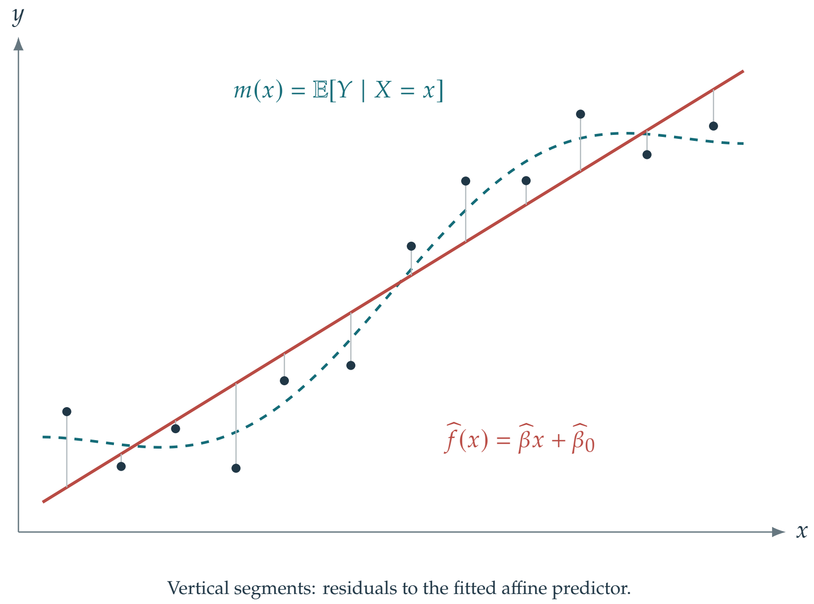


Figure 13.12. Linear regression.

where $(\mathbf{x}, \mathbf{y})$ has law π and the formula applies where the denominator is positive.

If $\mathcal{X}$ and $\mathcal{Y}$ are discrete, let $\pi_{x,y}$ denote the probability of $(\mathbf{x} = x, \mathbf{y} = y)$. Then

$$\forall\, x \in \mathcal{X}, \quad \bar{f}(x) = \frac{\sum_y y\pi_{x,y}}{\sum_y \pi_{x,y}}$$

with no prescribed value when the marginal of π on $\mathcal{X}$ vanishes at x.

Replacing π by the empirical distribution averages responses at each observed input but leaves predictions unspecified elsewhere. Generalization therefore requires assumptions on regularity or model complexity.

Penalized linear models. A linear model constrains f to have the form $f(x, \beta) = \langle x,\, \beta\rangle$ with parameters $\beta \in \mathcal{B} = \mathbb{R}^p$. Affine predictors are included through the identity $\langle x,\, \beta\rangle + \beta_0 = \langle (x, 1),\, (\beta, \beta_0)\rangle$, by appending a constant coordinate to x to form $(x, 1)$. We therefore use the linear notation below.

Under the square loss, the regularized ERM (13.11) is conveniently rewritten as

$$\hat{\beta} \in \operatorname*{argmin}_{\beta \in \mathcal{B}} \frac{1}{2}\langle \hat{C}\beta,\, \beta\rangle - \langle \hat{u},\, \beta\rangle + \lambda_n J(\beta) \tag{13.14}$$

where we introduced the empirical second-moment matrix (equal to the covariance (13.1) for centered data) and cross-moment vector

$$\hat{C} \stackrel{\text{def.}}{=} \frac{1}{n} X^* X = \frac{1}{n} \sum_{i=1}^n x_i x_i^* \quad \text{and} \quad \hat{u} \stackrel{\text{def.}}{=} \frac{1}{n} \sum_{i=1}^n y_i x_i = \frac{1}{n} X^* y \in \mathbb{R}^p.$$

As $n \to +\infty$, suitable moment assumptions on π give the following almost sure limits by the law of large numbers:

$$\hat{C} \to C \stackrel{\text{def.}}{=} \mathbb{E}(\mathbf{x}\mathbf{x}^*) \quad \text{and} \quad \hat{u} \to u \stackrel{\text{def.}}{=} \mathbb{E}(\mathbf{y}\mathbf{x}). \tag{13.15}$$

For appropriately chosen $\lambda_n \to 0$, the estimator can converge as $n \to +\infty$ to the population parameter

$$\bar{\beta} \in \underset{\beta}{\operatorname{argmin}} \ \{J(\beta) \ ; \ C\beta = u\}.$$

Problem (13.14) is equivalent to the regularized resolution of inverse problems (9.9), with $X/\sqrt{n}$ in place of Φ and $y/\sqrt{n}$ as the observation vector. Random design introduces sampling error into the operator as well: $\hat{C}$ estimates C. The typical scale is $1/\sqrt{n}$, in the sense that

$$\mathbb{E}(\|\hat{C} - C\|) = O(n^{-1/2}) \quad \text{and} \quad \mathbb{E}(\|\hat{u} - u\|) = O(n^{-1/2}),$$

assuming $\mathbb{E}(\mathbf{y}^4) < +\infty$ and $\mathbb{E}(\|\mathbf{x}\|^4) < +\infty$, which give finite second moments for $\mathbf{x}\mathbf{x}^*$ and $\mathbf{x}\mathbf{y}$. Using a linear predictor does not require a correctly specified model $\mathbf{y} = \langle \mathbf{x}, \beta \rangle + w$ with noise w independent of $\mathbf{x}$. The aim is to estimate the best linear predictor $\bar{\beta}$.

Extensions of Theorems 9.4, 11.9, and 11.11 can account for covariance-estimation error and yield prediction bounds of the form

$$\mathbb{E}(|\langle \hat{\beta}, \mathbf{x} \rangle - \langle \bar{\beta}, \mathbf{x} \rangle|^2) = O(n^{-\kappa})$$

and estimation rates of the form

$$\mathbb{E}(\|\hat{\beta} - \bar{\beta}\|^2) = O(n^{-\kappa'}),$$

under suitable source conditions involving C and u. Since the noise level is roughly $n^{-\frac{1}{2}}$, the ideal cases are when $\kappa = \kappa' = 1$, which is the so-called linear rate regime. Support-recovery guarantees can also be derived by extending Theorem 11.15. We now focus on quadratic regularization, a standard regression method. Sparse penalties such as ℓ^1 are useful when feature selection or a sparse model is appropriate; their statistical benefit depends on the data and assumptions. In imaging inverse problems, sparse regularization expresses prior structure of the object being reconstructed.

Ridge regression (quadratic penalization). For $J = \|\cdot\|^2/2$, the estimator (13.14) is obtained in closed form as

$$\hat{\beta} = (X^*X + n\lambda_n \mathrm{Id}_p)^{-1} X^* y = (\hat{C} + \lambda_n \mathrm{Id}_p)^{-1} \hat{u}. \tag{13.16}$$

This is often called ridge regression in the literature. The Woodbury identity gives the equivalent expression

$$\hat{\beta} = X^*(XX^* + n\lambda_n \mathrm{Id}_n)^{-1} y. \tag{13.17}$$

When $n \gg p$, formula (13.16) requires solving the smaller, $p \times p$ system. When $p \gg n$, formula (13.17) uses an $n \times n$ system and is preferable; it also extends to infinite-dimensional feature spaces.

Convergence to the minimum-norm population solution $\bar{\beta} = C^+ u$ requires control of both the regularization bias and sampling error. The condition $\lambda_n \to 0$ alone is insufficient when small eigenvalues amplify noise. Here is one elementary bound that also applies in a Hilbert feature space.

Theorem 13.6. *Assume $\|\mathbf{x}\| \leqslant \kappa$ almost surely, $\mathbb{E}(\mathbf{y}^2) < \infty$, and $u = C\bar{\beta}$. Suppose*

$$\bar{\beta} = C^\gamma z, \qquad \|z\| \leqslant \rho, \qquad 0 < \gamma \leqslant 1. \tag{13.18}$$

Set $B^2 = \kappa^2\big(\mathbb{E}(\mathbf{y}^2) + \kappa^2 \|\bar{\beta}\|^2\big)$. For every $\lambda > 0$,

$$\mathbb{E}\|\hat{\beta} - \bar{\beta}\|^2 \leqslant 2\rho^2 \lambda^{2\gamma} + \frac{4B^2}{n\lambda^2}.$$

If $B, \rho > 0$, choosing $\lambda_n = (B/(\rho\sqrt{n}))^{1/(\gamma+1)}$ gives

$$\mathbb{E}\|\hat{\beta} - \bar{\beta}\|^2 \leqslant 6\rho^{2/(\gamma+1)} B^{2\gamma/(\gamma+1)} n^{-\gamma/(\gamma+1)}. \tag{13.19}$$

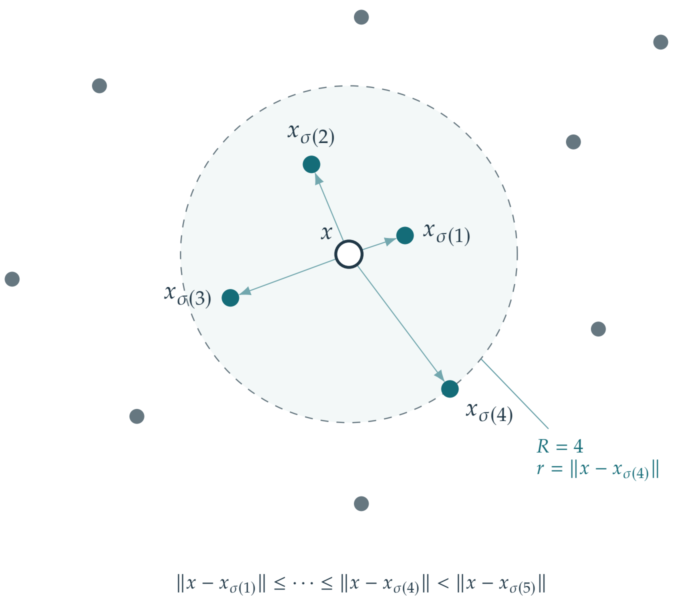


Figure 13.13. Nearest neighbors.

Proof. Let $\beta_\lambda = (C + \lambda \mathrm{Id})^{-1} u$. Spectral calculus gives $\|\beta_\lambda - \bar{\beta}\| \leqslant \rho \lambda^\gamma$ and $\|\beta_\lambda\| \leqslant \|\bar{\beta}\|$. Moreover,

$$\hat{\beta} - \beta_\lambda = (\hat{C} + \lambda \mathrm{Id})^{-1}\big((\hat{u} - u) - (\hat{C} - C)\beta_\lambda\big).$$

The inverse has norm at most $1/\lambda$. Independence and the second-moment bound yield $\mathbb{E}\|\hat{\beta} - \beta_\lambda\|^2 \leqslant 2B^2/(n\lambda^2)$. Combining the two errors with $\|a + b\|^2 \leqslant 2\|a\|^2 + 2\|b\|^2$ proves the first claim; the stated choice of λ_n gives the second. $\square$

In finite dimensions, a source representation exists for every $\bar{\beta} \in \mathrm{Im}(C)$, but its radius ρ depends on the spectrum and on the target. In infinite dimensions, it is an additional regularity assumption. The bound displays no explicit feature dimension, although κ, B, and ρ may depend on it. Standard ridge regularization has bias saturation beyond source exponent $\gamma = 1$; stronger source assumptions alone do not justify extending this estimate to $\gamma = 2$.

13.4 Supervised Learning: Classification

For classification, the labels are discrete: $y_i \in \mathcal{Y} = \{1, \ldots, k\}$. We study two standard methods: nearest neighbors and logistic classification. Both methods provide useful baselines for assessing more elaborate models. Nearest-neighbor methods also apply to regression.

13.4.1 Nearest-Neighbor Classification

The R-nearest-neighbor method (R-NN) classifies an input using the labels of its R nearest training observations. Increasing R reduces sensitivity to individual noisy labels, but excessive smoothing can obscure class boundaries. For consistency, a standard asymptotic choice has $R = R_n \to \infty$ and $R_n/n \to 0$. Nearest-neighbor classification is a useful baseline, especially when the feature dimension p is small.

The prediction $\hat{f}(x) \in \mathcal{Y}$ is the most frequent label among those R observations, with a fixed rule for breaking ties. The method is nonparametric: its representation depends on the training sample rather than on a fixed-dimensional parameter vector.

First compute the Euclidean distance from the query x to each training observation x_i. Sorting the distances generates an indexing σ (a permutation of $\{1, \ldots, n\}$) such that

$$\|x - x_{\sigma(1)}\| \leqslant \|x - x_{\sigma(2)}\| \leqslant \ldots \leqslant \|x - x_{\sigma(n)}\|.$$

For a given R, one can compute the "local" histogram of classes around x

$$h_\ell(x) \overset{\text{def.}}{=} \frac{1}{R} \#\{i \in \{1, \ldots, R\} : y_{\sigma(i)} = \ell\}.$$

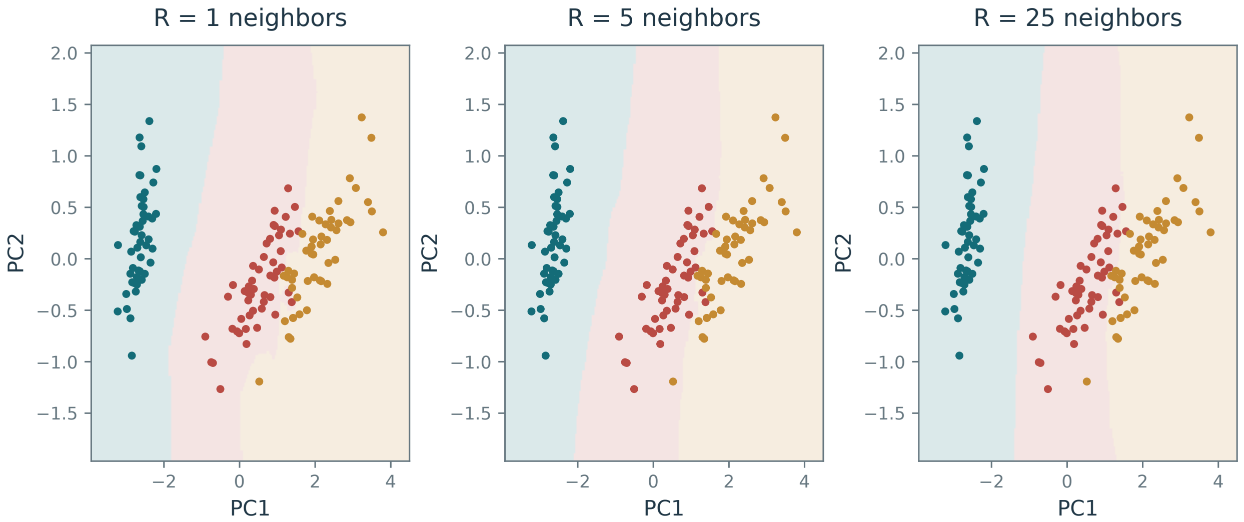


Figure 13.14. Classification boundaries produced by R-nearest neighbors.

The predicted class at x is an index attaining the largest histogram value:

$$\hat{f}(x) \in \underset{\ell}{\operatorname{argmax}}\ h_\ell(x).$$

Choose R by validation or cross-validation. For classification, the validation risk (13.13) typically uses the 0–1 loss, giving the fraction of misclassified observations

$$R_{\bar{n}} \overset{\text{def.}}{=} \frac{1}{\bar{n}} \sum_{j=1}^{\bar{n}} \delta(\bar{y}_j - \hat{f}(\bar{x}_j))$$

where $\delta(0) = 0$ and $\delta(s) = 1$ if $s \neq 0$. The method also applies to features in a general metric space in place of $\mathbb{R}^p$. Fast nearest-neighbor search can avoid explicitly sorting every distance.

Figure 13.14 shows the predicted class regions $\left\{x\ ;\ \hat{f}(x) = \ell\right\}$, for $\ell = 1, \ldots, k$, in a two-dimensional projection of the Iris dataset. Increasing R leads to smoother class boundaries.

13.4.2 Binary Logistic Classification

Logistic classification is a widely used baseline for binary and multiclass prediction. It outputs a probability for each class, providing a richer prediction than a class label alone. The accuracy of these probabilities must still be assessed on held-out data. The standard name for this model is logistic regression, even when the prediction task is classification.

Support vector machines provide another common approach. Their hinge loss is nonsmooth, and their scores are not probabilities without an additional calibration step. The choice between these methods depends on the task and evaluation criterion.

For the binary formulas below, use labels $y_i \in \mathcal{Y} = \{-1, 1\}$. In logistic classification, the prediction $f(\cdot, \beta) \in [0, 1]$ is a probability for the positive class rather than a class label. We retain the notation f for this probability predictor.

Approximate risk minimization. A linear score $\langle x, \beta \rangle$ defines the classifier $\operatorname{sign}(\langle x, \beta \rangle)$, with either class assigned when the score is zero. The empirical 0–1 objective counts classification errors, leading to the minimization problem

$$\min_\beta \sum_{i=1}^{n} \ell_0(-y_i \langle x_i, \beta \rangle) \tag{13.20}$$

where $\ell_0 = 1_{[0,+\infty)}$ counts zero margins as errors. Away from ties, misclassification corresponds to $\langle x_i, \beta \rangle$ and y_i having different signs, so that in this case $\ell_0(-y_i \langle x_i, \beta \rangle) = 1$ (and 0 otherwise for correct classification).

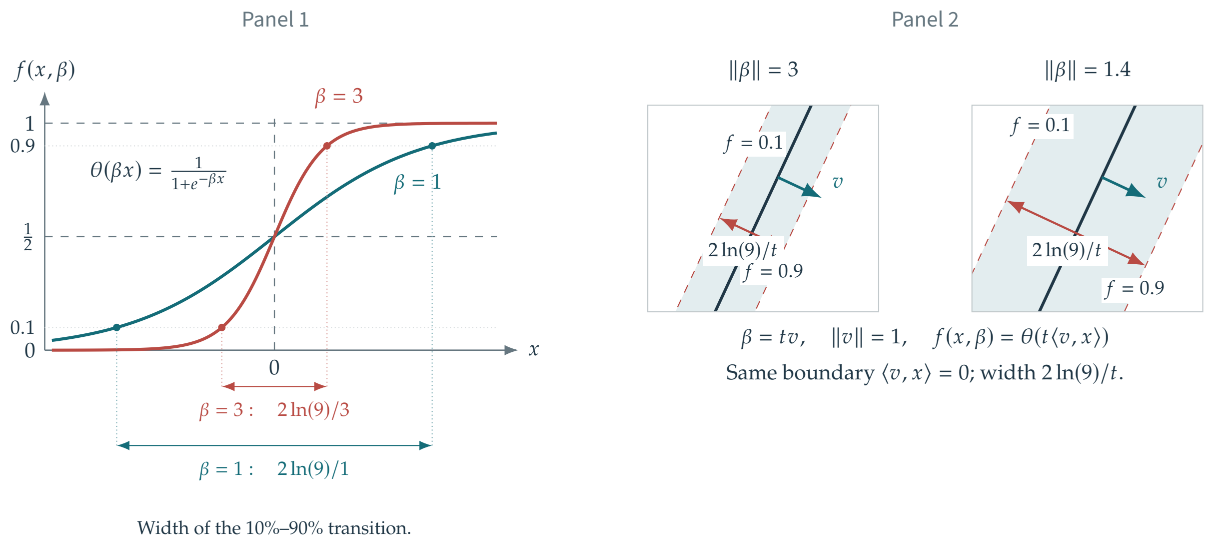


Figure 13.15. Logistic classification in one and two dimensions, showing how $\|\beta\|$ controls the width of the probability transition.

The loss ℓ_0 is nonconvex, and solving (13.20) globally is NP-hard in general. Convex upper bounds on ℓ_0 provide tractable surrogate objectives.

Two common surrogates are

$$\ell(u) = (1+u)_+ \quad \text{and} \quad \ell(u) = \log(1+\exp(u))/\log(2)$$

These are, respectively, the nonsmooth hinge loss used in support vector machines and the smooth logistic loss. The $1/\log(2)$ is just a constant which makes $\ell_0 \leqslant \ell$. AdaBoost uses the exponential loss $\ell(u) = e^u$. Least squares uses $\ell(u) = (1+u)^2$. Its growth for large negative margins makes it a poor pointwise approximation of ℓ_0, although it can still give useful classifiers. Unregularized hinge-loss minimization is equivalent to a linear program with nonnegative slack variables:

$$\min_{u \geqslant 0, \beta} \left\{ \sum_i u_i \; ; \; y_i \langle x_i, \beta \rangle \geqslant 1 - u_i \text{ for all } i \right\}.$$

Logistic loss probabilistic interpretation. Logistic regression applies a sigmoid to the linear score from Section 13.3.1. The resulting probability predictor is nonlinear. The predicted probability of class +1 is

$$f(x,\beta) \stackrel{\text{def.}}{=} \theta(\langle x, \beta\rangle) \quad \text{where} \quad \theta(s) \stackrel{\text{def.}}{=} \frac{e^s}{1+e^s} = (1+e^{-s})^{-1}, \tag{13.21}$$

which is often called the “logit” model. Despite its simplicity, a linear classifier can be effective when the feature dimension p is large. With $s = \langle x, \beta\rangle$, the predicted probability of class -1 is $1 - f(x,\beta) = \theta(-s)$. The identity $\theta(-s) = 1 - \theta(s)$ makes the two probabilities sum to one. It does not require balanced class frequencies; an intercept can account for unequal prior frequencies.

For a nonzero parameter, $\beta/\|\beta\|$ determines the normal direction of the separating hyperplane, while $1/\|\beta\|$ controls the width of the probability transition. As $\|\beta\| \to +\infty$ along a fixed direction, the transition approaches a sharp decision boundary.

The predictor $f(x,\beta)$ is also a single-layer perceptron with a logistic (sigmoid) activation; see Chapter 17.

For conditionally independent labels $y_i \in \{-1,+1\}$, write $\bar{y}_i = (y_i+1)/2 \in \{0,1\}$, $s_i = \langle x_i, \beta\rangle$, and $p_i = \theta(s_i)$. The likelihood is

$$\mathbb{P}(\mathbf{y} = y_i \mid \mathbf{x} = x_i) = p_i^{\bar{y}_i}(1-p_i)^{1-\bar{y}_i}.$$

Thus the negative log-likelihood is

$$\sum_i \{-\bar{y}_i \log p_i - (1-\bar{y}_i)\log(1-p_i)\} = \sum_i \log(1+\exp(-y_i s_i)).$$

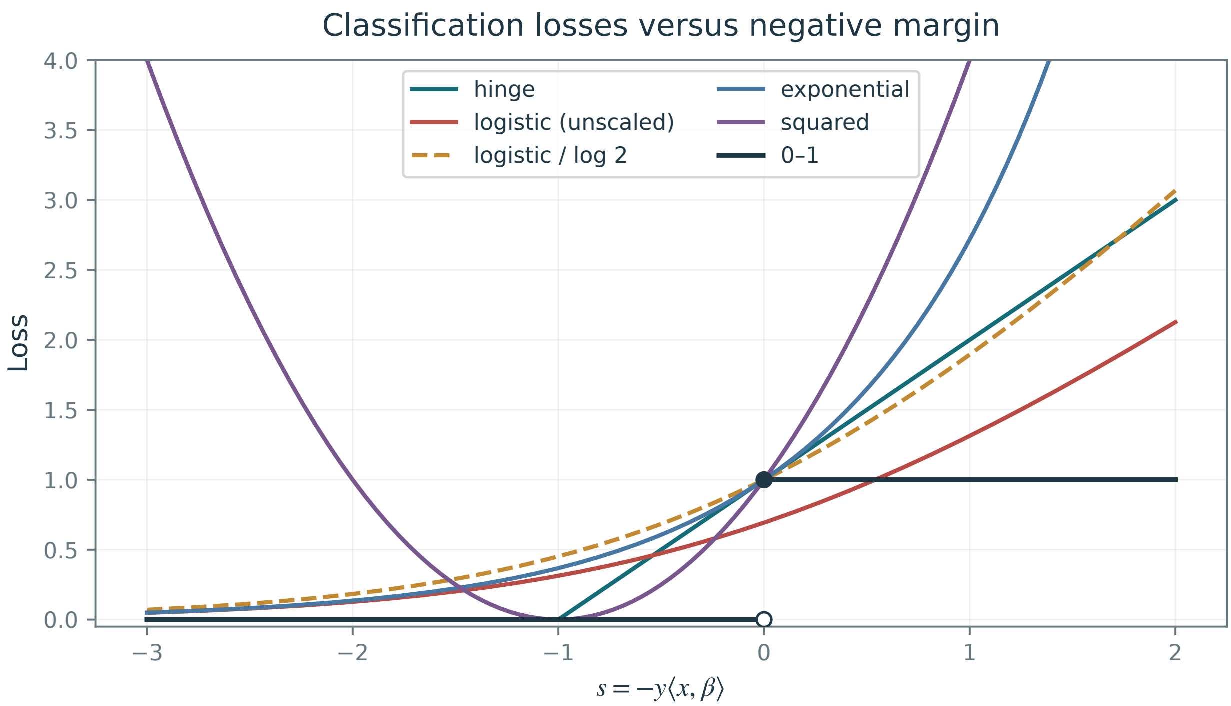


Figure 13.16. Classification losses as functions of the negative margin $s = -y\langle x, \beta\rangle$, with zero margin counted as an error. The unscaled logistic loss need not upper-bound the 0–1 loss; division by log 2 gives the upper-bound normalization used above.

After division by the sample size, this gives the empirical-risk objective (13.11) with logistic loss:

$$\hat{\beta} \in \underset{\beta\in\mathbb{R}^p}{\operatorname{argmin}}\ E(\beta) = \frac{1}{n}\sum_{i=1}^{n} L(\langle x_i, \beta\rangle, y_i) \tag{13.22}$$

where the logistic loss reads

$$L(s, y) \overset{\text{def.}}{=} \log(1 + \exp(-sy)). \tag{13.23}$$

Problem (13.22) is a smooth convex minimization. If X has full column rank, E is strictly convex, so it has at most one minimizer. A finite minimizer need not exist for separable data. A positive quadratic penalty guarantees existence and uniqueness.

Figure 13.16 compares the 0–1, logistic, and hinge losses.

Gradient descent method. Write the objective as

$$E(\beta) = \mathcal{L}(X\beta, y) \quad \text{where} \quad \mathcal{L}(s, y) = \frac{1}{n}\sum_i L(s_i, y_i),$$

Its gradient is

$$\nabla E(\beta) = X^* \nabla\mathcal{L}(X\beta, y) \quad \text{where} \quad \nabla\mathcal{L}(s, y) = -\frac{y}{n} \odot \theta(-y \odot s),$$

where $\odot$ is the pointwise multiplication operator, i.e. `.*` in Matlab. Once $\beta^{(\ell=0)} \in \mathbb{R}^p$ is initialized (for instance at 0_p), one step of gradient descent (14.13) reads

$$\beta^{(\ell+1)} = \beta^{(\ell)} - \tau_\ell \nabla E(\beta^{(\ell)}).$$

Figure 13.17 illustrates the method on samples from a Gaussian mixture, whose overlap is controlled by an offset ω. Overlaying the data on the probability map highlights the separating hyperplane $\{x\ ;\ \langle \beta, x\rangle = 0\}$.

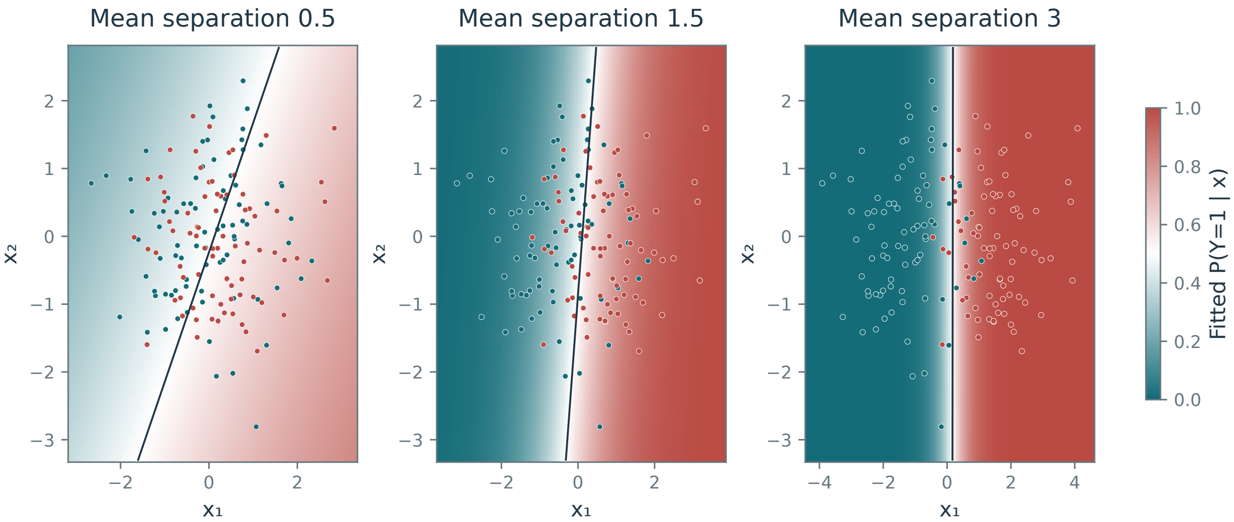


Figure 13.17. Effect of class separation on predicted classification probabilities.

13.4.3 Multiclass Logistic Classification

For multiclass logistic classification, introduce one weight vector for each of the k classes. Collect the vectors $\beta = (\beta_\ell)_{\ell=1}^k$ as columns of $\beta \in \mathbb{R}^{p\times k}$.

For an input $x \in \mathbb{R}^p$, convert the class scores into probabilities using the softmax model:

$$f(x,\beta) = \left(\frac{e^{\langle x,\beta_\ell\rangle}}{\sum_m e^{\langle x,\beta_m\rangle}}\right)_\ell \tag{13.24}$$

The vector $f(x,\beta) \in [0,1]^k$ gives the probabilities that x belongs to each class, and satisfies $\sum_\ell f(x,\beta)_\ell = 1$.

Estimate β by maximizing the log-likelihood over the training sample:

$$\max_{\beta\in\mathbb{R}^{p\times k}} \frac{1}{n}\sum_{i=1}^n \log(f(x_i,\beta)_{y_i})$$

where $y_i \in \mathcal{Y} = \{1,\dots,k\}$ is the label of observation x_i.

This is conveniently rewritten as

$$\min_{\beta\in\mathbb{R}^{p\times k}} \mathcal{E}(\beta) \stackrel{\text{def.}}{=} \frac{1}{n}\left(\sum_i \mathrm{LSE}(X\beta)_i - \langle X\beta,\, D\rangle\right)$$

where $D \in \{0,1\}^{n\times k}$ is the one-hot class matrix

$$D_{i,\ell} = \begin{cases} 1 & \text{if} \quad y_i = \ell, \\ 0 \text{otherwise}. \end{cases}$$

and LSE is the log-sum-exp operator

$$\mathrm{LSE}(S)_i = \log\left(\sum_\ell \exp(S_{i,\ell})\right), \qquad \mathrm{LSE}(S) \in \mathbb{R}^n.$$

For $k = 2$, identify the first class with label $+1$ and the second with label -1. The model then agrees with binary logistic classification in Section 13.4.2, with parameter $\beta_1 - \beta_2$. A single weight vector therefore suffices in that case.

Direct evaluation of log-sum-exp can overflow when $S_{i,\ell}$ is large. A stable evaluation subtracts the largest entry in each row before exponentiation, using the identity

$$\mathrm{LSE}(S + a) = \mathrm{LSE}(S) + a$$

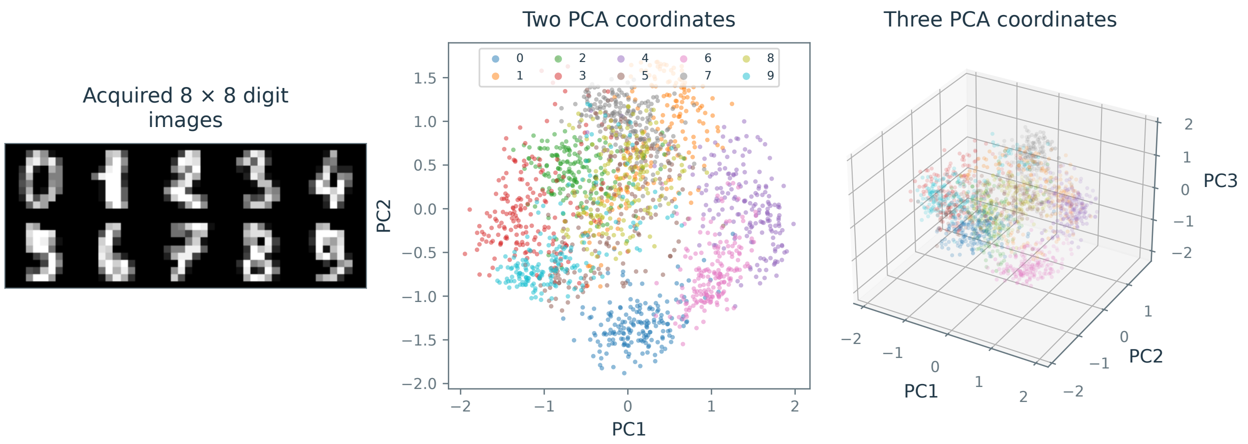


Figure 13.18. Digit images and their two- and three-dimensional PCA projections.

for a rowwise constant a. This shift is especially useful for well-separated classes, whose parameter vectors β_ℓ may grow large.

The gradient of the sum of the rowwise log-sum-exp values is the softmax operator

$$\nabla_S \sum_i \mathrm{LSE}(S)_i = \mathrm{SM}(S) \overset{\text{def.}}{=} \left(\frac{e^{S_{i,\ell}}}{\sum_m e^{S_{i,m}}} \right)$$

As with log-sum-exp, subtract the largest entry in each row before evaluating the exponentials.

Once the matrix D is computed, the gradient of $\mathcal{E}$ is computed as

$$\nabla \mathcal{E}(\beta) = \frac{1}{n} X^*(\mathrm{SM}(X\beta) - D).$$

The objective $\mathcal{E}$ can then be minimized by gradient descent.

To illustrate the method, we use a dataset of n images of size $p = 8 \times 8$, representing digits from 0 to 9 (so there are $k = 10$ classes). Figure 13.18 displays a few representative examples as well as 2-D and 3-D PCA projections. Figure 13.19 visualizes the fitted class probabilities $f(x, \hat{\beta})$ by assigning colors on a regular image grid.

13.5 Kernel Methods

Linear models may fail to capture the geometry needed for a regression or classification task. Moreover, inputs such as strings and graphs need not belong to a vector space.

Kernel methods introduce a feature map into a richer space and apply linear prediction there. This produces nonlinear regression and classification rules while retaining linear-system solvers and convex optimization. The kernel trick expresses the computations through similarities between the n observations, without constructing the feature vectors explicitly. Such methods are nonparametric when the number of fitted coefficients grows with n, allowing the model to become more expressive as data accumulate.

Algorithms that use the observations x_i only through inner products can often be kernelized. Examples include linear regression, nearest-neighbor methods, SVMs, logistic classification, and PCA. We introduce the construction through ridge regression and then extend it to other losses and distance-based methods.

13.5.1 Feature Map and Kernels

Let $\varphi : \mathcal{X} \to \mathcal{H}$ be a feature map into a real Hilbert space $\mathcal{H}$, which may be infinite dimensional. The input space $\mathcal{X}$ need not be a vector space. When $\mathcal{X} = \mathbb{R}^p$, the choice $\varphi(x) = x$ recovers ordinary linear methods. For regression, we consider predictions which are linear functions over this lifted space, i.e. of the form $x \mapsto \langle \varphi(x), \beta \rangle_{\mathcal{H}}$ for some weight vector $\beta \in \mathcal{H}$. For binary classification, one uses $x \mapsto \mathrm{sign}(\langle \varphi(x), \beta \rangle_{\mathcal{H}})$.

For scalar inputs ($p = 1$), the feature map $\varphi(x) = (1, x, x^2, \ldots, x^k) \in \mathbb{R}^{k+1}$ gives polynomial regression. Multivariate monomials extend this construction to higher-dimensional inputs. Let $\Phi = (\varphi(x_i)^*)_{i=1}^n$ be the feature matrix, whose rows are $\varphi(x_i)$. For $\mathcal{H} = \mathbb{R}^{\bar{p}}$, it is an ordinary matrix $\Phi \in \mathbb{R}^{n \times \bar{p}}$. In an infinite-dimensional feature space, the same notation represents a linear operator.

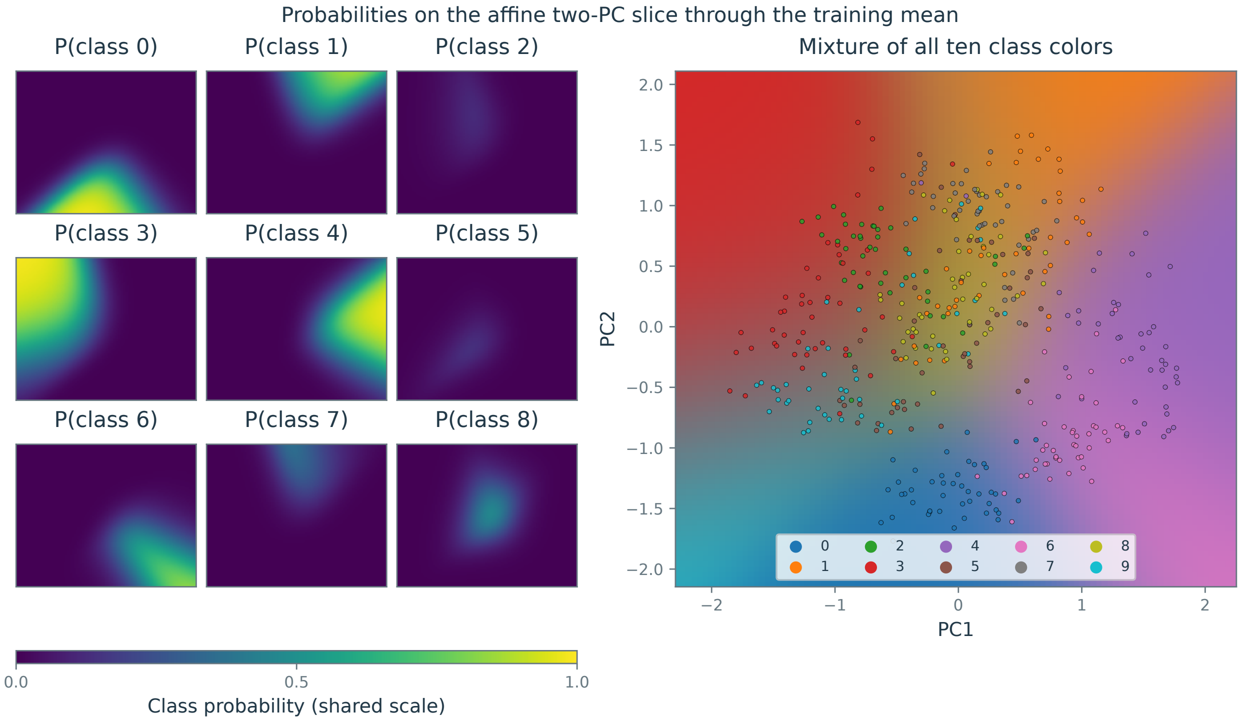


Figure 13.19. Digit classification. Left: the first nine class probabilities on an affine two-dimensional PCA slice through the training mean. Right: the mixture of all ten class colors, with projected observations overlaid.

For $\lambda > 0$, consider the regularized empirical-risk problem

$$\min_{\beta \in \mathcal{H}} \sum_{i=1}^{n} \ell(y_i, \langle \varphi(x_i), \beta \rangle) + \frac{\lambda}{2} \|\beta\|_{\mathcal{H}}^2 = \mathcal{L}(\Phi\beta, y) + \frac{\lambda}{2} \|\beta\|_{\mathcal{H}}^2 \tag{13.25}$$

Start with regression under the squared loss $\ell(y, z) = \frac{1}{2}(y - z)^2$:

$$\beta^\star = \underset{\beta \in \mathcal{H}}{\operatorname{argmin}} \|\Phi\beta - y\|_{\mathbb{R}^n}^2 + \lambda \|\beta\|_{\mathcal{H}}^2.$$

which is the ridge regression problem studied in Section 13.3.1. Setting the gradient to zero gives

$$\beta^\star = (\Phi^*\Phi + \lambda \mathrm{Id}_{\mathcal{H}})^{-1} \Phi^* y.$$

Direct computation may be impractical or impossible when $\mathcal{H}$ is infinite dimensional. The following Woodbury identity converts the problem into a finite system. This is the feature-space counterpart of (13.17), with Φ replacing X and λ replacing $n\lambda_n$.

Proposition 13.7. *One has*

$$(\Phi^*\Phi + \lambda \mathrm{Id}_{\mathcal{H}})^{-1} \Phi^* = \Phi^* (\Phi\Phi^* + \lambda \mathrm{Id}_{\mathbb{R}^n})^{-1}.$$

Proof. Denoting $U = (\Phi^*\Phi + \lambda \mathrm{Id}_{\mathcal{H}})^{-1}\Phi^*$ and $V = \Phi^*(\Phi\Phi^* + \lambda \mathrm{Id}_{\mathbb{R}^n})^{-1}$, one has

$$(\Phi^*\Phi + \lambda \mathrm{Id}_{\mathcal{H}}) U = \Phi^*$$

and

$$(\Phi^*\Phi + \lambda \mathrm{Id}_{\mathcal{H}}) V = (\Phi^*\Phi\Phi^* + \lambda \Phi^*)(\Phi\Phi^* + \lambda \mathrm{Id}_{\mathbb{R}^n})^{-1} = \Phi^*(\Phi\Phi^* + \lambda \mathrm{Id}_{\mathbb{R}^n})(\Phi\Phi^* + \lambda \mathrm{Id}_{\mathbb{R}^n})^{-1} = \Phi^*.$$

Since $(\Phi^*\Phi + \lambda \mathrm{Id}_{\mathcal{H}})$ is invertible, one obtains the result. □

The identity gives the alternative representation

$$\beta^\star = \Phi^* c^\star \quad \text{where} \quad c^\star \stackrel{\text{def.}}{=} (K + \lambda \mathrm{Id}_{\mathbb{R}^n})^{-1} y$$

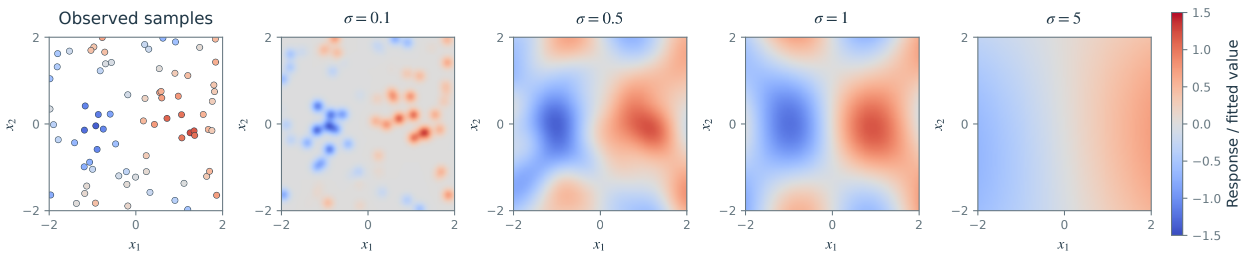


Figure 13.20. Regression using a Gaussian kernel.

where the empirical kernel matrix is

$$K = \Phi\Phi^* = (\kappa(x_i, x_j))_{i,j=1}^n \in \mathbb{R}^{n\times n} \quad \text{where} \quad \kappa(x, x') \stackrel{\text{def.}}{=} \langle \varphi(x), \varphi(x')\rangle.$$

The coefficients $c^\star \in \mathbb{R}^n$ depend on the inputs only through κ, without explicit computation of $\varphi(x)$. If κ is inexpensive to evaluate, forming K takes $O(n^2)$ kernel evaluations. A dense direct solve costs $O(n^3)$ operations; an iterative solve costs $O(n^2)$ per matrix-vector product, with the iteration count determined by conditioning and accuracy.

The fitted predictor at a new input $x \in \mathcal{X}$ also uses only kernel evaluations:

$$\langle \beta^\star, \varphi(x)\rangle_{\mathcal{H}} = \langle \sum_i c_i^\star \varphi(x_i), \varphi(x)\rangle_{\mathcal{H}} = \sum_i c_i^\star \kappa(x, x_i). \tag{13.26}$$

13.5.2 Kernel Design

One can also start directly from a kernel $\kappa(x, x')$ rather than an explicit feature map φ. This is often convenient because kernels can be designed through their geometric properties and combined, for example by addition. The required condition is that every Gram matrix $(\kappa(x_i, x_j))_{i,j}$ be symmetric positive semidefinite. Such kernels are commonly called positive definite kernels; the condition is equivalent to the existence of a Hilbert-space feature map generating the kernel. The underlying input space need not be Euclidean.

When using the linear kernel $\kappa(x, y) = \langle x, y\rangle$, one retrieves the linear models studied in the previous section, and the lifting is trivial $\varphi(x) = x$. A family of popular kernels are polynomial ones, $\kappa(x, x') = (\langle x, x'\rangle + c)^a$ for $a \in \mathbb{N}^*$ and $c > 0$, which corresponds to a lifting in finite dimension. For instance, for $a = 2$ and $p = 2$, one has a lifting in dimension 6

$$\kappa(x, x') = \left(x_1x_1' + x_2x_2' + c\right)^2 = \langle \varphi(x), \varphi(x')\rangle \quad \text{where} \quad \varphi(x) = (x_1^2, x_2^2, \sqrt{2}x_1x_2, \sqrt{2c}x_1, \sqrt{2c}x_2, c)^* \in \mathbb{R}^6.$$

A widely used choice on Euclidean spaces is the Gaussian kernel

$$\kappa(x, y) \stackrel{\text{def.}}{=} e^{-\frac{\|x-y\|^2}{2\sigma^2}}. \tag{13.27}$$

The bandwidth $\sigma > 0$ controls the locality of the model and is typically tuned by cross-validation. The Gaussian kernel has the infinite-dimensional feature map $\varphi(x)(t) = (2/(\pi\sigma^2))^{p/4}\exp(-\|x-t\|^2/\sigma^2) \in L^2(\mathbb{R}^p)$. Another related popular kernel is the Laplacian kernel $\exp(-\|x-y\|/\sigma)$. More generally, Bochner's theorem states that a continuous translation-invariant kernel $\kappa(x, x') = k(x - x')$ on $\mathbb{R}^p$ is positive definite if and only if k is the Fourier transform of a finite nonnegative measure. If this measure has an integrable density, a feature map can be built from the square root of that density. When k is integrable and its Fourier transform is an integrable function, the condition is $\hat{k} \geqslant 0$; strict positivity is not necessary. Figure 13.20 shows an example of regression using a Gaussian kernel.

Kernel on non-Euclidean spaces. On a general input space $\mathcal{X}$, a real kernel κ must produce a symmetric positive semidefinite Gram matrix $K = (\kappa(x_i, x_j))_{i,j}$ for every finite collection of inputs. Equivalently, there is a Hilbert space $\mathcal{H}$ and a feature map $\varphi : \mathcal{X} \to \mathcal{H}$ such that $\kappa(x, x') = \langle \varphi(x), \varphi(x')\rangle_{\mathcal{H}}$.

For instance, if $\mathcal{X} = \mathcal{P}(\mathcal{M})$ is the set of measurable sets of finite measure in a measure space $(\mathcal{M}, \mu)$, one can define $\kappa(A, B) = \mu(A \cap B)$ which corresponds to the lifting $\varphi(A) = 1_A$ so that $\langle \varphi(A), \varphi(B)\rangle = \int 1_A(x)1_B(x)\mathrm{d}\mu(x) = \mu(A \cap B)$. Kernels can also be defined on strings and graphs, although their construction is more involved.

When $\kappa(x,x) > 0$ for every input under consideration, the kernel can be normalized to have a unit diagonal. For a translation-invariant kernel on $\mathbb{R}^p$, the diagonal equals the constant $k(0)$, so normalization only requires division by $k(0) > 0$. This corresponds to replacing $\varphi(x)$ by $\varphi(x)/\|\varphi(x)\|_{\mathcal{H}}$ and thus replacing $\kappa(x,x')$ by $\frac{\kappa(x,x')}{\sqrt{\kappa(x,x)}\sqrt{\kappa(x',x')}}$.

13.5.3 General Case

The least-squares result extends to other losses under the assumptions below. With a squared Hilbert-space norm penalty, the minimizer belongs to a subspace generated by the data, of dimension at most n. This representer theorem reduces the problem to finitely many coefficients even when the feature space $\mathcal{H}$ is infinite dimensional.

Proposition 13.8. *Assume $\lambda > 0$ and the losses $\ell(y_i,\cdot)$ are finite, nonnegative, and convex. The solution $\beta^\star \in \mathcal{H}$ of*

$$\min_{\beta\in\mathcal{H}} \sum_{i=1}^{n} \ell(y_i, \langle\varphi(x_i),\, \beta\rangle) + \frac{\lambda}{2}\|\beta\|_{\mathcal{H}}^2 = \mathcal{L}(\Phi\beta, y) + \frac{\lambda}{2}\|\beta\|_{\mathcal{H}}^2 \tag{13.28}$$

is unique and can be written as

$$\beta = \Phi^* c^\star = \sum_i c_i^\star \varphi(x_i) \in \mathcal{H} \tag{13.29}$$

where $c^\star \in \mathbb{R}^n$ is a solution of

$$\min_{c\in\mathbb{R}^n} \mathcal{L}(Kc, y) + \frac{\lambda}{2}\langle Kc,\, c\rangle_{\mathbb{R}^n} \tag{13.30}$$

where we defined

$$K \overset{\text{def.}}{=} \Phi\Phi^* = (\langle\varphi(x_i),\, \varphi(x_j)\rangle_{\mathcal{H}})_{i,j=1}^n \in \mathbb{R}^{n\times n}.$$

Proof. Let $V = \operatorname{Span}\{\varphi(x_i) : 1 \leqslant i \leqslant n\}$ and decompose $\beta = \beta_V + \beta_\perp$ orthogonally. Predictions on the data depend only on β_V, while $\|\beta\|^2 = \|\beta_V\|^2 + \|\beta_\perp\|^2$. Since $\lambda > 0$, a minimizer has $\beta_\perp = 0$. The restriction to the finite-dimensional space V is continuous, coercive, and strictly convex, hence has a unique minimizer. Substituting $\beta = \Phi^* c$ gives (13.30); the coefficient vector c need not be unique if K is singular. □

Equation (13.29) places the minimizer in a subspace of dimension at most n, spanned by the observed feature vectors $\varphi(x_i)$. The potentially infinite-dimensional problem (13.28) becomes the finite-dimensional problem (13.30). A dense first-order iteration costs $O(n^2)$ once the kernel matrix is available. A low-rank approximation $K \approx \tilde{\Phi}^*\tilde{\Phi}$ can reduce this cost further by introducing a smaller approximate feature space.

For classification applications, one can use for ℓ a hinge or a logistic loss function, and then the decision boundary is computed following (13.26) using kernel evaluations only as

$$\operatorname{sign}(\langle\beta^\star,\, \varphi(x)\rangle_{\mathcal{H}}) = \operatorname{sign}(\sum_i c_i^\star \kappa(x, x_i)).$$

Figure 13.21 illustrates a nonlinear decision boundary in two dimensions. The decision rule is linear in the feature space $\mathcal{H}$; evaluating it through the nonlinear feature map produces a curved boundary in the input space.

Nearest-neighbor classification (Section 13.4.1) and regression can likewise use distances in feature space, computed entirely from the kernel:

$$\|\varphi(x_i) - \varphi(x_j)\|_{\mathcal{H}}^2 = \kappa(x_i, x_i) + \kappa(x_j, x_j) - 2\kappa(x_i, x_j).$$

13.6 Probably Approximately Correct Learning

This section introduces Probably Approximately Correct (PAC) learning, which studies generalization guarantees under explicit complexity and sampling assumptions. Distribution-free bounds avoid further restrictions on the data distribution. The main reference (and in particular the proofs of the mentioned results) for this section is the book “Foundations of Machine Learning” by Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar https://cs.nyu.edu/~mohri/mlbook/.

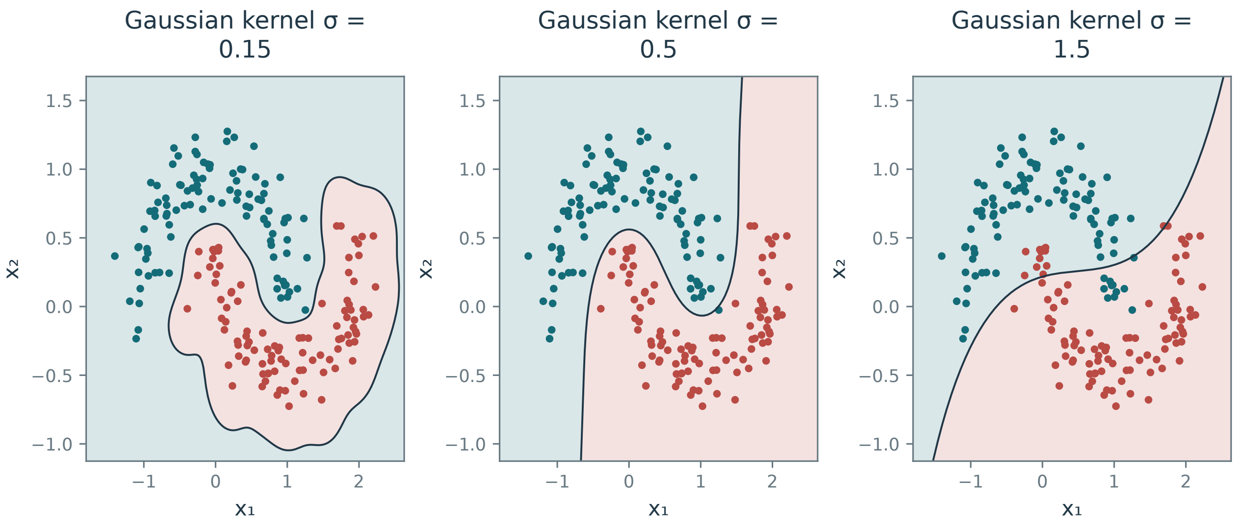


Figure 13.21. Nonlinear classification using a Gaussian kernel.

Assume the observations $\mathcal{D} \overset{\text{def.}}{=} (x_i, y_i)_{i=1}^n$ are independent and identically distributed copies of a random pair (X, Y) taking values in $\mathcal{X} \times \mathcal{Y}$. From this sample, we seek a predictor $f : \mathcal{X} \to \mathcal{Y}$ with risk close to the smallest achievable value:

$$L(f) \overset{\text{def.}}{=} \mathbb{E}(\ell(Y, f(X))) = \int_{\mathcal{X}\times\mathcal{Y}} \ell(y, f(x)) \mathrm{d}\mathbb{P}_{X,Y}(x, y)$$

where $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ is some loss function. To estimate this minimizer, we select a class of functions $\mathcal{F}$ and minimize the empirical risk

$$\hat{f} \in \underset{f\in\mathcal{F}}{\operatorname{argmin}}\ \hat{L}(f) \overset{\text{def.}}{=} \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, f(x_i)).$$

We assume a measurable minimizer exists. It is random because it depends on the sample $\mathcal{D}$.

Example 13.9 (Regression and classification). In regression, $\mathcal{Y} = \mathbb{R}$ and the most common choice of loss is $\ell(y, z) = (y - z)^2$. For binary classification, $\mathcal{Y} = \{-1, +1\}$ and the ideal choice is the 0-1 loss $\ell(y, z) = 1_{y\neq z}$. For many useful classes $\mathcal{F}$, minimizing the empirical 0–1 risk is computationally difficult. A common alternative uses a surrogate loss of the form $\ell(y, z) = \Gamma(-yz)$ where Γ is a convex function upper-bounding $1_{\mathbb{R}^+}$, which makes empirical risk minimization convex when the real-valued score class is convex.

PAC learning seeks a bound on the excess risk $L(\hat{f}) - \inf(L) \geqslant 0$ that holds with probability at least $1 - \delta$. The bound depends on the sample size n and confidence parameter δ. For the excess risk to vanish, the approximation class must become sufficiently rich while its statistical complexity remains controlled. Quantitative approximation rates require distributional assumptions, although some consistency results hold without them.

13.6.1 Nonparametric Models and Calibration

If $\mathcal{F}$ contains all measurable functions, a minimizer $f^\star$ of L is called a Bayes estimator: it achieves the smallest possible population risk.

Risk decomposition Denoting

$$\alpha(z|x) \overset{\text{def.}}{=} \mathbb{E}_Y(\ell(Y, z)|X = x) = \int_{\mathcal{Y}} \ell(y, z) \mathrm{d}\mathbb{P}_{Y|X}(y|x)$$

the average loss of predicting $z \in \mathcal{Y}$ at input $x \in \mathcal{X}$, the risk decomposes as

$$L(f) = \int_{\mathcal{X}} \left[\int_{\mathcal{Y}} \ell(y, f(x)) \mathrm{d}\mathbb{P}_{Y|X}(y|x) \right] \mathrm{d}\mathbb{P}_X(x) = \int_{\mathcal{X}} \alpha(f(x)|x) \mathrm{d}\mathbb{P}_X(x)$$

Consequently, $f^\star$ can be chosen separately at each input x by solving

$$f^\star(x) = \operatorname*{argmin}_z \alpha(z|x).$$

Example 13.10 (Regression). For squared-loss regression with $\mathbb{E}(Y^2) < \infty$, one has $f^\star(x) = \mathbb{E}(Y|X = x) = \int_{\mathcal{Y}} y \mathrm{d}\mathbb{P}_{Y|X}(y|x)$, which is the conditional expectation.

Example 13.11 (Classification). For classification applications, where $\mathcal{Y} = \{-1, 1\}$, it is convenient to introduce $\eta(x) \stackrel{\text{def.}}{=} \mathbb{P}_{Y|X}(y = 1|x) \in [0, 1]$. If the two classes are separable, then $\eta(x) \in \{0, 1\}$ for $\mathbb{P}_X$-almost every x. For the 0-1 loss $\ell(y, z) = 1_{y \neq z} = 1_{(0,+\infty)}(-yz)$, one may choose $f^\star = \operatorname{sign}(2\eta - 1)$, with either class selected at a tie and $L(f^\star) = \mathbb{E}_X(\min(\eta(X), 1 - \eta(X)))$. The conditional probability η is unknown and must be estimated from $\mathcal{D}$. Direct 0–1 optimization is computationally difficult for many useful classes $\mathcal{F}$. For a margin loss $\ell(y, z) = \Gamma(-yz)$, an unconstrained Bayes score satisfies

$$f^\star(x) = \operatorname*{argmin}_z \Big(\alpha(z|x) = \eta(x)\Gamma(-z) + (1 - \eta(x))\Gamma(z) \Big),$$

Thus the optimal score is a function of $\eta(x)$. This argmin notation presumes attainment; for logistic loss and $\eta \in \{0, 1\}$, the infimum is approached only as the score tends to the appropriate infinity.

Calibration in the classification setup We would like the classifier $\operatorname{sign}(f^\star)$ to agree with a Bayes classifier for the 0–1 loss, namely $\operatorname{sign}(2\eta - 1)$ away from ties. This property is called classification calibration. It does not require the scores $f^\star$ and $2\eta - 1$ themselves to agree. The following result characterizes calibration for finite convex margin losses.

Proposition 13.12. *A loss ℓ associated to a finite, nonnegative convex Γ is calibrated if and only if Γ is differentiable at 0 and $\Gamma'(0) > 0$.*

Both hinge and logistic loss satisfy this calibration criterion. Let L_Γ be the risk for $\ell(y, z) = \Gamma(-yz)$ and let $L_{0\text{–}1}$ be classification risk, with a fixed tie-breaking convention. Quantitative calibration bounds have the form

$$0 \leqslant L_{0\text{–}1}(\operatorname{sign} f) - \inf L_{0\text{–}1} \leqslant \Psi(L_\Gamma(f) - \inf L_\Gamma) \tag{13.31}$$

for some increasing function $\Psi : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying $\Psi(0) = 0$. Such a control ensures in particular that if $f^\star$ minimizes L_Γ, its induced classifier has Bayes-optimal 0–1 risk. At $\eta = 1/2$, either class is optimal; away from ties, its sign agrees almost surely with that of $2\eta - 1$. For hinge loss, the calibration bound holds with $\Psi(r) = r$. For logistic loss, it holds with $\Psi(s) = \sqrt{2s}$.

13.6.2 PAC bounds

Approximation–estimation decomposition. For a class $\mathcal{F}$ of functions, the excess risk of the empirical estimator

$$\hat{f} \in \operatorname*{argmin}_{f \in \mathcal{F}} \hat{L}(f)$$

decomposes into a random estimation error and a deterministic approximation error:

$$L(\hat{f}) - \inf L = \Big[L(\hat{f}) - \inf_{\mathcal{F}} L \Big] + \mathcal{A}(\mathcal{F}) \quad \text{where} \quad \mathcal{A}(\mathcal{F}) \stackrel{\text{def.}}{=} \Big[\inf_{\mathcal{F}} L - \inf L \Big]. \tag{13.32}$$

This separates the effects of sampling and model approximation.

Enlarging a nested class $\mathcal{F}$ reduces approximation error but typically increases the estimation-error bound. Choose the class size as a function of n to balance these effects and avoid overfitting.

Approximation error. A quantitative approximation bound requires assumptions on $f^\star$. No-free-lunch results rule out a uniform distribution-free approximation rate over all measurable targets. The following example illustrates such a bound.

Example 13.13 (Linearly parameterized functionals). A popular class of functions consists of linearly parameterized maps of the form

$$f(x) = f_w(x) = \langle \varphi(x), w \rangle_{\mathcal{H}}$$

where $\varphi : \mathcal{X} \to \mathcal{H}$ maps inputs to a Hilbert feature space. For $\mathcal{X} = \mathbb{R}^p$, the identity map $\varphi(x) = x$ recovers ordinary linear models. One can also consider for instance polynomial features, $\varphi(x) = (1, x_1, \dots, x_p, x_1^2, x_1x_2, \dots)$, giving rise to polynomial regression and polynomial classification boundaries. Restrict the parameters to the ball $\mathcal{F} = \{f_w \,;\, \|w\|_{\mathcal{H}} \leqslant R\}$. Suppose $f^\star = f_{w^\star}$ for some $w^\star \in \mathcal{H}$, the loss $\ell(y, \cdot)$ is uniformly Q-Lipschitz, and $\mathbb{E}\|\varphi(X)\|_{\mathcal{H}} < \infty$. Projecting $w^\star$ onto the radius-R ball gives

$$\mathcal{A}(\mathcal{F}) \leqslant Q\mathbb{E}(\|\varphi(X)\|_{\mathcal{H}}) \max(\|w^\star\|_{\mathcal{H}} - R, 0).$$

Remark 13.14 (Connections with RKHS). Note that this lifting actually corresponds to using functions f in a reproducing kernel Hilbert space, denoting

$$\|f\|_k \overset{\text{def.}}{=} \inf_{w \in \mathcal{H}} \{\|w\|_{\mathcal{H}} \,;\, f = f_w\}$$

with associated kernel $k(x, x') \overset{\text{def.}}{=} \langle \varphi(x), \varphi(x') \rangle_{\mathcal{H}}$. We use the feature-map representation directly below.

Estimation error. Concentration inequalities control sampling fluctuations, while a bound on the complexity of $\mathcal{F}$ makes this control uniform over the model class.

The estimation error is controlled by uniform deviations between L and $\hat{L}$. Let $g \in \mathcal{F}$ attain $\inf_{\mathcal{F}} L$, assuming for simplicity that such a minimizer exists. Then

$$L(\hat{f}) - \inf_{\mathcal{F}} L = \left[L(\hat{f}) - \hat{L}(\hat{f})\right] + \left[\hat{L}(\hat{f}) - \hat{L}(g)\right] + \left[\hat{L}(g) - L(g)\right] \leqslant 2 \sup_{f \in \mathcal{F}} |\hat{L}(f) - L(f)| \tag{13.33}$$

since $\hat{L}(\hat{f}) - \hat{L}(g) \leqslant 0$. Define the two one-sided deviations

$$\Delta_+(\mathcal{D}) = \sup_{f \in \mathcal{F}} (L(f) - \hat{L}(f)), \qquad \Delta_-(\mathcal{D}) = \sup_{f \in \mathcal{F}} (\hat{L}(f) - L(f)).$$

The preceding decomposition gives the sharper bound $L(\hat{f}) - \inf_{\mathcal{F}} L \leqslant \Delta_+ + \Delta_-$. Assume all suprema below are measurable (or use outer expectations).

Proposition 13.15 (McDiarmid concentration)**.** *Assume $0 \leqslant \ell(Y, f(X)) \leqslant \ell_\infty$ almost surely, uniformly over $f \in \mathcal{F}$. With probability at least $1 - \delta$, both signs satisfy*

$$\Delta_\pm(\mathcal{D}) - \mathbb{E}_{\mathcal{D}} \Delta_\pm(\mathcal{D}) \leqslant \ell_\infty \sqrt{\frac{\log(2/\delta)}{2n}}. \tag{13.34}$$

To bound $\mathbb{E}_{\mathcal{D}} \Delta_\pm(\mathcal{D})$, we need to control the complexity of $\mathcal{F}$. VC dimension gives one approach, but for norm-constrained linear models it can yield loose bounds that depend on the ambient dimension. Rademacher complexity gives a finer measure for a class $\mathcal{G}$ of functions from $\mathcal{X} \times \mathcal{Y}$ to $\mathbb{R}$:

$$\mathcal{R}_n(\mathcal{G}) \overset{\text{def.}}{=} \mathbb{E}_{\varepsilon, \mathcal{D}}\left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(x_i, y_i)\right]$$

where ε_i, independent of the data, are independent Rademacher random variables (i.e. $\mathbb{P}(\varepsilon_i = \pm 1) = 1/2$). The quantity $\mathcal{R}_n(\mathcal{G})$ depends on the distribution of (X, Y).

Apply this complexity measure to the loss class $\mathcal{G} = \ell[\mathcal{F}]$, defined by

$$\ell[\mathcal{F}] \overset{\text{def.}}{=} \{(x, y) \in \mathcal{X} \times \mathcal{Y} \mapsto \ell(y, f(x)) \,;\, f \in \mathcal{F}\},$$

Symmetrization gives the following bounds.

Proposition 13.16. *One has*

$$\mathbb{E}\Delta_\pm(\mathcal{D}) \leqslant 2\mathcal{R}_n(\ell[\mathcal{F}]).$$

If ℓ is Q-Lipschitz with respect to its second variable, one furthermore has $\mathcal{R}_n(\ell[\mathcal{F}]) \leqslant Q\mathcal{R}_n(\mathcal{F})$ (here the class of functions only depends on x).

Combining (13.32), (13.33), and Propositions 13.15 and 13.16 gives the following result.

Theorem 13.17. *Assume $\ell(y,\cdot)$ is uniformly Q-Lipschitz and $0 \leqslant \ell \leqslant \ell_\infty$ uniformly over the class. Then, with probability at least $1-\delta$*

$$0 \leqslant L(\hat{f}) - \inf L \leqslant \ell_\infty \sqrt{\frac{2\log(2/\delta)}{n}} + 4Q\mathcal{R}_n(\mathcal{F}) + \mathcal{A}(\mathcal{F}). \tag{13.35}$$

Example 13.18 (Linear models). In the case where $\mathcal{F} = \{\langle\varphi(\cdot),\, w\rangle_{\mathcal{H}}\ ;\ \|w\| \leqslant R\}$ where $\|\cdot\|$ is some norm on $\mathcal{H}$, one has

$$\mathcal{R}_n(\mathcal{F}) = \frac{R}{n}\mathbb{E}_{\varepsilon,\mathcal{D}}\|\sum_i \varepsilon_i\varphi(x_i)\|_*$$

where $\|\cdot\|_*$ is the dual norm, defined by

$$\|u\|_* \overset{\text{def.}}{=} \sup_{\|w\|\leqslant 1}\ \langle u,\, w\rangle_{\mathcal{H}}.$$

When $\|\cdot\| = \|\cdot\|_{\mathcal{H}}$ is the Hilbert norm, the expression simplifies further: $\|u\|_* = \|u\|_{\mathcal{H}}$ and

$$\mathcal{R}_n(\mathcal{F}) \leqslant \frac{R\sqrt{\mathbb{E}(\|\varphi(X)\|^2_{\mathcal{H}})}}{\sqrt{n}}.$$

The bound has no explicit dependence on feature dimension and also applies to infinite-dimensional RKHS feature maps with finite second moment. For fixed R and fixed loss constants, the estimation bound in (13.35) decays as $1/\sqrt{n}$; the approximation error remains a separate term. The radius R controls the balance between approximation and estimation. In practice, it can be selected by cross-validation.

Example 13.19 (Application to SVM classification). The 0–1 loss is not Lipschitz in a real-valued score. For a margin $\rho > 0$, use the bounded ramp loss

$$\Gamma_\rho(s) = \min\{1, \max\{0, 1 + s/\rho\}\}.$$

It is $1/\rho$-Lipschitz and satisfies

$$1_{[0,+\infty)}(s) \leqslant \Gamma_\rho(s) \leqslant \max\{1 + s/\rho, 0\}.$$

The left inequality counts score ties as errors, so it also bounds the risk of any fixed tie-breaking classifier. For $\mathcal{F}_R = \{f_w : \|w\|_{\mathcal{H}} \leqslant R\}$, the one-sided concentration and symmetrization bounds imply, simultaneously for every $f \in \mathcal{F}_R$, with probability at least $1-\delta$,

$$L_{0\text{–}1}(\operatorname{sign} f) \leqslant \frac{1}{n}\sum_i \max\{1 - y_i f(x_i)/\rho, 0\} + \frac{2R}{\rho}\sqrt{\frac{\mathbb{E}\|\varphi(X)\|^2_{\mathcal{H}}}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

This bound applies to a data-dependent predictor, including an empirical hinge-loss minimizer in the ball. It does not require minimizing the nonconvex ramp loss. In practice, the norm constraint is often replaced by a penalty:

$$\min_w \frac{1}{n}\sum_i \max\{1 - y_i f_w(x_i), 0\} + \lambda\|w\|^2_{\mathcal{H}}, \qquad \lambda > 0. \tag{13.36}$$

This is the kernel support vector machine.

Remark 13.20 (Resolution using the kernel trick). The kernel trick reduces problems such as (13.36), of the form

$$\min_{w\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^n \ell(y_i, \langle w,\, \varphi(x_i)\rangle) + \lambda\|w\|^2_{\mathcal{H}} \tag{13.37}$$

to optimization over n coefficients, even when the feature space is infinite dimensional. The representer theorem gives $w^\star = \sum_{i=1}^n c_i^\star\varphi(x_i)$ for some $c^\star \in \mathbb{R}^n$: an orthogonal component would leave the predictions unchanged and strictly increase the positive norm penalty. Substituting this representation into (13.37) reduces the data dependence to the empirical kernel matrix $K = (k(x_i, x_j) \overset{\text{def.}}{=} \langle\varphi(x_i),\, \varphi(x_j)\rangle)_{i,j=1}^n \in \mathbb{R}^{n\times n}$ since it reads

$$\min_{c\in\mathbb{R}^n} \frac{1}{n}\sum_{i=1}^n \ell(y_i, (Kc)_i) + \lambda\langle Kc,\, c\rangle$$

From any optimal $c^\star$ of this convex program (which need not be strictly convex when K is singular), the estimator is evaluated as $f_{w^\star}(x) = \langle w^\star,\, \varphi(x)\rangle = \sum_i c_i^\star k(x_i, x)$.

CHAPTER 14

Optimization & Machine Learning: Smooth Optimization

Training a model often reduces to minimizing a differentiable objective, so the geometry of that objective determines how optimization algorithms progress. Starting from regression and classification, we develop convexity, derivatives, and optimality conditions, then derive gradient descent and practical step-size rules. Quadratic examples guide the convergence analysis, which extends to smooth convex objectives and explains the effects of conditioning, strong convexity, and acceleration.

14.1 Motivation in Machine Learning

14.1.1 Unconstrained Optimization

Throughout most of this chapter, we consider unconstrained convex optimization problems of the form

$$\inf_{x \in \mathbb{R}^p} f(x), \tag{14.1}$$

and seek algorithms that approximate a minimizer, when one exists, with a low cost per iteration. The first-order methods studied here use gradient information. We write

$$\operatorname*{argmin}_x f(x) \overset{\text{def.}}{=} \{x \in \mathbb{R}^p \; ; \; f(x) = \inf f\},$$

for the set of minimizers of f, which may contain several points or be empty: $\operatorname{argmin} f = \emptyset$. When a minimizer exists, we write the optimization problem as

$$\min_{x \in \mathbb{R}^p} f(x). \tag{14.2}$$

In learning, $f(x)$ is typically an empirical risk for regression or classification, and p is the number of model parameters. For a linear model, the training sample is $(a_i, y_i)_{i=1}^n$ with feature vectors $a_i \in \mathbb{R}^p$. Let $A \in \mathbb{R}^{n \times p}$ be the matrix with feature vectors a_i as its rows.

14.1.2 Regression

For real-valued responses $y_i \in \mathbb{R}$, the least-squares objective is

$$f(x) = \frac{1}{2} \sum_{i=1}^n (y_i - \langle x, a_i \rangle)^2 = \frac{1}{2} \|Ax - y\|^2, \tag{14.3}$$

as illustrated in Figure 14.1. Here $\langle u, v \rangle = \sum_{i=1}^p u_i v_i$ is the canonical inner product in $\mathbb{R}^p$ and $\|\cdot\|^2 = \langle \cdot, \cdot \rangle$.

14.1.3 Classification

For classification, $y_i \in \{-1, 1\}$, in which case

$$f(x) = \sum_{i=1}^n \ell(-y_i \langle x, a_i \rangle) = L(-\operatorname{diag}(y) A x) \tag{14.4}$$

where ℓ is a smooth convex surrogate for the 0-1 loss $\mathbf{1}_{[0,\infty)}$, counting zero margin as an error. For instance, $\ell(u) = \log(1 + \exp(u))/\log 2$ satisfies $\ell(u) \geqslant \mathbf{1}_{[0,\infty)}(u)$, with equality at $u = 0$. The normalization makes this upper bound tight at the decision boundary. The matrix $\operatorname{diag}(y) \in \mathbb{R}^{n \times n}$ is the diagonal matrix with y_i along the diagonal (see Figure 14.1, panel 3). The separable loss function $L : \mathbb{R}^n \to \mathbb{R}$ is, for $z \in \mathbb{R}^n$, $L(z) = \sum_i \ell(z_i)$.

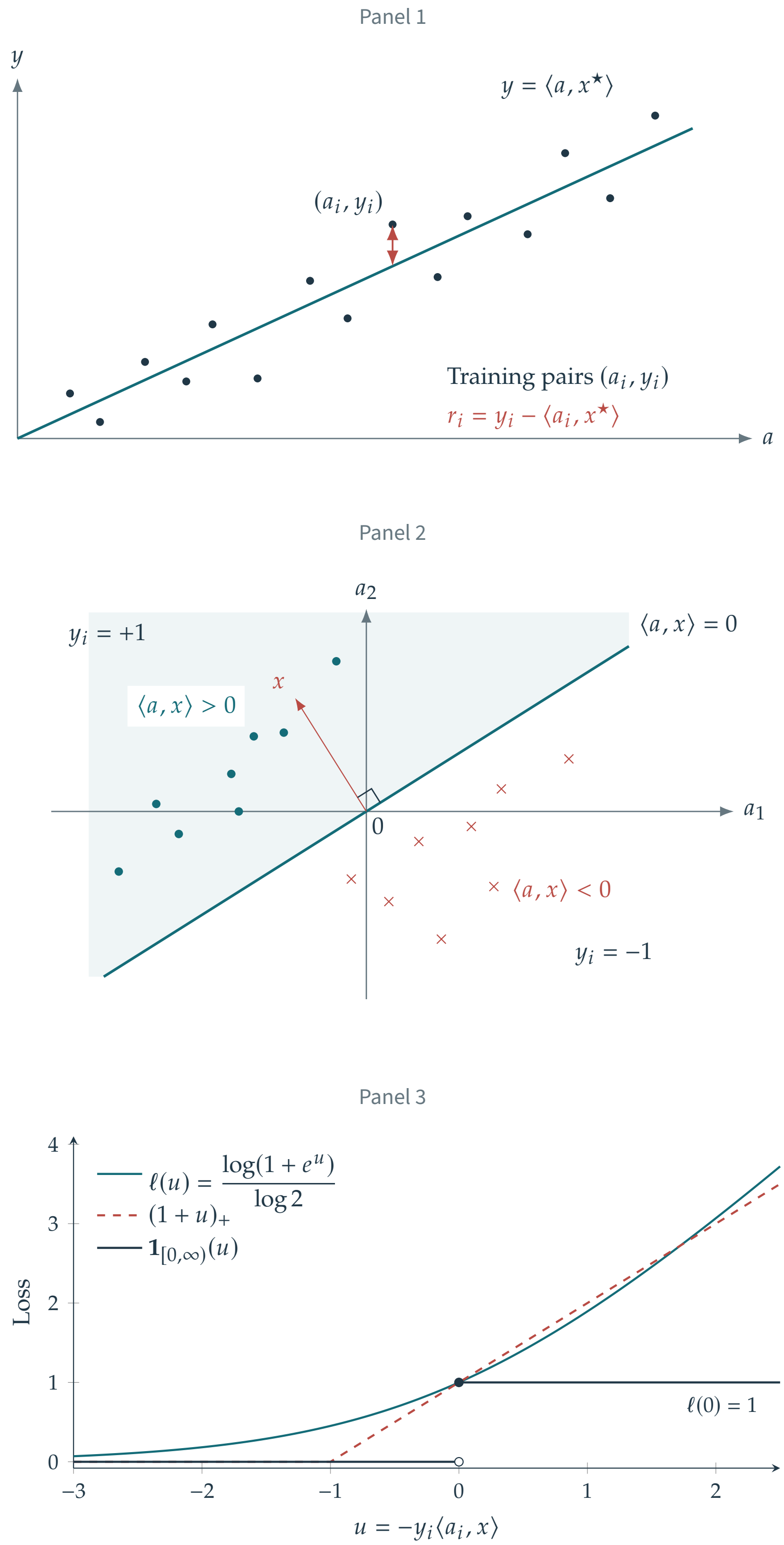


Figure 14.1. Panels 1 and 2: linear regression and a linear classifier. Panel 3: the 0-1 loss and its normalized logistic and hinge upper bounds, both tight at zero margin.

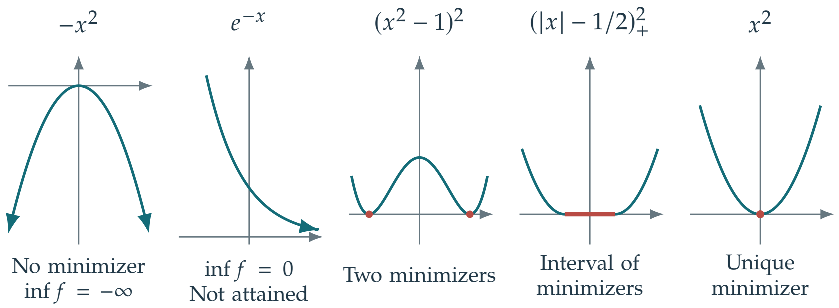


Figure 14.2. Left: no minimizer. Middle: multiple minimizers. Right: a unique minimizer.

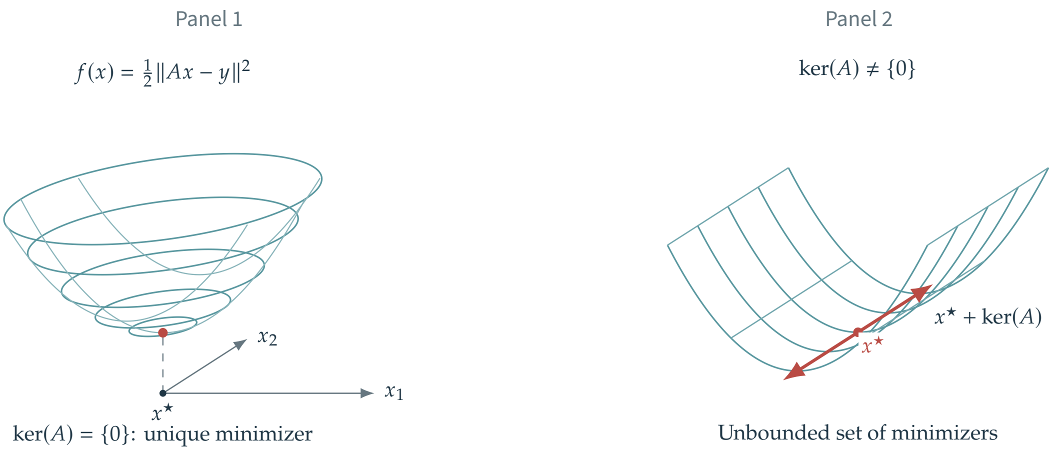


Figure 14.3. Coercivity condition for least squares.

14.2 Basics of Convex Analysis

14.2.1 Existence of Solutions

Problem (14.1) need not have a minimizer. An objective f may be unbounded below: for $f(x) = -x^2$, the infimum is $-\infty$. The objective f can also have a finite unattained infimum: for $f(x) = e^{-x}$, it is $\inf f = 0$.

A continuous function attains its minimum on a nonempty compact set $\Omega \subset \mathbb{R}^p$. In finite dimensions, compactness means that Ω is closed and bounded. Lower semicontinuity suffices for attainment, but we use continuity here. For unconstrained minimization, coercivity provides a compact set to which the search can be restricted: $f(x) \to +\infty$ as $\|x\| \to +\infty$. For any $x_0 \in \mathbb{R}^p$, consider the sublevel set

$$\Omega = \{x \in \mathbb{R}^p \; ; \; f(x) \leqslant f(x_0)\}$$

Coercivity makes this set bounded, and continuity of f makes it closed. For a continuous convex function, coercivity is in fact equivalent to a nonempty bounded minimizer set. Convexity matters: $f(x) = \min(1, x^2)$ has a unique minimizer but is not coercive.

Example 14.1 (Least squares). The least-squares objective $f(x) = \frac{1}{2}\|Ax - y\|^2$ is coercive exactly when $\ker(A) = \{0\}$, meaning the design has full column rank. If $\ker(A) \neq \{0\}$ and $x^\star$ is a minimizer, then $x^\star + u$ is also a minimizer for every $u \in \ker(A)$, so the minimizer set is unbounded. Conversely, if $\ker(A) = \{0\}$, we will show later that the minimizer is unique, see Fig. 14.3. For classification, strict convexity of ℓ and injectivity of A imply strict convexity of the objective, but do not ensure coercivity or existence. Separable logistic regression is a counterexample.

14.2.2 Convexity

Convexity is central to optimization because every local minimizer is global, and many convex problems admit efficient algorithms. A function is convex if, for every pair $(x, y) \in (\mathbb{R}^p)^2$,

$$\forall\, t \in [0, 1], \quad f((1-t)x + ty) \leqslant (1-t)f(x) + tf(y) \tag{14.5}$$

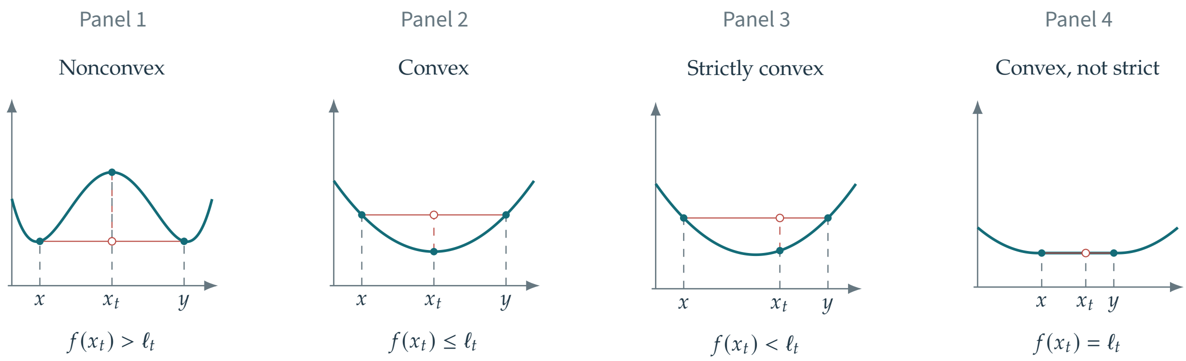


Figure 14.4. Convex and nonconvex functions; strictly convex and convex but not strictly convex functions. Here $x_t = (1-t)x + ty$ and $\ell_t = (1-t)f(x) + tf(y)$, with $0 < t < 1$.

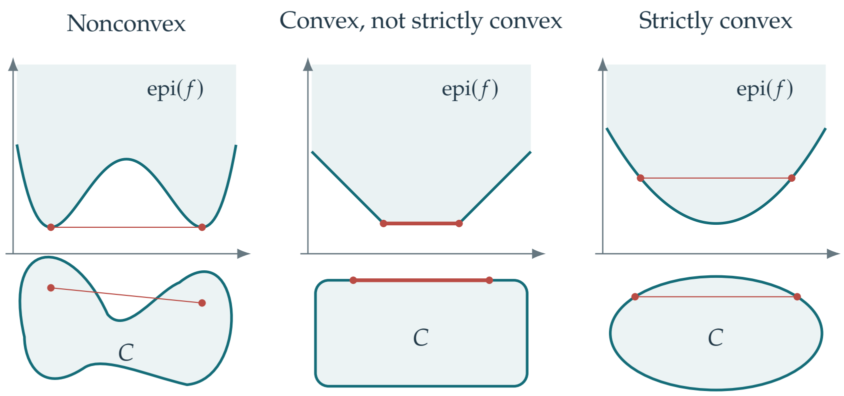


Figure 14.5. Comparison of convex functions $f : \mathbb{R}^p \to \mathbb{R}$ (for $p = 1$) and convex sets $C \subset \mathbb{R}^p$ (for $p = 2$).

The graph lies below each secant, and above each tangent when differentiable; see Figure 14.4. If $x^\star$ is a local minimizer of a convex f, then $x^\star$ is a global minimizer, i.e. $x^\star \in \operatorname{argmin} f$.

Convexity is preserved by several useful operations. If f and g are convex and a, b are positive, then $af + bg$ and $\max(f, g)$ are convex. If $g : \mathbb{R}^q \to \mathbb{R}$ is convex and $B \in \mathbb{R}^{q \times p}, b \in \mathbb{R}^q$, then the affine composition $f(x) = g(Bx + b)$ is convex. Thus the squared loss (14.3) is convex, since $\|\cdot\|^2/2$ is a sum of convex squares. Similarly, convexity of ℓ implies convexity of L and of the classification objective (14.4).

Strict convexity. When f is convex, one can strengthen the condition (14.5) and impose that the inequality is strict for distinct x, y and $t \in]0, 1[$ (see Figure 14.4, panels 3 and 4), i.e.

$$\forall t \in]0, 1[, \quad f((1-t)x + ty) < (1-t)f(x) + tf(y). \tag{14.6}$$

A minimizer $x^\star$ is then unique if it exists. Otherwise, two minimizers $x_1^\star \neq x_2^\star$ would satisfy $f(\frac{x_1^\star + x_2^\star}{2}) < f(x_1^\star)$ by strict convexity, contradicting optimality.

Example 14.2 (Least squares). For the quadratic loss function $f(x) = \frac{1}{2}\|Ax - y\|^2$, strict convexity is equivalent to $\ker(A) = \{0\}$. Its Hessian is $\partial^2 f(x) = A^\top A$, which is positive definite exactly when A is injective, since $\langle A^\top A z, z\rangle = \|Az\|^2$. Positive definiteness implies strict convexity. Conversely, a nonzero vector in $\ker(A)$ gives a line along which f is constant, ruling out strict convexity.

14.2.3 Convex Sets

A set $\Omega \subset \mathbb{R}^p$ is convex if every pair $(x, y) \in \Omega^2$ satisfies $(1-t)x + ty \in \Omega$ for all $t \in [0, 1]$. A function f is convex exactly when its epigraph $\operatorname{epi}(f) \overset{\text{def.}}{=} \{(x, t) \in \mathbb{R}^{p+1} \; ; \; t \geqslant f(x)\}$ is a convex set.

Remark 14.3 (Convexity of the minimizer set). A minimizer $x^\star$ need not be unique; see Figure 14.3. For convex f, the set $\operatorname{argmin}(f)$ is itself convex. Indeed, if $x_1^\star$ and $x_2^\star$ are minimizers, so that in particular

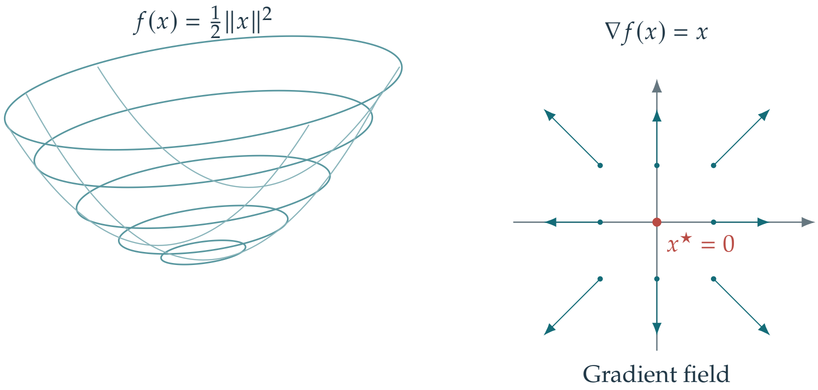


Figure 14.6. The gradient as a vector field.

$f(x_1^\star) = f(x_2^\star) = \min(f)$, then $f((1-t)x_1^\star + tx_2^\star) \leqslant (1-t)f(x_1^\star) + tf(x_2^\star) = f(x_1^\star) = \min(f)$, so that $(1-t)x_1^\star + tx_2^\star$ is itself a minimizer. Figure 14.5 shows convex and non-convex sets.

14.3 Derivative and gradient

14.3.1 Gradient

When all coordinate derivatives of f exist, define

$$\nabla f(x) \stackrel{\text{def.}}{=} \left(\frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_p}\right)^\top \in \mathbb{R}^p$$

as the gradient vector. The map $\nabla f : \mathbb{R}^p \to \mathbb{R}^p$ is a vector field, and its coordinate derivatives are defined by

$$\frac{\partial f(x)}{\partial x_k} \stackrel{\text{def.}}{=} \lim_{\eta \to 0} \frac{f(x + \eta\delta_k) - f(x)}{\eta}$$

where $\delta_k = (0, \dots, 0, 1, 0, \dots, 0)^\top \in \mathbb{R}^p$ is the kth canonical basis vector.

Existence of the coordinate derivatives, hence of $\nabla f(x)$, does not imply differentiability of f. Differentiability of f at x requires

$$f(x + \varepsilon) = f(x) + \langle \varepsilon, \nabla f(x) \rangle + o(\|\varepsilon\|). \tag{14.7}$$

The remainder notation $R(\varepsilon) = o(\|\varepsilon\|)$ means decay faster than the perturbation ε as it tends to 0: $\frac{R(\varepsilon)}{\|\varepsilon\|} \to 0$ as $\varepsilon \to 0$. Coordinate derivatives concern the behavior of f along the axes, whereas differentiability requires the expansion for every sequence $\varepsilon \to 0$. For example, $f(x) = \frac{2x_1x_2(x_1+x_2)}{x_1^2+x_2^2}$ with $f(0) = 0$ has both coordinate derivatives equal to zero at the origin. Yet $f(t,t) = 2t$, so the linear expansion with zero gradient fails.

The vector $\nabla f(x)$ is uniquely determined by (14.7). To prove differentiability of f and compute $\nabla f(x)$, it therefore suffices to establish an expansion

$$f(x + \varepsilon) = f(x) + \langle \varepsilon, g \rangle + o(\|\varepsilon\|),$$

which identifies $\nabla f(x) = g$.

For differentiable functions, convexity is equivalent to the graph lying above every tangent affine function.

Proposition 14.4. *If f is differentiable, then*

$$f \text{ convex} \quad \Leftrightarrow \quad \forall (x, x'),\ f(x) \geqslant f(x') + \langle \nabla f(x'), x - x' \rangle.$$

Proof. One can write the convexity condition as

$$f((1-t)x + tx') \leqslant (1-t)f(x) + tf(x') \quad \Longrightarrow \quad \frac{f(x + t(x'-x)) - f(x)}{t} \leqslant f(x') - f(x)$$

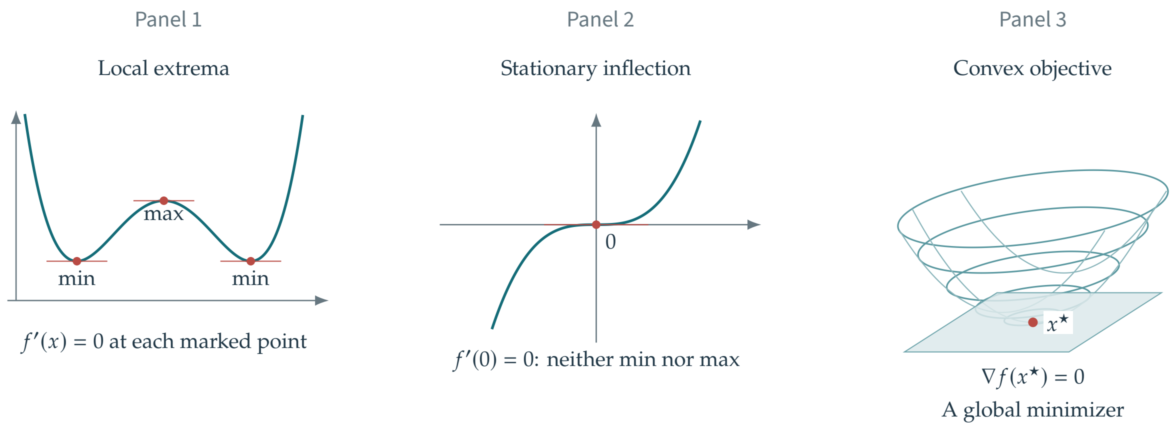


Figure 14.7. Local extrema (panel 1), a stationary inflection point (panel 2), and a global minimizer (panel 3).

hence, taking the limit $t \to 0$ one obtains

$$\langle \nabla f(x), x' - x \rangle \leqslant f(x') - f(x).$$

Conversely, set $x_t \overset{\text{def.}}{=} (1-t)x + tx'$. Apply the supporting-affine-function inequality at x_t to x and x':

$$\begin{aligned} f(x) &\geqslant f(x_t) + \langle \nabla f(x_t), x - x_t \rangle = f(x_t) + t\langle \nabla f(x_t), x - x' \rangle \\ f(x') &\geqslant f(x_t) + \langle \nabla f(x_t), x' - x_t \rangle = f(x_t) - (1-t)\langle \nabla f(x_t), x - x' \rangle, \end{aligned}$$

Multiplying the first inequality by $1-t$ and the second by t, then adding, gives

$$(1-t)f(x) + tf(x') \geqslant f(x_t).$$

□

14.3.2 First Order Conditions

A vanishing gradient is necessary for local optimality, and will also guide the algorithms developed below.

Proposition 14.5. *If f is differentiable and $x^\star$ is a local minimizer, meaning that $f(x^\star) \leqslant f(x)$ throughout some neighborhood of $x^\star$, then*

$$\nabla f(x^\star) = 0.$$

Proof. Fix a direction u and let $\varepsilon \downarrow 0$. Local optimality and differentiability give

$$f(x^\star) \leqslant f(x^\star + \varepsilon u) = f(x^\star) + \varepsilon\langle \nabla f(x^\star), u \rangle + o(\varepsilon) \quad \Longrightarrow \quad \langle \nabla f(x^\star), u \rangle \geqslant o(1) \quad \Longrightarrow \quad \langle \nabla f(x^\star), u \rangle \geqslant 0.$$

So applying this for u and $-u$ in the previous equation shows that $\langle \nabla f(x^\star), u \rangle = 0$ for all u, and hence $\nabla f(x^\star) = 0$. □

Note that the converse is not true in general, since one might have $\nabla f(x) = 0$ but x is not a local minimum. For instance $x = 0$ for $f(x) = -x^2$ (here x is a maximizer) or $f(x) = x^3$ (here x is neither a maximizer nor a minimizer; it is a stationary inflection point), see Fig. 14.7. Stationarity alone therefore neither certifies a local minimum nor guarantees convergence of a numerical method to that point. Both depend on the local geometry. Convexity of f makes stationarity sufficient as well as necessary.

Proposition 14.6. *If f is convex and $x^\star$ is a local minimizer, then $x^\star$ is a global minimizer. If f is differentiable and convex,*

$$x^\star \in \underset{x}{\operatorname{argmin}}\, f(x) \quad \Longleftrightarrow \quad \nabla f(x^\star) = 0.$$

Proof. Fix x and choose $0 < t < 1$ so that $tx + (1-t)x^\star$ lies in the neighborhood where $x^\star$ is minimal. Local optimality and convexity then give

$$f(x^\star) \leqslant f(tx + (1-t)x^\star) \leqslant tf(x) + (1-t)f(x^\star) \quad \Longrightarrow \quad f(x^\star) \leqslant f(x)$$

proving that $x^\star$ is a global minimizer.

For the second part, we already saw in Proposition 14.5 the $\Rightarrow$ implication. We assume that $\nabla f(x^\star) = 0$. Since the graph of f is above its tangent by convexity (as stated in Proposition 14.4),

$$f(x) \geqslant f(x^\star) + \langle \nabla f(x^\star),\, x - x^\star \rangle = f(x^\star).$$

□

For a differentiable convex objective, minimization is therefore equivalent to solving $\nabla f(x) = 0$, a system of p equations in p unknowns. An explicit solution is often unavailable, but the equation still characterizes the minimizers $x^\star$.

14.3.3 Least Squares

The gradient of the least-squares objective (14.3) follows by expanding the squared norm:

$$\begin{aligned} f(x+\varepsilon) &= \frac{1}{2}\|Ax - y + A\varepsilon\|^2 = \frac{1}{2}\|Ax-y\|^2 + \langle Ax - y,\, A\varepsilon\rangle + \frac{1}{2}\|A\varepsilon\|^2 \\ &= f(x) + \langle \varepsilon,\, A^\top(Ax-y)\rangle + o(\|\varepsilon\|). \end{aligned}$$

We used $\|A\varepsilon\|^2 = o(\|\varepsilon\|)$ and the transpose $A^\top$. The transpose exchanges rows and columns, $A^\top = (A_{j,i})_{i=1,\dots,n}^{j=1,\dots,p}$. Its essential role in gradient calculations is the adjoint identity

$$\forall (u,v) \in \mathbb{R}^p \times \mathbb{R}^n, \quad \langle Au,\, v\rangle_{\mathbb{R}^n} = \langle u,\, A^\top v\rangle_{\mathbb{R}^p}.$$

Computing gradients of functions involving linear operators requires such a transposition step. This computation shows that

$$\nabla f(x) = A^\top(Ax - y). \tag{14.8}$$

Hence every minimizer $x^\star$ of $f(x)$ satisfies the normal equations $(A^\top A)x^\star = A^\top y$. If $A^\star A \in \mathbb{R}^{p\times p}$ is invertible, then f has a single minimizer, namely

$$x^\star = (A^\top A)^{-1}A^\top y. \tag{14.9}$$

Thus $x^\star$ depends linearly on y. The operator $(A^\top A)^{-1}A^\star$ is the Moore–Penrose pseudoinverse of A; an ordinary inverse is unavailable when $p \neq n$. The condition that $A^\top A$ is invertible is equivalent to $\ker(A) = \{0\}$, since

$$A^\top A x = 0 \quad\Longrightarrow\quad \|Ax\|^2 = \langle A^\top A x,\, x\rangle = 0 \quad\Longrightarrow\quad Ax = 0.$$

When $n < p$, the system is underdetermined and injectivity is impossible. If $n \geqslant p$ and the entries of A have a joint density, then $\ker(A) = \{0\}$ almost surely. In general, injectivity is equivalent to the feature vectors a_i spanning $\mathbb{R}^p$.

14.3.4 Link with PCA

Assume the feature vectors are centered, $\sum_i a_i = 0$. Otherwise, replace each a_i by $a_i - m$, where $m \overset{\text{def.}}{=} n^{-1}\sum_i a_i$ is the empirical mean. Write $C \overset{\text{def.}}{=} A^\top A$. Then C/n is the empirical covariance matrix of the point cloud. With $a_i = (a_{i,1}, \dots, a_{i,p})^\top$ and $A = (a_{i,j})_{i,j}$, its entries are

$$\forall (k,\ell) \in \{1,\dots,p\}^2, \quad \frac{C_{k,\ell}}{n} = \frac{1}{n}\sum_{i=1}^n a_{i,k}a_{i,\ell}.$$

In particular, $C_{k,k}/n$ is the variance along the axis k. More generally, for any unit vector $u \in \mathbb{R}^p$, $\langle Cu,\, u\rangle/n \geqslant 0$ is the variance along the axis u.

For instance, in dimension $p = 2$,

$$\frac{C}{n} = \frac{1}{n}\begin{pmatrix} \sum_{i=1}^n a_{i,1}^2 & \sum_{i=1}^n a_{i,1}a_{i,2} \\ \sum_{i=1}^n a_{i,1}a_{i,2} & \sum_{i=1}^n a_{i,2}^2 \end{pmatrix}.$$

Since C is symmetric, choose an orthogonal matrix $U = (u_1, \dots, u_p)$ whose columns are eigenvectors of C/n. If $(C/n)u_k = \lambda_k u_k$, then $C/n = U\,\mathrm{diag}(\lambda_k)U^\top$ and $U^{-1} = U^\top$. In these coordinates, the quadratic form is a weighted sum of squares:

$$\frac{1}{n}\langle Cx,\, x\rangle = \langle U\,\mathrm{diag}(\lambda_k)U^\top x,\, x\rangle = \langle \mathrm{diag}(\lambda_k)(U^\top x),\, (U^\top x)\rangle = \sum_{k=1}^p \lambda_k \langle x,\, u_k\rangle^2. \tag{14.10}$$

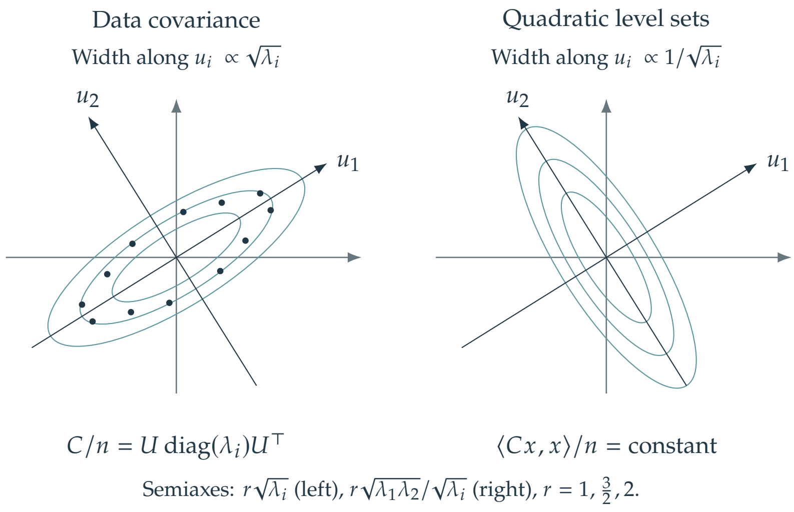


Figure 14.8. Left: point clouds $(a_i)_i$ and their principal directions. Right: the quadratic part of $f(x)$.

Here $(U^\top x)_k = \langle x, u_k\rangle$ is the coordinate k of x in the basis U. Since $\langle Cx, x\rangle = \|Ax\|^2$, this shows that all eigenvalues are nonnegative.

Order the eigenvalues as $\lambda_1 \geqslant \lambda_2 \geqslant \ldots \geqslant \lambda_p$. Projection of a_i onto the leading m eigenvectors gives principal component analysis (PCA), the optimal linear reconstruction in squared error. It is used for compression, dimensionality reduction, and visualization in two dimensions ($m = 2$) or three dimensions ($m = 3$).

When C is positive definite, its covariance ellipsoid has principal axes $(u_k)_k$ and semiaxes proportional to the standard deviations $\sqrt{\lambda_k}$. This ellipsoid closely describes a Gaussian cloud with density proportional to $\exp(-\frac{n}{2}\langle C^{-1}a, a\rangle)$. For the quadratic objective $\frac{1}{2}\langle Cx, x\rangle$ in $f(x)$, the level ellipsoid $\left\{x\,;\,\frac{1}{n}\langle Cx, x\rangle \leqslant 1\right\}$ has the same principal axes but reciprocal widths $1/\sqrt{\lambda_k}$. Figure 14.8 shows the reciprocal axis widths in dimension $p = 2$. If C is singular, the covariance ellipsoid is degenerate and the quadratic sublevel sets are unbounded along $\ker(C)$.

14.3.5 Classification

For the classification objective (14.4), assume L is differentiable and apply (14.7) at $-\mathrm{diag}(y)Ax$:

$$\begin{aligned} f(x+\varepsilon) &= L(-\mathrm{diag}(y)Ax - \mathrm{diag}(y)A\varepsilon) \\ &= L(-\mathrm{diag}(y)Ax) + \langle \nabla L(-\mathrm{diag}(y)Ax),\, -\mathrm{diag}(y)A\varepsilon\rangle + o(\|\mathrm{diag}(y)A\varepsilon\|). \end{aligned}$$

Using the fact that $o(\|\mathrm{diag}(y)A\varepsilon\|) = o(\|\varepsilon\|)$, one obtains

$$\begin{aligned} f(x+\varepsilon) &= f(x) + \langle \nabla L(-\mathrm{diag}(y)Ax),\, -\mathrm{diag}(y)A\varepsilon\rangle + o(\|\varepsilon\|) \\ &= f(x) + \langle -A^\top \mathrm{diag}(y)\nabla L(-\mathrm{diag}(y)Ax),\, \varepsilon\rangle + o(\|\varepsilon\|), \end{aligned}$$

Using $(AB)^\top = B^\top A^\top$ and $\mathrm{diag}(y)^\top = \mathrm{diag}(y)$ gives

$$\nabla f(x) = -A^\top \mathrm{diag}(y)\nabla L(-\mathrm{diag}(y)Ax).$$

Since $L(z) = \sum_i \ell(z_i)$, its gradient is $\nabla L(z) = (\ell'(z_i))_{i=1}^n$. For the normalized logistic loss, $\ell'(u) = e^u/((1+e^u)\log 2)$. Evaluated at the negative margin $u = -y_i\langle x, a_i\rangle$, this scaled sigmoid is large for incorrectly classified observations and weights their contribution to the gradient.

14.3.6 Chain Rule

More generally, for $f(x) = g(Bx)$ with $B \in \mathbb{R}^{q\times p}$ and $g : \mathbb{R}^q \to \mathbb{R}$,

$$f(x+\varepsilon) = g(Bx + B\varepsilon) = g(Bx) + \langle \nabla g(Bx),\, B\varepsilon\rangle + o(\|B\varepsilon\|) = f(x) + \langle \varepsilon,\, B^\top \nabla g(Bx)\rangle + o(\|\varepsilon\|),$$

which shows that

$$\nabla(g\circ B) = B^\top \circ \nabla g \circ B \tag{14.11}$$

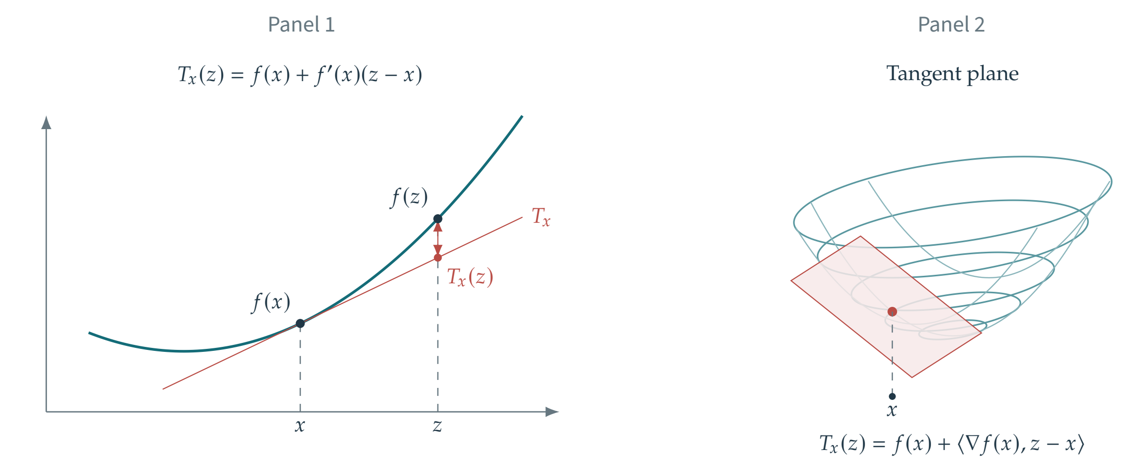


Figure 14.9. Panels 1 and 2: first-order Taylor approximations in one and two dimensions. Panels 3 and 4: the gradient is orthogonal to the level set; the diagram illustrates the proof.

where "$\circ$" denotes the composition of functions.

Nonlinear compositions require the differential. For $F : \mathbb{R}^p \to \mathbb{R}^q$, its differential at x is the linear map $\partial F(x) : \mathbb{R}^p \to \mathbb{R}^q$. We represent it by its Jacobian matrix, also denoted $\partial F(x)$, with dimensions $\partial F(x) \in \mathbb{R}^{q \times p}$. Writing $F(x) = (F_1(x), \dots, F_q(x))$, the Jacobian entries are

$$\forall\, (i, j) \in \{1, \dots, q\} \times \{1, \dots, p\}, \quad [\partial F(x)]_{i,j} \overset{\text{def.}}{=} \frac{\partial F_i(x)}{\partial x_j}.$$

The map F is differentiable at x when it admits the expansion

$$F(x + \varepsilon) = F(x) + [\partial F(x)](\varepsilon) + o(\|\varepsilon\|). \tag{14.12}$$

Here $[\partial F(x)](\varepsilon)$ denotes matrix-vector multiplication. The matrix in this expansion is unique, so the expansion can also be used to compute the differential.

For the special case $q = 1$, i.e. if $f : \mathbb{R}^p \to \mathbb{R}$, then the differential $\partial f(x) \in \mathbb{R}^{1 \times p}$ and the gradient $\nabla f(x) \in \mathbb{R}^{p \times 1}$ are linked by equating the Taylor expansions (14.12) and (14.7)

$$\forall\, \varepsilon \in \mathbb{R}^p, \quad [\partial f(x)](\varepsilon) = \langle \nabla f(x),\, \varepsilon \rangle \quad \Leftrightarrow \quad \partial f(x) = \nabla f(x)^\top.$$

The differential satisfies the following chain rule

$$\partial(G \circ H)(x) = [\partial G(H(x))] \times [\partial H(x)]$$

where "$\times$" is the matrix product. For instance, if $H : \mathbb{R}^p \to \mathbb{R}^q$ and $G = g : \mathbb{R}^q \mapsto \mathbb{R}$, then $f = g \circ H : \mathbb{R}^p \to \mathbb{R}$ and one can compute its gradient as follows

$$\nabla f(x) = (\partial f(x))^\top = ([\partial g(H(x))] \times [\partial H(x)])^\top = [\partial H(x)]^\top \times [\partial g(H(x))]^\top = [\partial H(x)]^\top \times \nabla g(H(x)).$$

When $H(x) = Bx$ is linear, one recovers formula (14.11).

14.4 Gradient Descent Algorithm

14.4.1 Steepest Descent Direction

The Taylor expansion (14.7) approximates f near x by an affine function:

$$f(z) = T_x(z) + o(\|x - z\|) \quad \text{where} \quad T_x(z) \overset{\text{def.}}{=} f(x) + \langle \nabla f(x),\, z - x \rangle,$$

Figure 14.9 illustrates the approximation. First-order methods use f through its local model T_x.

A nonzero gradient $\nabla f(x)$ points uphill locally, so $-\nabla f(x)$ is a descent direction. At a fixed point x, examine f along the ray

$$\tau \in \mathbb{R}^+ = [0, +\infty[\longmapsto f(x - \tau \nabla f(x)) \in \mathbb{R}.$$

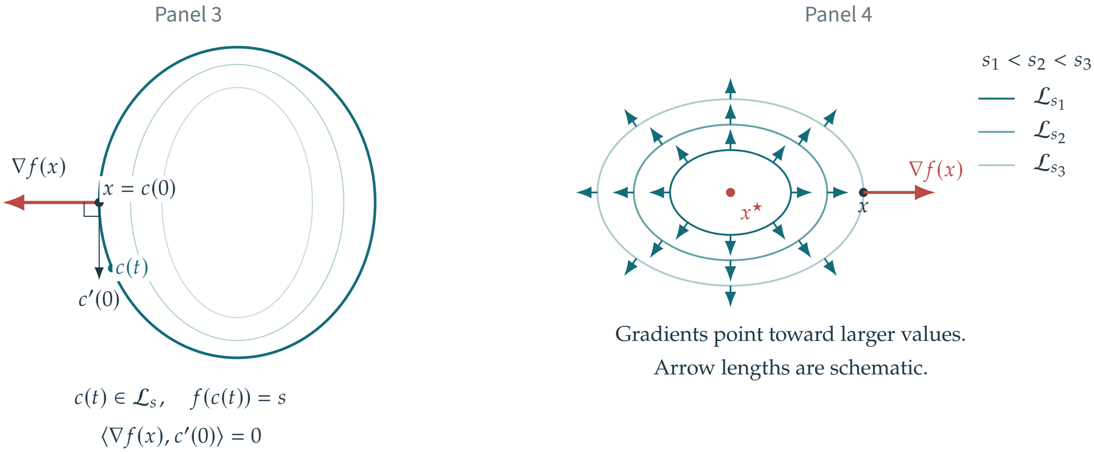


Figure 14.9 (continued). Panels 1 and 2: first-order Taylor approximations in one and two dimensions. Panels 3 and 4: the gradient is orthogonal to the level set; the diagram illustrates the proof.

If f is differentiable at x, one has

$$f(x - \tau\nabla f(x)) = f(x) - \tau\langle \nabla f(x), \nabla f(x)\rangle + o(\tau) = f(x) - \tau\|\nabla f(x)\|^2 + o(\tau).$$

If $\nabla f(x) = 0$, then x is stationary and is a global minimizer when f is convex. Otherwise, for sufficiently small τ,

$$f(x - \tau\nabla f(x)) < f(x)$$

so the step from x to $x - \tau\nabla f(x)$ decreases the objective.

Remark 14.7 (Orthogonality to level sets). A level set of f contains points with the same value of f. For $s \in \mathbb{R}$, define

$$\mathcal{L}_s \overset{\text{def.}}{=} \{x \,;\, f(x) = s\}.$$

Assume f is continuously differentiable near x and $\nabla f(x) \neq 0$. Then x is a regular point of the level set $\mathcal{L}_{f(x)}$. The gradient is normal to its tangent space and points toward increasing values of f; see Figure 14.9, panels 3 and 4. Take a smooth curve through x in $\mathcal{L}_s$, parameterized by $t \in \mathbb{R} \mapsto c(t)$ with $c(0) = x$. The function $h(t) \overset{\text{def.}}{=} f(c(t))$ has constant value $h(t) = s$ because $c(t)$ lies in the level set. Thus $h'(t) = 0$. Expanding at $t = 0$ also gives

$$h(t) = f(c(0) + tc'(0) + o(t)) = h(0) + t\langle c'(0), \nabla f(c(0))\rangle + o(t)$$

i.e. $h'(0) = \langle c'(0), \nabla f(x)\rangle = 0$, so that $\nabla f(x)$ is orthogonal to the tangent $c'(0)$ of the curve c, which lies in the tangent plane of $\mathcal{L}_s$ (as shown in Figure 14.9, panel 3). Since the curve c is arbitrary, the whole tangent plane is thus orthogonal to $\nabla f(x)$.

Remark 14.8 (Local optimal descent direction). Assume f is continuously differentiable near x and $\nabla f(x) \neq 0$. Among displacements of length r, every minimizing direction approaches the normalized negative gradient as $r \downarrow 0$:

$$\frac{1}{r}\underset{\|u\|=r}{\operatorname{argmin}}\, f(x+u) \xrightarrow{r\to 0} -\frac{\nabla f(x)}{\|\nabla f(x)\|}.$$

To verify the sign, let u_r minimize $f(x+u)$ on the radius-r sphere. Compare it with $-r\nabla f(x)/\|\nabla f(x)\|$. The uniform first-order remainder gives

$$\langle \nabla f(x), u_r/r\rangle \leqslant -\|\nabla f(x)\| + o(1).$$

Since $\|u_r/r\| = 1$, equality in the limiting Cauchy–Schwarz bound forces the stated direction.

14.4.2 Gradient Descent

Starting from $x_0 \in \mathbb{R}^p$, gradient descent iterates

$$x_{k+1} \overset{\text{def.}}{=} x_k - \tau_k \nabla f(x_k) \tag{14.13}$$

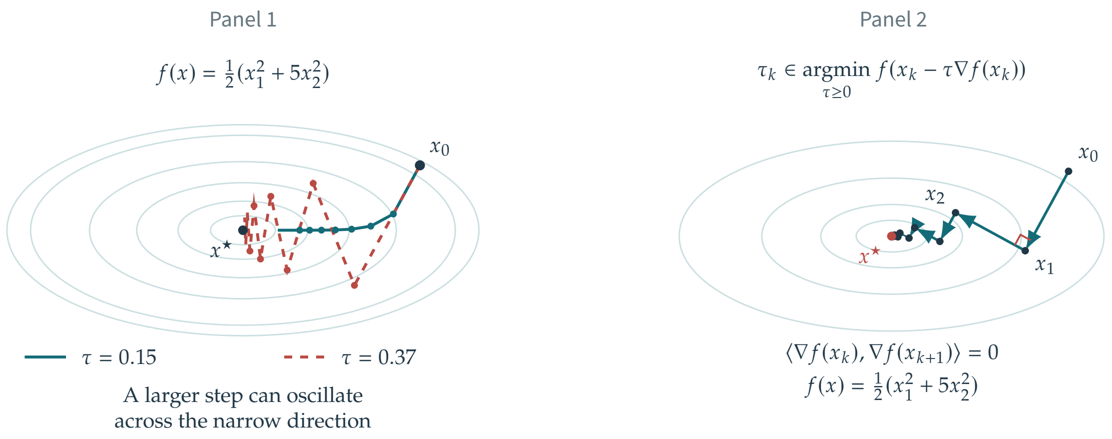


Figure 14.10. Effect of the step size τ on gradient descent (panel 1) and exact line search (panel 2).

The parameter $\tau_k > 0$ is the step size, or learning rate. A sufficiently small τ_k decreases f at a nonstationary iterate. Choosing τ_k balances stable descent against progress per step. One may fix $\tau_k = \tau$ or adapt τ_k at each iteration; see Figure 14.10.

Remark 14.9 (Greedy choice). Exact line search chooses the best step along the current direction, although computing it can be expensive:

$$\tau_k \in \operatorname*{argmin}_{\tau \geqslant 0} h(\tau) \overset{\text{def.}}{=} f(x_k - \tau \nabla f(x_k)).$$

The line-search objective $h(\tau)$ is scalar. Its derivative follows from a Taylor expansion of h:

$$h(\tau + \delta) = f(x_k - \tau \nabla f(x_k) - \delta \nabla f(x_k)) = f(x_k - \tau \nabla f(x_k)) - \delta \langle \nabla f(x_k - \tau \nabla f(x_k)), \nabla f(x_k) \rangle + o(\delta).$$

Note that at $\tau = \tau_k$, $\nabla f(x_k - \tau \nabla f(x_k)) = \nabla f(x_{k+1})$ by definition of x_{k+1} in (14.13). If the current gradient is nonzero and the minimum is attained, an optimal step satisfies $\tau_k > 0$, since small positive steps decrease f. The first-order condition is therefore

$$h'(\tau_k) = -\langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle = 0.$$

Thus successive gradients, and hence successive descent directions, are orthogonal; see Figure 14.10.

Remark 14.10 (Armijo rule). Instead of finding the optimal τ, Armijo backtracking seeks an admissible τ that produces sufficient decrease. Given $0 < \alpha < 1$ (values below $1/2$ are customary, especially for Newton-type methods), one considers a τ to be valid for a descent direction d_k (for instance $d_k = -\nabla f(x_k)$) if it satisfies

$$f(x_k + \tau d_k) \leqslant f(x_k) + \alpha \tau \langle d_k, \nabla f(x_k) \rangle \tag{14.14}$$

For small τ, one has $f(x_k + \tau d_k) = f(x_k) + \tau \langle d_k, \nabla f(x_k) \rangle + o(\tau)$, so that, assuming d_k is a valid descent direction (i.e $\langle d_k, \nabla f(x_k) \rangle < 0$), condition (14.14) will always be satisfied for τ small enough (if f is convex, the set of allowable τ is of the form $[0, \tau_{\max}]$). Start from a positive trial step and repeatedly reduce it by $\tau \leftarrow \beta \tau$, with $0 < \beta < 1$, until (14.14) holds. This procedure is called backtracking line search.

14.5 Convergence Analysis

14.5.1 Quadratic Case

Convergence analysis for the quadratic case. We first analyze gradient descent for the quadratic objective

$$f(x) = \frac{1}{2}\|Ax - y\|^2 = \frac{1}{2}\langle Cx, x \rangle - \langle x, b \rangle + \text{cst} \quad \text{where} \quad \begin{cases} C \overset{\text{def.}}{=} A^\top A \in \mathbb{R}^{p \times p}, \\ b \overset{\text{def.}}{=} A^\top y \in \mathbb{R}^p. \end{cases}$$

We already saw that in (14.9) if $\ker(A) = \{0\}$, which is equivalent to C being invertible, then there exists a single global minimizer $x^\star = (A^\top A)^{-1} A^\top y = C^{-1} b$.

The quadratic $\frac{1}{2}\langle Cx, x \rangle - \langle x, b \rangle$ is convex exactly when the symmetric matrix C is positive semidefinite, as follows from the spectral decomposition (14.10).

Proposition 14.11. *Let $f(x) = \frac{1}{2}\langle Cx, x\rangle - \langle b, x\rangle$, where C is symmetric and its eigenvalues lie in $[\mu, L]$ with $0 < \mu \leqslant L$. If the step sizes satisfy*

$$0 < \tau_{\min} \leqslant \tau_\ell \leqslant \tau_{\max} < \frac{2}{L}$$

then there exists $0 \leqslant \tilde{\rho} < 1$ such that

$$\|x_k - x^\star\| \leqslant \tilde{\rho}^k \|x_0 - x^\star\|. \tag{14.15}$$

The smallest uniform one-step contraction bound based only on μ and L is obtained with the constant step

$$\tau_\ell = \frac{2}{L+\mu} \quad \Longrightarrow \quad \tilde{\rho} \overset{\text{def.}}{=} \frac{L-\mu}{L+\mu} = 1 - \frac{2\varepsilon}{1+\varepsilon} \quad \textit{where} \quad \varepsilon \overset{\text{def.}}{=} \mu/L. \tag{14.16}$$

Proof. One iterate of gradient descent reads

$$x_{k+1} = x_k - \tau_k(Cx_k - b).$$

The minimizer $x^\star$ satisfies $Cx^\star = b$, so

$$x_{k+1} - x^\star = x_k - x^\star - \tau_k C(x_k - x^\star) = (\mathrm{Id}_p - \tau_k C)(x_k - x^\star).$$

If $S \in \mathbb{R}^{p\times p}$ is a symmetric matrix, one has

$$\|Sz\| \leqslant \|S\|_{\mathrm{op}} \|z\| \quad \text{where} \quad \|S\|_{\mathrm{op}} \overset{\text{def.}}{=} \max_k |\lambda_k(S)|,$$

where $\lambda_k(S)$ are the eigenvalues of S and $\sigma_k(S) \overset{\text{def.}}{=} |\lambda_k(S)|$ are its singular values. Indeed, S can be diagonalized in an orthogonal basis U, so that $S = U\,\mathrm{diag}(\lambda_k(S))U^\top$, and $S^\top S = S^2 = U\,\mathrm{diag}(\lambda_k(S)^2)U^\top$ so that

$$\begin{aligned} \|Sz\|^2 &= \langle S^\top S z, z\rangle = \langle U\,\mathrm{diag}(\lambda_k^2)U^\top z, z\rangle = \langle \mathrm{diag}(\lambda_k^2)U^\top z, U^\top z\rangle \\ &= \sum_k \lambda_k^2 (U^\top z)_k^2 \leqslant \max_k(\lambda_k^2)\|U^\top z\|^2 = \max_k(\lambda_k^2)\|z\|^2. \end{aligned}$$

Applying this to $S = \mathrm{Id}_p - \tau C$ gives the contraction factor as a function of the step τ:

$$h(\tau) \overset{\text{def.}}{=} \|\mathrm{Id}_p - \tau C\|_{\mathrm{op}} = \max_j |1 - \tau\lambda_j(C)| \leqslant H(\tau) \overset{\text{def.}}{=} \max\big(|1-\tau L|, |1-\tau\mu|\big).$$

If μ and L are the extreme eigenvalues, equality holds and Figure 14.11 plots $h = H$. In general, $H(\tau) < 1$ for $0 < \tau < 2/L$. Continuity on $[\tau_{\min}, \tau_{\max}]$ gives a common bound $\tilde{\rho} < 1$, proving (14.15). The bound H is minimized at $\tau^\star = \frac{2}{L+\mu}$, giving

$$H(\tau^\star) = \left|1 - \frac{2L}{L+\mu}\right| = \frac{L-\mu}{L+\mu}.$$

□

Note that when the inverse condition number $\xi \overset{\text{def.}}{=} \mu/L \ll 1$ is small (which is the typical setup for ill-posed problems), then the contraction constant appearing in (14.16) scales like

$$\tilde{\rho} \sim 1 - 2\xi. \tag{14.17}$$

We also use ε below for this inverse condition number. The condition number is the ratio of the largest to the smallest singular value, and its minimum value is 1, attained by orthogonal matrices.

The geometric decay $O(\rho^k)$ in (14.15) is called linear convergence in optimization. The bound is global because it holds for every k, rather than only sufficiently large k.

If $\ker(A) \neq \{0\}$, then C has zero eigenvalues and the minimizer set is infinite. A linear rate still follows because the increments $x_{k+1} - x_k$ are orthogonal to $\ker(A)$, while the kernel component remains equal to that of x_0. The preceding proof applies on the orthogonal complement, with μ replaced by the smallest positive eigenvalue of C. This eigenvalue may be very small, giving a contraction factor close to one. The spectral argument also relies on the quadratic structure. We therefore give a different analysis that provides a sublinear bound on the objective value.

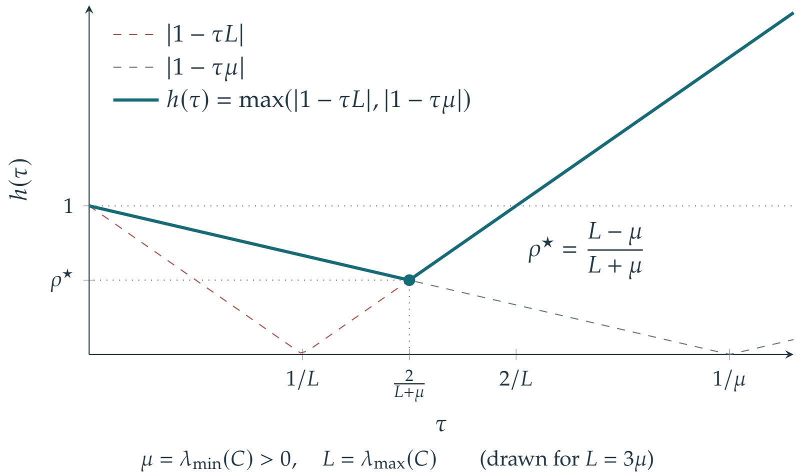


Figure 14.11. Contraction constant $h(\tau)$ for a quadratic function.

Proposition 14.12. *Let* $f(x) = \frac{1}{2}\langle Cx, x\rangle - \langle b, x\rangle$ *with* $C \succeq 0$, $b \in \mathrm{Im}(C)$, *and eigenvalues at most* $L > 0$. *For a constant step* $0 < \tau_k = \tau < 1/L$ *and* $k \geqslant 1$,

$$f(x_k) - f(x^\star) \leqslant \frac{\mathrm{dist}(x_0, \operatorname{argmin} f)^2}{\tau 8k}.$$

where

$$\mathrm{dist}(x_0, \operatorname{argmin} f) \stackrel{\text{def.}}{=} \min_{x^\star \in \operatorname{argmin} f} \|x_0 - x^\star\|.$$

Proof. We have $Cx^\star = b$ for any minimizer $x^\star$ and $x_{k+1} = x_k - \tau(Cx_k - b)$ so that as before

$$x_k - x^\star = (\mathrm{Id}_p - \tau C)^k (x_0 - x^\star).$$

Now one has

$$\frac{1}{2}\langle C(x_k - x^\star), x_k - x^\star\rangle = \frac{1}{2}\langle Cx_k, x_k\rangle - \langle Cx_k, x^\star\rangle + \frac{1}{2}\langle Cx^\star, x^\star\rangle$$

and we have $\langle Cx_k, x^\star\rangle = \langle x_k, Cx^\star\rangle = \langle x_k, b\rangle$ and also $\langle Cx^\star, x^\star\rangle = \langle x^\star, b\rangle$ so that

$$\frac{1}{2}\langle C(x_k - x^\star), x_k - x^\star\rangle = \frac{1}{2}\langle Cx_k, x_k\rangle - \langle x_k, b\rangle + \frac{1}{2}\langle x^\star, b\rangle = f(x_k) + \frac{1}{2}\langle x^\star, b\rangle.$$

Note also that

$$f(x^\star) = \frac{1}{2}\langle Cx^\star, x^\star\rangle - \langle x^\star, b\rangle = \frac{1}{2}\langle x^\star, b\rangle - \langle x^\star, b\rangle = -\frac{1}{2}\langle x^\star, b\rangle.$$

This shows that

$$\frac{1}{2}\langle C(x_k - x^\star), x_k - x^\star\rangle = f(x_k) - f(x^\star).$$

This thus implies

$$f(x_k) - f(x^\star) = \frac{1}{2}\langle (\mathrm{Id}_p - \tau C)^k C (\mathrm{Id}_p - \tau C)^k (x_0 - x^\star), x_0 - x^\star\rangle \leqslant \frac{\sigma_{\max}(M_k)}{2}\|x_0 - x^\star\|^2$$

where we have denoted

$$M_k \stackrel{\text{def.}}{=} (\mathrm{Id}_p - \tau C)^k C (\mathrm{Id}_p - \tau C)^k.$$

Since $x^\star$ can be chosen arbitrarily, one can replace $\|x_0 - x^\star\|$ by $\mathrm{dist}(x_0, \operatorname{argmin} f)$. One has, for any ℓ, the following bound

$$\sigma_\ell(M_k) \leqslant \max_{0 \leqslant s \leqslant L} s(1 - \tau s)^{2k} \leqslant \frac{1}{\tau 4k}$$

because the eigenvalues of M_k have the form $s(1 - \tau s)^{2k}$ for eigenvalues s of C. Setting $t = \tau s \in [0, 1]$, use

$$\forall t \in [0, 1], \quad (1 - t)^{2k} t \leqslant \frac{1}{4k}.$$

Indeed, one has

$$(1-t)^{2k}t \leqslant (e^{-t})^{2k}t = \frac{1}{2k}(2kt)e^{-2kt} \leqslant \frac{1}{2k}\sup_{u\geqslant 0} ue^{-u} = \frac{1}{2ek} \leqslant \frac{1}{4k}.$$

□

14.5.2 General Case

We now extend the convergence analysis to general smooth convex functions. The Hessian replaces the constant matrix C of a quadratic objective.

Hessian. For a twice continuously differentiable function, the Hessian matrix is

$$(\partial^2 f)(x) = \left(\frac{\partial^2 f(x)}{\partial x_i \partial x_j}\right)_{1\leqslant i,j\leqslant p} \in \mathbb{R}^{p\times p}.$$

Each entry differentiates the ith component of the gradient with respect to the jth coordinate. Equality of mixed partial derivatives makes $\partial^2 f(x)$ symmetric.

Twice differentiability of f at x implies the Taylor expansion

$$f(x+\varepsilon) = f(x) + \langle \nabla f(x),\, \varepsilon\rangle + \frac{1}{2}\langle \partial^2 f(x)\varepsilon,\, \varepsilon\rangle + o(\|\varepsilon\|^2). \tag{14.18}$$

Thus f has a local quadratic approximation near x. The Hessian is the unique symmetric matrix in this expansion. If an expansion has the form below with a symmetric matrix H,

$$f(x+\varepsilon) = f(x) + \langle \nabla f(x),\, \varepsilon\rangle + \frac{1}{2}\langle H\varepsilon,\, \varepsilon\rangle + o(\|\varepsilon\|^2).$$

comparison with (14.18) identifies $\partial^2 f(x) = H$, avoiding separate computation of all p^2 partial derivatives. Equivalently, differentiate the gradient:

$$\nabla f(x+\varepsilon) = \nabla f(x) + [\partial^2 f(x)](\varepsilon) + o(\|\varepsilon\|)$$

where $[\partial^2 f(x)](\varepsilon) \in \mathbb{R}^p$ is the product of the Hessian $\partial^2 f(x)$ with ε.

A twice differentiable function f on $\mathbb{R}^p$ is convex exactly when, at every x, the Hessian $\partial^2 f(x)$ is positive semidefinite. Positive definiteness everywhere implies strict convexity of f, but is not necessary: x^4 is strictly convex on $\mathbb{R}$ although its second derivative vanishes at $x = 0$.

For a quadratic objective $f(x) = \frac{1}{2}\langle Cx,\, x\rangle - \langle x,\, u\rangle$, the gradient $\nabla f(x) = Cx - u$ gives the constant Hessian $\partial^2 f(x) = C$. For the classification function, one has

$$\nabla f(x) = -A^\top \operatorname{diag}(y)\nabla L(-\operatorname{diag}(y)Ax).$$

and thus

$$\begin{aligned}\nabla f(x+\varepsilon) &= -A^\top \operatorname{diag}(y)\nabla L(-\operatorname{diag}(y)Ax - \operatorname{diag}(y)A\varepsilon)\\ &= \nabla f(x) - A^\top \operatorname{diag}(y)[\partial^2 L(-\operatorname{diag}(y)Ax)](-\operatorname{diag}(y)A\varepsilon) + o(\|\varepsilon\|)\end{aligned}$$

Since $\nabla L(u) = (\ell'(u_i))$ one has $\partial^2 L(u) = \operatorname{diag}(\ell''(u_i))$. This means that

$$\partial^2 f(x) = A^\top \operatorname{diag}(y) \times \operatorname{diag}(\ell''(-\operatorname{diag}(y)Ax)) \times \operatorname{diag}(y)A.$$

This matrix is symmetric positive semidefinite when ℓ is convex, because ℓ'' is nonnegative.

Remark 14.13 (Second order optimality condition). The Hessian can classify a stationary point $x^\star$ with $\nabla f(x^\star) = 0$. If $\partial^2 f(x^\star)$ is positive definite, then $x^\star$ is a strict local minimizer. Positive semidefiniteness of $\partial^2 f(x^\star)$ alone does not classify the point; for example, x^3 on $\mathbb{R}$ has a vanishing Hessian at its stationary inflection point. Conversely, if $x^\star$ is a local minimum, then $\partial^2 f(x^\star)$ is positive semidefinite.

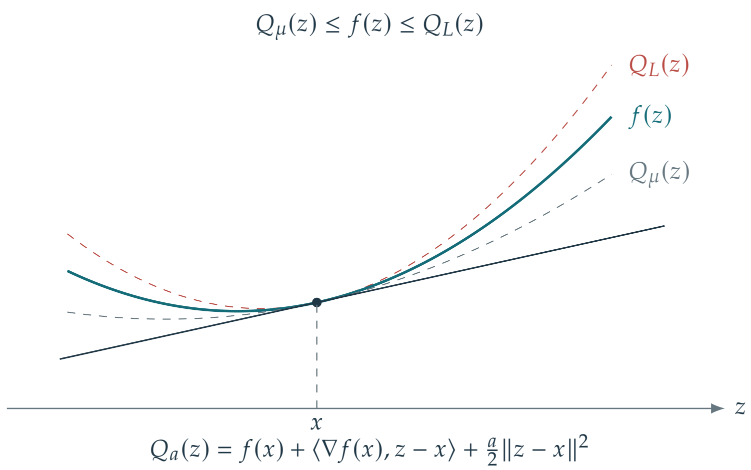


Figure 14.12. Quadratic upper and lower bounds for a smooth strongly convex function.

Remark 14.14 (Second order algorithms). The Hessian also yields second-order methods such as Newton's method. These can converge faster locally than gradient descent, at a greater cost per iteration. A variable-metric gradient step has the form

$$x_{k+1} = x_k - H_k \nabla f(x_k)$$

where $H_k \in \mathbb{R}^{p \times p}$ is symmetric positive definite. Gradient descent uses $H_k = \tau_k \mathrm{Id}_p$; Newton's method uses $H_k = [\partial^2 f(x_k)]^{-1}$ when the Hessian is positive definite. Note that

$$f(x_{k+1}) = f(x_k) - \langle H_k \nabla f(x_k),\, \nabla f(x_k) \rangle + o(\|H_k \nabla f(x_k)\|).$$

Positive definiteness of H_k ensures that, if x_k is nonstationary with $\nabla f(x_k) \neq 0$, then $\langle H_k \nabla f(x_k),\, \nabla f(x_k) \rangle > 0$. Scaling H_k sufficiently small therefore guarantees descent, $f(x_{k+1}) < f(x_k)$. We do not develop second-order algorithms further here.

For convergence analysis, uniform Hessian bounds play the role of the spectral bounds on C in the quadratic case. We require these bounds on $\partial^2 f(x)$ at every x. Integrating these local bounds yields global inequalities that support an analysis similar to the quadratic one.

Smoothness and strong convexity. For $L > 0$, quantify the smoothness of f by requiring an L-Lipschitz gradient:

$$\forall\, (x, x') \in (\mathbb{R}^p)^2, \quad \|\nabla f(x) - \nabla f(x')\| \leqslant L\|x - x'\|. \tag{$\mathcal{R}_L$}$$

A linear convergence guarantee for the iterates follows from a uniform lower curvature bound. We impose this by assuming that f is μ-strongly convex

$$\forall\, (x, x') \in (\mathbb{R}^p)^2, \quad \langle \nabla f(x) - \nabla f(x'),\, x - x' \rangle \geqslant \mu\|x - x'\|^2. \tag{$\mathcal{S}_\mu$}$$

For C^2 functions, these conditions are equivalent to uniform Hessian bounds.

Proposition 14.15. *Conditions ($\mathcal{R}_L$) and ($\mathcal{S}_\mu$) imply*

$$\forall\, (x, x'), \quad f(x') + \langle \nabla f(x'),\, x - x' \rangle + \frac{\mu}{2}\|x - x'\|^2 \leqslant f(x) \leqslant f(x') + \langle \nabla f(x'),\, x - x' \rangle + \frac{L}{2}\|x - x'\|^2. \tag{14.19}$$

If f is of class C^2, conditions ($\mathcal{R}_L$) and ($\mathcal{S}_\mu$) are equivalent to

$$\forall\, x, \quad \mu \mathrm{Id}_p \preceq \partial^2 f(x) \preceq L \mathrm{Id}_p \tag{14.20}$$

where $\partial^2 f(x) \in \mathbb{R}^{p \times p}$ is the Hessian and $\preceq$ denotes the semidefinite order:

$$A \preceq B \quad \Longleftrightarrow \quad \forall\, u \in \mathbb{R}^p, \quad \langle Au,\, u \rangle \leqslant \langle Bu,\, u \rangle.$$

Proof. We prove (14.19), using Taylor expansion with integral remainder

$$f(x') - f(x) = \int_0^1 \langle \nabla f(x_t),\, x' - x\rangle \mathrm{d}t = \langle \nabla f(x),\, x' - x\rangle + \int_0^1 \langle \nabla f(x_t) - \nabla f(x),\, x' - x\rangle \mathrm{d}t$$

where $x_t \overset{\text{def.}}{=} x + t(x' - x)$. Using Cauchy–Schwarz, and then the smoothness hypothesis ($\mathcal{R}_L$)

$$f(x') - f(x) \leqslant \langle \nabla f(x),\, x' - x\rangle + \int_0^1 L\|x_t - x\|\|x' - x\|\mathrm{d}t \leqslant \langle \nabla f(x),\, x' - x\rangle + L\|x' - x\|^2 \int_0^1 t\mathrm{d}t$$

which is the desired upper-bound. Using directly ($\mathcal{S}_\mu$) gives

$$f(x') - f(x) = \langle \nabla f(x),\, x' - x\rangle + \int_0^1 \langle \nabla f(x_t) - \nabla f(x),\, \frac{x_t - x}{t}\rangle \mathrm{d}t \geqslant \langle \nabla f(x),\, x' - x\rangle + \mu \int_0^1 \frac{1}{t}\|x_t - x\|^2 \mathrm{d}t$$

which gives the desired result since $\|x_t - x\|^2/t = t\|x' - x\|^2$. □

Equation (14.19) places the objective between a lower quadratic tangent model from strong convexity and an upper quadratic tangent model from smoothness.

Condition (14.20) places every eigenvalue of $\partial^2 f(x)$ in $[\mu, L]$. The upper bound is also equivalent to $\|\partial^2 f(x)\|_{\mathrm{op}} \leqslant L$ where $\|\cdot\|_{\mathrm{op}}$ is the operator norm, i.e. the largest singular value. In the special case of a quadratic function of the form $\frac{1}{2}\langle Cx,\, x\rangle - \langle b,\, x\rangle$ (with C symmetric positive semidefinite), $\partial^2 f(x) = C$ is constant, so that $[\mu, L]$ can be chosen to be the range of the eigenvalues of C.

Convergence analysis. For a general smooth convex function, gradient descent admits a sublinear objective bound. Strong convexity gives a linear rate for the iterates. Other structural assumptions can also yield faster rates, but are outside this analysis. Without strong convexity, the minimizer need not be unique.

Theorem 14.16. *If f is convex, satisfies ($\mathcal{R}_L$), and has a minimizer, assume there are constants $\tau_{\min}$, $\tau_{\max}$ such that*

$$0 < \tau_{\min} \leqslant \tau_\ell \leqslant \tau_{\max} < \frac{2}{L},$$

then x_k converges to a solution $x^\star$ of (14.1) and there exists $C > 0$ such that

$$f(x_k) - f(x^\star) \leqslant \frac{C}{k+1}. \tag{14.21}$$

If furthermore f is μ-strongly convex, then there exists $0 \leqslant \rho < 1$ such that $\|x_k - x^\star\| \leqslant \rho^k \|x_0 - x^\star\|$.

Proof. We first derive the objective bound and the strong-convexity estimate for the step $1/L$. We then explain convergence for the full step-size range in the theorem. Take $\tau_\ell = 1/L$. The L-smoothness property implies (14.19), which reads

$$f(x_{k+1}) \leqslant f(x_k) + \langle \nabla f(x_k),\, x_{k+1} - x_k\rangle + \frac{L}{2}\|x_{k+1} - x_k\|^2.$$

Using the fact that $x_{k+1} - x_k = -\frac{1}{L}\nabla f(x_k)$, one obtains

$$f(x_{k+1}) \leqslant f(x_k) - \frac{1}{L}\|\nabla f(x_k)\|^2 + \frac{1}{2L}\|\nabla f(x_k)\|^2 \leqslant f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2 \tag{14.22}$$

Thus the objective sequence $(f(x_k))_k$ is nonincreasing. By convexity

$$f(x_k) + \langle \nabla f(x_k),\, x^\star - x_k\rangle \leqslant f(x^\star)$$

and plugging this in (14.22) shows

$$f(x_{k+1}) \leqslant f(x^\star) - \langle \nabla f(x_k),\, x^\star - x_k\rangle - \frac{1}{2L}\|\nabla f(x_k)\|^2 \tag{14.23}$$

$$= f(x^\star) + \frac{L}{2}\left(\|x_k - x^\star\|^2 - \|x_k - x^\star - \frac{1}{L}\nabla f(x_k)\|^2\right) \tag{14.24}$$

$$= f(x^\star) + \frac{L}{2}\left(\|x_k - x^\star\|^2 - \|x^\star - x_{k+1}\|^2\right). \tag{14.25}$$

Summing these inequalities for $\ell = 0, \ldots, k$, one obtains

$$\sum_{\ell=0}^{k} f(x_{\ell+1}) - (k+1)f(x^\star) \leqslant \frac{L}{2}\left(\|x_0 - x^\star\|^2 - \|x^{(k+1)} - x^\star\|^2\right)$$

Since $f(x_{k+1})$ is nonincreasing, $\sum_{\ell=0}^{k} f(x_{\ell+1}) \geqslant (k+1)f(x^{(k+1)})$. Hence

$$f(x^{(k+1)}) - f(x^\star) \leqslant \frac{L\|x_0 - x^\star\|^2}{2(k+1)}$$

which gives (14.21) for $C \stackrel{\text{def.}}{=} L\|x_0 - x^\star\|^2$.

If we now assume f is μ-strongly convex, then, using $\nabla f(x^\star) = 0$, one has $\frac{\mu}{2}\|x^\star - x\|^2 \leqslant f(x) - f(x^\star)$ for all x. Rearranging (14.25) gives

$$\frac{\mu}{2}\|x_{k+1} - x^\star\|^2 \leqslant f(x_{k+1}) - f(x^\star) \leqslant \frac{L}{2}\left(\|x_k - x^\star\|^2 - \|x^\star - x_{k+1}\|^2\right),$$

and hence

$$\|x_{k+1} - x^\star\| \leqslant \sqrt{\frac{L}{L+\mu}}\|x_k - x^\star\|, \tag{14.26}$$

which is the desired result.

For the general step-size range, use the cocoercivity inequality for a convex function with an L-Lipschitz gradient:

$$\|\nabla f(x) - \nabla f(y)\|^2 \leqslant L\langle \nabla f(x) - \nabla f(y),\, x - y\rangle.$$

It follows by applying the descent bound to $f - \langle \nabla f(y), \cdot\rangle$, whose minimum is at y, and then adding the inequality with x and y exchanged. Set $g_k = \nabla f(x_k)$ and $a = \tau_{\min}(1 - L\tau_{\max}/2) > 0$. The descent bound and cocoercivity give

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\geqslant a\|g_k\|^2, \\ \|x_{k+1} - x^\star\|^2 &\leqslant \|x_k - x^\star\|^2 - \frac{2a}{L}\|g_k\|^2. \end{aligned}$$

The second inequality makes the iterates bounded and their distance to every minimizer nonincreasing. Summation gives $g_k \to 0$. Every accumulation point is therefore a minimizer; the nonincreasing distance to such a point proves convergence of the entire sequence.

For the objective rate, write $d_k = f(x_k) - f(x^\star)$ and $r_k = \|x_k - x^\star\|$. Convexity and the update imply

$$d_k \leqslant \frac{r_k^2 - r_{k+1}^2}{2\tau_{\min}} + \frac{\tau_{\max}}{2a}(d_k - d_{k+1}).$$

Summing from 0 to k and using monotonicity of d_k proves $(k+1)d_k \leqslant r_0^2/(2\tau_{\min}) + \tau_{\max}d_0/(2a)$. Under strong convexity, the same distance calculation gives

$$r_{k+1}^2 \leqslant \big(1 - \mu\tau_{\min}(2 - L\tau_{\max})\big)r_k^2,$$

whose factor belongs to $[0, 1)$. This establishes all claims for the stated step sizes. □

For poor conditioning, $\varepsilon \ll 1$, the rate (14.26) has the same qualitative dependence as the quadratic rate (14.17):

$$\sqrt{\frac{L}{L+\mu}} = (1+\varepsilon)^{-\frac{1}{2}} \sim 1 - \frac{1}{2}\varepsilon.$$

14.5.3 Acceleration

For a convex function with an L-Lipschitz gradient, ordinary gradient descent has a worst-case $O(1/k)$ objective bound. Accelerated methods combine a gradient step with extrapolation. Starting from $y_0 = x_0$ and $0 < s \leqslant 1/L$, one example is

$$x_{k+1} = y_k - s\nabla f(y_k), \qquad y_{k+1} = x_{k+1} + \beta_k(x_{k+1} - x_k), \qquad \beta_k = \frac{k}{k+3}.$$

This is a Nesterov-type scheme. It differs from the heavy-ball method, which evaluates the gradient at x_k and adds momentum to that step. The accelerated scheme satisfies

$$f(x_k) - f(x^\star) = O\left(\frac{\|x_0 - x^\star\|^2}{sk^2}\right).$$

This is a worst-case guarantee; it does not ensure that every individual trajectory is faster. If strong convexity is known, appropriately tuned acceleration or restart can give a linear rate.

A formal continuous-time limit explains the momentum schedule. Set $\tau = \sqrt{s}$ and $t = k\tau$. Since $\beta_{k-1} = (k-1)/(k+2)$,

$$\frac{x_{k+1} - 2x_k + x_{k-1}}{\tau^2} + \frac{3}{(k+2)\tau}\frac{x_k - x_{k-1}}{\tau} + \nabla f(y_k) = 0.$$

If the interpolated iterates converge smoothly as $\tau \to 0$, the limit solves

$$x''(t) + \frac{3}{t}x'(t) + \nabla f(x(t)) = 0, \qquad x(0) = x_0, \quad x'(0) = 0.$$

The term $3x'(t)/t$ is a time-dependent friction. Its decay permits inertial motion while retaining an accelerated energy bound. The coefficient 3 is not the only admissible choice: related dynamics with friction α/t for $\alpha \geqslant 3$ also admit $O(t^{-2})$ objective estimates under the usual convexity assumptions.

CHAPTER 15

Optimization & Machine Learning: Advanced Topics

Reliable and efficient learning requires control of model complexity, economical use of large datasets, and accurate derivative calculations. We study regularization through ridge regression and sparsity, followed by stochastic gradient methods that reduce the cost of individual updates. The final section derives automatic differentiation on computational graphs and applies it to feedforward networks, recurrent models, and optimization procedures.

15.1 Regularization

When the sample size n is small relative to the model dimension p, regularization helps control the complexity of empirical risk minimization.

15.1.1 Penalized Least Squares

For simplicity, consider regularized least-squares regression:

$$\min_{x\in\mathbb{R}^p} f_\lambda(x) \overset{\text{def.}}{=} \frac{1}{2}\|Ax-y\|^2 + \lambda R(x) \tag{15.1}$$

where $R(x)$ is the regularizer and $\lambda \geqslant 0$ the regularization parameter. The regularizer expresses prior structure of the weight vector x, such as small magnitude or sparsity. The strength λ can be selected by cross-validation.

We assume that R is finite, nonnegative, lower semicontinuous, and coercive, i.e. $R(x) \longrightarrow +\infty$ as $\|x\| \longrightarrow +\infty$. The following proposition shows that as $\lambda \downarrow 0$, regularization selects among the exact solutions when $Ax = y$ is feasible. This is particularly useful when $\ker(A) \neq \{0\}$ and there are infinitely many exact solutions.

Proposition 15.1. *Assume $Ax = y$ is feasible and $\lambda_k > 0$ tends to zero. If x_{λ_k} minimizes f_{λ_k}, then this sequence is bounded, and every accumulation point $x^\star$ is a solution of the constrained optimization problem*

$$\min_{Ax=y} R(x). \tag{15.2}$$

Proof. Choose an arbitrary feasible point x_0, so $Ax_0 = y$. Optimality of x_{λ_k} gives

$$\frac{1}{2}\|Ax_{\lambda_k} - y\|^2 + \lambda_k R(x_{\lambda_k}) \leqslant \lambda_k R(x_0). \tag{15.3}$$

Nonnegativity of the terms implies $R(x_{\lambda_k}) \leqslant R(x_0)$. Coercivity of the regularizer makes the sequence $(x_{\lambda_k})_k$ bounded, since R has bounded sublevel sets. The residual also satisfies $\|Ax_{\lambda_k} - y\|^2 \leqslant 2\lambda_k R(x_0)$, so every accumulation point obeys $Ax^\star = y$. Along a subsequence converging to $x^\star$, lower semicontinuity gives $R(x^\star) \leqslant R(x_0)$. Since x_0 was arbitrary, $x^\star$ solves (15.2). □

15.1.2 Ridge Regression

Ridge regression uses the quadratic penalty $R(x) = \frac{1}{2}\|x\|^2_{\mathbb{R}^p}$. For $\lambda > 0$, the objective in (15.1) has a unique minimizer:

$$x_\lambda \overset{\text{def.}}{=} \underset{x\in\mathbb{R}^p}{\operatorname{argmin}}\, f_\lambda(x) = \frac{1}{2}\|Ax-y\|^2_{\mathbb{R}^n} + \frac{\lambda}{2}\|x\|^2_{\mathbb{R}^p}.$$

One has

$$\nabla f_\lambda(x) = A^\top(Ax-y) + \lambda x, \qquad \nabla f_\lambda(x_\lambda) = 0$$

Thus x_λ depends linearly on y and can be computed by solving a linear system. Two equivalent expressions are available.

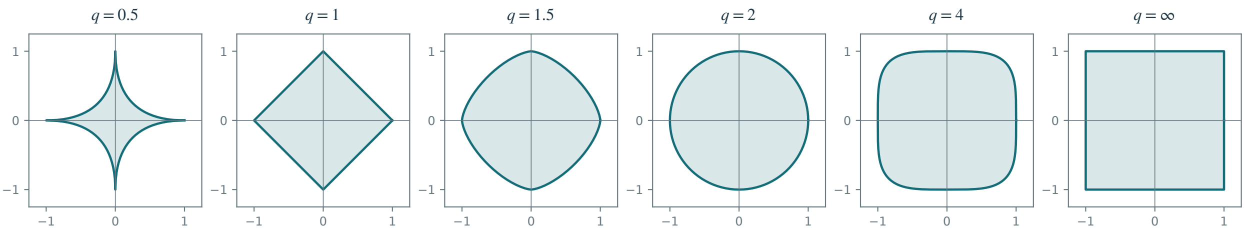


Figure 15.1. ℓ^q balls $\{x \; ; \; \sum_k |x_k|^q \leqslant 1\}$ for varying q.

Proposition 15.2. *One has*

$$x_\lambda = (A^\top A + \lambda \mathrm{Id}_p)^{-1} A^\top y, \tag{15.4}$$

$$= A^\top (A A^\top + \lambda \mathrm{Id}_n)^{-1} y. \tag{15.5}$$

Proof. Denoting $B \stackrel{\text{def.}}{=} (A^\top A + \lambda \mathrm{Id}_p)^{-1} A^\top$ and $C \stackrel{\text{def.}}{=} A^\top (A A^\top + \lambda \mathrm{Id}_n)^{-1}$, one has $(A^\top A + \lambda \mathrm{Id}_p) B = A^\top$ while

$$(A^\top A + \lambda \mathrm{Id}_p) C = (A^\top A + \lambda \mathrm{Id}_p) A^\top (A A^\top + \lambda \mathrm{Id}_n)^{-1} = A^\top (A A^\top + \lambda \mathrm{Id}_n)(A A^\top + \lambda \mathrm{Id}_n)^{-1} = A^\top.$$

Since $A^\top A + \lambda \mathrm{Id}_p$ is invertible, this gives the desired result. □

The systems can be solved directly, for example by Cholesky factorization, or iteratively, for example by conjugate gradients. The latter exploits the quadratic structure more effectively than ordinary gradient descent.

For $n > p$, use the smaller system (15.4); for $n < p$, use (15.5).

Pseudo-inverse. As $\lambda \downarrow 0$, x_λ converges to the minimum-norm least-squares solution $A^+ y$. If $Ax = y$ is feasible, this is, by (15.2)

$$\underset{Ax=y}{\mathrm{argmin}} \; \|x\|.$$

If A has full column rank, so $\ker(A) = \{0\}$ and $n \geqslant p$, $A^\top A \in \mathbb{R}^{p \times p}$ is an invertible matrix, and $(A^\top A + \lambda \mathrm{Id}_p)^{-1} \to (A^\top A)^{-1}$, so that

$$x_0 = A^+ y \quad \text{where} \quad A^+ \stackrel{\text{def.}}{=} (A^\top A)^{-1} A^\top.$$

If A has full row rank, so $\ker(A^\top) = \{0\}$ and $n \leqslant p$, the alternative formula is

$$x_0 = A^+ y \quad \text{where} \quad A^+ \stackrel{\text{def.}}{=} A^\top (A A^\top)^{-1}.$$

When $n = p$ and A is invertible, both formulas reduce to $A^+ = A^{-1}$. For any rank, the singular value decomposition (SVD) gives the Moore–Penrose pseudoinverse by inverting the nonzero singular values and leaving the zero ones unchanged.

15.1.3 Lasso

The Lasso uses an ℓ^1 penalty:

$$R(x) = \|x\|_1 \stackrel{\text{def.}}{=} \sum_{k=1}^{p} |x_k|.$$

The penalty promotes sparsity in the solutions x_λ of

$$x_\lambda \in \underset{x \in \mathbb{R}^p}{\mathrm{argmin}} \; f_\lambda(x) = \frac{1}{2} \|Ax - y\|^2_{\mathbb{R}^n} + \lambda \|x\|_1$$

The solutions x_λ may be nonunique, and typically have many zero coefficients. Figure 15.1 illustrates the geometry of ℓ^q balls. They concentrate around the coordinate axes as $q \to 0$, favoring sparse solutions, but become nonconvex for $q < 1$.

Sparsity can encode a prior on the unknown, as in imaging, or select a small subset of predictive features. Feature selection can make a model easier to interpret and cheaper to evaluate. Whether Lasso or ridge regression performs better depends on sparsity, correlations, noise, and the prediction task.

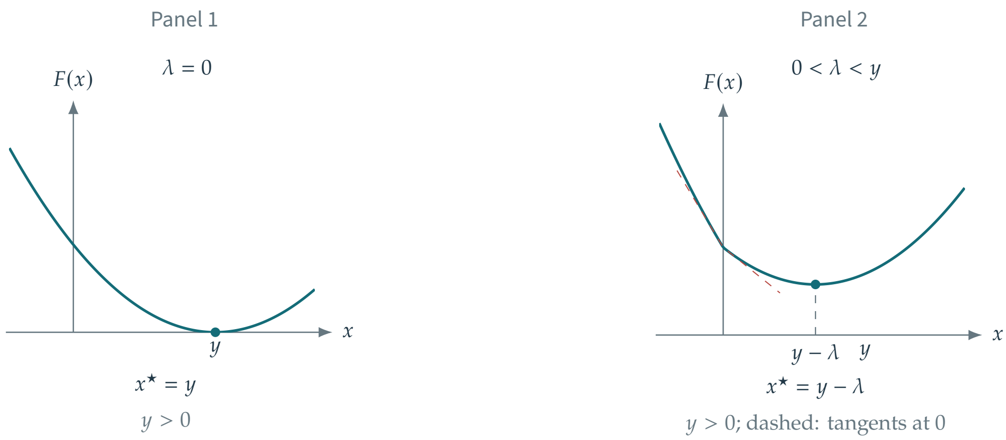


Figure 15.2. Effect of λ on the penalized scalar objective $F(x) \stackrel{\text{def.}}{=} \frac{1}{2}(x-y)^2 + \lambda|x|$.

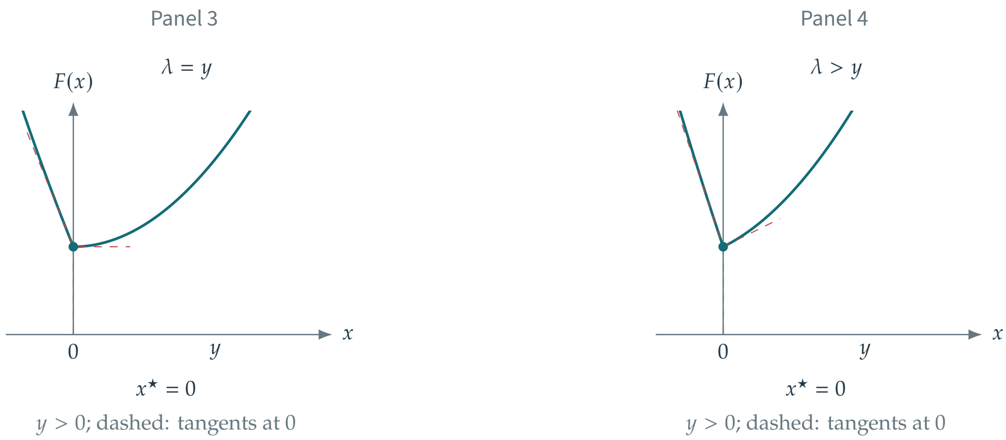


Figure 15.2 (continued). Effect of λ on the penalized scalar objective $F(x) \stackrel{\text{def.}}{=} \frac{1}{2}(x-y)^2 + \lambda|x|$.

The objective f_λ is convex, but the ℓ^1 penalty is nonsmooth, so ordinary gradient descent does not apply at all points. Section 15.1.4 develops an appropriate modification of gradient descent. A closed-form solution x_λ is generally unavailable, but the case of an orthogonal design matrix A reduces to coordinatewise thresholding.

Proposition 15.3. *When $n = p$ and $A = \mathrm{Id}_n$, one has*

$$\underset{x\in\mathbb{R}^p}{\operatorname{argmin}} \frac{1}{2}\|x-y\|^2 + \lambda\|x\|_1 = S_\lambda(y) \quad \textit{where} \quad S_\lambda(x) = (\operatorname{sign}(x_k)\max(|x_k|-\lambda, 0))_k$$

Proof. The separable form $f_\lambda(x) = \sum_k[\frac{1}{2}(x_k-y_k)^2 + \lambda|x_k|]$ reduces the problem to minimizing the scalar function $x \in \mathbb{R} \mapsto \frac{1}{2}(x-y)^2 + \lambda|x|$. Figure 15.2 illustrates this minimization graphically. First suppose $y \geqslant 0$. On $x < 0$, $F'(x) = x - y - \lambda < 0$, so no minimizer is negative. On $x > 0$, $F'(x) = x - y + \lambda$ vanishes at $x = y - \lambda$ when $y > \lambda$. Otherwise, the one-sided derivatives at zero have opposite weak signs, making zero the minimizer. The negative case follows by the symmetry $x \mapsto -x$. □

The nonlinear map S_λ is soft thresholding.

15.1.4 Iterative Soft Thresholding

We derive an iterative algorithm by minimizing a quadratic surrogate. We aim at minimizing

$$f(x) \stackrel{\text{def.}}{=} \frac{1}{2}\|y - Ax\|^2 + \lambda\|x\|_1$$

For a fixed reference point x', introduce the surrogate with parameter $\tau > 0$:

$$f_\tau(x, x') \stackrel{\text{def.}}{=} f(x) - \frac{1}{2}\|Ax - Ax'\|^2 + \frac{1}{2\tau}\|x - x'\|^2.$$

The surrogate touches the objective, $f_\tau(x, x) = f(x)$. Its additional quadratic term is

$$K(x, x') \stackrel{\text{def.}}{=} -\frac{1}{2}\|Ax - Ax'\|^2 + \frac{1}{2\tau}\|x - x'\|^2 = \frac{1}{2}\langle\left(\frac{1}{\tau}\mathrm{Id}_p - A^\top A\right)(x - x'),\, x - x'\rangle.$$

The term $K(x, x')$ is nonnegative when $\lambda_{\max}(A^\top A) \leqslant 1/\tau$, equivalently $\tau \leqslant 1/\|A\|_{\mathrm{op}}^2$, where $\|A\|_{\mathrm{op}} = \sigma_{\max}(A)$ is the operator norm. Thus $f_\tau(x, x')$ is a valid surrogate:

$$f(x) \leqslant f_\tau(x, x'), \quad f_\tau(x, x) = f(x), \quad \text{and} \quad f(\cdot) - f_\tau(\cdot, x') \text{ is smooth.}$$

The surrogate is strongly convex in its first argument, so its minimizer is unique. Define the iteration

$$x_{k+1} \stackrel{\text{def.}}{=} \underset{x}{\operatorname{argmin}}\, f_\tau(x, x_k) \tag{15.6}$$

which ensures by construction that

$$f(x_{k+1}) \leqslant f(x_k).$$

Proposition 15.4. *The iterates x_k defined by (15.6) satisfy*

$$x_{k+1} = S_{\lambda\tau}\left(x_k - \tau A^\top(Ax_k - y)\right) \tag{15.7}$$

Here $S_\lambda(x) = (s_\lambda(x_m))_m$ applies the scalar soft-thresholding map $s_\lambda(r) = \operatorname{sign}(r)\max(|r| - \lambda, 0)$ coordinatewise.

Proof. One has

$$\begin{aligned}
f_\tau(x, x') &= \frac{1}{2}\|Ax - y\|^2 - \frac{1}{2}\|Ax - Ax'\|^2 + \frac{1}{2\tau}\|x - x'\|^2 + \lambda\|x\|_1 \\
&= C + \frac{1}{2}\|Ax\|^2 - \frac{1}{2}\|Ax\|^2 + \frac{1}{2\tau}\|x\|^2 - \langle Ax,\, y\rangle + \langle Ax,\, Ax'\rangle - \frac{1}{\tau}\langle x,\, x'\rangle + \lambda\|x\|_1 \\
&= C + \frac{1}{2\tau}\|x\|^2 + \langle x,\, -A^\top y + A^\top Ax' - \frac{1}{\tau}x'\rangle + \lambda\|x\|_1 \\
&= C' + \frac{1}{\tau}\left(\frac{1}{2}\|x - (x' - \tau A^\top(Ax' - y))\|^2 + \tau\lambda\|x\|_1.\right)
\end{aligned}$$

Proposition 15.3 shows that the minimizer of $f_\tau(x, x')$ is thus indeed $S_{\lambda\tau}(x' - \tau A^\top(Ax' - y))$ as claimed. □

Equation (15.7) defines iterative soft thresholding. For $A \neq 0$, the surrogate argument applies when $0 < \tau \leqslant 1/\|A\|_{\mathrm{op}}^2$. Convergence to a Lasso solution holds for a fixed step in the larger range $0 < \tau < 2/\|A\|_{\mathrm{op}}^2$. If $A = 0$, every positive step is admissible.

15.2 Stochastic Optimization

We study stochastic gradient methods for large datasets of size n and for objectives defined as expectations.

15.2.1 Minimizing Sums and Expectations

Many learning objectives are finite averages:

$$\min_{x \in \mathbb{R}^p} f(x) \stackrel{\text{def.}}{=} \frac{1}{n}\sum_{i=1}^{n} f_i(x) \tag{15.8}$$

or expectations:

$$\min_{x \in \mathbb{R}^p} f(x) \stackrel{\text{def.}}{=} \mathbb{E}_{\mathbf{z}\sim\pi}(f(x, \mathbf{z})) = \int_{\mathcal{Z}} f(x, z)\mathrm{d}\pi(z). \tag{15.9}$$

A finite average is an expectation under the uniform measure $\pi = \frac{1}{n}\sum_{i=1}^{n}\delta_i$, with $f(x, i) = f_i(x)$. Conversely, drawing $(z_i)_i$ i.i.d. from π and setting $f_i(x) = f(x, z_i)$ gives an empirical approximation of (15.9). Under integrability, the empirical objective converges pointwise by the law of large numbers. Convergence of minima or minimizers requires additional assumptions.

For example, empirical risk minimization with linear predictors uses

$$f_i(x) = \ell(\langle a_i,\, x\rangle, y_i) \quad \text{and} \quad f(x, z) = \ell(\langle a,\, x\rangle, y) \tag{15.10}$$

for $z = (a, y) \in \mathcal{Z} = (\mathcal{A} = \mathbb{R}^p) \times \mathcal{Y}$ (typically $\mathcal{Y} = \mathbb{R}$ or $\mathcal{Y} = \{-1, +1\}$ for regression and classification), where ℓ is some loss function. We illustrate the methods with binary logistic classification, using

$$L(s, y) \stackrel{\text{def.}}{=} \log(1 + \exp(-sy)). \tag{15.11}$$

The same computational approach extends to general parametric models, including deep neural networks.

The stochastic methods below apply both to finite sums (15.8) and to expectations (15.9), under the corresponding unbiasedness and moment assumptions. We present them for finite sums. When n is large, the aim is to make each iteration inexpensive without evaluating every summand.

When the summands $f_i(x)$ are similar, a single gradient can approximate the average well: $\nabla f_i \approx \nabla f$. Evaluating ∇f_i is typically n times cheaper than forming the full gradient. Stochastic optimization is nevertheless not always faster than batch gradient descent. For moderate n, a deterministic method may use the available computation more effectively. When n is too large for a full pass to be practical, stochastic methods divide the work into small updates and begin improving the model immediately. Mini-batches also permit parallel computation.

15.2.2 Batch Gradient Descent (BGD)

Batch gradient descent (BGD), analyzed in Section 14.4, iterates

$$x_{k+1} = x_k - \tau_k \nabla f(x_k)$$

For a convex objective with an L-Lipschitz gradient and a minimizer, the condition $0 < \tau_{\min} \leqslant \tau_k \leqslant \tau_{\max} < 2/L$ ensures convergence. Strong convexity strengthens this to a linear rate.

For a finite-sum objective, the gradient is

$$\nabla f(x) = \frac{1}{n} \sum_{i=1}^{n} \nabla f_i(x) \tag{15.12}$$

Its cost is typically $O(np)$ when each ∇f_i can be evaluated in time linear in p.

For an ERM summand of the form (15.10), Taylor expansion gives

$$\begin{aligned} f_i(x + \varepsilon) &= \ell(\langle a_i,\, x\rangle + \langle a_i,\, \varepsilon\rangle, y_i) = \ell(\langle a_i,\, x\rangle, y_i) + \ell'(\langle a_i,\, x\rangle, y_i)\langle a_i,\, \varepsilon\rangle + o(\|\varepsilon\|) \\ &= f_i(x) + \langle \ell'(\langle a_i,\, x\rangle, y_i) a_i,\, \varepsilon\rangle + o(\|\varepsilon\|), \end{aligned}$$

Here $\ell'(s, y)$ denotes differentiation with respect to the score s, with the response y fixed. Consequently,

$$\nabla f_i(x) = \ell'(\langle a_i,\, x\rangle, y_i) a_i. \tag{15.13}$$

For the logistic loss, one has

$$L'(s, y) = -y \frac{e^{-sy}}{1 + e^{-sy}}.$$

15.2.3 Stochastic Gradient Descent (SGD)

For very large n, computing the full gradient ∇f as in (15.12) is prohibitive. SGD replaces the full gradient by the gradient of one summand f_i, with i sampled uniformly. This gives an unbiased estimate:

$$\mathbb{E}_{\mathbf{i}} \nabla f_{\mathbf{i}}(x) = \nabla f(x) \tag{15.14}$$

where $\mathbf{i}$ is a random variable distributed uniformly in $\{1, \ldots, n\}$.

Starting from x_0, stochastic gradient descent (SGD) iterates

$$x_{k+1} = x_k - \tau_k \nabla f_{i(k)}(x_k)$$

where, for each iteration index k, $i(k)$ is drawn independently of previous draws, uniformly from $\{1, \ldots, n\}$. The iterates x_{k+1} are random. We therefore study convergence to a minimizer of f in a probabilistic sense, such as mean square or probability, together with the corresponding rates.

A batch-gradient step costs $O(np)$, whereas an SGD step costs $O(p)$. This makes SGD attractive when n is large and repeated passes through the data are costly. Redundancy between observations can sometimes yield accurate predictions after only $k \ll n$ updates, before all observations have been used.

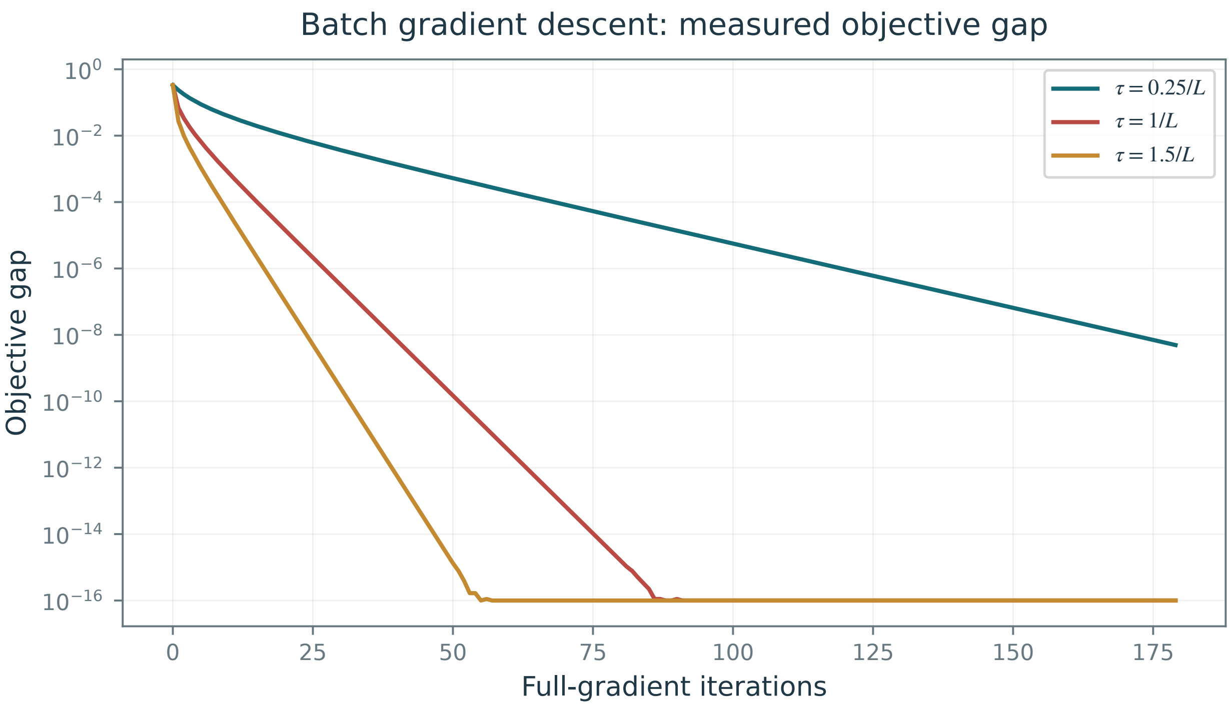


Figure 15.3. Objective-error evolution for batch gradient descent in logistic classification.

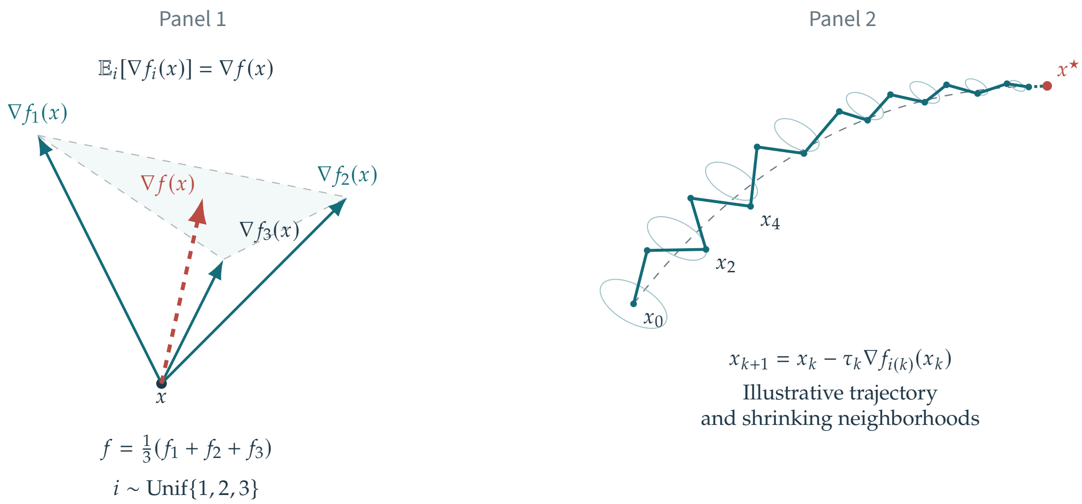


Figure 15.4. Unbiased stochastic gradient estimate (panel 1) and a schematic SGD trajectory (panel 2).

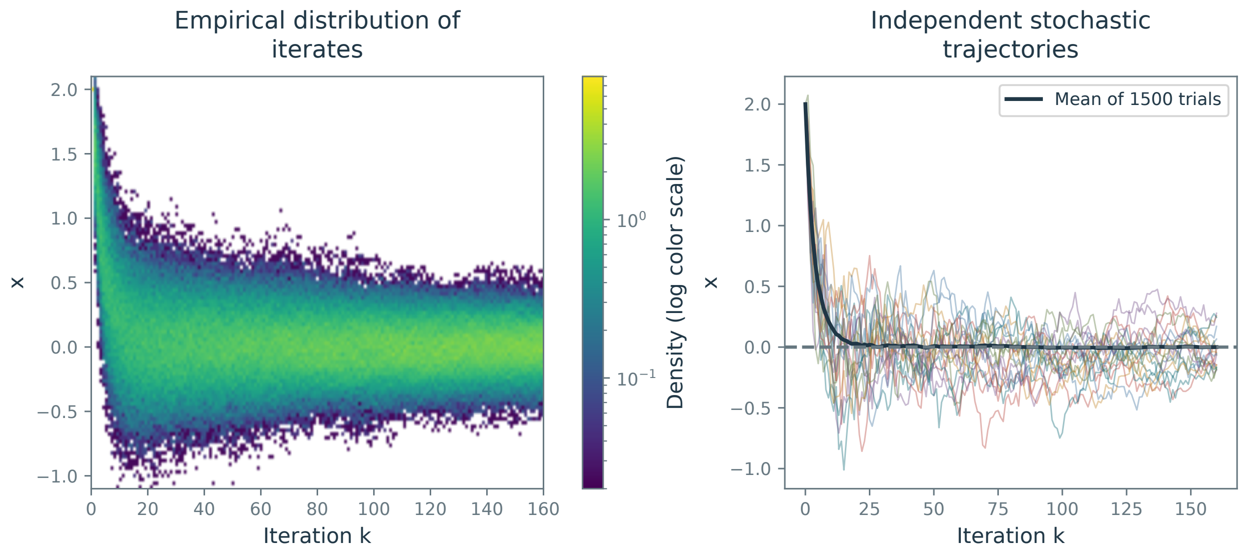


Figure 15.5. Repeated SGD runs for $k \mapsto x_k \in \mathbb{R}$. Left: the distribution of iterates at each iteration. Right: individual trajectories.

The step-size schedule τ_k must balance two requirements: it usually needs to decrease to suppress stochastic noise, but not so quickly that the method stops making progress before reaching a minimizer.

A common schedule has asymptotic behavior $\tau_k \sim k^{-1}$ as $k \to +\infty$. One such schedule is

$$\tau_k \overset{\text{def.}}{=} \frac{\tau_0}{1 + k/k_0} \tag{15.15}$$

where $\tau_0 > 0$ sets the initial step and $k_0 > 0$ controls how long the steps remain approximately constant.

Figure 15.5 illustrates minimization of $f_1(x) + f_2(x)$ for $x \in \mathbb{R}$, with $f_1(x) = (x-1)^2$ and $f_2(x) = (x+1)^2$. The distribution of x_k concentrates around the minimizer $x^\star = 0$. The initial iterate x_0 is uniform on $[-1/2, 1/2]$.

The following theorem gives a mean-square convergence rate. The second-moment assumption is imposed along the iterates, not uniformly on all of $\mathbb{R}^p$: a globally strongly convex function cannot have a globally bounded gradient.

Theorem 15.5. *Let f be differentiable and μ-strongly convex, with minimizer $x^\star$. Let $\mathcal{F}_k$ contain the initial point and all randomness before the kth update, so x_k is $\mathcal{F}_k$-measurable. Assume the stochastic gradient G_k satisfies*

$$\mathbb{E}(G_k \mid \mathcal{F}_k) = \nabla f(x_k), \qquad \mathbb{E}(\|G_k\|^2 \mid \mathcal{F}_k) \leqslant C^2.$$

For $x_{k+1} = x_k - \tau_k G_k$, $\tau_k = 1/(\mu(k+2))$, and deterministic x_0,

$$\mathbb{E}\|x_k - x^\star\|^2 \leqslant \frac{R}{k+2}, \qquad R = \max\{2\|x_0 - x^\star\|^2, C^2/\mu^2\}. \tag{15.16}$$

If f additionally has an L-Lipschitz gradient, then $\mathbb{E}(f(x_k) - f(x^\star)) \leqslant LR/(2(k+2))$.

Proof. Strong convexity and $\nabla f(x^\star) = 0$ imply

$$\langle \nabla f(x_k),\, x_k - x^\star \rangle \geqslant \mu \|x_k - x^\star\|^2. \tag{15.17}$$

Expand the squared update and take a conditional expectation, then average over the history. With $e_k = \mathbb{E}\|x_k - x^\star\|^2$,

$$e_{k+1} \leqslant (1 - 2\mu\tau_k) e_k + \tau_k^2 C^2. \tag{15.18}$$

The claim holds at $k = 0$ by the definition of R. Set $m = k + 2 \geqslant 2$. Since $1 - 2/m \geqslant 0$, the induction hypothesis gives

$$e_{k+1} \leqslant \left(1 - \frac{2}{m}\right) \frac{R}{m} + \frac{R}{m^2} = \frac{R(m-1)}{m^2} \leqslant \frac{R}{m+1}.$$

The objective bound follows from L-smoothness at the minimizer. $\square$

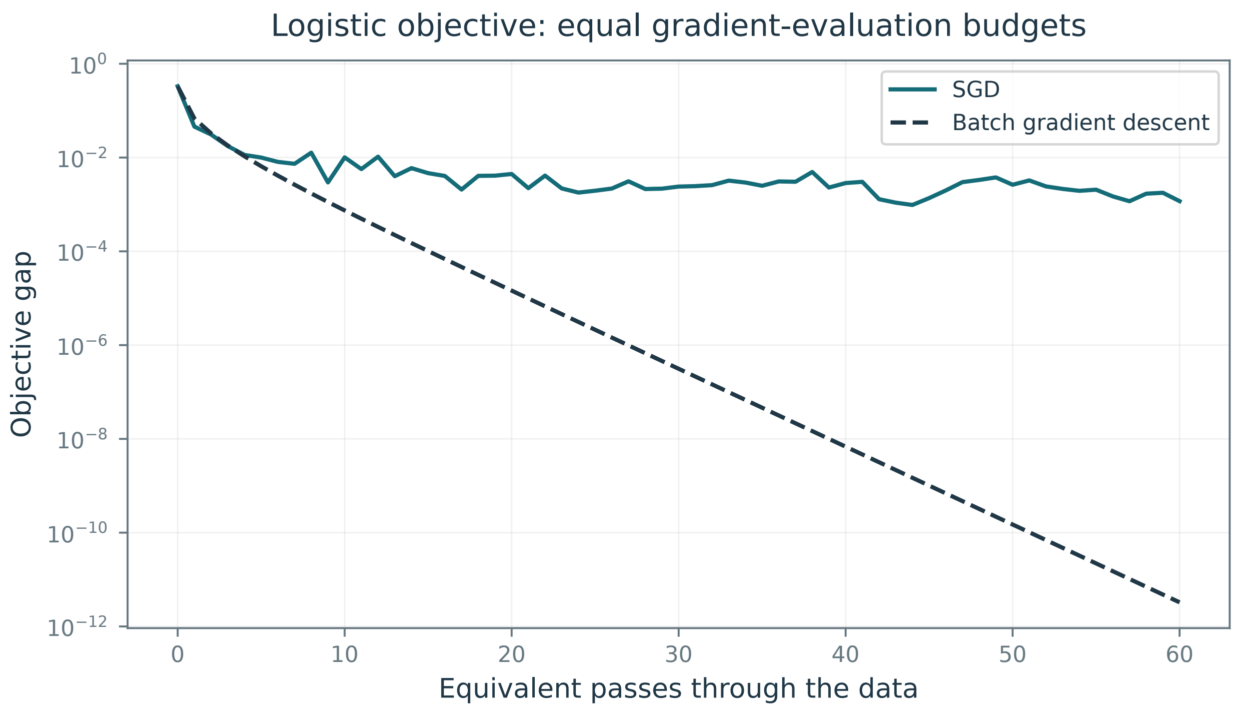


Figure 15.6. Objective-error evolution for SGD in logistic classification; the dashed curve shows batch gradient descent.

The preceding SGD bound, and the averaging scheme below, give sublinear rates even when f is strongly convex. By contrast, batch gradient descent has a linear rate under the assumptions of Theorem 14.16.

Figure 15.6 compares the objective values $f(x_k)$. The black dashed curve shows batch gradient descent, with the iteration axis rescaled to reflect the n-fold cost of a batch update.

15.2.4 Stochastic Gradient Descent with Averaging (SGA)

Diminishing steps can slow SGD as τ_k approaches zero. Averaging reduces fluctuations in the estimate. Run SGD on auxiliary iterates $(\tilde{x}_k)_k$,

$$\tilde{x}_{k+1} = \tilde{x}_k - \tau_k \nabla f_{i(k)}(\tilde{x}_k)$$

and return their Cesàro average:

$$x_k \overset{\text{def.}}{=} \frac{1}{k} \sum_{\ell=1}^{k} \tilde{x}_\ell.$$

This gives stochastic gradient descent with averaging (SGA), also called averaged SGD.

The average can be maintained without storing the full trajectory, using the recursive update

$$x_{k+1} = \frac{1}{k+1} \tilde{x}_{k+1} + \frac{k}{k+1} x_k.$$

A typical step-size schedule is

$$\tau_k \overset{\text{def.}}{=} \frac{\tau_0}{1 + \sqrt{k/k_0}}.$$

These steps decrease more slowly, at rate $k^{-1/2}$.

Averaging can reduce sensitivity to the tuning parameters (k_0, τ_0), although their choice still affects performance.

Bach proves that for logistic classification, averaging can adapt to local curvature under suitable regularity and step-size assumptions. This is a problem-dependent result, rather than a universal improvement of every constant over SGD.

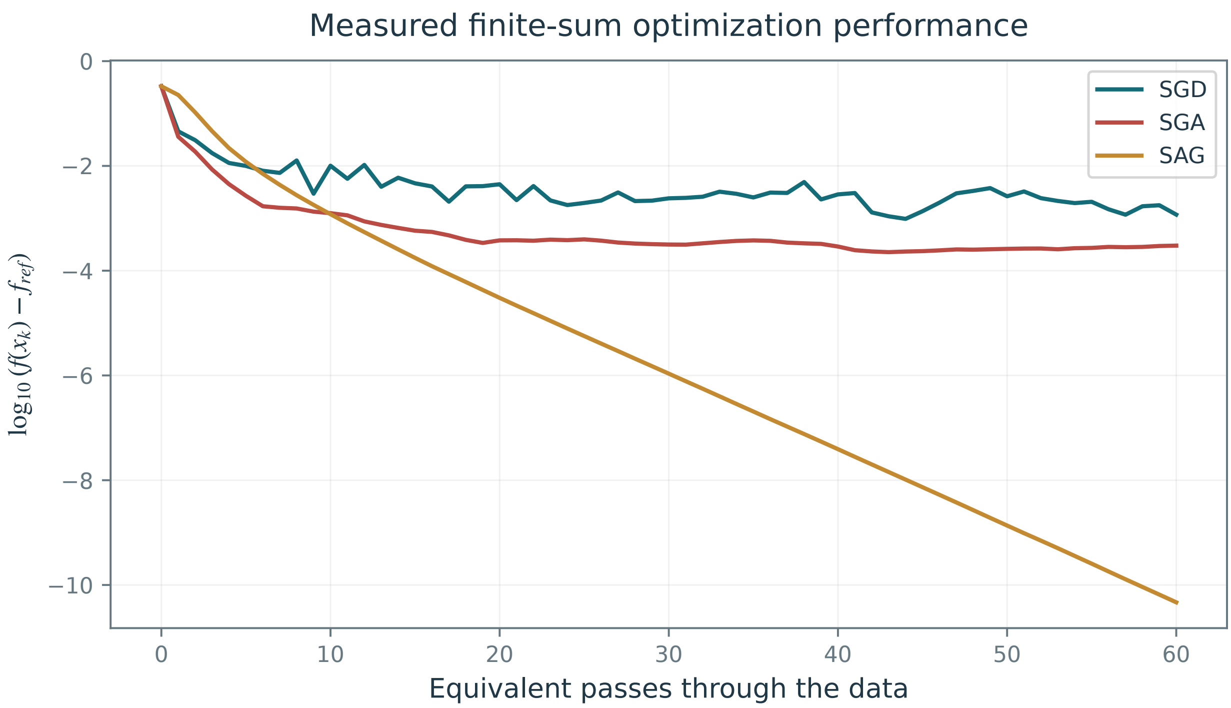


Figure 15.7. Objective gaps for SGD, averaged SGD (SGA), and SAG, displayed on logarithmic axes.

15.2.5 Stochastic Averaged Gradient Descent (SAG)

For a finite dataset with n samples of total size $n \times p$, one can trade additional memory for reduced gradient noise by storing past gradient evaluations. This yields the stochastic averaged gradient (SAG) method.

Store the latest gradient of each summand in a table $(G^i)_{i=1}^n$, requiring $O(np)$ memory. Maintain its average g as an approximation to the full gradient.

Initialize $G^i = 0$ for all i and $g = 0$, so that $g = n^{-1} \sum_i G^i$ is the stored average. At each iteration, update the table and then x:

$$x_{k+1} = x_k - \tau g \quad \text{where} \quad \begin{cases} h \leftarrow \nabla f_{i(k)}(x_k), \\ g \leftarrow g + \frac{1}{n}(h - G^{i(k)}), \\ G^{i(k)} \leftarrow h. \end{cases}$$

Because most entries were evaluated at earlier iterates, the table average is generally biased conditional on the current history. Its convergence analysis therefore differs from the unbiased-SGD proof above. SAG uses a fixed step size τ. As for batch gradient descent, choose τ on the scale of $1/L$, where L bounds the Lipschitz constants of the individual gradients ∇f_i; the admissible numerical constant depends on the convergence theorem.

For smooth convex finite sums with a minimizer and a suitable step size, SAG admits an $O(1/k)$ expected objective bound for averaged iterates. Furthermore, in the presence of strong convexity (for instance after adding a positive quadratic penalty to logistic regression), it has a linear convergence rate, i.e.

$$\mathbb{E}(f(x_k)) - f(x^\star) = O\left(\rho^k\right),$$

for some $0 < \rho < 1$.

These improvements exploit the finite-sum structure: SAG stores information about each of the n summands. SGD and averaged SGD also extend beyond finite n to general expectations (15.9).

Figure 15.7 shows a comparison of SGD, SGA and SAG.

15.3 Automatic Differentiation

Gradient evaluation $\nabla f(x)$ is the main operation in batch and stochastic gradient methods. For linear models and shallow networks, explicit formulas usually reduce it to matrix-vector products. For deeper

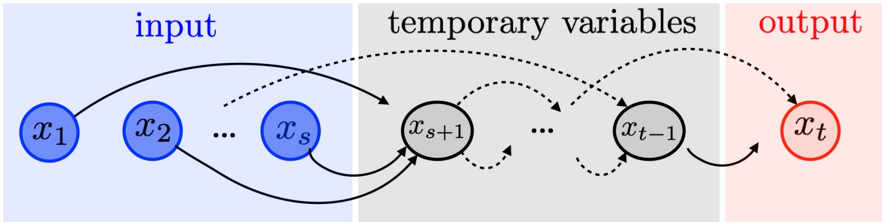


Figure 15.8. A computational graph.

or more elaborate computations, expanded derivative formulas become unwieldy and can duplicate work. Automatic differentiation organizes the chain rule recursively to reuse intermediate results.

15.3.1 Finite Differences and Symbolic Calculus

For $f : \mathbb{R}^p \to \mathbb{R}$, we want to evaluate $\nabla f : \mathbb{R}^p \mapsto \mathbb{R}^p$. A finite-difference approximation with a small parameter $\varepsilon > 0$ is

$$\frac{1}{\varepsilon}(f(x+\varepsilon\delta_1) - f(x), \dots, f(x+\varepsilon\delta_p) - f(x))^\top \approx \nabla f(x)$$

This requires $p+1$ evaluations of f. Here $\delta_k = (0,\dots,0,1,0,\dots,0)$ is the coordinate vector with its 1 at position k. For large p, repeated evaluations are expensive. Finite differences also balance truncation error against floating-point cancellation as ε decreases. Reverse-mode automatic differentiation computes the full gradient at a cost typically proportional to one evaluation of f. Like symbolic differentiation, automatic differentiation applies exact derivative rules, subject to floating-point error. It organizes those rules around the operations of the program that evaluates the function, preserving reuse of intermediate computations.

15.3.2 Computational Graphs

Let $x = (x_1, \dots, x_s)$ be the inputs to a program evaluating f. Number its intermediate variables $x_{s+1}, \dots, x_t$, with output $x_t = f(x)$, and let $x_k \in \mathbb{R}^{n_k}$. We seek the Jacobian blocks $\frac{\partial f(x)}{\partial x_k} \in \mathbb{R}^{n_t \times n_k}$ for $k = 1, \dots, s$. The scalar case $n_k = 1$ is easiest to follow; the formulas also apply to vector variables if the matrix products retain the displayed order. The total input dimension is $p = \sum_{k=1}^s n_k$.

Represent the computation as a sequence of elementary maps:

$$\forall\, k = s+1, \dots, t, \quad x_k = f_k(x_1, \dots, x_{k-1})$$

Each f_k depends only on earlier variables; see Figure 15.8. The directed acyclic graph (DAG) connects each argument used by f_k to its output x_k. The indexing is a topological order: every edge points from a smaller to a larger index. Evaluating $f(x)$ is a forward traversal of the graph. Automatic differentiation differentiates the supplied computational graph. It does not by itself choose an efficient implementation of the original formula; the derivative computation benefits from the efficiency and reuse already present in the forward program.

15.3.3 Forward Mode of Automatic Differentiation

Forward mode propagates derivatives of each intermediate variable with respect to a chosen input block, here x_1. For scalar inputs, repeating the traversal for all p coordinates gives the full derivative. The formulas below also propagate a block Jacobian when $n_1 > 1$.

Initialize the input derivatives by

$$\frac{\partial x_1}{\partial x_1} = \mathrm{Id}_{n_1 \times n_1}, \quad \frac{\partial x_2}{\partial x_1} = 0_{n_2 \times n_1}, \dots, \quad \frac{\partial x_s}{\partial x_1} = 0_{n_s \times n_1},$$

These are one and zero for scalar inputs. Propagate the derivatives using

$$\forall\, k = s+1, \dots, t, \quad \frac{\partial x_k}{\partial x_1} = \sum_{\ell \in \mathrm{parent}(k)} \left[\frac{\partial x_k}{\partial x_\ell}\right] \times \frac{\partial x_\ell}{\partial x_1} = \sum_{\ell \in \mathrm{parent}(k)} \frac{\partial f_k}{\partial x_\ell}(x_1, \dots, x_{k-1}) \times \frac{\partial x_\ell}{\partial x_1}.$$

The notation "parent(k)" denotes the nodes $\ell < k$ of the graph that are connected to k, see Figure 15.9, left. The stored quantities are the derivatives $\frac{\partial x_\ell}{\partial x_1}$. The symbol $\times$ denotes matrix multiplication in the vector-valued

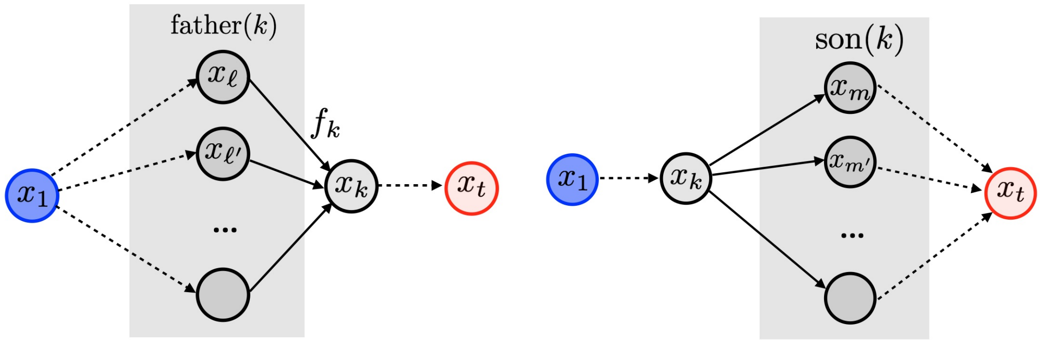


Figure 15.9. Derivative dependencies in forward mode (left) and reverse mode (right).

case. The brackets $[\ldots]$ indicate a local derivative: $\frac{\partial x_k}{\partial x_\ell}$ is the derivative of the elementary map f_k with respect to that argument. It can be evaluated when needed, assuming the local derivative is available in closed form.

Suppose the local derivative operations $\frac{\partial f_k}{\partial x_\ell}$ have comparable costs, as when the variable dimensions n_k are uniformly bounded, and each node has a bounded number of parents. Computing the full Jacobian coordinate by coordinate requires p traversals, each with cost comparable to a forward evaluation. This becomes expensive for large p.

Simple example. We consider the function

$$f(x, y) = y\log(x) + \sqrt{y\log(x)} \tag{15.19}$$

on the open domain $x > 0$, $y \log x > 0$, where all elementary derivatives exist. Its computational graph is displayed in Figure 15.10. The iterations of the forward mode to compute the derivative with respect to x read

$$\begin{aligned}
&\frac{\partial x}{\partial x} = 1, \quad \frac{\partial y}{\partial x} = 0 \\
&\frac{\partial a}{\partial x} = \left[\frac{\partial a}{\partial x}\right]\frac{\partial x}{\partial x} = \frac{1}{x}\frac{\partial x}{\partial x} && \{x \mapsto a = \log(x)\} \\
&\frac{\partial b}{\partial x} = \left[\frac{\partial b}{\partial a}\right]\frac{\partial a}{\partial x} + \left[\frac{\partial b}{\partial y}\right]\frac{\partial y}{\partial x} = y\frac{\partial a}{\partial x} + 0 && \{(y, a) \mapsto b = ya\} \\
&\frac{\partial c}{\partial x} = \left[\frac{\partial c}{\partial b}\right]\frac{\partial b}{\partial x} = \frac{1}{2\sqrt{b}}\frac{\partial b}{\partial x} && \{b \mapsto c = \sqrt{b}\} \\
&\frac{\partial f}{\partial x} = \left[\frac{\partial f}{\partial b}\right]\frac{\partial b}{\partial x} + \left[\frac{\partial f}{\partial c}\right]\frac{\partial c}{\partial x} = 1\frac{\partial b}{\partial x} + 1\frac{\partial c}{\partial x} && \{(b, c) \mapsto f = b + c\}
\end{aligned}$$

A second forward pass gives the derivative with respect to y:

$$\begin{aligned}
&\frac{\partial x}{\partial y} = 0, \quad \frac{\partial y}{\partial y} = 1 \\
&\frac{\partial a}{\partial y} = \left[\frac{\partial a}{\partial x}\right]\frac{\partial x}{\partial y} = 0 && \{x \mapsto a = \log(x)\} \\
&\frac{\partial b}{\partial y} = \left[\frac{\partial b}{\partial a}\right]\frac{\partial a}{\partial y} + \left[\frac{\partial b}{\partial y}\right]\frac{\partial y}{\partial y} = 0 + a\frac{\partial y}{\partial y} && \{(y, a) \mapsto b = ya\} \\
&\frac{\partial c}{\partial y} = \left[\frac{\partial c}{\partial b}\right]\frac{\partial b}{\partial y} = \frac{1}{2\sqrt{b}}\frac{\partial b}{\partial y} && \{b \mapsto c = \sqrt{b}\} \\
&\frac{\partial f}{\partial y} = \left[\frac{\partial f}{\partial b}\right]\frac{\partial b}{\partial y} + \left[\frac{\partial f}{\partial c}\right]\frac{\partial c}{\partial y} = 1\frac{\partial b}{\partial y} + 1\frac{\partial c}{\partial y} && \{(b, c) \mapsto f = b + c\}
\end{aligned}$$

Dual numbers. Dual numbers provide a convenient implementation of the forward pass. They form a real algebra whose elements have the form $x + \varepsilon x'$ where ε is a symbol obeying the rule that $\varepsilon^2 = 0$. Here

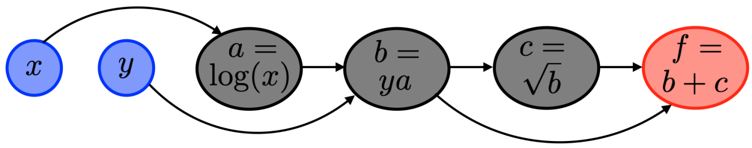


Figure 15.10. Example of a simple computational graph.

$(x, x') \in \mathbb{R}^2$ and x' is intended to store a derivative with respect to some input variable. Multiplication and, when $x \neq 0$, inversion are given by

$$(x + \varepsilon x')(y + \varepsilon y') = xy + \varepsilon(xy' + yx') \quad \text{and} \quad \frac{1}{x + \varepsilon x'} = \frac{1}{x} - \varepsilon \frac{x'}{x^2}.$$

For polynomial or rational f, the arithmetic rules imply

$$f(x + \varepsilon) = f(x) + \varepsilon f'(x).$$

For another differentiable elementary function f, define its action on dual numbers by

$$f(x + \varepsilon x') \overset{\text{def.}}{=} f(x) + \varepsilon f'(x)x'.$$

This definition gives

$$(f \circ g)(x + \varepsilon) = f(g(x)) + \varepsilon f'(g(x))g'(x)$$

which is the chain rule. More generally, for $f(x_1, \ldots, x_s)$ built from these elementary operations,

$$f(x_1 + \varepsilon, x_2, \ldots, x_s) = f(x_1, \ldots, x_s) + \varepsilon \frac{\partial f}{\partial x_1}(x_1, \ldots, x_s)$$

the dual-number evaluation computes $\frac{\partial f}{\partial x_1}(x_1, \ldots, x_s)$ by forward-mode differentiation. The other coordinates are treated identically.

15.3.4 Reverse Mode of Automatic Differentiation

Reverse mode computes the derivatives $\frac{\partial x_t}{\partial x_k}$ of the output with respect to the intermediate variables. For a scalar output and many input coordinates, this avoids the repeated forward traversals required to compute each $\frac{\partial x_k}{\partial x_i}$ separately.

Initialize the derivative of the output node by

$$\frac{\partial x_t}{\partial x_t} = \mathrm{Id}_{n_t \times n_t},$$

and apply the following recursion from the last node to the first:

$$\forall\, k = t-1, t-2, \ldots, 1, \quad \frac{\partial x_t}{\partial x_k} = \sum_{m \in \mathrm{son}(k)} \frac{\partial x_t}{\partial x_m} \times \left[\frac{\partial x_m}{\partial x_k}\right] = \sum_{m \in \mathrm{son}(k)} \frac{\partial x_t}{\partial x_m} \times \frac{\partial f_m(x_1, \ldots, x_{m-1})}{\partial x_k}.$$

The set $\mathrm{son}(k)$ contains the children $m > k$ that depend directly on x_k, as shown in Figure 15.9, right. Contributions from every child must be added.

Back-propagation. For a scalar output $x_t \in \mathbb{R}$, we have $\frac{\partial x_t}{\partial x_k} = [\nabla_{x_k} f(x)]^\top \in \mathbb{R}^{1 \times n_k}$. The reverse recursion can then be written for gradient vectors:

$$\forall\, k = t-1, t-2, \ldots, 1, \quad \nabla_{x_k} f(x) = \sum_{m \in \mathrm{son}(k)} \left(\frac{\partial f_m(x_1, \ldots, x_{m-1})}{\partial x_k}\right)^\top (\nabla_{x_m} f(x)).$$

Here $\left(\frac{\partial f_m(x_1,\ldots,x_{m-1})}{\partial x_k}\right)^\top \in \mathbb{R}^{n_k \times n_m}$ is the adjoint Jacobian of f_m. This adjoint recursion is called backpropagation and is the standard form used in machine learning.

For a scalar output ($n_t = 1$), reverse mode is usually the preferred way to compute the full gradient. It requires access to the intermediate variables $(x_k)_{k=s+1}^t$, which can be costly to store for a large graph. Checkpointing trades additional computation for reduced storage.

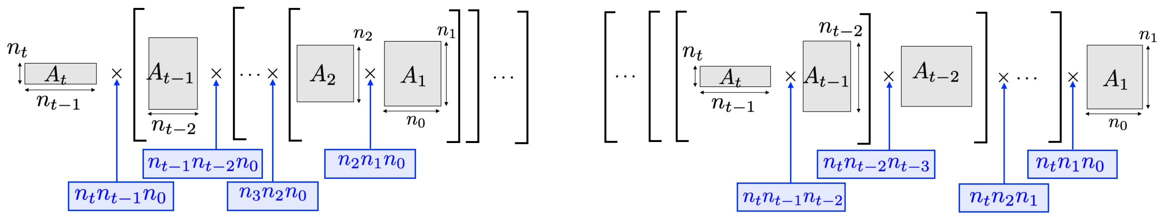


Figure 15.11. Matrix multiplication costs for forward mode (left) and reverse mode (right) along a chain.

Simple example. For the two-input example $f(x,y)$ in (15.19), reverse mode gives

$$\begin{aligned}
\frac{\partial f}{\partial f} &= 1 && \\
\frac{\partial f}{\partial c} &= \frac{\partial f}{\partial f}\left[\frac{\partial f}{\partial c}\right] = \frac{\partial f}{\partial f}1 && \{c \mapsto f = b + c\} \\
\frac{\partial f}{\partial b} &= \frac{\partial f}{\partial c}\left[\frac{\partial c}{\partial b}\right] + \frac{\partial f}{\partial f}\left[\frac{\partial f}{\partial b}\right] = \frac{\partial f}{\partial c}\frac{1}{2\sqrt{b}} + \frac{\partial f}{\partial f}1 && \{b \mapsto c = \sqrt{b}, b \mapsto f = b + c\} \\
\frac{\partial f}{\partial a} &= \frac{\partial f}{\partial b}\left[\frac{\partial b}{\partial a}\right] = \frac{\partial f}{\partial b}y && \{a \mapsto b = ya\} \\
\frac{\partial f}{\partial y} &= \frac{\partial f}{\partial b}\left[\frac{\partial b}{\partial y}\right] = \frac{\partial f}{\partial b}a && \{y \mapsto b = ya\} \\
\frac{\partial f}{\partial x} &= \frac{\partial f}{\partial a}\left[\frac{\partial a}{\partial x}\right] = \frac{\partial f}{\partial a}\frac{1}{x} && \{x \mapsto a = \log(x)\}
\end{aligned}$$

A single reverse traversal computes both derivatives with respect to x and y, whereas coordinatewise forward mode requires two passes.

15.3.5 Feed-forward Compositions

For a feedforward chain, the computed function is the composition

$$f = f_t \circ f_{t-1} \circ \ldots \circ f_2 \circ f_1 \tag{15.20}$$

for functions $f_k : \mathbb{R}^{n_{k-1}} \to \mathbb{R}^{n_k}$.

The forward evaluation starts from $x_0 = x \in \mathbb{R}^{n_0}$ and computes

$$\forall\, k = 1, \ldots, t, \quad x_k = f_k(x_{k-1})$$

giving the output $f(x) = x_t$.

Let $A_k \stackrel{\text{def.}}{=} \partial f_k(x_{k-1}) \in \mathbb{R}^{n_k \times n_{k-1}}$ be the Jacobian of the local map. The chain rule gives

$$\partial f(x) = A_t \times A_{t-1} \times \ldots A_2 \times A_1.$$

Forward mode multiplies the Jacobians from right to left, while reverse mode multiplies them from left to right

$$\begin{aligned}
\partial f(x) &= A_t \times (A_{t-1} \times (\ldots \times (A_3 \times (A_2 \times A_1)))), \\
\partial f(x) &= ((((A_t \times A_{t-1}) \times A_{t-2}) \times \ldots) \times A_2) \times A_1.
\end{aligned}$$

We note that the computation of the product $A \times B$ of $A \in \mathbb{R}^{n \times p}$ with $B \in \mathbb{R}^{p \times q}$ requires $O(npq)$ arithmetic operations with the standard dense algorithm. As shown on Figure 15.11, the arithmetic costs of the two parenthesizations are proportional to

$$n_0 \sum_{k=1}^{t-1} n_k n_{k+1} \quad \text{and} \quad n_t \sum_{k=0}^{t-2} n_k n_{k+1}$$

For roughly comparable intermediate widths, these costs favor reverse mode when $n_t \ll n_0$. A scalar learning objective typically has $n_t = 1$, making reverse mode a natural choice; the exact cost comparison depends on all the intermediate dimensions.

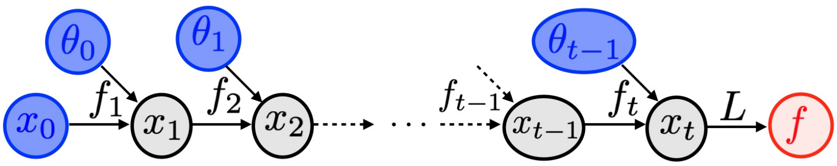


Figure 15.12. Computational graph for a feedforward architecture.

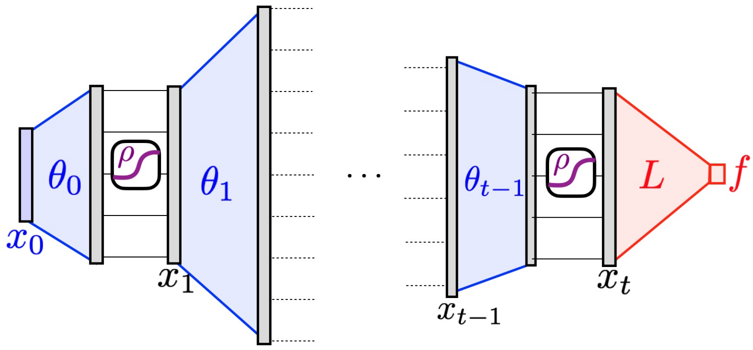


Figure 15.13. Multilayer-perceptron parameterization.

15.3.6 Feed-forward Architecture

A parameterized feedforward architecture, such as a neural network, has the form

$$\forall\, k = 1, \dots, t, \quad x_k = f_k(x_{k-1}, \theta_{k-1}) \tag{15.21}$$

where θ_{k-1} is the parameter vector for a layer and the input $x_0 \in \mathbb{R}^{n_0}$ is fixed. The objective is

$$f(\theta) \overset{\text{def.}}{=} L(x_t) \tag{15.22}$$

Here $L : \mathbb{R}^{n_t} \to \mathbb{R}$ is a loss, such as squared prediction error or logistic loss, and $\theta = (\theta_k)_{k=0}^{t-1}$ collects the parameters. Figure 15.12 shows the computational graph.

Reverse mode computes the gradient of f by propagating derivatives through the variables (x_k, θ_k). Initialize

$$\nabla_{x_t} f = \nabla L(x_t)$$

and recurse from $k = t$ down to 1:

$$z_{k-1} = [\partial_x f_k(x_{k-1}, \theta_{k-1})]^\top z_k \quad \text{and} \quad \nabla_{\theta_{k-1}} f = [\partial_\theta f_k(x_{k-1}, \theta_{k-1})]^\top (\nabla_{x_k} f) \tag{15.23}$$

where $z_k \overset{\text{def.}}{=} \nabla_{x_k} f(\theta)$ is the gradient with respect to x_k.

Multilayer perceptrons. For a fully connected network without biases, use

$$\forall\, x_{k-1} \in \mathbb{R}^{n_{k-1}}, \quad f_k(x_{k-1}, \theta_{k-1}) = \rho(\theta_{k-1} x_{k-1}) \tag{15.24}$$

where $\theta_{k-1} \in \mathbb{R}^{n_k \times n_{k-1}}$ contains the weights and ρ acts coordinatewise; see Figure 15.13. The derivative formulas below apply where ρ is differentiable. At a ReLU kink, a chosen derivative convention defines the computed update, but the classical derivative need not exist. For an adjoint vector $z_k \in \mathbb{R}^{n_k}$, typically $\nabla_{x_k} f$,

$$\begin{cases} [\partial_x f_k(x_{k-1}, \theta_{k-1})]^\top (z_k) = \theta_{k-1}^\top w_k z_k, \\ [\partial_\theta f_k(x_{k-1}, \theta_{k-1})]^\top (z_k) = (w_k z_k) x_{k-1}^\top \end{cases} \quad \text{where} \quad w_k \overset{\text{def.}}{=} \operatorname{diag}(\rho'(\theta_{k-1} x_{k-1})).$$

Link with adjoint state method. A residual form of (15.21) can be interpreted as a time discretization of an ordinary differential equation. Keep the state dimension fixed, $n_k = n$, and interpret x_k as an approximation

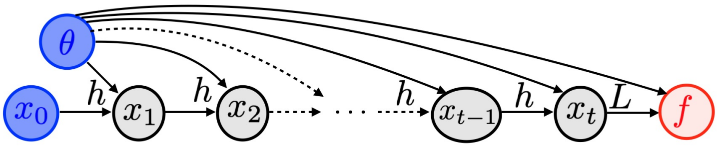


Figure 15.14. Computational graph for a recurrent architecture.

of a continuous trajectory $x(t)$ at time $t = k\tau$. The formal continuum limit takes $\tau \downarrow 0$ with $k\tau$ approaching a fixed time. Impose the residual structure

$$f_k(x_{k-1}, \theta_{k-1}) = x_{k-1} + \tau u(x_{k-1}, \theta_{k-1}, k\tau) \tag{15.25}$$

with a parameterized vector field $u(x, \theta, t) \in \mathbb{R}^n$. Formally, as $\tau \to 0$, this approaches the nonlinear ODE

$$\dot{x}(t) = u(x(t), \theta(t), t) \tag{15.26}$$

with $x(t = 0) = x_0$.

Writing $z(t) = \nabla_{x(t)} f(\theta)$ for the adjoint, the discrete relations (15.23) become a linear backward ODE:

$$\dot{z}(t) = -[\partial_x u(x(t), \theta(t), t)]^\top z(t) \quad \text{and} \quad \nabla_{\theta(t)} f(\theta) = [\partial_\theta u(x(t), \theta(t), t)]^\top z(t).$$

The adjoint has terminal condition $z(T) = \nabla L(x(T))$. The parameter derivative is interpreted as the L^2 functional gradient with respect to the time-dependent control. Its discrete normalization is $\frac{1}{\tau}\nabla_{\theta_{k-1}} f \to \nabla_{\theta(t)} f(\theta)$.

15.3.7 Recurrent Architectures

A recurrent architecture reuses the parameter $\theta = \theta_k$ and map $f_k = h$ at every step of (15.21):

$$\forall\, k = 1, \dots, t, \quad x_k = h(x_{k-1}, \theta). \tag{15.27}$$

Consider a scalar objective

$$f(\theta) = L(x_t, \theta)$$

The loss now depends explicitly on θ, extending (15.22). Figure 15.14 shows the corresponding graph.

Backpropagation gives

$$\nabla_{x_{k-1}} f = [\partial_x h(x_{k-1}, \theta)]^\top \nabla_{x_k} f \quad \text{and} \quad \nabla_\theta f = \nabla_\theta L(x_t, \theta) + \sum_k [\partial_\theta h(x_{k-1}, \theta)]^\top \nabla_{x_k} f. \tag{15.28}$$

Similarly, with $h(x, \theta) = x + \tau u(x, \theta)$, the formal limit $(k, k\tau) \to (+\infty, t)$ gives a forward nonlinear ODE with a time-independent vector field:

$$\dot{x}(t) = u(x(t), \theta)$$

and the following linear backward adjoint equation, for $f(\theta) = L(x(T), \theta)$

$$\dot{z}(t) = -[\partial_x u(x(t), \theta)]^\top z(t) \quad \text{and} \quad \nabla_\theta f(\theta) = \nabla_\theta L(x(T), \theta) + \int_0^T [\partial_\theta u(x(t), \theta)]^\top z(t) \mathrm{d}t. \tag{15.29}$$

with terminal condition $z(T) = \nabla_x L(x(T), \theta)$.

Residual recurrent networks. A residual recurrent network uses the update

$$h(x, \theta) = x + W_2^\top \rho(W_1 x)$$

as illustrated in Figure 15.15. Here $\theta = (W_1, W_2) \in (\mathbb{R}^{q\times n})^2$ contains the weights and ρ is a pointwise activation. Increasing the hidden width q enlarges the class of representable residual maps. When $W_2 = -\tau W_1$ and $\rho = \psi'$, this update is a gradient step for $\mathcal{E}(x, \theta) = \sum_i \psi((W_1 x)_i)$. Repeating it gives an iterative implementation of the argmin layer (15.31), provided the iterations converge to a minimizer. The Jacobians $\partial_\theta h$ and $\partial_x h$ follow from the layer formulas in the multilayer-perceptron discussion.

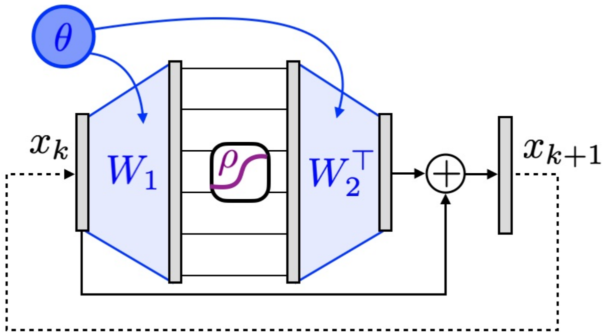


Figure 15.15. Recurrent residual perceptron parameterization.

Mitigating Memory Requirements. Backpropagation requires access to the iterates $(x_k)_{k=0}^t$, whose storage can dominate memory use. Checkpointing stores selected states and reruns parts of the forward computation to reconstruct missing values. Recursive schedules can use $O(\log t)$ stored states with an $O(\log t)$ computational overhead.

If the forward computation is invertible, stored states can sometimes be reconstructed instead. Assume maps g_k satisfy

$$x_k = g_k(x_{k+1}, \dots, x_t).$$

These inverse updates may amplify numerical errors, so algebraic invertibility alone does not ensure stable reconstruction. In practice, the inverse can also depend on a few extra variables, in particular on the input values $(x_0, \dots, x_s)$.

For example, split the continuous state as $x(t) = (r(t), s(t))$ and suppose the vector field in (15.26) has the separated form $u((r,s), \theta, t) = (F(s, \theta, t), G(r, \theta, t))$. An invertible staggered update is

$$r_{k+1} = r_k + \tau F(s_k, \theta_k, \tau k) \quad \text{and} \quad s_{k+1} = s_k + \tau G(r_{k+1}, \theta_{k+1/2}, \tau(k+1/2)).$$

In exact arithmetic, these updates can be inverted as

$$s_k = s_{k+1} - \tau G(r_{k+1}, \theta_{k+1/2}, \tau(k+1/2)). \quad \text{and} \quad r_k = r_{k+1} - \tau F(s_k, \theta_k, \tau k).$$

Fixed point maps If h is continuous and the iterates converge to $x^\star(\theta)$, their limit is a fixed point:

$$x^\star(\theta) = h(x^\star(\theta), \theta).$$

Backpropagation differentiates the finite computation $f(\theta) = L(x_t, \theta)$. For the limiting objective $f^\star(\theta) = L(x^\star(\theta), \theta)$, the implicit function theorem gives an alternative:

$$\nabla f^\star(\theta) = [\partial x^\star(\theta)]^\top (\nabla_x L(x^\star(\theta), \theta)) + \nabla_\theta L(x^\star(\theta), \theta). \tag{15.30}$$

Assume h is continuously differentiable and $\mathrm{Id} - \partial_x h$ is invertible at the fixed point. The implicit function theorem then gives

$$\partial x^\star(\theta) = \left(\mathrm{Id} - \partial_x h(x^\star(\theta), \theta)\right)^{-1} \partial_\theta h(x^\star(\theta), \theta).$$

In practice, one replaces in these formulas $x^\star(\theta)$ by x_t, which approximates the derivative of the converged objective $f^\star$, rather than the derivative of the finite unrolling f. This approach replaces storage of the iteration history by a linear solve at the approximate equilibrium.

Argmin layers An argmin layer maps a parameter θ to a minimizer $x(\theta)$ of a parameterized objective. Assume a unique minimizer, or specify a branch when several exist:

$$x(\theta) = \underset{x}{\operatorname{argmin}}\, \mathcal{E}(x, \theta).$$

One way to compute the layer is gradient descent, initialized here at $x_0 = 0$:

$$x_{k+1} = x_k - \tau \nabla \mathcal{E}(x_k, \theta). \tag{15.31}$$

This has the form (15.25) when using the vector field $u(x, \theta) = -\nabla_x \mathcal{E}(x, \theta)$.

Assume $\mathcal{E}$ is twice continuously differentiable near the selected minimizer and its Hessian with respect to x is invertible. The implicit function theorem gives a locally differentiable minimizer branch; applying (15.30) to its stationarity equation yields

$$\nabla f^\star(\theta) = -\left(\frac{\partial^2 \mathcal{E}}{\partial x \partial \theta}(x^\star(\theta), \theta)\right)^\top \left(\frac{\partial^2 \mathcal{E}}{\partial x^2}(x^\star(\theta), \theta)\right)^{-1} (\nabla_x L(x^\star(\theta), \theta)) + \nabla_\theta L(x^\star(\theta), \theta)$$

If $f(\theta)$ is the minimized value itself, $f(\theta) = \mathcal{E}(x^\star(\theta), \theta)$, then $L = \mathcal{E}$. Formula (15.30) simplifies because $\nabla_x L(x^\star(\theta), \theta) = 0$, giving

$$\nabla f^\star(\theta) = \nabla_\theta L(x^\star(\theta), \theta). \tag{15.32}$$

This is the envelope theorem, also associated with Danskin's theorem.

Sinkhorn's algorithm Let $a \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$ be strictly positive probability vectors, let $C \in \mathbb{R}^{n \times m}$ be a cost matrix, and let $\varepsilon > 0$. Entropic optimal transport minimizes

$$\sum_{ij} C_{ij} P_{ij} + \varepsilon \sum_{ij} P_{ij} (\log P_{ij} - 1)$$

over $P \geqslant 0$ with row sums a and column sums b. Set $K_{ij} = e^{-C_{ij}/\varepsilon} > 0$. Sinkhorn's alternating updates, initialized with positive scaling vectors, are

$$u_{k+1} = \frac{a}{K v_k}, \qquad v_{k+1} = \frac{b}{K^\top u_{k+1}}.$$

All divisions are componentwise. The second update uses the newly computed u_{k+1}; this sequential dependence is part of Sinkhorn's alternating scheme.

The regularized transport value is the maximum of the dual objective

$$\mathcal{E}(u, v; a, b) = \varepsilon \left(\langle a, \log u\rangle + \langle b, \log v\rangle - \langle u, Kv\rangle\right).$$

Sinkhorn performs alternating maximization of this objective. A finite number of iterations defines a differentiable recurrent map on positive inputs. Its Jacobian products can be computed by chaining the two scaling steps. For example, for $u = a/(Kv)$,

$$\partial_v u = -\operatorname{diag}\left(\frac{a}{(Kv)^2}\right) K, \qquad \partial_a u = \operatorname{diag}\left(\frac{1}{Kv}\right),$$

and analogous formulas apply to the v update, with the dependence through the newly computed u included.

At convergence, the envelope theorem gives marginal derivatives represented by the dual potentials

$$\nabla_a f^\star = \varepsilon \log u^\star, \qquad \nabla_b f^\star = \varepsilon \log v^\star.$$

The scalings have the invariance $(u, v) \mapsto (cu, v/c)$, so these potentials are defined up to opposite additive constants. On the probability simplices, derivatives act on zero-sum perturbations and are therefore independent of this gauge. An implicit differentiation formula must fix the gauge before inverting a Jacobian; the unmodified fixed-point Jacobian is singular in that direction.

CHAPTER 16

Shallow Learning

Networks with one hidden layer provide a setting in which to ask which functions neural models can represent and how their size controls approximation error. These questions distinguish expressive power from the difficulty of fitting a model to data. We derive the training gradients, prove universal approximation, and develop quantitative bounds through Barron's theorem, a representation by measures, and the Frank–Wolfe algorithm.

16.1 Multilayer Perceptrons

A multilayer perceptron (MLP) with one hidden layer has two trainable layers: a hidden layer and an output layer. Models with no hidden layer recover the linear predictors studied earlier. Deeper MLPs compose several layers; their derivatives are efficiently computed by the automatic differentiation methods of the multilayer-perceptron discussion.

16.1.1 Networks of Arbitrary Depth

We first define a network of arbitrary depth representing a map $f_\theta : \mathbb{R}^d \to \mathbb{R}^{d'}$. Starting from $x_0 = x \in \mathbb{R}^{d_0} = \mathbb{R}^d$, the network computes, for $s = 0, \dots, S-1$

$$x_{s+1} = \sigma(W_s x_s + b_s)$$

where $W_s \in \mathbb{R}^{d_{s+1} \times d_s}$ and $b_s \in \mathbb{R}^{d_{s+1}}$. The output is $f_\theta(x) = x_S$, with $d_S = d'$ and parameters $\theta = ((W_s, b_s))_{s=0}^{S-1}$. The final activation can be omitted to obtain a linear output layer. The activation σ is a nonlinear, nonpolynomial function applied componentwise: $\sigma(z) = (\sigma(z_i))_i$.

Common activations include the sigmoid functions

$$\sigma(r) = \frac{e^r}{1+e^r} \quad \text{and} \quad \sigma(r) = \frac{1}{\pi}\operatorname{atan}(r) + \frac{1}{2}$$

and the rectified linear unit (ReLU), $\sigma(r) = \max(r, 0)$. Sigmoids are bounded, whereas ReLU is unbounded and does not saturate on the positive half-line: its derivative is one there and zero on the negative half-line. The positive 1-homogeneity of ReLU allows weights to be rescaled, which is useful in proofs that restrict their directions to a unit sphere. At zero, ReLU is not differentiable. Implementations choose a derivative convention, while mathematical arguments must account for the kink.

For increasing width to yield universal approximation at fixed depth, the activation σ must be nonpolynomial. Otherwise, f_θ has degree at most m^S for an activation of degree m and fixed depth S, so increasing the width cannot produce a dense class of continuous functions. The choice $\sigma = \mathrm{Id}$ remains useful for matrix factorization and supervised learning, but its input-output map is affine (linear when the biases vanish).

16.1.2 Two-layer MLPs

We now specialize to networks with one hidden layer and a linear output layer:

$$f_\theta(x) := \sum_{k=1}^{n} u_k \sigma(\langle v_k, x\rangle + b_k), \quad \forall x \in \mathbb{R}^d, \tag{16.1}$$

Here σ is the activation, and the neuron parameters are $\theta_k = (u_k \in \mathbb{R}^{d'}, v_k \in \mathbb{R}^d, b_k \in \mathbb{R})$ for $k = 1, \dots, n$. We usually take $d' = 1$ for simplicity, so the output is scalar.

In practice, neural networks are trained by gradient methods. For example, consider a quadratic loss

$$\min_\theta E(\theta) := \frac{1}{2}\int \|f_\theta(x) - y\|^2 \mathrm{d}\rho(x,y) \tag{16.2}$$

where ρ is the joint distribution of the inputs and outputs. Taking ρ to be the empirical distribution gives the training loss. Gradient descent, or its stochastic counterpart, updates the parameters by

$$\theta_{t+1} = \theta_t - \tau_t \nabla E(\theta_t).$$

Gradient computation Omitting the biases b_k, the network has the matrix form $f_\theta(x) = U\sigma(V^\top x)$, where $U \in \mathbb{R}^{d' \times n}$ and $V \in \mathbb{R}^{d \times n}$. For a training set with N observations, collect the inputs as the columns of $X = (x_i)_{i=1}^N \in \mathbb{R}^{d \times N}$ and the outputs as the columns of $Y = (y_i)_{i=1}^N \in \mathbb{R}^{d' \times N}$. Here N counts observations and n counts hidden neurons. All matrix norms and inner products in this computation are Frobenius norms and inner products. Omitting the constant factor $1/N$, training with squared loss gives

$$\min_{U,V} E(U,V) := \frac{1}{2}\|U\sigma(V^\top X) - Y\|^2.$$

Set $Z := \sigma(V^\top X)$, the matrix obtained by applying the feature map $x \to \sigma(V^\top x)$ to the data. With these features fixed, training U is the least-squares problem $\frac{1}{2}\|UZ - Y\|^2$, whose gradient is

$$\nabla_U E(U,V) = (UZ - Y)Z^\top.$$

Assume σ is differentiable at the entries of $V^\top X$. To differentiate with respect to V, expand the objective in a perturbation D and set $R := U\sigma(V^\top X) - Y$ and $S := \sigma'(V^\top X)$:

$$E(U, V + \varepsilon D) = \frac{1}{2}\|R + \varepsilon U[S \odot (D^\top X)] + o(\varepsilon)\|^2 = E(U,V) + \varepsilon\langle R, H\rangle + o(\varepsilon)$$

where $H = U[S \odot (D^\top X)]$, the product $A \odot B = (A_{i,j}B_{i,j})$ is coordinatewise, and

$$\langle R, H\rangle = \langle R, U[S \odot (D^\top X)]\rangle = \langle D^\top X, (U^\top R) \odot S\rangle = \langle X^\top D, (R^\top U) \odot S^\top\rangle$$

which leads to

$$\nabla_V E(U,V) = X[(R^\top U) \odot S^\top].$$

For deeper networks, writing a fully expanded derivative becomes cumbersome and can duplicate computations. Backpropagation organizes the same chain rule into an efficient reverse pass.

16.2 L^∞ Nonquantitative Universal Approximation

If σ is a sigmoid function, George Cybenko's theorem, later refined by Kurt Hornik, Maxwell Stinchcombe, and Halbert White, demonstrates that the functions f_θ can approximate any continuous function uniformly on a compact domain. Uniform approximation is stronger than the mean-square approximation used in the training loss (16.2).

Proposition 16.1. *Let $K \subset \mathbb{R}^d$ be compact, and let σ be nondecreasing, not necessarily continuous, with*

$$\lim_{s\to-\infty} \sigma(s) = 0 \quad \text{and} \quad \lim_{s\to+\infty} \sigma(s) = 1,$$

Then, for any continuous function f on K and any $\varepsilon > 0$, there exist n and parameters $(\theta_k)_{k=1}^n$ such that

$$\sup_{x\in K} |f(x) - f_\theta(x)| \leqslant \varepsilon.$$

The theorem establishes universal approximation by two-layer networks. It gives no quantitative bound on the number of neurons n needed for accuracy ε, nor an efficient procedure for finding the parameters of f_θ. Cybenko [15] proved the result by duality. We present the more constructive argument of Hornik and collaborators, based on Stone–Weierstrass approximation by trigonometric functions. This density argument avoids relying on uniform convergence of Fourier partial sums, which need not hold for an arbitrary continuous function.

Proof. First establish density for networks with cosine activation; we will then approximate each cosine using the prescribed sigmoid σ. Consider the function space:

$$A := \left\{ \sum_{k=1}^n u_k \cos(\langle v_k, x\rangle + b_k) : n \in \mathbb{N}, (u_k, b_k, v_k)_k \right\}.$$

This space is an algebra: the product-to-sum identity expresses a product of cosines as a sum of two cosines. It contains constants and separates points. Indeed, for $x \neq x'$, choose v so that $\langle v, x' - x\rangle = \pi$ and set $b = -\langle v, x\rangle$; the resulting cosine takes values 1 and -1 at these points. The Stone–Weierstrass theorem therefore implies that A is dense in $C(K)$.

It remains to approximate each univariate cosine on a bounded interval by finite linear combinations of σ. We give an argument that applies to any continuous function q on $[a, b]$. Choose a fine partition $a = t_0 < \cdots < t_Q = b$ and set $\Delta_j = q(t_j) - q(t_{j-1})$. Uniform continuity makes every $|\Delta_j|$ and the oscillation of q on each interval small. The staircase

$$q(t_0) + \sum_{j=1}^{Q} \Delta_j 1_{[t_j, +\infty)}(s)$$

approximates q uniformly, up to this oscillation. Replace the step functions by $\sigma(M(s - t_j))$. Choose disjoint small neighborhoods of the finitely many t_j. Outside these neighborhoods, convergence to the steps is uniform as $M \to \infty$; inside any one neighborhood, only one transition contributes a potentially large discrepancy, bounded by the small jump $|\Delta_j|$. Since $0 \leqslant \sigma \leqslant 1$, the resulting approximation is uniform. A constant is itself a neuron with zero inner weight and an appropriately scaled nonzero activation value.

Apply this construction to each cosine in a finite approximation of f, choosing the individual errors small enough after weighting by the coefficients. Substitution of $s = \langle v_k, x\rangle + b_k$ gives a finite network with uniform error at most ε. □

16.3 L^2 Quantitative Approximation (Barron's Theorem)

We now study L^2 approximation, as measured by the loss (16.2). To isolate approximation error, assume noiseless observations $y = f(x)$ for a target f, with inputs x distributed according to $\rho(x)$. We also assume that ρ is supported in a ball of radius R. Without additional regularity, no uniform approximation rate follows: the error can decay arbitrarily slowly. Barron introduced a function class for which the approximation rate has an exponent independent of the ambient dimension.

16.3.1 Barron Space

For an integrable function f, this section uses the following normalization of the Fourier transform at $\xi \in \mathbb{R}^d$:

$$\hat{f}(\xi) \triangleq (2\pi)^{-d} \int_{\mathbb{R}^d} f(x) e^{-\mathrm{i}\langle \xi, x\rangle} \mathrm{d}x.$$

The Barron space [1] is the class of functions for which the Fourier seminorm

$$\|f\|_B \triangleq \int_{\mathbb{R}^d} \|\xi\| |\hat{f}(\xi)| \mathrm{d}\xi$$

is finite. Throughout this section, f denotes its continuous Fourier-inversion representative, with the convention $f(x) = \int \hat{f}(\xi) e^{\mathrm{i}\langle \xi, x\rangle} \mathrm{d}\xi$. For integrable f, finiteness of the displayed seminorm ensures that $\hat{f}$ is integrable: it is bounded near the origin, and its first moment controls the tail. This representative matters when the sampling measure has atoms. Constants can be handled separately; $|f(0)| + \|f\|_B$ removes the ambiguity due to constants in the corresponding Fourier-measure space. This Fourier-defined class should be distinguished from activation-dependent Barron spaces. One has

$$\|f\|_B = \int_{\mathbb{R}^d} \|\widehat{\nabla f}(\xi)\| \mathrm{d}\xi,$$

which relates the Fourier seminorm to first-order regularity. The following examples illustrate the seminorm and its dependence on the function class.

- *Gaussians*: for $f(x) = e^{-\|x\|^2/2}$, one has $\|f\|_B = \mathbb{E}\|Z\| \leqslant \sqrt{d}, \quad Z \sim \mathcal{N}(0, \mathrm{Id}_d)$
- *Ridge functions*: for $f(x) = \psi(\langle x, b\rangle + c)$, the Fourier transform may be a measure supported on a line rather than an integrable density. In the Fourier-measure extension of the seminorm,

  $$\|f\|_B \leqslant \|b\| \int_{\mathbb{R}} |u| \, |\hat{\psi}(u)| \mathrm{d}u.$$

  A sufficient condition is that ψ is compactly supported and belongs to $C^{2,\delta}(\mathbb{R})$ for some $\delta > 0$. Smoothness alone without decay is not enough for this Fourier integral.

- *Sobolev functions*: if $s > d/2 + 1$, the Cauchy–Schwarz inequality gives

$$\|f\|_B \leqslant C(d,s)\|f\|_{H^s}, \qquad \|f\|_{H^s}^2 \overset{\text{def.}}{=} \int_{\mathbb{R}^d} (1 + \|\xi\|^2)^s |\hat{f}(\xi)|^2 \mathrm{d}\xi.$$

  Indeed, $\int \|\xi\|^2 (1 + \|\xi\|^2)^{-s} \mathrm{d}\xi$ is finite precisely when $s > d/2 + 1$. For integer s, this squared Sobolev norm is equivalent, up to Fourier-normalization constants, to the sum of the squared L^2 norms of all weak derivatives of order at most s. This is a sufficient condition; the Fourier-measure class also contains ridge functions outside these ambient-space Sobolev classes.

16.3.2 Barron's Theorem

The main result is as follows.

Theorem 16.2 (Barron [1]). *Assume ρ is a probability measure supported on $B(0, R)$, f is the continuous Fourier-inversion representative described above with $\|f\|_B < \infty$, and σ is a nondecreasing sigmoid with limits 0 and 1. Allow an output bias c_0, so the approximation is $c_0 + f_\theta$. For every $n \geqslant 1$,*

$$\inf_{c_0,\theta} \|c_0 + f_\theta - f\|_{L^2(\rho)} \leqslant \frac{2R\|f\|_B}{\sqrt{n}}.$$

The infimum can be restricted to networks with $\sum_{k=1}^n |u_k| \leqslant 2R\|f\|_B$. In particular, for every $\varepsilon > 0$, a finite network attains the displayed bound plus ε.

The exponent $1/2$ in the approximation rate does not depend on the dimension. The constant $\|f\|_B$ can still depend on it: the Gaussian example gives growth of order $\sqrt{d}$, whereas the bound for a fixed ridge profile depends on the ridge direction and profile.

16.3.3 Mean field representation.

Absorb a factor n into the outer coefficients u_k to express the same network as an average:

$$f_\theta(x) := \frac{1}{n} \sum_{k=1}^n \varphi(x, \omega_k),$$

where $\theta = (\omega_k)_{k=1}^n$, $\omega_k = (u_k, v_k, b_k) \in \mathbb{R}^{d'} \times \mathbb{R}^{d+1}$ and $\varphi(x, \omega) := u\sigma(\langle v, x\rangle + b)$. Introducing the empirical measure:

$$\hat{\mu} := \frac{1}{n} \sum_{k=1}^n \delta_{\omega_k},$$

this neural network can be expressed as an integral:

$$f_\theta(x) := \int_\Omega \varphi(x, \omega)\, \mathrm{d}\hat{\mu}(\omega)$$

where $\Omega \subset \mathbb{R}^{d'} \times \mathbb{R}^{d+1}$ is the admissible parameter set. A uniform bound on u will be essential for the approximation estimate. The representation is linear in the measure μ. It therefore extends naturally from empirical measures to general probability measures μ.

We now restrict to scalar outputs, $d' = 1$. The Fourier condition yields a bounded dictionary representation in a closure sense. This distinction matters: the assumptions do not automatically provide an exact probability measure on finite sigmoid parameters.

Proposition 16.3. *Let $M = 2R\|f\|_B$. On $K \subset B(0, R)$, the function $f - f(0)$ belongs to the closed convex hull, in $L^2(\rho)$, of the centered neurons*

$$\varphi_\omega(x) = u\big(\sigma(\langle v,\, x\rangle + b) - \sigma(b)\big), \qquad \omega = (u, v, b), \quad |u| \leqslant M.$$

Every dictionary element has $L^2(\rho)$ norm at most M. For general measures μ on parameters, write

$$\Phi(\mu)(x) = \int \varphi_\omega(x) \mathrm{d}\mu(\omega). \tag{16.3}$$

The proposition asserts membership in the closure of such averages, rather than existence of one exact average.

Proof sketch. Fourier inversion gives

$$f(x) - f(0) = \int \big(\cos(\langle \xi, x\rangle + \Theta(\xi)) - \cos\Theta(\xi)\big)|\hat{f}(\xi)| \mathrm{d}\xi,$$

where Θ is the phase of $\hat{f}$. For $a = R\|\xi\|$ and $|t| \leqslant a$,

$$\cos(t + \Theta) - \cos\Theta = \int_{-a}^{a} -\sin(s + \Theta)\big(1_{t\geqslant s} - 1_{0\geqslant s}\big)\mathrm{d}s.$$

The total variation of these coefficients is at most $2a$. After integration over ξ, it is therefore at most $2R\|f\|_B = M$. Step functions are limits of dilated sigmoids; choosing thresholds away from atoms of the projected data distribution, dominated convergence gives the same closed convex hull in $L^2(\rho)$. A zero atom absorbs any unused coefficient mass. Centering the neurons accounts for an output bias when the sum is rewritten in the form (16.1). □

For the next argument, write $g = f - f(0)$. First suppose $g = \Phi(\mu)$ exactly for a probability measure over this bounded dictionary. The same bound extends to g in its closed convex hull by approximating g arbitrarily closely before sampling.

16.3.4 Probabilistic proof

Draw independent parameters $\omega_1, \dots, \omega_n$ with law μ and set $g_n = n^{-1}\sum_i \varphi_{\omega_i}$. In the Hilbert space $L^2(\rho)$, independence yields

$$\mathbb{E}g_n = g, \qquad \mathbb{E}\|g_n - g\|^2 = \frac{1}{n}\left(\mathbb{E}\|\varphi_\omega\|^2 - \|g\|^2\right) \leqslant \frac{M^2}{n}.$$

Consequently at least one realization has squared error at most M^2/n. For a target in the closed convex hull, choose an average g_ε with $\|g - g_\varepsilon\| \leqslant \varepsilon$ and apply the same argument. The resulting network has error at most $M/\sqrt{n} + \varepsilon$. Each outer coefficient has magnitude at most M/n, so their absolute sum is at most M. This proves the approximation theorem, including its coefficient bound.

16.3.5 Proof by optimization

A deterministic argument uses n Frank–Wolfe steps and its $O(1/n)$ objective convergence rate. For the centered target $g = f - f(0)$, minimize over probability measures $\mathcal{P}(\Omega)$:

$$\inf_{\mu\in\mathcal{P}(\Omega)} E(\mu) := \frac{1}{2}\int_K (\Phi(\mu)(x) - g(x))^2 \,\mathrm{d}\rho(x), \tag{16.4}$$

where ρ is the probability measure supported on K. This optimization problem is infinite-dimensional.

Ordinary network training instead restricts μ to an empirical measure and descends in the neuron parameters. This parameterization is nonconvex. Chizat and Bach relate suitable mean-field limits to Wasserstein gradient flows and prove global convergence under additional structural and initialization assumptions. Such results do not imply that every sufficiently wide finite network, initialized from an arbitrary density, avoids all nonglobal stationary points.

The deterministic approximation argument below uses convex optimization over measures. Its linear minimization oracle is a global optimization problem over a single neuron's parameters and may itself be computationally difficult in high dimensions. Since the parameter domain need not be compact, exact oracle minimizers may fail to exist; approximate oracles or closure arguments then replace exact minimization.

First order variations. We derive the algorithm by linearizing the objective on measures equipped with the total variation norm. The same construction applies to differentiable convex objectives on a Banach space. We write integration as a pairing between a function and a measure:

$$\langle f, \mu\rangle := \int f(x)\mathrm{d}\mu(x).$$

Let $\mu + \varepsilon\eta$ be a small perturbation of μ, where η is a finite signed measure. The first variation $\nabla E(\mu)$ represents the derivative through the expansion

$$E(\mu + \varepsilon\eta) = E(\mu) + \varepsilon\langle\nabla E(\mu), \eta\rangle + o(\varepsilon),$$

Thus $\nabla E(\mu)$ represents the Fréchet derivative of E. For the present quadratic objective,

$$E(\mu + \varepsilon\eta) = \frac{1}{2}\int_K (\Phi(\mu)(x) - g(x) + \varepsilon\Phi(\eta)(x))^2 \, \mathrm{d}\rho(x),$$

which expands to:

$$E(\mu + \varepsilon\eta) = E(\mu) + \varepsilon \int_K \Phi(\eta)(x)\,(\Phi(\mu)(x) - g(x)) \, \mathrm{d}\rho(x) + O(\varepsilon^2).$$

Rewriting this in terms of φ, we find:

$$\nabla E(\mu)(\omega) = \int_K \varphi(x,\omega)\,(\Phi(\mu)(x) - g(x)) \, \mathrm{d}\rho(x).$$

It is a bounded function of ω; continuity follows when the activation and parameterization are continuous.

Frank-Wolfe algorithm. Frank–Wolfe minimizes a differentiable convex function over a convex subset of a Banach space:

$$\min_{\mu \in C} E(\mu).$$

Each step linearizes the objective $E(\mu)$. Choose an initial measure μ_0, for example a Dirac mass. At iteration k, update with step size τ_k:

$$\mu_{k+1} = (1 - \tau_k)\mu_k + \tau_k \nu_k^*,$$

where ν_k^* minimizes the linearized objective:

$$\nu_k^* \in \arg\min_{\nu \in \mathcal{P}(\Omega)} \langle \nabla E(\mu_k),\, \nu \rangle$$

This linear minimization step is called an oracle. On a finite parameter grid it amounts to comparing finitely many values. Over a continuous neuron parameter space Ω, it is generally a nonconvex optimization problem. Here $C = \mathcal{P}(\Omega)$ carries the total variation norm $\|\mu\|_{\mathrm{TV}} = |\mu|(\Omega)$, the measure analogue of the L^1 norm. In this special case, a key property of the algorithm is that the solution ν_k^* can be taken as a Dirac measure when the minimum over Ω is attained since, denoting $g_k := \nabla E(\mu_k)$,

$$\nu_k^* = \delta_{\omega_k^*}, \quad \text{where } \omega_k^* \in \arg\min_{\omega \in \Omega} g_k(\omega).$$

This holds because for any $\nu \in \mathcal{P}(\Omega)$:

$$\int g_k(\omega)\,\mathrm{d}\nu(\omega) \geqslant \min(g_k),$$

and equality is achieved when $\nu = \delta_{\omega_k^*}$. Therefore, if μ_0 is initialized as a Dirac measure, each iteration of the algorithm ensures that μ_k remains a sum of at most $k+1$ Dirac masses.

Convergence Rate The next theorem gives the Frank–Wolfe convergence rate. In our case, $\|\cdot\| = \|\cdot\|_{\mathrm{TV}}$ is the total variation norm of measures. Bounded functions paired with these measures carry the supremum norm $\|\cdot\|_\infty$. Throughout this argument, L^∞ denotes the pointwise supremum norm, so values at atoms are retained. We first establish the quadratic upper bound implied by a Lipschitz gradient.

Lemma 16.4. *Let C be convex, and let $E : C \to \mathbb{R}$ be differentiable with L-Lipschitz gradient with respect to the norm $\|\cdot\|$. That is, for all $\mu, \nu \in C$,*

$$\|\nabla E(\mu) - \nabla E(\nu)\|_* \leqslant L\|\nu - \mu\|,$$

where $\|\cdot\|_*$ denotes the dual norm of $\|\cdot\|$. Then, for all $\mu, \nu \in C$:

$$E(\nu) \leqslant E(\mu) + \langle \nabla E(\mu), \nu - \mu\rangle + \frac{L}{2}\|\nu - \mu\|^2.$$

Proof. By the fundamental theorem of calculus, we can express $E(\nu)$ as:

$$E(\nu) = E(\mu) + \int_0^1 \langle \nabla E(\mu + t(\nu - \mu)), \nu - \mu \rangle \, dt.$$

Adding and subtracting $\nabla E(\mu)$ inside the integrand:

$$E(\nu) = E(\mu) + \langle \nabla E(\mu), \nu - \mu \rangle + \int_0^1 \langle \nabla E(\mu + t(\nu - \mu)) - \nabla E(\mu), \nu - \mu \rangle \, dt.$$

Using the L-Lipschitz property of the gradient, we bound the difference:

$$\|\nabla E(\mu + t(\nu - \mu)) - \nabla E(\mu)\|_* \leqslant Lt\|\nu - \mu\|.$$

Substitute this bound into the integral:

$$\left| \int_0^1 \langle \nabla E(\mu + t(\nu - \mu)) - \nabla E(\mu), \nu - \mu \rangle \, dt \right| \leqslant \int_0^1 Lt\|\nu - \mu\|^2 \, dt.$$

Evaluate the integral:

$$\int_0^1 Lt \, dt = \frac{L}{2}.$$

Thus:

$$E(\nu) \leqslant E(\mu) + \langle \nabla E(\mu), \nu - \mu \rangle + \frac{L}{2}\|\nu - \mu\|^2.$$

□

Theorem 16.5. *Let C be a bounded convex set of diameter $D = \sup_{\mu,\nu \in C} \|\mu - \nu\|$, and let E be convex with L-Lipschitz derivative in the corresponding dual norm. Assume the linear minimization oracle is attained. With steps $\tau_k = 2/(k+2)$ for $k \geqslant 0$, the Frank–Wolfe iterates satisfy, for $k \geqslant 1$,*

$$E(\mu_k) - E^* \leqslant \frac{2LD^2}{k+2}, \qquad E^* = \inf_{\mu \in C} E(\mu).$$

Proof. Write $h_k = E(\mu_k) - E^*$ and $g_k = \min_{\nu \in C} \langle \nabla E(\mu_k), \nu - \mu_k \rangle$. Convexity implies $g_k \leqslant -h_k$. Lemma 16.4 therefore gives

$$h_{k+1} \leqslant h_k + \tau_k g_k + \frac{L}{2}\tau_k^2 D^2 \leqslant (1 - \tau_k)h_k + \frac{L}{2}\tau_k^2 D^2. \tag{16.5}$$

Since $\tau_0 = 1$, $h_1 \leqslant LD^2/2 \leqslant 2LD^2/3$, establishing the base case. If $h_k \leqslant 2LD^2/(k+2)$, then

$$h_{k+1} \leqslant \frac{2LD^2(k+1)}{(k+2)^2} \leqslant \frac{2LD^2}{k+3},$$

because $(k+1)(k+3) \leqslant (k+2)^2$. This proves the claim by induction. □

For the objective E in (16.4), the proposition below gives $L \leqslant M^2$ with $M = 2R\|f\|_B$. The total variation diameter of probability measures is $D = 2$, and $\|f\|_B$ denotes the Barron seminorm of the target f. For the centered target $g = f - f(0)$, Proposition 16.3 gives $\inf E = 0$, even if the infimum is not attained by a measure on finite sigmoid parameters. The rate therefore concerns approximation of g. When the linear oracle attains its minimum, Frank–Wolfe constructs a measure μ_k supported on at most $k+1$ points, with objective error

$$E(\mu_k) = O\left(\frac{1}{k}\right).$$

Thus each iteration adds at most one neuron and gives an $O(k^{-1/2})$ bound on the L^2 error. If the oracle is not attained, choosing an $O(1/k)$-accurate linear oracle adds an $O(1/k^2)$ term to the descent recurrence and preserves this order of approximation.

Proposition 16.6. *The first variation of the objective in* (16.4) *is*

$$\nabla E(\mu)(\omega) = \int_K \varphi(x,\omega)\,(\Phi(\mu)(x) - g(x))\;\mathrm{d}\rho(x),$$

where $\Phi(\mu)(x) = \int_\Omega \varphi(x,\omega)\,\mathrm{d}\mu(\omega)$ *and* $\varphi(x,\omega) = u\big(\sigma(\langle v, x\rangle + b) - \sigma(b)\big)$ *with* $|u| \leqslant M$. *This first variation is L-Lipschitz from the total variation norm to the supremum norm: for any* $\mu, \mu' \in \mathcal{P}(\Omega)$*,*

$$\|\nabla E(\mu) - \nabla E(\mu')\|_\infty \leqslant L\|\mu - \mu'\|_{TV},$$

where $L = M^2$.

Proof. The difference of $\nabla E(\mu)$ and $\nabla E(\mu')$ is:

$$\nabla E(\mu)(\omega) - \nabla E(\mu')(\omega) = \int_K \varphi(x,\omega)\,(\Phi(\mu)(x) - \Phi(\mu')(x))\;\mathrm{d}\rho(x).$$

$$\Phi(\mu)(x) - \Phi(\mu')(x) = \int_\Omega \varphi(x,\omega)\,\mathrm{d}(\mu - \mu')(\omega).$$

Substitute the above into the expression for $\nabla E(\mu)$:

$$\nabla E(\mu)(\omega) - \nabla E(\mu')(\omega) = \int_K \varphi(x,\omega)\left(\int_\Omega \varphi(x,\omega')\,\mathrm{d}(\mu - \mu')(\omega')\right)\mathrm{d}\rho(x).$$

Using Fubini's theorem, introducing $k(\omega,\omega') := \int_K \varphi(x,\omega)\varphi(x,\omega')\,\mathrm{d}\rho(x)$,

$$\nabla E(\mu)(\omega) - \nabla E(\mu')(\omega) = \int_\Omega k(\omega,\omega')\mathrm{d}(\mu - \mu')(\omega').$$

Take the L^∞ norm with respect to ω:

$$\|\nabla E(\mu) - \nabla E(\mu')\|_\infty = \sup_{\omega\in\Omega}\left|\int_\Omega k(\omega,\omega')\mathrm{d}(\mu - \mu')(\omega')\right|.$$

Using the triangle inequality:

$$\|\nabla E(\mu) - \nabla E(\mu')\|_\infty \leqslant \|k\|_{L^\infty(\Omega\times\Omega)}\|\mu - \mu'\|_{\mathrm{TV}}.$$

One has

$$\|k\|_{L^\infty(\Omega\times\Omega)} = \sup_{(\omega,\omega')\in\Omega^2} |\int_K \varphi(x,\omega)\varphi(x,\omega')\mathrm{d}\rho(x)| \leqslant \|\varphi\|^2_{L^\infty(K\times\Omega)} \leqslant M^2.$$

$\square$

CHAPTER 17

Deep Learning

Deep networks learn representations through successive transformations of the input. Their architecture controls which structures they can exploit and how efficiently they can be trained, making it essential to understand the operations within each layer. We develop fully connected and convolutional models and their gradients, then introduce residual connections, normalization, attention, and wavelet scattering.

17.1 Deep Architectures

17.1.1 Deep Network Structure

Deep learning constructs estimators $f(x,\beta)$ by composing parameterized maps. The hidden layers compute a feature representation of the input, and the output layer applies a linear or generalized linear predictor to those features. Unlike a fixed kernel feature map, this representation is learned from the training data.

Feedforward architectures. A feedforward chain has the structure introduced in (15.20), with the loss $\mathcal{L}$ omitted. Starting from $x_0 = x$, compute

$$x_{\ell+1} = f_\ell(x_\ell, \beta_\ell),$$

so that $x_L = f(x,\beta)$ with

$$f(\cdot,\beta) = f_{L-1}(\cdot,\beta_{L-1}) \circ f_{L-2}(\cdot,\beta_{L-2}) \circ \ldots \circ f_0(\cdot,\beta_0)$$

where $\beta = (\beta_0, \ldots, \beta_{L-1})$ collects the parameters, and each layer has the dimensions

$$f_\ell(\cdot,\beta_\ell) : \mathbb{R}^{n_\ell} \to \mathbb{R}^{n_{\ell+1}}.$$

We focus on feedforward chains. More general architectures use branched computational graphs or share parameters across recurrent steps.

The parameters β are fitted by empirical risk minimization (13.11), typically using the stochastic methods of Section 15.2. The resulting objective is generally nonconvex, so Theorem 15.5 does not apply. Under suitable smoothness, noise, and step-size assumptions, one can instead bound measures of stationarity. Initialization and optimization dynamics can also provide implicit regularization.

For chain architectures, reverse mode computes the gradient of the ERM loss (15.13) as described in Section 15.3.5. For convolutional layers, it gives the backpropagation equations (17.3) below. More general computational graphs require the reverse pass to accumulate contributions from every branch and every use of a shared parameter; Section 15.3.4 explains this general procedure.

Deep MLP. A common computational block $f_\ell(\cdot,\beta_\ell)$ consists of

– an affine map, $B_\ell \cdot + b_\ell$ with a matrix $B_\ell \in \mathbb{R}^{\tilde{n}_\ell \times n_\ell}$ and a vector $b_\ell \in \mathbb{R}^{\tilde{n}_\ell}$ parametrized (in most cases linearly) by β_ℓ,

– a nonlinearity independent of β_ℓ, given by $\rho_\ell : \mathbb{R}^{\tilde{n}_\ell} \to \mathbb{R}^{n_{\ell+1}}$

which we write as

$$\forall\, x_\ell \in \mathbb{R}^{n_\ell}, \quad f_\ell(x_\ell,\beta_\ell) = \rho_\ell(B_\ell x_\ell + b_\ell) \in \mathbb{R}^{n_{\ell+1}}. \tag{17.1}$$

In a fully connected layer, every entry of B_ℓ and b_ℓ is trainable, so $(B_\ell, b_\ell) = \beta_\ell$. A common choice is a pointwise activation ρ_ℓ, written as $\rho_\ell(z) = (\tilde{\rho}_\ell(z_k))_k$ with $\tilde{\rho}_\ell : \mathbb{R} \to \mathbb{R}$ nonlinear. Such an activation preserves dimension, so $n_{\ell+1} = \tilde{n}_\ell$. Standard examples are the rectified linear unit (ReLU), $\tilde{\rho}_\ell(s) = \max(s, 0)$, and the sigmoid, $\tilde{\rho}_\ell(s) = \theta(s) = (1 + e^{-s})^{-1}$.

Interleaving affine maps with nonlinear activations produces increasingly expressive functions $f(\cdot,\beta)$.

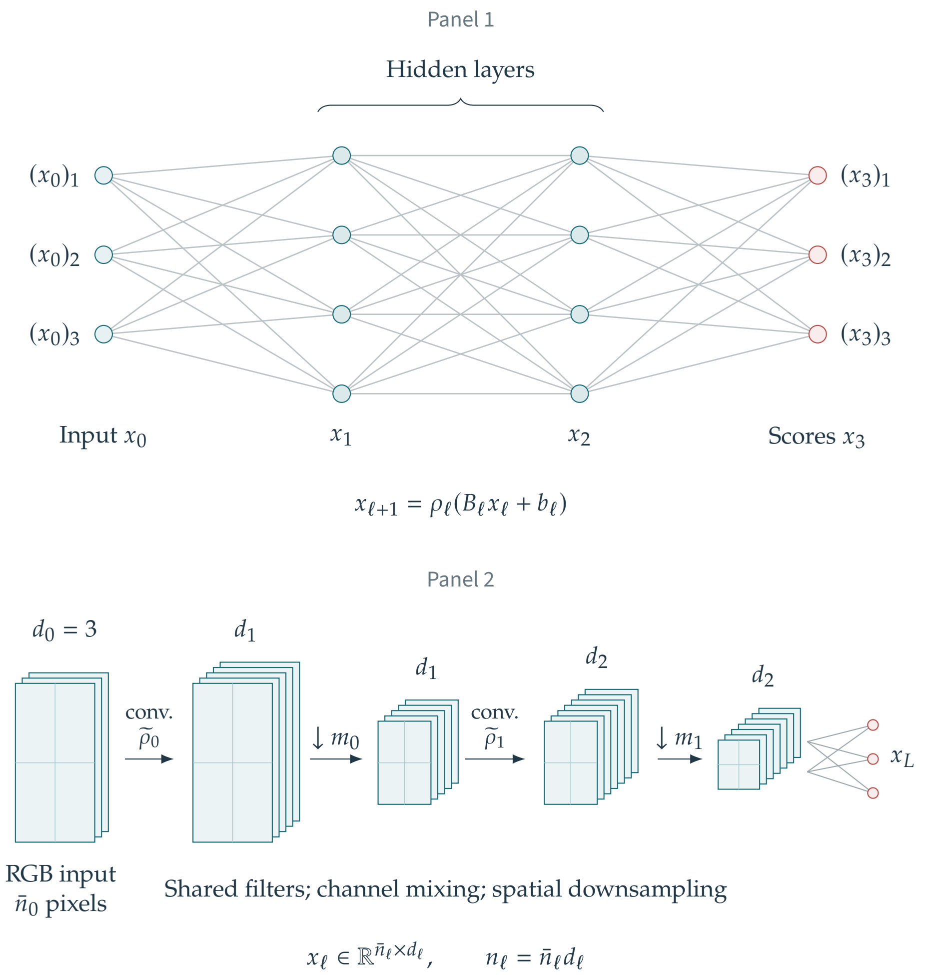


Figure 17.1. Panel 1: fully connected network. Panel 2: convolutional neural network.

Training the parameters $\beta = (B_\ell, b_\ell)_\ell$ means minimizing (13.11), for example by stochastic gradient descent. Automatic differentiation computes the gradient at a cost proportional to a forward evaluation, namely $O(\sum_\ell n_\ell \bar{n}_\ell)$ for dense layers with pointwise activations. The chain structure gives the backpropagation formula (15.23).

For regression tasks, one usually uses an affine output layer with a squared ℓ^2 loss L. An output nonlinearity is appropriate when the target is constrained, for example to nonnegative values. For classification, a multiclass logistic map (13.24) converts the final scores into class probabilities.

For high-dimensional inputs such as images, fully connected layers can require prohibitively many parameters. They also ignore spatial structure and symmetries. Convolutional architectures exploit this structure, as detailed in Section 17.1.3.

17.1.2 Perceptron and Shallow Models

The logistic classifiers from Sections 13.4.2 and 13.4.3 are simple instances of this network formulation.

The binary logistic model (13.21) is a one-layer ($L = 1$) instance of (17.1). Omitting the bias, its components are

$$B_0 x = \langle x, \beta\rangle \quad \text{and} \quad \tilde{\rho}_0(u) = \theta(u).$$

The network $f(x, \beta) = \theta(\langle x, \beta\rangle)$ can include a bias by appending a constant coordinate to x. For labels $y \in \{0, 1\}$, it is trained with binary cross-entropy:

$$L(t, y) = -\log(t^y(1-t)^{1-y}) = -y\log(t) - (1-y)\log(1-t).$$

The resulting empirical-risk objective is convex in the parameters.

For K classes, compute the scores $B_0 x = (\langle x, \beta_k\rangle)_{k=1}^K$ and apply the normalized exponential map

$$f(x, \beta) = \mathcal{N}((\exp(\langle x, \beta_k\rangle))_k) \quad \text{where} \quad \mathcal{N}(u) = \frac{u}{\sum_k u_k}$$

For target vectors y on the probability simplex, use the cross-entropy loss

$$L(t, y) = -\sum_{k=1}^{K} y_k \log(t_k).$$

17.1.3 Convolutional Neural Networks

Signals, images, and videos carry spatial structure that can guide the architecture. Convolutional layers respect translations of the spatial grid, subject to boundary conditions, and reuse the same filters at every location.

At depth ℓ, a convolutional network arranges the activation vector $x_\ell \in \mathbb{R}^{n_\ell}$ as an array in $\mathbb{R}^{\bar{n}_\ell \times d_\ell}$, with $\bar{n}_\ell$ spatial positions and d_ℓ channels, so $n_\ell = \bar{n}_\ell d_\ell$. The spatial positions typically form a one-, two-, or three-dimensional grid. For an RGB image, $\bar{n}_0$ is the number of pixels and $d_0 = 3$. Write $x_\ell = ((x_\ell)_r[i])_{r,i}$, where $i \in \{1, \dots, \bar{n}_\ell\}$ indexes the spatial position and $r \in \{1, \dots, d_\ell\}$ the channel.

We require the linear map $B_\ell : \mathbb{R}^{\bar{n}_\ell \times d_\ell} \to \mathbb{R}^{\bar{n}_\ell \times d_{\ell+1}}$ to be translation equivariant. With periodic boundary conditions, this is equivalent to a sum of convolutions across the input channels. The number of output channels may differ from the number of input channels.

Proposition 17.1. *Let $B : \mathbb{R}^{n \times d} \to \mathbb{R}^{n \times d'}$ be linear, with n spatial positions and periodic boundary conditions. The map $x = (x_r[i])_{r,i} \mapsto Bx$ is translation equivariant, meaning*

$$\forall\, \tau, \quad B(x[\cdot - \tau]) = (Bx)[\cdot - \tau]$$

(the same shift is applied to every channel) if and only if there exist filters $\psi_{r,s} \in \mathbb{R}^n$, for $1 \leqslant r \leqslant d'$ and $1 \leqslant s \leqslant d$, such that

$$\forall\, r = 1, \dots, d', \quad (Bx)_r = \sum_{s=1}^{d} \psi_{r,s} \star x_s \tag{17.2}$$

$$\text{i.e.} \quad (Bx)_r[i] = \sum_{s=1}^{d} \sum_{j} \psi_{r,s}[i-j] x_s[j]$$

where $\star$ denotes periodic discrete convolution on the spatial grid, with all spatial indices interpreted periodically.

Proof. Periodic convolution commutes with spatial shifts, so (17.2) defines an equivariant operator. Conversely, suppose B is equivariant. Let δ^s be the impulse at spatial position $i = 0$ in channel s: $(\delta^s)_{s'}[i] = 1_{s=s'} 1_{i=0}$. Define its response by $\psi_{r,s} \overset{\text{def.}}{=} (B\delta^s)_r$. Every input decomposes into shifted impulses:

$$x = \sum_{j,s} x_s[j] \delta^s[\cdot - j],$$

so that by linearity and translation equivariance

$$(Bx)_r = \sum_{j,s} x_s[j](B\delta^s[\cdot - j])_r = \sum_{j,s} x_s[j](B\delta^s)_r[\cdot - j] = \sum_{j,s} x_s[j]\psi_{r,s}[\cdot - j] = \sum_{s} \psi_{r,s} \star x_s$$

Thus each filter is the corresponding impulse response. The adjoint convolution uses the reversed filter, as in (17.3) below. □

The proposition parameterizes an equivariant map B_ℓ by filters $(\psi_\ell)_{r,s}$ indexed by output channel r and input channel s. Writing $x_\ell = ((x_\ell)_s)_{s=1}^{d_\ell}$ for the input channels, the map becomes

$$\forall\, r \in \{1, \dots, d_{\ell+1}\}, \quad (B_\ell x_\ell)_r = \sum_{s=1}^{d_\ell} (\psi_\ell)_{r,s} \star (x_\ell)_s,$$

and the bias is spatially constant to preserve translation equivariance: $(b_\ell)_r[i] = (\tilde{b}_\ell)_r$ for every spatial position i, with one trainable scalar $(\tilde{b}_\ell)_r$ per output channel. This is weight sharing: the same filters ψ_ℓ act at every spatial position.

A pointwise nonlinearity is followed, when desired, by downsampling to reduce the spatial dimension and computational cost. The filtering/downsampling structure resembles the fast wavelet transform. Let m_ℓ be the downsampling factor, commonly $m_\ell = 1$ (no reduction) or $m_\ell = 2$ (a factor of two in each spatial direction). Then

$$\rho_\ell(u) = (\tilde{\rho}_\ell(u_s[m_\ell \cdot]))_{s=1\ldots,d_{\ell+1}} .$$

Max-pooling replaces subsampling by local maxima, for example over groups of m_ℓ successive values in one dimension. Strided convolutions provide another way to reduce spatial resolution. These operations preserve equivariance only to shifts compatible with the sampling stride.

Composing local layers enlarges a unit's receptive field: the set of input locations that can affect it. Its actual sensitivity within that set depends on the parameters, nonlinearities, and input. Multiple channels allow different features to be represented, from local edges to more elaborate patterns.

After spatial downsampling, a classifier can use fully connected output layers or aggregate the spatial features before the final prediction.

The trainable parameters are the filters and biases. To describe backpropagation, consider a single channel per layer, omit biases and downsampling, and use periodic convolution. The forward pass computes

$$z_\ell = \psi_\ell \star x_\ell, \qquad x_{\ell+1} = \rho_\ell(z_\ell), \qquad \ell = 0, \ldots, L-1.$$

For the loss $\mathcal{E}(\beta) = \mathcal{L}(x_L, y)$, let $a_\ell = \nabla_{x_\ell}\mathcal{E}$ denote the activation gradient. Initialize $a_L = \nabla_1 \mathcal{L}(x_L, y)$ and, for $\ell = L-1, \ldots, 0$, compute

$$q_\ell = \rho'_\ell(z_\ell) \odot a_{\ell+1}, \qquad a_\ell = \bar{\psi}_\ell \star q_\ell, \qquad \nabla_{\psi_\ell}\mathcal{E} = \bar{x}_\ell \star q_\ell. \tag{17.3}$$

Here $\bar{v}[i] = v[-i]$ denotes reversal, and $\odot$ denotes coordinatewise multiplication. The activation and filter gradients are distinct. For a filter with restricted support, retain only the trainable coefficients of the filter gradient. A downsampling layer contributes its adjoint upsampling operator to the reverse pass.

These equations are an instance of reverse-mode automatic differentiation. For nonsmooth activations such as ReLU, implementations choose a derivative convention at the kink; away from such points, the ordinary chain rule applies. More general computational graphs, including shared weights and recurrent connections, require accumulation of all contributions as described in Section 15.3.4.

17.1.4 Advanced Architectures

Residual Networks Residual networks add skip connections at selected layers ℓ:

$$f_\ell(x_\ell, \beta_\ell) = x_\ell + \tilde{f}_\ell(x_\ell, \beta_\ell)$$

The residual map $\tilde{f}_\ell(\cdot, \beta_\ell) : \mathbb{R}^{n_\ell} \to \mathbb{R}^{n_\ell}$ preserves the dimension. Such connections help stabilize the training of very deep networks and admit an interpretation as explicit Euler steps for an ordinary differential equation. A common residual block uses a bottleneck:

$$\tilde{f}_\ell(x_\ell, \beta_\ell) = C_\ell^\top \rho_\ell(B_\ell x_\ell)$$

(ignoring biases), where $\beta_\ell = (B_\ell, C_\ell) \in \mathbb{R}^{m_\ell \times n_\ell} \times \mathbb{R}^{m_\ell \times n_\ell}$. Choosing $m_\ell \ll n_\ell$ reduces the parameter count and forces the residual map to pass through a lower-dimensional representation.

Batch normalization Batch normalization [22] standardizes activations during training. For one feature with values $z_1, \ldots, z_m$ in a mini-batch, define

$$\bar{z} = \frac{1}{m}\sum_{i=1}^m z_i, \qquad s^2 = \frac{1}{m}\sum_{i=1}^m (z_i - \bar{z})^2, \qquad \tilde{z}_i = \gamma \frac{z_i - \bar{z}}{\sqrt{s^2 + \varepsilon}} + \beta.$$

The stabilizer $\varepsilon > 0$ is fixed, while γ and β are learned. During training, the output depends on the whole mini-batch. At inference, stored estimates of the mean and variance are used. For convolutional layers, statistics are usually shared across spatial positions within each channel.

Transformer Networks A self-attention layer [33] acts on a sequence $x = (x_i)_{i=1}^n$ with $x_i \in \mathbb{R}^d$. A single scaled dot-product attention head maps

$$x_k \longmapsto \sum_{i=1}^{n} a_{ki} V x_i, \qquad a_{ki} = \frac{\exp(\langle Q x_k,\, K x_i \rangle / \sqrt{d_k})}{\sum_{j=1}^{n} \exp(\langle Q x_k,\, K x_j \rangle / \sqrt{d_k})}.$$

Here $Q, K \in \mathbb{R}^{d_k \times d}$ and $V \in \mathbb{R}^{d_v \times d}$ are the learned query, key, and value matrices. For each query index k, the weights a_{ki} are nonnegative and sum to one over i. Dense attention computes all pairs of scores, so its cost is quadratic in the sequence length for fixed feature dimensions. Transformer blocks combine several attention heads with residual connections, normalization, and positionwise feedforward maps. Positional information is required to distinguish sequence order; without it, unmasked self-attention is permutation equivariant.

17.1.5 Scattering Transform

The scattering transform, introduced by Mallat and collaborators, is a convolutional architecture with fixed wavelet filters, modulus nonlinearities, and local averaging. It is a nonlinear extension of the wavelet transform. Its filters are chosen from the geometry of signals and images rather than learned from data. Under appropriate assumptions on the wavelets and deformations, scattering has stability guarantees with respect to small diffeomorphisms. Scattering coefficients can also be used as fixed features before a learned predictor.

CHAPTER 18

Convex Analysis

Convexity turns local optimality conditions into certificates of global solutions, even when objectives have corners or encode constraints. This provides a common language for understanding regularization and designing optimization methods. We develop subgradients and normal cones, study convex conjugates and their relation to smoothness, and derive dual problems and optimality conditions.

We write $f(x)$ for the objective and x for the variable to be optimized. The main references are [13, 4].

18.1 Basics of Convex Analysis

We consider minimization problems of the form

$$\min_{x\in\mathcal{H}} f(x) \tag{18.1}$$

over the finite-dimensional Hilbert space $\mathcal{H} \overset{\text{def.}}{=} \mathbb{R}^N$, with the canonical inner product $\langle\cdot,\cdot\rangle$. Many of these results extend to infinite-dimensional Hilbert spaces, with additional hypotheses where compactness is used.

The objective $f : \mathcal{H} \longrightarrow \bar{\mathbb{R}} \overset{\text{def.}}{=} \mathbb{R} \cup \{+\infty\}$ is convex. Allowing f to take the value $+\infty$ encodes constraints directly in the objective; its effective domain is

$$\mathrm{dom}(f) \overset{\text{def.}}{=} \{x \,;\, f(x) < +\infty\}.$$

For a set $C \subset \mathcal{H}$, define the indicator function

$$\iota_C(x) \overset{\text{def.}}{=} \begin{cases} 0 & \text{if} \quad x \in C, \\ +\infty & \text{otherwise.} \end{cases}$$

18.1.1 Convex Sets and Functions

A set $\Omega \subset \mathcal{H}$ is convex if

$$\forall (x,y,t) \in \Omega^2 \times [0,1], \quad (1-t)x + ty \in \Omega.$$

A function is convex if

$$\forall (x,y,t) \in \mathcal{H}^2 \times [0,1], \quad f((1-t)x+ty) \leqslant (1-t)f(x) + tf(y) \tag{18.2}$$

Equivalently, its epigraph $\{(x,r) \in \mathcal{H}\times\mathbb{R} \,;\, r \geqslant f(x)\}$ is convex. The inequality is interpreted in the extended real line $\bar{\mathbb{R}}$, with the convention $0\cdot(+\infty) = 0$ at the endpoints $t = 0, 1$. The function f is strictly convex if the inequality in (18.2) is strict whenever $x, y \in \mathrm{dom}(f)$ are distinct and $0 < t < 1$. A set Ω is convex if and only if ι_Ω is a convex function.

Throughout the chapter, convex functions f are assumed proper, meaning $\mathrm{dom}(f) \neq \emptyset$, and lower semicontinuous (lsc), meaning that for every $x \in \mathcal{H}$,

$$\liminf_{y\to x} f(y) \geqslant f(x).$$

For a convex function, lower semicontinuity is equivalent to the epigraph $\mathrm{epi}(f)$ being closed. We denote by $\Gamma_0(\mathcal{H})$ the set of proper convex lsc functions.

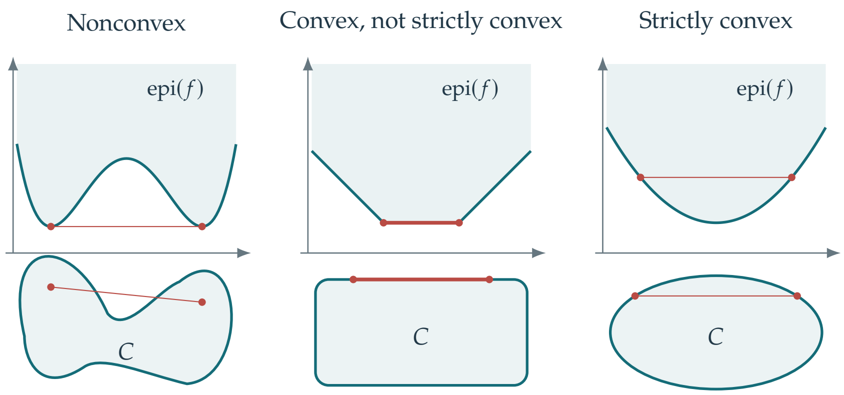


Figure 18.1. Convexity and strict convexity for functions and sets.

18.1.2 First-Order Conditions

Existence of minimizers. We first address existence. Lower semicontinuity and coercivity provide a useful sufficient condition.

Proposition 18.1. *If f is proper, lower semicontinuous, and coercive (i.e. $f(x) \to +\infty$ as $\|x\| \to +\infty$), then there exists a minimizer $x^\star$ of f.*

Proof. Choose $x_0 \in \mathrm{dom}(f)$. The sublevel set $K = \{x : f(x) \leqslant f(x_0)\}$ is nonempty, closed by lower semicontinuity, and bounded by coercivity. It is therefore compact in finite dimensions. A lower semicontinuous function attains its infimum on K, at some $x^\star$. Since $f(x) > f(x_0) \geqslant f(x^\star)$ outside K, this point is a global minimizer. □

This compactness argument is known as the direct method in the calculus of variations. For f in $\Gamma_0(\mathcal{H})$, the minimizer set $\operatorname{argmin} f$ is closed and convex, and every local minimizer is global. Strict convexity of f implies that there is at most one minimizer.

Subdifferential. The subdifferential at x of f is

$$\partial f(x) \overset{\text{def.}}{=} \{u \in \mathcal{H}^* \,;\, \forall y \in \mathcal{H}, f(y) \geqslant f(x) + \langle u,\, y - x\rangle\}, \qquad x \in \mathrm{dom}(f).$$

We set $\partial f(x) = \emptyset$ outside $\mathrm{dom}(f)$. Here $\mathcal{H}^* = \mathbb{R}^N$ denotes the space of dual vectors. The inner product identifies the dual space with $\mathcal{H}$, but distinguishing primal and dual variables remains useful. The duality pairing itself is canonical; the identification of a linear functional with a vector depends on the inner product. The subdifferential $\partial f(x)$ consists of the slopes u of supporting affine functions $f(x) + \langle u,\, z - x\rangle$ that lie below f.

At an interior point of $\mathrm{dom}(f)$, the function f is differentiable exactly when its subdifferential consists of the gradient alone:

$$\partial f(x) = \{\nabla f(x)\}.$$

Multiple elements of $\partial f(x)$ indicate distinct supporting slopes of f at x.

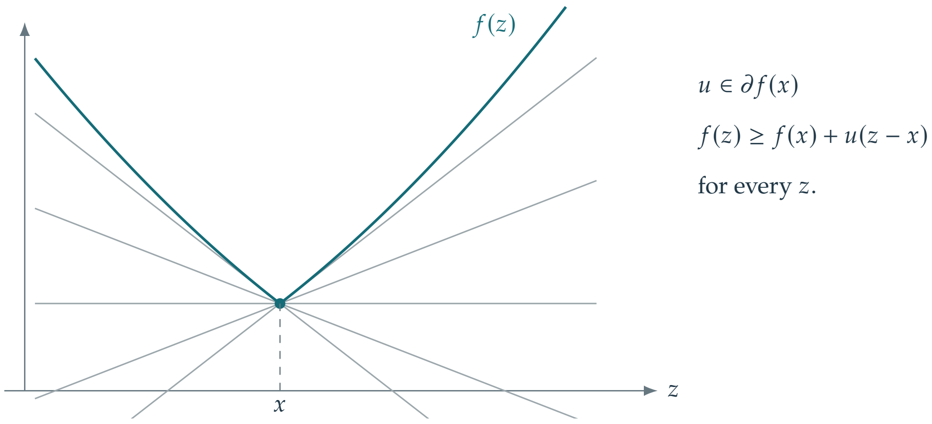
The subdifferential can also be empty at a boundary point of the effective domain. For example, $f(x) = -\sqrt{1 - x^2}$ on $[-1, 1]$, extended by $+\infty$ outside this interval, has no subgradient at $x = \pm 1$.

Since $\partial f(x) \subset \mathcal{H}^*$ is an intersection of half-spaces, it is a closed convex set. Thus $\partial f : \mathcal{H} \mapsto 2^{\mathcal{H}^*}$ is a set-valued operator, also written $\partial f : \mathcal{H} \hookrightarrow \mathcal{H}^*$.

Remark 18.2 (Maximally monotone operator). The subdifferential ∂f is monotone: setting $U = \partial f$ gives

$$\forall\, (u, v) \in U(x) \times U(y), \quad \langle y - x,\, v - u\rangle \geqslant 0.$$

For single-valued maps on the real line, monotonicity means being nondecreasing. Subdifferentials can also be shown to be maximally monotone, meaning that their graphs are not properly contained in the graph of another monotone operator. Conversely, a monotone map need not be a subdifferential; an example is $(x, y) \mapsto (-y, x)$. Many results extend from subdifferentials to arbitrary maximally monotone operators. We restrict attention to subdifferentials here.



Several affine functions support the graph at this kink.

Figure 18.2. The subdifferential

For the absolute value $f(x) = |x|$, write $\partial f(x) = \partial|\cdot|(x)$. Its subdifferential is

$$\partial|\cdot|(x) = \begin{cases} \{-1\} & \text{if} \quad x < 0, \\ \{1\} & \text{if} \quad x > 0, \\ [-1, 1] & \text{if} \quad x = 0. \end{cases}$$

First-order conditions. The subdifferential gives a necessary and sufficient optimality condition.

Proposition 18.3. *$x^\star$ is a minimizer of f if and only if $0 \in \partial f(x^\star)$.*

Proof. One has

$$x^\star \in \operatorname{argmin} f \quad \Leftrightarrow \quad \big(\forall y, f(x^\star) \leqslant f(y) + \langle 0,\, x^\star - y\rangle\big) \quad \Leftrightarrow \quad 0 \in \partial f(x^\star).$$

□

Subdifferential calculus. Calculus rules simplify subdifferential computations. For a separable function $f(x_1, \dots, x_K) = \sum_{k=1}^K f_k(x_k)$, the subdifferential is the Cartesian product

$$\partial f(x_1, \dots, x_K) = \partial f_1(x_1) \times \dots \times \partial f_K(x_K).$$

Applying this rule to the ℓ^1 norm $\|x\|_1 = \sum_{k=1}^N |x_k|$ gives

$$\partial\|\cdot\|_1(x) = \prod_{k=1}^N \partial|\cdot|(x_k)$$

a hyperrectangle. With $I = \operatorname{supp}(x)$, the condition $u \in \partial\|\cdot\|_1(x)$ is equivalent to

$$u_I = \operatorname{sign}(x_I) \quad \text{and} \quad \|u_{I^c}\|_\infty \leqslant 1.$$

A sum rule requires a domain qualification. If one function is finite and continuous at a point in the domain of the other, then for every $x \in \operatorname{dom}(f) \cap \operatorname{dom}(g)$,

$$\partial(f+g)(x) = \partial f(x) \oplus \partial g(x) = \{u + v \;;\; (u, v) \in \partial f(x) \times \partial g(x)\}$$

where $\oplus$ denotes the Minkowski sum. In particular, if f is differentiable at x, then

$$\partial(f+g)(x) = \nabla f(x) + \partial g(x) = \{\nabla f(x) + v \;;\; v \in \partial g(x)\}\,.$$

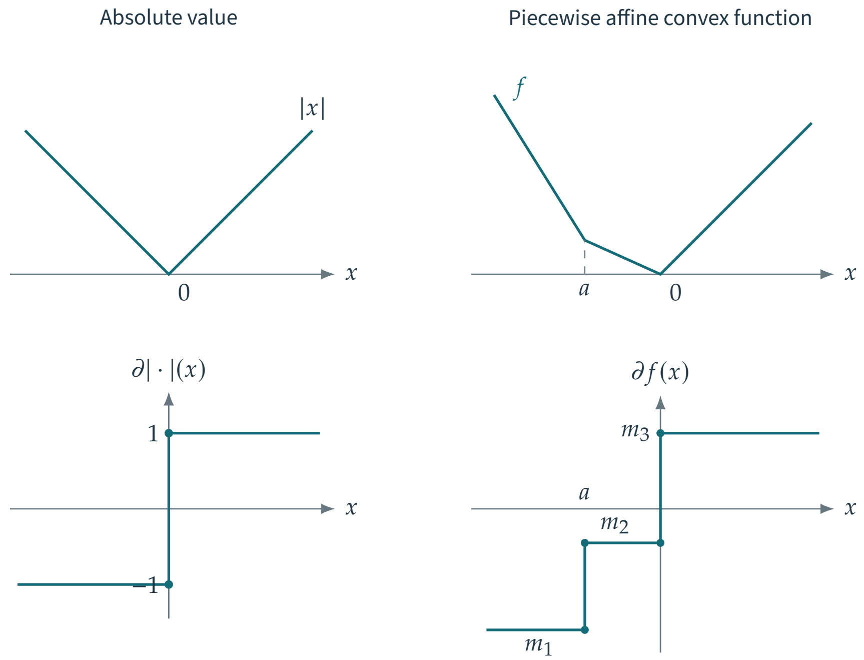


Figure 18.3. Subdifferential of the absolute value and a piecewise affine convex function.

Positive scaling gives

$$\forall\, \lambda > 0, \quad \partial(\lambda f)(x) = \lambda(\partial f(x)).$$

General compositions need not preserve convexity, so no unrestricted chain rule is available. Composition with a linear map does preserve convexity. For $A \in \mathbb{R}^{P\times N}$ and $f \in \Gamma_0(\mathbb{R}^P)$, assume $\operatorname{Im}(A) \cap \operatorname{relint}(\operatorname{dom}(f)) \neq \emptyset$. Then $f \circ A \in \Gamma_0(\mathbb{R}^N)$ and

$$\partial(f \circ A)(x) = A^*(\partial f)(Ax) \overset{\text{def.}}{=} \{A^* u \;;\; u \in \partial f(Ax)\}\,.$$

Remark 18.4 (Application to the Lasso). For $F(x) = \frac{1}{2}\|Ax - y\|^2 + \lambda\|x\|_1$ with $\lambda > 0$, the sum rule gives

$$0 \in A^*(Ax - y) + \lambda\partial\|\cdot\|_1(x).$$

Writing $r = y - Ax$, optimality is equivalent to $(A^* r)_i = \lambda \operatorname{sign}(x_i)$ on $\operatorname{supp}(x)$ and $|(A^* r)_i| \leqslant \lambda$ elsewhere. These conditions are useful both for analysis and for checking a numerical solution.

Normal cone. For a nonempty closed convex set C, the subdifferential of its indicator is the normal cone:

$$\forall\, x \in C, \quad \partial\iota_C(x) = \mathcal{N}_C(x) \overset{\text{def.}}{=} \{v \;;\; \forall\, z \in C, \langle z - x,\, v\rangle \leqslant 0\}\,.$$

Outside the set, $x \notin C$ implies $\partial\iota_C(x) = \emptyset$. For an affine space $C = a + \mathcal{V}$ with direction space $\mathcal{V} \subset \mathcal{H}$, the normal cone at each of its points is $\mathcal{N}_C(x) = \mathcal{V}^\perp$, the orthogonal complement of $\mathcal{V}$. More generally, at an interior point $x \in \operatorname{int}(C)$ of C, we have $\mathcal{N}_C(x) = \{0\}$. At a boundary point, the normal cone describes the supporting directions of C at x.

The normal cone expresses first-order optimality for the constrained problem

$$\min_{x\in C} f(x)$$

If f is finite and continuous at a point of C, the sum rule gives the optimality condition

$$0 \in \partial f(x) + \partial\iota_C(x) \quad \Leftrightarrow \quad \exists \xi \in \partial f(x), -\xi \in \mathcal{N}_C(x) \quad \Leftrightarrow \quad \partial f(x) \cap (-\mathcal{N}_C(x)) \neq \emptyset.$$

If f is differentiable, it reads $-\nabla f(x) \in \mathcal{N}_C(x)$.

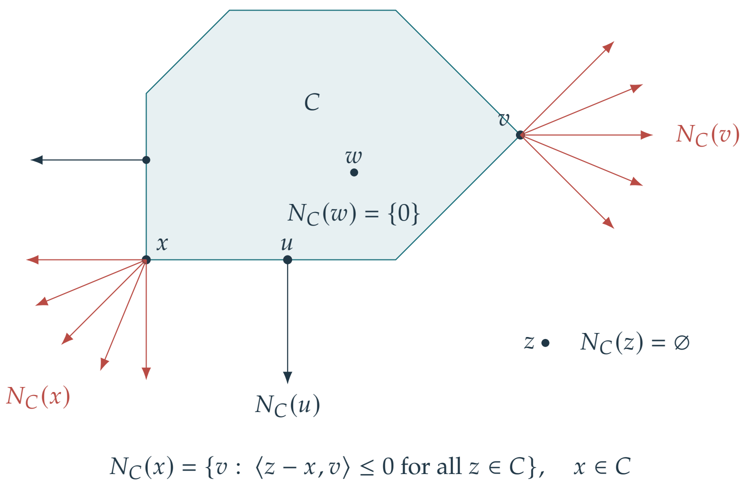


Figure 18.4. Normal cones

18.2 Legendre–Fenchel Transform

The Legendre–Fenchel transform provides a dual representation of a convex function. It makes many Lagrange-duality calculations systematic by expressing inner minimizations through conjugates.

18.2.1 Legendre Transform

For $f \in \Gamma_0(\mathcal{H})$, we define its Legendre–Fenchel transform, or convex conjugate, by

$$f^*(u) \overset{\text{def.}}{=} \sup_x \langle x,\, u\rangle - f(x). \tag{18.3}$$

As a supremum of continuous affine functions, f^* is convex and lower semicontinuous; in fact, $f^\star \in \Gamma_0(\mathcal{H}^*)$. The following biconjugacy theorem recovers the original function.

Theorem 18.5. *One has*

$$\forall\, f \in \Gamma_0(\mathcal{H}), \quad (f^*)^* = f.$$

Even when f is not convex, f^* is convex. If f admits an affine minorant, f^{**} is its lower semicontinuous convex envelope: the largest lower semicontinuous convex function below f.

The subdifferentials of f and f^* satisfy the following inverse relation.

Proposition 18.6. *We have $\partial f^* = (\partial f)^{-1}$, where the inverse of a set-valued map is defined in* (19.15). *The Fenchel–Young inequality states that*

$$\forall\,(x, y), \quad \langle x,\, y\rangle \leqslant f(x) + f^*(y).$$

Equality is characterized by

$$\langle x,\, y\rangle = f(x) + f^*(y) \quad \Leftrightarrow \quad x \in \partial f^*(y) \quad \Leftrightarrow \quad y \in \partial f(x).$$

Proposition 18.7. *For $1 \leqslant p, q \leqslant \infty$ with $1/p + 1/q = 1$,*

$$(\iota_{\|\cdot\|_p \leqslant 1})^* = \|\cdot\|_q \quad \textit{and} \quad (\|\cdot\|_q)^* = \iota_{\|\cdot\|_p \leqslant 1}$$

Here are useful examples and transformation rules.

Proposition 18.8. *If A is symmetric positive definite and $f(x) = \frac{1}{2}\langle Ax,\, x\rangle - \langle b,\, x\rangle$, then*

$$f^*(u) = \frac{1}{2}\langle A^{-1}(u + b),\, u + b\rangle.$$

In particular, $f = \|\cdot\|^2/2$ satisfies $f^* = f$. For $\lambda > 0$,

$$[f(\cdot - z)]^* = f^* + \langle z,\, \cdot\rangle, \qquad [f + \langle z,\, \cdot\rangle]^* = f^*(\cdot - z), \qquad (\lambda f)^* = \lambda f^*(\cdot/\lambda).$$

Proof. The maximizer in $f^*(u) = \sup_x\{\langle u+b,\, x\rangle - \frac{1}{2}\langle Ax,\, x\rangle\}$ satisfies $Ax = u+b$. Substitution of $x = A^{-1}(u+b)$ gives the formula. The translation and scaling identities follow directly by changing variables in (18.3). ◻

18.2.2 Legendre Transform and Smoothness

The Legendre–Fenchel transform exchanges smoothness and strong convexity. It inverts subdifferentials; when f is twice continuously differentiable with positive definite Hessian and $y = \nabla f(x)$, the inverse function theorem yields

$$\partial^2 f^*(y) = (\partial^2 f(x))^{-1}.$$

The next statement gives a version of this correspondence that does not require second derivatives.

Proposition 18.9. *For $f \in \Gamma_0(\mathcal{H})$ and $L > 0$,*

$$f \text{ is differentiable on } \mathcal{H} \text{ with } L\text{-Lipschitz gradient} \quad \Longleftrightarrow \quad f^* \text{ is } 1/L\text{-strongly convex}.$$

Here strong convexity means that $f^* - \|\cdot\|^2/(2L)$ is convex, including when f^* is extended-valued.

This suggests a useful smoothing operation. The infimal convolution is

$$(f \otimes g)(x) \stackrel{\text{def.}}{=} \inf_{y+y'=x} \{f(y) + g(y')\}.$$

For proper convex functions, whenever this infimum defines a proper function, it is convex and satisfies

$$(f \otimes g)^* = f^* + g^*.$$

Moreover, $(f+g)^* = f^* \otimes g^*$ under the qualification $\operatorname{relint}(\operatorname{dom}(f)) \cap \operatorname{relint}(\operatorname{dom}(g)) \neq \emptyset$; without a qualification, a lower semicontinuous closure can be necessary.

For $\mu > 0$, the Moreau–Yosida regularization of $f \in \Gamma_0(\mathcal{H})$ is

$$f_\mu \stackrel{\text{def.}}{=} f \otimes \left(\frac{1}{2\mu}\|\cdot\|^2\right) = \left(f^* + \frac{\mu}{2}\|\cdot\|^2\right)^*. \tag{18.4}$$

The function inside the conjugate is μ-strongly convex, so f_μ has a $1/\mu$-Lipschitz gradient.

For the absolute value, the Moreau–Yosida regularization is

$$(|\cdot|_\mu)(x) = \begin{cases} \frac{1}{2\mu}x^2 & \text{if} \quad |x| \leqslant \mu, \\ |x| - \frac{\mu}{2} & \text{if} \quad |x| > \mu. \end{cases}$$

Compare this with the earlier regularization $\sqrt{x^2 + \mu^2}$, which lies above the absolute value.

18.3 Convex Duality

A dual problem depends on the chosen primal formulation. A change of variables or a different representation of the constraints can produce a different dual problem.

18.3.1 Lagrange Duality: Affine Constraints

Begin with affine equality constraints:

$$\min_{Ax=y} f(x)$$

where $A \in \mathbb{R}^{P\times p}$, $y \in \mathbb{R}^P$, and $f : \mathbb{R}^p \to \mathbb{R}$ is convex and finite everywhere, hence continuous.

Assume that the affine constraint is feasible. Introducing $u \in \mathbb{R}^P$ gives the Lagrangian $\mathcal{L}(x, u) = f(x) + \langle Ax - y,\, u\rangle$. Minimization over x yields

$$F(u) = -f^*(-A^*u) - \langle y,\, u\rangle.$$

Because f is finite and continuous everywhere, feasibility is a sufficient qualification for strong duality:

$$\inf_{Ax=y} f(x) = \sup_{u\in\mathbb{R}^P} \{-f^*(-A^*u) - \langle y,\, u\rangle\}.$$

A primal-dual optimal pair satisfies $Ax^\star = y$ and $-A^*u^\star \in \partial f(x^\star)$.

18.3.2 Lagrange Duality: General Case

We consider a minimization of the form

$$p^\star = \inf_{x \in \mathbb{R}^N} \{f(x)\,;\; Ax = y \quad \text{and} \quad g(x) \leqslant 0\} \tag{18.5}$$

for a continuous convex function $f : \mathcal{H} \to \mathbb{R}$, a matrix $A \in \mathbb{R}^{P \times N}$ and a function $g : \mathcal{H} \to \mathbb{R}^Q$ such that each coordinate $g_i : \mathcal{H} \to \mathbb{R}$ is continuous and convex. In the standard formulation of a convex program, equality constraints are affine. Problems with convex inequality constraints can be written in the form (18.5). The representation is not unique, and different formulations lead to different dual problems.

For simplicity, f is assumed finite and continuous on its full domain $\text{dom}(f) = \mathbb{R}^N$. The theory extends to more general $\text{dom}(f)$, with additional domain qualifications. Here all constraints are expressed through $Ax = y$ and $g(x) \leqslant 0$.

Conic duality extends this construction from nonnegativity $x \geqslant 0$ to positive semidefiniteness $X \succeq 0$ of a matrix X, and more generally to membership in a convex cone.

We use the following fact

$$\sup_{u \in \mathbb{R}^P} \langle r,\, u\rangle = \begin{cases} 0 & \text{if} \quad r = 0, \\ +\infty & \text{if} \quad r \neq 0, \end{cases} \qquad \text{and} \quad \sup_{v \in \mathbb{R}^Q_+} \langle s,\, v\rangle = \begin{cases} 0 & \text{if} \quad s \leqslant 0, \\ +\infty & \text{otherwise}, \end{cases}$$

to encode the constraints $r = Ax - y = 0$ and $s = g(x) \leqslant 0$.

Define the Lagrangian by

$$\mathcal{L}(x, u, v) \stackrel{\text{def.}}{=} f(x) + \langle Ax - y,\, u\rangle + \langle g(x),\, v\rangle.$$

The primal value in (18.5) is therefore

$$p^\star = \inf_x \sup_{u \in \mathbb{R}^P, v \in \mathbb{R}^Q_+} \mathcal{L}(x, u, v).$$

The dual objective is $F(u, v) \stackrel{\text{def.}}{=} \inf_x \mathcal{L}(x, u, v)$. Exchanging the order of optimization gives the dual value

$$d^\star = \sup_{(u,v) \in \mathbb{R}^P \times \mathbb{R}^Q_+} F(u, v). \tag{18.6}$$

The dual objective F is concave because it is an infimum of affine functions. The dual problem therefore maximizes a concave objective.

Weak duality states that every feasible dual value is a lower bound on the primal optimum.

Proposition 18.10. *One always has, for all $(u, v) \in \mathbb{R}^P \times \mathbb{R}^Q_+$,*

$$F(u, v) \leqslant p^\star \quad \Longrightarrow \quad d^\star \leqslant p^\star.$$

Proof. Since $g(x) \leqslant 0$ and $v \geqslant 0$, one has $\langle g(x),\, v\rangle \leqslant 0$, and since $Ax = y$, one has $\langle Ax - y,\, u\rangle = 0$, so that

$$F(u, v) \leqslant \mathcal{L}(x, u, v) \leqslant f(x) \quad \text{for every feasible } x.$$

Taking the infimum over feasible x gives $F(u, v) \leqslant p^\star$, and taking the supremum over feasible multipliers yields $d^\star \leqslant p^\star$. □

The next theorem gives a constraint qualification that guarantees equality of the primal and dual values.

Theorem 18.11. *If*

$$\exists x_0 \in \mathbb{R}^N, \quad Ax_0 = y \quad \textit{and} \quad g(x_0) < 0, \tag{18.7}$$

then $p^\star = d^\star$. If $p^\star$ is finite, the dual supremum is attained. Furthermore, $x^\star$ and $(u^\star, v^\star)$ solve (18.5) and (18.6), respectively, if and only if

$$Ax^\star = y, \quad g(x^\star) \leqslant 0, \quad v^\star \geqslant 0 \tag{18.8}$$

$$0 \in \partial f(x^\star) + A^* u^\star + \sum_i v_i^\star \partial g_i(x^\star) \tag{18.9}$$

$$\forall\, i, \quad v_i^\star g_i(x^\star) = 0 \tag{18.10}$$

The existence of x_0 is Slater's constraint qualification; the strict inequality $g(x_0) < 0$ holds componentwise. Other, weaker qualifications are possible.

Condition (18.8) states primal and dual feasibility. Condition (18.9) expresses optimality of $\mathcal{L}(x, u, v)$ with respect to x. Together with feasibility, condition (18.10) expresses maximality of $\mathcal{L}(x^\star, u, v)$ over the admissible multipliers (u, v). Together these are the Karush–Kuhn–Tucker (KKT) conditions. Under the stated qualification, they are necessary and sufficient for primal-dual optimality.

Complementary slackness, $v_i^\star g_i(x^\star) = 0$, relates a multiplier to its constraint: if $g_i(x^\star) < 0$, the constraint is inactive and $v_i^\star = 0$; if $v_i^\star > 0$, the constraint must be active, $g_i(x^\star) = 0$.

For an affine inequality $g_i(x) = \langle x,\, h_i\rangle - c_i \leqslant 0$, strict feasibility $g_i(x_0) < 0$ may be replaced by ordinary feasibility $\langle x_0,\, h_i\rangle \leqslant c_i$ in the qualification.

If $\mathrm{dom}(f)$ is a proper subset of $\mathbb{R}^N$, the objective itself encodes constraints in addition to those written with $\leqslant$. The qualification then also requires $x_0 \in \mathrm{relint}(\mathrm{dom}(f))$.

Theorem 18.11 extends the Lagrange-multiplier conditions to convex problems with inequalities. Inequality constraints introduce nonnegative multipliers v. Convexity ensures that the KKT conditions are sufficient, and the stated constraint qualification ensures their necessity.

As an example, consider projection onto a nonempty affine space $\{x : Ax = y\}$:

$$p^\star = \min_{Ax=y} \frac{1}{2}\|x-z\|^2 = \max_u F(u), \qquad F(u) \overset{\text{def.}}{=} \min_x \left\{\frac{1}{2}\|x-z\|^2 + \langle Ax-y,\, u\rangle\right\}.$$

The minimizing x satisfies $x - z + A^* u = 0$, hence $x = z - A^* u$. Therefore

$$F(u) = -\frac{1}{2}\|A^* u\|^2 + \langle u,\, Az - y\rangle.$$

An optimal multiplier solves $AA^* u^\star = Az - y$. Since $y \in \mathrm{Im}(A)$, a solution is $u^\star = (AA^*)^+(Az - y)$, where $^+$ denotes the Moore–Penrose inverse. If A has full row rank, $(AA^*)^+ = (AA^*)^{-1}$. The unique projection is

$$x^\star = \mathrm{Proj}_{A\cdot=y}(z) = z - A^+(Az - y) = (\mathrm{Id} - A^+A)z + A^+ y. \tag{18.11}$$

A further example, the Lasso, is treated in Section 11.1.5.

18.3.3 Fenchel–Rockafellar Duality

Conjugates often give explicit expressions for the Lagrange dual of an objective f. Fenchel–Rockafellar duality is a particularly useful example.

We consider the following structured minimization problem

$$p^\star = \inf_x f(x) + g(Ax). \tag{18.12}$$

Introducing an auxiliary variable gives

$$\inf_{y=Ax} f(x) + g(y).$$

We can then form the primal-dual problem

$$\inf_{(x,y)} \sup_u f(x) + g(y) + \langle Ax - y,\, u\rangle. \tag{18.13}$$

Under the domain qualification in Theorem 18.12, one can exchange the infimum and supremum to obtain the dual problem

$$d^\star = \sup_u \inf_{(x,y)} f(x) + g(y) + \langle Ax - y,\, u\rangle \tag{18.14}$$

$$= \sup_u \left\{\left(\inf_x \langle x,\, A^* u\rangle + f(x)\right) + \left(\inf_y -\langle y,\, u\rangle + g(y)\right)\right\}. \tag{18.15}$$

This gives the Fenchel–Rockafellar dual problem. The following theorem states a sufficient domain qualification for strong duality.

Theorem 18.12 (Fenchel–Rockafellar). *Let $f \in \Gamma_0(\mathbb{R}^N)$, $g \in \Gamma_0(\mathbb{R}^P)$, and $A \in \mathbb{R}^{P\times N}$. If*

$$0 \in \operatorname{relint}(\operatorname{dom}(g)) - A \operatorname{relint}(\operatorname{dom}(f)) \tag{18.16}$$

then strong duality holds

$$\inf_x f(x) + g(Ax) = \inf_x \sup_u \mathcal{L}(x,u) = \sup_u \inf_x \mathcal{L}(x,u) = \sup_u \; -f^*(-A^*u) - g^*(u) \tag{18.17}$$

$$\textit{where} \quad \mathcal{L}(x,u) \stackrel{\text{def.}}{=} f(x) + \langle Ax,\, u\rangle - g^*(u).$$

A pair $(x^\star, u^\star)$ is primal-dual optimal if and only if

$$-A^*u^\star \in \partial f(x^\star) \quad \textit{and} \quad Ax^\star \in \partial g^*(u^\star). \tag{18.18}$$

Condition (18.16) is the constraint qualification ensuring that one can exchange the infimum and supremum in (18.17). It is the relative-interior qualification corresponding to the equality-constrained reformulation (18.14). The relations (18.18) express first-order optimality in the primal variable x and dual variable u for the saddle function $\mathcal{L}$. They can be collected in matrix notation as

$$0 \in \begin{pmatrix} \partial f & A^* \\ -A & \partial g^* \end{pmatrix} \begin{pmatrix} x^\star \\ u^\star \end{pmatrix}.$$

CHAPTER 19

Nonsmooth Convex Optimization

Many recovery and learning problems combine smooth losses with constraints or penalties that are not differentiable. Solving them efficiently requires understanding both the cost of an iteration and the conditions for convergence. We begin with subgradient, projection, and barrier methods, then develop proximal maps and splitting algorithms, including ADMM and primal–dual schemes.

The main references for this chapter are [10, 11, 4]. Further background is provided in [25, 3, 2].

We consider a general convex optimization problem

$$\min_{x \in \mathcal{H}} f(x) \tag{19.1}$$

where $\mathcal{H} = \mathbb{R}^p$ is a finite-dimensional Euclidean space, and seek algorithms with a low cost per iteration. The first-order methods below use gradients, subgradients, or proximal maps. Unless stated otherwise, functions are proper, lower semicontinuous, and convex, and a minimizer is assumed to exist.

19.1 Descent Methods

We extend the gradient method of Section 14.4 to nonsmooth objectives and constraints.

19.1.1 Gradient Descent

The optimization problem (9.26) is an unconstrained problem of the form (19.1) with a smooth objective: $f : \mathcal{H} \to \mathbb{R}$ is continuously differentiable with Lipschitz gradient. Here $\nabla f(x) \in \mathcal{H}$ is the gradient of the objective with respect to its variable x. It is distinct from the spatial gradient ∇x of a signal or image, which is a vector field. The functional gradient satisfies the first-order expansion

$$f(x+r) = f(x) + \langle \nabla f(x), r \rangle_{\mathcal{H}} + O(\|r\|_{\mathcal{H}}^2)$$

The Lipschitz-gradient assumption gives the quadratic remainder $O(\|r\|_{\mathcal{H}}^2)$; differentiability alone gives $o(\|r\|_{\mathcal{H}})$. Examples of gradient computations appear in Section 9.5.3.

For such a function, the gradient descent algorithm is defined as

$$x^{(\ell+1)} \overset{\text{def.}}{=} x^{(\ell)} - \tau_\ell \nabla f(x^{(\ell)}). \tag{19.2}$$

The step size $\tau_\ell > 0$ balances stability against progress per iteration.

19.1.2 Subgradient Method

For a nonsmooth objective f, replace the gradient in (19.2) by a subgradient:

$$x^{(\ell+1)} \overset{\text{def.}}{=} x^{(\ell)} - \tau_\ell g^{(\ell)} \quad \text{where} \quad g^{(\ell)} \in \partial f(x^{(\ell)}). \tag{19.3}$$

A subgradient step need not decrease the objective. Convergence generally requires diminishing step sizes, as the example $f(x) = |x|$ illustrates. This can make the method slow. When the objective admits a useful decomposition, the proximal methods developed below can exploit it more effectively.

Theorem 19.1. *Assume f is convex, a minimizer $x^\star$ exists, and the chosen subgradients satisfy $\|g^\ell\| \leqslant G$. If $\sum_\ell \tau_\ell = +\infty$ and $\sum_\ell \tau_\ell^2 < +\infty$, then the iterates converge to a minimizer. More generally,*

$$\min_{0 \leqslant \ell < N} \{f(x^\ell) - f(x^\star)\} \leqslant \frac{\|x^0 - x^\star\|^2 + G^2 \sum_{\ell<N} \tau_\ell^2}{2 \sum_{\ell<N} \tau_\ell}.$$

For a prescribed horizon N and positive bounds D, G, the constant step $\tau_\ell = D/(G\sqrt{N})$ gives a bound $DG/\sqrt{N}$ when $\|x^0 - x^\star\| \leqslant D$. A step $1/(\ell+1)$ satisfies the convergence conditions but does not, by itself, give an $O(N^{-1/2})$ bound.

19.1.3 Projected Gradient Descent

We consider a constrained optimization problem

$$\min_{x\in C} f(x) \tag{19.4}$$

where $C \subset \mathbb{R}^S$ is a nonempty closed convex set and $f : \mathbb{R}^S \to \mathbb{R}$ is convex and continuously differentiable. The convergence theorem below additionally assumes a Lipschitz gradient.

To impose the constraint, follow the gradient step (19.2) by projection:

$$x^{(\ell+1)} \overset{\text{def.}}{=} \mathrm{Proj}_C\left(x^{(\ell)} - \tau_\ell \nabla f(x^{(\ell)})\right). \tag{19.5}$$

Here Proj_C is the orthogonal projection onto C, defined by

$$\mathrm{Proj}_C(x) = \operatorname*{argmin}_{x'\in C} \|x - x'\|$$

The projection is unique because C is nonempty, closed, and convex. Projected gradient retains analogous convergence guarantees under the following assumptions.

Theorem 19.2. *Assume f is convex with L-Lipschitz gradient, C is nonempty, closed, and convex, and a constrained minimizer exists. For a fixed $0 < \tau < 2/L$, projected gradient descent converges to a constrained minimizer. If $0 < \tau \leqslant 1/L$, then for $\ell \geqslant 1$,*

$$f(x^\ell) - f(x^\star) \leqslant \frac{\|x^0 - x^\star\|^2}{2\tau\ell}.$$

If f is μ-strongly convex, the iterates converge linearly.

Proof. This is the special case $g = \iota_C$ of Theorem 19.14. In the strongly convex case, the gradient step is a contraction for the admissible fixed step sizes, and composition with the nonexpansive projector preserves that property. □

The main practical difficulty in (19.5) is evaluating the projection. For several sets arising in ℓ^1 minimization, it can be computed efficiently.

19.2 Interior Point Methods

We briefly describe interior-point methods, which often require relatively few but expensive iterations. They extend Newton-type methods to problems whose constraints have a suitable barrier representation, such as nonnegativity of vectors or positive definiteness of matrices.

To illustrate the main idea, we consider the following problem

$$\min_{x\in\mathbb{R}^d, Ax\leqslant y} f(x) \tag{19.6}$$

for $A \in \mathbb{R}^{m\times d}$. Related formulations replace the vector Ax by a matrix and the componentwise inequality $\leqslant$ by a positive-semidefinite constraint.

Example 19.3 (Lasso). The Lasso problem (applied to the regression problem $Bw \approx b$ for $B \in \mathbb{R}^{n\times p}$)

$$\min_{w\in\mathbb{R}^p} \frac{1}{2}\|Bw - b\|^2 + \lambda\|w\|_1 \tag{19.7}$$

with $\lambda > 0$ can be recast as (19.6) by introducing nonnegative variables $x = (x_-, x_+) \in \mathbb{R}^{2p}$ (so that $d = 2p$) and $w = x_+ - x_-$ with $(x_+, x_-) \geqslant 0$, so that $f(x) = \frac{1}{2}\|B(x_+ - x_-) - b\|^2 + \lambda\langle x, \mathbb{1}\rangle$, $A = -\mathrm{Id}_{2p}$, and $y = 0$ (so that $m = 2p$).

Example 19.4 (Dual of the Lasso). Alternatively, solve the dual of (19.7):

$$\min_{\|B^\top q\|_\infty\leqslant 1} f(q) = \frac{\lambda}{2}\|q\|^2 - \langle q, b\rangle \tag{19.8}$$

so that one can use $A = \begin{pmatrix} B^\top \\ -B^\top \end{pmatrix}$ and the constraint right-hand side is $\mathbb{1}_{2p}$ (so that $d = n$ and $m = 2p$).

Interior-point methods approximate (19.6) by introducing a logarithmic barrier:

$$\min_{x\in\mathbb{R}^d,\ Ax<y} f_t(x) \stackrel{\text{def.}}{=} f(x) - \frac{1}{t}\operatorname{Log}(y - Ax) \tag{19.9}$$

where

$$\operatorname{Log}(u) \stackrel{\text{def.}}{=} \sum_i \log(u_i)$$

The function $-\operatorname{Log}$ is strictly convex and diverges at the boundary of the positive orthant. Assuming strict feasibility and existence of central-path minimizers $x(t)$, the objective values $f(x(t))$ approach the original optimal value as $t \longrightarrow +\infty$.

For a fixed t, assume f_t is twice continuously differentiable with positive definite Hessian along the iterates. Apply Newton's method to (19.9), using a line search such as Armijo backtracking (14.14) to choose $0 < \tau_\ell \leqslant 1$ while maintaining $Ax^{(\ell)} < y$:

$$x^{(\ell+1)} \stackrel{\text{def.}}{=} x^{(\ell)} - \tau_\ell[\partial^2 f_t(x^{(\ell)})]^{-1}\nabla f_t(x^{(\ell)}) \tag{19.10}$$

The gradient and Hessian are

$$\nabla f_t(x) = \nabla f(x) + \frac{1}{t}A^\top \frac{1}{y - Ax} \quad \text{and} \quad \partial^2 f_t(x) = \partial^2 f(x) + \frac{1}{t}A^\top \operatorname{diag}\left(\frac{1}{(y - Ax)^2}\right) A.$$

A natural stopping criterion for the inner Newton iterations is

$$\langle[\partial^2 f_t(x^{(\ell)})]^{-1}\nabla f_t(x^{(\ell)}),\ \nabla f_t(x^{(\ell)})\rangle < \frac{\varepsilon}{2}.$$

The barrier method approximately minimizes f_t through (19.10), tracing the central path $t \mapsto x(t)$ for increasing parameters $t = t_k = \mu^k t_0$ with $\mu > 1$. A warm start makes this continuation strategy efficient: initialize the Newton iterations (19.10) for $x(t_k)$ at the preceding solution $x(t_{k-1})$. Along the central path, the logarithmic barrier gives $f(x(t_k)) - f(x^\star) \leqslant m/t_k$, where m is the number of scalar constraints. To reach error ε, take $k = 0, \dots, K$ with

$$\frac{m}{t_K} = \frac{m}{t_0\mu^K} \leqslant \varepsilon.$$

Thus $O(|\log(\varepsilon)|)$ barrier-parameter updates suffice for accuracy ε. The total cost also depends on the number of inner Newton iterations (19.10), whose control requires further assumptions on f.

The relevant complexity theory uses self-concordant barriers. A convex function φ on a line is self-concordant when

$$|\varphi^{(3)}(s)| \leqslant 2\varphi''(s)^{3/2}.$$

The logarithmic barrier has this property. For a linear objective, the appropriately scaled objective $tf - \operatorname{Log}(y - Ax)$ is self-concordant; arbitrary positive rescaling does not preserve the same self-concordance constant. With a suitable path-following neighborhood and barrier-parameter schedule, the number of Newton steps can be bounded polynomially in the problem parameters and logarithmically in $1/\varepsilon$. The bound depends on the barrier parameter (equal to m for this logarithmic barrier), not only on the requested accuracy. A fixed number of Newton steps per outer iteration cannot be asserted for an arbitrary update factor $\mu > 1$.

19.3 Proximal Algorithm

For a nonsmooth objective f, an explicit gradient step may be undefined. An implicit step remains available even when f is nonsmooth.

19.3.1 Proximal Map

For a step size $\tau > 0$, define the proximal map by

$$\forall\, x, \quad \operatorname{Prox}_{\tau f}(x) \stackrel{\text{def.}}{=} \operatorname*{argmin}_{x'} \frac{1}{2}\|x - x'\|^2 + \tau f(x'). \tag{19.11}$$

It balances a decrease in f against a quadratic penalty for moving away from x. Since $f \in \Gamma_0(\mathcal{H})$, the objective $\frac{1}{2}\|x - \cdot\|^2 + \tau f$ is strongly convex and coercive, so $\operatorname{Prox}_{\tau f}$ is well defined and single-valued.

For an indicator $f = \iota_C$, the proximal map is the projection $\operatorname{Prox}_{\iota_C} = \operatorname{Proj}_C$. It therefore generalizes projection to convex functions. For each fixed input, the proximal point is also the projection onto a suitable sublevel set of f. Like an orthogonal projection, the proximal map is nonexpansive.

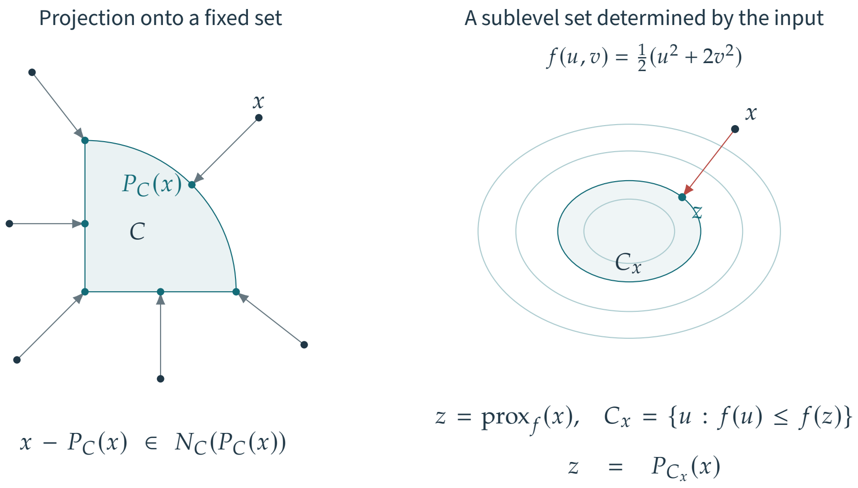


Figure 19.1. Proximal map and projection map.

Proposition 19.5. *One has* $\|\mathrm{prox}_f(x) - \mathrm{prox}_f(y)\| \leqslant \|x - y\|$.

Examples Several common proximal maps have closed forms.

Proposition 19.6. *One has*

$$\mathrm{Prox}_{\frac{\tau}{2}\|\cdot\|^2}(x) = \frac{x}{1+\tau}, \quad \textit{and} \quad \mathrm{Prox}_{\tau\|\cdot\|_1}(x) = \mathcal{S}^1_\tau(x), \tag{19.12}$$

where the soft-thresholding is defined as

$$\mathcal{S}^1_\tau(x) \stackrel{\text{def.}}{=} (S_\tau(x_i))_{i=1}^p \quad \textit{where} \quad S_\tau(r) \stackrel{\text{def.}}{=} \mathrm{sign}(r)(|r| - \tau)_+,$$

(see also (10.5)). For $A \in \mathbb{R}^{P\times N}$*, one has*

$$\mathrm{Prox}_{\frac{\tau}{2}\|A\cdot - y\|^2}(x) = (\mathrm{Id}_N + \tau A^*A)^{-1}(x + \tau A^* y). \tag{19.13}$$

Proof. The proximal map of $\|\cdot\|_1$ was derived informally in Proposition 10.1. For a rigorous proof, use separability to reduce to a scalar input r. The optimality condition for its proximal point z is $0 \in z - r + \tau\partial|\cdot|(z)$. The candidate $z = 0$ is optimal precisely when $0 \in -r + [-\tau, \tau]$, or equivalently $r \in [-\tau, \tau]$. For $z \neq 0$, optimality requires $z = r - \tau\,\mathrm{sign}(z)$. When $r \notin [-\tau, \tau]$, the candidate $z = r - \tau\,\mathrm{sign}(r)$ has the same sign as r, so it is the unique minimizer. For the quadratic case

$$z = \mathrm{Prox}_{\frac{\tau}{2}\|A\cdot - y\|^2}(x) \quad \Leftrightarrow \quad z - x + \tau A^*(Az - y) = 0 \quad \Leftrightarrow \quad (\mathrm{Id}_N + \tau A^*A)z = x + \tau A^* y.$$

□

For some nonconvex functions, the proximal minimization also has solutions, but they may be nonunique. For example, $\mathrm{Prox}_{\tau\|\cdot\|_0}$ is hard thresholding at $\sqrt{2\tau}$; see Proposition 10.1. At $|x_i| = \sqrt{2\tau}$, both 0 and x_i are minimizers.

19.3.2 Basic Properties

The following identities simplify proximal computations.

Proposition 19.7. *One has*

$$\mathrm{Prox}_{f+\langle y,\cdot\rangle} = \mathrm{Prox}_f(\cdot - y), \quad \mathrm{Prox}_{f(\cdot - y)} = y + \mathrm{Prox}_f(\cdot - y).$$

If $f(x) = \sum_{k=1}^K f_k(x_k)$ *for* $x = (x_1, \ldots, x_K)$ *is separable, then*

$$\mathrm{Prox}_{\tau f}(x) = (\mathrm{Prox}_{\tau f_k}(x_k))_{k=1}^K. \tag{19.14}$$

Proof. The optimality condition for $z = \operatorname{Prox}_{f+\langle y,\cdot\rangle}(x)$ is $0 \in z - x + \partial f(z) + y$, or $x - y \in z + \partial f(z)$. Thus $z = \operatorname{Prox}_f(x-y)$. For the translated function, substitute $w = z - y$ in $\min_z\{\frac{1}{2}\|z-x\|^2 + f(z-y)\}$ to obtain $z = y + \operatorname{Prox}_f(x-y)$. Separability follows by minimizing each coordinate block independently. □

Composition with a linear map having orthonormal rows admits an explicit formula.

Proposition 19.8. *If $A \in \mathbb{R}^{P\times N}$ satisfies $AA^* = \mathrm{Id}_P$, then*

$$\operatorname{Prox}_{f\circ A} = A^* \circ \operatorname{Prox}_f \circ A + \mathrm{Id}_N - A^*A.$$

In particular, if A is orthogonal, then $\operatorname{Prox}_{f\circ A} = A^* \circ \operatorname{Prox}_f \circ A$.

19.3.3 Related Concepts

Link with the subdifferential. For a set-valued map $U : \mathcal{H} \hookrightarrow \mathcal{G}$, we define the inverse set-valued map $U^{-1} : \mathcal{G} \hookrightarrow \mathcal{H}$ by

$$h \in U^{-1}(g) \quad\Longleftrightarrow\quad g \in U(h) \tag{19.15}$$

The proximal map is the resolvent of the subdifferential.

Proposition 19.9. *One has* $\operatorname{Prox}_{\tau f} = (\mathrm{Id} + \tau\partial f)^{-1}$.

Proof. The optimality condition gives the equivalences

$$z = \operatorname{Prox}_{\tau f}(x) \Leftrightarrow 0 \in z - x + \tau\partial f(z) \Leftrightarrow x \in (\mathrm{Id} + \tau\partial f)(z) \Leftrightarrow z = (\mathrm{Id} + \tau\partial f)^{-1}(x)$$

The final relation is an equality because the inverse is single-valued here. □

This resolvent interpretation connects proximal algorithms with the theory of maximally monotone operators.

Link with duality. Moreau decomposition relates the proximal maps of a function and its conjugate.

Theorem 19.10 (Moreau decomposition)**.** *One has*

$$\operatorname{Prox}_{\tau f} = \mathrm{Id} - \tau \operatorname{Prox}_{f^*/\tau}(\cdot/\tau).$$

Thus a tractable proximal map for f gives one for f^*, and conversely. For example, Proposition 18.7 gives the following expression for soft thresholding

$$\operatorname{Prox}_{\tau\|\cdot\|_1}(x) = x - \tau \operatorname{Proj}_{\|\cdot\|_\infty \leqslant 1}(x/\tau) = x - \operatorname{Proj}_{\|\cdot\|_\infty \leqslant \tau}(x) \quad \text{where} \quad \operatorname{Proj}_{\|\cdot\|_\infty \leqslant \tau}(x) = \min(\max(x, -\tau), \tau).$$

The minimum and maximum in this clipping formula act componentwise.

For an indicator $f = \iota_C$ of a closed convex cone C,

$$(\iota_C)^* = \iota_{C^\circ} \quad \text{where} \quad C^\circ \stackrel{\text{def.}}{=} \{y \;;\; \forall\, x \in C, \langle x,\, y\rangle \leqslant 0\} \tag{19.16}$$

where C° is the polar cone. This conjugacy relation (19.16) is specific to cones: for a general convex set C, the conjugate of its indicator need not be an indicator. Equation (19.16) yields Moreau polar decomposition:

$$x = \operatorname{Proj}_C(x) +^\perp \operatorname{Proj}_{C^\circ}(x)$$

The notation $+^\perp$ indicates that the two summands are orthogonal. For a linear subspace $C = V$, this reduces to the orthogonal decomposition $\mathbb{R}^p = V \oplus^\perp V^\perp$.

Link with Moreau–Yosida regularization. A proximal step is also a gradient step on the Moreau–Yosida envelope f_μ of f from (18.4).

Proposition 19.11. *One has*

$$\operatorname{Prox}_{\mu f} = \mathrm{Id} - \mu\nabla f_\mu.$$

19.4 Proximal Gradient Algorithms

Splitting algorithms exploit a decomposition of the objective into terms whose proximal maps can be computed separately. Different decompositions of the same objective produce different splitting algorithms. A useful decomposition balances the cost of the proximal maps against the convergence of the resulting iteration. Step sizes and other parameters also need to be chosen; theoretical convergence bounds and line-search strategies can guide their selection.

19.4.1 Proximal Point Algorithm

One has the following equivalence

$$x^\star \in \operatorname{argmin} f \quad \Leftrightarrow \quad 0 \in \partial f(x^\star) \quad \Leftrightarrow \quad x^\star \in (\mathrm{Id} + \tau\partial f)(x^\star) \tag{19.17}$$

$$\Leftrightarrow \quad x^\star = (\mathrm{Id} + \tau\partial f)^{-1}(x^\star) = \mathrm{Prox}_{\tau f}(x^\star). \tag{19.18}$$

This shows that being a minimizer of f is equivalent to being a fixed point of $\mathrm{Prox}_{\tau f}$. This fixed-point characterization suggests the proximal point iterations

$$x^{(\ell+1)} \stackrel{\text{def.}}{=} \mathrm{Prox}_{\tau_\ell f}(x^{(\ell)}). \tag{19.19}$$

The proximal point method does not require an upper stability bound on the step size. Steps must nevertheless be large enough in aggregate; the following sufficient condition is convenient.

Theorem 19.12. *If $0 < \tau_{\min} \leqslant \tau_\ell \leqslant \tau_{\max} < +\infty$, then $x^{(\ell)} \to x^\star$ for a minimizer of f.*

This implicit step (19.19) should be compared with a gradient descent step (19.2)

$$x^{(\ell+1)} \stackrel{\text{def.}}{=} (\mathrm{Id} - \tau_\ell \nabla f)(x^{(\ell)}).$$

The implicit resolvent $(\mathrm{Id} + \tau_\ell \partial f)^{-1}$ replaces the explicit map $\mathrm{Id} - \tau_\ell \nabla f$. For small τ_ℓ and smooth f, the two agree to first order. The implicit step also remains well defined for nonsmooth objectives and converges under the assumptions above, without the upper step-size restriction of explicit gradient descent. This resembles the greater stability of implicit Euler integration. The computational tradeoff is that a general proximal map may be as difficult to evaluate as the original minimization problem.

19.4.2 Forward–Backward Splitting

For a general objective, evaluating $\mathrm{Prox}_{\gamma f}$ may be as difficult as solving the original problem. A tractable alternative exploits a decomposition into a smooth term and a term with an accessible proximal map:

$$\min_x \mathcal{E}(x) \stackrel{\text{def.}}{=} f(x) + g(x) \tag{19.20}$$

where $g \in \Gamma_0(\mathcal{H})$ can be nonsmooth, whereas f is convex and has a Lipschitz gradient.

For this structure, the fixed-point argument (19.17) becomes

$$x^\star \in \operatorname{argmin} f + g \quad \Leftrightarrow \quad 0 \in \nabla f(x^\star) + \partial g(x^\star) \quad \Leftrightarrow \quad x^\star - \tau \nabla f(x^\star) \in (\mathrm{Id} + \tau \partial g)(x^\star)$$

$$\Leftrightarrow \quad x^\star = (\mathrm{Id} + \tau \partial g)^{-1} \circ (\mathrm{Id} - \tau \nabla f)(x^\star).$$

This fixed point suggests the following algorithm, known as forward–backward splitting

$$x^{(\ell+1)} \stackrel{\text{def.}}{=} \mathrm{Prox}_{\tau_\ell g}\left(x^{(\ell)} - \tau_\ell \nabla f(x^{(\ell)})\right). \tag{19.21}$$

Derivation using surrogate functionals. A quadratic surrogate gives another derivation of the algorithm and helps explain its convergence.

Replace the objective $\mathcal{E}(x)$ at the current iterate by a surrogate $\mathcal{E}(x, x^{(\ell)})$ that is easier to minimize, and define

$$x^{(\ell+1)} \stackrel{\text{def.}}{=} \operatorname*{argmin}_x \mathcal{E}(x, x^{(\ell)}). \tag{19.22}$$

Require the surrogate to majorize the objective and agree with it at the current point:

$$\mathcal{E}(x) \leqslant \mathcal{E}(x, x') \quad \text{and} \quad \mathcal{E}(x, x) = \mathcal{E}(x) \tag{19.23}$$

and $\mathcal{E}(x) - \mathcal{E}(x, x')$ should be a smooth function. Property (19.23) ensures that $\mathcal{E}$ does not increase along the iterations:

$$\mathcal{E}(x^{(\ell+1)}) \leqslant \mathcal{E}(x^{(\ell)})$$

For proximal-gradient surrogates with a fixed $0 < \tau < 1/L$, sufficient decrease and the optimality conditions imply that every accumulation point is a minimizer. The majorization and touching conditions alone do not establish this conclusion for arbitrary surrogate schemes.

To construct a surrogate for (19.20), use the quadratic upper bound (14.19) supplied by the L-Lipschitz gradient of f:

$$f(x) \leqslant f(x') + \langle \nabla f(x'),\, x - x' \rangle + \frac{L}{2}\|x - x'\|^2,$$

so that for $0 < \tau < \frac{1}{L}$, the function

$$\mathcal{E}(x, x') \stackrel{\text{def.}}{=} f(x') + \langle \nabla f(x'),\, x - x' \rangle + \frac{1}{2\tau}\|x - x'\|^2 + g(x) \tag{19.24}$$

satisfies the surrogate conditions (19.23). Minimizing this surrogate is exactly a proximal-gradient step.

Proposition 19.13. *The update* (19.22) *for the surrogate* (19.24) *is exactly* (19.21)*.*

Proof. This follows from the fact that

$$\langle \nabla f(x'),\, x - x' \rangle + \frac{1}{2\tau}\|x - x'\|^2 = \frac{1}{2\tau}\|x - (x' - \tau \nabla f(x'))\|^2 + \text{cst}.$$

□

Convergence of FB. The majorization argument uses $\tau \leqslant 1/L$, whereas convergence of the iterates holds on the larger interval $0 < \tau < 2/L$.

Theorem 19.14. *Let f be convex with L-Lipschitz gradient and $g \in \Gamma_0(\mathcal{H})$, and assume* $\operatorname{argmin}(f + g) \neq \emptyset$*. For a fixed step $0 < \tau < 2/L$, the forward–backward iterates converge to a minimizer. If $0 < \tau \leqslant 1/L$, then*

$$(f + g)(x^\ell) - \min(f + g) \leqslant \frac{\|x^0 - x^\star\|^2}{2\tau\ell}, \qquad \ell \geqslant 1.$$

If f is μ-strongly convex, the iterates converge linearly. In particular, for $0 < \tau \leqslant 1/L$, $\|x^{\ell+1} - x^\star\| \leqslant (1 - \tau\mu)\|x^\ell - x^\star\|$.

Taking $g = \iota_C$ in (19.21) recovers projected gradient descent (19.5), because the proximal map of an indicator is the projection onto the constraint set.

Applying (19.21) efficiently requires a tractable proximal map $\operatorname{Prox}_{\tau g}$. A closed form is available for several useful penalties, including the ℓ^1 penalty used by ISTA in Section 10.3.3.

19.5 Primal-Dual Algorithms

Convex duality, developed in Section 18.3, suggests new algorithms and allows existing methods to be applied to dual formulations.

19.5.1 Forward–Backward Splitting on the Dual

As a first example, apply a familiar method to the Fenchel–Rockafellar problem (18.12):

$$p^\star = \inf_x f(x) + g(Ax), \tag{19.25}$$

Assume additionally that f is μ-strongly convex, and, for simplicity, that f and g are finite and continuous everywhere. Even if f is smooth, applying forward–backward splitting directly to the primal problem requires $\operatorname{Prox}_{\tau(g \circ A)}$. This map can be difficult to evaluate despite having a simple $\operatorname{Prox}_{\tau g}$. Proposition 19.8 gives an explicit formula in the special case $AA^* = \mathrm{Id}$.

Example 19.15 (TV denoising). Chambolle [9] developed this approach for total variation denoising:

$$\min_x \frac{1}{2}\|y - x\|^2 + \lambda\|\nabla x\|_{1,2} \tag{19.26}$$

where $\nabla x \in \mathbb{R}^{N\times d}$ is the discrete gradient of a signal ($d = 1$) or image ($d = 2$) with N spatial samples. The mixed ℓ^1–ℓ^2 norm sums the Euclidean norms of the vectors $v_i \in \mathbb{R}^d$:

$$\|v\|_{1,2} \stackrel{\text{def.}}{=} \sum_i \|v_i\|.$$

Here

$$f = \frac{1}{2}\|\cdot - y\|^2 \quad \text{and} \quad g = \lambda\|\cdot\|_{1,2}$$

Thus f is strongly convex with constant $\mu = 1$, and $A = \nabla$ is the discrete gradient operator.

Example 19.16 (Support Vector Machine). We consider a supervised classification problem with data $(a_i \in \mathbb{R}^p, y_i \in \{-1, +1\})_{i=1}^n$. With regularization parameter $\lambda > 0$, the support vector machine objective is

$$\min_{x\in\mathbb{R}^p} \sum_i \ell(-y_i\langle x,\, a_i\rangle) + \frac{\lambda}{2}\|x\|^2 = \frac{\lambda}{2}\|x\|^2 + g(Ax) \tag{19.27}$$

where $\ell(s) = \max(0, s + 1)$ is the hinge loss and the notation is $A_{i,j} = -y_i a_i[j]$ and $g(u) = \sum_i \ell(u_i)$. It corresponds to the split (19.25) using $f(x) = \frac{\lambda}{2}\|x\|^2$.

Continuity supplies the qualification for Fenchel–Rockafellar duality, Theorem 18.12, giving

$$p^\star = \sup_u \; -g^*(u) - f^*(-A^*u). \tag{19.28}$$

Strong convexity of f with constant μ makes the conjugate f^* differentiable with $1/\mu$-Lipschitz gradient. Apply forward–backward splitting (19.21) to the negative dual objective:

$$u^{(\ell+1)} = \operatorname{Prox}_{\tau g^*}\left(u^{(\ell)} + \tau A\nabla f^*(-A^*u^{(\ell)})\right).$$

Choose a fixed step $0 < \tau < 2/L$, where $L > 0$ bounds the Lipschitz constant of the smooth dual gradient; $L = \|A\|^2/\mu$ is valid when $A \neq 0$. The fixed-step assumption ensures that Theorem 19.14 applies.

Once a dual maximizer $u^\star$ is computed, equivalently a minimizer of the negative dual objective, the primal-dual relations (18.18) recover the unique primal minimizer:

$$-A^*u^\star \in \partial f(x^\star) \quad \Leftrightarrow \quad x^\star \in (\partial f)^{-1}(-A^*u^\star) = \partial f^*(-A^*u^\star) \quad \Leftrightarrow \quad x^\star = \nabla f^*(-A^*u^\star)$$

using differentiability of f^*.

Example 19.17 (TV denoising). In the particular case of the TV denoising problem (19.26), one has

$$g^* = \iota_{\|\cdot\|_{\infty,2}\leqslant\lambda} \quad \text{where} \quad \|v\|_{\infty,2} \stackrel{\text{def.}}{=} \max_i \|v_i\|$$

$$f^\star(h) = \frac{1}{2}\|h\|^2 + \langle h,\, y\rangle,$$

The dual, expressed as a minimization, is therefore

$$\min_{\|v\|_{\infty,2}\leqslant\lambda} \frac{1}{2}\|-\nabla^*v + y\|^2.$$

Projected gradient descent, a special case of forward–backward splitting, gives

$$\operatorname{Prox}_{\tau g^*}(u) = \left(\frac{u_i}{\max\{1, \|u_i\|/\lambda\}}\right)_i \quad \text{and} \quad \nabla f^\star(h) = h + y.$$

Furthermore, one has $\mu = 1$ and $A^*A = -\Delta$ for the usual negative-semidefinite discrete Laplacian. With unit grid spacing, standard forward differences satisfy $\|A\|^2 \leqslant 4d$; equality depends on the boundary conditions and grid size.

Example 19.18 (Support Vector Machine). For the SVM problem (19.27), one has

$$\forall\, u \in \mathbb{R}^n, \quad g^\star(u) = \sum_i \ell^\star(u_i)$$

where

$$\ell^\star(t) = \sup_s\, ts - \max(0, s+1) = -t + \sup_s\, ts - \max(0, s) = -t + \iota_{[0,1]}(t)$$

because the conjugate of $\max(0, s)$ is the indicator of its subdifferential at zero, namely $[0, 1]$. Also, one has $f(x) = \frac{\lambda}{2}\|x\|^2$ so that

$$f^\star(z) = \lambda(\|\cdot\|^2/2)(z/\lambda) = \lambda \frac{\|z/\lambda\|^2}{2} = \frac{\|z\|^2}{2\lambda}.$$

The dual problem reads

$$\min_{0 \leqslant v \leqslant 1} \; -\langle \mathbb{1}, v\rangle + \frac{\|A^\top v\|^2}{2\lambda}$$

which can be solved by forward–backward splitting, here equivalent to projected gradient descent. One thus has

$$\mathrm{Prox}_{\tau g^*}(u) = \underset{v \in [0,1]^n}{\mathrm{argmin}} \left\{ \frac{1}{2}\|u - v\|^2 - \tau\langle \mathbb{1}, v\rangle \right\} = \mathrm{Proj}_{[0,1]^n}(u + \tau\mathbb{1}) = \min\{\max\{u + \tau\mathbb{1}, 0\}, \mathbb{1}\}.$$

19.5.2 Douglas–Rachford Splitting

We consider here the structured minimization problem

$$\min_{x \in \mathbb{R}^p} f(x) + g(x), \tag{19.29}$$

Unlike the setting of Section 19.4.2, f need not be smooth. We assume that the proximal maps of both f and g are tractable.

Example 19.19 (Constrained Lasso). Douglas–Rachford splitting applies, for example, to the noiseless constrained Lasso problem (10.11):

$$\min_{Ax=y} \|x\|_1$$

Take $f = \iota_{C_y}$ with $C_y \overset{\text{def.}}{=} \{x \;;\; Ax = y\}$, and $g = \|\cdot\|_1$. As noted in Section 10.3.1, this problem is equivalent to a linear program. The proximal map of g is soft thresholding (19.12). The proximal map of f is projection onto the affine space C_y, obtained by solving the linear system in (18.11). This projection is particularly efficient for inpainting and for deconvolution operators diagonalized by the Fourier transform.

The Douglas–Rachford iterations read

$$\tilde{x}^{(\ell+1)} \overset{\text{def.}}{=} \left(1 - \frac{\mu}{2}\right)\tilde{x}^{(\ell)} + \frac{\mu}{2}\,\mathrm{rProx}_{\tau g}(\mathrm{rProx}_{\tau f}(\tilde{x}^{(\ell)})) \quad \text{and} \quad x^{(\ell+1)} \overset{\text{def.}}{=} \mathrm{Prox}_{\tau f}(\tilde{x}^{(\ell+1)}), \tag{19.30}$$

where the reflected proximal map is

$$\mathrm{rProx}_{\tau f}(x) = 2\,\mathrm{Prox}_{\tau f}(x) - x.$$

Assume $0 \in \partial f(x^\star) + \partial g(x^\star)$ for some $x^\star$ (for example, a minimizer exists and the relative interiors of the domains intersect). Then, for any $\tau > 0$, any $0 < \mu < 2$, and any $\tilde{x}_0$, the shadow iterates $x^{(\ell)}$ converge to a minimizer of $f + g$, which may depend on the initialization.

Interchanging f and g gives another valid iteration.

More than two functions. A product-space formulation treats K functions $(f_k)_k$ symmetrically:

$$\min_x \sum_k f_k(x) = \min_{X=(x_1,\ldots,x_K)} f(X) + g(X) \quad \text{where} \quad f(X) = \sum_k f_k(x_k) \quad \text{and} \quad g(X) = \iota_\Delta(X)$$

where $\Delta = \{X \;;\; x_1 = \ldots = x_K\}$ is the diagonal. The proximal operator of g is

$$\mathrm{Prox}_{\tau g}(X) = \mathrm{Proj}_\Delta(X) = (\bar{x}, \ldots, \bar{x}) \quad \text{where} \quad \bar{x} = \frac{1}{K}\sum_k x_k$$

The proximal map of f separates into the maps of $(f_k)_k$ by (19.14). Douglas–Rachford then applies through (19.30).

Handling a linear operator. For an objective of the form (19.25), introduce an auxiliary variable:

$$\inf_x f_1(x) + f_2(Ax) = \inf_{z=(x,y)} f(z) + g(z) \quad \text{where} \quad \begin{cases} f(z) = f_1(x) + f_2(y) \\ g(z) = \iota_C(x,y), \end{cases}$$

where $C = \{(x,y)\ ;\ Ax = y\}$. Douglas–Rachford splitting applies because the proximal map of f separates into those of f_1 and f_2 by (19.14). The proximal map of g is projection onto C. The following proposition gives two equivalent formulas, allowing a choice between systems in the input and output spaces of A.

Proposition 19.20. *One has*

$$\mathrm{Proj}_C(x,y) = (x - A^*\tilde{y}, y + \tilde{y}) = (\tilde{x}, A\tilde{x}) \quad \textit{where} \quad \begin{cases} \tilde{y} \overset{\text{def.}}{=} (\mathrm{Id}_P + AA^*)^{-1}(Ax - y) \\ \tilde{x} \overset{\text{def.}}{=} (\mathrm{Id}_N + A^*A)^{-1}(A^*y + x). \end{cases} \tag{19.31}$$

Proof. Minimizing $\frac{1}{2}\|z - x\|^2 + \frac{1}{2}\|Az - y\|^2$ gives $(\mathrm{Id}_N + A^*A)z = x + A^*y$, hence $z = \tilde{x}$. Alternatively, the Lagrange multiplier for $Az = w$ satisfies $(\mathrm{Id}_P + AA^*)\tilde{y} = Ax - y$, with $z = x - A^*\tilde{y}$ and $w = y + \tilde{y}$. □

Remark 19.21 (Inverting structured linear operators). Projection computations often require the inverse of a matrix of the form AA^*, A^*A, $\mathrm{Id}_P + AA^*$ or $\mathrm{Id}_N + A^*A$ (see for instance (19.31)). The structure of the operator can make this inexpensive. For inpainting, AA^* is diagonal. For deconvolution or partial Fourier measurements such as MRI, A^*A is diagonalized by the Fourier transform. When inversion is expensive, primal-dual methods can replace it by applications of A and A^*, trading cheaper iterations for a potentially larger iteration count.

19.5.3 Alternating Direction Method of Multipliers

Douglas–Rachford splitting, introduced in Section 19.5.2, can minimize $f + g \circ A$ without strong convexity of f. Its direct application requires the proximal maps $\mathrm{Prox}_{\tau f}$ and $\mathrm{Prox}_{\tau g \circ A}$. The alternating direction method of multipliers (ADMM) is related to Douglas–Rachford splitting on the dual problem (19.28). This formulation uses $\mathrm{Prox}_{\tau g^*}$, computable from $\mathrm{Prox}_{\tau g}$ by Moreau decomposition, and $\mathrm{Prox}_{\tau(f^* \circ A^*)}$. The primal updates derived below express these operations as alternating minimizations.

For $A \in \mathbb{R}^{n \times p}$, introduce $y = Ax$ as in (18.13) and scale the multiplier by $\gamma > 0$. Define

$$\mathcal{L}((x,y),z) \overset{\text{def.}}{=} f(x) + g(y) + \gamma\langle z,\ Ax - y\rangle.$$

The primal problem becomes

$$\inf_{x \in \mathbb{R}^p} f(x) + g(Ax) = \inf_{y = Ax} f(x) + g(y) = \inf_{x \in \mathbb{R}^p, y \in \mathbb{R}^n} \sup_{z \in \mathbb{R}^n} \mathcal{L}((x,y),z). \tag{19.32}$$

Add a quadratic penalty for violating $Ax = y$ to obtain the augmented Lagrangian

$$\mathcal{L}_\gamma((x,y),z) \overset{\text{def.}}{=} \mathcal{L}((x,y),z) + \frac{\gamma}{2}\|Ax - y\|^2 = f(x) + g(y) + \gamma\langle z,\ Ax - y\rangle + \frac{\gamma}{2}\|Ax - y\|^2.$$

Replacing $\mathcal{L}$ by $\mathcal{L}_\gamma$ in (19.32) preserves the constrained problem: the supremum over z enforces $Ax = y$, where the added penalty vanishes.

The ADMM method then updates

$$y^{(\ell+1)} \overset{\text{def.}}{=} \underset{y}{\mathrm{argmin}}\ \mathcal{L}_\gamma((x^{(\ell)}, y), z^{(\ell)}), \tag{19.33}$$

$$x^{(\ell+1)} \overset{\text{def.}}{=} \underset{x}{\mathrm{argmin}}\ \mathcal{L}_\gamma((x, y^{(\ell+1)}), z^{(\ell)}), \tag{19.34}$$

$$z^{(\ell+1)} \overset{\text{def.}}{=} z^{(\ell)} + Ax^{(\ell+1)} - y^{(\ell+1)}. \tag{19.35}$$

For comparison, the method of multipliers jointly minimizes the augmented Lagrangian in x and y, then updates the multiplier:

$$(x^{(\ell+1)}, y^{(\ell+1)}) \overset{\text{def.}}{=} \underset{x,y}{\mathrm{argmin}}\ \mathcal{L}_\gamma((x,y), z^{(\ell)}),$$

$$z^{(\ell+1)} \overset{\text{def.}}{=} z^{(\ell)} + Ax^{(\ell+1)} - y^{(\ell+1)},$$

ADMM instead updates x and y alternately, which often makes the individual subproblems tractable.

Step (19.35) is gradient ascent in the multiplier z with step $1/\gamma$. Assume for this calculation that f is differentiable. Combining the x and z updates enforces $\nabla f(x^{(\ell+1)}) + \gamma A^\top z^{(\ell+1)} = 0$, the stationarity relation in x. Indeed, the optimality condition for (19.34) reads

$$0 = \nabla_x \mathcal{L}_\gamma((x^{(\ell+1)}, y^{(\ell+1)}), z^{(\ell)}) = \nabla f(x^{(\ell+1)}) + \gamma A^\top z^{(\ell)} + \gamma A^\top(Ax^{(\ell+1)} - y^{(\ell+1)}) = \nabla f(x^{(\ell+1)}) + \gamma A^\top z^{(\ell+1)}.$$

Step (19.33) is a proximal step, since

$$\begin{aligned} y^{(\ell+1)} &= \operatorname*{argmin}_y g(y) + \gamma\langle z^{(\ell)}, Ax^{(\ell)} - y\rangle + \frac{\gamma}{2}\|Ax^{(\ell)} - y\|^2 \\ &= \operatorname*{argmin}_y \frac{1}{2}\|y - Ax^{(\ell)} - z^{(\ell)}\|^2 + \frac{1}{\gamma}g(y) = \operatorname{Prox}_{g/\gamma}(Ax^{(\ell)} + z^{(\ell)}). \end{aligned}$$

Step (19.34) uses a quadratic penalty composed with A:

$$x^{(\ell+1)} \in \operatorname*{argmin}_x \left\{\frac{1}{2}\|Ax - y^{(\ell+1)} + z^{(\ell)}\|^2 + \frac{1}{\gamma}f(x)\right\} \overset{\text{def.}}{=} \operatorname{Prox}^A_{f/\gamma}(y^{(\ell+1)} - z^{(\ell)}).$$

Here $\operatorname{Prox}^A_{f/\gamma}(u)$ denotes the minimizer set of $\frac{1}{2}\|Ax - u\|^2 + f(x)/\gamma$; it can be set-valued when A is not injective. Under the usual qualification for this subproblem, if

$$p = \operatorname{Prox}_{\gamma(f^*\circ A^*)}(\gamma u),$$

then every minimizing x satisfies $Ax = u - p/\gamma$ and $A^* p \in \partial f(x)$. If A has full column rank, this gives

$$\operatorname{Prox}^A_{f/\gamma}(u) = A^+\left(u - \frac{1}{\gamma}\operatorname{Prox}_{\gamma(f^*\circ A^*)}(\gamma u)\right).$$

For noninjective A, applying A^+ alone need not recover a minimizer; the component in $\ker(A)$ must also minimize f.

Example 19.22 (Constrained TV recovery). We consider the problem of minimizing a TV norm under affine constraints

$$\min_{Bx=y} \|\nabla x\|_1$$

which is of the form $f + g \circ A$ when defining $A = \nabla$, $g = \|\cdot\|_1$ and $f = \iota_{B\cdot=y}$. The proximal map of g is block soft thresholding for isotropic TV (componentwise soft thresholding for anisotropic TV), while the "twisted" proximal operator is

$$\operatorname{Prox}^A_{f/\gamma}(u) = \operatorname*{argmin}_{Bx=y} \frac{1}{2}\|Ax - u\|^2 = \operatorname*{argmin}_x \sup_w \left\{\frac{1}{2}\|\nabla x - u\|^2 + \langle w, Bx - y\rangle\right\}$$

Introducing the multiplier w for $Bx = y$ gives first-order conditions for $(x^\star, w^\star)$ in the form of a linear system:

$$\begin{pmatrix} \nabla^\top\nabla & B^\top \\ B & 0 \end{pmatrix}\begin{pmatrix} x^\star \\ w^\star \end{pmatrix} = \begin{pmatrix} \nabla^\top u \\ y \end{pmatrix}$$

Here $\Delta = -\nabla^\top\nabla$ is the discrete Laplacian.

19.5.4 Primal-Dual Splitting

If neither $\operatorname{Prox}_{\tau g\circ A}$ nor $\operatorname{Prox}_{\tau f^\star\circ A^*}$ is tractable, a primal-dual method can work directly with the saddle problem. Its efficiency depends on the available operator evaluations, proximal maps, and conditioning. For the structured problem (19.25), biconjugacy $g = (g^*)^*$ gives the saddle formulation

$$\inf_x f(x) + g(Ax) = \inf_x f(x) + \sup_u \langle Ax, u\rangle - g^*(u) = \sup_u \inf_x f(x) + \langle Ax, u\rangle - g^*(u). \tag{19.36}$$

We assume a primal-dual saddle point exists, so strong duality holds; strong convexity of f is not required.

Example 19.23 (Total variation regularization of inverse problems). For example, TV regularization for the inverse problem $\mathcal{K}x = y$ leads to

$$\min_x \frac{1}{2}\|y - \mathcal{K}x\|^2 + \lambda\|\nabla x\|_{1,2},$$

where $A = \nabla$, $f(x) = \frac{1}{2}\|y - \mathcal{K}x\|^2$, and $g = \lambda\|\cdot\|_{1,2}$. For this split, the proximal map of f requires solving a system involving $\mathrm{Id} + \tau\mathcal{K}^*\mathcal{K}$, while the proximal map of g^* is a pointwise projection. If that linear solve is costly, use instead $Ax = (\mathcal{K}x, \nabla x)$, $f = 0$, and $g(z, v) = \frac{1}{2}\|y - z\|^2 + \lambda\|v\|_{1,2}$. The two blocks of g^* then have simple proximal maps, so each iteration uses only $\mathcal{K}, \mathcal{K}^*, \nabla$, and ∇^*.

A standard primal-dual algorithm, described in [11], initializes $\tilde{x}^0 = x^0$ and updates

$$\begin{aligned}
z^{(\ell+1)} &= \mathrm{Prox}_{\sigma g^*}\big(z^{(\ell)} + \sigma A\tilde{x}^{(\ell)}\big),\\
x^{(\ell+1)} &= \mathrm{Prox}_{\tau f}\big(x^{(\ell)} - \tau A^* z^{(\ell+1)}\big),\\
\tilde{x}^{(\ell+1)} &= x^{(\ell+1)} + \theta(x^{(\ell+1)} - x^{(\ell)}).
\end{aligned}$$

For $\theta = 1$, $\sigma, \tau > 0$, and $\sigma\tau\|A\|^2 < 1$, the primal and dual iterates converge to a saddle point, provided one exists. The value $\theta = 0$ is not covered by this general convergence guarantee.

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